

Exact Arithmetic in Polynomial Optimization: Values, Optimizers, and Certificates

Abstract

A polynomial optimization problem can be asked for approximate values, approximate optimizers, exact decisions about optimizers, or exact descriptions of optimizers and optimality certificates. For convex quadratic programs all four are available in polynomial time. We study higher degrees, for explicit rational input in the bit model, under convexity hypotheses stated for each result. For globally convex polynomials of unary degree, polynomial-time value approximation is known; we add an effective error bound and approximation of one fixed optimizer to precision 2^{-q} in time polynomial in the input length L , the degree, and q . Convexity only on a polytope suffices for cubics, whereas for quartics convex only on a box a single constant-accuracy optimizer query can decide PosSLP, the sign problem of integer circuits. For explicit globally strongly convex polynomials of unary degree, every sign or equality test of a polynomial at the minimizer reduces to one PosSLP instance. For quartics with a checkable strict convexity certificate the order tests are PosSLP-complete, also for coordinates of a rational minimizer with known minimum value; equality has only the upper bound. For globally strongly convex quartics over rational polyhedra, exact comparison lies in $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$, structured box and network-flow problems reduce to one PosSLP instance per predicate, and small nonlinear dimension admits ordinary fixed-parameter algorithms. The coordinates of points isolated by rational convex polynomial inequalities are exactly the real algebraic numbers with one real conjugate. For certified strongly convex quartics of size polynomial in the dimension, the degree of such coordinates, the bit length of rational optimizers and of interior Gram matrices, and the degree of the least coefficient field of sum-of-squares certificates can grow exponentially with the dimension, although each exhibited family has a polynomial-size description in another format, such as a shared rational circuit or a root circuit. Further results bound common-field algebraic witnesses for nonconvex quadratic constraints with few independent Hessians. For second-order cone systems over one explicitly represented real number field, exact feasibility and witness recovery take polynomial time when both the integer dimension and the dimension of the span of the continuous Hessians of the squared cone residuals are fixed. For cubics on product boxes that are convex in the residual variables but possibly nonconvex in a core of dimension k , one random tilt of the core yields the full selected optimizer of the tilted instance at every precision, correct on every draw, with expected work $k^{O(k)}(1 + \Lambda/\sigma)^k \text{poly}(L + q)$, where σ is the tilt width and Λ is an upper bound on the coordinate second derivatives in the core.

Keywords. polynomial optimization; convex optimization; exact arithmetic; PosSLP; sums of squares; algebraic degree; bit complexity; mixed-integer nonlinear programming.

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1 Introduction

Solvers for mixed-integer nonlinear programs repeatedly solve convex subproblems and act on the answers. The optimal value of a relaxation decides whether a node of the search tree can be pruned. The location of an optimizer guides branching and rounding, and whether a bound or constraint is active at that optimizer informs the choice of branching variables. A certificate of optimality is what an independent checker verifies. In floating-point arithmetic all of these quantities are approximations. Exact and proof-producing solvers need exact versions of them. For linear and convex quadratic subproblems the exact versions are available: a convex quadratic program has a rational optimizer of polynomial bit length, and it can be computed in polynomial time [71]. From degree four on, a globally convex objective can have a unique optimizer that is irrational, and the different outputs come apart.

This paper studies polynomial optimization problems with explicitly encoded rational data. Most results assume convexity of one of three kinds, stated in each theorem: global convexity or strong convexity of the objective, convexity on the feasible set only, or, in Section 11, convexity in a subset of the variables. Appendix L also treats nonconvex quadratic constraints and cone constraints with explicit algebraic data. We ask which of the following outputs can be computed or represented efficiently, and what makes the others hard:

- (i) the optimal value to a requested precision 2^{-q} ;
- (ii) an optimizer to precision 2^{-q} , preferably the same optimizer at every precision;
- (iii) an exact decision about the optimizer, such as the sign of a given polynomial at the optimizer, the comparison of the optimal value with a rational threshold, or whether a constraint is active;
- (iv) an exact description of the optimizer, or of a certificate of its optimality.

Section 2 defines these outputs, the input encodings, and the computational models. Throughout, L denotes the total binary length of the input, n the number of continuous variables, k a structural parameter specified in each statement, q the number of requested bits of precision, f the objective, and p the selected optimizer, which is the unique optimizer when the objective is strongly convex and the feasible set is convex.

Output (i) is well understood for globally convex polynomials. Slot, Steurer, and Wiedmer consider globally convex polynomials with rational coefficients written in binary and exponent vectors written in unary. On a nonempty rational polyhedron such a polynomial is either unbounded below or attains its minimum at a point of norm at most $2^{\text{poly}(L)}$ [111, Theorem 1.1], and its optimal value, when finite, can be approximated to objective gap 2^{-q} in time polynomial in the input length and q [111, Corollary 1.2]. The norm bound does not bound the bit length of the coordinates of an optimizer, which can be irrational. In arXiv version 2511.03440v1 of November 5, 2025, two questions about convex quartics are marked open: whether every rational polyhedron that contains a point with $f(x) \leq 0$ also contains such a rational point of polynomial bit length, and the complexity of deciding exactly whether such a point exists. All comparisons with that paper below refer to this version. For outputs (iii) and (iv) the arithmetic benchmark is the problem PosSLP [6]: given a division-free arithmetic circuit with constants 0 and 1, decide whether the integer it computes is positive. Such an integer can have exponentially many bits. PosSLP lies in the counting hierarchy [6, Theorem 1.4], and the Square Root Sum problem is decidable in polynomial time with a PosSLP oracle [6, Section 1.4]. Whether PosSLP is in P, or is NP-hard, is open [22]. Exact feasibility of semidefinite programs is already at least as hard as PosSLP [113].

The four outputs form a ladder: each step up needs an ingredient that the step below does not supply. Our results identify these ingredients and the classes of problems in which they are available or provably missing.

- *From values to points.* An effective error bound is needed, that is, a bound on the distance to the optimal set in terms of the objective gap, with a constant computable from the input, together with a rule that selects one optimizer. Global convexity supplies such a bound at every degree, and convexity on a polytope supplies one for cubics. For quartics that are convex only on a box, a single constant-accuracy point already decides Square Root Sum or PosSLP (Section 1.1).
- *From points to exact decisions.* A separation bound for nonzero values is needed, together with a compact way to reach the precision that it demands. For explicit globally strongly convex objectives of unary degree, every exact comparison at the minimizer reduces to one PosSLP instance, and already for certified quartics the order comparisons are PosSLP-hard (Section 1.2). Linear constraints add a search for the active face, and integer variables add a discrete search (Section 1.3).
- *From exact decisions to exact descriptions.* Separate degree, height, and field arguments are needed, and the size of a description depends on its format. Expanded rational coordinates, coefficient lists of minimal polynomials, Gram matrices, and sum-of-squares certificates can be exponentially long in the dimension for certified inputs of polynomial size, while the exhibited families have short descriptions in compact formats such as circuits (Sections 1.5 and 1.6).

The central result is Theorem 5.1: for explicit quartics with a checkable strict convexity certificate, exact order comparisons of the minimum value, and of a minimizer coordinate, with a rational threshold are PosSLP-complete. Together with Theorem 4.4, it shows that for explicit globally strongly convex polynomials of unary degree, exact order comparisons at the minimizer are as hard as PosSLP and no harder; for equality only the upper bound is known. Table 1 summarizes the results, and Section 1.8 separates them from prior work.

1.1 From values to points

Global convexity gives an effective conversion from values to points. Theorem 3.4 treats an explicitly sparse rational polynomial f that is convex on $\mathbb{R}^n$, of degree at most D , and an arbitrary rational polyhedron P . In time polynomial in L and D one can decide whether P is empty and whether f is unbounded below on P . Otherwise the minimum f_* is attained on a set S , and one can compute rationals R and Γ of bit length $\text{poly}(L, D)$ such that the minimum-norm optimizer satisfies $\|p\| \leq R$ and

$$\text{dist}(x, S) \leq \Gamma (f(x) - f_*)^{1/D} \quad (x \in P, f(x) - f_* \leq 1). \quad (1)$$

Consequently this fixed optimizer can be approximated to precision 2^{-q} in time $\text{poly}(L, D, q)$. Nonunique optima, singular Hessians, and unbounded polyhedra are allowed. Global error bounds for convex polynomials are known [81, 82, 93, 119], and we do not claim the form of the exponent as new. Our addition is a constant that is computable from the input and has polynomial bit length on arbitrary rational polyhedra, together with the fixed-selector output that it makes possible. With a supplied strong convexity modulus the conversion from values to points is immediate; the theorem matters when the objective has flat directions or several minimizers.

Convexity on the feasible set alone suffices for cubics. Theorem 3.7 treats a cubic that is convex on a bounded rational polytope, possibly of lower dimension, but not necessarily on $\mathbb{R}^n$. It proves

(1) with exponent $1/4$ and a constant of polynomial bit length, and approximates both the optimizer set and its minimum-norm point in time $\text{poly}(L + q)$. A cubic that is convex on all of $\mathbb{R}^n$ is a quadratic, so the domain-convex case is the one that matters for degree three. The univariate cubic t^3 on $[0, 1]$ shows that no exponent larger than $1/3$ is possible.

For quartics, convexity on the feasible set is not enough. Section 3 constructs rational quartics that are convex on a box and have a unique minimizer, for which a single approximation of the minimizer to accuracy $1/4$ decides the Square Root Sum problem or PosSLP. For Square Root Sum the constructed quartics have all coefficients other than the constant term bounded by an absolute constant, and their interaction graphs, which join two variables when some monomial contains both, have treewidth at most two. A deterministic polynomial-time point algorithm for this class would therefore put the corresponding problem in P. Unconditionally, there are explicit box-convex quartics in $n + 1$ variables, easy to solve by other means, for which every error bound of the form (1) with exponent ϑ has a constant Γ with $\log_2 \Gamma \geq \vartheta 2^{n+2}$, and for which isotropic regularization approximates the optimizer to accuracy $1/2$ only with a parameter whose denominator has more than 2^{n+2} bits. These instances have input length $L = \Theta(n^2)$, so the two quantities are exponential in the dimension and of order $2^{\Omega(\sqrt{L})}$: superpolynomial, but not exponential, in L . The same section separates finer output contracts. For a strongly convex cubic on a box, deciding whether a designated coordinate of the optimizer equals its lower bound is Square Root Sum-hard, although Theorem 3.7 approximates the optimizer in polynomial time. There are also box-convex quartics for which some optimizer is easy to approximate, while approximating the minimum-norm optimizer to constant accuracy decides Square Root Sum.

The point problem also changes when part of the objective is nonconvex. Section 11 considers a cubic $F(v, z)$ on $[0, 1]^k \times [0, 1]^m$ that is convex in the residual variables z for each value of the core variables v , but possibly nonconvex in v . Residual values at rational cores are easy to compute, but approximations to an optimal core do not determine the optimal residual: in the cubic $v^2(2 - z)$ on the unit square, the minimum-norm residual optimizer is 0 at $v = 0$ and 1 at every $v > 0$. We perturb only the core, by a linear term $\gamma^\top v$ drawn once from the uniform law on a finite rational grid in $[-\sigma, \sigma]^k$. For every precision q , an evaluator returns the lexicographically first optimal core of the perturbed objective together with its minimum-norm residual optimizer, within 2^{-q} , and every returned point is correct, whatever the draw. The expected work is $k^{O(k)}(1 + \Lambda/\sigma)^k \text{poly}(L + q)$, where $\partial^2 F / \partial v_i^2 \leq \Lambda$ along each core coordinate on the product box, and one random factor governs all precisions. Beyond a certified approximation of the selected core, the proof certifies which core coordinates lie at their bounds, by comparing slightly tilted problems, certifies margins of the relevant minors by lattice basis reduction, and then completes the residual by regularization. For every fixed degree, the same section also shows the following. Certified values and the selected core are available under residual convexity on a product box, or under a supplied convexifier on a polytope that couples core and residual variables; a convexifier is a rational $\alpha \geq 0$ such that adding $\frac{\alpha}{2} \|v\|^2$ makes the objective convex there. A supplied convexifier also yields full points, provided that the convexified objective is convex on the polytope and has degree three, or is convex on all of $\mathbb{R}^{k+m}$. Quadratic objectives permit exact rational optimizer and value recovery, by a polynomial height bound and rational reconstruction (Corollary 11.4). A positive semidefinite quadratic plus positive even powers of rational affine forms gives a checkable subclass of the global convexifier hypothesis (Corollary 11.6). With $\alpha = 0$, that is, for objectives that are jointly convex, the expected work in these results is polynomial, with no factor $k^{O(k)}$ and no numerical dependence on σ . For residual quartics no full-point guarantee of this form is expected: such a guarantee, applied after adding a dummy core to the box-convex quartic reductions above, would give a Las Vegas algorithm with expected polynomial running time for Square Root Sum and for PosSLP (Remark 11.14).

1.2 From points to exact decisions

Exact decisions require arithmetic beyond approximation. Theorem 4.4 treats explicit polynomials f and h of unary degree together with a rational $\mu > 0$ such that $\nabla^2 f \succeq \mu I$ on $\mathbb{R}^n$; the curvature bound is either a promise or derived from a checked certificate. For each relation $\bowtie \in \{<, \leq, =, \neq, \geq, >\}$, a deterministic polynomial-time algorithm, which makes no oracle calls, constructs one integer arithmetic circuit whose value is positive exactly when $h(p) \bowtie 0$. No gap between $h(p)$ and zero is assumed. The construction obtains a starting point of polynomial bit length from convex value approximation, stores polynomially many exact Newton steps as a shared rational circuit, and uses real quantifier elimination to show that a nonzero value $h(p)$ has magnitude at least $2^{-2^{e(L)}}$ for a fixed polynomial e . The Newton iterates reach this precision because their error squares at each step. Shifted comparisons handle the six relations, and division is removed by tracking numerators and positive denominators. Section 4 proves the same reduction for observables at the unique zero of a globally strongly monotone polynomial map of unary degree whose Jacobian need not be symmetric. The gradients of certified quartics are such cubic maps, so order tests are PosSLP-hard already for these gradient maps and complete for the certified cubic-map class. Equality retains only the upper bound. Cubic maps are the first nontrivial case, because a strongly monotone quadratic map is affine.

The matching lower bound holds for explicitly certified quartics. A *full Hessian Gram* for a quartic f is a rational symmetric matrix A with $v^\top \nabla^2 f(x) v = (v, x \otimes v)^\top A (v, x \otimes v)$; when $A \succ 0$ it certifies global strong convexity and can be verified in polynomial time (Section 2). Theorem 5.1 proves that, for explicit rational quartics supplied with a positive definite full Hessian Gram, strict and weak comparisons of the minimum value with a rational threshold, and of a coordinate of the minimizer with a rational threshold, are PosSLP-complete under polynomial-time many-one reductions. For the value comparisons the hard instances have nonzero minima. In the same class, nonnegativity, membership in the cone of real sums of squares, and existence of a positive definite rational polynomial Gram matrix are PosSLP-complete, and membership in the cone of rational sums of squares is PosSLP-hard. Equality tests have the upper bound of Theorem 4.4 and no matching lower bound. Section 5 shows further that coordinate order comparisons remain PosSLP-complete when the minimizer is rational and lies in $[-1, 1]^n$, the minimum is known to be zero, and $n + 1$ short rational square factors are supplied. Rationality of the minimizer is then a promise of the problem, not a property the reduction recognizes.

In particular, deciding whether $f(x) \leq 0$ has a real solution, that is, whether $f_* \leq 0$, is PosSLP-complete for certified quartics. The exact feasibility question for convex quartics that is marked open in the arXiv version of [111] is therefore PosSLP-hard, already without constraints. Corollary 5.9 gives the same hardness over a halfspace and over the unit cube, and for certified quartics on arbitrary rational polyhedra Section 6 gives the upper bound $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$. For convex quartics that are not strongly convex, no upper bound relative to PosSLP is known.

The main tool of the lower bounds is Lemma 7.6. Let rational quadratics $r_1, \dots, r_m$ vanish at a point $a \in \mathbb{R}^n$ with well-conditioned gradients there, and let a rational quadratic G vanish at a , have a positive definite quadratic part, and have a sufficiently small gradient at a . Then

$$F = \left(\frac{G}{t\nu}\right)^2 + \sum_{j=1}^m \left(\frac{r_j}{\nu}\right)^2$$

is a rational sum of $m + 1$ squares, has unique zero a , satisfies $\nabla^2 F \succeq \frac{3}{2}I$ on $\mathbb{R}^n$, and has a positive definite rational full Hessian Gram, for suitable rationals t and ν determined by explicit quantitative bounds on the data. The Gram matrix is computed directly from the rational square factors: canonical Hessian Gram matrices of quadratic squares are exactly covariant under translation, so

once the factors are fixed, the certificate requires no further approximation of the point a , which need not even be known. Squaring the residuals alone does not give convexity: $x^2 + x^4 + \varepsilon(y - x^2)^2$ has second derivative -2 in x at $(0, 1/\varepsilon)$. Section 5 applies the lemma to two simulations of integer arithmetic. The first uses cube roots of numbers near one, in which an integer V is encoded by the leading term $V\delta^d$ of a tiny analytic signal. The second uses products of rational unit quaternions, which keeps every coordinate of the realized point rational.

Section 4 also proves a compilation result that several constrained results use. Consider an integer circuit of size S with Boolean inputs, gates $+$, $-$, $\times$, and threshold gates $v \mapsto [v > 0]$, whose output is Boolean. In polynomial time it can be converted into a division-free integer circuit of size $O(S^2)$, without threshold gates, whose value is positive exactly when the original output is one. Each threshold gate is replaced by a rational approximation of the sign function: scaled maps $x \mapsto 2\sigma x/(\sigma + x^2)$ first compress the range, and the inverse Newton iteration $x \mapsto 2x/(1 + x^2)$ then refines the sign. The exact gate values are integers of absolute value at most 2^{2^S} , so distinct values are at least one apart, and a single error budget keeps every approximation within $1/4$ of its exact value; numerators with positive denominators remove the divisions. A padded interpreter for query descriptions extends the construction to every deterministic polynomial-time computation with a PosSLP oracle, which therefore reduces to one PosSLP instance (Theorem 4.10). We use this to state deterministic oracle algorithms as single-instance reductions. The construction compiles one output bit at a time and does not apply to nondeterministic or randomized oracle algorithms.

1.3 Constraints and integer variables

With linear constraints the optimizer lies on an unknown face, and finding that face is a search problem separate from the arithmetic. Section 6 treats globally strongly convex quartics on rational polyhedra and, where stated, strongly monotone cubic variational inequalities. For an arbitrary rational polyhedron, exact comparison of the value, a coordinate, or an explicit observable lies in $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$: the full active set is a unique certificate, and the normal-cone condition is verified by a linear program whose objective is given by a circuit. Whether the comparison lies in P^{PosSLP} , equivalently whether one PosSLP instance suffices, is open. The section removes the search in four ways.

- For globally strongly convex objectives of unary degree, a supplied lower bound on all nonzero optimal slacks lets ordinary approximation recover the active set, after which one PosSLP instance decides each relation (Proposition 4.9 in Section 4).
- If the cubic and quartic parts of f depend on only k linear forms, exact comparison and recovery of the full active set take ordinary deterministic time $F(k)L^C$, with $F(k) = 2^{O(k)}$ and an absolute constant C . The quadratic part, the number of constraints, and their rank are unrestricted. The optimizer is returned as an affine image of the unique stationary point of a strongly convex quartic in at most k variables.
- For objectives of fixed degree on boxes whose Hessian interactions form a forest, have nonpositive off-diagonal entries, or have a positive definite comparison matrix, and on network-flow polyhedra with separable objectives, exact constrained Newton steps can be executed with polynomially many arithmetic operations. Combined with the compilation theorem, each exact predicate about the optimizer, and each polynomial-size Boolean combination of such predicates, reduces to one PosSLP instance (Corollary 6.15). The full active set is recovered by a polynomial list of such instances, prepared without adaptivity.

- If the constraint normals have rank r , a Las Vegas algorithm with a PosSLP oracle runs in expected time $(r + 1)^{O(r)} L^C$.

Integer variables add a second search. Let the quartic objective be strongly convex in all variables jointly, and let t variables be integer. An ordinary algorithm running in time $2^{O(t \log(t+1))} L^C$ lists candidate integer blocks that include every optimal block when the continuous fibers are unconstrained; there are at most 2^t optimal blocks. Exact selection among the candidates then uses PosSLP queries that can be prepared in advance. Under arbitrary mixed linear constraints the list takes ordinary time $a(t) L^C$ for a computable function a , and the methods above combine to give an ordinary algorithm in time $F(k + t) L^C$, or a Las Vegas algorithm in expected time $F(t, r) L^C$ with a PosSLP oracle. Thus discrete search needs only ordinary precision, and exact arithmetic is needed only to select among the candidates. This selection cannot be avoided: deciding the value of a single binary variable at the unique optimum remains PosSLP-hard even when the optimal value is known, the optimizer is rational and bounded, and the continuous constraints have rank one.

1.4 Quadratic and algebraic arithmetic interfaces

Quadratic constraints change the arithmetic even when they are convex. Appendix J studies positive semidefinite quadratic constraints whose Hessians span a space of dimension h . The optimal value and a canonical optimizer lie in one number field of degree at most $\max_{0 \leq s \leq \min(h, n)} 2^s \binom{n}{s}$, and the degree bound is sharp. For fixed h , exact feasibility and common-field point recovery take ordinary polynomial time, but expanded minimal-polynomial output admits no bound $F(h) L^C$ with an absolute exponent C . The rational-witness boundary is also sharp: nonempty systems of Hessian span at most two have rational points of polynomial bit length, whereas three ellipsoids can intersect at a single irrational point. The appendix also treats one-field optimizer certificates, rational positive-aggregate infeasibility certificates, and linear and convex quadratic programs with rational matrices and affine data in an explicitly represented real number field. These results explain which parts of the higher-degree account already arise from quadratic constraints.

Appendix L extends the description and input questions in two directions. For rational quadratic constraints that need not be convex, with Hessian span h , a feasible point has a common-field algebraic description of degree and total length $L^{O(h+1)}$. This feasibility bound follows from the quadratic-map sampling theorem of Grigoriev and Pasechnik [48, Theorem 1.2] and an elementary argument on the minimal face of the affine constraints. The appendix controls boxed optimal values, finite infima that need not be attained, and minimum-norm optimizers when attainment holds. It keeps affine constraints in the optimization formulas: an affine inequality that is inactive at an optimizer can still change the global optimal value when deleted. For fixed h , feasibility belongs to NP, and the status, finite value, and an attained optimizer can be recovered in FP^{NP} . These are exact arithmetic guarantees, rather than ordinary polynomial-time algorithms for nonconvex optimization.

The second extension allows cone data in one explicitly represented number field $K \subseteq \mathbb{R}$, with its real embedding specified. With fixed integer dimension t and a fixed dimension h for the span over K of the continuous Hessians of the squared cone residuals, exact feasibility and witness recovery take ordinary polynomial time, without a supplied box or a Slater condition. The continuous witness is written in the joint field generated by K and its coordinates. For a continuous system its minimum-norm point has absolute field degree and total description length $L^{O(h+1)}$; in the mixed-integer case the algorithm first finds an integer assignment and then applies this result to its exact continuous fiber. Known algebraic thresholds can therefore be used exactly in a later cone problem. The extension concerns explicit algebraic input, sign and rounding operations, and

common-field output; its sampling and fixed-dimensional integer ingredients are prior methods. Exact second-order cone feasibility with unrestricted h is PosSLP-hard (Remark 2.18); in those gate constructions h can grow with the circuit size. The polynomial-time bound here fixes both t and h .

1.5 Exact optimizers

Exact optimizers can be irrational even in very small, well-conditioned instances. With

$$A = 12599x^2 - 10000xy + 7937y^2 - 15874x - 12599y + 20000,$$

the integer quartic

$$F(x, y) = A^2 + 10000[(x^2 - y)^2 + (y^2 - 2x)^2]$$

satisfies $\nabla^2 F \succeq 4124 I$ on $\mathbb{R}^2$, has a positive definite integer full Hessian Gram, and has the single zero $(2^{1/3}, 2^{2/3})$ (Proposition 7.2). Hence the constraint $F \leq 0$ is satisfiable but has no rational solution. This answers negatively the rational-witness question for convex quartics that is marked open in the arXiv version of [111]. It does not settle the complexity of the decision, since the point has a short algebraic description; Section 1.2 shows that the decision is PosSLP-hard. A univariate convex quartic with rational minimum has a rational minimizer [111, Lemma C.3], so two variables are needed.

Theorem 7.4 identifies which numbers occur. A real algebraic number is a coordinate of a point isolated by finitely many rational convex polynomial inequalities if and only if its minimal polynomial has exactly one real root; the same numbers are the coordinates of unique minimizers of rational convex polynomials on $\mathbb{R}^n$, and of single real zeros of rational polynomials. Conversely, each such irrational number is a coordinate of the unique zero of a single rational, strongly convex quartic that is a sum of squares and carries a positive definite rational full Hessian Gram; the quartic is constructed in polynomial time from the dense minimal polynomial. For the prescribed point of consecutive powers of the number, $(d+1)/2$ variables suffice and are necessary, where d is the degree. The restriction to isolated points matters: minimizing $-x$ subject to $x^2 \leq 2$ has the unique optimizer $\sqrt{2}$, whose minimal polynomial has two real roots.

Certified strong convexity does not bound the algebraic degree. Section 7 constructs integer quartics in n variables with $O(n^2)$ monomials, coefficients of $O(\log n)$ bits, $n+1$ integer square factors, and a positive definite integer full Hessian Gram, whose unique zero p satisfies

$$[\mathbb{Q}(p) : \mathbb{Q}] = [\mathbb{Q}(p_1) : \mathbb{Q}] = d_n = \frac{2^{n+1} - (-1)^{n+1}}{3}.$$

The Hessian condition number at the zero is $O(n^2)$. After a rational translation, the minimal polynomial of the first coordinate has all d_n+1 coefficients nonzero. Even with the full Hessian Gram counted, the input length is $L = O(n^4 \log n)$, so dense and sparse coefficient lists of this minimal polynomial have length $2^{\Omega(n)} = 2^{\Omega((L/\log L)^{1/4})}$: exponential in the dimension and superpolynomial, but not exponential, in L . Every coordinate of the zero is nevertheless an integer power of $2^{1/d_n}$ whose exponent has $O(n)$ bits, so one real root gate, whose index d_n is written in binary, followed by $O(n)$ multiplications and divisions computes the whole optimizer. In the other direction, if a rational sum of squares of quadratics has a unique real zero with positive definite Hessian there, the degree of the field generated by the coordinates of the zero is odd and at most $2^n - 1$. Global convexity improves the bound to $2^n - 3$ for $n \geq 3$ and $2^n - 5$ for $n \geq 4$, and the exact maxima for $n = 2, 3, 4, 5$ are 3, 5, 11, 21, attained by the construction above. Theorem 7.9 proves that a globally convex polynomial of degree exactly four that is a sum of squares and has a zero with positive definite Hessian needs at least $n+1$ squares, even with real coefficients; the bound is sharp and our

constructions attain it. Auxiliary variables also matter: some algebraic numbers whose minimal polynomials have $O(b)$ bits require degree $2^{\Omega(b)}$ in every rational convex univariate realization, while two-variable quartics of polynomial size realize them.

Exact rational optimizers and rational feasible points can be long. Theorem 8.5 gives strongly convex quartics in $2(k+1)$ variables, with short rational square factors and a full positive definite Hessian Gram, whose rational minimizer lies in a fixed box and has last coordinates with reduced denominator exactly 5^{2^k} . A rescaled variant has a minimizer whose last denominators are divisible by 5^{2^k} and a Hessian condition number at the minimizer that is polynomial in k ; the two properties are proved for these two related families, not for one instance. Section 8 also gives certified quartics in n variables whose zero sublevel set is compact with nonempty interior, yet every rational point of it needs $\Omega(n2^{n/2})$ denominator bits. These bounds are exponential in the dimension, hence superpolynomial in the input length, which is polynomial in n . They concern expanded rational output. The long rational minimizers above are values of polynomial-size circuits built by repeated squaring, and when the minimum of an instance of Theorem 4.4 is negative, the Newton circuit of that theorem gives a strictly feasible point as a polynomial-size shared rational circuit (Section 4). Expanded formats can also be long for well-conditioned instances. For quartics that are strongly convex on a box, with bounded curvature and a path as interaction graph, Section 3 shows that the minimal polynomial of an optimizer coordinate can have a number of nonzero coefficients exponential in the dimension, and that every rational box certifying the optimizer inside the feasible box can need endpoints whose bit length is exponential in the dimension. With interaction graphs of treewidth two, the same holds for rational boxes on which the sign of an active multiplier is certified. In each of these families a short recurrence describes the optimizer exactly.

1.6 Exact certificates

Certificates of optimality have their own arithmetic. Let f be a quartic with a positive definite full Hessian Gram and positive minimum. A positive definite rational polynomial Gram matrix of total bit length $\text{poly}(L)2^{O(n)}$ exists, and Section 8 shows that $\Omega(n2^{n/2})$ bits can be necessary, even though the same instances have short singular rational Gram matrices. Theorem 8.12 constructs, in deterministic polynomial time and without any sign oracle, a polynomial-size shared rational circuit for a Gram matrix of f that is positive definite exactly when the minimum is positive. Deciding whether this matrix is positive definite is therefore equivalent to deciding whether the minimum is positive, which is PosSLP-complete in this class. With a positive definite full Hessian Gram, the optimal order-two moment matrix is unique, and the field $\mathbb{Q}(p)$ generated by the coordinates of the minimizer is the least field over which a maximal-rank optimal Gram matrix, or a nonzero matrix exposing the face of optimal Gram matrices, can be written. For the rational minimizers of large height above, the optimal moment matrix, every maximal-rank rational optimal Gram matrix, and every nonzero rational exposing matrix have entries whose bit length is exponential in the dimension, while short optimal Gram matrices of lower rank remain.

Rational certificates at the optimum can fail to exist. Section 9 gives a ternary integer quartic $F_\star$ (Example 9.6) with $\nabla^2 F_\star \succeq I$, a positive definite rational full Hessian Gram, and minimum zero, for which a sum-of-squares decomposition, or a positive semidefinite Gram matrix, with entries in a real field E exists exactly when $2^{1/5} \in E$. Three variables are the minimum under this certificate hypothesis, and $F_\star + \eta$ is a rational sum of squares for every rational $\eta > 0$. For every prime $\ell \geq 5$ there is an analogous integer quartic in $(\ell+1)/2$ variables, with coefficients and Hessian Gram entries of $O(\log \ell)$ bits, whose least field is $\mathbb{Q}(2^{1/\ell})$; it is constructed in time polynomial in ℓ . A family of quartics in $3k$ variables, of size polynomial in k , has least field $\mathbb{Q}(2^{1/5^k})$; in every algebraic certificate some individual coefficient has degree at least 5^k . For this family, compact formats avoid

the cost: the certificate coefficients are outputs of one polynomial-size circuit with k real fifth-root gates, and adjoining the coordinates of the minimizer as auxiliary variables gives a polynomial-size rational certificate modulo rational quadratic equations that these coordinates satisfy. Section 9 also proves a sufficient condition for rational descent. Suppose that some rational globally convex quartic that is a rational sum of squares vanishes at a point p with positive definite Hessian there, that the rational polynomials of degree at most two vanishing at p span a space of dimension $n + 1$, and that their pairwise products span the rational polynomials of degree at most four that vanish at p together with their gradients. Then every rational globally convex quartic with minimum zero at p and positive definite Hessian there is a rational sum of squares. For the cyclic points of Section 7, the dimension and product-independence hypotheses hold for $n \geq 4$. Whether the products exhaust the stationary quartics for all these points remains open.

The format of a certificate changes what exists and how large it is. Section 10 shows that the explicit ternary quartic $F_\star$ has a short exact rational certificate as a sum of squares of rational functions with the common denominator $1 + \|x\|^2$, and that the same holds for the tower quartics, whose polynomial certificates need a coefficient of degree 5^k , after a polynomial-size increase of a scale parameter; two is the least possible denominator degree. For a scaled family, every fixed power $(1 + \|x\|^2)^N$ fails as a multiplier for large scale. The least admissible power, which is finite, grows as $\Omega(\log L / \log \log L)$, while a multiplier adapted to the scale keeps the certificate of size $O(L)$. For a sum $f(x) + g(y, z)$ of strongly convex quartics in disjoint variables with zero minima, a short joint rational sum of squares can coexist with the absence of any rational certificate separated by blocks, and the separated least field can be $\mathbb{Q}(2^{1/5^k})$. After a certified quartic with a tiny positive minimum is added to the second block, separated rational certificates exist but need $\Omega(k2^k)$ bits. Theorem 7.15 supplies the tool that makes the second block strongly convex: from a quartic with zero minimum and a positive definite full Hessian Gram, it constructs a polynomial-size certified strongly convex quartic whose unique zero is the optimizer together with all products of its coordinates.

1.7 Summary of results

Table 1 lists the main results by output. In the model column, “ordinary” means a deterministic algorithm in the bit model without an oracle. “One instance” means a polynomial-time many-one reduction to PosSLP, and “complete” means that, in addition, PosSLP reduces to the problem. “Conditional” marks a reduction from Square Root Sum or PosSLP to an approximation task.

Table 1: Main results by output. Promised structure is part of the setting; L is the input length, D the degree, q the requested precision, Λ an upper bound on the coordinate second derivatives in the core, and σ the tilt width.

Output	Setting	Result and model	Where
<i>Values and points</i>			
Fixed optimizer	Globally convex, explicit sparse, degree D ; rational polyhedron	Error exponent $1/D$, constant of $\text{poly}(L, D)$ bits; time $\text{poly}(L, D, q)$; ordinary	Thm. 3.4
Fixed optimizer	Cubic, convex on a bounded rational polytope	Exponent $1/4$, constant of $\text{poly}(L)$ bits; time $\text{poly}(L + q)$; ordinary	Thm. 3.7
Some optimizer, accuracy $1/4$	Quartic convex on a box, unique optimizer	Decides Square Root Sum (treewidth at most two) or PosSLP; conditional	Thm. 3.12
Active bound	Strongly convex cubic on a box	Decides Square Root Sum, while points are easy; conditional	Prop. 3.10

Output	Setting	Result and model	Where
Full selected optimizer	Residual-convex cubic on a product box, one random core tilt	Expected $k^{O(k)}(1 + \Lambda/\sigma)^k \text{poly}(L + q)$, correct for every draw	Thm. 11.9
Values and fixed core	Fixed degree; residual convexity on a product box, or a supplied convexifier on a coupled polytope	One finite core-noise law; every draw correct; one expected-work factor for all precisions	Thm. 11.2
Exact optimizer and value	Quadratic in these core-noise models	Rational output of polynomial length; expected parameterized work	Cor. 11.4
Full selected optimizer	Supplied convexifier; cubic on a polytope or globally convex after convexification	Expected parameterized work; ordinary polynomial expected work when jointly convex	Thm. 11.5, Cor. 11.3
<i>Exact decisions</i>			
Sign or equality of $h(p)$	Explicit f, h , unary degree, $\nabla^2 f \succeq \mu I$ supplied	One instance per relation, no gap assumed	Thm. 4.4
Same, at a zero of a map	Strongly monotone polynomial map, unary degree	One instance per relation; order tests complete for certified cubic maps, hard already for gradient maps; equality only upper	Thm. 4.7
Order tests of value and coordinates; nonnegativity; real SOS	Quartic with checked full Hessian Gram $A \succ 0$	Complete; rational SOS membership hard; equality only upper	Thm. 5.1
Coordinate order	As above, rational minimizer in $[-1, 1]^n$, minimum known to be 0	Complete	Thm. 5.2
Feasibility of $f \leq 0$	Quartic with checked $A \succ 0$, on a halfspace or the unit cube	PosSLP-hard; in $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$	Cor. 5.9, Thm. 6.2
Value, coordinate, observable	Strongly convex quartic on a rational polyhedron	$\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$	Thm. 6.2
Same	Strongly convex polynomial on a rational polyhedron, supplied gap for nonzero slacks	One instance per relation; active set by ordinary approximation	Prop. 4.9
Same, and active set	Nonlinear dimension k	Ordinary $F(k)L^C$	Thm. 6.6
Any Boolean predicate	Fixed degree; forest, nonpositive off-diagonal, or comparison-matrix Hessian on a box; separable network flow	One instance	Cor. 6.15
Value, coordinate, observable	Constraint rank r	Las Vegas $(r + 1)^{O(r)}L^C$ with PosSLP oracle	Thm. 6.16
Optimal integer blocks	t integer variables	Ordinary candidate lists in FPT time; exact selection by PosSLP queries	Thms. 6.18, 6.20
One binary decision	Known optimal value, rational optimizer, rank one	PosSLP-hard	Cor. 6.24

Output	Setting	Result and model	Where
<i>Exact optimizers</i>			
Possible coordinates	Points isolated by rational convex polynomial inequalities	Exactly the numbers with one real conjugate; one certified quartic realizes each, in polynomial time	Thm. 7.4
Coordinate field degree	Certified convex rational SOS quartics, unique nondegenerate zero	Degree $d_n = (2^{n+1} - (-1)^{n+1})/3$ with $O(\log n)$ -bit integer data; at most $2^n - 5$ for $n \geq 4$; exact maxima for $n \leq 5$	Thms. 7.8, 7.11
Number of squares	Convex quartic SOS, nondegenerate zero	At least $n + 1$ real squares; sharp	Thm. 7.9
Rational minimizer	Certified quartics in n variables	Last coordinates with denominator exactly $5^{2^{n/2-1}}$	Thm. 8.5
Rational feasible points	Certified quartics in n variables, sublevel set with interior	Every rational point needs $\Omega(n2^{n/2})$ bits	Thm. 8.3
<i>Exact certificates</i>			
Interior Gram matrix	Positive minimum, full Hessian Gram	$\text{poly}(L)2^{O(n)}$ bits suffice, $\Omega(n2^{n/2})$ can be needed; shared rational circuit in polynomial time	Thms. 8.11, 8.12
Moment and exposing matrices	Full Hessian Gram	Unique moment optimum; least field $\mathbb{Q}(p)$; long entries for long rational minimizers	Thm. 8.6, Cor. 8.10
SOS coefficient field	Certified quartics with minimum 0	Least field $\mathbb{Q}(2^{1/\ell})$ in $(\ell + 1)/2$ variables; $\mathbb{Q}(2^{1/5^k})$ in $3k$ variables, with a coefficient of degree $\geq 5^k$; polynomial-size root-circuit formats for this family	Thms. 9.4, 9.9, 9.10, 9.11
Certificate format	Rational functions, radial multipliers, block separation	Quadratic common denominators suffice, also for the tower after a scale increase; growing radial order; separation fails or costs $\Omega(k2^k)$ bits	Cor. 10.4, Thms. 10.7, 10.8, 10.10, 10.11
<i>Further arithmetic interfaces</i>			
Feasible point, finite infimum, attained optimizer	Rational quadratics, not necessarily convex; constraint Hessian span h	Common-field algebraic descriptions of degree and total length $L^{O(h+1)}$; finite infima may be unattained; fixed- h feasibility in NP and exact output in FP^{NP}	Thms. L.2, L.7, L.8
Cone feasibility and witnesses	Explicit common real number field; integer dimension t ; continuous Hessians of squared cone residuals span h directions over that field	Ordinary polynomial time for fixed t, h ; continuous point in a joint field with degree and total length $L^{O(h+1)}$ for continuous input; no box or Slater hypothesis	Thms. L.9, L.11

1.8 Relation to prior work

Convex polynomial optimization and error bounds. Slot, Steurer, and Wiedmer establish, for globally convex polynomials with exponents written in unary on rational polyhedra, the dichotomy between unboundedness and attainment together with a bound on the norm of an optimizer whose logarithm is polynomial in the input length [111, Theorem 1.1], and polynomial-time value approximation [111, Corollary 1.2]. Section 3 uses the same structural facts, which rest on their

analysis and on a tangent quadratic bound from the proof of Lemma 5 of Ahmadi, Chaudhry, and Zhang [5], and rederives attainment and the radius with explicit rational constants. Global error bounds for convex polynomials, also relative to polyhedral constraints, are studied in [81, 82, 93, 119]; the bounds of [81, 82, 93] are Hölder-type estimates whose constants depend on the function. The addition of Theorem 3.4 is the quantitative selector: an error-bound constant computed from the input, of polynomial bit length on arbitrary rational polyhedra, evaluation of sparse polynomials without expanding affine substitutions, and a regularization schedule that approximates one fixed optimizer at every precision. The form of the error exponent is not claimed as new. The distinction between approximating the residual of a solution and approximating the solution itself, and the hardness of the latter for equilibria, are due to Etessami and Yannakakis [38]; Section 3 shows the same phenomenon for minimizers of polynomials that are convex on a box. Recognizing convexity is NP-hard for quartics on $\mathbb{R}^n$ [4] and for cubics on a box [2]. Convexity is therefore a promise in our theorems, unless a certificate is supplied and checked.

Exact arithmetic, Newton circuits, and PosSLP. Allender, Bürgisser, Kjeldgaard-Pedersen, and Miltersen [6] introduce PosSLP and show that polynomial-time constant-free real computation on Boolean inputs corresponds to P^{PosSLP} [6, Proposition 1.1]; divisions are removed there by computing numerators and denominators separately, and Newton iteration gives short circuits that approximate square roots [6, Section 1.4]. Etessami, Stewart, and Yannakakis [39, Appendix C] reduce threshold comparisons at least fixed points of probabilistic polynomial systems to one PosSLP instance, by storing polynomially many Newton steps in a circuit and combining them with an algebraic separation bound. Our upper reduction uses this architecture for a different class of inputs, globally strongly convex explicit polynomials and strongly monotone polynomial maps. The class-specific steps are a polynomial-bit starting point obtained from convex value approximation, a separation bound that holds even when the complex critical set of the objective has positive dimension, the treatment of equality, and the handling of supplied certificates. For the compilation theorem the nearest prior work is the oracle characterization of P^{PosSLP} through constant-free real computation [6, 22] and specific many-one reductions such as that of [39]. These works state adaptive (Turing) relationships or particular reductions, and the sign iteration and division elimination are classical numerical tools. Section 4 gives the explicit $O(S^2)$ compilation with its error budget and interpreter in full; we make no priority claim for it. Bürgisser and Jindal [22] study the hardness of PosSLP itself; no unconditional NP-hardness of PosSLP is known, so our PosSLP-hardness results are not NP-hardness results.

Exact conic optimization. Tarasov and Vyalii [113, Theorems 3 and 4] reduce PosSLP to exact semidefinite feasibility. They compare the values of two circuits built from the constant 1 by the operations $x + y$ and $x^2/2$, represent each value as the optimum of a semidefinite program, and use duality. Gate constraints on two-by-two matrices are rotated second-order cone constraints, and standard conic duality then gives the same hardness for exact second-order cone feasibility (Remark 2.18). Hardness of exact convex optimization in general is therefore not new. Our lower bounds hold in a much smaller class: a single explicit quartic objective, globally strongly convex with a checkable strict certificate and without constraints, for which Theorem 4.4 supplies the matching upper bound. Squaring the residuals of the gate equations does not preserve convexity, so this restriction requires the realization of Lemma 7.6.

Algebraic degree and irrational optimizers. Nie and Ranestad [94] compute the algebraic degree of generic polynomial optimization problems. Such counts concern generic data, and an

optimizer of a particular instance can have much smaller degree. Irrational singletons defined by convex polynomials were known before. In the arXiv version of [111], Appendix C gives a univariate convex sextic whose minimum zero is attained at a single irrational point. Bienstock, Del Pia, and Hildebrand [17, Example 2.1] give a cubic whose nonpositive sublevel set within a rational rectangle is the single point $(2^{1/3}, 2^{2/3})$; its Hessian is positive definite on that rectangle, so the cubic is convex there, but no cubic is convex on all of $\mathbb{R}^2$. Our degree results concern specific certified convex classes. They characterize which numbers occur, construct a certified strongly convex quartic for each of them from its dense minimal polynomial, bound the degree from above, determine the maxima in dimensions at most five, and exhibit degree exponential in the dimension with coefficients of logarithmic bit length.

Rational sums of squares. Scheiderer [107] constructs rational forms that are sums of squares over $\mathbb{R}$ but not over $\mathbb{Q}$, shows that for every number field with a real embedding some rational form is a sum of squares over $\mathbb{R}$ but not over that field [107, Corollary 2.11], and characterizes the nonnegative rational ternary quartic forms that are not sums of squares over $\mathbb{Q}$ [107, Theorem 4.1]. Sums of squares descend from totally real number fields to $\mathbb{Q}$ [56, Theorem 1.4], [107, Theorem 1.2]; the least fields $\mathbb{Q}(2^{1/\ell})$ and $\mathbb{Q}(2^{1/5^k})$ in our examples are not totally real, so this descent does not apply to them. Laplagne [75, Section 3] gives rational quartic forms that are sums of squares over $\mathbb{Q}(2^{1/3})$ but not over $\mathbb{Q}$, one of them positive away from the origin; convexity plays no role in those constructions. Chua, Plaumann, Sinn, and Vinzant [25] record that rational positive semidefinite Gram matrices and rational sums of squares are equivalent, and that the analogous statement fails over general ordered fields. Peyrl and Parrilo [100] round interior numerical Gram matrices to exact rational certificates. A sum-of-squares-convex polynomial that is stationary at a zero is a real sum of squares, by integrating its Hessian certificate [3, 53, 76]. Our field results add global strong convexity with a strict rational convexity certificate, attained minimum zero, exact least fields for both sums of squares and Gram matrices, field degree 5^k for inputs of size polynomial in k together with a lower bound for individual coefficients and polynomial-size root-circuit certificates for that family, and a sufficient condition for rational descent.

Size of certificates. Algorithms that produce exact rational sum-of-squares certificates have bit bounds that depend on a supplied eigenvalue margin or on the existence of a rational certificate [27, 45, 86, 100, 106], and bit complexity lower bounds are known for sum-of-squares proofs from constraints [101]. Our size bounds concern unconstrained certificates for a single short certified quartic, and each is stated for a specific format: interior Gram matrices, maximal-rank optimal Gram matrices, exposing matrices, prescribed multipliers, or certificates separated by blocks. They identify certified instances where interior certificates are necessarily long and where rational certificates at the optimum do not exist. Kojima, Kim, and Waki [67] discuss real sums of squares respecting a block separation; our block results concern rational coefficients and their size.

Constraints and integer variables. The ingredients of Section 6 include essential-variable extraction [24], approximation algorithms for objectives with a low-rank nonlinear part [63, 89], quadratic convergence of exact proximal Newton steps [77], exact algorithms for structured box-constrained and network-flow quadratic programs [98, 117], violator spaces [44], convex integer minimization in fixed dimension [55], integer feasibility algorithms that query separation oracles only at integer points [8, 11, 54], and lattice basis reduction [78]. Our additions there are the exact arithmetic interfaces that these methods need for algebraic optimizers and their compositions. For example, the integer-query theorems are not applied directly to the algebraic value of a continuous

fiber; Section 6 replaces them by its own approximate projected-gradient and integer-only cutting argument.

Quadratic sampling and algebraic cone input. The feasibility bound in Appendix L is a polyhedral consequence of Grigoriev and Pasechnik’s component-sampling theorem [48, Theorem 1.2]. Its sampling architecture is prior work. The appendix gives separate proofs for finite infima and attained minimum-norm optimizers, using ordered regularization and perturbation limits and controlling one joint field for the selected limiting tuple. Its cone algorithm uses that arithmetic interface together with the integer-witness method of Khachiyan and Porkolab [65, Theorem 1.1] and rational mixed-integer linear feasibility at fixed integer dimension [79, Section 5]. The extension specifies one real embedding for explicit algebraic input, preserves exact feasibility through rational rounding and cone lifts, and returns continuous coordinates in a field containing the input field. These input, formula, sign, rounding, and output guarantees are distinct from the established sampling and integer algorithms on which they rely.

1.9 Scope of the claims

The hardness results for continuous convex polynomial families are reductions from PosSLP or Square Root Sum. They do not establish NP-hardness or unconditional running-time lower bounds. Appendix L also proves strong NP-hardness of nonconvex quadratic feasibility and attainment at a supplied zero infimum, even with one constraint-Hessian direction. Upper bounds relative to PosSLP are not ordinary polynomial-time algorithms, since the constructed circuits compute integers with exponentially many bits. PosSLP does not describe polynomial optimization in general: our upper bounds require explicit input of unary degree and a global strong convexity bound, or one of the constrained structures of Section 6. Three questions about exact decisions remain open: whether exact comparison for globally strongly convex quartics over arbitrary rational polyhedra lies in P^{PosSLP} , equivalently reduces to one PosSLP instance, rather than only in $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$; whether exact comparison at minimizers of convex quartics that are not strongly convex has any upper bound relative to PosSLP; and whether equality tests are PosSLP-hard. Lower bounds on output size are stated for specific expanded formats and do not exclude compact descriptions. Most of them concern families whose input length is polynomial in the dimension and are exponential in the dimension, hence superpolynomial in the input length; none of these is claimed to be exponential in the input length. Promised properties, such as convexity, strong convexity, a unique real root, a rational optimizer, a zero minimum, or a slack gap, are not recognized by our algorithms unless a certificate is supplied and checked.

1.10 Organization

Section 2 fixes encodings, promises and certificates, output contracts, and computational models, and proves a few elementary facts used throughout. Section 3 studies value-to-point conversion. Sections 4 and 5 prove the upper and lower bounds for exact decisions, and Section 6 extends them to constraints and integer variables. Sections 7 and 8 study the algebraic degree and the bit length of exact optimizers and of interior, moment, and exposing matrices. Sections 9 and 10 study coefficient fields and formats of sum-of-squares certificates. Section 11 treats cubic recourse with a nonconvex core. Section 12 discusses consequences for exact optimization and open questions.

The main sections state results, explain why they matter, and outline the proofs. The complete proofs for Sections 3–11 are in Appendices C, A, B, D, E, F, G, H, and I, respectively. Appendix J

compares the results with convex quadratically constrained problems, where the number of independent constraint Hessians controls the arithmetic. Appendix K records examples that mark the limits of the methods: degenerate quartics, information-based precision limits, penalty representations, and facial reduction. Appendix L treats two settings outside the convex classes above: quadratic constraints with few independent Hessians, and cone constraints with bounded integer dimension over an explicitly represented real number field. It develops common-field witness, finite-value, attainment, and algebraic-input contracts, as summarized in Section 1.4.

Readers interested mainly in complexity can read Sections 2, 4, 5, and 6; readers interested in certificates, Sections 2 and 7–10; and readers interested in solver design, Sections 3, 6, 11, and 12.

2 Models and conventions

This section fixes the input encodings, the promises and certificates that may accompany an input, the output contracts, and the computational models. Later sections use these definitions without restating them. The section also proves a few elementary facts that are used throughout.

2.1 Notation

For $x \in \mathbb{R}^n$, $\|x\|$ is the Euclidean norm, and $\|x\|_1$ and $\|x\|_\infty$ are the ℓ_1 and maximum norms. For a set $S \subseteq \mathbb{R}^n$, $\text{dist}(x, S) = \inf_{s \in S} \|x - s\|$. The space of real symmetric $m \times m$ matrices is Sym_m ; $A \succeq 0$ and $A \succ 0$ mean positive semidefinite and positive definite, and $\lambda_{\min}(A)$ is the least eigenvalue. The Kronecker product $x \otimes v \in \mathbb{R}^{n^2}$ has coordinates $x_i v_j$, ordered lexicographically by (i, j) .

The following symbols are reserved. The total binary input length is L (Definition 2.4); the number of continuous variables is n ; k is a structural parameter defined in each statement; q is a number of requested bits of precision, and $\varepsilon = 2^{-q}$. The objective is f , the feasible set P , the optimal value $f_* = \inf_P f$, and the set of optimizers S . The selected optimizer is p ; it is the unique optimizer when f is strictly convex on a convex feasible set, and otherwise, for example in mixed-integer problems with several optimal points, the selection rule is stated. An explicit polynomial whose sign or value at p is requested is an *observable* and is denoted by h . Lower curvature bounds are denoted by μ and upper curvature or Lipschitz bounds by Λ . The degree of the input polynomials is D . We write $\text{poly}(\cdot)$ for a polynomial whose exponent is an absolute constant, C for an absolute constant, and $F(k)$, $a(t)$ for computable functions of the parameters shown. For a point p with algebraic coordinates, $\mathbb{Q}(p) = \mathbb{Q}(p_1, \dots, p_n)$ is its coordinate field.

2.2 Input encodings

Definition 2.1 (Rational data). A rational number a/b with $b \geq 1$ and $\gcd(a, b) = 1$ is encoded by a and b in binary. Its bit length is $\lceil \log_2(|a| + 1) \rceil + \lceil \log_2(b + 1) \rceil + 1$, and the bit length of a vector or matrix is the sum of the bit lengths of its entries.

Definition 2.2 (Explicit polynomials and degree). An *explicit* (or explicitly sparse) polynomial $f = \sum_{\alpha} f_{\alpha} x^{\alpha} \in \mathbb{Q}[x_1, \dots, x_n]$ is given by the list of its nonzero coefficients f_{α} together with their exponent vectors $\alpha \in \mathbb{N}^n$. Its degree is $\deg f = \max\{|\alpha| : f_{\alpha} \neq 0\}$. When the degree is not fixed, we use the numerical degree $D = \max\{2, \deg f\}$, or the maximum over all input polynomials. A bound of the form $\text{poly}(L, D)$ is polynomial in the input length when the degree is at most polynomial in L , for example when exponents are written in unary; we then speak of *unary degree*.

The *interaction graph* of a polynomial has one vertex for each variable and an edge between two variables when a nonzero monomial contains both. For a polynomial, this is equivalent to the

corresponding off-diagonal entry of its Hessian being a nonzero polynomial. The graph records which variables interact; bounds on its treewidth or a supplied forest are stated where used.

Definition 2.3 (Polyhedra). A rational polyhedron is $P = \{x \in \mathbb{R}^n : Bx \leq b\}$ with $B \in \mathbb{Q}^{m \times n}$ and $b \in \mathbb{Q}^m$; an equality is written as two inequalities. A box is $\{x : l \leq x \leq u\}$ with $l_i \in \mathbb{Q} \cup \{-\infty\}$ and $u_i \in \mathbb{Q} \cup \{+\infty\}$. A network-flow polyhedron is $\{x \in \mathbb{R}^E : Ax = b, l \leq x \leq u\}$, where A is the node-arc incidence matrix of a directed graph with arc set E .

Definition 2.4 (Input length). The input length L is the total bit length of all input data: the polynomials (objective, constraints, observables), the polyhedron, thresholds, supplied numbers such as curvature bounds, slack gaps, or noise widths, and supplied certificates. The number of variables is part of the input, so $n \leq L$, and we assume $L \geq 2$.

Quantities computed from the input in polynomial time have bit length $\text{poly}(L)$, so bounds stated in terms of L remain valid when such quantities are added to the input.

Remark 2.5 (Encodings not covered). Three other encodings are excluded unless a statement says otherwise. First, a polynomial given by an arithmetic circuit can have exponentially many monomials and exponentially large degree. Second, exponents written in binary allow degree exponential in L ; our bounds that are polynomial in D are then not polynomial in L . Third, expansion is not free under substitution: $(x_1 + \dots + x_n)^D$ has $\binom{n+D-1}{D}$ monomials, so a rational affine substitution $x = x_0 + Vu$ can make a short polynomial of growing degree long. Algorithms for variable degree therefore evaluate such compositions instead of expanding them. At fixed degree, expansion changes the length only polynomially.

2.3 Promises and certificates of convexity

A function f is *convex* on a convex set K if $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$ for all $x, y \in K$ and $\lambda \in [0, 1]$. It is *globally convex* if it is convex on $\mathbb{R}^n$, equivalently $\nabla^2 f(x) \succeq 0$ for all x . For $\mu > 0$ it is *globally μ -strongly convex* if $\nabla^2 f(x) \succeq \mu I$ for all $x \in \mathbb{R}^n$. A map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is *globally μ -strongly monotone* if $(T(x) - T(y))^\top (x - y) \geq \mu \|x - y\|^2$ for all x, y ; for a polynomial map this is equivalent to $\frac{1}{2}(J_T(x) + J_T(y))^\top \succeq \mu I$ for all x, y , where J_T is the Jacobian. A solution of the variational inequality defined by T on a closed convex set K is a point $p \in K$ with $T(p)^\top (y - p) \geq 0$ for all $y \in K$; for $T = \nabla f$ these are the minimizers of f on K .

These properties enter in two ways. As a *promise*, the property is assumed and not checked; an algorithm for a promise problem may answer arbitrarily on inputs that violate it. Recognizing convexity is NP-hard already for quartics on $\mathbb{R}^n$ [4] and for cubics on a box [2], so in general the promise cannot be checked in polynomial time unless $P = NP$. As a *checked certificate*, the input contains data that prove the property and can be verified in polynomial time. The input is then an ordinary language, and a malformed certificate is mapped to a fixed answer.

Definition 2.6 (Hessian Gram matrices). Let $f \in \mathbb{Q}[x_1, \dots, x_n]$ have degree at most four, and put $w(x, v) = (v, x \otimes v) \in \mathbb{R}^{n+n^2}$. A *Hessian Gram matrix* of f is a matrix $A \in \text{Sym}_{n+n^2}$ such that

$$v^\top \nabla^2 f(x) v = w(x, v)^\top A w(x, v)$$

as polynomials in (x, v) . It is a *full positive definite Hessian Gram* if $A \succ 0$. For higher degree the same definition is used with a vector $w(x, v)$ of monomials that are linear in v and include $v_1, \dots, v_n$ among them.

A rational Hessian Gram matrix is checked in polynomial time by comparing coefficients and by rational symmetric Gaussian elimination. Hessian Gram matrices are not unique, since several entries contribute to the same monomial of the biform.

Lemma 2.7 (Determinant–trace margin). *Let $A \in \text{Sym}_m$ be rational and positive definite, and put*

$$\rho(A) = \frac{\det A}{(\text{tr } A)^{m-1}}.$$

Then $A \succeq \rho(A)I$. The rational number $\rho(A)$ is computable in time polynomial in the bit length of A and has polynomial bit length.

Proof. Let $\lambda_1 \leq \dots \leq \lambda_m$ be the eigenvalues. Each is at most $\text{tr } A$, so $\det A = \lambda_1 \prod_{i \geq 2} \lambda_i \leq \lambda_1 (\text{tr } A)^{m-1}$. The determinant of a rational matrix is computed in polynomial time with polynomial bit length by fraction-free elimination, and the power $(\text{tr } A)^{m-1}$ has bit length at most m times that of $\text{tr } A$. $\square$

Lemma 2.8 (Consequences of a Hessian Gram). *Let A be a Hessian Gram matrix of a quartic f in n variables.*

- (a) *If $A \succeq \mu I$ for some $\mu > 0$, then $\nabla^2 f(x) \succeq \mu I$ for all $x \in \mathbb{R}^n$. In particular, a full positive definite Hessian Gram certifies global strong convexity with the rational modulus $\rho(A)$ of Lemma 2.7.*
- (b) *Let $T \in \mathbb{Q}^{n \times n}$ be invertible, $c \in \mathbb{Q}^n$, and $g(y) = f(Ty + c)$. Then $S^\top AS$ is a Hessian Gram matrix of g , where*

$$S = \begin{pmatrix} T & 0 \\ c \otimes T & T \otimes T \end{pmatrix}$$

is invertible. Hence a full positive definite Hessian Gram is preserved by invertible rational affine changes of variables.

- (c) *If $A \succ 0$, then the homogeneous quartic part f_4 of f satisfies $f_4(x) > 0$ for all $x \neq 0$.*

Proof. (a) Since v is a subvector of $w = w(x, v)$, $v^\top \nabla^2 f(x) v = w^\top A w \geq \mu \|w\|^2 \geq \mu \|v\|^2$. (b) We have $v^\top \nabla^2 g(y) v = (Tv)^\top \nabla^2 f(Ty + c)(Tv)$ and $w(Ty + c, Tv) = (Tv, (T \otimes T)(y \otimes v) + (c \otimes T)v) = Sw(y, v)$. The matrix S is block lower triangular with invertible diagonal blocks T and $T \otimes T$. (c) Let Q be the block of A indexed by $x \otimes v$. Comparing the terms of degree two in x in the defining identity gives $v^\top \nabla^2 f_4(x) v = (x \otimes v)^\top Q(x \otimes v)$. Setting $v = x$ and using Euler's identity $x^\top \nabla^2 f_4(x) x = 12f_4(x)$ gives $12f_4(x) = (x \otimes x)^\top Q(x \otimes x) > 0$ for $x \neq 0$, because $Q \succ 0$. $\square$

Remark 2.9 (Hessian Gram and polynomial Gram matrices). A full positive definite Hessian Gram is stronger than global strong convexity. For example, $f(x, y) = x^2 + y^2 + y^4$ satisfies $\nabla^2 f \succeq 2I$, but its quartic part y^4 vanishes at $(1, 0)$, so by Lemma 2.8(c) it has no full positive definite Hessian Gram. A sum $f(x) + g(y)$ in disjoint blocks of variables has none either: if v_i is a direction coordinate of the y -block, the monomial $x_j^2 v_i^2$ has coefficient zero in the Hessian biform, and in $w^\top A w$ it arises only from the diagonal entry indexed by the coordinate $x_j v_i$ of w , which must therefore vanish. A Hessian Gram certifies convexity, not nonnegativity. A full positive definite Hessian Gram is compatible with minimum zero at an irrational point, as for the quartic of Section 1.5, whereas a positive definite polynomial Gram matrix on the full monomial vector of Definition 2.19, which contains 1, implies strict positivity at every real point. A positive definite Gram on a custom polynomial vector can coexist with zeros when all entries of that vector vanish there, as in Section 10. The Hessian basis $w(x, v) = (v, x \otimes v)$ instead contains every v_i , so it cannot vanish when $v \neq 0$; this is what makes its positive definiteness a global curvature certificate.

Lemma 2.10 (Strong convexity). *Let f be continuously differentiable and globally μ -strongly convex, and let $K \subseteq \mathbb{R}^n$ be nonempty, closed, and convex. Then f attains its minimum over K at a unique point p , and for every $x \in K$*

$$f(x) - f(p) \geq \frac{\mu}{2} \|x - p\|^2, \quad \|x - p\| \leq \frac{2}{\mu} \|\nabla f(x)\|.$$

If $K = \mathbb{R}^n$, then also $\|x - p\| \leq \|\nabla f(x)\| / \mu$.

Proof. For $x_0 \in K$, $f(x) \geq f(x_0) + \nabla f(x_0)^\top (x - x_0) + \frac{\mu}{2} \|x - x_0\|^2$, so the sublevel sets of f on K are compact and the minimum is attained; strict convexity makes the minimizer unique. First order optimality gives $\nabla f(p)^\top (x - p) \geq 0$, and Taylor's formula with $\nabla^2 f \succeq \mu I$ gives the first inequality. For the second, $0 \geq f(p) - f(x) \geq \nabla f(x)^\top (p - x) + \frac{\mu}{2} \|p - x\|^2 \geq -\|\nabla f(x)\| \|p - x\| + \frac{\mu}{2} \|p - x\|^2$. If $K = \mathbb{R}^n$, then $\nabla f(p) = 0$, and $\mu \|x - p\|^2 \leq (\nabla f(x) - \nabla f(p))^\top (x - p) \leq \|\nabla f(x)\| \|x - p\|$. $\square$

Lemma 2.11 (Low degree).

- (a) *A globally convex polynomial of degree at most three has degree at most two. Consequently, if a globally convex polynomial of degree at most three attains its minimum on a rational polyhedron, then it has a rational minimizer whose bit length is polynomial in the input length, computable in polynomial time [71].*
- (b) *A globally monotone polynomial map $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ of degree at most two is affine. If it is strongly monotone and has rational coefficients, its unique zero is rational and is computed by rational linear algebra.*

Proof. (a) Write $f = f_0 + f_1 + f_2 + f_3$ by homogeneous degree. For fixed u , the function $x \mapsto u^\top \nabla^2 f(x) u$ is affine and nonnegative on $\mathbb{R}^n$, hence constant. Thus $u^\top \nabla^2 f_3(x) u = 0$ for all x and u , and Euler's identity gives $6f_3(x) = x^\top \nabla^2 f_3(x) x = 0$. (b) Monotonicity is equivalent to $v^\top J_T(x) v \geq 0$ for all x and v . For fixed v this function of x is affine and nonnegative, hence constant. Let U be the homogeneous quadratic part of T ; then $v^\top J_U(x) v = 0$ for all x, v , that is, $\partial_i U_j + \partial_j U_i = 0$ for all i, j . Put $a_{kij} = \partial_k \partial_i U_j$, a constant tensor symmetric in its first two indices. Differentiating the identity gives $a_{kij} = -a_{kji}$. A tensor that is symmetric in its first two indices and antisymmetric in its last two vanishes: $a_{kij} = -a_{kji} = -a_{jki} = a_{jik} = a_{ijk} = -a_{ikj} = -a_{kij}$. Hence all second derivatives of U vanish, so $U = 0$ and $T(x) = Ax + c$. Strong monotonicity gives $\frac{1}{2}(A + A^\top) \succ 0$, so A is invertible and $p = -A^{-1}c$ is rational. $\square$

Lemma 2.11 explains the degrees in this paper. For globally convex objectives on rational polyhedra, degree four is the first degree that permits a unique irrational optimizer. A convex cubic problem on a rational polyhedron can have a unique irrational optimizer only if the objective is convex on the feasible set but not on $\mathbb{R}^n$. Nonunique optimal sets can contain irrational points already for quadratics; for example, $f(x, y) = x^2$ is minimized at every point $(0, y)$.

2.4 Output contracts

Let f and P be given, with f_* finite and attained, and let $S = \arg \min_P f$. The outputs studied in this paper are the following.

Definition 2.12 (Values). A *value enclosure* of precision q consists of a rational number ℓ and a feasible rational point $x \in P$ with $\ell \leq f_* \leq f(x) \leq \ell + 2^{-q}$. A *value-gap point* is a feasible rational x with $f(x) - f_* \leq 2^{-q}$.

Definition 2.13 (Points). A *set approximation* of precision q is a feasible rational x with $\text{dist}(x, S) \leq 2^{-q}$. A *fixed selector* consists of a point $p \in S$ determined by the instance, independently of q , and an algorithm that on input q returns a feasible rational x_q with $\|x_q - p\| \leq 2^{-q}$. For convex problems the default selected point is the point of S of minimum Euclidean norm in the original coordinates. The sequence (x_q) is a Cauchy name of p ; it need not give a short exact description of p .

Definition 2.14 (Exact predicates). Let h be an explicit rational polynomial (the observable). An *exact comparison* asks whether $h(p) \bowtie 0$, for a relation $\bowtie \in \{<, \leq, =, \neq, \geq, >\}$. Value thresholds ($h = f - r$), coordinate thresholds ($h = x_j - r$), and constraint slacks ($h = b_i - B_i x$) are special cases. The *active set* of p is the set of constraints with zero slack. A *Boolean predicate* is a polynomial-size Boolean combination of exact comparisons. We call $<$ and $>$ strict, $\leq$ and $\geq$ weak, and $=$ and $\neq$ equality relations.

Definition 2.15 (Exact representations). An exact representation of p is one of the following.

- (a) An *implicit* representation: a rational affine map $t \mapsto u + Mt$ and an explicit polynomial system with a unique real solution t_* , such that $p = u + Mt_*$.
- (b) A *circuit* representation of a rational point: a shared rational circuit (Definition 2.16) whose value is p .
- (c) A *root-circuit* representation, when available: a shared rational circuit with the explicitly defined real-root gates below. This format includes radical expressions but does not represent every algebraic point.
- (d) An *algebraic* representation: for each coordinate, or for a primitive element of $\mathbb{Q}(p)$ together with the coordinates written as polynomials in it, the minimal polynomial over $\mathbb{Q}$ and an isolating interval. The minimal polynomial is given either as a *dense list* of all its coefficients or as a *sparse list* of its nonzero coefficients with their exponents.
- (e) An *expanded rational* representation: reduced fractions for all coordinates.

These outputs form a hierarchy only in part. A fixed selector gives set approximations and, by continuity, value-gap points; the converse fails, because a set approximation may approach different optimizers at different precisions. No approximation of fixed precision decides an exact predicate, since $h(p)$ may be nonzero but smaller than any fixed bound. An exact decision procedure need not produce a short representation, and a short representation in one format need not be short in another: the number $2^{1/d}$ has the two-term minimal polynomial $T^d - 2$, while $1 + 2^{1/d}$ has the minimal polynomial $(T - 1)^d - 2$, all of whose $d + 1$ coefficients are nonzero.

2.5 Computational models

An *ordinary* algorithm is a deterministic algorithm in the bit model, without an oracle. A *Las Vegas* algorithm is randomized, returns a correct answer whenever it halts, halts with probability one, and is analyzed by its expected running time on each input. A parameterized algorithm is *fixed-parameter tractable* (FPT) if its running time is at most $F(k)L^C$ with F computable and C an absolute constant; a bound $L^{F(k)}$ is not of this form.

Definition 2.16 (Arithmetic circuits). An *integer arithmetic circuit* is a finite directed acyclic graph whose sources are labeled by the constants 0 or 1, and whose other nodes are labeled by $+$,

$-$, or $\times$ and have two ordered inputs; one node is the output. Its size is the number of nodes, and nodes may feed several later nodes. A circuit of size s can compute an integer with about 2^s bits. A *rational circuit* may in addition use rational constants written in binary and division by nonzero values. It is evaluated as a pair of integer circuits (N, M) with $M > 0$, representing N/M : sums and products follow the usual fraction rules, and division by a value $N_2/M_2 \neq 0$ uses

$$\frac{N_1/M_1}{N_2/M_2} = \frac{N_1 M_2 N_2}{M_1 N_2^2},$$

which keeps the denominator positive without any sign test (Lemma 4.2). A *shared rational circuit* for a vector or matrix is a single circuit with several outputs. Such circuits have rational values. A *root circuit* may also have a gate $a \mapsto \sqrt[d]{a}$ with the positive integer d written in binary: it selects the unique real root when d is odd and the nonnegative real root when d is even, in the latter case requiring $a \geq 0$. A circuit with algebraic inputs instead uses explicitly represented algebraic numbers as sources. The representation of each such source is part of its size; a rational circuit applied to an unspecified irrational source is not an exact description.

Definition 2.17 (PosSLP and reductions). PosSLP is the set of integer arithmetic circuits whose value is positive [6]. A problem *reduces to one* PosSLP *instance* if a deterministic polynomial-time algorithm maps each instance x to a circuit C_x with x a yes-instance exactly when $C_x \in \text{PosSLP}$; for a promise problem, the equivalence is required on instances that satisfy the promise. The problem is *PosSLP-hard* if PosSLP reduces to it by such a polynomial-time many-one reduction, and *PosSLP-complete* if both hold. P^{PosSLP} is the class of problems decided in deterministic polynomial time by an oracle machine whose queries are circuit descriptions. $\text{UP}^{\text{PosSLP}}$ is the class of problems decided by a nondeterministic polynomial-time oracle machine with exactly one accepting computation on each yes-instance and none on each no-instance, and $\text{coUP}^{\text{PosSLP}}$ is the class of their complements. Las Vegas and FPT algorithms with a PosSLP oracle are defined analogously.

The sign of a rational circuit (N, M) is the sign of N , so comparing a rational circuit with zero is one PosSLP instance. Since $1 - N > 0$ exactly when $N \leq 0$, strict and weak comparisons, and a problem and its complement, reduce to each other. PosSLP lies in the counting hierarchy, the class P^{PosSLP} coincides with the class of Boolean problems decided in polynomial time by constant-free real machines, and Square Root Sum lies in P^{PosSLP} [6]. Section 4 proves that every problem in P^{PosSLP} reduces to one PosSLP instance; we therefore state deterministic oracle algorithms as single-instance reductions. This conversion does not apply to $\text{UP}^{\text{PosSLP}}$, to Las Vegas algorithms, or to algorithms whose running time is FPT rather than polynomial.

These notions should be read with care. PosSLP-hardness is hardness relative to the sign problem of integer circuits. It is not NP-hardness, and it does not exclude a polynomial-time algorithm unconditionally; whether PosSLP is in P, or is NP-hard, is open [22]. Conversely, a reduction to one PosSLP instance is not an ordinary algorithm: the circuit computes an integer whose binary expansion can have exponentially many bits.

Remark 2.18 (Exact conic feasibility). Exact feasibility of second-order cone programs is PosSLP-hard. This follows from the gate constructions of Tarasov and Vyalıy [113] and standard conic duality, as follows. By their basis equivalence, PosSLP reduces to deciding $A > B$ for the values A, B of two circuits that start from the constant 1 and use only the operations $x + y$ and $x^2/2$; all gate values are positive. For one such circuit, introduce a variable x_i for every gate other than the input, substitute 1 for the input, and impose $x_i \geq x_j + x_k$ for an addition gate and

$$\begin{pmatrix} 2x_i & x_j \\ x_j & 1 \end{pmatrix} \succeq 0, \quad \text{that is,} \quad 2x_i \geq x_j^2,$$

for a squaring gate. By induction in topological order, every feasible point dominates the gate values, because addition is monotone and squaring is monotone on positive numbers; the gate values themselves are feasible. Hence the minimum of the output variable is the circuit value, and raising each variable above its bound in topological order gives a strictly feasible point. The cone is a product of half-lines and two-by-two positive semidefinite cones, which is self-dual, so the dual program has the same form and attains the same value. Now require a dual feasible point for A with objective value d , a primal feasible point for B with objective value e , and $\begin{pmatrix} d-e & 1 \\ 1 & s \end{pmatrix} \succeq 0$. Weak duality gives $d \leq A$ and $e \geq B$, and the last constraint is satisfiable exactly when $d > e$; attainment gives the converse. The system is therefore feasible exactly when $A > B$. Finally, $\begin{pmatrix} a & b \\ b & c \end{pmatrix} \succeq 0$ if and only if $\|(2b, a - c)\| \leq a + c$, so the system is a second-order cone program of polynomial size. This consequence of [113] is recorded for comparison only; the lower bounds of this paper concern a much smaller class.

2.6 Certificates of nonnegativity

Definition 2.19 (Gram matrices and sums of squares). Let f have degree at most four, and let $z(x)$ be the vector of the $\binom{n+2}{2}$ monomials of degree at most two, with the constant first. A *polynomial Gram matrix* of f is a real symmetric matrix Q with $f = z^\top Q z$. Let $E \subseteq \mathbb{R}$ be a field. The polynomial f is a *sum of squares (SOS)* over E if $f = \sum_i g_i^2$ with $g_i \in E[x]$, and it has a *positive semidefinite Gram matrix* over E if some polynomial Gram matrix with entries in E is positive semidefinite. A positive definite polynomial Gram matrix is called *interior*.

An SOS over E gives a positive semidefinite Gram matrix over E ; the converse holds for $E = \mathbb{Q}$ by Lemma 2.20, but not for every real field [25], so field statements for the two kinds of certificate need separate proofs. In an SOS representation of a quartic, every g_i has degree at most two, because the highest-degree squares cannot cancel. If $f(p) = 0$, every positive semidefinite Gram matrix Q satisfies $Qz(p) = 0$, so f has no interior Gram matrix; every square factor then vanishes at p .

Lemma 2.20 (Rational squares). *Every positive rational u/v , with positive integers u and v , is a sum of at most $2\lceil \log_2(uv + 1) \rceil$ squares of rationals, computable in time polynomial in its bit length. Consequently, a rational positive semidefinite Gram matrix of f yields, in polynomial time, rational polynomials g_i with $f = \sum_i g_i^2$ and total bit length polynomial in that of the matrix.*

Proof. Write $u/v = uv/v^2$ and expand $uv = \sum_j 2^{e_j}$ in binary. Each term $2^{2h}/v^2$ is the square $(2^h/v)^2$, and each term $2^{2h+1}/v^2$ is the sum of two copies of that square. For the Gram matrix, rational symmetric Gaussian elimination without square roots writes $Q = \sum_i d_i \ell_i \ell_i^\top$ with rationals $d_i \geq 0$ and rational vectors ℓ_i ; when a pivot vanishes, positive semidefiniteness forces its row and column to vanish, and the row is skipped. Then $f = \sum_i d_i (\ell_i^\top z)^2$, and each d_i is expanded as above. The entries of the factors have polynomial bit length, as for fraction-free elimination. $\square$

Lemma 2.20 avoids integer factorization and four-square algorithms; it may use more rational squares than the rank of Q , so it says nothing about the least number of rational squares.

Definition 2.21 (Other certificate formats).

- (a) A polynomial f is a *sum of squares of rational functions with common denominator* $d \in \mathbb{Q}[x] \setminus \{0\}$ if $d^2 f = \sum_i g_i^2$ with $g_i \in \mathbb{Q}[x]$. A *multiplier certificate* is an identity $mf = \sum_i g_i^2$ with m a polynomial that is positive on $\mathbb{R}^n$; a *radial multiplier* is $m = (1 + \|x\|^2)^N$. The degree of a multiplier and the degree of a common denominator are different quantities.

- (b) For $f(x) + g(y)$ in disjoint blocks of variables, a *block-separated certificate* over E consists of a constant $c \in E$ and certificates over E for $f + c$ and for $g - c$, each with its own monomial vector, including its own constant monomial.
- (c) The *order-two moment relaxation* of a quartic f minimizes $\sum_{\alpha} f_{\alpha} y_{\alpha}$ over sequences $(y_{\alpha})_{|\alpha| \leq 4}$ with $y_0 = 1$ and moment matrix $M_2(y) = (y_{\alpha+\beta})_{|\alpha|, |\beta| \leq 2} \succeq 0$. Its dual is the problem of maximizing γ such that $f - \gamma$ has a positive semidefinite polynomial Gram matrix. When f_* is attained in the dual, the positive semidefinite polynomial Gram matrices of $f - f_*$ are the *optimal Gram matrices*. A nonzero positive semidefinite matrix Z with $\langle Z, Q \rangle = 0$ for every real optimal Gram matrix Q is an *exposing matrix* for the set of optimal Gram matrices.

2.7 Size measures

For an algebraic number α , the degree is $[\mathbb{Q}(\alpha) : \mathbb{Q}]$, and the minimal polynomial is normalized to a primitive integer polynomial with positive leading coefficient. For a point p , every coordinate satisfies $[\mathbb{Q}(p_j) : \mathbb{Q}] \leq [\mathbb{Q}(p) : \mathbb{Q}]$, so a lower bound on the degree of one coordinate also bounds the degree of the joint field $\mathbb{Q}(p)$. The converse fails: the joint field can have much larger degree than every individual coordinate, so a lower bound on the joint degree need not force any single coordinate to have comparable degree. The bit length of a rational point is the sum of the bit lengths of its reduced coordinates.

For input data in a number field, Appendix L uses one common field $K = \mathbb{Q}(\alpha)$ with a specified real embedding. Its defining polynomial, an isolating interval selecting α , and the coefficient lists expressing all data in the basis $1, \alpha, \dots, \alpha^{[K:\mathbb{Q}]-1}$ are part of the input. A witness may require a larger field $K(p)$; its defining data and all coordinate lists are charged to the output size. Separate short descriptions of individual coordinates need not give a short dense description of their joint field.

Most of our lower-bound families have input length polynomial in the number of variables n , or in a family index k , and output size exponential in n or k . Such bounds are superpolynomial in L , but they are not of the form $2^{\Omega(L)}$, and we do not state them that way.

2.8 Imported results

Besides standard linear algebra, the proofs rely on the external results listed in Table 2. Each is stated with its exact hypotheses where it is used.

Table 2: External results used in the proofs, and where their precise statements appear.

Result	Used for	Stated in
Weak convex optimization by the ellipsoid method with separation oracles; exact rational linear and convex quadratic programming [49, 71]	Value approximation and starting points	Sec. 3
One-block real quantifier elimination with degree and coefficient-height bounds [12]	Separation of nonzero values	Secs. 4, 6
The rounded central-cut ellipsoid method [49]	Starting points for monotone maps	Sec. 4
Linear programming with an operation count polynomial in the encoding of the rational constraint matrix and independent of the objective and right-hand-side encoding lengths [49]	Normal-cone verification	Sec. 6

Result	Used for	Stated in
Exact separable quadratic network flow with a strongly polynomial operation count [117]; exact constrained Newton convergence [77]	Structured constrained problems	Sec. 6
Violator-space sampling; integer feasibility with separation queries at integer points; convex integer minimization in fixed dimension	Rank and integer parameters	Sec. 6
Certified isolation of real and complex roots of univariate polynomials	Singleton constructions	Sec. 7
Bézout bounds for isolated solutions; Cayley–Bacharach and residual intersection; classification of varieties of low degree; Brouwer degree	Degree bounds and number of squares	Sec. 7
Rational points of bounded height in open semialgebraic sets	Interior Gram matrices	Sec. 8
Nonnegative ternary quartic forms that are not rational sums of squares [107]	Minimal dimension of the field obstruction	Sec. 9
Lattice basis reduction [78]; two-block quantifier elimination with separate dependence on coefficient height	Core faces, margins, and exact fallback	Sec. 11
Multihomogeneous Bézout bounds; Hilbert irreducibility; algebraic-number recognition	Quadratic comparisons	App. J
Effective Łojasiewicz-type inequalities for semialgebraic functions	Exact penalty representations	App. K
Sampling over quadratic maps with degree and height bounds [48, Theorem 1.2]	Common-field witnesses and finite infima	App. L
Integer-point bounds for convex first-order sets; rational mixed-integer linear feasibility at fixed integer dimension [65, 79]	Algebraic cone feasibility	App. L

3 From values to points

Convex optimization algorithms are value algorithms at heart. The ellipsoid method and its relatives return a feasible point whose objective value is close to the optimum; they say little about where the optimizers are. Many uses in mixed-integer nonlinear optimization need more. Branching and variable fixing act on coordinates of an optimizer. A decomposition method that passes the optimizer of a subproblem to another stage needs the same optimizer at every requested accuracy; otherwise successive refinements may move between different optimal points. Exact certification needs labels, such as the set of active constraints, that no finite accuracy settles. This section asks when value accuracy can be converted into point accuracy in the ordinary bit model, and it identifies where the conversion fails.

Output contracts. We minimize f over a rational polyhedron P , with optimal value f_* and optimal set S , and use the output contracts of Section 2.4. The requested precision is q bits, and $\varepsilon = 2^{-q}$; running times are polynomial in q , not in $1/\varepsilon$. A *value enclosure* (Definition 2.12) is a feasible rational x with a rational ℓ such that $\ell \leq f_* \leq f(x) \leq \ell + \varepsilon$. A *set approximation* (Definition 2.13) is a feasible rational x with $\text{dist}(x, S) \leq \varepsilon$; approximations returned for different q may approach different optimizers. A *fixed selector* approximates one optimizer p , determined by the instance alone, at every precision; our selector is the optimizer of minimum Euclidean norm in the original coordinates. Beyond these numerical outputs are exact predicates, such as whether a

coordinate of the optimizer equals its bound (Definition 2.14), exact representations, such as the minimal polynomial of an optimizer coordinate (Definition 2.15), and certificates, such as a rational box that encloses the optimizer (Proposition 3.15).

A fixed selector gives set approximations. A set approximation x gives the value enclosure $[f(x) - \Lambda\varepsilon, f(x)]$ whenever Λ bounds $\|\nabla f\|$ on a convex region containing x and its nearest optimizer. Exact predicates and representations are formats that no amount of numerical precision supplies by itself; exact comparisons are the subject of Section 3.4 and of Theorem 4.4. Polynomials are explicit: each nonzero coefficient is listed with its exponent vector. Convexity is a promise in the sense of Section 2.3.

Results. Lemma 3.3 reduces the fixed selector to an *effective error modulus*: a bound $\text{dist}(x, S) \leq \Gamma(f(x) - f_*)^{1/d}$ whose exponent and constant are known and whose constant Γ has polynomially many bits. With such a modulus and a bound R on the norm of the selected optimizer, a value algorithm becomes a fixed-selector algorithm at the price of $O(d(1 + q + \log \Gamma + \log R))$ value bits. The section determines when an effective modulus exists.

- For an explicit sparse polynomial convex on all of $\mathbb{R}^n$ and an arbitrary rational polyhedron, Theorem 3.4 decides emptiness and unboundedness, proves attainment, computes a radius for the minimum-norm optimizer and an effective modulus with exponent $1/D$, and approximates that optimizer to 2^{-q} in time polynomial in L , the numerical degree D , and q .
- A polynomial of degree three that is convex only on a bounded polytope also has an effective modulus, with exponent $1/4$ (Theorem 3.7).
- For quartics convex only on a box the conversion fails. Unconditionally, for an explicit family, converting values into points through any error modulus needs a number of value bits exponential in the dimension, and isotropic regularization needs a parameter whose bit length is exponential in the dimension (Proposition 3.13). Conditionally, one constant-accuracy point decides Square Root Sum or PosSLP (Theorem 3.12).
- Even where points are easy, exact labels (Proposition 3.10), the minimum-norm selector among several optimizers (Proposition 3.11), expanded algebraic output (Proposition 3.14), and rational box certificates (Proposition 3.15) can be substantially harder than a point. Table 3 summarizes the picture.

Relation to prior work. Weak convex optimization in the bit model is classical [49], and Lemma 3.1 applies it with an exact feasibility repair. For polynomials convex on $\mathbb{R}^n$ and rational polyhedra, Slot et al. [111, Theorem 1.1 and Corollary 1.2] proved that the infimum is either $-\infty$ or attained at a point whose norm has a polynomially bounded logarithm, and that unboundedness can be detected and the optimal value approximated in polynomial time. Their structure theory separates the directions in which f is invariant from those in which it changes linearly, reading them from the coefficients of the gradient, and it bounds the remaining part of f below by a strongly convex quadratic; the underlying tangent quadratic bound appears in the proof of Lemma 5 of Ahmadi et al. [5]. We use the same structural facts. We rederive attainment and the radius because the point theorem needs explicit rational constants for sparse inputs of variable degree, and because transformed polynomials must be evaluated rather than expanded (Remark 3.6). Hölder error bounds for convex polynomials are classical: Li [81, Theorem 4.2 and Corollary 4.1] and Li [82, Theorem 1] prove global bounds with exponent $1/((D-1)^n + 1)$ and a constant whose existence is shown; Yang [119] also studies error bounds for convex polynomials, and we make no claim about the

exponents or constants obtained there. We do not claim the exponent $1/D$ as new. The contribution of Theorem 3.4 is a constant computed from the input with polynomially many bits, valid on the whole, possibly unbounded, polyhedron, together with its use to approximate one fixed optimizer at every precision. For degree two, Kozlov et al. [71] compute exact rational optimizers, a stronger output. The distinction between approximating the residual of a solution and approximating the solution itself, and the hardness of the latter for equilibria, is due to Etessami and Yannakakis [38]; Theorem 3.12 exhibits the same phenomenon for minimizers of convex polynomials.

3.1 Three tools

The first tool turns explicit convex objectives into value enclosures. It assumes only convexity on the feasible set and exact evaluation of values and subgradients.

Lemma 3.1 (Convex value interface). *Let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ with rational A, b , and let $\varrho \geq 1$ be a rational number with $P \subseteq [-\varrho, \varrho]^n$.*

- (a) *Let $\varphi : P \rightarrow \mathbb{R}$ be continuous and convex. Suppose that rationals $W \geq \max\{1, \sup_P |\varphi|\}$ and $G \geq 1$ are given, together with an algorithm that, for every rational $x \in P$, returns the rational value $\varphi(x)$ and a rational vector $s(x)$ with $\|s(x)\| \leq G$ and $\varphi(x') \geq \varphi(x) + s(x)^\top(x' - x)$ for all $x' \in P$, in time bounded by a fixed polynomial in $\Delta + \langle x \rangle$; here Δ is the encoding length of A, b, ϱ, W, G and of the description of φ . Then there is an algorithm that, given a rational $\eta > 0$, either reports $P = \emptyset$ or returns a rational $\hat{x} \in P$ and a rational ℓ with*

$$\ell \leq \min_P \varphi \leq \varphi(\hat{x}) \leq \ell + \eta.$$

Its running time is bounded by a polynomial in $\Delta + \langle \eta \rangle$ that depends only on the polynomial bounding the evaluation time.

- (b) *In particular, let $\varphi(x) = f(Tx + t) + \tau \|x\|^2$, where f is an explicit sparse polynomial of numerical degree at most D , the matrix T and vector t are rational, $\tau \geq 0$ is rational, and φ is convex on P . Then the conclusion of (a) holds in time polynomial in D and the encoding lengths of $A, b, \varrho, f, T, t, \tau, \eta$. The composition $f(Tx + t)$ is evaluated at query points and never expanded.*

The proof in Appendix C.1 reduces P to its affine hull by linear programming, places a rational ball inside the reduced polytope, applies the weak optimization theorem of Grötschel et al. [49, Corollary (4.2.7)] to a capped epigraph, and repairs the approximately feasible output by an explicit homothety toward the center of the ball. The returned point is exactly feasible, also when P is lower dimensional. For part (b), Lemma C.1 supplies exact values and gradients, and the bounds W and G , by evaluating the sparse polynomial; this matters when the degree grows, because expanded compositions can have superpolynomially many monomials (Remark 3.6). For polynomials convex on $\mathbb{R}^n$ and unbounded polyhedra, value approximation is due to Slot et al. [111]; Theorem 3.4(c) recovers it through Lemma 3.1 after a radius computation.

The second tool is a Hoffman bound with an explicit constant [57]. Its right-hand sides may be irrational. This matters because the optimal set of a convex problem is a slice $\{x \in P : Nx = Np\}$ with a known rational matrix N but an unknown, possibly irrational, right-hand side Np .

Lemma 3.2 (Hoffman bound with explicit constant). *Let $A \in \mathbb{R}^{k \times s}$, $T \in \mathbb{R}^{l \times s}$, $b \in \mathbb{R}^k$, and $t \in \mathbb{R}^l$, and let $Q = \{x : Ax \leq b\}$ and $Z = \{x \in Q : Tx = t\}$ with $Z \neq \emptyset$. Let $C \geq 1$ bound the absolute values of the entries of A and T , and let $\mu \in (0, 1]$ be such that every linearly independent family*

of rows of A and T has a maximal square submatrix with determinant of absolute value at least μ . Then

$$\text{dist}(x, Z) \leq \frac{(sC)^{s-1}}{\mu} \|Tx - t\| \quad (x \in Q).$$

For integer matrices A and T the determinant condition holds with $\mu = 1$.

The third tool converts an effective error modulus into a fixed selector. It is a quantitative form of Tikhonov regularization, which is known to select the minimum-norm optimizer in the limit; the schedule below makes the rate explicit and charges only the bit length of the parameter.

Lemma 3.3 (Fixed minimum-norm selector). *Let $P \subseteq \mathbb{R}^n$ be nonempty, closed, and convex, and let $\varphi : P \rightarrow \mathbb{R}$ be convex and continuous with nonempty optimal set S and optimal value φ_* . Let p be the point of S of minimum norm. Suppose that $d \geq 2$, $\Gamma \geq 1$, and $R \geq 1$ satisfy $\|p\| \leq R$ and*

$$\text{dist}(x, S) \leq \Gamma(\varphi(x) - \varphi_*)^{1/d} \quad (x \in P, \varphi(x) - \varphi_* \leq 1).$$

For $\varepsilon \in (0, 1]$ let $\tau = \varepsilon^{2d-2}/(4 \cdot 8^{d-1} R^d \Gamma^d)$. Then $\Phi_\tau = \varphi + \tau \|\cdot\|^2$ has a unique minimizer x_τ on P , with $\|x_\tau\| \leq \|p\|$ and $\|x_\tau - p\| \leq \varepsilon/2$, and every $x \in P$ with $\Phi_\tau(x) - \Phi_\tau(x_\tau) \leq \tau \varepsilon^2/4$ satisfies $\|x - p\| \leq \varepsilon$. If every point of S has norm at most R , the factor 4 in τ may be replaced by 2.

The optimal set S need not be bounded; the projection of x_τ onto S may lie far outside the ball of radius R , and the factor 4 accounts for it. Since $\log_2(1/\tau) = (2d-2)q + O(d) + d \log_2 R + d \log_2 \Gamma$, the selector costs polynomially many value bits exactly when $\log \Gamma$ and $\log R$ are polynomial. The remaining question in this section is when they are.

3.2 Globally convex sparse polynomials

Theorem 3.4 (Global point oracle). *Let $f \in \mathbb{Q}[x_1, \dots, x_n]$ be an explicit sparse polynomial that is convex on $\mathbb{R}^n$, and let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ with rational A and b . Let L be the encoding length of (f, A, b) and $D = \max\{2, \deg f\}$. There is a deterministic algorithm with the following properties. All running times are bounded by polynomials, with absolute exponents, in the indicated quantities, uniformly in D .*

- (a) *In time polynomial in L, D it decides whether P is empty and, if not, whether f is unbounded below on P . In the unbounded case it returns rational $a \in P$ and d with $Ad \leq 0$ and $f(a + td) = f(a) - t$ for all $t \geq 0$.*
- (b) *Otherwise f attains its minimum f_* on P , and the optimal set S has a unique point p of minimum Euclidean norm. In time polynomial in L, D the algorithm computes rationals $R \geq 1$ and $\Gamma \geq 1$, with $\log R + \log \Gamma$ bounded by a polynomial in L, D , such that $\|p\| \leq R$ and*

$$\text{dist}(x, S) \leq \Gamma(f(x) - f_*)^{1/D} \quad (x \in P, f(x) - f_* \leq 1).$$

- (c) *For every integer $q \geq 0$ it computes, in time polynomial in L, D, q , a rational $x_q \in P$ with $\|x_q - p\| \leq 2^{-q}$. For every further integer $q_f \geq 0$ it can return, in time polynomial in $L, D, q + q_f$, also a rational ℓ with $\ell \leq f_* \leq f(x_q) \leq \ell + 2^{-q_f}$.*

The bounds are polynomial in the numerical degree D . They are polynomial in the input length when exponents are written in unary, or more generally when D is polynomially bounded in L . The error bound holds on the whole polyhedron P , which may be unbounded, and not only on a bounded part of it. The selector p depends only on the instance, so the outputs for different q approximate the same point. The decision in (a), attainment, and a norm bound of polynomial bit length for some optimizer were proved by Slot et al. [111] for inputs with unary degrees; we rederive them with explicit constants. The effective modulus and the fixed selector are the new outputs.

Proof outline. The complete proof is in Appendix C.2. It has four steps.

Step 1: a computable quadratic minorant. For every a, x , univariate interpolation of f along the line through a and x gives

$$f(x) \geq f(a) + \nabla f(a)^\top (x - a) + \frac{(x - a)^\top \nabla^2 f(a) (x - a)}{\kappa_D}, \quad \kappa_D = (D + 1)D^{D+2}.$$

Averaging over a in the unit cube yields $f(x) \geq m_0 + g^\top x + x^\top Mx / \kappa_D$, where $M = \int \nabla^2 f(a) da$ and the other coefficients are rational and computed monomial by monomial. The kernel of M is the common kernel of all Hessians. With Π the projection onto the range of M and $w = -(I - \Pi)\nabla f(0)$,

$$f(x) = f(\Pi x) - w^\top x \quad (x \in \mathbb{R}^n),$$

so f is a function of Πx plus a linear term in the remaining directions.

Step 2: boundedness and a radius. One linear system, $w = A^\top \lambda + Mz$ with $\lambda \geq 0$, decides boundedness. If it is infeasible, Farkas' lemma gives a recession direction of P along which f decreases linearly. If it is feasible, $f(x) \geq \beta_0 - \nu \|\Pi x\| + \mu_0 \|\Pi x\|^2$ on P for explicit rationals β_0 , $\nu \geq 0$, and $\mu_0 > 0$. On a sublevel set this bounds $\|\Pi x\|$, and together with convexity at the origin it bounds $w^\top x$ on both sides; the constraints $Ax \leq b$ alone give only one side. Points with equal values of Mx and $w^\top x$ have equal objective values, so Lemma 3.2 replaces every point of the sublevel set by an equally good feasible point of norm less than an explicit R . This proves attainment and $\|p\| < R$ without assuming either.

Step 3: an effective modulus. Let $x \in P$ have gap $E \leq 1$ and let $h(u) = f(p+u) - f(p) - \nabla f(p)^\top u$, which is convex and nonnegative on $\mathbb{R}^n$. On the segment from p to x we have $0 \leq h \leq E$, and interpolation extrapolates this bound along the whole line. For a point c of the unit cube, convexity of h at the midpoint of $2(c - p)$ and $2t(x - p)$ bounds $h(c - p + t(x - p))$ for $0 \leq t \leq E^{-1/D}$; differentiating at $t = 0$ gives

$$|\nabla f(c)^\top (x - p)| \leq C_* E^{1/D} \quad (c \in [0, 1]^n),$$

with an explicit C_* that depends on $\|p\|$ but not on x . The polynomial $z \mapsto \nabla f(z)^\top (x - p)$ has coefficient vector $N(x - p)$, where the rows of the rational matrix N are read from the monomials of ∇f . A tensor interpolation inequality bounds these coefficients by $\beta_{D-1}^n C_* E^{1/D}$, where $\beta_k = (k+1)2^k k^k$; the interpolation grid appears only in the proof. At zero gap this shows $S = \{x \in P : Nx = Np\}$, and Lemma 3.2 gives Γ .

Step 4: selection. Lemma 3.3 with $d = D$ gives a parameter τ with polynomially many bits. The regularized minimizer has norm at most $\|p\| < R$, so it is also the minimizer over the bounded polytope $P \cap [-R, R]^n$, to which Lemma 3.1 applies.

Corollary 3.5 (Bounded polytopes). *Suppose in addition that P is bounded and a rational $\varrho \geq 1$ with $P \subseteq [-\varrho, \varrho]^n$ is given. Then parts (b) and (c) of Theorem 3.4 hold with $R = \max\{1, n\varrho\}$ and the constant Γ of Step 3, without Steps 1 and 2, and the selector may use $\tau = \varepsilon^{2D-2} / (2 \cdot 8^{D-1} R^D \Gamma^D)$. In particular, for every fixed degree bound, the minimum-norm optimizer of a globally convex polynomial on a bounded rational polytope can be approximated to 2^{-q} in time polynomial in $L + q$.*

Remark 3.6 (Representation and degree). The algorithms evaluate f at rational points, also after affine changes of coordinates, and never expand $f(Tx + t)$. Expansion can be exponentially long in the dimension even when the degree is written in unary: for $n = 2r \geq 4$, the polynomial $f(y) = \sum_i y_i^{2r}$, convex on $\mathbb{R}^n$, has n monomials, whereas $f(Uy)$ for the rational orthogonal matrix $U = I - \frac{2}{n}\mathbf{1}\mathbf{1}^\top$ has at least 2^r monomials (Appendix C.2). The theorem is polynomial in the

numerical degree D . With exponents written in binary, D can be exponential in L , and we claim no bound polynomial in $\log D$. The theorem also does not apply to polynomials given by arithmetic circuits: the matrix N of Step 3 is read from the monomial list.

The exponent $1/D$ cannot be improved in general. A polynomial of odd degree at least three is not convex on $\mathbb{R}^n$, because its restriction to a suitable line has a second derivative of odd degree, which takes negative values. For even D , $f(t) = t^D$ on $P = \mathbb{R}$ has $\text{dist}(t, S) = (f(t) - f_*)^{1/D}$.

3.3 Cubics convex on a polytope

A polynomial of degree at most three that is convex on $\mathbb{R}^n$ is quadratic (Lemma 2.11), and for such polynomials exact convex quadratic programming already returns rational optimizers [71]. Genuine cubics are convex only on part of space, and convexity on the feasible polytope is the natural promise; for cubics on boxes it is strongly NP-hard to recognize [2, Theorem 2.3]. The next theorem shows that this weaker promise still yields an effective modulus.

Theorem 3.7 (Cubics convex on a polytope). *Let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ be a nonempty bounded rational polytope and let $f \in \mathbb{Q}[x_1, \dots, x_n]$ be an explicit polynomial of degree at most three that is convex on P . Let L be the encoding length of (f, A, b) and $S = \arg \min_P f$. In time polynomial in L one can compute a rational $\Gamma_P \geq 1$ with $\log \Gamma_P$ polynomial in L such that*

$$\text{dist}(x, S) \leq \Gamma_P (f(x) - f_*)^{1/4} \quad (x \in P, f(x) - f_* \leq 1).$$

For every integer $q \geq 0$, in time polynomial in $L + q$ one can compute a rational $x \in P$ with $\text{dist}(x, S) \leq 2^{-q}$, and also a rational $x_q \in P$ with $\|x_q - p\| \leq 2^{-q}$, where p is the point of S of minimum Euclidean norm. No uniqueness of the optimizer, strong convexity, or interiority of the optimizer is assumed, and P may be lower dimensional.

Two facts specific to degree three drive the proof. The Hessian $H(u) = \nabla^2 \varphi(u)$ of a cubic φ is affine in u , and the third derivative is a constant symmetric tensor. The first fact lets an interior reference point control every Hessian on the polytope. The second links the curvature at that point to the curvature at an optimizer and a feasible point. The recourse analysis reuses the following two lemmas with a box in place of the polytope.

Lemma 3.8 (Cubic transverse estimate). *Let φ be a real polynomial of degree at most three on $\mathbb{R}^m$, let $\Omega \subseteq \mathbb{R}^m$ be convex with $\nabla^2 \varphi \succeq 0$ on Ω , let v minimize φ over Ω , and let $u \in \Omega$. Put $d = u - v$ and $E = \varphi(u) - \varphi(v)$. For every $c \in \mathbb{R}^m$ and all Λ, ρ_1, ρ_2 with $\Lambda \geq \max\{\|\nabla^2 \varphi(u)\|, \|\nabla^2 \varphi(v)\|\}$, $\rho_1 \geq \|c - v\|$, and $\rho_2 \geq \|d\|$,*

$$(d^\top \nabla^2 \varphi(c) d)^2 \leq 12\Lambda(\rho_1 + \rho_2)^2 E.$$

Lemma 3.9 (Cubic optimal slice). *Let φ and Ω be as in Lemma 3.8, let $c \in \Omega$, and suppose that $K = \ker \nabla^2 \varphi(c) \subseteq \ker \nabla^2 \varphi(u)$ for every $u \in \Omega$. Let $g_c = \nabla \varphi(c)$ and let Π_K be the orthogonal projection onto $K^\perp$. If v minimizes φ over Ω , then*

$$\arg \min_{\Omega} \varphi = \{w \in \Omega : \nabla^2 \varphi(c)(w - v) = 0, g_c^\top(w - v) = 0\}.$$

If moreover $\Lambda \geq \sup_{\Omega} \|\nabla^2 \varphi\|$ and $R_{\Omega} \geq \sup_{u \in \Omega} \|u - c\|$ are finite, then $|g_c^\top(u - v)| \leq \varphi(u) - \varphi(v) + \Lambda R_{\Omega} \|\Pi_K(u - v)\|$ for every $u \in \Omega$.

The gradient row in Lemma 3.9 is necessary: along the common Hessian kernel a cubic can still change linearly.

Proof outline. The proof is in Appendix C.3. Linear programming finds the affine hull of P , a rational parametrization $x = x_0 + Vu$, and a ball of radius r_0 inside the reduced polytope P' , which lies in a ball of radius R_0 about the same center. This reduction must come first: convexity on a lower-dimensional polytope does not make the ambient Hessian positive semidefinite. Reflection through the center and affinity of the Hessian give $0 \preceq H(u) \preceq (1 + R_0/r_0)H_c$ on P' , where H_c is the Hessian at the center. Hence $\ker H_c$ is the common kernel, it is rational, and individual boundary Hessians may only have larger kernels. Lemma 3.8 gives $E \geq (d^\top H_c d)^2 / (108\Lambda R_0^2)$, which bounds the displacement transverse to $\ker H_c$ by a multiple of $E^{1/4}$ once the least positive eigenvalue of H_c is bounded below by its principal minors. Lemma 3.9 describes S as a slice with the rational matrix $(\frac{H_c}{g_c})$, and Lemma 3.2 gives Γ_P . Lemmas 3.3 and 3.1 then give both outputs. The regularizer is the original norm $\|x\|^2$, not the norm of the reduced coordinates u , which would select a different optimizer in general.

The exponent $1/4$ suffices for polynomial dependence on q . The cubic t^3 on $[0, 1]$ has $\text{dist}(t, S) = (f(t) - f_*)^{1/3}$, so no exponent larger than $1/3$ is possible; whether $1/3$ is attainable with polynomial-bit constants in all dimensions is open.

3.4 Where the conversion fails

The positive results use global convexity or degree three. The constructions below show that neither hypothesis can be dropped for quartics, and that outputs beyond a set approximation can be hard even when set approximations are easy. They use two standard arithmetic decision problems. *Square Root Sum* asks, for positive integers $a_1, \dots, a_n$ and an integer B , whether $\sum_i \sqrt{a_i} \leq B$. *PosSLP* asks whether the integer computed by a straight-line program with constants $0, 1$ and operations $+, -, \times$ is positive. Allender et al. [6] show that Square Root Sum is decidable in polynomial time with a PosSLP oracle and that both problems lie in the counting hierarchy; by Theorem 4.10(b), that oracle algorithm can be replaced by a polynomial-time many-one reduction to one PosSLP instance. Whether either problem lies in P is open. The statements below are reductions and conditional consequences, not NP-hardness results. Their structural bounds refer to the *interaction graph* of a polynomial, whose vertices are the variables, two variables being adjacent when some monomial with a nonzero coefficient contains both.

Proposition 3.10 (Exact active bounds for strongly convex cubics). *There is a polynomial-time algorithm that maps every Square Root Sum instance either to its answer or to an explicit rational cubic F on a rational box X with a distinguished coordinate $\theta \in [0, 1]$ such that*

- (i) $\nabla^2 F \succeq \frac{1}{16}I$ on X , so F has a unique minimizer z^* ;
- (ii) $\theta^* = 0$ if and only if $\sum_i \sqrt{a_i} \leq B$, and otherwise $0 < \theta^* < 1$;
- (iii) the box widths and the absolute values of the coefficients of F are bounded by an absolute constant, the diagonal entries of $\nabla^2 F$ are at most 5 on X , and the interaction graph of F has treewidth at most two.

A point within 2^{-q} of z^* is computable in time polynomial in $L + q$.

Thus exact active-bound recognition for uniformly strongly convex cubics with bounded data and treewidth two is at least as hard as Square Root Sum, including the equality case, although Theorem 3.7 and even plain strong convexity give points. The base of the construction is a complete binary averaging tree whose cubic leaves $u_i^3/3 - c_i u_i$ select normalized square roots and whose squared residuals propagate their average to the root; a sign test on the root then activates or deactivates the distinguished coordinate.

Proposition 3.11 (Selection versus some optimizer). *There is a polynomial-time algorithm that maps every Square Root Sum instance either to its answer or to an explicit rational quartic F' on a rational box, convex on that box, with the same treewidth and coefficient bounds, such that its optimal set is a segment $\{z^*\} \times [0, 1]$ if $\sum_i \sqrt{a_i} \leq B$ and the single point $(z^*, 1)$ otherwise. The point $(z^*, 1)$ is optimal in both cases and can be approximated to 2^{-q} in time polynomial in $L + q$, whereas the last coordinate of the minimum-norm optimizer is 0 in the first case and 1 in the second.*

A set approximation is therefore easy for this family, while the minimum-norm selector at accuracy $1/4$, or the dimension of the optimal set, decides Square Root Sum. Requiring minimum norm strengthens the output contract. The next theorem shows that for quartics convex only on a box even a set approximation at constant accuracy can carry the arithmetic difficulty.

Theorem 3.12 (Constant-accuracy points for box-convex quartics).

- (a) *There is a polynomial-time algorithm that maps every Square Root Sum instance either to its answer or to an explicit rational quartic F on $[0, 1]^N$, convex on $[0, 1]^N$, with a unique minimizer z^* whose designated coordinate is 1 if $\sum_i \sqrt{a_i} > B$ and 0 if $\sum_i \sqrt{a_i} < B$; equality does not occur in the produced instances. The interaction graph of F has treewidth at most two, every coefficient other than the constant term has absolute value bounded by an absolute constant, and the diagonal Hessian entries are at most 20 on $[0, 1]^N$.*
- (b) *There is a polynomial-time algorithm that maps every straight-line program, with output A , to an explicit rational quartic F on $[0, 1]^N$, convex on $[0, 1]^N$, with a unique minimizer whose designated coordinate is 1 if $A > 0$ and 0 if $A \leq 0$. All coefficients of F have absolute value bounded by an absolute constant.*
- (c) *Consequently, if some deterministic algorithm returns, for every explicit rational quartic convex on a rational box with a unique minimizer, a point within Euclidean distance $1/4$ of the minimizer in time polynomial in the input length, then Square Root Sum and PosSLP lie in P; for Square Root Sum the instances of (a) suffice. If the algorithm is always correct and runs in expected polynomial time, both problems have Las Vegas algorithms with expected polynomial running time.*

Both constructions place two independent strongly convex sign tests side by side, one activated when the arithmetic quantity is positive and the other when it is negative, and couple their activation amplitudes $\theta_{\pm}$ to a new coordinate $y \in [0, 1]$ by $\frac{1}{192}[\theta_+^2(y-1)^2 + \theta_-^2 y^2]$. The coupling is not convex by itself, but the curvature of the sign tests absorbs its negative part, so the whole objective is convex on the box. The coupling vanishes at the right endpoint and forces every minimizer to it, however small the active amplitude is. For Square Root Sum, an equality $\sum_i \sqrt{a_i} = B$ forces every radicand to be a perfect square, by a trace argument in the multiquadratic field (Lemma C.14), so equality is decided by integer square roots. For PosSLP, gate values are encoded as bounded pairs (u, v) with u/v equal to the integer value, the gate equations are squared with geometric weights that make the objective strongly convex on its box, and the output signal $(2u - v)/4$ has the sign of $2A - 1 \neq 0$. The final instances are convex on their boxes but not on $\mathbb{R}^N$, and, before any rescaling, their curvature in the direction of y at the minimizer is $\frac{1}{96}((\theta_+^*)^2 + (\theta_-^*)^2)$, which can be extremely small. So the theorem is consistent with Theorem 3.4, with Theorem 3.7, with polynomial-time value enclosures from Lemma 3.1, and with algorithms whose running time depends on a supplied growth constant. Etessami and Yannakakis [38] proved PosSLP-hardness of strong approximation of equilibrium coordinates, and Tarasov and Vyalı [113] reduced arithmetic circuit comparison to exact semidefinite feasibility; Theorem 3.12 transfers the first phenomenon to minimizers of polynomials convex on a box.

The last result of this subsection is unconditional, but it concerns the value-to-point conversion rather than every algorithm.

Proposition 3.13 (Value tolerance and isotropic regularization). *For $n \geq 1$ let*

$$F_n(x, y) = 8 \sum_{i=1}^n x_i^2 - \frac{x_1}{4} - \sum_{i=2}^n x_{i-1}^2 x_i + x_n^2 (y - 1)^2 \quad \text{on } [0, 1]^{n+1}.$$

- (a) F_n is a quartic with $O(n)$ monomials and constant-size rational coefficients, convex on $[0, 1]^{n+1}$, with the unique minimizer $z^* = (x^*, 1)$, where $0 < x_n^* \leq 2^{-2^{n+1}}$.
- (b) The feasible point $(x^*, 0)$ has objective gap at most $2^{-2^{n+2}}$ and distance one from z^* . Hence every error bound $\text{dist}(z, S) \leq \Gamma(F_n(z) - F_{n,*})^\vartheta$ valid for gaps at most one has $\log_2 \Gamma \geq \vartheta 2^{n+2}$.
- (c) For $\lambda > 0$ let z_λ minimize $F_n + \lambda \|z\|^2$ on $[0, 1]^{n+1}$. If $\|z_\lambda - z^*\| \leq \frac{1}{2}$, then $\lambda \leq 2^{-2^{n+2}}$, so the denominator of a rational λ has more than 2^{n+2} bits.
- (d) Nevertheless $F_n(z_\lambda) - F_{n,*} \leq (n+1)\lambda$ for every $\lambda > 0$, and a point within 2^{-q} of z^* is computable in time polynomial in $n+q$.

With every exponent vector written in full (Definition 2.2), the input has length $L = \Theta(n^2)$. The quantities bounded below in (b) and (c), $(\log_2 \Gamma)/\vartheta$ and the bit length of λ , are exponential in n , hence of the form $2^{\Omega(\sqrt{L})}$: superpolynomial, but not exponential, in L . Part (b) says that a value tolerance with fewer than 2^{n+2} bits does not certify any point accuracy below one. Every error modulus of this family has $(\log_2 \Gamma)/\vartheta \geq 2^{n+2}$, so the value precision $\vartheta^{-1}(q + \log_2 \Gamma)$ needed to convert values into points through it is exponential in n whatever the exponent; for exponents bounded below by an inverse polynomial, the constant itself has a number of bits exponential in n . Theorem 3.7 therefore does not extend to quartics convex on a box, and the obstruction is in the modulus itself, not in its proof. Part (c) shows that plain Tikhonov selection, which works in Lemma 3.3 because Γ is short, needs a parameter whose bit length is exponential in n here. Part (d) shows that this family is nonetheless easy for both values and points: setting $y = 1$ leaves a uniformly strongly convex cubic. The difficulty is created by the method, not by the instance.

3.5 Representations and certificates

A procedure that returns rational approximations at every precision describes an optimizer compactly, but it is not an exact representation. The next two results show that the usual exact formats can have length exponential in the dimension, hence superpolynomial in the input length, even for strongly convex, uniformly conditioned instances whose interaction graph is a path (or has treewidth two, for the multiplier rectangles), where points are cheap.

Proposition 3.14 (Expanded algebraic output on a conditioned path). *For $n \geq 2$ let $\Psi_n(x) = \sum_{i=1}^{n-1} (x_i^2 - x_{i+1} - 2)^2 + (x_n - 1)^2$ on $[1, 2]^n$.*

- (a) Ψ_n has the unique minimizer a with $a_n = 1$ and $a_i = \sqrt{a_{i+1} + 2}$. On $[1, 2]^n$ it satisfies $\Psi_n(x) \geq \frac{1}{4} \|x - a\|^2$ and $\partial_{ii} \Psi_n \leq 38$. Its interaction graph is a path, and it has $O(n)$ monomials with constant-size coefficients.
- (b) The minimal polynomial of a_1 over $\mathbb{Q}$ has degree 2^{n-1} and exactly $2^{n-2} + 1$ nonzero coefficients.
- (c) In time $O(n)$ one can compute a box $B = \prod_i [\ell_i, u_i] \subseteq [1, 2]^n$ with dyadic endpoints of $O(1)$ bits that contains a and on which $\nabla^2 \Psi_n \succeq \frac{1}{32} I$.

On the box B , Ψ_n is a strongly convex quartic with a uniformly conditioned path structure, and Lemma 3.1 with strong convexity gives points within 2^{-q} of a in time polynomial in $n + q$. The recurrence $a_i = \sqrt{a_{i+1} + 2}$ describes the whole optimizer with $n - 1$ additions and $n - 1$ square-root gates. Yet a dense defining polynomial of a_1 has more than 2^{n-1} coefficients, and the minimal polynomial written as a sparse list has $2^{n-2} + 1$ terms, while the input, with every exponent vector written in full, has length $\Theta(n^2)$. The support bound concerns the minimal polynomial; it does not exclude a sparser non-minimal multiple. Theorem 7.4 and the related results on exact optimizers treat other formats.

Proposition 3.15 (Rectangular certificates). *For $n \geq 1$ let $X = [0, \frac{1}{2}]^{n+1}$, let*

$$\Xi_n(x) = \left(x_0 - \frac{1}{4}\right)^2 + \sum_{i=1}^n \left(x_i - \frac{x_{i-1}^2}{4}\right)^2,$$

and let $m = 4 \cdot 2^n - 2$.

- (a) Ξ_n has the unique minimizer a with $a_i = 2^{-(4 \cdot 2^i - 2)}$, which lies in the interior of X . On X , $\nabla^2 \Xi_n \succeq \frac{5}{8}I$, $\Xi_n(x) \geq \frac{9}{16} \|x - a\|^2$, all diagonal Hessian entries are at most $\frac{35}{16}$, and the interaction graph is a path. Every rectangle with rational endpoints that contains a and lies in the open box $(0, \frac{1}{2})^{n+1}$ has an endpoint with denominator at least 2^m .
- (b) Let $\tilde{\Xi}_n(x, y) = \Xi_n(x) + y^2 + \frac{y}{4}(x_n - \frac{1}{8}x_{n-1}^2)$ on $X \times [0, \frac{1}{2}]$. It has the unique minimizer $(a, 0)$, satisfies $\nabla^2 \tilde{\Xi}_n \succeq \frac{11}{32}I$ and $\tilde{\Xi}_n \geq \frac{135}{256}(\|x - a\|^2 + y^2)$, has diagonal Hessian entries at most $\frac{35}{16}$ and treewidth two, and its active lower bound $y = 0$ has $\partial_y \tilde{\Xi}_n(a, 0) = a_n/8 > 0$. Every rectangle $B \subseteq X$ with rational endpoints that contains a and satisfies $\partial_y \tilde{\Xi}_n(x, 0) \geq 0$ for all $x \in B$ has an endpoint with denominator at least 2^m .
- (c) Both optimizers have exact descriptions of size $O(n)$: the recurrence $a_0 = \frac{1}{4}$, $a_i = \frac{1}{4}a_{i-1}^2$ as an arithmetic circuit, with optimality certified by a contraction argument on the closed box X and, in (b), by the displayed lower bound.

The two rectangle formats are natural for interval methods: a box strictly inside the feasible region that encloses the optimizer, and a box on which the sign of an active multiplier is certified uniformly. Under uniform conditioning, fixed degree, and treewidth at most two, both can require endpoints whose bit length is exponential in n , while correlated implicit certificates, which keep the residual equations, stay short. These are lower bounds for specified formats, not for exact optimization in general; in particular, boxes that touch the boundary of X and certificates of interval-Newton type that use the residual equations are not excluded.

3.6 Summary

Table 3 collects the results by output contract. Values are available in all three settings under a convexity promise. The route from values to set approximations goes through an effective error modulus. It exists under global convexity and for cubics convex on a polytope, but not in general for quartics convex on a box, where even constant-accuracy points are as hard as Square Root Sum or PosSLP. A fixed selector needs, in addition, a selection rule; the minimum-norm rule costs nothing extra when the modulus is effective (Lemma 3.3), but it can be hard where a set approximation is easy (Proposition 3.11). Exact labels and exact representations are different problems: they can be hard, or long, for strongly convex cubics and quartics whose points are easy. Exact comparisons under global strong convexity are treated in Theorem 4.4.

Output	Convex on $\mathbb{R}^n$, sparse, degree D	Cubic, convex on a polytope	Quartic, convex on a box
Value enclosure	polynomial (Theorem 3.4)	polynomial (Lemma 3.1)	polynomial (Lemma 3.1)
Effective modulus	exponent $1/D$, polynomial bits (Theorem 3.4)	exponent $1/4$, polynomial bits (Theorem 3.7)	none for some families: every modulus has $(\log \Gamma)/\vartheta \geq 2^{n+2}$ (Proposition 3.13)
Set approximation	polynomial in L, D, q	polynomial in $L + q$	decides Square Root Sum or PosSLP at accuracy $1/4$ (Theorem 3.12)
Minimum-norm selector	polynomial in L, D, q	polynomial in $L + q$	decides Square Root Sum where some optimizer is easy (Proposition 3.11)
Exact active label	an exact comparison, not decided by points (Theorem 4.4 treats the strongly convex unconstrained case)	decides Square Root Sum, even with uniform strong convexity (Proposition 3.10)	as for cubics
Expanded output	not implied by points	not implied by points	minimal polynomials and rectangle endpoints exponential in n for strongly convex instances of treewidth at most two (Propositions 3.14 and 3.15)

Table 3: Output contracts for convex polynomial minimization in the bit model. “Polynomial” means time polynomial in L, D, q in the global convex column and in L, q in the fixed-degree columns, under the stated promises. Hardness entries are reductions, not unconditional lower bounds, except for the bit counts of moduli and formats.

4 Exact comparisons with one sign test

Branch-and-bound methods for mixed-integer nonlinear programs make exact decisions. A node is pruned when its relaxation value is at least the incumbent value. Bound activity at a relaxation optimizer informs branching decisions. Two subproblems are ranked by their optimal values. Approximate values and points do not settle such decisions by themselves. For the classes studied here, a nonzero difference can be as small as $2^{-2^{\text{poly}(L)}}$, so a sign certified by approximation can require exponentially many printed bits.

For globally strongly convex explicit polynomials, this section reduces every order or equality comparison of a polynomial observable at the minimizer to the sign of one integer arithmetic circuit, that is, to one instance of PosSLP. The circuit is built in polynomial time without oracle calls. The lower bounds of Section 5 show that the order comparisons are also PosSLP-hard for certified quartics. In that class the order comparisons are therefore equivalent to PosSLP under polynomial-time many-one reductions.

The proof has four parts. An ordinary algorithm computes a starting point with polynomially many bits. Newton’s method, run as a shared rational circuit, reaches doubly exponential accuracy with polynomially many gates. A real-algebraic separation bound, used only in the analysis, determines how much accuracy is needed. Finally, the observable is shifted by half of the separation gap and division is removed without changing signs. The same method gives an upper bound for

zeros of strongly monotone maps that are not gradients, for polyhedra on which a gap between zero and nonzero optimal slacks is supplied, and, with an elementary separation bound, for the certified cube-root circuits of Section 5. The last subsection compiles an adaptive polynomial-time computation with exact sign tests into a single sign test.

4.1 Conventions

We use the input and circuit models of Section 2. The following points are used repeatedly.

- (i) *Input.* Rational numbers are written in binary. An explicit sparse polynomial is the list of its nonzero coefficients together with their exponent vectors, written in unary, and the dimension n is written in unary as well; a polynomial map $T = (T_1, \dots, T_n)$ is the list of its components. We write $L \geq 2$ for the total binary length of the input, including the curvature bound, the observable, thresholds, constraints and certificates. Thus every printed numerator and denominator has absolute value at most 2^L , every listed polynomial has at most L terms, and the dimension n and the numerical degree D are at most L . If an encoding only guarantees $D \leq \text{poly}(L)$, the results hold after replacing L by that polynomial. Sparse polynomials with binary exponents and no polynomial degree bound are not covered: evaluating them at a short rational point can already produce exponentially many bits.
- (ii) *Circuits.* An *integer circuit* is a directed acyclic graph whose sources are the constants 0 and 1 and whose other nodes are binary $+$, $-$ and $\times$ gates; its size is its number of nodes. A *rational circuit* also allows rational constants written in binary and division by nonzero values. A circuit for a vector has several output nodes, and all nodes may be shared. Circuit values can have exponentially many bits; they are never expanded. PosSLP asks whether the output of an integer circuit is positive.
- (iii) *Reductions.* A problem *reduces to one* PosSLP *instance* if a deterministic polynomial-time algorithm, making no oracle calls, maps every input to an integer circuit whose output is positive exactly when the answer is yes. For a promise problem this is required only on inputs satisfying the promise. A supplied curvature bound is a promise and is not checked. The certificate formats below are checked in polynomial time; an invalid certificate is mapped to the circuit 0, so these formats define ordinary languages.
- (iv) *Comparisons.* We write $\bowtie$ for one of the relations $>, \geq, <, \leq, =, \neq$. A rational threshold is absorbed into the observable: the value comparison $\min f \bowtie r$ uses $h = f - r$, and the coordinate comparison $p_j \bowtie r$ uses $h = X_j - r$.

4.2 Three tools

The first tool says how close to zero a nonzero algebraic quantity defined by a short real formula can be. It is used only in proofs; the reductions never perform quantifier elimination. For an integer N , its bit size is $\lceil \log_2(|N| + 1) \rceil$.

Lemma 4.1 (Algebraic separation). *There is an absolute integer constant $c_0 \geq 1$ with the following property. Let $\Phi(y, z)$ be a quantifier-free formula in real variables $y \in \mathbb{R}^s$, $s \geq 1$, and $z \in \mathbb{R}$, that is, a Boolean combination of conditions $P(y, z) > 0$, $P(y, z) = 0$ and $P(y, z) < 0$, where every polynomial $P \in \mathbb{Z}[y, z]$ that occurs has total degree at most $d \geq 2$ and coefficients of bit size at most $\tau \geq 1$. Suppose that the set $\{z \in \mathbb{R} : \exists y \in \mathbb{R}^s \Phi(y, z)\}$ consists of a single point α . Then α is algebraic, and the following hold.*

- (a) Polynomial-size formulas. If $s, d, \tau \leq L^c$ for integers $L \geq 2$ and $c \geq 1$, then $\alpha = 0$ or $|\alpha| \geq 2^{-2^e}$ with $e = c(c_0 + 2)L^{c+1}$.
- (b) Explicit form. If $\alpha \neq 0$, then $|\alpha| \geq 2^{-2\tau d^{c_0 s}}$. For fixed d , the exponent is linear in τ , with a factor that depends only on s .

The proof in Appendix A.2 applies a one-block quantifier-elimination theorem with bounds on degrees and coefficient bit sizes [12, Theorem 2.27]. Some polynomial of the eliminated formula must vanish at α , because otherwise the formula would define a neighborhood of α ; an elementary root bound then gives the estimate. The lemma does not require the defining system to have finitely many complex solutions. For instance, $f(x, y) = (x^2 + y^2)^2 + x^2 + y^2$ satisfies $\nabla^2 f \succeq 2I$ on $\mathbb{R}^2$, yet its complex critical locus contains the curve $x^2 + y^2 = -1/2$. Only the real projection matters. Form (a) is used for problems with polynomially many variables. Form (b) keeps the dependence on τ linear and the dependence on s single exponential for fixed d , which is what the parameterized algorithms of Section 6 need.

The second tool removes division from a rational circuit. The rule $(N_a/Q_a)/(N_b/Q_b) = (N_a Q_b)/(Q_a N_b)$ can produce a negative denominator, which would reverse the sign of the final numerator; detecting this needs a sign test. Squaring the divisor's numerator avoids the problem.

Lemma 4.2 (Division elimination with positive denominators). *Let C be a rational circuit with σ nodes whose rational constants have total bit length b . In time polynomial in $\sigma + b$, and without sign tests, one can construct an integer circuit of size $O(\sigma + b)$ that contains, for every node v of C , two nodes N_v, Q_v with the following property. If every division gate of C divides by a nonzero value, then $Q_v > 0$ and the value of v equals N_v/Q_v for every node v . In that case the value of v is positive if and only if $N_v > 0$, so the sign of any node of C is decided by one PosSLP instance.*

Tracking numerators and denominators is standard in PosSLP reductions [6, 39]. Division by a node with numerator N_b and denominator Q_b is implemented as

$$\frac{N_a/Q_a}{N_b/Q_b} = \frac{N_a Q_b N_b}{Q_a N_b^2}, \quad (2)$$

whose denominator is positive whenever $Q_a > 0$ and $N_b \neq 0$. The full proof is in Appendix A.3. We state the lemma once because every result in this section and several in Sections 6 and 8 use it.

The third tool is a short circuit for a very accurate approximation of the optimizer. It is uniform in the observable: one circuit serves every observable up to a stated encoding length. This uniformity is used when many observables are tested at the same point, as in the active-set certificates of Section 6 and the circuit Grams of Theorem 8.12.

Lemma 4.3 (Shared Newton point). *Let an input of length L consist of $n \geq 1$, a rational $\mu > 0$, and either*

- (a) *an explicit sparse polynomial $f \in \mathbb{Q}[X_1, \dots, X_n]$ with the promise $\nabla^2 f(x) \succeq \mu I$ for all $x \in \mathbb{R}^n$, in which case put $T = \nabla f$; or*
- (b) *an explicit sparse polynomial map $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ with the promise $\langle T(x) - T(y), x - y \rangle \geq \mu \|x - y\|^2$ for all $x, y \in \mathbb{R}^n$.*

Then T has a unique zero p . Given in addition an integer $L' \geq L$, written in unary, one can construct in deterministic time polynomial in L' , without oracle calls, a shared rational circuit with

outputs $\hat{x} \in \mathbb{Q}^n$ in which every division is by a nonzero value, such that the following holds for every explicit sparse polynomial $h \in \mathbb{Q}[X_1, \dots, X_n]$ of encoding length at most L' :

$$h(p) = 0 \quad \text{or} \quad |h(p)| \geq g_{L'}, \quad |h(\hat{x}) - h(p)| \leq g_{L'}/8, \quad g_{L'} := 2^{-2e(L')},$$

where $e(L') := (c_0 + 3)L'^2$ and c_0 is the constant of Lemma 4.1.

The circuit consists of an ordinary rational starting point x_0 with $\|x_0 - p\| \leq \mu/(4\Lambda)$, where $\Lambda = 2^{10L'^2}$ bounds the relevant derivatives, followed by $e(L') + 1$ exact Newton steps

$$x_{t+1} = x_t - J_T(x_t)^{-1}T(x_t).$$

In case (a), the starting point comes from polynomial-time convex value optimization (Lemma 3.1) and strong convexity. In case (b) no objective is available, and we use the ellipsoid method with the cuts $\{y : T(x)^\top(y - x) \leq 0\}$; every such cut keeps a fixed ball around p , which forces the method to reach a point with small residual (Lemma A.5). The Newton error contracts as $\|x_{t+1} - p\| \leq (\Lambda/2\mu) \|x_t - p\|^2$, so after $e(L') + 1$ steps it is at most $2^{-6 \cdot 2^{e(L')}}$. Each step solves a linear system whose matrix has a positive definite symmetric part; Gaussian elimination without pivoting has positive pivots, so no sign test is needed. Sharing the nodes of earlier steps keeps the size polynomial, although the expanded coordinates of $\hat{x}$ can have exponentially many bits. The proof is in Appendices A.4 and A.5.

4.3 Strongly convex objectives

Theorem 4.4 (Exact comparison at a strongly convex minimizer). *Let the input consist of $n \geq 1$, explicit sparse polynomials $f, h \in \mathbb{Q}[X_1, \dots, X_n]$, a relation $\bowtie$, and one of the following.*

- (a) *A rational $\mu > 0$, with the promise $\nabla^2 f(x) \succeq \mu I$ for every $x \in \mathbb{R}^n$.*
- (b) *A Hessian certificate: an explicit list $w(X, v) = (w_1, \dots, w_m)$ of distinct monomials of the form $X^\beta v_i$ that contains each of $v_1, \dots, v_n$, and a rational symmetric $m \times m$ matrix M . The certificate is valid if $M \succ 0$ and $v^\top \nabla^2 f(X) v = w(X, v)^\top M w(X, v)$ as polynomials in (X, v) .*

Let p be the unique minimizer of f on $\mathbb{R}^n$. Deciding $h(p) \bowtie 0$ reduces to one PosSLP instance: in case (a) on all promised inputs, and in case (b) on all inputs, where the language consists of the inputs with a valid certificate and $h(p) \bowtie 0$. No lower bound on $|h(p)|$ is assumed.

Proof outline. Apply Lemma 4.3(a) with $L' = L$ and put $\hat{\alpha} = h(\hat{x})$ and $g = g_L$, computed by $e(L)$ squarings of $1/2$. Since $\alpha = h(p)$ is zero or at least g in absolute value, and $|\hat{\alpha} - \alpha| \leq g/8$, each rational circuit in the following table is positive exactly when its relation holds. Its absolute value is at least $3g/8$ in the four order rows and at least $15g^2/64$ in the two equality rows, so it is never zero.

Relation	Circuit	Relation	Circuit
$\alpha > 0$	$\hat{\alpha} - g/2$	$\alpha < 0$	$-\hat{\alpha} - g/2$
$\alpha \geq 0$	$\hat{\alpha} + g/2$	$\alpha \leq 0$	$-\hat{\alpha} + g/2$
$\alpha = 0$	$g^2/4 - \hat{\alpha}^2$	$\alpha \neq 0$	$\hat{\alpha}^2 - g^2/4$

Lemma 4.2 turns the selected rational circuit into an integer circuit with the same sign. For case (b), the identity is checked by sparse coefficient comparison, $M \succ 0$ by rational symmetric elimination, and then $\mu = \det M / (\text{tr } M)^{m-1}$ is a rational lower bound for the least eigenvalue of M (Lemma 2.7). Since w contains every v_i , $v^\top \nabla^2 f(X) v \geq \mu \|w\|^2 \geq \mu \|v\|^2$. Case (a) then applies with L replaced by the polynomially larger length of the input together with μ . Details are in Appendix A.7. $\square$

The theorem decides the sign of $\min f - r$ and of $p_j - r$, including the equality cases, with no promise that these numbers are nonzero. For quartics with the full basis $w = (v, X \otimes v)$, the strict and weak comparisons are PosSLP-hard (Section 5), and Theorem 5.1 records the resulting classification. Equality has only the upper bound: none of the lower reductions in this paper produces equality instances. The certificate in (b) is a sufficient format, not a recognition algorithm for strong convexity. Degree four is the first degree at which the theorem has content: a globally convex polynomial of degree at most three is quadratic (Lemma 2.11), and then p and $h(p)$ are obtained by rational linear algebra.

For certified quartics, nonnegativity of f and membership in the cone of real sums of squares both mean $\min f \geq 0$, and the existence of a positive definite polynomial Gram matrix means $\min f > 0$; these equivalences are recorded with Theorem 5.1, and Theorem 8.12 gives the corresponding circuit Gram. The three certificate questions therefore inherit the one-instance upper bound. Rational sum-of-squares membership is different: when $\min f = 0$ it is not determined by the sign of the minimum (Section 9), and no matching upper bound is claimed for it.

Corollary 4.5 (Comparing two minima). *Let $f_1 \in \mathbb{Q}[X_1, \dots, X_{n_1}]$ and $f_2 \in \mathbb{Q}[Y_1, \dots, Y_{n_2}]$ each satisfy the hypotheses of Theorem 4.4, both with promises (a) or both with certificates (b). For each relation $\bowtie$, deciding $\min f_1 \bowtie \min f_2$ reduces to one PosSLP instance.*

Proof. Use disjoint variables and put $F(X, Y) = f_1(X) + f_2(Y)$ and $h(X, Y) = f_1(X) - f_2(Y)$. The Hessian of F is block diagonal, so $\nabla^2 F \succeq \min\{\mu_1, \mu_2\}I$. With certificates, check both, map invalid inputs to the circuit 0, and take $\mu_i = \det M_i / (\text{tr } M_i)^{m_i-1}$. The minimizer of F is the pair of minimizers, and there h equals $\min f_1 - \min f_2$. Apply Theorem 4.4(a) to F and h . $\square$

The proof needs only a curvature bound for the sum. A block-separated sum need not have a positive definite Hessian Gram on the full joint basis, and none is required.

Proposition 4.6 (Circuit witnesses for strict inequalities). *Under the hypotheses of Theorem 4.4(a), the point $\hat{x}$ of Lemma 4.3 with $L' = L$ is a rational vector with a shared circuit of size polynomial in L , constructed without oracle calls and without knowing any sign. If $h(p) > 0$, then $h(\hat{x}) > 0$. In particular, if $\min f < 0$ then $f(\hat{x}) < 0$.*

Proof. If $h(p) > 0$, then $h(p) \geq g_L$ and $h(\hat{x}) \geq 7g_L/8 > 0$. Apply this to $h = -f$ for the last claim. $\square$

Thus every strictly feasible instance $\{f < 0\} \neq \emptyset$ has a short feasible witness in the shared-circuit format, built by the same algorithm on every input. Three limits matter. Checking that $f(\hat{x}) < 0$ is again a PosSLP question. The expanded numerators and denominators of $\hat{x}$ can be exponentially long, and Section 8 shows that for some strictly feasible quartics every rational feasible point has exponentially many bits. When $\min f = 0$, the set $\{f \leq 0\}$ is the single point p , which can be irrational (Theorem 7.4); then no rational witness exists at all.

4.4 Zeros of strongly monotone polynomial maps

The same comparison holds at the zero of a polynomial map that is not a gradient. Such maps describe equilibria of games and variational models without a potential. The only new ingredient is the starting point, because there is no objective whose value can be minimized.

Theorem 4.7 (Zeros of strongly monotone maps). *Let the input consist of $n \geq 1$, an explicit sparse polynomial map $T = (T_1, \dots, T_n)$, an explicit sparse polynomial h , a relation $\bowtie$, and one of the following.*

- (a) A rational $\mu > 0$, with the promise $\langle T(x) - T(y), x - y \rangle \geq \mu \|x - y\|^2$ for all $x, y \in \mathbb{R}^n$.
- (b) A certificate (w, M) as in Theorem 4.4(b), valid if $M \succ 0$ and $v^\top J_T(X) v = w(X, v)^\top M w(X, v)$ as polynomials.

Then T has a unique zero p , and deciding $h(p) \bowtie 0$ reduces to one PosSLP instance, in the same sense as in Theorem 4.4.

The Jacobian J_T need not be symmetric; the certificate controls only its symmetric part, which suffices because $\langle T(x) - T(y), x - y \rangle = \int_0^1 (x - y)^\top J_T(y + \theta(x - y))(x - y) d\theta$. Existence of the zero is proved, not assumed. The proof (Appendix A.7) uses Lemma 4.3(b); the Newton steps solve the nonsymmetric system directly by elimination without pivoting.

Example 4.8 (Maps that are not gradients). For a rational skew-symmetric matrix A and rational b , the cubic map $T(x) = (1 + \|x\|^2)x + Ax - b$ has $J_T(x) = (1 + \|x\|^2)I + 2xx^\top + A$, hence $v^\top J_T(x)v = \|v\|^2 + \|x \otimes v\|^2 + 2(x^\top v)^2$. A positive definite certificate on $(v, x \otimes v)$ is $I \oplus (I + 2dd^\top)$, where d selects the coordinates $x_i v_i$. If $A \neq 0$, then J_T is not symmetric and T is not a gradient. The antisymmetric part of the Jacobian can also vary with x . In two variables, let

$$T(x, y) = (1 + x^2 + y^2)(x, y) + (x^2 y / 2, 0).$$

With $z = (xv_1, xv_2, yv_1, yv_2)$ one checks $v^\top J_T v = \|v\|^2 + z^\top Q_0 z$ for

$$Q_0 = \begin{pmatrix} 3 & 1/4 & 1/2 & 2 \\ 1/4 & 1 & 0 & 0 \\ 1/2 & 0 & 1 & 0 \\ 2 & 0 & 0 & 3 \end{pmatrix},$$

whose leading principal minors are 3, 47/16, 43/16 and 65/16. Here $(J_T)_{12} - (J_T)_{21} = x^2/2$, so T is not the sum of a gradient and a constant linear map.

For cubic maps with the full basis $w = (v, X \otimes v)$, the gradients of the certified quartics of Section 5 are instances of Theorem 4.7(b). Hence strict and weak comparisons at zeros of certified cubic maps are PosSLP-complete, while equality again has only the upper bound. Cubic degree is the first degree with content: a strongly monotone polynomial map of degree at most two is affine (Lemma 2.11), and its zero is rational. Constrained variational inequalities with such maps are treated in Section 6.

4.5 Polyhedra with a supplied slack gap

On a polyhedron the minimizer lies on an unknown face. This paper gives no deterministic polynomial-time method with a PosSLP oracle that finds this face for general strongly convex quartics; Section 6 discusses that open case and several structured classes. The difficulty disappears when the input bounds the nonzero optimal slacks away from zero. Then ordinary approximation finds the full active set, and the remaining problem is an unconstrained comparison on an affine subspace.

Proposition 4.9 (Supplied slack gap). *Let the input consist of explicit sparse $f, h \in \mathbb{Q}[X_1, \dots, X_n]$, a rational $\mu > 0$ with the promise $\nabla^2 f \succeq \mu I$ on $\mathbb{R}^n$, a rational polyhedron $P = \{x : Ax \leq b\}$ with rows $a_i^\top$, a rational $\delta > 0$, and a relation $\bowtie$. When $P \neq \emptyset$, let p_P be the minimizer of f on P , and assume the promise that every slack $b_i - a_i^\top p_P$ is either 0 or at least δ . Then deciding whether $P \neq \emptyset$ and $h(p_P) \bowtie 0$ reduces to one PosSLP instance. The reduction also computes the set of active rows at p_P by an ordinary polynomial-time algorithm.*

The proof (Appendix A.8) decides emptiness by linear programming, obtains a feasible rational point whose distance to p_P is at most $\delta/(8 \max_i \|a_i\|_1)$ from convex value optimization, and reads off the active set S as the rows with slack below $\delta/2$. If $d \in \ker A_S$, then $p_P \pm \theta d \in P$ for small $\theta > 0$, so $\nabla f(p_P)^\top d = 0$; hence p_P minimizes f on the affine space $\{x : A_S x = b_S\}$. In a chart of this space with an identity block, the restricted function keeps the modulus μ , and the Newton machinery applies. The chart is never expanded into monomials, so unary degree is allowed. No multiplier bound, strict complementarity or Slater point is needed. The slack gap is a promise that the algorithm does not check, and it is a real restriction: Lemma 4.1 only guarantees nonzero slacks of size $2^{-2^{\text{poly}(L)}}$, which cannot be printed in polynomially many bits. The proposition gives no deterministic P^{PosSLP} bound for general polyhedra.

4.6 Nested cube roots

The lower bounds of Section 5 pass through a language of certified cube-root circuits, in which every gate is the real cube root of an affine combination of earlier gates and their squares, and supplied positive intervals certify every gate. Its comparison problem has an upper bound by the same scheme with two elementary substitutions. For a circuit with k gates, multiplication by a common denominator turns the gate values into algebraic integers in a field of degree at most 3^k , and all conjugates of the gate values are bounded by 2^{3L} ; the norm of the scaled difference between the output gate ξ_o and the threshold r then separates $\xi_o - r$ from zero by $2^{-2^{6L}}$. Newton's iteration for the cube root, started at the upper end of the certified range, replaces the warm start. Theorem 5.6 states the resulting classification: strict and weak comparisons are PosSLP-complete, and equality has the upper bound only. The upper bound is a tailored instance of the classical approximation-and-separation argument, not a new result about general radical circuits.

4.7 One sign test for adaptive sign tests

Several algorithms in this paper make many exact comparisons, and later comparisons depend on earlier answers; examples are the exact quadratic programming steps of Section 6. Such an algorithm is a polynomial-time Turing reduction to PosSLP. The next theorem converts it into a reduction to one instance.

Theorem 4.10 (Compiling sign tests).

- (a) *Let C be a circuit with Boolean input nodes $u_1, \dots, u_m$, constants 0 and 1, binary $+$, $-$ and $\times$ gates, and unary threshold gates $H(v) = 1$ if $v > 0$ and $H(v) = 0$ otherwise. Let $S \geq 1$ be its number of nodes. Suppose that for every $u \in \{0, 1\}^m$ all node values are integers and the output is 0 or 1. In time polynomial in S , without oracle calls, one can construct an integer circuit C' of size $O(S^2)$ with the same input nodes and no threshold gates such that $C'(u) > 0$ exactly when $C(u) = 1$, for every $u \in \{0, 1\}^m$.*
- (b) *Let Π be a decision or promise problem decided by a deterministic oracle Turing machine with oracle PosSLP that halts within a polynomial number of steps on every input and for all oracle answers. Then Π reduces to one PosSLP instance. Consequently, P^{PosSLP} is the class of problems that are polynomial-time many-one reducible to PosSLP.*
- (c) *In particular, from integer circuits $C_1, \dots, C_k$ and a Boolean circuit β with k inputs one can construct in polynomial time one integer circuit whose output is positive exactly when $\beta([C_1 > 0], \dots, [C_k > 0]) = 1$.*

The proof is in Appendix A.9. In (a), all exact values have absolute value at most 2^{2^S} , and distinct integers are at least one apart. Each threshold gate is replaced by a rational map that is within $2^{-2^{2^{S+4}}}$ of the correct bit whenever its input is within $1/4$ of the exact integer. The map first shrinks the magnitude from $[1, 2^{2^{S+2}}]$ to $[1, 2]$ by the scaled steps $x \mapsto 2\sigma x/(\sigma + x^2)$ with $\sigma = 2^{2^{S+2}}, 2^{2^{S+1}}, \dots, 2^{2^1}$, and then applies $x \mapsto 2x/(1 + x^2)$, the classical inverse Newton iteration for the sign function, $2S + 5$ times. An absolute-error induction over the original gates shows that the errors never exceed $1/4$; all denominators are positive, so the final pair P/Q gives the integer output $2P - Q$. For (b), the oracle machine is unrolled into such a circuit. Each possible query is evaluated by a padded universal interpreter: the query string is parsed by a Boolean circuit, each instruction slot selects its operands by sums of earlier slot values multiplied by address indicators, and malformed queries evaluate to the answer no. The interpreter enumerates addresses, not answer transcripts, so its size is polynomial although later queries depend on earlier answers. Part (c) is the special case of a nonadaptive query list.

The theorem changes the form of reductions; it gives no polynomial-time algorithm for PosSLP. It applies to deterministic computations with a worst-case time bound. It does not remove randomness from Las Vegas algorithms, does not turn the nondeterministic classes $\text{UP}^{\text{PosSLP}}$ or $\text{NP}^{\text{PosSLP}}$ into deterministic ones, and compiles one output bit at a time; a vector output such as an active set is a polynomial list of separate instances. The exact separation of distinct integers is essential, and the proof uses the full lengths of the compressor and refinement schedules; no claim is made for shorter schedules.

The main application in this paper concerns constrained problems. For strongly convex objectives on boxes with structured Hessians and for separable objectives on network-flow polyhedra, Section 6 decides exact predicates at the constrained minimizer by deterministic algorithms that solve exact quadratic subproblems on shared circuits and ask adaptive sign queries. Their warm starts are ordinary computations, and the number of later operations is bounded by a polynomial in the input length that does not depend on the expanded sizes of the circuit values. Part (b) therefore compiles each such predicate into one PosSLP instance (Corollary 6.15). The randomized constraint-rank algorithm of Theorem 6.16 is not covered, because its running time is bounded only in expectation.

4.8 Relation to prior work

The combination of Newton approximation by short arithmetic circuits, an algebraic separation bound and numerator-denominator division elimination is established. Allender, Bürgisser, Kjeldgaard-Pedersen and Miltersen showed that Square Root Sum lies in P^{PosSLP} by a polynomial-time Turing reduction and characterized P^{PosSLP} as the Boolean part of constant-free polynomial-time real computation [6]. Etessami, Stewart and Yannakakis gave polynomial-time many-one reductions to PosSLP for threshold comparisons of least fixed points of probabilistic polynomial systems with the same architecture [39, Appendix C]. Exact semidefinite feasibility has comparable arithmetic-circuit reductions [113]. The warm start for strongly convex objectives uses the elementary estimate $\|p\| \leq \|\nabla f(0)\|/\mu$ to fix an explicit box, and then applies the polynomial-time convex value interface of Lemma 3.1, which rests on the ellipsoid method. For merely convex polynomial programs on unbounded domains, Slot et al. [111] give radius and objective-gap approximation bounds; see Section 3. The warm start for monotone maps uses the rounded central-cut ellipsoid method of Grötschel et al. [49, Section 3.2].

This section proves a one-instance upper bound for explicit globally strongly convex objectives and strongly monotone maps of polynomially bounded degree, for arbitrary polynomial observables and all six relations, including equality, together with checked certificate languages. The lower bounds of Section 5 give matching PosSLP completeness for strict and weak value and coordinate

comparisons on certified quartics, and for coordinate comparisons on their certified cubic gradient maps. Equality has only the upper bound. The probabilistic polynomial systems of Etessami, Stewart and Yannakakis form a different input class, and no reduction between the two classes is used here. For Theorem 4.10, the rational sign iteration and its scaled variants are classical numerical tools, and the oracle characterization of P^{PosSLP} is known; the statement proved here is the uniform compilation of adaptive integer sign tests into one instance, with the error budget and interpreter given in full. This compilation also converts the Square Root Sum oracle upper bound into a one-instance upper bound. We make no priority claim for it.

5 Lower reductions: certified quartics carry the full sign problem

Theorem 4.4 turns every exact order or equality test at the minimizer of an explicitly given, globally strongly convex polynomial into one PosSLP instance. This section shows that the order tests are no easier, even in a setting that removes every familiar source of difficulty: a single quartic on all of $\mathbb{R}^n$, supplied with a rational positive definite Hessian Gram matrix that certifies uniform strong convexity and can be checked in polynomial time. The tests stay PosSLP-hard when the optimal value is promised to be nonzero. The coordinate tests stay PosSLP-hard when the optimizer is promised to be rational, to lie in $[-1, 1]^n$, and to have the known optimal value zero.

These are the exact decisions that a branch-and-bound method for mixed-integer nonlinear optimization makes at a node: whether the bound of a node relaxation lies below an incumbent value, and on which side of a branching threshold an optimal coordinate lies. Certified strong convexity, the absence of constraints, and a rational optimizer each remove a known obstacle to exact computation, but none of them removes the arithmetic one. An ordinary polynomial-time algorithm for any of the comparisons below would place PosSLP in P. Allender et al. [6] show that PosSLP lies in the counting hierarchy and that P^{PosSLP} is the Boolean part of polynomial-time computation over the reals with constants 0 and 1 in the Blum–Shub–Smale model; whether PosSLP lies in P is open. The results here concern exact comparisons only. Approximate optimal values and approximate optimizers of the same quartics are computable in polynomial time (Theorem 3.4).

Mechanism. Every lower bound below has three steps.

- (1) A *sign compiler* converts an integer straight-line program into a circuit whose gate values stay bounded and whose designated output has the sign of the program’s output. The compilers never represent the integer values themselves, which can have doubly exponential magnitude. They carry only signs, through real signals that can be as small as $2^{-2^{cT}}$ for a program with T operations.
- (2) A *realization* places the list of all gate values at the unique zero of one rational quartic. A gate-specific exposing quadratic supplies positive curvature in every direction at that zero, and the assembly Lemma 7.6 combines it with the gate residuals into a globally strongly convex sum of squares with a rational positive definite Hessian Gram.
- (3) For value comparisons, a *cubic perturbation* transfers the designated sign to the sign of the optimal value without losing the Hessian certificate.

We use two compilers. Certified cube-root circuits (Section 5.3) give hardness of exact feasibility, of optimization over a fixed cube, and of the sign of an unconstrained minimum. Circuits of rational unit quaternions (Section 5.6) keep every gate value rational and give coordinate hardness with a rational optimizer.

5.1 Statements

PosSLP asks whether the output V of a division-free integer straight-line program with constants $0, 1$ and operations $+, -, \times$ is positive. Programs and all circuits below are directed acyclic graphs: a gate may be used by any number of later gates, and sizes count gates of the shared graph, never an expanded formula or expanded fractions. The integer $W = 2V - 1$ is never zero, and $W > 0$ exactly when $V > 0$; replacing V by $1 - V$ reverses the answer. All reductions are deterministic polynomial-time many-one reductions, and L denotes total binary input length.

A *certified quartic* on $\mathbb{R}^n$ is a pair (f, M) , where $f \in \mathbb{Q}[x_1, \dots, x_n]$ has degree at most four and M is a rational symmetric positive definite matrix of order $n + n^2$ such that

$$v^\top \nabla^2 f(x) v = (v, x \otimes v)^\top M (v, x \otimes v) \quad \text{identically in } (x, v) \in \mathbb{R}^n \times \mathbb{R}^n, \quad (3)$$

with $(x \otimes v)_{(k,j)} = x_k v_j$ (Definition 2.6). Coefficient comparison and rational symmetric elimination check this format in polynomial time. The matrix certifies $\nabla^2 f \succeq \lambda_{\min}(M)I$, and $\det M / (\text{tr } M)^{n+n^2-1}$ is a positive rational lower bound on $\lambda_{\min}(M)$ of polynomial bit length (Lemma 2.7). Hence f has a unique minimizer p and optimal value $f^* = f(p)$.

Theorem 5.1 (Certified quartic classification). *Consider inputs consisting of a certified quartic (f, M) on $\mathbb{R}^n$, a rational threshold r and, for coordinate tests, an index $j \in \{1, \dots, n\}$. Inputs violating the format are no-instances.*

- (i) *Each of the eight tests $f^* < r$, $f^* \leq r$, $f^* > r$, $f^* \geq r$, $p_j < r$, $p_j \leq r$, $p_j > r$, $p_j \geq r$ is PosSLP-complete.*
- (ii) *Each of the tests $f^* = r$ and $p_j = r$ reduces to PosSLP. No matching lower bound is asserted.*
- (iii) *Each of the following properties is PosSLP-complete: $f \geq 0$ on $\mathbb{R}^n$; f is a sum of squares of real polynomials; f has a rational positive definite Gram matrix on the vector of all monomials of degree at most two. Being a sum of squares of rational polynomials is PosSLP-hard.*
- (iv) *The lower bounds for the value tests in (i) and for all properties in (iii) hold already for instances with $r = 0$, $M \succeq I$ and $f^* \neq 0$. The lower bounds for the coordinate tests hold under the promises of Theorem 5.2.*

The upper bounds are Theorem 4.4; the content of this section is the lower bounds. Part (iii) concerns three different certificate notions. In the certified class, nonnegativity and real sums of squares are equivalent, and a rational positive definite polynomial Gram exists exactly when $f^* > 0$. Rational sums of squares at $f^* = 0$ are a different matter: Example 9.6 is a certified quartic with minimum zero that is a sum of squares over a real field E exactly when $2^{1/5} \in E$. No upper bound for rational sum-of-squares membership is claimed.

Quartics are the lowest degree at which these lower bounds can hold. A globally convex polynomial of degree at most three is quadratic (Lemma 2.11), and a strongly convex rational quadratic has a rational minimizer computable by rational linear algebra.

Theorem 5.2 (Rational optimizer and known value). *There is a polynomial-time many-one reduction from PosSLP to the test $p_j > 0$ for certified quartics, whose output instances (F, M, j) on $\mathbb{R}^N$ satisfy all of the following promises:*

- (i) $p \in \mathbb{Q}^N \cap [-1, 1]^N$ and $p_j \neq 0$;
- (ii) $F^* = 0$, and F is supplied as a sum of $N + 1$ squares of rational quadratic polynomials;

(iii) $\nabla^2 F \succeq \frac{3}{2}I$ on $\mathbb{R}^N$.

Consequently each of the four coordinate order tests at threshold zero remains PosSLP-complete on the instances satisfying these promises.

Rationality of the optimizer is a promise satisfied by the constructed instances; the reduction neither recognizes nor needs to recognize it. The theorem shows that exact field membership does not supply an exact algorithm: the optimizer is rational, its value is known, and the objective comes with short rational square factors, yet a single coordinate sign still carries an arbitrary PosSLP instance. Rationality also gives no bound on representation size: in the constructed instances one coordinate of p is a nonzero rational of absolute value below $2^{-2^{cN}}$ for an absolute constant $c > 0$ (Remark 5.14), so its expanded fraction has exponentially many bits. Theorem 8.5 gives unconditional families of bounded rational optimizers with long expanded encodings and additional conditioning properties. Corollary 6.24 uses the instances of Theorem 5.2 to show that one optimal binary decision remains PosSLP-hard to extract when the optimal value is known in advance.

Relation to earlier hardness results. Exact convex feasibility was already known to capture arithmetic circuit comparison: Tarasov and Vyalıy [113] reduce PosSLP to exact feasibility of semidefinite programs. Bounded arithmetic circuits also capture PosSLP: Etessami and Yannakakis [38, Lemma 5] reduce it, with a circuit of linear size over $\{+, \times, /\}$ whose gate values all lie in $(0, 1)$, to an order comparison of that circuit’s output. The contribution here is the restriction to one explicitly expanded quartic with a checked strict Hessian certificate and no constraints, together with the two sign compilers it requires. The analytic compiler avoids products of independent values, which the quartic realization cannot accept; the quaternion compiler keeps every gate value exactly rational and bounded. Commutators of near-identity rotations are a classical tool for multiplying small rotation angles, used in the Solovay–Kitaev algorithm for approximate synthesis in $SU(2)$ [28], and identity testing for shared circuits over linear groups has also been studied [70]. Those works concern approximate synthesis and identity testing, respectively; they do not decide the sign of a designated rational coordinate in a fixed compact group. Our use requires exact rational gates and error control below a doubly exponentially small nonzero signal. Version 1 of Slot et al. [111] proves polynomial-time approximation for convex polynomial programming and leaves the complexity of the exact decision problem $\exists x \in P : f(x) \leq 0$ for convex quartics open. Corollary 5.9 shows that this decision is PosSLP-hard already when P is a halfspace or the unit cube and f is a certified quartic. It does not determine the complexity class of that constrained problem: Theorem 4.4 concerns unconstrained minimization, and Section 6 treats constrained exact comparison.

5.2 Two obstacles

The following two facts explain the shape of the constructions. The first shows why the compilers carry only signs through bounded signals. The second shows why the realizations need a gate-specific exposing quadratic rather than squared gate equations.

Proposition 5.3 (Radius and naive lifting).

- (i) If $F: \mathbb{R}^n \rightarrow \mathbb{R}$ is twice continuously differentiable with $\nabla^2 F \succeq I$ and minimizer p , then $\|p\| \leq \|\nabla F(0)\|$. In particular, a rational polynomial whose coefficients have bit length at most b has a minimizer of norm at most $\sqrt{n} 2^b$.
- (ii) For every $\varepsilon > 0$ the nonnegative quartic $H(x, y) = x^2 + x^4 + \varepsilon(y - x^2)^2$ has the unique zero $(0, 0)$, at which the residual map $(x, y - x^2)$ has the identity Jacobian, but $\partial_x^2 H(0, y) = 2 - 4\varepsilon y$ is negative for $y > 1/(2\varepsilon)$.

Proof. For (i), strong monotonicity of ∇F gives $(\nabla F(0) - \nabla F(p))^\top(0 - p) \geq \|p\|^2$; since $\nabla F(p) = 0$, the Cauchy–Schwarz inequality gives the bound. The linear coefficients of F form $\nabla F(0)$. For (ii), the zero set is clear, and $\partial_x^2[\varepsilon(y - x^2)^2] = -4\varepsilon y + 12\varepsilon x^2$. $\square$

Part (i) rules out realizations that keep unscaled integer values: a program with m squarings computes 2^{2^m} , whereas a polynomial-size normalized quartic has a minimizer of singly exponential norm. Part (ii) shows that a nonsingular Jacobian at the intended zero does not make squared gate residuals globally convex: at $x = 0$ the squared residual term $\varepsilon(y - x^2)^2$ contributes $-4\varepsilon y$ to $\partial_x^2 H$, which is unbounded below as y grows. The realizations below control such negative curvature with one additional square, that of a gate-specific exposing quadratic whose quadratic part is positive definite; after scaling, this square makes the whole sum uniformly strongly convex on the whole space (Lemma 7.6). This is a feature of our construction, not a necessary condition for convexity: $(x^2 - y^2)^2 + (2xy)^2 = (x^2 + y^2)^2$ is convex, although both squared quadratics have indefinite quadratic parts.

5.3 Sign compiler I: certified cube-root circuits

Definition 5.4 (Certified odd-root circuits). A *certified odd-root circuit* with k gates consists of odd integers $d_i = 2n_i - 1 \geq 3$ written in unary, rational coefficients c_i and a_{ije} for $1 \leq j < i \leq k$ and $1 \leq e \leq n_j$, and rational boxes $[\underline{\xi}_i, \bar{\xi}_i]$ not containing zero, such that for every i

$$c_i + \sum_{j < i} \sum_{e=1}^{n_j} a_{ije} R_{j,e} \subseteq [\underline{\xi}_i^{d_i}, \bar{\xi}_i^{d_i}], \quad R_{j,e} = \{s^e : s \in [\underline{\xi}_j, \bar{\xi}_j]\}, \quad (4)$$

where the left side is a Minkowski sum of intervals. The circuit defines reals $\xi_1, \dots, \xi_k$ by

$$\xi_i^{d_i} = \rho_i(\xi) := c_i + \sum_{j < i} \sum_{e=1}^{n_j} a_{ije} \xi_j^e, \quad (5)$$

taking the unique real d_i -th root. It is a *cube-root circuit* if every $d_i = 3$; then each radicand is a rational affine combination of earlier roots and their squares.

The test (4), which we call the *direct interval test*, uses exact rational arithmetic and unary exponents, so it is checkable in polynomial time. By induction over i , it implies $\xi_i \in [\underline{\xi}_i, \bar{\xi}_i]$ for every gate: if all earlier roots lie in their boxes, then $\rho_i(\xi)$ lies in the left side of (4), and the real d_i -th root is increasing. Radicand coefficients may have either sign. A product $\xi_j \xi_l$ of two different earlier gates is not available; this restriction is what makes a quartic realization possible, and it is the reason a new compiler is needed.

Theorem 5.5 (Cube-root sign compiler). *Given an integer straight-line program with output V , one can compute in polynomial time a certified cube-root circuit with boxes $[1 - w_i, 1 + w_i]$, $0 < w_i \leq 10^{-9}$, and two distinct gates o and e such that*

$$\xi_o \neq 1, \quad \text{sign}(\xi_o - 1) = \text{sign}(2V - 1), \quad |\xi_o - 1| \leq 2^{-29}, \quad 0 < (\xi_e - 1)^2 \leq \frac{1}{8} |\xi_o - 1|. \quad (6)$$

Every radicand is a rational constant, an affine function $1 + \sum_j \lambda_j (\xi_j - 1)$ with $\sum_j |\lambda_j| \leq 6$, or the square gate $3\xi_j^2 - 6\xi_j + 4 = 1 + 3(\xi_j - 1)^2$.

The proof is in Appendix B.4. Its gadgets act on small real signals x , stored as roots $\xi = 1 + x$. Put

$$\phi(z) = (1 + 3z)^{1/3} - 1, \quad \text{sq}(z) = \phi(z^2), \quad \theta(z) = \frac{1}{2} [\text{sq}(\phi(z)) + \text{sq}(\phi(-z))],$$

$$\text{mul}(x, y) = \phi\left(\frac{1}{4}[\theta(x + y) - \theta(x - y)]\right).$$

One root gate computes ϕ of an affine combination of signals, and one square gate computes sq ; mul uses nine root gates. The function θ is even with $\theta(z) = z^2 + O(z^4)$, so $\text{mul}(x, y) = xy + O(|(x, y)|^4)$. The decisive property is exact: $\text{mul}(x, 0) = \text{mul}(0, y) = 0$. A Cauchy estimate then gives the relative error bound $|\text{mul}(x, y) - xy| \leq 2^{24} |xy| (|x| + |y|)$, which stays proportional to the product even when the two inputs have very different magnitudes. Each integer gate value V_i is carried by a pair of signals of a common order d_i in a tiny parameter δ , with leading coefficients V_i and 1; addition cross-multiplies by the denominator signals, so that it only ever adds signals of equal order. All error bounds are absolute multiples of δ^{d_i+1} and never divide by a leading coefficient, so cancellation and zero intermediate values need no special treatment. The parameter δ is the output of $2T + 5$ chained square gates starting from the printed rational $1000^{-O(T)}$; it is never printed itself. The boxes are verified directly, independently of the analytic estimates, by Lemma B.3.

For completeness of the picture, the language produced by the compiler is itself PosSLP-complete.

Theorem 5.6 (Certified cube-root comparison). *For a certified cube-root circuit with positive boxes, a gate o , and a rational r , each of the tests $\xi_o > r$, $\xi_o \geq r$, $\xi_o < r$, $\xi_o \leq r$ is PosSLP-complete, and $\xi_o = r$ reduces to PosSLP. Malformed inputs, including inputs failing the direct interval test, are no-instances.*

The lower bounds follow from Theorem 5.5 with $r = 1$, after replacing V by $1 - V$ for the tests $<$ and $\leq$. The upper bound, proved in Appendix B.5, uses the same architecture as Theorem 4.4 with a more elementary separation bound: after clearing denominators every gate value is an algebraic integer of degree at most 3^k , so a field norm bounds $|\xi_o - r|$ away from zero, and polynomially many exact Newton steps for each cube root reach that accuracy in a shared rational circuit.

5.4 Quartic realization of certified root circuits

Theorem 5.7 (Signed odd-root realization). *There is a deterministic polynomial-time algorithm that, given a certified odd-root circuit, computes signs $\sigma_i \in \{-1, 1\}$, a rational $\kappa \geq 1$, and a rational quartic F in the $N = \sum_i n_i$ variables $X = (X_{i,e})_{1 \leq i \leq k, 1 \leq e \leq n_i}$ with the following properties.*

- (i) F is supplied as a sum of $N + 1$ squares of rational quadratic polynomials, and $\nabla^2 F \succeq \frac{3}{2}I$ on $\mathbb{R}^N$.
- (ii) $F^{-1}(0) = \{p\}$, where $p_{i,e} = (\sigma_i \kappa \xi_i)^e$. In particular $\xi_i = \sigma_i p_{i,1} / \kappa$.
- (iii) A rational positive definite Hessian Gram matrix for F in the sense of (3) is computed as part of the output.

Here σ_i is the sign of the box of gate i , and $\kappa = \max\{1, \max_i 1 / \min(|\underline{\xi}_i|, |\bar{\xi}_i|)\}$.

The proof is in Appendix B.6. After the normalization $\alpha_i = \sigma_i \kappa \xi_i \geq 1$, the coordinates $X_{i,e}$ stand for the powers α_i^e , $e \leq n_i$, and the N residuals

$$X_{i,1}X_{i,e} - X_{i,e+1} \quad (1 \leq e < n_i), \quad X_{i,n_i-1}X_{i,n_i} - b_i(X), \quad b_i(X) = c'_i + \sum_{j < i, e} a'_{ije} X_{j,e},$$

have p as their unique real zero, with a block-triangular Jacobian whose determinant has absolute value $\prod_i d_i \alpha_i^{d_i-1} \geq 1$. For one gate with root α and $d = 2n - 1$, put $\ell_j = X_{j+1} - \alpha X_j$ for $0 \leq j < n$,

with $X_0 = 1$. A tridiagonal positive definite form $P_\alpha = \ell^\top T_\alpha \ell$ restricts to the power curve $X_j = t^j$ as

$$P_\alpha(t, t^2, \dots, t^n) = (t - \alpha)(t^d - \alpha^d),$$

and the gate's exposing quadratic is

$$E_i^* = P_{\alpha_i}(X_i) + (\alpha_i - X_{i,1})(b_i(X) - \alpha_i^{d_i}).$$

It vanishes to second order at p , its quadratic part is positive definite on the gate's own coordinates, and its only coupling to earlier gates is the quadratic cross term $-u_{i,1} \sum_{j,e} a'_{ije} u_{j,e}$ in centered coordinates $u = X - p$. Geometrically decreasing gate weights make the weighted sum $\sum_i \rho^{i-1} E_i^*$ positive definite in all directions, including signed couplings and arbitrary reuse of earlier gates. The coefficients of E_i^* involve the algebraic number α_i , but E_i^* is an explicit combination, with coefficients polynomial in α_i , of rational quadratics vanishing at p . Replacing α_i by a certified rational approximation in those coefficients therefore keeps the zero exactly while making the gradient at p as small as required. Lemma 7.6 then supplies the quartic and its certificate. No minimal polynomial of a gate value, or of the field generated by all gate values, is ever expanded; that field has degree at most $\prod_i d_i$.

Remark 5.8 (Monotone cube-root circuits). If $d_i = 3$, $c_i \geq 1$, and every radicand uses only first powers with nonnegative coefficients $a_{ij1} \geq 0$, boxes need not be part of the input. With $A = 2 + \sum_i (c_i + \sum_j a_{ij1})$, the boxes $[1, A]$ pass the direct interval test, because each radicand interval lies in $[1, A(c_i + \sum_j a_{ij1})] \subseteq [1, A^2] \subseteq [1, A^3]$. Theorem 5.7 then applies with $\kappa = 1$.

5.5 Feasibility, a fixed cube, and the sign of the minimum

Apply Theorem 5.7 to the circuit of Theorem 5.5. All boxes are positive, so every $\sigma_i = 1$ and $\kappa = 1/(1 - \max_i w_i) \in [1, 1 + 2 \cdot 10^{-9}]$. The unique zero has $p_{i,1} = \kappa \xi_i$ and $p_{i,2} = \kappa^2 \xi_i^2$, all in $(0, 4)$.

Corollary 5.9 (Exact feasibility and a fixed cube). *For the quartic F just described, with designated output gate o :*

- (i) *The set $\{X : F(X) \leq 0, X_{o,1} \geq \kappa\}$ is nonempty exactly when $V > 0$, and it is then a single point.*
- (ii) *Let $X = \Delta Y + \kappa e_{o,1}$, where Δ is diagonal with entry $8 - \kappa$ at $(o, 1)$ and 8 elsewhere, and put $\tilde{F}(Y) = F(\Delta Y + \kappa e_{o,1})$. Then $\tilde{F}$ is a sum of $N + 1$ squares of rational quadratics, has a rational positive definite Hessian Gram, satisfies $\nabla^2 \tilde{F} \succeq 54I$, and*

$$\min_{Y \in [0,1]^N} \tilde{F}(Y) = 0 \iff V > 0;$$

otherwise the minimum over the cube is positive.

Proof. Since $F \geq 0$ vanishes only at p , the first set is $\{p\}$ when $p_{o,1} = \kappa \xi_o \geq \kappa$, that is, when $\xi_o \geq 1$, and empty otherwise. By (6), $\xi_o \geq 1$ holds exactly when $\xi_o > 1$, that is, when $2V - 1 > 0$. For (ii), the affine map sends $[0, 1]^N$ onto the box with $X_{o,1} \in [\kappa, 8]$ and all other coordinates in $[0, 8]$. All coordinates of p lie in $(0, 4)$, so this box contains p exactly when $p_{o,1} \geq \kappa$. If it does, the minimum is $F(p) = 0$. If not, F is positive on the compact box. Composition with an affine map preserves the rational square factors, and Lemma 2.8 transforms the Hessian Gram by an invertible rational congruence. Finally $\nabla^2 \tilde{F}(Y) = \Delta \nabla^2 F(X) \Delta \succeq \frac{3}{2} \cdot 36 I$ because $8 - \kappa > 6$. $\square$

Part (i) is a degenerate exact feasibility problem: its feasible set is empty or a single point, so it has no Slater point. Part (ii) separates that degeneracy from the domain: the cube has the interior point $(\frac{1}{2}, \dots, \frac{1}{2})$, and only the threshold $\tilde{F} \leq 0$ is degenerate. In the no case of part (ii), the positive minimum is not shown to be at least $2^{-\text{poly}(L)}$.

The unconstrained value test needs a different idea, since an unconstrained minimum of a sum of squares is never negative. The auxiliary gate e of Theorem 5.5 supplies a nonzero signal whose square is at most one eighth of the output signal's absolute value. A cubic tilt built from the two signals encodes the output sign in the minimum.

Theorem 5.10 (Sign of an unconstrained minimum). *There is a polynomial-time many-one reduction from PosSLP to the test $f^* < 0$ for certified quartics, whose output instances (G, M_G) satisfy $M_G \succeq I$ and $G^* \neq 0$. More precisely, $V > 0$ implies $G^* < 0$, and $V \leq 0$ implies $G^* > 0$.*

Proof. Let F , p , κ and the gates $o \neq e$ be as above, and let M be the rational positive definite Hessian Gram of F , of order $N + N^2$. Put

$$u(X) = X_{e,1} - \kappa, \quad v(X) = X_{o,1} - \kappa, \quad u_0 = u(p) = \kappa(\xi_e - 1), \quad v_0 = v(p) = \kappa(\xi_o - 1).$$

By (6) and $1 \leq \kappa < 2$,

$$u_0 \neq 0, \quad v_0 \neq 0, \quad \text{sign } v_0 = \text{sign}(2V - 1), \quad |v_0| \leq \frac{1}{8}, \quad u_0^2 \leq \frac{\kappa}{8} |v_0| \leq \frac{1}{2} |v_0|. \quad (7)$$

The cubic $-u^2v$ has $z^\top \nabla^2(-u^2v)(X)z = 2\kappa z_a^2 + 4\kappa z_a z_b - 2X_b z_a^2 - 4X_a z_a z_b$, where $a = (e, 1)$ and $b = (o, 1)$ are distinct indices. On the basis $(z, X \otimes z)$ this biform has the rational symmetric Gram matrix B whose only nonzero entries are

$$B_{z_a, z_a} = 2\kappa, \quad B_{z_a, z_b} = B_{z_b, z_a} = 2\kappa, \quad B_{z_a, X_b z_a} = B_{X_b z_a, z_a} = -1, \quad B_{z_b, X_a z_a} = B_{X_a z_a, z_b} = -2,$$

and $\|B\| \leq \|B\|_F = (12\kappa^2 + 10)^{1/2} < 8$. Let $\mu = \det M / (\text{tr } M)^{N+N^2-1} \leq \lambda_{\min}(M)$ and $\lambda = \lceil 9/\mu \rceil$. Then

$$G = \lambda F - u^2v, \quad M_G = \lambda M + B \succeq (\lambda\mu - 8)I \succeq I,$$

so (G, M_G) is a certified quartic with $\nabla^2 G \succeq I$; the cubic tilt leaves the quartic part unchanged. All data have polynomial bit length.

Since $F \geq 0$ and $F(p) = 0$, we have $\nabla F(p) = 0$, hence

$$G(p) = -u_0^2 v_0, \quad \nabla G(p) = -2u_0 v_0 e_a - u_0^2 e_b, \quad \|\nabla G(p)\|^2 = 4u_0^2 v_0^2 + u_0^4.$$

If $V > 0$, then $v_0 > 0$ and $G^* \leq G(p) < 0$. If $V \leq 0$, write $t = -v_0 > 0$. Strong convexity with modulus one gives $G(X) \geq G(p) + \nabla G(p)^\top (X - p) + \frac{1}{2} \|X - p\|^2 \geq G(p) - \frac{1}{2} \|\nabla G(p)\|^2$, so

$$G^* \geq u_0^2 t - 2u_0^2 t^2 - \frac{1}{2} u_0^4 = u_0^2 (t - 2t^2 - \frac{1}{2} u_0^2) \geq u_0^2 (t - \frac{t}{4} - \frac{t}{4}) = \frac{1}{2} u_0^2 t > 0,$$

using $t \leq \frac{1}{8}$ and $u_0^2 \leq \frac{t}{2}$ from (7). $\square$

The perturbation can move the minimizer to an irrational point; no rationality statement is made for G . The positive value in the no case is at least $\frac{1}{2} u_0^2 |v_0|$, a quantity of doubly exponentially small size, and no better lower bound is claimed. The perturbation argument uses only two distinct affine signals satisfying (7) at the zero of a certified sum of squares; the compiler is what produces such signals with polynomial-size data.

5.6 Sign compiler II: rational unit quaternions

Write a quaternion as $u = (u_0, \vec{u})$ with scalar part u_0 and vector part $\vec{u} = (u_1, u_2, u_3)$. The product is $uw = (u_0w_0 - \vec{u} \cdot \vec{w}, u_0\vec{w} + w_0\vec{u} + \vec{u} \times \vec{w})$, the conjugate is $\bar{u} = (u_0, -\vec{u})$, and $|uw| = |u||w|$ for the Euclidean norm. For a unit quaternion, $u^{-1} = \bar{u}$. Let $\mathbf{1} = (1, 0, 0, 0)$ and $\mathbf{i} = (0, 1, 0, 0)$.

Definition 5.11 (Rational unit-quaternion circuits). A *rational unit-quaternion circuit* with k gates lists, in topological order, gates of three kinds: an explicitly given rational unit quaternion; the product $g_a g_b$ of two earlier gates, where $a = b$ is allowed; or the conjugate $\bar{g}_a$ of an earlier gate. Every gate value is a rational unit quaternion. The input lists the generators and the operations, not the expanded coordinates of gate values.

Theorem 5.12 (Quaternion sign compiler). *Let $\mathbf{c} = \frac{1}{2}(1, 1, 1, 1)$ and $\mathbf{w}_\vartheta = (\frac{1-\vartheta^2}{1+\vartheta^2}, \frac{2\vartheta}{1+\vartheta^2}, 0, 0)$ with $\vartheta = 2^{-20}$. Given an integer straight-line program with output V , one can compute in polynomial time a rational unit-quaternion circuit whose generators are among $\mathbf{1}, \mathbf{i}, \mathbf{c}, \mathbf{w}_\vartheta$, together with an output gate whose value u satisfies*

$$u_1 \neq 0, \quad \text{sign } u_1 = \text{sign}(2V - 1).$$

Consequently, deciding whether a specified coordinate of a specified gate of a rational unit-quaternion circuit is positive is PosSLP-complete.

The proof is in Appendix B.7. Two conjugations do the work. Conjugation by $\mathbf{i}$ is $\iota(u) = (u_0, u_1, -u_2, -u_3)$, and conjugation by $\mathbf{c}$ permutes the axes cyclically, $\text{rot}(u) = (u_0, u_3, u_1, u_2)$. For a unit quaternion, the product

$$\text{pr}(u) = u \iota(u) = (1 - 2u_1^2, 2u_0u_1, 2u_1u_3, -2u_1u_2)$$

keeps the selected coordinate, doubled, and suppresses the other vector coordinates to second order. The commutator satisfies the exact identity $[u, w] - \mathbf{1} = uw\bar{u}\bar{w} - \mathbf{1} = 2(0, \vec{u} \times \vec{w}) \bar{u}\bar{w}$, so near the identity it multiplies small rotation vectors. Accordingly, $\text{add}(u, w) = \text{pr}(uw)$ adds two signals and $\text{prod}(u, w) = \text{pr}(\text{rot}([u, \text{rot}(w)]))$ multiplies them, up to the constant factors two and four and errors of relative order δ . Iterating $\text{prod}(g, g)$ from $\mathbf{w}_\vartheta$ produces an exact rational quaternion whose selected coordinate δ satisfies $0 < \delta < 2^{-16 \cdot 2^j}$ after j steps. The same numerator–denominator homogenization as in the cube-root compiler handles orders and cancellation; here the common leading factors are four for products and eight for sums. The upper bound is elementary: four integer numerator circuits and one positive common denominator per gate represent every gate value exactly.

Proposition 5.13 (Realization of quaternion circuits). *Given a rational unit-quaternion circuit with k gates and values $p_1, \dots, p_k$, one can compute in polynomial time a rational quartic F in the $N = 4k$ variables $X = (X_1, \dots, X_k)$, $X_i \in \mathbb{R}^4$, such that F is supplied as a sum of $N + 1$ squares of rational quadratics, $\nabla^2 F \succeq \frac{3}{2}I$, $F^{-1}(0) = \{p\}$ with $p = (p_1, \dots, p_k)$, and a rational positive definite Hessian Gram for F is part of the output. The coordinates of p are preserved exactly; in particular $p \in \mathbb{Q}^N \cap [-1, 1]^N$.*

The proof is in Appendix B.8. The residuals are $X_i - p_i$ for a constant gate, $X_i - X_a X_b$ and $X_i - \bar{X}_a$. With $q_i = |X_i|^2 - 1$, the exposing quadratic of a product gate is $q_i - 2\langle p_i, X_i - X_a X_b \rangle - q_a - q_b$, which equals $|X_i - p_i|^2 - |X_a - p_a \bar{X}_b|^2$ because $p_i \bar{p}_b = p_a$ for unit quaternions; the identity also holds when $a = b$. Weights 16^{1-i} absorb the negative terms of all later gates. Only the coefficients p_i multiplying the residuals are rounded, so the zero stays exactly at the rational gate list.

Proof of Theorem 5.2. Apply Theorem 5.12 and then Proposition 5.13, and let j index the first vector coordinate of the output gate. The unique zero of F is its unique minimizer, so $F^* = 0$ and p is the rational list of gate values, which lies in $[-1, 1]^N$ because every gate is a unit quaternion. The selected coordinate satisfies $p_j \neq 0$ and $p_j > 0$ exactly when $V > 0$. The remaining promises are part of Proposition 5.13. This proves the reduction to $p_j > 0$ and, since $p_j \neq 0$, also to $p_j \geq 0$. Replacing V by $1 - V$ gives the tests $p_j < 0$ and $p_j \leq 0$. The upper bounds follow from Theorem 4.4 with the observable x_j . $\square$

Remark 5.14 (Length of the rational optimizer). In Theorem 5.2, the generator gate after $2T + 2$ iterations is one of the gates, so its selected coordinate is a coordinate of p . It is a nonzero rational of absolute value below $2^{-64 \cdot 4^T}$ (Lemma B.11), so its reduced denominator has more than $64 \cdot 4^T$ bits. The construction has $N = O(T)$ variables, so this length is exponential in N . The reduction never computes this fraction.

5.7 Proof of the classification

Proof of Theorem 5.1. Validity. A reduction first checks the format (3) by coefficient comparison and rational elimination. On an invalid input it outputs a fixed program with output -1 , which is a no-instance of PosSLP. All further statements concern valid inputs.

Upper bounds in (i) and (ii). A valid certified quartic satisfies the hypotheses of Theorem 4.4, with the checked Hessian Gram as curvature certificate. Applying it to the observable $h = f - r$, for which $h(p) = f^* - r$, handles the five relations between f^* and r , and $h = x_j - r$ handles those between p_j and r ; each yields one PosSLP instance.

Lower bounds in (i). Theorem 5.10 maps an instance with output V to a certified quartic with $r = 0$, $M_G \succeq I$, $G^* \neq 0$, and $V > 0 \iff G^* < 0$. Because $G^* \neq 0$, the same instance also decides $G^* \leq 0$. Applied to the program with output $1 - V$, which is positive exactly when $V \leq 0$, it gives $V > 0 \iff G^* > 0 \iff G^* \geq 0$. The coordinate tests are Theorem 5.2.

Part (iii). For a valid certified quartic, $f - f^*$ is a sum of squares of real polynomials by Taylor integration of the Hessian Gram at the stationary point p (Lemma 8.1). Hence $f \geq 0$, $f^* \geq 0$ and membership in the real sum-of-squares cone are equivalent. If f has a positive definite Gram matrix Q on the monomial vector $m(x)$ of degree at most two, then $f(x) \geq \lambda_{\min}(Q) \|m(x)\|^2 \geq \lambda_{\min}(Q) > 0$, because $m(x)$ contains the constant monomial; hence $f^* > 0$. Conversely, if $f^* > 0$, a rational positive definite Gram exists by Theorem 8.11. Thus these three properties are the tests $f^* \geq 0$, $f^* \geq 0$ and $f^* > 0$, which reduce to PosSLP by the upper bounds in (i). For the lower bounds, apply Theorem 5.10 to the program with output $1 - V$. If $V > 0$, its output quartic has $G^* > 0$; it is nonnegative, real SOS, has a rational positive definite polynomial Gram, and is a sum of squares of rational polynomials, because rational elimination of a positive definite rational Gram gives positive rational weights and every positive rational is a sum of rational squares (Lemma 2.20). If $V \leq 0$, then $G^* < 0$, and G has none of these properties.

Part (iv). The instances used above have exactly the stated additional properties. $\square$

5.8 Related comparisons

Sums of numbers with one real conjugate. A different source problem is also encoded by one quartic and one affine inequality. It concerns algebraic numbers given by dense minimal polynomials, rather than circuits, and its relation to PosSLP and to the Square Root Sum problem is not settled here.

Proposition 5.15 (One-real-conjugate sums). *Let $P_1, \dots, P_m \in \mathbb{Q}[T]$ be given by dense coefficient lists, each irreducible with exactly one real root α_i , and let $c_1, \dots, c_m, b \in \mathbb{Q}$. In polynomial time one can compute a rational quartic F , supplied as a sum of squares of rational quadratics with $\nabla^2 F \succeq I$, and a rational affine function ℓ , such that $\{F \leq 0\}$ is a single point and*

$$\{F \leq 0, \ell \leq b\} \neq \emptyset \iff \sum_{i=1}^m c_i \alpha_i \leq b.$$

The comparison $\sum_i c_i \alpha_i \leq b$, and likewise $<$ and $=$, reduces to PosSLP.

Proof. For each i with $\deg P_i > 1$, Theorem 7.4 constructs, in time polynomial in the dense encoding of P_i , a rational quartic F_i on its own block of variables, given as a sum of squares of rational quadratics, with $\nabla^2 F_i \succeq I$ and a unique zero whose first coordinate is α_i . For a rational root use the one-variable quartic $(z - \alpha_i)^2 + (z - \alpha_i)^4$. Put $F = \sum_i F_i$ on disjoint blocks and $\ell = \sum_i c_i z_{i,1}$, where $z_{i,1}$ is the first coordinate of block i . Every summand is nonnegative, so $F \leq 0$ forces every block to its zero, and the affine row then tests the sum. The Hessian is block diagonal with blocks at least I . For an empty list, use $F(z) = z^2 + z^4$ and $\ell = 0$. Finally, F has the supplied curvature bound $\nabla^2 F \succeq I$, and the observable $\ell - b$ at its unique minimizer is $\sum_i c_i \alpha_i - b$; Theorem 4.4 reduces each relation of that observable to zero to one PosSLP instance. $\square$

Signed sums of real cube roots of integers are the special case $P_i = T^3 - a_i$. Such a cubic is irreducible unless a_i is a perfect cube, which binary search detects in polynomial time; negative radicands only change the sign of c_i . Unlike Theorem 5.1, the proposition is an upper bound for its source problem together with a convex encoding of it. We do not know whether PosSLP reduces to the comparison of sums of one-real-conjugate numbers, and none of our results implies such a reduction. Square Root Sum is not covered by this construction: every coordinate of a singleton defined by rational convex polynomial inequalities has exactly one real conjugate (Theorem 7.4), so no such singleton has a coordinate $\sqrt{2}$. This obstruction concerns value-preserving encodings only; it does not exclude other reductions. The block sum has a rational positive semidefinite Hessian Gram, obtained by embedding the block certificates. When $m \geq 2$, so that F has at least two nonempty blocks of variables, it has no positive definite Hessian Gram on the full basis (Remark 2.9). With a single block, and for the empty list, F is itself a quartic of Theorem 7.4, so it has a positive definite rational Hessian Gram; for the empty list, $z^2 + z^4$ has the Hessian Gram $\text{diag}(2, 12)$ on (v, zv) . The reduction to PosSLP in the proof uses the supplied curvature bound $\nabla^2 F \succeq I$, which holds in every case, and needs no positive definite Hessian Gram.

A compact convex program with the same optimizer. Placing the values of a certified root circuit at the unique optimizer of *some* compact convex program is much easier than Theorem 5.7.

Proposition 5.16 (Box-convex baseline). *Let a certified odd-root circuit have boxes with $\xi_i \geq 1$, which the normalization in Theorem 5.7 always achieves, and let $A \geq 1$ bound every $\bar{\xi}_i$ and the sum of the absolute values of all coefficients. In the variables X_i and $Y_{i,e}$, $2 \leq e \leq n_i$, with $Y_{i,1} = X_i$, consider the compact convex set defined by the boxes $X_i \in [\xi_i, \bar{\xi}_i]$, $Y_{i,e} \in [\xi_i^e, \bar{\xi}_i^e]$ and the inequalities*

$$X_i^{d_i} \leq c_i + \sum_{j < i, e} a_{ije} Y_{j,e}, \quad X_i^e \leq Y_{i,e} \quad (2 \leq e \leq n_i).$$

Order the variables gate by gate, each root before its powers, and set $y_t = X_i$ or $y_t = -Y_{i,e}$ accordingly. With $C = NA^N$ and $W_t = (2C + 2)^{N-t}$, the circuit point $X_i = \xi_i$, $Y_{i,e} = \xi_i^e$ is the unique maximizer of $\sum_t W_t y_t$ over this set. All data have polynomial bit length.

The proof is in Appendix B.9. The program has constraints of the circuit’s own degrees, needs the boxes to be convex, and represents the circuit only as the optimizer of a linear objective. The realization of Theorem 5.7 adds what the comparisons in Theorem 5.1 need: a single globally strongly convex polynomial of degree four, without constraints, with the circuit as its unique zero and a checked rational Hessian certificate.

5.9 Scope

The lower bounds are conditional statements about exact computation: a polynomial-time algorithm for any problem in Theorem 5.1(i) or (iii) would give one for PosSLP. They imply neither NP-hardness nor a superpolynomial lower bound, since neither is known for PosSLP, and they give no lower bound for approximate value or point computation. Equality tests have only the upper bound of Theorem 4.4; the constructed instances deliberately avoid equality. Every hard instance encodes its sign by a doubly exponentially small signal, so no inverse-polynomial separation of the hard instances from the threshold is asserted. In particular, the results say nothing about algorithms that may err when the optimal value or coordinate lies within $2^{-\text{poly}(L)}$ of the threshold. The quartics are explicitly expanded with polynomially many monomials; the lower bounds do not depend on succinct or circuit-encoded input. They concern unconstrained optimization, a halfspace, or a fixed cube. Exact comparison over general rational polyhedra is treated in Section 6.

6 Linear constraints and integer variables

Branch-and-bound methods for mixed-integer nonlinear programming solve constrained continuous relaxations, and the decisions they take are exact comparisons. A node is discarded when its bound is at least the incumbent value, a constraint is either binding or slack, and an integer assignment is either optimal or not. For a globally strongly convex objective without constraints, Theorem 4.4 reduces every such comparison at the optimizer to one PosSLP instance. This section asks what rational linear constraints and integer variables add to that picture.

The answer has three parts.

Arbitrary polyhedra. The set of all constraints that are tight at the unique optimizer is a certificate of polynomial length, and it is the only certificate that our verifier accepts. Exact comparisons therefore lie in $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$ (Theorem 6.2). We do not know how to find the active set deterministically in polynomial time with a PosSLP oracle. Section 6.7 explains what a deterministic method would have to supply.

Structure. Three different kinds of structure remove the guess. If the cubic and quartic part of the objective depends on only k linear forms, exact comparison needs no oracle and is fixed-parameter tractable in k (Theorem 6.6). If every Taylor quadratic program can be solved exactly with polynomially many arithmetic operations, independently of the bit lengths of its data, then exact Newton steps give a deterministic polynomial-time algorithm with a PosSLP oracle (Theorem 6.12). Boxes with forest-structured or Stieltjes Hessians and separable polynomial network flows are examples, and for them each exact predicate reduces to a single PosSLP instance (Corollary 6.15). If the constraint normals span a space of small dimension r , a Las Vegas algorithm with a PosSLP oracle runs in expected time $(r+1)^{O(r)}L^C$ (Theorem 6.16).

Integer variables. The search over integer blocks needs only ordinary rational computation at polynomial precision. It produces a list that contains every optimal integer block (Theorems 6.18 and 6.20). Exact arithmetic is needed only to compare the continuous fiber minima of the listed blocks. Knowing the optimal value does not remove that arithmetic: one binary decision can carry a PosSLP-hard sign (Corollary 6.24).

Additional structure	Computation	Output	Result
none (arbitrary rational polyhedron)	$\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$	sign of $h(p)$	Thm. 6.2
supplied lower bound on nonzero optimal slacks	one PosSLP instance	sign of $h(p)$	Prop. 4.9
nonlinear dimension k	ordinary, $F(k)L^C$	sign, active set, implicit p	Thm. 6.6
exact Taylor QP with polynomial arithmetic cost	$\mathbf{P}^{\text{PosSLP}}$	sign, active set	Thm. 6.12
box with forest, Stieltjes, or comparison-matrix Hessians; separable flows	one PosSLP instance per predicate	sign, one active bound	Cor. 6.13– 6.15
constraint rank r	Las Vegas with PosSLP oracle, expected $(r+1)^{O(r)}L^C$	sign, implicit p	Thm. 6.16
t integer variables, unconstrained continuous fibers	ordinary list in $2^{O(t \log(t+1))}L^C$; nonadaptive PosSLP selection	every optimal integer block	Thm. 6.18, Cor. 6.19
t integer variables, mixed linear constraints	ordinary list in $F(t)L^C$	list containing every optimal block	Thm. 6.20
same, with nonlinear dimension k	ordinary, $F(k+t)L^C$	optimal block, value signs	Cor. 6.21
same, with continuous constraint rank r	Las Vegas with PosSLP oracle, expected $F(t, r)L^C$	first optimal block, implicit p	Thm. 6.22
one binary variable, rank one, known optimal value	PosSLP-hard	the optimal bit	Cor. 6.24

Table 4: Exact comparison at the optimizer of a globally strongly convex explicit quartic (or the solution of a strongly monotone cubic variational inequality) under linear constraints and integer variables. Here F denotes a computable function and C an absolute constant. “Implicit p ” means a rational affine chart together with an explicit polynomial system whose unique real solution determines p .

Table 4 summarizes the results. All of them assume global strong convexity, or global strong monotonicity in the variational-inequality versions, with a supplied rational modulus. The established ingredients are credited where they are used: polyhedral optimality and Farkas’ lemma, essential-variable extraction, the quadratic convergence of exact constrained Newton steps, exact box and flow quadratic programming, linear programming with an arithmetic count independent of the objective and right-hand-side lengths, Clarkson-type sampling for violator spaces, and integer feasibility with integer-point separation oracles. What this section adds to them is the exact-arithmetic analysis that connects them: separation bounds that hold uniformly over the unknown active face, circuit simulations whose cost does not depend on the length of expanded rational numbers, and oracle constructions whose answers never require an exact fiber value.

6.1 Setting

Unless stated otherwise, $f \in \mathbb{Q}[x_1, \dots, x_n]$ is an explicitly encoded polynomial of degree at most four, and a rational $\mu > 0$ is supplied with the promise

$$\nabla^2 f(x) \succeq \mu I \quad (x \in \mathbb{R}^n). \quad (8)$$

The feasible set is an explicit rational polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ with $A \in \mathbb{Q}^{m \times n}$, rows $a_i^\top$, and $b \in \mathbb{Q}^m$. Rows may be zero, repeated, or redundant; P may be unbounded or lower-dimensional; an equality is written as two inequalities. An *observable* is an explicitly encoded $h \in \mathbb{Q}[x]$ of degree at most four, and $\bowtie$ ranges over $\{<, \leq, =, \neq, \geq, >\}$. Value and coordinate comparisons $f(p) \bowtie \tau$ and $p_j \bowtie \tau$ with $\tau \in \mathbb{Q}$ are the observables $h = f - \tau$ and $h = x_j - \tau$. We write $L \geq 2$ for the total binary length of all these data, including μ , h , thresholds, and any supplied certificate.

When $P \neq \emptyset$, (8) makes f coercive, so f has a unique minimizer p on P . Its *active set* is $S(p) = \{i : a_i^\top p = b_i\}$, the complete set of tight rows rather than a selected support. Emptiness of P is decided by rational linear programming before any question about p is asked, and value comparisons use the convention $\min \emptyset = +\infty$.

Several results hold for variational inequalities without a potential. Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an explicitly encoded polynomial map of degree at most three, with a supplied rational $\mu > 0$ and the promise

$$(T(x) - T(y))^\top (x - y) \geq \mu \|x - y\|^2 \quad (x, y \in \mathbb{R}^n). \quad (9)$$

A solution of $\text{VI}(T, P)$ is a point $p \in P$ with $T(p)^\top (x - p) \geq 0$ for every $x \in P$. For $T = \nabla f$, condition (9) is equivalent to (8), and the solutions of $\text{VI}(\nabla f, P)$ are the minimizers of f on P .

The curvature conditions are promises; nothing below recognizes them. If the input instead carries a checked certificate, namely a positive definite full Hessian Gram for f or the Jacobian Gram of Theorem 4.7 for T , then an invalid certificate is rejected before anything else is done. Each statement below then becomes a statement about an ordinary language, with a positive rational modulus derived from the certificate. The nonlinear dimension theorem uses a weaker checkable format (Remark 6.8).

The structural parameters of this section are the nonlinear dimension k (Section 6.3), the rank r of the continuous constraint normals (Section 6.5), and the number t of integer variables (Section 6.6). An algorithm is *fixed-parameter tractable* (FPT) in a parameter κ if its running time is at most $F(\kappa)L^C$ for a computable function F and an absolute constant C . A constant that depends on the parameter never enters the exponent of L .

The following geometric facts are used throughout. Their proofs are elementary and are given in Appendix D.1. Index sets of rows are denoted by J ; A_J and b_J are the corresponding submatrix and subvector.

Lemma 6.1 (Polyhedral optimality and affine charts). *Let $P \neq \emptyset$, let T satisfy (9), and for $x \in P$ write $S(x) = \{i : a_i^\top x = b_i\}$.*

- (a) $\text{VI}(T, P)$ has exactly one solution p .
- (b) A point $x \in P$ solves $\text{VI}(T, P)$ if and only if $-T(x) \in \text{cone}\{a_i : i \in S(x)\}$. In that case there are a set $J \subseteq S(x)$ of linearly independent rows, so $|J| \leq \text{rank } A$, and a vector $\lambda \in \mathbb{R}_{\geq 0}^J$ with $T(x) + A_J^\top \lambda = 0$.
- (c) For every set J of linearly independent rows, rational linear algebra computes in polynomial time $\bar{x} \in \mathbb{Q}^n$ and $Z \in \mathbb{Q}^{n \times (n - |J|)}$ with

$$\{x : A_J x = b_J\} = \{\bar{x} + Zy : y \in \mathbb{R}^{n - |J|}\}, \quad Z^\top Z \succeq I.$$

Their bit lengths are bounded by one polynomial in L for all J . The explicit cubic map $U_J(y) = Z^T T(\bar{x} + Zy)$ satisfies (9) with the same μ . For $T = \nabla f$ it is the gradient of $g_J(y) = f(\bar{x} + Zy)$, and $\nabla^2 g_J \succeq \mu I$.

- (d) If $A_J p = b_J$ and $T(p) \in \text{span}\{a_i : i \in J\}$, then $p = \bar{x} + Zy_J$, where y_J is the unique zero of U_J . This applies to every support J given by (b) at $x = p$, and to every row basis J of $A_{S(p)}$.

When $|J| = n$, the matrix Z is empty and the chart is the single rational point $\bar{x}$; we then read every statement about U_J and its zero as a statement about this point. Part (d) is what makes a guessed or computed active set useful: it converts a constrained problem into the unconstrained zero of an explicit strongly monotone map, to which Theorems 4.4 and 4.7 apply. Part (c) needs no conditioning assumption, because the identity block in Z gives $Z^T Z \succeq I$.

6.2 Arbitrary polyhedra: a unique active-set certificate

Lemma 6.1(b) suggests a certificate. Guess an independent support J , compute the zero y_J of U_J , and check primal feasibility, nonnegativity of the multipliers

$$\lambda_J(y) = -(A_J A_J^T)^{-1} A_J T(\bar{x} + Zy), \quad (10)$$

and stationarity $T(\bar{x} + Zy) + A_J^T \lambda_J(y) = 0$ at $y = y_J$. Every tested quantity is an explicit polynomial of degree at most three evaluated at the zero of a strongly monotone cubic map, so Theorem 4.7 decides each test with PosSLP. This puts exact comparison in $\text{NP}^{\text{PosSLP}} \cap \text{coNP}^{\text{PosSLP}}$, but the certificate is not unique: with degenerate multipliers many supports pass. The complete active set $S(p)$ is unique. Verifying it, however, requires showing that $-T(p)$ lies in the cone of the active normals without knowing which multipliers to use. That is a linear program whose objective is the algebraic vector $T(p)$.

Theorem 6.2 (Unambiguous certificates). *Let T satisfy (9), let $P = \{x : Ax \leq b\}$ be an explicit rational polyhedron, let h be an observable, and let $\bowtie \in \{<, \leq, =, \neq, \geq, >\}$. Consider the promise problem whose yes-instances satisfy $P \neq \emptyset$ and $h(p) \bowtie 0$, where p is the unique solution of $\text{VI}(T, P)$. It belongs to $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$. In particular, for an explicit quartic f satisfying (8), every exact value, coordinate, and observable comparison at the minimizer of f on P belongs to $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$. With a checked Hessian or Jacobian Gram in place of the bare modulus, the same holds for the ordinary languages in which invalid certificates are rejected.*

Proof idea. The machine first rejects invalid certificates and decides $P \neq \emptyset$ by linear programming. It then guesses one bit per row, that is, a set $S \subseteq \{1, \dots, m\}$. Deterministically it chooses a row basis J of A_S , the chart of Lemma 6.1(c), and the zero $p_S = \bar{x} + Zy_J$ of U_J . It accepts the optimality part of the certificate exactly when two conditions hold: the slack $b_i - a_i^T p_S$ is zero for $i \in S$ and positive for $i \notin S$; and

$$T(p_S)^T v \geq 0 \quad \text{for every } v \in Q_S, \quad Q_S = \{v : A_S v \leq 0, -1 \leq v_j \leq 1 \ (j \leq n)\}. \quad (11)$$

Every slack is an affine observable at the zero of U_J . The test (11) is the delicate step. Every vertex v of Q_S has coordinates of polynomial bit length, so all vertex observables $v^T T(\bar{x} + Zy)$ have one common encoding bound. One shared Newton circuit $\hat{y}$ for y_J (Lemma 4.3) therefore approximates all of them to within one eighth of their common separation bound. A linear-programming algorithm whose arithmetic count depends only on the encoded constraint matrix, simulated on rational circuits with PosSLP comparisons, minimizes the rational circuit objective $T(\bar{x} + Z\hat{y})^T v$ over Q_S . Its optimal vertex is recovered as printed rational numbers, and one exact comparison of its true observable

decides (11). A passing set S forces $p_S = p$ and $S = S(p)$, and $S(p)$ passes. Thus exactly one guess passes, and the machine finally tests $h(p_S) \bowtie 0$, or its complement for the coUP side. Appendix D.2 gives the complete proof, including the circuit-objective linear program (Lemma D.3). $\square$

Remark 6.3 (What the theorem does and does not give). Guessing an independent support instead of the full mask gives the simpler $\text{NP}^{\text{PosSLP}} \cap \text{coNP}^{\text{PosSLP}}$ bound described above (Lemma D.2). Enumerating all supports is deterministic but takes $\sum_{s \leq n} \binom{m}{s}$ steps. By Allender et al. [6], PosSLP lies in the counting hierarchy, hence so do all predicates of Theorem 6.2. The theorem does not reduce the constrained problem to one PosSLP instance, and the adaptive compilation of Theorem 4.10(b) does not apply, because the guess of the active set is not removed. Nor does ordinary approximation find the active set in general: nothing in the separation bound prevents a nonzero slack at p from being doubly exponentially small in L , and an approximation to polynomially many bits cannot distinguish such a slack from zero.

If the input also supplies a margin that separates the active rows from the inactive ones, ordinary approximation finds the active set, and the oracle is needed only for one final comparison. Proposition 4.9 states this: when every nonzero optimal slack is at least a supplied rational $\delta > 0$, an exactly feasible rational approximation of the minimizer determines every slack to within $\delta/8$, thresholding at $\delta/2$ returns $S(p)$, and Theorem 4.4, applied on the chart of a row basis of $A_{S(p)}$, yields one PosSLP instance per relation. No lower bound on nonzero multipliers is needed. The promise is substantial and is not verified, and the separation bound for algebraic numbers does not supply a margin of polynomial bit length.

6.3 Few nonlinear directions

Many models are quadratic except for a few higher-degree terms in a few linear combinations of the variables, for example a strongly convex quadratic plus a quartic penalty $(a^\top x)^4$, or a quartic risk term in a few factors. The constraints of such a model may be numerous and of full rank. The next theorem shows that the algebraic difficulty of the optimizer then lives in few dimensions, and that exact comparison needs no oracle.

Definition 6.4 (Nonlinear dimension). Write $f = f_{\leq 2} + f_3 + f_4$, where f_j is homogeneous of degree j and $f_{\leq 2}$ has degree at most two. The *nonlinear kernel* of f is

$$K_f = \{v \in \mathbb{R}^n : D_v f_3 \equiv 0 \text{ and } D_v f_4 \equiv 0\},$$

where $D_v g = \sum_i v_i \partial_i g$ and $\equiv$ denotes equality of polynomials. The *nonlinear dimension* of f is $k = n - \dim K_f$.

Lemma 6.5 (Essential directions). (a) Let W be the rational matrix with rows indexed by the monomials x^β of degrees two and three, whose entry $W_{\beta,i}$ is the coefficient of x^β in $\partial_i(f_3 + f_4)$. Then $K_f = \ker W$. Hence $k = \text{rank } W$, and rational matrices $U \in \mathbb{Q}^{k \times n}$ with $\ker U = K_f$ and $V \in \mathbb{Q}^{n \times k}$ with $UV = I_k$ are computable in polynomial time.

(b) $f_3(x) + f_4(x) = \theta(Ux)$ for all x , where $\theta(u) = f_3(Vu) + f_4(Vu)$.

(c) The nonlinear dimension does not change under an invertible rational affine change of variables.

Part (b) is the essential-variable construction; for homogeneous forms it is Carlini [24, Proposition 1]. Here it is applied separately to the cubic and quartic parts, and part (c) shows that the resulting parameter is intrinsic even though translation mixes these parts.

Theorem 6.6 (Exact comparison in few nonlinear directions). *Let f satisfy (8) and have nonlinear dimension k , let $P = \{x : Ax \leq b\}$ be an explicit rational polyhedron, and let h be an observable. There are a function $F(k) = 2^{O(k)}$ and an absolute constant C such that a deterministic ordinary algorithm, using no oracle, runs in time $F(k)L^C$ and either reports $P = \emptyset$ or*

- (i) *decides whether $h(p)$ is negative, zero, or positive;*
- (ii) *computes the active set $S(p)$; and*
- (iii) *computes $\bar{c} \in \mathbb{Q}^n$ and $D \in \mathbb{Q}^{n \times s}$ of full column rank $s \leq k$ such that $p = \bar{c} + Du_*$, where u_* is the unique zero of the gradient of the explicit quartic $\varphi(u) = f(\bar{c} + Du)$ on $\mathbb{R}^s$, and $\nabla^2 \varphi \succeq \mu D^\top D \succeq \nu I$ with the rational modulus $\nu = \mu \det(D^\top D) / \text{tr}(D^\top D)^{s-1} > 0$ when $s \geq 1$.*

The data in (iii) have bit length at most $T(L)$ for a polynomial T that does not depend on k . The number of constraints, their rank, the ambient dimension, and the rank of the quadratic part are unrestricted.

Proof idea. The proof uses the unknown active set only to establish a bound. Let J be a row basis of $A_{S(p)}$ and $\bar{x} + Zy$ its chart. By Lemma 6.1(d), p is the unconstrained minimizer of $f(\bar{x} + Zy)$. Split y along $\ker(UZ)$ and a complement. On the kernel directions the cubic and quartic parts are constant, so f is a strongly convex quadratic there and can be minimized in closed form. What remains is a strongly convex quartic φ in $s = \text{rank}(UZ) \leq k$ variables, with data of bit length $T(L)$ for every possible J . The formula $\exists u (\nabla \varphi(u) = 0 \wedge z = h(\bar{c} + Du))$ has s quantified variables and defines the singleton $\{h(p)\}$. The coefficient bound in one-block quantifier elimination is linear in the input coefficient length, with a factor depending only on the number of quantified variables (Lemma 4.1). Hence $h(p) \neq 0$ implies $|h(p)| \geq 2^{-G(k)T(L)}$ with $G(k) = 2 \cdot 4^{c_0 k}$, where c_0 is the absolute constant of that lemma. An exactly feasible rational point with objective error $2^{-O(G(k)T(L))}$ is within distance $2^{-G(k)T(L)} / (8B)$ of p , where B bounds $\|\nabla h\|$, and is computable in ordinary time $F(k)L^C$ (Lemma 3.1). Comparing its rational observable value with $\pm 2^{-G(k)T(L)-1}$ decides the sign exactly. The slacks give $S(p)$, and then the elimination above is carried out on the now known set. Appendix D.3 gives the details. $\square$

The algorithm prints $G(k)T(L)$ accuracy bits. This number is FPT in k , but for unconstrained objectives of nonlinear dimension n it is exponential in n , and Theorem 4.4 is then the better bound. The two results measure different sources of difficulty.

Example 6.7. Let $f(x) = \frac{1}{2}x^\top Qx + c^\top x + (a^\top x)^4$ with rational $Q \succeq \mu I$, $a \neq 0$, and c . Then $K_f = a^\perp$ and $k = 1$. Theorem 6.6 decides every exact comparison at the minimizer over an arbitrary rational polyhedron in ordinary polynomial time, although Q may be dense, the constraint matrix may have rank n , and the polyhedron may have exponentially many faces. The optimizer is an explicit affine function of the unique real root of one univariate cubic equation.

Remark 6.8 (A checkable curvature certificate). A rational positive semidefinite matrix G_0 with $v^\top (\nabla^2 f(x) - \mu I)v = w^\top G_0 w$, where $w = (v, x \otimes v)$, certifies (8). Coefficient matching and rational semidefiniteness testing check it in polynomial time. A positive definite full Hessian Gram is the wrong format here: if $k < n$, it cannot exist. Indeed, for $0 \neq e \in K_f$ the Hessian $\nabla^2 f(se)$ does not depend on s , whereas a positive definite Gram would give $v^\top \nabla^2 f(se)v \geq c(1 + s^2 \|e\|^2) \|v\|^2$ for some $c > 0$. We do not claim that every globally strongly convex quartic has a certificate of either kind.

Remark 6.9 (Related parameters). Low-rank optimization usually bounds the rank of the whole objective [89] and asks for approximation. Kannan and Rademacher [63] treat a convex function

plus a polynomial perturbation in k variables, again with an approximation guarantee whose cost grows with the reciprocal accuracy. Here only the cubic and quartic part is restricted, the quadratic part and the constraints are arbitrary, the output is exact including zero, and the exponent of L does not depend on k . The ingredients that make this possible are strong convexity, which turns objective accuracy into point accuracy and eliminates the quadratic directions, and the linear dependence of the elimination bound on coefficient length.

6.4 Exact constrained Newton steps

For an unconstrained problem, a Newton step is a linear solve, so exact Newton steps can be carried out on shared rational circuits; this is the mechanism behind Theorem 4.4. With constraints, a Newton step is a quadratic program over P . Its convergence is classical. The difficulty is arithmetic: the Taylor data of the j -th step can have bit length exponential in j , so a quadratic-programming algorithm whose running time is polynomial in the bit length of its data is of no use. What is needed is an algorithm whose number of arithmetic operations does not depend on the bit length of the Hessian and linear cost.

Definition 6.10 (Arithmetic Taylor-QP solver). Let $\mathcal{C}$ be a class of instances (f, P) with f explicit, satisfying (8), and $P \neq \emptyset$. An *arithmetic Taylor-QP solver* for $\mathcal{C}$ is a deterministic algorithm with the following properties. It receives the explicit data of an instance in $\mathcal{C}$ and rational $H \in \mathbb{Q}^{n \times n}$, $c \in \mathbb{Q}^n$ such that $H = \nabla^2 f(x)$ and $c = \nabla f(x) - \nabla^2 f(x)x$ for some $x \in P$. It returns the exact minimizer of $\frac{1}{2}y^\top Hy + c^\top y$ over P . Its operations on numbers are addition, subtraction, multiplication, division by a nonzero number, and comparison, applied to the entries of H and c , to earlier results, and to rational constants computed from the explicit data in time polynomial in L . The choice of each operation is determined by the outcomes of earlier comparisons through a control that runs in time polynomial in L , and the number of operations is at most a polynomial in L for all admissible (H, c) , independently of their bit lengths.

A dependence on logarithms of numerical bounds that are themselves of polynomial bit length is allowed; a dependence on the bit lengths of H or c is not. A quadratic-programming algorithm that is polynomial only in the bit length of its data does not satisfy the definition.

Lemma 6.11 (Constrained Newton step). *Let f be twice continuously differentiable with $\nabla^2 f \succeq \mu I$, let P be a nonempty closed convex set with minimizer p of f , and let $x \in P$. Let Λ be a Lipschitz constant of $\nabla^2 f$, in operator norm, on a convex set containing x and p . Then the minimizer*

$$N(x) = \arg \min_{y \in P} \left\{ \nabla f(x)^\top (y - x) + \frac{1}{2} (y - x)^\top \nabla^2 f(x) (y - x) \right\}$$

satisfies $\|N(x) - p\| \leq \frac{\Lambda}{2\mu} \|x - p\|^2$.

Proof. Write $y = N(x)$ and $H = \nabla^2 f(x)$. The first-order conditions of the two convex problems give $(\nabla f(x) + H(y - x))^\top (p - y) \geq 0$ and $\nabla f(p)^\top (y - p) \geq 0$. Add them and put $\rho = \nabla f(p) - \nabla f(x) - H(p - x)$. Then

$$\mu \|y - p\|^2 \leq (y - p)^\top H (y - p) \leq \rho^\top (y - p) \leq \|\rho\| \|y - p\|.$$

Taylor's formula gives $\rho = \int_0^1 (\nabla^2 f(x + \vartheta(p - x)) - H)(p - x) d\vartheta$, so $\|\rho\| \leq \frac{\Lambda}{2} \|x - p\|^2$. $\square$

This is the exact proximal Newton estimate of Lee et al. [77, Theorem 3.4], with the indicator function of P as the nonsmooth term. We record the proof because it shows a point that matters here: the estimate never identifies the optimal face. It holds when the Taylor model and the true problem have different active sets, and when optimal multipliers vanish.

Theorem 6.12 (Transfer from arithmetic QP solvers). *Fix $D \geq 2$. Let $\mathcal{C}$ be a class of instances (f, P) with f explicit of degree at most D satisfying (8) and P a nonempty explicit rational polyhedron, and suppose that $\mathcal{C}$ has an arithmetic Taylor-QP solver. For every explicit observable h of degree at most D , each predicate $h(p) \bowtie 0$ is decided in deterministic polynomial time with a PosSLP oracle, and $S(p)$ is computed in the same way. If $f(p) < \tau$ for a rational τ , the computation also yields a point $x \in P$ with $f(x) < \tau$, given by a shared rational circuit of polynomial size.*

Proof idea. An ordinary convex-optimization call gives an exactly feasible rational warm start x_0 with $\|x_0 - p\| \leq 1/(2\kappa)$, where $\kappa = \max\{1, \Lambda/(2\mu)\}$ and Λ is an explicit Hessian-Lipschitz bound on a ball that contains all iterates. Lemma 6.11 gives $\|x_j - p\| \leq \kappa^{-1}2^{-2^j}$ for the exact iterates $x_{j+1} = N(x_j)$. Each iterate is computed by the solver, with every number held as a shared numerator-denominator circuit and every comparison answered by PosSLP. The Karush–Kuhn–Tucker system of the original instance, extended by $z = h(x)$, defines the singleton $\{h(p)\}$, so Lemma 4.1 gives $h(p) = 0$ or $|h(p)| \geq 2^{-2^{e_D(L)}}$ for a polynomial e_D that depends only on D . After polynomially many steps a final comparison of $h(x_j)$ with $\pm 2^{-2^{e_D(L)-1}}$ decides the predicate. The comparisons inside the solver depend on earlier oracle answers, so the procedure is a Turing reduction. Appendix D.4 gives the complete argument. $\square$

The theorem isolates the arithmetic interface; it does not assert that arbitrary convex quadratic programs have such solvers. We now give two classes that do.

For a symmetric matrix H , its *comparison matrix* $M(H)$ has $M(H)_{ii} = H_{ii}$ and $M(H)_{ij} = -|H_{ij}|$ for $i \neq j$.

Corollary 6.13 (Structured boxes). *Let $P = \{x : \ell \leq x \leq u\}$ with $\ell_i \in \mathbb{Q} \cup \{-\infty\}$ and $u_i \in \mathbb{Q} \cup \{+\infty\}$, and let f be explicit of fixed degree D and satisfy (8). Suppose that one of the following holds for every $x \in P$:*

- (i) *every pair $i \neq j$ with $\partial_i \partial_j f(x) \neq 0$ is an edge of a supplied forest on $\{1, \dots, n\}$;*
- (ii) *$\partial_i \partial_j f(x) \leq 0$ for all $i \neq j$;*
- (iii) *$M(\nabla^2 f(x))$ is positive definite.*

Then for every explicit observable h of degree at most D , the predicates $h(p) \bowtie 0$ are in $\mathbf{P}^{\text{PosSLP}}$, and so is the set of active bounds. The number of active bounds and the dimension are unrestricted, and zero multipliers are allowed.

Conditions (i) and (ii) imply (iii). For (ii), $M(H) = H$. For (i), choosing signs along each tree gives a diagonal sign matrix Σ with $M(H) = \Sigma H \Sigma$. The proof (Appendix D.4.1) first truncates infinite bounds without changing p (Lemma D.6). It then shows that every positive definite H with $M(H) \succ 0$ admits a rational vector $v \geq \mathbf{1}$, computable by rational linear algebra and sign tests, such that $H_{SS}^{-1} v_S \geq 0$ for every nonempty S (Lemma D.7). With such a vector, a parametric principal pivoting method in the style of Pang and Han [98, Algorithm I] solves the box quadratic program with at most $2n$ pivots and $O(n^4)$ arithmetic operations (Lemma D.8). We include complete proofs of both lemmas so that the operation count is explicit. Condition (i) can be checked from the coefficients of f when the box is full-dimensional. Conditions (ii) and (iii) are promises.

For any graph G , rational $w_{ij} \geq 0$, and rational c ,

$$f(x) = \frac{\mu}{2} \|x\|^2 + \sum_{ij \in E(G)} w_{ij} (x_i - x_j)^4 + c^\top x$$

has Hessian μI plus a graph Laplacian with nonnegative weights $12w_{ij}(x_i - x_j)^2$, so it satisfies (ii) on every box. The graph may be dense, the box constraints have rank n , and the nonlinear dimension can be $n - 1$. For example, if G is a star with center 1 and positive weights, the forms $x_1 - x_j$ are linearly independent, so their cubes are linearly independent and K_f is spanned by the all-ones vector. Neither Theorem 6.6 nor Theorem 6.16 then gives a polynomial bound. Replacing each difference by $x_i + x_{i+1}$ along a path gives a tridiagonal Hessian with positive off-diagonal entries, covered by (i) but not by (ii).

Corollary 6.14 (Separable network flows). *Let $G = (V, E)$ be a directed graph with incidence matrix A_G , and let $P = \{x \in \mathbb{R}^E : A_G x = b, \ell \leq x \leq u\}$ with rational b and with $\ell_e \in \mathbb{Q} \cup \{-\infty\}$, $u_e \in \mathbb{Q} \cup \{+\infty\}$. Let $f(x) = \sum_{e \in E} f_e(x_e)$ with explicit $f_e \in \mathbb{Q}[s]$ of fixed degree D and $f_e'' \geq \mu$ on $\mathbb{R}$. Then for every explicit observable h of degree at most D , the predicates $h(p) \bowtie 0$ are in $\mathbf{P}^{\text{PosSLP}}$, and so is the set of active bounds. The graph, the dimension, and the constraint rank are unrestricted; parallel arcs, redundant balance equations, and fixed arcs are allowed.*

The Taylor model of a separable objective is a separable strictly convex quadratic flow problem with arc costs $\frac{1}{2}f_e''(x_e)y_e^2 + (f_e'(x_e) - f_e''(x_e)x_e)y_e$. The strongly polynomial algorithm of Végh [117, Theorem 20] solves it exactly with a number of arithmetic operations and comparisons that depends only on the size of the graph. Appendix D.4.2 states the precise interface and proves that the artificial arc costs used by such algorithms can be chosen from the circuit data. For $D \leq 4$ the curvature promise is checkable: $f_e'' - \mu$ is a univariate quadratic. The observable h may couple many arcs; only the objective must be separable.

The deterministic oracle algorithms of Corollaries 6.13 and 6.14 make adaptive queries. The compilation theorem for deterministic PosSLP computations removes the adaptivity.

Corollary 6.15 (One sign instance for structured classes). *Under the hypotheses of Theorem 6.12, and in particular of Corollary 6.13 or Corollary 6.14, there is a deterministic polynomial-time algorithm that maps each instance, observable h , and relation $\bowtie$ to one integer arithmetic circuit Φ such that, on every promised instance, $\Phi > 0$ if and only if $h(p) \bowtie 0$. The same holds for each predicate “bound i is active at p ” and for every polynomial-size Boolean combination of such predicates.*

Proof. The algorithm of Theorem 6.12 is a deterministic oracle Turing machine whose queries are integer circuits. Its running time and the lengths of its queries are bounded by one polynomial in L that does not depend on the expanded values of the circuits: the warm start is an ordinary computation, the number of Newton steps and of solver operations per step is polynomial in L , and each query is a circuit that the machine has written. A clock and syntax checks make the machine total on all inputs without changing it on promised ones. Theorem 4.10(b) compiles such a machine into one circuit. Appendix D.4.3 checks these hypotheses in detail. $\square$

The corollary concerns one requested predicate at a time. The set of active bounds is a vector of bits; it is obtained from a nonadaptive list of compiled instances, one per bound, not from one circuit. The structured classes are not claimed to be PosSLP-hard, and no ordinary polynomial-time algorithm is claimed. Theorem 6.2 and the Las Vegas Theorem 6.16 are not covered: the first guesses its certificate, and the second uses randomness with an expected rather than a worst-case time bound. For parameterized compositions, for example the candidate lists of Section 6.6 with structured fibers, the same compilation gives a many-one reduction whose construction time is FPT in the parameter, not polynomial when the parameter grows.

6.5 Few constraint directions

A third parameter is the rank $r = \text{rank } A$ of the constraint normals. A polyhedron can have very many constraints whose normals lie in a space of small dimension, for example many inequalities on a few linear forms of the variables. By Lemma 6.1(b) the solution is then determined by at most r active rows, but enumerating the candidate supports takes $m^{O(r)}$ steps, with r in the exponent of the input length. Random sampling in the style of Clarkson removes r from that exponent.

Theorem 6.16 (Constraint rank). *Let T satisfy (9), let $P = \{x : Ax \leq b\}$ be an explicit rational polyhedron with $r = \text{rank } A$, and let h be an observable. There is a Las Vegas algorithm with a PosSLP oracle that decides whether $P = \emptyset$ and otherwise decides any predicate $h(p) \bowtie 0$ at the unique solution p of $\text{VI}(T, P)$. Its expected running time is $(r + 1)^{O(r)} L^C$ for an absolute constant C , every oracle query has length polynomial in L , and every answer it returns is correct. It also returns a set J of at most r linearly independent rows such that, in the notation of Lemma 6.1, $p = \bar{x} + Zy_J$ with y_J the unique zero of the explicit map U_J . For $T = \nabla f$ this covers exact value, coordinate, and observable comparisons for strongly convex quartic optimization.*

Proof idea. After checking that P is nonempty, every subset G of the rows defines a nonempty polyhedron P_G with a unique solution p_G of $\text{VI}(T, P_G)$. The map $V(G) = \{i : a_i^\top p_G > b_i\}$ defines a violator space in the sense of Gärtner et al. [44]. Locality follows from uniqueness, so no potential function is needed. Applying Lemma 6.1(b) to an inclusion-minimal basis shows that every basis consists of linearly independent rows, so the combinatorial dimension is at most r . The sampling algorithms need only violation tests whose first argument has at most r elements. Such a test enumerates the at most 2^r independent subsets of its argument and checks each with the support test of Lemma D.2, using Theorem 4.7. Clarkson’s algorithms, as analyzed for violator spaces by Gärtner et al. [44], use an expected $O(rm + r^{O(r)})$ such tests. Appendix D.5 gives the complete proof, including the binary implementation of the weighted sampling. $\square$

The parameter is the rank of all constraint normals, not the size of the final active support; the sampling analysis bounds the bases of all subproblems. A box has rank n , so the theorem does not make box-constrained problems easy. The oracle is essential even for $r = 0$: unconstrained exact comparison is already PosSLP-hard (Theorem 5.1). Randomness affects only the running time. We do not claim a deterministic algorithm with this parameter dependence, and the compilation of Corollary 6.15 does not apply to this algorithm.

6.6 Integer variables

We now add t integer variables. Consider

$$\min\{f(z, y) : z \in \mathbb{Z}^t, y \in \mathbb{R}^n, (z, y) \in P\}, \quad (12)$$

where $f \in \mathbb{Q}[z, y]$ is explicit of degree at most four with a supplied rational $\mu > 0$ and the promise $\nabla^2 f(z, y) \succeq \mu I_{t+n}$ on all of $\mathbb{R}^{t+n}$. We treat two kinds of feasible sets: $P = Q \times \mathbb{R}^n$ with an explicit rational polyhedron $Q \subseteq \mathbb{R}^t$ (unconstrained continuous fibers), and $P = \{(z, y) : Az + By \leq d\}$ with explicit rational A, B, d (mixed linear constraints). An *optimal integer block* is the z -part of an optimal solution. For z in the projection of P onto the z -coordinates, write $g(z) = \min\{f(z, y) : (z, y) \in P\}$ for its fiber minimum; z is *feasible* if it is an integer point of this projection. When (12) is feasible, coercivity on the closed set $(\mathbb{Z}^t \times \mathbb{R}^n) \cap P$ gives an optimal solution, each fiber has a unique minimizer, and g is μ -strongly convex on the projection of P .

The fiber values $g(z)$ are algebraic numbers, typically irrational. An integer optimization method that compares them, or bisects on the objective, needs exact arithmetic at every step. Our search

avoids exact values altogether. At each queried integer point it computes, at polynomial precision, a rational inequality that is violated by the query and satisfied by every other integer point whose fiber value is no larger. The following lemma is the source of all such cuts.

Lemma 6.17 (Integer cut). *Let $z \in \mathbb{Z}^t$ be feasible and let $c_z \in \mathbb{Q}^t$ satisfy*

$$g(w) - g(z) \geq c_z^\top(w - z) + \frac{\mu}{4}\|w - z\|^2 \quad (13)$$

for every feasible $w \in \mathbb{Z}^t \setminus \{z\}$. If $c_z = 0$, then z is the unique optimal integer block. Otherwise the inequality $c_z^\top(x - z) \leq -\mu/4$ is violated by z and satisfied by every feasible integer $w \neq z$ with $g(w) \leq g(z)$, in particular by every optimal integer block other than z .

Proof. If $c_z = 0$, (13) gives $g(w) > g(z)$ for every feasible integer $w \neq z$. Otherwise $g(w) \leq g(z)$ and $\|w - z\| \geq 1$ give $c_z^\top(w - z) \leq -\frac{\mu}{4}\|w - z\|^2 \leq -\frac{\mu}{4}$. $\square$

Such a cut may remove points of the continuous sublevel set $\{x : g(x) \leq g(z)\}$, including nonintegral points of smaller value. It is therefore not a separation oracle for that convex set. We use integer feasibility algorithms whose oracle is queried only at integer points and must separate a fixed closed convex set from each rejected query (Theorem D.12). The proofs fix the target set after the fact: the empty set for the running-time bound, and a single omitted optimal block for completeness.

Theorem 6.18 (Candidate lists with unconstrained fibers). *Let $P = Q \times \mathbb{R}^n$ with an explicit rational polyhedron $Q \subseteq \mathbb{R}^t$, possibly unbounded or lower-dimensional. A deterministic ordinary algorithm, using no oracle, either certifies $Q \cap \mathbb{Z}^t = \emptyset$ or outputs a finite list $\mathcal{Z} \subseteq Q \cap \mathbb{Z}^t$ that contains every optimal integer block. Its running time and the total encoding length of $\mathcal{Z}$ are at most $2^{O(t \log(t+1))} L^C$, with C independent of t and n . There are at most 2^t optimal integer blocks, and this bound is attained.*

Proof idea. Fixed-dimensional integer linear feasibility gives a feasible block, and strong convexity gives a radius R , of bit length $2^{O(t)} L^{O(1)}$, that contains every optimal block. The fiber minimum g is differentiable with $\nabla g(z) = \nabla_z f(z, y(z))$, where $y(z)$ is the fiber minimizer. An ordinary approximation of $y(z)$ at polynomial precision gives c_z with $\|c_z - \nabla g(z)\| \leq \mu/4$, and strong convexity then gives (13). The integer-query feasibility algorithm is run with these cuts and is never told that a point is feasible. Every point it queries in $Q \cap [-R, R]^t$ is recorded. Viewed against the empty target set, the run is valid, which bounds its length. If an optimal block w were never queried, the same run would be a valid run for the target $\{w\}$, and the algorithm would wrongly report that $\{w\}$ has no integer point. The parity bound follows because two optimal blocks with the same parity have an integer midpoint with strictly smaller value. Appendix D.6 gives the details. $\square$

The list separates discrete search from exact arithmetic. The search uses ordinary rational computation at polynomial precision, and its length is FPT in t with an exponent of L that is independent of both t and n . Exact arithmetic is needed only to compare the finitely many fiber minima of the list. Two fiber minima are compared through one observable: $g(z) - g(z')$ is the value of $f(z, y) - f(z', y')$ at the minimizer of the separable strongly convex quartic $f(z, y) + f(z', y')$, so no comparison of two independently represented algebraic numbers is needed.

Fixed-dimensional integer minimization from first-order information is classical; the closest predecessor is the method of Oertel et al. [96], whose oracle contract differs from the integer-point separation interface used here. The integer feasibility theorem and the idea of avoiding bisection on the objective by keeping visited candidates are imported [8, 11, 54]. Those results do not by

themselves optimize an algebraic fiber value. What is added here is an implementation of the integer-point interface for the fiber minimum from rational data of polynomial precision, the proof that every optimal block is listed, and the extension to mixed linear constraints without a constraint qualification.

Corollary 6.19 (Exact selection by nonadaptive queries). *In the setting of Theorem 6.18, let $\tau \in \mathbb{Q}$.*

- (a) *Each predicate $\min \bowtie \tau$ for problem (12) is a Boolean combination of the predicates $g(z) \bowtie' \tau$ with $z \in \mathcal{Z}$ and $\bowtie' \in \{<, \leq\}$. Each of these reduces to one PosSLP instance by Theorem 4.4, and all instances are built from $\mathcal{Z}$ before any oracle answer is known. By Theorem 4.10(c), the predicate therefore has a many-one reduction to a single PosSLP instance, computable in time $2^{O(t \log(t+1))} L^C$.*
- (b) *The set of all optimal integer blocks is computed in time $2^{O(t \log(t+1))} L^C$ with at most $|\mathcal{Z}|^2$ nonadaptive PosSLP queries. For an optimal block z_* , the continuous part of the optimal solution is the unique real solution of $\nabla_y f(z_*, y) = 0$.*

In particular, for each fixed t , exact optimization of (12) with unconstrained fibers is in $\mathbf{P}^{\text{PosSLP}}$, and each predicate $\min \bowtie \tau$ has a polynomial-time many-one reduction to PosSLP.

The construction time in (a) is FPT; it is polynomial only for bounded t . The oracle cannot be removed unless $\text{PosSLP} \in \mathbf{P}$, even for fixed t : the case $t = 0$ is unconstrained exact comparison, which is PosSLP-hard (Theorem 5.1), and new integer variables can be added without changing the optimal value. With the padding of Lemma D.14 the instances even keep a positive definite full Hessian Gram, so the hardness holds in the certified format for every fixed $t \geq 1$. Corollary 6.19 returns the complete optimal set; a sequential tournament with $|\mathcal{Z}| - 1$ adaptive comparisons returns one optimal block.

Theorem 6.20 (Candidate lists with mixed linear constraints). *Let $P = \{(z, y) : Az + By \leq d\}$ with explicit rational data. A deterministic ordinary algorithm, using no oracle, either certifies that (12) is infeasible or outputs a finite list of feasible integer blocks that contains every optimal integer block. Its running time and the total length of the list are at most $F(t)L^C$ for a computable F and an absolute constant C . The polyhedron may be unbounded or lower-dimensional, with redundant or degenerate constraints; no Slater condition and no bound on Lagrange multipliers is assumed. There are at most 2^t optimal integer blocks.*

Proof idea. Fibers may now be infeasible, and g need not be differentiable. At an integer query with an infeasible fiber, a rational Farkas certificate gives a cut that is valid for the projection of P . At a query z with a feasible fiber, let $\hat{y}$ be an exactly feasible rational fiber point and $\lambda \geq 0$ a rational multiplier vector. Joint strong convexity and the multiplier inequality give

$$g(w) - g(z) \geq c_z^\top(w - z) + \frac{\mu}{2}\|w - z\|^2 - E, \quad c_z = \nabla_z f(z, \hat{y}) + A^\top \lambda,$$

for every feasible w , where $E = \lambda^\top \sigma + \|\nabla_y f(z, \hat{y}) + B^\top \lambda\|^2 / (2\mu)$ and $\sigma = d - Az - B\hat{y} \geq 0$. If $E \leq \mu/4$, this is (13). To make E small without a constraint qualification, choose $\hat{y}$ close to the fiber minimizer by an ordinary convex approximation. A true optimal multiplier λ_* exists by Lemma 6.1(b), and complementary slackness gives the identity $\lambda_*^\top \sigma = \nabla_y f(z, y_z)^\top (\hat{y} - y_z)$, where y_z is the fiber minimizer. This bounds the multiplier term without bounding $\|\lambda_*\|$. The best rational multiplier then comes from an exact convex quadratic program in λ . Appendix D.7 gives the details. $\square$

The sharper parameter function $2^{O(t \log(t+1))}$ of Theorem 6.18 is not claimed here; it depends on the initial mixed-integer feasibility step. Exact selection among the listed blocks is a separate problem, because the fibers are now constrained. Each of the constrained results of this section gives a selection method, with its own computational model.

Corollary 6.21 (Mixed selection in few nonlinear directions). *In the setting of Theorem 6.20, let k be the nonlinear dimension of f as a polynomial on $\mathbb{R}^{t+n}$. A deterministic ordinary algorithm, using no oracle, decides each predicate $\min \bowtie \tau$, returns every optimal integer block, and represents the corresponding continuous optimizers as in Theorem 6.6, in time $F(k+t)L^C$.*

Theorem 6.22 (Mixed constraints and constraint rank). *In the setting of Theorem 6.20, let $r = \text{rank } B$. A Las Vegas algorithm with a PosSLP oracle either detects infeasibility or returns the lexicographically smallest optimal integer block z_* together with a representation $y_* = \bar{y} + Zu_*$ of the optimal continuous part, where u_* is the unique zero of an explicit strongly monotone cubic map. It also decides each predicate $\min \bowtie \tau$, and can return every optimal integer block. Its expected running time is $F(t, r)L^C$ for a computable F and an absolute constant C , and every returned answer is correct.*

The fibers of a pair of candidates are compared on their product, whose constraint matrix $\text{diag}(B, B)$ has rank $2r$, by Theorem 6.16; each fiber of f has the curvature bound μ , and the nonlinear dimension of a product of fibers is at most $2k$. Appendix D.7 proves both compositions. The rank of A does not enter: its effect is absorbed by the ordinary list.

Without any structure on the fibers, the list still reduces the mixed problem to the general constrained bound.

Corollary 6.23 (Mixed unambiguous certificates). *In the setting of Theorem 6.20, the language of instances that are feasible with $\min \bowtie \tau$ is decided by an unambiguous nondeterministic machine with a PosSLP oracle, and so is its complement, both in time $F(t)L^C$. In particular, for each fixed t these predicates are in $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$.*

The machine computes the list, guesses one active set for each listed fiber, and verifies all of them as in Theorem 6.2. Exactly one guess passes for each fiber. The fiber minima are then the unconstrained minima of explicit quartics on charts, which are compared as above (Appendix D.7).

Knowing the optimal value does not make the integer decision easy. In the following problem there are two candidate integer blocks, the optimal value is known in advance, the optimizer is rational and bounded, and the continuous constraint normals have rank one; nevertheless, the optimal bit is PosSLP-hard.

Corollary 6.24 (A hard binary decision at a known optimum). *There is a polynomial-time map from integer straight-line programs to instances*

$$\min \left\{ H(x, z) = F(x) + (z - \tfrac{1}{2})^2 : x \in \mathbb{R}^N, z \in \mathbb{Z}, 0 \leq z \leq 1, z - 1 \leq x_j \leq z \right\}, \quad (14)$$

where F is a rational quartic with $N+1$ rational quadratic square factors and a positive definite rational full Hessian Gram, such that: the optimal value is $1/4$; the optimal solution (x_*, z_*) is unique, with $x_* \in \mathbb{Q}^N \cap [-1, 1]^N$; $\nabla^2 H \succeq \frac{3}{2}I$; the continuous constraint normals have rank one; and $z_* = 1$ exactly when the program's output is positive. Adding $\epsilon z(z-1)\|x\|^2 + \frac{1}{4}z^2(z-1)^2$ with a suitable rational $\epsilon > 0$ changes no feasible objective value and gives the objective a positive definite rational full Hessian Gram on all $N+1$ variables.

Subtracting $\frac{1}{4}$ from H gives an equivalent instance with known optimal value zero; the optimal solution, the Hessian, and the full Hessian Gram matrices do not change.

The proof (Appendix D.8) starts from Theorem 5.2: the designated coordinate p_j of the rational optimizer of F is nonzero, and the constraints admit $x = p$ exactly for the bit $z = \mathbf{1}[p_j > 0]$. A polynomial-time algorithm that returns the optimal integer block of (14) would therefore decide PosSLP. The gap between the two fiber values lies between $\frac{3}{4}p_j^2$ and $\frac{\Lambda_F}{2}p_j^2$, where Λ_F bounds $\|\nabla^2 F\|$ on $[-1, 1]^N$. No inverse-polynomial lower bound on $|p_j|$ is assumed, and Lemma 4.1 guarantees only a doubly exponentially small one. That separation bound gives no guarantee that fiber values approximated to polynomially many bits certify the bit. The Las Vegas oracle algorithm of Theorem 6.22 cannot be replaced by an ordinary polynomial-time algorithm, even for $t = r = 1$, unless PosSLP $\in$ P.

6.7 The open case: arbitrary polyhedra and unrestricted structure

For an arbitrary rational polyhedron, unrestricted nonlinear dimension, and unrestricted constraint rank, the strongest bound proved here is Theorem 6.2. Exact comparison is PosSLP-hard already without constraints. Whether it is in P^{PosSLP} is open; by Theorem 4.10(b) this is the same as asking for a polynomial-time many-one reduction to one PosSLP instance. We record what the methods of this section need and why three natural routes fall short. These are limitations of particular proofs, not lower bounds.

Exact quadratic programming with circuit Hessians. Lemma 6.11 does not depend on the polyhedron. Theorem 6.12 would therefore settle the general case if every Taylor quadratic program over an arbitrary rational polyhedron could be solved exactly with a number of arithmetic operations that is polynomial in the encoding of A and b and independent of the encoding of the Hessian and the linear cost. The classical polynomial-time algorithm for convex quadratic programming [71] is polynomial in the bit length of all data, which grows exponentially along the Newton iteration. Granot and Skorin-Kapov [47] remove the dependence on the linear cost and the right-hand side, but not on the Hessian. The linear program with a circuit objective in the proof of Theorem 6.2 handles a varying linear objective over a fixed polytope, not a varying Hessian. We know of no algorithm with the required interface for general polyhedra and Hessians; the sufficient condition is not claimed to be necessary.

Penalties. One could try to remove the constraints by a penalty and apply Theorem 4.4. For $\min_{x \geq 0} \frac{1}{2}(x+1)^2$, with minimizer 0, the squared-hinge penalty $\frac{1}{2}(x+1)^2 + \frac{\lambda}{2} \max\{0, -x\}^2$ has minimizer $-1/(1+\lambda)$. Reaching distance γ requires $\lambda \geq \gamma^{-1} - 1$. At the separation scale $\gamma = 2^{-2^{\text{poly}(L)}}$ allowed by Lemma 4.1, the penalty coefficient needs exponentially many printed bits, and the penalty is piecewise quadratic rather than polynomial. A short circuit for λ does not help, because Theorem 4.4 accepts explicitly encoded polynomials and starts from an ordinary warm start of polynomial bit length.

Accuracy-driven iteration. An interior-point or ellipsoid method with iteration count polynomial in $\log(1/\varepsilon)$ needs exponentially many iterations at $\varepsilon = 2^{-2^{\text{poly}(L)}}$. Storing ε as a circuit does not shorten the iteration. A rapid refinement phase with a polynomial number of steps, which exact Newton steps provide when the subproblems can be solved, is the missing ingredient.

The positive results of this section supply such refinement or avoid it in different ways: a small nonlinear dimension makes the separation bound singly exponential, so that ordinary approximation suffices; structured quadratic programs make exact Newton steps arithmetic; low constraint rank confines the combinatorial search to small subproblems. None of them covers a general strongly convex quartic over a general polyhedron.

7 Algebraic singletons and the degree of exact optimizers

The previous sections ask how hard it is to compare an exact optimal value or coordinate with a rational threshold. This section asks a prior question: what kind of number can an exact optimizer be, and how large must an expanded description of it be? We study the simplest setting in which the answer is not obvious. The objective f is an unconstrained rational quartic, globally strongly convex, certified convex by rational data, and with minimum value zero. Its unique minimizer p is then the unique real zero of f , and the zero sublevel set $\{f \leq 0\}$ is the singleton $\{p\}$.

The answers are as follows.

- (i) Already in two variables the zero can be irrational, so a single strongly convex quartic inequality with integer coefficients can have no rational solution (Proposition 7.2).
- (ii) The coordinates that occur are exactly the real algebraic numbers with one real conjugate. The same class arises for every singleton defined by rational convex inequalities, for every unique minimizer of a rational convex polynomial, and even for every rational polynomial with a single real zero (Theorem 7.4). A number of degree d is realized in $(d+1)/2$ variables, in time polynomial in its dense minimal polynomial; this count is optimal for consecutive power coordinates (Theorem 7.7).
- (iii) The degree can be exponential in the number of variables while all coefficients have $O(\log n)$ bits (Theorem 7.8). The construction uses $n+1$ integer squares, the least possible number even over $\mathbb{R}$ (Theorem 7.9).
- (iv) For a rational sum of quadratic squares with a single real zero, at which the Hessian is positive definite, the field degree of the zero is at most $2^n - 1$; global convexity improves this to $2^n - 3$ for $n \geq 3$ and $2^n - 5$ for $n \geq 4$. For $n = 5$ the sharp bound is 21. The construction in (iii) is therefore extremal for $n \leq 5$ (Theorem 7.11).
- (v) Univariate convex realizations can require exponential degree, although an auxiliary variable or a compressed circuit avoids that cost (Theorem 7.14).
- (vi) The optimizer of a certified quartic can be realized together with all of its quadratic products, again by a polynomial-size certified quartic (Theorem 7.15).

These results matter for exact output. A rational convexity certificate does not produce rational feasible points, does not bound the algebraic degree of the optimizer by a constant, and does not make dense minimal-polynomial output short. They also supply the construction (Lemma 7.6) that the lower reductions of Section 5 and the height and field constructions of Sections 8 and 9 use to place a prescribed point at the zero of a certified quartic. None of these results is a lower bound for exact decision problems; they concern which exact outputs exist and how long their expanded forms must be. Complete proofs are in Appendix E.

7.1 Hessian Gram matrices and output formats

In this section a *quadratic* is a polynomial of degree at most two. For $x, v \in \mathbb{R}^n$ put $w(x, v) = (v, x \otimes v) \in \mathbb{R}^{n+n^2}$, where $x \otimes v$ has entries $x_k v_j$ indexed by ordered pairs (k, j) . As in Section 8, a symmetric matrix A is a *full Hessian Gram* of a polynomial f of degree at most four if $v^\top \nabla^2 f(x) v = w(x, v)^\top A w(x, v)$ for all x, v . If $A \succeq \lambda I$ then $\nabla^2 f \succeq \lambda I$ everywhere, since $\|w(x, v)\| \geq \|v\|$. A full Hessian Gram certifies the Hessian; it is not a Gram matrix of f itself. At a zero of f , no polynomial Gram matrix of f can be positive definite, while a full Hessian Gram can be.

Sums of squares of quadratics carry a canonical full Hessian Gram. For a quadratic $g(x) = g_0 + b^\top x + x^\top T x$ with T symmetric, define the $n \times n^2$ matrix $\Delta(b, T)$ and the symmetric matrices $\Xi(T)$ and $\Gamma(g)$ by

$$\begin{aligned} \Delta(b, T)_{i,(k,j)} &= 2b_k T_{ij} + 4b_i T_{kj}, & \Xi(T) &= 8 \operatorname{vec}(T) \operatorname{vec}(T)^\top + 4 T \otimes T, \\ \Gamma(g) &= \begin{pmatrix} 2bb^\top + 4g_0 T & \Delta(b, T) \\ \Delta(b, T)^\top & \Xi(T) \end{pmatrix}. \end{aligned} \quad (15)$$

Here $\operatorname{vec}(T)_{(k,j)} = T_{kj}$ and $(T \otimes T)_{(k,j),(k',j')} = T_{kk'} T_{jj'}$. The map $g \mapsto \Gamma(g)$ is quadratic in the coefficients of g ; let $\hat{\Gamma}(g, g')$ be its symmetric bilinear polarization. For a polynomial written as $f = \sum_{k,l} S_{kl} g_k g_l$ with quadratics g_k and a symmetric matrix S , the *canonical Hessian Gram* of this representation is $\Gamma_S(g) = \sum_{k,l} S_{kl} \hat{\Gamma}(g_k, g_l)$. For a weighted sum of squares $f = \sum_k c_k g_k^2$ it is $\sum_k c_k \Gamma(g_k)$.

Lemma 7.1 (Canonical Hessian Grams). *Let $f = \sum_{k,l} S_{kl} g_k g_l$ as above.*

- (a) $v^\top \nabla^2 f(x) v = w(x, v)^\top \Gamma_S(g) w(x, v)$ for all $x, v \in \mathbb{R}^n$.
- (b) Covariance. For $c \in \mathbb{R}^n$ let $\Psi_c = \begin{pmatrix} I_n & 0 \\ -c \otimes I_n & I_{n^2} \end{pmatrix}$, where $(c \otimes I_n)v = c \otimes v$. Then, as an identity of matrices, $\Gamma_S(g) = \Psi_c^\top \Gamma_S(g(\cdot + c)) \Psi_c$.
- (c) Consequently, if the canonical Gram of the recentred representation satisfies $\Gamma_S(g(\cdot + c)) \succeq \lambda I$, then $\Gamma_S(g) \succeq \lambda(1 + \|c\|)^{-2} I$.

Part (b) is the reason the constructions below need no second approximation. Positivity is proved for the Gram centred at the common zero p of the square factors, where they have no constant terms. Covariance then shows that the matrix $\Gamma_S(g)$, computed directly from the rational square factors, is the congruent image of that positive definite matrix. It is rational, positive definite, and of polynomial bit length, and no approximation of p is needed to obtain it.

Output formats. For an algebraic point p we write $\mathbb{Q}(p) = \mathbb{Q}(p_1, \dots, p_n)$ and call $[\mathbb{Q}(p) : \mathbb{Q}]$ the *field degree* of p . A coordinate can be output by the *dense* coefficient list of its minimal polynomial or by an *ordinary sparse* list of its nonzero terms (Definition 2.15). The rational circuits of Definition 2.16 compute only rational numbers, so for irrational coordinates we also use *root circuits*: rational circuits that may in addition contain gates $z \mapsto z^{1/k}$, returning the positive real k -th root of a positive input, with the integer $k \geq 2$ written in binary. A coordinate of degree D has a dense minimal-polynomial list with $D + 1$ entries. The field degree of p bounds the degree of every coordinate, but a coordinate need not attain it, and degree alone says nothing about sparse lists or root circuits. The lower bounds below always name the coordinate and the format they concern.

7.2 An integer quartic with an irrational zero

Proposition 7.2 (Two variables suffice). *Let $q_1 = x^2 - y$, $q_2 = y^2 - 2x$,*

$$A = 12599x^2 - 10000xy + 7937y^2 - 15874x - 12599y + 20000, \quad F = A^2 + (100q_1)^2 + (100q_2)^2.$$

Then $F \in \mathbb{Z}[x, y]$ has coefficients of absolute value less than 2^{30} , and:

- (a) $\nabla^2 F(x, y) \succeq 4124 I$ for all $(x, y) \in \mathbb{R}^2$;

- (b) $F^{-1}(0) = \{F \leq 0\} = \{(2^{1/3}, 2^{2/3})\}$, so $\{F \leq 0\}$ contains no rational point;
- (c) the canonical Hessian Gram Γ_F of the displayed three squares is a positive definite integer 6×6 matrix, printed in (53);
- (d) the biform $v^\top(\nabla^2 F(x) - 4096I)v$ has the explicit representation (54) as a positive integer combination of squares of polynomials with rational coefficients.

The construction starts from the three quadratics q_1, q_2 and $q_3 = (x - y)^2 - 2x - y + 4$, which vanish at $p = (r, r^2)$, $r = 2^{1/3}$. The real combination $(2r - 1)q_1 + (r^2 - 1)q_2 + q_3$ equals $(x - r, y - r^2) \begin{pmatrix} 2r & -1 \\ -1 & r^2 \end{pmatrix} (x - r, y - r^2)^\top$, a positive definite quadratic form centred at p ; the rational combination $A = 7599q_1 + 2937q_2 + 5000q_3$ approximates 5000 times it. Thus A vanishes at p exactly, has a positive definite quadratic part, and has a very small gradient at p . Squaring A supplies curvature far from p ; the two small squares supply curvature near p . Part (a) is proved by an explicit global estimate, part (c) by Lemma 7.1 and a Schur complement bound, and part (d) records a direct rational certificate of a slightly weaker modulus.

In version 1 of [111], Table 1 leaves open whether exact feasibility of globally convex rational quartic inequalities always has a rational witness of polynomial bit length; Appendix C there gives a convex sextic with an irrational unique zero. Proposition 7.2 answers the quartic witness question negatively in two variables. It says nothing about the complexity of the decision problem: the point has the short algebraic description $(2^{1/3}, 2^{2/3})$.

7.3 Which algebraic numbers occur

Lemma 7.3 (One real conjugate). *Let $\{p\} \subseteq \mathbb{R}^n$ be a singleton that is either the real zero set of finitely many polynomials with rational coefficients, or the set $\{x : f_i(x) \leq 0, 1 \leq i \leq s\}$ for finitely many rational convex polynomials f_i . Then p has real algebraic coordinates, the field $K = \mathbb{Q}(p)$ has exactly one embedding into $\mathbb{R}$, $[K : \mathbb{Q}]$ is odd, and the minimal polynomial of every coordinate p_j has exactly one real root.*

For the zero-set case, every real embedding of K maps p to another real common zero, hence fixes p . In the convex case the inequalities active at p already define $\{p\}$, and the same argument applies to them. The passage from the joint field to a single coordinate uses that $[K : \mathbb{Q}(p_j)]$ is odd, so every real embedding of $\mathbb{Q}(p_j)$ extends to a real embedding of K .

Theorem 7.4 (Coordinates of rational convex singletons). *For a real number α the following are equivalent.*

- (i) α is algebraic and its minimal polynomial over $\mathbb{Q}$ has exactly one real root.
- (ii) For some n , some $f \in \mathbb{Q}[x_1, \dots, x_n]$ has a single real zero, and α is one of its coordinates.
- (iii) For some n , finitely many rational convex polynomials f_i on $\mathbb{R}^n$ satisfy $\{x : f_i(x) \leq 0 \forall i\} = \{p\}$ with α a coordinate of p .
- (iv) For some n , α is a coordinate of the unique minimizer of a rational convex polynomial on $\mathbb{R}^n$ that has exactly one minimizer.
- (v) For some n , α is a coordinate of the unique zero of a rational quartic f on $\mathbb{R}^n$ that is a sum of $n + 1$ squares of rational quadratics, satisfies $\nabla^2 f \succeq I$ on $\mathbb{R}^n$, and has a positive definite rational full Hessian Gram.

If (i) holds and α has degree $d \geq 3$, then (v) holds with $n = (d + 1)/2$ and zero $(\alpha, \alpha^2, \dots, \alpha^n)$, and f , its square factors, and its Gram matrix are computed from the dense coefficient list of any rational multiple of the minimal polynomial in time polynomial in that list's binary length. If α is rational, (v) holds with $n = 1$ and $f = (x - \alpha)^2 + (x - \alpha)^4$.

The theorem says that convexity, certified convexity, degree four, and a sum of squares representation cost nothing in the class of reachable coordinates: they reach exactly the numbers that any rational polynomial with a single real zero can reach. The restriction to one real conjugate concerns singletons and unique unconstrained minimizers. It does not apply to constrained optimizers: $-x$ has the unique minimizer $\sqrt{2}$ on the rational convex set $\{x : x^2 \leq 2\}$, and $\sqrt{2}$ has two real conjugates. In particular, no singleton defined by rational convex inequalities has $\sqrt{2}$ as a coordinate, which is why Square Root Sum comparisons are not encoded by such singletons (Section 5).

Several convex quadratic inequalities suffice when one prefers quadratics to one quartic. The following corollary realizes power coordinates of degree d with three rational ellipsoids, so the Hessian span of a convex quadratic system with an irrational singleton can be three while the degree of the singleton is unbounded.

Corollary 7.5 (Three rational ellipsoids). *Let $P \in \mathbb{Q}[T]$ be irreducible of degree $d \geq 3$ with exactly one real root α , given by its dense coefficient list. In time polynomial in the length of that list one can compute rational quadratics g_1, g_2, g_3 in $d - 1$ variables, each with positive definite Hessian, such that $\{x : g_i(x) \leq 0, i = 1, 2, 3\} = \{(\alpha, \alpha^2, \dots, \alpha^{d-1})\}$. Each set $\{g_i \leq 0\}$ is a full-dimensional ellipsoid.*

7.4 A generic quantitative construction

The realization results of this section, and those of Sections 5, 8 and 9, use one assembly step. A point p is described by rational quadratics r_j vanishing at it with a well-conditioned Jacobian, and by one rational quadratic g vanishing at p whose quadratic part is positive definite and whose gradient at p is small. Such a g usually comes from an irrational positive semidefinite *exposing* quadratic that vanishes to second order at p ; approximating its coefficients inside the rational space of quadratics that vanish at p keeps $g(p) = 0$ exactly and makes $\nabla g(p)$ as small as required.

Lemma 7.6 (Quantitative quartic realization). *Let $p \in \mathbb{R}^n$, and let $g, r_1, \dots, r_m \in \mathbb{Q}[x_1, \dots, x_n]$ be polynomials of degree at most two that vanish at p . Write*

$$g(p + u) = \ell^\top u + u^\top H u, \quad r_j(p + u) = b_j^\top u + u^\top T_j u,$$

with symmetric H, T_j . Let μ, Λ, β, ν be positive rationals with

$$H \succeq \mu I, \quad \|H\| \leq \Lambda, \quad \|T_j\| \leq 1, \quad \|b_j\| \leq \beta \ (1 \leq j \leq m), \quad \sum_{j=1}^m b_j b_j^\top \succeq \nu^2 I. \quad (16)$$

Let $t > 0$ be rational, put $\varepsilon = t^2$, and suppose

$$\varepsilon \leq \min \left\{ 1, \frac{\mu^2}{2m}, \frac{\nu^2 \mu^2}{36 n (\Lambda + m\beta)^2} \right\}, \quad \|\ell\| \leq \varepsilon. \quad (17)$$

Then $f = (g/(t\nu))^2 + \sum_{j=1}^m (r_j/\nu)^2$ has the following properties.

- (a) $f \in \mathbb{Q}[x]$ has degree exactly four, is a sum of $m + 1$ squares of rational quadratics vanishing at p , and its quartic form is positive definite.

- (b) $\nabla^2 f(x) \succeq \frac{3}{2}I$ for every $x \in \mathbb{R}^n$. Hence p is the unique zero and the unique minimizer of f .
- (c) The canonical Hessian Gram Γ_f of the displayed representation is a rational full Hessian Gram of f , and $\Gamma_f \succeq \gamma I$ with

$$\gamma = \frac{\min\{3/2, 2\mu^2/(\varepsilon\nu^2)\}}{((1+\theta)^2+1)(1+\|p\|)^2}, \quad \theta = \frac{3\sqrt{n}\varepsilon(\Lambda+m\beta)}{\mu^2}.$$

Its entries are fixed polynomial expressions in the coefficients of $g, r_1, \dots, r_m$ and in t, ν ; computing it uses $O(m(n+n^2)^2)$ rational operations on numbers of the input bit length.

The factor n in the last bound of (17) is used only for part (c); part (b) holds without it (Remark E.2). The proof computes the canonical Gram centred at p , where it has three blocks: a constant block at least $2\varepsilon\nu^2 I$, a quartic block at least $2\mu^2 I$ even though the individual $T_j \otimes T_j$ can be indefinite, and a small cross block. A Schur complement estimate gives positivity, and covariance transfers it to the rational Gram. Squaring an indefinite quadratic does not produce a convex function, and the estimate retains the corresponding negative terms. Positive definite Hessian Gram matrices and Schur-complement arguments for SOS-convex regularization are established techniques [5]; the additional feature here is that a prescribed, possibly irrational, zero is kept exactly while all coefficients stay rational and of polynomial size. The hypotheses force p to be algebraic: n of the residuals have a nonsingular Jacobian at p , so a small rational box isolates p among their real common zeros, and the transfer argument in the proof of Lemma 7.3 applies. The construction, however, uses no algebraic representation of p , and p need not be known or rational. The constructions in Sections 5, 8, and 9 apply the lemma at the gate values of certified odd-root circuits and of rational unit-quaternion circuits, at points given by cube-root chains and by repeated complex squaring, and at points with coordinates in towers of radicals.

7.5 Power coordinates and the dimension count

Theorem 7.7 (Realization in power coordinates). *Let $P \in \mathbb{Q}[T]$ be irreducible of degree $d \geq 3$ with exactly one real root α , given by its dense list of rational coefficients of total binary length L , and let n be an integer with $(d+1)/2 \leq n \leq d-1$. A deterministic algorithm running in time polynomial in L computes rational quadratics $g, r_1, \dots, r_n$ in n variables and positive rationals $t, \mu, \Lambda, \beta, \nu$ that satisfy the hypotheses of Lemma 7.6 at $p = (\alpha, \alpha^2, \dots, \alpha^n)$. Hence $F_P = (g/(t\nu))^2 + \sum_{j=1}^n (r_j/\nu)^2$ is a sum of $n+1$ squares of rational quadratics, $\nabla^2 F_P \succeq \frac{3}{2}I$, $F_P^{-1}(0) = \{p\}$, and the canonical Hessian Gram of F_P is a positive definite rational matrix; all outputs have bit length polynomial in L .*

Conversely, if $f \in \mathbb{Q}[x_1, \dots, x_k]$ has degree at most four, is nonnegative on $\mathbb{R}^k$, and has the unique real zero $(\alpha, \alpha^2, \dots, \alpha^k)$, then $k \geq (d+1)/2$.

The algorithm first replaces P by the monic minimal polynomial P_1 of α ; the input may be any rational multiple of it, including a negative one. The exposing quadratic comes from the nonnegative univariate polynomial $(T - \alpha)P_1(T)(1 + T^2)^e$, where $e = n - (d+1)/2$ raises the degree to $2n$: its positive factor $P_1(T)(1 + T^2)^e/(T - \alpha)$ is a product of real quadratics with positive definite 2×2 Gram matrices, which a recursion compresses onto consecutive monomials. Multiplying by the difference matrix for $T - \alpha$ produces a positive semidefinite matrix Q_* of order $n+1$ whose kernel is spanned by $(1, \alpha, \dots, \alpha^n)$. Its entries involve all complex roots, so the algorithm approximates the roots with certified precision, forms a rational approximation, and projects it exactly onto the rational space of matrices that represent multiples of P . Root separation from the integer discriminant keeps every required precision at polynomially many bits. The residuals are $x_1 x_j - x_{j+1}$

and one quadratic version of $P_1(x_1) = 0$, whose Jacobian determinant at p is a positive multiple of $P'_1(\alpha)$.

The lower bound concerns the prescribed consecutive-power point only. A degree d number has other coordinate encodings, and we do not know whether some of them need fewer variables. Irreducibility and the single real root are promises; the algorithm does not test them.

7.6 Exponential degree with small coefficients

Theorem 7.7 uses about $d/2$ variables for degree d . Far fewer variables suffice for special numbers. Let $n \geq 2$, $m = n + 1$, $\eta = (-1)^m$, and

$$d_n = \frac{2^m - \eta}{3} = \frac{2^{n+1} - (-1)^{n+1}}{3}, \quad e_i = \frac{(-2)^i - 1}{3} \quad (0 \leq i \leq n), \quad p_i = 2^{e_i/d_n}.$$

The sequence d_n begins 3, 5, 11, 21, 43, 85. With $x_0 = 1$ and indices modulo m , the point $p = (p_1, \dots, p_n)$ is the unique real common zero of the m cyclic binomials $x_i^2 - 2^{\theta_i} x_{i+1} x_{i+2}$, where $\theta_i = 0$ except $\theta_{m-2} = \eta$ and $\theta_{m-1} = -\eta$. The weights p_i^{-2} combine these binomials into the anchored cycle energy $\frac{1}{2} \sum_i (x_i/p_i - x_{i+1}/p_{i+1})^2$, an exposing quadratic whose curvature and Jacobian are controlled by inverse-polynomial quantities.

Theorem 7.8 (Small integer quartics with exponentially large field degree). *For every $n \geq 2$ a deterministic algorithm running in time polynomial in n computes integer quadratics $s_0, \dots, s_n$ in n variables, with coefficients of $O(\log(n+1))$ bits, such that $F_n = \sum_{i=0}^n s_i^2$ satisfies:*

- (a) $\nabla^2 F_n \succeq I$ on $\mathbb{R}^n$, and the canonical Hessian Gram of $\sum_i s_i^2$ is a positive definite integer matrix;
- (b) the unique zero of F_n is $p \in (1/2, 2)^n$, and $[\mathbb{Q}(p) : \mathbb{Q}] = [\mathbb{Q}(p_1) : \mathbb{Q}] = d_n$;
- (c) F_n has $O(n^2)$ monomials, each with an $O(\log(n+1))$ -bit coefficient;
- (d) F_n is not a sum of fewer than $n+1$ squares of real polynomials;
- (e) the condition number of $\nabla^2 F_n(p)$ is at most $65600 n^2$;
- (f) after the substitution $x_1 \mapsto x_1 - 1$, the first coordinate of the unique zero is $1 + 2^{-1/d_n}$, whose primitive minimal polynomial $2(T-1)^{d_n} - 1$ has all $d_n + 1$ coefficients nonzero and total binary length $\Theta(d_n^2)$.

The field degree is exponential in the dimension, while the whole instance is computed in time polynomial in n . Even with dense exponent vectors and the full canonical Hessian Gram of order $n + n^2$ included, the certified input has binary length $L = O(n^4 \log n)$. The dense and ordinary sparse minimal-polynomial lists in part (f) therefore have length $2^{\Omega(n)} = 2^{\Omega((L/\log L)^{1/4})}$, which is superpolynomial, though not exponential, in L . No algorithm that writes such a list for this coordinate runs in time polynomial in L . Short output in other formats is not excluded. Each coordinate $p_i = 2^{e_i/d_n}$ is determined by its rational exponent e_i/d_n , of $O(n)$ bits. With $a = 2^{1/d_n}$, the relation $e_{i+1} = -2e_i - 1$ gives $p_1 = 1/a$ and $p_{i+1} = 1/(ap_i^2)$, so one root gate with the $O(n)$ -bit index d_n , applied to 2, followed by $O(n)$ multiplications and divisions, is a root circuit for the whole optimizer. Part (e) shows that the large degree does not come from ill conditioning: the Hessian at the optimizer has condition number polynomial in n . This is a statement at the optimizer, not a uniform smoothness bound on a neighbourhood. Part (f) needs the translation, because $p_1 = 2^{-1/d_n}$ itself has the two-term minimal polynomial $2T^{d_n} - 1$. The grounded determinant d_n of the directed

square cycle and its exponential growth are classical; see [84]. The supporting construction of the proof also gives $n + 1$ rational ellipsoids with the same singleton intersection.

The number of squares in part (d) is optimal for a general reason.

Theorem 7.9 (Minimum number of squares). *Let $f \in \mathbb{R}[x_1, \dots, x_n]$ be globally convex of degree exactly four, and suppose $f = \sum_{i=1}^m g_i^2$ with $g_i \in \mathbb{R}[x]$, $f(p) = 0$, and $\nabla^2 f(p) \succ 0$ for some $p \in \mathbb{R}^n$. Then $m \geq n + 1$. The bound is attained for every n by $(\sum_i x_i^2)^2 + \sum_i x_i^2$, by every quartic of Lemma 7.6 with $m = n$, and by the quartics F_n of Theorem 7.8.*

Each hypothesis is needed. The quartic $(1 + \|x\|^2)^2$ is globally strongly convex with one square and minimum one; $\|x\|^4$ has a degenerate zero and one square; the nonconvex $x^2 + (y - x^2)^2$ has two squares in two variables and Hessian $2I$ at its unique zero; a positive definite quadratic is a sum of n squares; and the sextic $(x_1 + x_1^3)^2 + \sum_{j \geq 2} x_j^2$ is globally strongly convex with n squares. The proof shows first that the leading quadratic part of the square factors does not depend on directions in which the quartic form vanishes, and that convexity forbids quadratic coupling to those directions; minimizing over them leaves a square quadratic map with one regular zero and a proper leading part. A proper even map has even topological degree, whereas a map with one regular zero has degree ± 1 .

The cyclic network is optimal among the binomial systems that its exposing strategy can use.

Proposition 7.10 (Optimality within binomial exposing networks). *Let $n \geq 2$, $m = n + 1$, and let $\mathcal{D}$ be a strongly connected directed multigraph on $\{0, \dots, n\}$, loops and parallel arcs allowed, with exactly two outgoing arcs $i \rightarrow a(i)$, $i \rightarrow b(i)$ at every vertex. Fix $x_0 = 1$ and nonzero rationals c_i , and consider the m equations $x_i^2 = c_i x_{a(i)} x_{b(i)}$, $0 \leq i \leq n$. If the system has exactly one real solution p , then p is algebraic and $[\mathbb{Q}(p) : \mathbb{Q}] \leq d_n$; if $\mathcal{D}$ is not Eulerian, then $[\mathbb{Q}(p) : \mathbb{Q}] \leq 2^{n-1} - 1$. Conversely, if m rational homogeneous quadratic binomials in $(X_0, \dots, X_n)$ vanish at a point $(1, p_1, \dots, p_n)$ with nonzero coordinates and some real linear combination of them is a positive semidefinite quadratic form with kernel spanned by that point, then after reordering and rational rescaling the binomials have the above form; in particular the field degree is at most d_n , and fewer than m binomials cannot have such a combination.*

The proof counts complex solutions as characters of the finite group $\mathbb{Z}^n / \Lambda$, where Λ is the row lattice of the exponent matrix; its order is the gcd g of the rooted spanning-tree counts. Uniqueness of the real solution is equivalent to g being odd. For Eulerian graphs, classical identities for the interlace polynomial of an Euler tour give $3g \leq 2^m - (-1)^m$ [9, 10]. The description of binomial solution sets by characters of a lattice quotient is standard [35]. The proposition restricts the method, not the problem: it does not bound the field degree of arbitrary rational sums of quadratic squares.

7.7 Upper bounds on the degree

How large can the field degree of the zero be? For quartics given as rational sums of quadratic squares, intersection theory bounds it in terms of n . After a rational change of coordinates and elimination of flat directions, global convexity leaves the square factors without common real zeros at infinity, which would otherwise spoil classical Cayley–Bacharach arguments.

Theorem 7.11 (Degree bounds for rational sums of quadratic squares). *Let $f = \sum_{j=1}^m r_j^2$ with $r_j \in \mathbb{Q}[x_1, \dots, x_n]$ of degree at most two, suppose that f has exactly one real zero p and that $\nabla^2 f(p) \succ 0$, and put $D = [\mathbb{Q}(p) : \mathbb{Q}]$. Then p is algebraic, D is odd, and:*

- (a) $D \leq 2^n - 1$;

- (b) if f is globally convex and $n \geq 3$, then $D \leq 2^n - 3$; if moreover $n \geq 4$, then $D \leq 2^n - 5$;
(c) if f is globally convex and $n = 5$, then $D \leq 21$.

Consequently, among globally convex quartics in this class the largest field degrees in dimensions 1, 2, 3, 4, 5 are exactly 1, 3, 5, 11, 21; for $n \geq 2$ they are attained by the quartics F_n of Theorem 7.8.

n	2	3	4	5	6
cyclic degree d_n	3	5	11	21	43
upper bound	3	5	11	21	59

Table 5: Field degrees of zeros of globally convex rational sums of quadratic squares with a nondegenerate zero. The maximum is known exactly for $n \leq 5$. For $n \geq 6$ the cyclic family gives $d_n = (2^{n+1} - (-1)^{n+1})/3$ and the upper bound is $2^n - 5$.

Part (a) selects n square factors with a nonsingular Jacobian at p ; all conjugates of p are simple isolated zeros of these n quadrics, so a weighted form of Bézout’s theorem bounds their number by 2^n , and oddness gives $2^n - 1$. For (b), global convexity lets us remove, by a rational change of coordinates that preserves the field and the rational squares, every direction in which the quartic form vanishes. Afterwards the square factors have no common real zero at infinity, although common complex zeros at infinity may remain. If the degree were $2^n - 1$ or $2^n - 3$, the conjugates of p would leave a residual of length one or three in a complete intersection of n rational linear combinations of the square factors, chosen generically for length three, and Cayley–Bacharach theorems [36] would make that residual contain a second real common zero. In five variables, the remaining residual lengths 1, 3, 5, 7, 9 are excluded by a parity argument for a Gorenstein pairing on the residual scheme, and positive-dimensional complex base components are excluded by a residual-intersection count over a list of low-degree real-point free curves and surfaces. These five-variable steps import precisely stated results of intersection theory [37, 43] and the classification of curves of minimal degree [34]; the contracts are recorded in Appendix E.9.

Rational square factors are essential to this argument. Without them, the gradient equations give the following weaker bound, and in two variables a classification of rational nonnegative ternary quartics recovers the sharp bound.

Proposition 7.12 (Bounds without square factors). *Let $f \in \mathbb{Q}[x_1, \dots, x_n]$ have degree at most four, be nonnegative on $\mathbb{R}^n$, and have exactly one real zero p , with $\nabla^2 f(p)$ nonsingular. Then p is algebraic and $[\mathbb{Q}(p) : \mathbb{Q}]$ is odd and at most $2 \cdot 3^{n-1} - 1$. If $n = 2$ and f is globally convex, then $[\mathbb{Q}(p) : \mathbb{Q}] \leq 3$.*

Lemma 7.13 (Rational descent in two variables). *Let $f \in \mathbb{Q}[x_1, x_2]$ have degree four, be nonnegative on $\mathbb{R}^2$, have exactly one real zero, and have a positive definite quartic form. Then f is a sum of squares of polynomials with rational coefficients. In particular this holds for every certified quartic in two variables with minimum zero.*

The lemma uses Scheiderer’s description of rational nonnegative ternary quartics that are not sums of rational squares [107]: such a form is a product of four complex lines in general position, and then it has at least two real projective zeros. The lemma is the reason why the three-variable rational descent obstruction of Section 9 cannot occur in two variables. Global convexity cannot be dropped from Theorem 7.11(b): Example E.10 in the appendix gives three rational quadratics in three variables whose sum of squares has a unique real zero of field degree $7 > 2^3 - 3$, with positive definite Hessian there.

7.8 Univariate realizations: degree, dimension, and format

A univariate rational polynomial can also have a single real zero at a number with one real conjugate; the square of its minimal polynomial does. Requiring convexity changes the cost.

Theorem 7.14 (Univariate convex realizations). *(a) A real number is the unique real zero of a rational globally strongly convex univariate polynomial if and only if it is algebraic with exactly one real conjugate.*

(b) Let $P \in \mathbb{Z}[T]$ be irreducible of degree $d \geq 1$, with positive leading coefficient, coefficients of absolute value at most 2^τ , and exactly one real root α . In time polynomial in $d + \tau$ one can compute $c \in \mathbb{Q}$, $\kappa \in \mathbb{Q}_{>0}$, and an integer $N \geq 2$, each of binary length $O(d^3(\tau + \log(d + 1)))$, such that $G = \kappa P^2(1 + (T - c)^2)^N$ satisfies $G'' \geq 1$ on $\mathbb{R}$ and $G^{-1}(0) = \{\alpha\}$. Thus G is computed by a rational arithmetic circuit with input T and size polynomial in $d + \tau$, while its degree $2d + 2N$ may be exponential and its value at the integer $2^{\tau+2}$ is at least 2^N .

(c) For $k \geq 1$ let $P_k = T^3 - 3T + 2 + 2^{-2k}$. It is irreducible, has a single real root α_k , and has binary length $O(k)$. Every rational globally convex univariate F whose only real zero is α_k satisfies $(\deg F)^2 > \frac{3}{4}2^k$. Every finite system of rational convex univariate inequalities with solution set $\{\alpha_k\}$ contains a polynomial of such degree. Every nonconstant rational globally convex J with a minimizer at α_k satisfies $(\deg J - 1)^2 > \frac{3}{4}2^k$.

(d) The two-variable quartic of Theorem 7.7 ($d = 3$, $n = 2$) has unique zero (α_k, α_k^2) and binary length polynomial in k . The quartic $J_k = T^4/4 - 3T^2/2 + (2 + 2^{-2k})T$ is globally quasiconvex, not convex, and has unique minimizer α_k .

Parts (a) and (b) multiply P^2 by $(1 + (T - c)^2)^N$; the exponent N must be chosen before the rational centre c , which must then lie very close to α (a fixed centre does not work in general). Convexification of positive polynomials by such multipliers is established [73]; the zero of P^2 is what forces the moving rational centre. Part (c) uses that two complex conjugates of α_k lie within 2^{-k} of the real point 1, at distance more than three from α_k : a convex polynomial vanishing at α_k must vanish at these conjugates, and Markov's inequality on the interval from α_k to their common real part forces large degree [20]. Thus, in one variable, exact convex realization has an unavoidable exponential degree for some $O(k)$ -bit inputs; one auxiliary variable, a compressed circuit, or quasiconvexity removes this cost. Part (b) supplies no cheap exact evaluation: expanded rational values of G at short rational arguments can be exponentially long.

7.9 Realizing the quadratic products of an optimizer

Some constructions need a certified quartic whose zero records not only an optimizer p but also all products $p_i p_j$. The next theorem provides one without computing p or any minimal polynomial. It is used later in the paper to make both blocks of a block-separated certificate obstruction strongly SOS-convex.

Theorem 7.15 (Quadratic graph realization). *Let $f \in \mathbb{Q}[x_1, \dots, x_n]$ have degree at most four and a rational full Hessian Gram $A \succ 0$, and suppose that $\min f = 0$; this is a promise and is not checked. Let p be the minimizer, $N = n + n(n + 1)/2$, and use variables (y, z) with $z = (z_{ij})_{1 \leq i < j \leq n}$. In deterministic time polynomial in the binary length L of (f, A) one can compute rational quadratics $g, r_1, \dots, r_N$ in (y, z) and positive rationals $t, \mu, \Lambda, \beta, \nu$ that satisfy the hypotheses of Lemma 7.6 at $w_* = (p, (p_i p_j)_{i < j})$. Hence $B = (g/(t\nu))^2 + \sum_{j=1}^N (r_j/\nu)^2$ is a sum of $N + 1$ rational squares, $B^{-1}(0) = \{w_*\}$, $\nabla^2 B \succeq \frac{3}{2}I$, and the canonical Hessian Gram of B is a positive definite rational matrix. No representation of B uses fewer than $N + 1$ real squares.*

The exposing quadratic is a quadratic lift of f itself. For a rational centre $\hat{p}$, the Taylor formula with integral remainder and the full Hessian Gram express $f(y)$ as a rational quadratic $g_{\hat{p}}(y, z)$ evaluated on the graph $z_{ij} = y_i y_j$. Hence $g_{\hat{p}}(w_*) = f(p) = 0$ exactly for every $\hat{p}$, which is what the assembly lemma needs. Its quadratic part is uniformly positive definite, and its gradient at w_* is at most a constant multiple of $\|\hat{p} - p\|$. A rational $\hat{p}$ close to p is computed by approximate convex minimization (Lemma 3.1) and strong convexity. The residuals are the graph relations $y_i y_j - z_{ij}$ and a quadratic version of $\nabla f(y)$, with Jacobian determinant $\det \nabla^2 f(p)$ at w_* .

7.10 Scope and open questions

The results of this section are statements about exact outputs and exact certificates. They give no lower bound for deciding feasibility, comparing values, or approximating the optimizer. In the explicit families above the optimizer also has a short description, as a radical expression, a root of the input polynomial, or the zero of the input quartic, and the comparison problems for certified quartics are covered by the upper bounds of Section 4. Dimension-exponential lower bounds on output length are superpolynomial in the input length of the families, not exponential in it.

Several questions remain open. The largest field degree of the zero of a globally convex rational sum of quadratic squares is unknown for $n \geq 6$; the cyclic family gives about $\frac{2}{3}2^n$ and Theorem 7.11 gives $2^n - 5$. The sharp bound for rational convex quartics without a rational square representation is unknown for $n \geq 3$; Proposition 7.12 gives $2 \cdot 3^{n-1} - 1$. We do not know the fewest variables needed for a degree d number in coordinate encodings other than consecutive powers, the optimal degree in Theorem 7.14(c), or the shortest sparse or circuit description of the univariate realizations.

8 Expanded outputs, heights, and Gram and moment certificates

The preceding sections compare exact decisions with approximation. This section asks a different question: once an exact object is known to exist, how long must its ordinary binary encoding be? The objects are a rational point of a strictly feasible convex body, a rational optimizer, and the rational matrices that certify optimality or positivity. These are optimal order-two moment matrices, optimal polynomial Gram matrices, matrices exposing the optimal Gram face, and positive definite Gram matrices.

All instances below are rational quartics supplied with a positive definite rational Hessian Gram on the full basis $(v, X \otimes v)$. Convexity is therefore certified by short rational data, the Hessian is bounded below by a positive multiple of the identity on all of $\mathbb{R}^n$, and the lower bounds do not come from an unrecognized or nearly degenerate convexity hypothesis. The answers depend on the representation and on the shape of the certificate:

- (i) a compact convex body with nonempty interior can have only rational points with $\Omega(n2^{n/2})$ bits (Theorem 8.3);
- (ii) a unique rational optimizer in $[-1, 1]^n$ can require $2^{n/2-1} \log_2 5$ denominator bits, even when the Hessian at the optimizer has polynomially bounded condition number (Theorem 8.5);
- (iii) a strict Hessian Gram makes the order-two moment relaxation exact with a unique rank-one optimum, and the optimizer field is then exactly the least field for maximal-rank optimal Grams and for nonzero positive semidefinite matrices exposing the optimal Gram face (Theorem 8.6);
- (iv) consequently, a long rational optimizer forces long entries in every maximal-rank rational optimal Gram, in the optimal moment matrix, and in every rational exposing matrix, while short optimal Grams of lower rank remain (Corollary 8.10);

- (v) positive definite rational polynomial Grams always exist with $\text{poly}(L)2^{O(n)}$ bits for strictly positive certified quartics, and $\Omega(n2^{n/2})$ bits are sometimes necessary although a short singular Gram exists (Theorem 8.11);
- (vi) for every certified quartic, a polynomial-size shared rational circuit, constructed without a sign oracle, gives a strictly feasible point when the minimum is negative and a positive definite Gram when the minimum is positive; over general certified quartics, validating either output is as hard as the exact decision itself (Proposition 4.6 and Theorem 8.12). The rational optimizers of (ii) are computed by short squaring circuits, and short circuits built on these also give the optimal moment matrix, the exposing matrices, and a maximal-rank optimal Gram. A circuit with rational constants and rational operations has rational values, so it cannot describe the irrational optimizers and certificates covered by (iii).

These results matter for exact certification in mixed-integer and polynomial optimization. A solver that must print an exact rational feasible point, an exact rational optimizer, or a strictly feasible rational Gram certificate can be forced to print exponentially many bits on small, perfectly certified convex instances. Approximation does not suffer from this effect. Shared rational circuits avoid it for strictly feasible points and positive definite Grams, whenever a certified quartic has them, and for the long rational optimizers constructed here. For the first two the price is validation: over general certified quartics, checking the circuit output is as hard as the exact decision. The certificate shape also matters. Methods that seek a maximal-rank or interior Gram, or that perform facial reduction while retaining every real feasible Gram, must reproduce the optimizer's arithmetic, whereas a lower-rank certificate can be short.

Three limits apply throughout. First, every lower bound concerns *expanded* rational encodings. None excludes short algebraic, circuit, or other implicit descriptions, and none is a lower bound on the complexity of a decision problem or of approximation. Second, the constructed families have total input length L_k bounded by a polynomial in the family parameter k , and their dimension is $n = \Theta(k)$. The lower bounds are exponential in n and hence superpolynomial in L_k , but they are not of the form $2^{\Omega(L_k)}$. Third, only the stated certificate shapes are covered. In particular, the lower bounds for Gram matrices concern maximal-rank, exposing, or positive definite Grams; the same instances have short rational Gram matrices of lower rank.

8.1 Encodings, heights, and Taylor Gram matrices

A rational number r is written in lowest terms as a/b with $b \geq 1$, and $\text{den}(r) = b$. Its height is $\text{ht}(r) = \log_2 \max\{|a|, b\}$; for $r = 0$ put $\text{ht}(r) = 0$. A rational vector or matrix has *entry height at most* B if every entry has height at most B . Any ordinary binary encoding of a rational vector uses at least $\sum_i \log_2 \text{den}(r_i)$ bits, so lower bounds on denominators are lower bounds on expanded output length. A shared rational circuit is a directed acyclic graph of rational arithmetic gates, as in Section 2; its size counts gates and the binary length of its constants. A circuit of size $O(k)$ can compute a rational whose reduced fraction has more than 2^k bits, for instance 2^{-2^k} by repeated squaring.

For $X \in \mathbb{R}^n$ let $z(X)$ be the vector of all monomials of degree at most two, with the constant monomial first; its length is $N = \binom{n+2}{2}$. A symmetric matrix Q is a (polynomial) Gram matrix of f if $f = z^\top Q z$. Put $w(X, v) = (v, X \otimes v) \in \mathbb{R}^{n+n^2}$. A symmetric matrix A of order $n + n^2$ is a *full Hessian Gram* of a polynomial f of degree at most four if

$$v^\top \nabla^2 f(X) v = w(X, v)^\top A w(X, v) \quad (X, v \in \mathbb{R}^n). \quad (18)$$

A *certified quartic* is a pair (f, A) with $f \in \mathbb{Q}[X]$ of degree at most four and A a rational full Hessian Gram with $A \succ 0$. Validity is checked in polynomial time by coefficient comparison and rational elimination. If $A \succ 0$ has order $s = n + n^2$, then the number $\mu_A = \det A / (\text{tr } A)^{s-1}$, denoted $\rho(A)$ in Lemma 2.7, is a positive rational of polynomial bit length with $A \succeq \mu_A I$, and $\nabla^2 f \succeq \mu_A I$ on $\mathbb{R}^n$ (Lemma 2.8(a)). Thus f is coercive and has a unique minimizer, denoted p . Polynomial Gram matrices and Hessian Gram matrices are different objects: the former represent f in the basis z , the latter represent its Hessian biform in the basis w .

The following integration lemma is the common source of all exact Gram matrices used in this paper. For $a \in \mathbb{R}^n$ write $d = X - a$ and define matrices $C_U(a), C_V(a)$ with $n + n^2$ rows and N columns by

$$(d, a \otimes d) = C_U(a) z(X), \quad (0, d \otimes d) = C_V(a) z(X).$$

Their entries are integer polynomials of degree at most two in the coordinates of a . For a symmetric matrix B of order $n + n^2$ define the *Taylor Gram*

$$\mathcal{T}_a(B) = \frac{1}{2}(C_U(a) + \frac{1}{3}C_V(a))^T B (C_U(a) + \frac{1}{3}C_V(a)) + \frac{1}{36}C_V(a)^T B C_V(a). \quad (19)$$

Lemma 8.1 (Taylor integration of a full Hessian Gram). *Let $f \in \mathbb{R}[X_1, \dots, X_n]$ have degree at most four and a real full Hessian Gram A . Fix $a \in \mathbb{R}^n$, and put $d = X - a$ and $b = \nabla f(a)$.*

(a) Identity. $f(X) = f(a) + b^T d + z(X)^T \mathcal{T}_a(A) z(X)$ identically. The identity rests on

$$\int_0^1 (1-t) (U + tV)^T A (U + tV) dt = \frac{1}{2}(U + V/3)^T A (U + V/3) + \frac{1}{36}V^T A V$$

for $U = (d, a \otimes d)$ and $V = (0, d \otimes d)$, whose scalar form is $\int_0^1 (1-t)(u + tv)^2 dt = \frac{1}{2}(u + v/3)^2 + (v/6)^2$.

(b) Positivity and the strict kernel. If $A \succeq 0$ then $\mathcal{T}_a(A) \succeq 0$. If $A \succ 0$ then $\ker \mathcal{T}_a(A) = \mathbb{R}z(a)$; equivalently, after the invertible change to the translated basis $(1, (d^\alpha)_{1 \leq |\alpha| \leq 2})$, the Taylor Gram is $0 \oplus J$ with J positive definite on the nonconstant monomials.

(c) Gradient completion. Suppose $A \succeq \mu I$ with $\mu > 0$. Put $\bar{A} = A - \mu \text{diag}(I_n, 0)$, $c_a = f(a) - \|b\|^2 / (2\mu)$, let $C_{\text{aff}}(a)$ be the coefficient matrix with $d + b/\mu = C_{\text{aff}}(a)z(X)$, and let e_0 select the constant monomial. Then $f = z^T \mathcal{Q}_a z$ for

$$\mathcal{Q}_a = \mathcal{T}_a(\bar{A}) + \frac{\mu}{2} C_{\text{aff}}(a)^T C_{\text{aff}}(a) + c_a e_0 e_0^T. \quad (20)$$

If $c_a \geq 0$ then $\mathcal{Q}_a \succeq 0$, and if $c_a > 0$ then $\mathcal{Q}_a \succ 0$.

(d) Fields. Let $E \subseteq \mathbb{R}$ be a field containing the coefficients of f , the entries of A , and the coordinates of a (and μ in part (c)). Then $C_U(a)$, $C_V(a)$, $C_{\text{aff}}(a)$, $\mathcal{T}_a(A)$, and $\mathcal{Q}_a$ have entries in E . In particular, at a critical point a of f with $A \succeq 0$, the matrix $\mathcal{T}_a(A)$ is a positive semidefinite Gram of $f - f(a)$ with entries in E .

(e) Squares. If $A = \sum_k s_k \ell_k \ell_k^T$ with $s_k \geq 0$, then, with U, V as in (a),

$$f - f(a) - b^T d = \sum_k s_k \left[2 \left(\frac{\ell_k^T (U + V/3)}{2} \right)^2 + \left(\frac{\ell_k^T V}{6} \right)^2 \right].$$

If $A \succeq 0$ is rational, the s_k and ℓ_k can be chosen rational with each s_k a sum of rational squares, so $f - f(a) - b^T d$ is a sum of squares of quadratics whose coefficients are rational polynomials of degree at most two in the coordinates of a . In particular, at a critical point a , $f - f(a)$ is a sum of squares over every subfield $E \subseteq \mathbb{R}$ that contains the coordinates of a .

The proof is in Appendix F.1. The integral identity is the classical argument that an SOS-convex polynomial minus its first-order Taylor expansion is a sum of squares [53, 76]. The lemma adds what the exact constructions need: an explicit Gram matrix, its exact kernel in the strict case, gradient completion with an explicit constant, and field bookkeeping.

Remark 8.2 (Gram fields and square factors). Part (d) concerns the field of *Gram entries*, and part (e) the field of *square factors*; the two need separate arguments. Over $\mathbb{Q}$ a positive semidefinite rational Gram yields a sum of squares of rational polynomials: rational $LDL^\top$ elimination gives nonnegative rational weights, and each positive rational weight is a sum of rational squares (Lemma 2.20). This is why part (e) needs a rational Hessian Gram but no rationality of a . Over a general real field E the same elimination gives a sum of squares with nonnegative weights in E , but an unweighted sum of squares with coefficients in E need not exist. For example, with $E = \mathbb{Q}(\sqrt{2}) \subset \mathbb{R}$, the polynomial $\sqrt{2}X_1^2$ has a positive semidefinite Gram over E , namely the matrix whose only nonzero entry is the diagonal entry $\sqrt{2}$ at the monomial X_1 , but an identity $\sqrt{2}X_1^2 = \sum_i q_i^2$ with $q_i \in E[X_1]$ is impossible: comparing the coefficients of X_1^2 and applying the conjugation $\sqrt{2} \mapsto -\sqrt{2}$ would make $-\sqrt{2}$ a sum of squares of real numbers. Field statements in this section are therefore always about Gram entries; least fields of square factorizations are treated in Section 9.

8.2 Strictly feasible bodies with only long rational points

A strictly feasible convex body always contains rational points, because it has an interior. The next theorem shows that it need not contain a short one, even when it is the zero sublevel set of a single strongly convex quartic with a short rational convexity certificate and lies in a fixed box.

Construction. Let $k \geq 1$, $n = 2k$, and $M_k = 1000^{k+3}$. A doubly exponentially small positive number is generated by k cube roots:

$$\delta_0 = M_k^{-1}, \quad \delta_i = (1 + 3\delta_{i-1}^2)^{1/3} - 1 \quad (1 \leq i \leq k). \quad (21)$$

Since $(1+t)^3 \geq 1+3t$ for $t \geq 0$, induction gives $0 < \delta_i \leq \delta_{i-1}^2$, hence $0 < \delta_k \leq M_k^{-2^k}$. With $\xi_i = 1 + \delta_i$ the gate equations are

$$\xi_1^3 = 1 + 3M_k^{-2}, \quad \xi_i^3 = 3\xi_{i-1}^2 - 6\xi_{i-1} + 4 \quad (2 \leq i \leq k). \quad (22)$$

Write the coordinates as $X = (x_1, y_1, \dots, x_k, y_k)$ and put $p = (\xi_1, \xi_1^2, \dots, \xi_k, \xi_k^2)$. Lemma F.1 in the appendix constructs, in time polynomial in k , a rational quartic F_k given as a sum of $n+1$ squares of rational quadratics, with $\nabla^2 F_k \succeq \frac{3}{2}I$, unique zero p , and a rational positive definite full Hessian Gram A_k , all of polynomial bit length. The number δ_k is never printed; it is the value at p of the rational affine function $u_k(X) = x_k - 1$. Let $\iota_k \in \mathbb{R}^{n+n^2}$ be the coordinate vector of v_{x_k} in the first block of $w(X, v)$, and put

$$\lambda_k = \left\lceil \frac{3}{\mu_{A_k}} \right\rceil, \quad g_k = \lambda_k F_k - u_k^2, \quad B_k = \lambda_k A_k - 2\iota_k \iota_k^\top. \quad (23)$$

The Hessian biform of u_k^2 is $2v_{x_k}^2$, so B_k is a full Hessian Gram of g_k , and $B_k \succeq 3I - 2I = I$. The integer λ_k can be exponentially large, but its binary length is polynomial in k .

Theorem 8.3 (Strict feasibility without short rational points). *For every $k \geq 1$ the certified quartic (g_k, B_k) of (23) in $n = 2k$ variables is computable in time polynomial in k , and its total binary length L_k is bounded by a polynomial in k . Let $S_k = \{X \in \mathbb{R}^n : g_k(X) \leq 0\}$.*

- (a) $\nabla^2 g_k \succeq I$ on $\mathbb{R}^n$. The set S_k is compact and strictly convex, has nonempty interior, and is contained in $(0.99, 1.01)^n$.
- (b) Every rational point $X \in S_k$, including boundary points, satisfies the following. If $X_1 = a/b$ with integers a and $b \geq 1$, then

$$\log_2 b > \frac{1}{3}((2^k - 2)(k + 3) \log_2 1000 - 4).$$

In particular $\log_2 b > k2^k = \frac{n}{2} 2^{n/2}$ for every $k \geq 2$, so every rational point of S_k has ordinary binary length greater than $\frac{n}{2} 2^{n/2}$.

- (c) A shared rational circuit of size polynomial in k for a point $\hat{x} \in \mathbb{Q}^n$ with $g_k(\hat{x}) < 0$ is computable in time polynomial in k , without sign tests (Proposition 4.6).

The proof is in Appendix F.3. Since $F_k(p) = 0$ and $\nabla F_k(p) = 0$, the point p has $g_k(p) = -\delta_k^2$ and $\|\nabla g_k(p)\| = 2\delta_k$. Strong convexity then confines every point of S_k , including boundary points, to the ball of radius $(2 + \sqrt{6})\delta_k < 5M_k^{-2^k}$ about p . The first coordinate of p is the irrational number ξ_1 , a root of the integer cubic $M_k^2 T^3 - (M_k^2 + 3)$. For a rational point with first coordinate a/b , the integer $M_k^2 a^3 - (M_k^2 + 3)b^3$ is nonzero, so its absolute value is at least one; the difference-of-cubes factorization bounds it by $16b^3 M_k^{2-2^k}$, which forces the denominator bound.

The argument needs no Diophantine approximation theorem: the irrational coordinate satisfies an explicit integer cubic with small coefficients, and a nonzero integer has absolute value at least one. The obstruction is precision near that coordinate, not a large coordinate magnitude, empty interior, or a degenerate convexity certificate. The body lies in a ball of radius below $5M_k^{-2^k}$ about p , so the theorem is also an explicit example in which strict feasibility supplies no quantitative Slater radius of polynomial bit length. The localization applies to every point of S_k , so the bound is not a statement about one particular rounded point.

Large rational witnesses for convex systems are classical. Khachiyan's repeated-squaring system $x_i \geq x_{i+1}^2$, $x_m \geq 2$, recalled by Pataki and Touzov [99], is a strictly feasible convex quadratic system whose rational solutions have exponentially many bits, because its coordinates are huge. Theorem 8.3 adds the restrictions of a single quartic inequality with a supplied strict rational full Hessian Gram, a compact full-dimensional feasible set, and bounded coordinates. In the other direction, a nonempty set defined by one strict integer polynomial inequality of degree d with coefficients of bit length τ contains a rational point with coordinates of bit length $\tau d^{O(n)}$ [14, Theorem 4.1.2]; see also [106, Proposition 2.5]. Applied to $\{g_k < 0\}$ this gives rational points with $\text{poly}(k)2^{O(k)}$ bits, so the exponential dependence on the dimension in Theorem 8.3 is necessary and attained up to the constants in the exponent. The irrational singleton examples of Section 7 have no rational feasible point at all; here rational points exist but are long.

8.3 Rational optimizers with exponentially long denominators

A rational optimizer can also be long. The construction powers a rational point of the unit circle. Repeated squaring of $(3 + 4i)/5$ never cancels a factor of five, so the denominators grow doubly exponentially while the coordinates stay in $[-1, 1]$.

Lemma 8.4 (Denominators of the circle chain). *For an integer $m \geq 1$ write $(3 + 4i)^m = A_m + iB_m$ with $A_m, B_m \in \mathbb{Z}$. Then $A_m \equiv 3$ and $B_m \equiv 4 \pmod{5}$. Consequently the real and imaginary parts of $((3 + 4i)/5)^m$ have reduced denominators exactly 5^m , and their absolute values are at most one.*

Proof. In $\mathbb{Z}[i]$ we have $(3 + 4i)^2 = -7 + 24i \equiv 3 + 4i$ modulo $5\mathbb{Z}[i]$. By induction $(3 + 4i)^m \equiv 3 + 4i$ for all $m \geq 1$. Since 5 divides neither 3 nor 4, the fractions $A_m/5^m$ and $B_m/5^m$ are in lowest terms. The modulus of $((3 + 4i)/5)^m$ is one. $\square$

Let $k \geq 0$ and $n = 2(k + 1)$, and write the coordinates as $X = (x_0, y_0, \dots, x_k, y_k)$. Put $z_j = ((3 + 4i)/5)^{2^j} = a_j + ib_j$. The rational quadratic residuals

$$r_0 = x_0 - \frac{3}{5}, \quad s_0 = y_0 - \frac{4}{5}, \quad r_j = x_j - x_{j-1}^2 + y_{j-1}^2, \quad s_j = y_j - 2x_{j-1}y_{j-1} \quad (1 \leq j \leq k) \quad (24)$$

have the unique common real zero $p = (a_j, b_j)_{j=0}^k$; they express complex squaring with coefficients of constant size. Every circle relation $x_j^2 + y_j^2 - 1$ also vanishes at p , and the circle relations supply the positive curvature that the residuals lack. Lemma F.2 in the appendix combines these quadratics, with coefficients approximated only where approximation preserves the exact zero, into a quartic f_k via Lemma 7.6. A rescaled version with circles of radius 2^{-j} controls the Hessian at the optimizer.

Theorem 8.5 (Bounded rational optimizers with long denominators). *For every integer $k \geq 0$ put $n = 2(k + 1)$. In time polynomial in k one can construct rational quartics f_k and $\tilde{f}_k$ in n variables, each given as a sum of $n + 1$ squares of rational quadratics and each supplied with a rational positive definite full Hessian Gram, all of total binary length bounded by a polynomial in k , with the following properties.*

- (a) $\nabla^2 f_k \succeq \frac{3}{2}I$ and $\nabla^2 \tilde{f}_k \succeq \frac{3}{2}I$ on $\mathbb{R}^n$. Both minima are zero and are attained only at points p and $\tilde{p}$, respectively.
- (b) $p \in \mathbb{Q}^n \cap [-1, 1]^n$, and the last two coordinates of p have reduced denominator exactly 5^{2^k} .
- (c) $\tilde{p} \in \mathbb{Q}^n$, $\|\tilde{p}\|^2 < \frac{4}{3}$, the last two coordinates of $\tilde{p}$ have reduced denominators divisible by 5^{2^k} , and

$$2I \preceq \nabla^2 \tilde{f}_k(\tilde{p}) \preceq (8(k + 1)^2 + 1)I.$$

Thus the condition number of the Hessian at the optimizer is at most $4(k + 1)^2 + \frac{1}{2}$.

Consequently, writing either optimizer as ordinary reduced fractions requires at least $2^k \log_2 5 = 2^{n/2-1} \log_2 5$ bits in one coordinate. This is exponential in n and superpolynomial in the input length, but it is not of the form $2^{\Omega(L_k)}$.

The proof is in Appendix F.4. Both optimizers have short exact descriptions: the shared circuit computing $((3 + 4i)/5)^{2^k}$ by k squarings, followed by a rescaling for $\tilde{p}$. They can also be approximated to any requested precision in polynomial time. The theorem therefore separates the size of an exact expanded rational output from both exact implicit output and approximation. It is also different from the rational-optimizer hardness of Theorem 5.2: that result concerns the cost of *comparing* a rational optimizer coordinate with zero, under a rationality promise, whereas Theorem 8.5 is an unconditional bound on the *length* of the optimizer itself.

Part (c) shows that a long rational optimizer is compatible with good local conditioning: the Hessian at $\tilde{p}$ has condition number $O(n^2)$. This is a statement about the Hessian at the optimizer only. The coefficients of $\tilde{f}_k$ have magnitudes up to $2^{O(k)}$, and no uniform smoothness bound on a fixed neighborhood of $\tilde{p}$, no bound on higher derivatives, and no bound on the conditioning of rational reconstruction is asserted. Multiplying f_k by a sufficiently large integer square normalizes its Hessian Gram to be at least the identity without changing p .

The norm and approximation guarantees for convex polynomial optimization of Slot, Steurer, and Wiedmer [111] are compatible with Theorem 8.5: they bound magnitudes and approximation

cost, not denominators. Jiang's exact algorithm for convex functions with rational minimizers [60] is parameterized by a bound on the common denominator of the minimizer; for f_k the least common denominator of the coordinates of p is 5^{2^k} , although the coordinates are bounded by one. Exponentially long rational local minimizers of nonconvex cubics, caused by large coordinate magnitudes, appear in Zhang's thesis [120, Example 2.5.3]. We do not claim that powering a rational point of the circle is new; the contribution is its realization as the unique optimizer of a globally strongly convex quartic with short rational square factors and a short strict Hessian certificate.

8.4 Exact moment relaxations and the arithmetic of optimal Grams

For a certified quartic f consider the unconstrained order-two moment relaxation and its sum-of-squares dual. For a sequence $y = (y_\alpha)_{|\alpha| \leq 4}$ let $\Lambda_y(f) = \sum_\alpha f_\alpha y_\alpha$ and $M_2(y) = (y_{\alpha+\beta})_{|\alpha|, |\beta| \leq 2}$, indexed like z . The programs are

$$(P) \quad \min\{\Lambda_y(f) : y_0 = 1, M_2(y) \succeq 0\}, \quad (S) \quad \max\{\gamma : f - \gamma = z^\top Q z, Q \succeq 0\}. \quad (25)$$

Their data are rational. Let p be the minimizer of f , $f_* = f(p)$, $e = z(p)$, and let $K = \mathbb{Q}(p_1, \dots, p_n) \subseteq \mathbb{R}$ be the field generated by the coordinates of p . An *optimal Gram* is a positive semidefinite Gram of $f - f_*$; the set of all of them is denoted $\mathcal{G}_*$, and the *optimal-level Gram system* is $f - f_* = z^\top Q z$, $Q \succeq 0$. A nonzero positive semidefinite Z *exposes* $\mathcal{G}_*$ if $\langle Z, Q \rangle = \text{tr}(ZQ) = 0$ for every $Q \in \mathcal{G}_*$. A facial-reduction step [19] restricts a semidefinite system to the face cut out by such a matrix of the form $\mathcal{A}^*(y)$, where $\mathcal{A}$ is the coefficient map of the system and y annihilates its right-hand side; the *singularity degree* is the least number of steps after which the restricted system has a feasible point in the relative interior of its face.

Theorem 8.6 (Moments, maximal-rank Grams, and exposing matrices). *Let (f, A) be a certified quartic in n variables. Put $r = \dim_{\mathbb{Q}} \text{span}_{\mathbb{Q}}\{p^\alpha : |\alpha| \leq 2\}$.*

- (a) *Maximal rank. The Taylor Gram $Q_* = \mathcal{T}_p(A)$ is an optimal Gram with entries in K and $\ker Q_* = \mathbb{R}e$. Every optimal Gram annihilates e , so $N - 1$ is the largest rank of an optimal Gram.*
- (b) *Exactness and uniqueness. Both programs in (25) attain the value f_* . The only optimal solution of (P) is $y_\alpha = p^\alpha$, that is, $M_2(y) = ee^\top$. The optimal pair $(ee^\top, Q_*)$ is strictly complementary, and both programs have strictly feasible points.*
- (c) *Least field of maximal-rank Grams. For a field $E \subseteq \mathbb{R}$, some optimal Gram of rank $N - 1$ has all entries in E if and only if $K \subseteq E$.*
- (d) *Rational optimal Grams. Every rational optimal Gram has rank at most $N - r$. If $p \notin \mathbb{Q}^n$ then $r \geq 2$, so no rational optimal Gram is strictly complementary to the optimal moment matrix.*
- (e) *Faces and exposing matrices. The smallest face of the positive semidefinite cone containing $\mathcal{G}_*$ is $\mathcal{F}_p = \{Q \succeq 0 : Qe = 0\}$, and Q_* lies in its relative interior. A nonzero positive semidefinite Z exposes $\mathcal{G}_*$ if and only if $Z = cee^\top$ with $c > 0$. The entries of every such Z generate a field containing K , and $ee^\top$ has entries in K . The matrix $ee^\top$ is a facial-reduction certificate for the optimal-level Gram system, one reduction step reaches $\mathcal{F}_p$, and the real singularity degree of that system is exactly one. If $p \notin \mathbb{Q}^n$, no nonzero rational positive semidefinite matrix exposes $\mathcal{G}_*$.*

The proof is in Appendix F.5. Its core is short. The Taylor Gram of Lemma 8.1 at the stationary point p has kernel exactly $\mathbb{R}e$. If y is optimal for (P), then $\text{tr}(Q_*M_2(y)) = \Lambda_y(f - f_*) = 0$; two positive semidefinite matrices with zero trace product have orthogonal ranges, so $M_2(y)$ is a multiple of $ee^\top$, and $y_0 = 1$ fixes the multiple. Every moment of degree at most four is an entry of $M_2(y)$. A maximal-rank Gram over E has a one-dimensional kernel over E , and normalizing its constant coordinate recovers e , hence p . The same kernel argument identifies every exposing matrix.

Exactness of low-order moment relaxations for SOS-convex problems, with extraction of the optimizer from first-order moments, is due to Lasserre [76, Theorems 2.6 and 3.3]. Theorem 8.6 uses the stronger strict Hessian certificate to determine *all* moments and the complete kernel of the optimal Gram face. The use of real zeros as Gram kernel vectors, and of their conjugates to obtain rational kernel relations, is established in Laplagne’s facial-reduction method for exact SOS decompositions [74, Proposition 3.2, Section 3.2]; part (d) is a dimension count for that mechanism. Algebraic, possibly irrational, exact solutions of degenerate semidefinite programs and their certification are also established [69]. The point of Theorem 8.6 is the precise identification of the field forced on each certificate shape, in a setting where the relaxation is exact, both programs are strictly feasible, strict complementarity holds over the reals, and the optimal Gram face is reached in one real facial-reduction step. Field statements concern Gram and moment entries (Remark 8.2).

Part (e) concerns matrices orthogonal to *all real* optimal Grams. It does not prevent a procedure that deliberately restricts attention to rational Gram candidates. Such a procedure may impose the rational kernel relations of part (d) and work in a smaller face that discards real optimal Grams; that is a different feasibility problem. The interpretation of $ee^\top$ as a facial-reduction certificate for a *rational* optimal-level system needs $f_* \in \mathbb{Q}$. If $f_* \notin \mathbb{Q}$, rational optimal Grams do not exist at all, because the constant coefficient of $f - f_*$ is irrational.

Remark 8.7 (The strict Hessian hypothesis). Strong convexity together with SOS-convexity does not suffice for part (b). The polynomial $f = X_1^2 + X_2^2 + X_2^4$ has $\nabla^2 f = \text{diag}(2, 2 + 12X_2^2) \succeq 2I$ and a diagonal SOS Hessian, with minimizer 0 and $f_* = 0$. The sequence with $y_0 = 1$, $y_{(4,0)} = t \geq 0$, and all other moments zero has $M_2(y) = \text{diag}(1, 0, 0, t, 0, 0)$ in the basis $(1, X_1, X_2, X_1^2, X_1X_2, X_2^2)$ and objective value zero, so (P) has infinitely many optimal solutions. This f has no positive definite full Hessian Gram: its Hessian biform has no $X_1^2v_1^2$ term, so the diagonal entry of a full Hessian Gram at the coordinate X_1v_1 must vanish. The hypothesis is sufficient but not necessary. The proof uses it only to obtain an optimal Gram with kernel exactly $\mathbb{R}e$; for instance $X_1^2 + X_2^2 + X_1^4 + X_2^4$ has no positive definite full Hessian Gram, by the same coefficient argument, but has the optimal Gram $0 \oplus I_2 \oplus \begin{pmatrix} 1 & 0 & -1/2 \\ 0 & 1 & 0 \\ -1/2 & 0 & 1 \end{pmatrix}$, whose kernel is $\mathbb{R}e$.

Remark 8.8 (Large fields). For the cyclic quartics of Theorem 7.8, which are certified quartics with minimum zero and optimizer field of degree $d_n = (2^{n+1} - (-1)^{n+1})/3$, parts (c) and (e) show that every maximal-rank optimal Gram and every exposing matrix of the optimal Gram face has entries generating a field of degree at least d_n over $\mathbb{Q}$. This is a statement about fields; it does not exclude short circuit or other implicit descriptions of these matrices.

Remark 8.9 (Nullvectors of one feasible matrix). Facial reduction preserves every real feasible point when it uses a vector in the common kernel of the feasible set. A maximum-rank feasible matrix has exactly this common kernel: if $G_0, G \succeq 0$ then $\ker(G_0 + G) = \ker G_0 \cap \ker G$, so a feasible G that did not vanish on $\ker G_0$ would give the feasible midpoint $(G_0 + G)/2$ a larger rank. This is why part (a) matters. A nullvector of an arbitrary feasible matrix does not suffice, even when it is rational. Proposition K.4 constructs a rational affine semidefinite family with a rational positive definite member and a singular feasible member with a rational nullvector. Restricting the family to

matrices that annihilate that vector leaves only a single irrational feasible matrix, removing every rational feasible point. The construction and full proof are given in Appendix K.2.

8.5 Heights forced by a long rational optimizer

When the optimizer is rational, Theorem 8.6 gives rational maximal-rank Grams and rational exposing matrices, but their entries must encode the optimizer. For the circle family this makes them long, although a short rational optimal Gram of smaller rank is part of the input.

Corollary 8.10 (Long maximal-rank, moment, and exposing certificates). *Let f_k be the quartic of Theorem 8.5 in $n = 2(k + 1)$ variables, with optimizer p and minimum zero, and let $N = \binom{n+2}{2}$ and $H_k = 2^k \log_2 5$.*

- (a) *The unique optimal moment matrix $ee^\top$ has an entry with reduced denominator 5^{2^k} .*
- (b) *Rational optimal Grams of rank $N - 1$ exist. If such a Gram has entry height at most B , then*

$$B \geq \frac{H_k - \log_2((N - 1)!)}{N^2 - 1} = \Omega(2^k/n^4),$$

and the sum of the binary lengths of all its entries is at least $H_k - \log_2((N - 1)!)$.

- (c) *If a nonzero rational positive semidefinite matrix exposing the optimal Gram face has entry height at most B , then $B \geq H_k/2$.*
- (d) *f_k has a rational optimal Gram of rank at most $n + 1 < N - 1$ and binary length polynomial in k , namely the Gram of its supplied square factors.*

The same statements hold for $\tilde{f}_k$, with “reduced denominator 5^{2^k} ” replaced by “reduced denominator divisible by 5^{2^k} ”.

The proof is in Appendix F.5. For (b), the principal submatrix of a maximal-rank optimal Gram on the nonconstant monomials is nonsingular and determines the remaining coordinates of $e = (1, w)$ by Cramer’s rule; a terminal coordinate of p appears in w , so its denominator divides a determinant bounded in terms of the entry heights. For (c), every exposing matrix is $cee^\top$, and a terminal coordinate of p is the ratio of two of its entries. These lower bounds are exponential in n and superpolynomial in the input length; part (d) shows that they cannot be extended to all optimal Grams.

The phenomenon of short low-rank feasible points coexisting with long maximal-rank feasible points is classical in semidefinite programming. In the homogenized Khachiyan system with blocks $\begin{pmatrix} x_1 & 2t \\ 2t & t \end{pmatrix}$ and $\begin{pmatrix} x_j & x_{j-1} \\ x_{j-1} & t \end{pmatrix}$ for $2 \leq j \leq k$ (compare [99]), the point $t = x_1 = \dots = x_{k-1} = 0$, $x_k = 1$ is a short feasible point of rank one. Every feasible point of full rank has $t > 0$, $x_1/t > 4$, and $x_j/t > (x_{j-1}/t)^2$, hence $x_k/t > 2^{2^k}$; if x_k and t have height at most B then $2^{2^B} \geq x_k/t$, so $B > 2^{k-1}$. Corollary 8.10 does not claim this general distinction as new. Its content is the realization of the distinction by the optimal Gram face of a short certified convex quartic with a bounded rational optimizer, where it concerns maximal-rank, moment, and exposing certificates simultaneously.

8.6 Interior Gram matrices

A polynomial has a positive definite Gram matrix only if it is strictly positive, because z contains the constant monomial. For certified quartics the converse holds: if $\min f > 0$, then by continuity

some rational center a satisfies $c_a > 0$ in Lemma 8.1(c), and $\mathcal{Q}_a$ is a rational positive definite Gram. Interior rational Grams are the certificates produced by methods that round an approximate interior solution and project it onto the rational coefficient equations [100]. The question is how long they must be.

Let $\|f\|_1$ denote the sum of the absolute values of the coefficients of f . Continuing the construction of Section 8.2, put

$$h_k = \lambda_k F_k + u_k^2, \quad \lambda_k A_k + 2\iota_k \iota_k^\top \succeq 3I, \quad (26)$$

which differs from g_k only in the sign of the square.

Theorem 8.11 (Interior Gram matrices).

- (a) Upper bound. *Let (f, A) be a certified quartic in n variables with $\min f > 0$, and let L be the total binary length of f and A . Then f has a rational positive definite Gram matrix of total binary length at most $\text{poly}(L) 2^{O(n)}$, and f is a sum of squares of rational quadratics of the same total length.*
- (b) A strictly positive family. *For every $k \geq 1$ the certified quartic $(h_k, \lambda_k A_k + 2\iota_k \iota_k^\top)$ of (26), in $n = 2k$ variables, is computable in time polynomial in k and has total binary length polynomial in k . It satisfies $\nabla^2 h_k \succeq 3I$, $h_k > 0$ on $\mathbb{R}^n$, and*

$$0 < m_k := \min_{\mathbb{R}^n} h_k < M_k^{-2^{k+1}}, \quad M_k = 1000^{k+3}.$$

It is given as $\lambda_k \sum_{j=1}^{n+1} F_{k,j}^2 + u_k^2$ with rational quadratics $F_{k,j}$, so it has a rational positive semidefinite Gram of rank at most $n + 2 < N$ and a rational sum-of-squares expression, both of binary length polynomial in k . It also has rational positive definite Gram matrices.

- (c) Lower bound. *Every rational positive definite Gram Q of h_k , with reduced upper-triangular entry denominators b_{ij} , satisfies*

$$\sum_{i \leq j} \log_2 b_{ij} > 2^k \log_2 M_k - \frac{N-1}{2} \log_2 T_k, \quad T_k = \max\{1, 15(n+1) \|h_k\|_1\},$$

where $\log_2 T_k$ is bounded by a polynomial in k . Hence every rational positive definite Gram of h_k has total denominator length $\Omega(k2^k) = \Omega(n2^{n/2})$, and some entry has $\Omega(2^k/k^3)$ denominator bits.

The proofs are in Appendix F.6. The lower bound rests on two facts. Every positive semidefinite Gram of h_k has trace at most T_k , because the second-moment matrix of z under the uniform measure on $[-1, 1]^n$ is at least $I/(15(n+1))$. Every positive definite Gram has smallest eigenvalue at most $m_k = \min h_k$: at a minimizer a , the Rayleigh quotient on $z(a)$ is $m_k / \|z(a)\|^2 \leq m_k$, since $z(a)$ contains the constant one. Hence $0 < \det Q < M_k^{-2^{k+1}} T_k^{N-1}$. On the other hand, the product of the squared upper-triangular denominators clears every denominator in the determinant expansion, so a positive rational determinant is at least the reciprocal of that product. A singular Gram has determinant zero, which is why the short Gram of rank $n + 2$ escapes the bound. The upper bound uses the strict-set rational sampling theorem of Basu, Pollack, and Roy [14, Theorem 4.1.2] to find a rational center with $c_a > 0$ and $\text{poly}(L) 6^{O(n)}$ coordinate bits, and then evaluates the explicit formula (20). It is a size bound for some interior certificate; it does not give an algorithm polynomial in L , and it is not claimed for strictly positive quartics without a supplied strict Hessian Gram.

Parts (a) and (c) show that exponential dependence on the dimension is necessary and sufficient for interior rational Grams in this certified class; the constants in the exponents do not match. The

two kinds of Gram matrix in the theorem should not be confused: the supplied Hessian Gram of h_k is short and has smallest eigenvalue at least three, while every interior polynomial Gram of h_k is long and has smallest eigenvalue below $M_k^{-2^{k+1}}$. This also explains the relation to the exact rounding theorem of Gärtner, Magron, and Vallentin [45, Corollary 1.3], which recovers an exact rational Gram when a positive lower bound on the smallest eigenvalue of some Gram is supplied as part of the input. For h_k every such bound is smaller than $M_k^{-2^{k+1}}$, so its encoding alone has more than $2^{k+1} \log_2 M_k$ bits, and by part (c) every rational positive definite Gram that any method prints has $\Omega(k2^k)$ bits. Compactness of Gram spectrahedra is known [25, Lemma 1.5]; the trace bound above is an explicit version in the monomial basis that the encoding argument needs. Exponentially large coefficients are known to be necessary in some constrained sum-of-squares proofs [95, 101]; the theorem concerns ordinary Gram matrices of a single globally positive quartic. Whether some family in this certified class forces long encodings of *every* rational positive semidefinite Gram remains open; part (b) shows that the family h_k does not.

8.7 Shared-circuit witnesses and interior Grams

For two of the objects above, the expanded lower bounds disappear when outputs may be shared rational circuits: strictly feasible points when the minimum is negative, and interior Gram matrices when it is positive. The Newton circuits of Section 4 approximate the optimizer to doubly exponential accuracy with polynomially many gates, and the separation bound tells how much accuracy suffices for a given polynomial observable. With the observable f itself this gives a circuit point in $\{f < 0\}$ whenever $\min f < 0$ (Proposition 4.6); for the families g_k of Theorem 8.3 such a point has polynomial circuit size, while its expanded coordinates need more than $k2^k$ bits. With a different observable the same circuit point yields interior Gram matrices, without deciding any sign.

Theorem 8.12 (Shared-circuit interior Grams). *There is a deterministic polynomial-time algorithm that receives a pair (f, A) , where $f \in \mathbb{Q}[X_1, \dots, X_n]$ has degree at most four and A is a rational symmetric matrix of order $n + n^2$, rejects the pair unless it is a certified quartic, and otherwise outputs, without any sign oracle, a symmetric $N \times N$ matrix Q whose entries are computed by one shared rational circuit of polynomial size, such that $f = z^\top Q z$ identically and*

$$Q \succ 0 \iff \min_{\mathbb{R}^n} f > 0.$$

Every division in the circuit is by a nonzero rational, on every certified input and regardless of the sign of $\min f$.

The proof is in Appendix F.7. Put $\mu = \mu_A$ and $h = f - \|\nabla f\|^2 / (2\mu)$, an explicit polynomial of degree at most six with $h(p) = f(p)$. The shared Newton point $\hat{x}$ of Lemma 4.3, whose accuracy covers every observable up to a stated encoding length, satisfies $|h(\hat{x}) - h(p)| \leq g/8$ for a gap $g > 0$ such that $h(p) \neq 0$ implies $|h(p)| \geq g$. The matrix Q is the completed Taylor Gram (20) at the exact rational center $\hat{x}$. Its constant is $c_{\hat{x}} = h(\hat{x})$, which is positive whenever $\min f > 0$; conversely, $Q \succ 0$ forces $f \geq \lambda_{\min}(Q) > 0$ because z contains the constant monomial.

Neither output is an expanded rational. For the family h_k of Theorem 8.11, the value of Q is a rational positive definite Gram of h_k , so its expanded encoding needs $\Omega(n2^{n/2})$ denominator bits, while the circuit has size polynomial in k . For these two outputs the circuit model removes the expanded-size obstruction, but it moves the cost to validation over general certified quartic inputs, as follows.

- (i) By Theorem 8.12, $Q \succ 0$ holds exactly when $\min f > 0$, and deciding $\min f > 0$ for certified quartics is PosSLP-hard (Theorem 5.1). Hence deciding positive definiteness of a matrix

given by a shared rational circuit is PosSLP-hard already for these outputs, and it is not in polynomial time unless PosSLP is.

- (ii) For the point $\hat{x}$ of Proposition 4.6, $f(\hat{x}) \geq \min f$, so $f(\hat{x}) < 0$ holds exactly when $\min f < 0$. Checking the sign of $f(\hat{x})$ is therefore as hard as deciding $\min f < 0$, which is PosSLP-complete for certified quartics.
- (iii) The identity $f = z^\top Q z$ holds by construction. We claim no deterministic polynomial-time procedure that verifies such an identity for a circuit matrix received without its construction.
- (iv) When $\min f > 0$, symmetric elimination applied to Q produces polynomial-size circuits for a weighted rational sum of squares with positive weights. Writing those weights as sums of rational squares by the binary method takes time polynomial in their *expanded* length, which can be exponential. We do not claim polynomial-size circuits for an unweighted rational sum of squares.

The construction requires the full strict Hessian Gram on $(v, X \otimes v)$: the positive definiteness step uses $\bar{A} \succeq \mu \text{diag}(0, I_{n^2})$ to supply every translated quadratic monomial. It needs no lower bound on the positive minimum, which may be doubly exponentially small, as for h_k .

The rational objects of the circle family also have short circuits. For f_k and $\tilde{f}_k$ of Theorem 8.5, k complex squarings compute the optimizer, followed for $\tilde{p}$ by multiplications with powers of two. The entries of $ee^\top$, which is the optimal moment matrix and, up to a positive factor, every exposing matrix, are monomials of degree at most four in the optimizer. The maximal-rank optimal Gram $\mathcal{T}_p(A)$ of Theorem 8.6(a) is given by the fixed formula (19) in the optimizer and the supplied Hessian Gram A . Hence the optimizer, $ee^\top$, and $\mathcal{T}_p(A)$ have circuits of size polynomial in k , although their expanded encodings are long (Theorem 8.5 and Corollary 8.10). None of this extends to irrational outputs. A circuit with rational constants and rational operations has a rational value. When the optimizer is irrational, such a circuit therefore cannot describe the optimizer, a maximal-rank optimal Gram, or a nonzero matrix exposing the optimal Gram face (Theorem 8.6(c),(e)). These objects need a different output model, for example circuits with root gates; this section makes no claim about such models.

8.8 Summary of representation costs

Table 6 collects the results. In every row the input is a certified quartic of total length polynomial in k , with $n = \Theta(k)$ variables and a rational positive definite Hessian Gram.

The lower bounds limit printed rational outputs only; the limits listed at the beginning of this section apply to every row. The circuit witness and the circuit Gram can be constructed in polynomial time. Their validation over general certified quartic inputs has the following complexity. Deciding whether the circuit witness $\hat{x}$ satisfies $f(\hat{x}) < 0$ is PosSLP-complete, and deciding whether the circuit Gram is positive definite is PosSLP-hard (Section 8.7, items (i) and (ii)). These statements concern the general input class, not the displayed families: since $\min g_k < 0$ and $\min h_k > 0$, both predicates are always true on those families.

9 Coefficient fields of exact sum-of-squares certificates

Let (f, A) be a certified quartic in the sense of Section 8.1 whose minimum value is zero, attained at p . Convexity of f is then certified by short rational data, and Lemma 8.1 turns the Hessian certificate into a sum of squares for f itself. The squares produced in this way have coefficients in the

Object	Expanded rational encoding	Short alternative
Point of a strictly feasible body (g_k)	every rational point has more than $\frac{n}{2}2^{n/2}$ bits for $n \geq 4$ (Thm. 8.3)	circuit witness of polynomial size (Prop. 4.6)
Unique rational optimizer ($f_k, \tilde{f}_k$)	$2^{n/2-1} \log_2 5$ denominator bits in one coordinate (Thm. 8.5)	squaring circuit of size $O(k)$; polynomial-time approximation
Optimal moment matrix (f_k)	an entry with denominator 5^{2^k} (Cor. 8.10)	same circuit
Maximal-rank optimal Gram (f_k)	entry height $\Omega(2^k/n^4)$, total length $\Omega(2^k)$ (Cor. 8.10)	circuit for $\mathcal{T}_p(A)$ from the circuit for p ; an optimal Gram of rank at most $n+1$ and polynomial length
Exposing matrix of the optimal Gram face (f_k)	entry height at least $2^{k-1} \log_2 5$ (Cor. 8.10)	$ee^\top$ from the circuit for p
Positive definite Gram (h_k)	total denominator length $\Omega(n2^{n/2})$; at most $\text{poly}(L)2^{O(n)}$ for every strictly positive certified quartic (Thm. 8.11)	circuit Gram of polynomial size (Thm. 8.12)
Positive semidefinite Gram or rational SOS (h_k)	polynomial length (rank at most $n+2$)	—

Table 6: Output size of exact objects for certified quartics.

field $\mathbb{Q}(p)$ generated by the coordinates of p , which can have large degree (Section 7). This section asks how much of that field an exact certificate of the optimal value must use. Three questions are natural.

- (i) Does a rational convexity certificate guarantee a sum of squares of *rational* polynomials for f ? This is the question of rational descent.
- (ii) If it does not, which real fields carry an exact sum-of-squares or positive semidefinite Gram certificate of f , and can the smallest such field be forced to be large while the input stays short?
- (iii) Which compact exact certificates remain available when every polynomial certificate needs a large field?

The answers matter for exact certification. Exact optimality certificates are typically obtained by rounding a numerical semidefinite solution to rational data and verifying a polynomial identity [62, 100]. When the optimal value is attained at an irrational point, such rounding must fail if no rational certificate exists, and an algebraic certificate is only useful if its field is of manageable size. The results below separate the certification of convexity from the certification of the optimal value, and they identify which certificate representations are long and which stay short.

The main results are the following.

- (a) For every prime $\ell \geq 5$ there is an integer quartic in $(\ell+1)/2$ variables with an integer positive definite full Hessian Gram; each of its coefficients and each Gram entry has $O(\log \ell)$ bits. A sum of squares or a positive semidefinite Gram of this quartic over a real field E exists exactly when $2^{1/\ell} \in E$ (Theorem 9.4). For $\ell = 5$ this is a ternary quartic, and three variables are the minimum for a failure of rational descent among certified quartics with minimum zero (Corollary 9.7); an explicit ternary instance has 31 monomials and coefficients of absolute value at most 448 (Example 9.6).

- (b) For every k there is a rational quartic in $3k$ variables, with input and full Hessian Gram of size polynomial in k , whose least real certificate field is $\mathbb{Q}(2^{1/5^k})$, of degree 5^k (Theorem 9.9). Every algebraic certificate even has one individual coefficient of degree at least 5^k (Theorem 9.10). Shared root circuits and rational certificates with auxiliary root variables are nevertheless of polynomial size (Theorem 9.11).
- (c) At the tower point, the rational quadratic relations, the span of their products, and the rational quartics with zero value and gradient can be determined explicitly. These descriptions yield an exact rational recognition algorithm (Theorem 9.12 and Corollary 9.13).
- (d) Conversely, if the rational quadratics vanishing at p form a space of the minimum possible dimension $n + 1$ and their products exhaust the stationary quartics, then every globally convex rational quartic with minimum zero at p and positive definite Hessian there is a rational sum of squares, provided some globally convex rational sum of squares of degree four vanishes at p with positive definite Hessian there (Theorem 9.14).

All statements concern polynomial sums of squares and polynomial Gram matrices of the attained value. Rational-function certificates, multipliers, and block-structured certificates are treated in Section 10. No result of this section is a lower bound for the complexity of a decision problem or for approximation.

9.1 Certificate models and basic tools

We use the conventions of Section 8.1: $z(X)$ is the vector of monomials of degree at most two with the constant monomial first, and $w(X, v) = (v, X \otimes v)$ is the full Hessian basis. For a polynomial P and a monomial m , $[m]P$ denotes the coefficient of m in P . For a subfield $E \subseteq \mathbb{R}$, a point $p \in \mathbb{R}^n$, and $d \geq 1$ put

$$\mathcal{I}_d^E(p) = \{g \in E[X]_{\leq d} : g(p) = 0\}, \quad \mathcal{J}_d^E(p) = \{g \in E[X]_{\leq d} : g(p) = 0, \nabla g(p) = 0\},$$

and let $\mathcal{W}^E(p)$ be the E -span of all products gh with $g, h \in \mathcal{I}_2^E(p)$. The product rule gives $\mathcal{W}^E(p) \subseteq \mathcal{J}_4^E(p)$.

Definition 9.1 (Certificates over a real field). Let $E \subseteq \mathbb{R}$ be a subfield and let $f \in \mathbb{R}[X]$ have degree at most four.

- (a) f is a *sum of squares over E* if $f = \sum_j g_j^2$ for finitely many $g_j \in E[X]$.
- (b) f has a *positive semidefinite Gram matrix over E* if $f = z^\top Q z$ for a symmetric matrix Q with entries in E that is positive semidefinite as a real matrix.
- (c) A subfield $E_0 \subseteq \mathbb{R}$ is the *least real field* for one of these certificate types if, for every subfield $E \subseteq \mathbb{R}$, a certificate of that type over E exists exactly when $E_0 \subseteq E$.

The field E is arbitrary: it need not be a number field and may contain transcendental elements. A sum of squares over E gives a positive semidefinite Gram matrix over E , namely the sum of the outer products of the coefficient vectors of the square factors. Over $\mathbb{Q}$ the converse holds, by rational elimination and the splitting of positive rationals into rational squares (Lemma 2.20). Over a general real field the converse fails: Remark 8.2 gives a positive semidefinite Gram over $\mathbb{Q}(\sqrt{2})$ of a polynomial that is not a sum of squares over $\mathbb{Q}(\sqrt{2})$; see also [25, Lemma 1.6 and Remark 1.7]. Every least-field statement below therefore proves necessity separately for square factorizations and for positive semidefinite Gram matrices, and proves sufficiency by exhibiting actual squares.

Lemma 9.2 (Basic tools). *Let $E \subseteq \mathbb{R}$ be a subfield and $p \in \mathbb{R}^n$.*

- (a) Degree. *If $f = \sum_j g_j^2$ with real g_j and $\deg f \leq 4$, then every g_j has degree at most two.*
- (b) Kernel restriction. *Suppose $\deg f \leq 4$ and $f(p) = 0$. Every factor of a real sum of squares of f vanishes at p . If $f = z^\top Q z$ with $Q \succeq 0$, then $Qz(p) = 0$. If moreover Q has entries in E and $g = (g_1, \dots, g_t)^\top$ lists a basis of $\mathcal{I}_2^E(p)$, then $f = g^\top S g$ for a positive semidefinite S with entries in E .*
- (c) Obstruction functionals. *Let φ be a real linear functional on $\mathbb{R}[X]_{\leq 4}$ with $\varphi(g^2) \geq 0$ for every $g \in \mathcal{I}_2^E(p)$. If $\deg f \leq 4$, $f(p) = 0$, and $\varphi(f) < 0$, then f is neither a sum of squares over E nor has a positive semidefinite Gram matrix over E .*
- (d) Odd radicals over real fields. *Let $D \geq 1$ be odd and $a = 2^{1/D} > 0$. The minimal polynomial of a over E is $T^r - a^r$ for some divisor r of D . In particular, if $D = \ell$ is prime and $a \notin E$, then $T^\ell - 2$ is irreducible over E ; and if $D = 5^k$ with $k \geq 1$ and $a \notin E$, then $[E(a) : E(a^5)] = 5$.*

The proof is in Appendix G.1. Part (d) is the reason why the least fields below can be described for *all* real fields, including transcendental ones: the only real root of unity of odd order is one, so the constant term of a factor of $T^D - 2$ over a real field is a power of a . Sufficiency always comes from Lemma 8.1(e): if f has a rational positive semidefinite full Hessian Gram and a zero p with $\nabla f(p) = 0$, then f is a sum of squares of quadratics whose coefficients are rational polynomials of degree at most two in the coordinates of p , hence a sum of squares over every subfield $E \subseteq \mathbb{R}$ with $p \in E^n$. The scalar identity $\int_0^1 (1-t)(U+tV)^2 dt = 2((U+V/3)/2)^2 + (V/6)^2$ behind it produces actual squares and introduces no square roots. For a rational $A \succ 0$ of order h we write $\mu_A = \det A / (\text{tr } A)^{h-1}$; then $A \succeq \mu_A I$ (Lemma 2.7).

9.2 Rational convexity certificates do not give rational descent

Let $\ell \geq 5$ be prime, put $s = (\ell - 1)/2$ and $n = s + 1$, and use variables $x_0, \dots, x_s$. Let $a = 2^{1/\ell}$ and

$$p = (a^s, a^{s+1}, \dots, a^{2s}) \in [1, 2]^n.$$

Its coordinates generate $\mathbb{Q}(a)$, since $a = 2/p_s$, a field of degree ℓ by Eisenstein's criterion. Define the rational linear functional

$$\varphi_\ell(P) = [x_s]P + [x_0^2]P. \quad (27)$$

Lemma 9.3 (A coefficient obstruction at p). *Let $E \subseteq \mathbb{R}$ be a subfield with $a \notin E$, and let $g = g_0 + \sum_i v_i x_i + \sum_{i \leq j} h_{ij} x_i x_j \in E[X]_{\leq 2}$ vanish at p . Then $v_s + h_{00} = 0$ and $\varphi_\ell(g^2) = v_0^2$.*

Proof. A monomial of degree at most two evaluates at p to 1, to a^{s+i} with $0 \leq i \leq s$, or to a^{2s+i+j} with $0 \leq i \leq j \leq s$. Reduce the exponents with $a^\ell = a^{2s+1} = 2$. The exponents $2s + i + j$ with $i + j \geq 1$ lie in $[2s + 1, 4s]$ and reduce to $2a^{i+j-1}$ with $i + j - 1 \leq 2s - 1$. Hence the only monomials contributing to a^{2s} are x_s and x_0^2 , and the coefficient of a^{2s} in $g(p)$ is $v_s + h_{00}$. By Lemma 9.2(d) the powers $1, a, \dots, a^{2s}$ are linearly independent over E , so $g(p) = 0$ forces $v_s + h_{00} = 0$. In g^2 the monomial x_s has coefficient $2g_0 v_s$ and x_0^2 has coefficient $v_0^2 + 2g_0 h_{00}$. Therefore $\varphi_\ell(g^2) = v_0^2 + 2g_0(v_s + h_{00}) = v_0^2$. $\square$

The lemma makes φ_ℓ nonnegative on squares of E -quadratics vanishing at p , but only those: $\varphi_\ell((1 - x_s)^2) = -2$. The construction therefore builds a strictly convex rational sum of squares from vanishing quadratics with no linear x_0 term, and subtracts the square of the vanishing quadratic $r = x_1 x_s - 2x_0$, which has $\varphi_\ell(r^2) = 4$. Appendix G.2 gives the construction and its quantitative convexity proof.

Theorem 9.4 (Prime least fields). *For every prime $\ell \geq 5$ one can construct, in time polynomial in ℓ , an integer quartic f_ℓ in the $n = (\ell + 1)/2$ variables $x_0, \dots, x_s$ and an integer positive definite full Hessian Gram of f_ℓ with the following properties.*

- (a) f_ℓ has $O(\ell^2)$ monomials, and its coefficients and the entries of its Hessian Gram have $O(\log \ell)$ bits.
- (b) $\nabla^2 f_\ell \succeq I$ on $\mathbb{R}^n$, $f_\ell(p) = 0$, and $f_\ell(x) \geq \frac{1}{2} \|x - p\|^2$. Thus p is the unique minimizer and $\min f_\ell = 0$.
- (c) $\varphi_\ell(f_\ell) = -4$.
- (d) For every subfield $E \subseteq \mathbb{R}$ the following are equivalent: $2^{1/\ell} \in E$; f_ℓ is a sum of squares over E ; f_ℓ has a positive semidefinite Gram matrix over E .

Proof of (d) from (a)–(c). If $a \in E$ then $p \in E^n$, and Lemma 8.1(e) applies, because f_ℓ has a rational positive definite Hessian Gram and a stationary zero at p ; a sum of squares over E gives a positive semidefinite Gram over E . If $a \notin E$, Lemma 9.3 shows $\varphi_\ell(g^2) = v_0^2 \geq 0$ for every $g \in \mathcal{I}_2^E(p)$, and Lemma 9.2(c), together with $\varphi_\ell(f_\ell) = -4$, excludes both certificate types. $\square$

For the Gram case the argument can be made entirely explicit: if $f_\ell = z^\top Q z$ with $Q \succeq 0$ over E , then the constant row of Q is a vanishing quadratic, so $Q_{1,x_s} + Q_{1,x_0^2} = 0$, and

$$\varphi_\ell(f_\ell) = 2Q_{1,x_s} + Q_{x_0,x_0} + 2Q_{1,x_0^2} = Q_{x_0,x_0} \geq 0.$$

No factorization of Q over E is used.

Theorem 9.4 says more than that rational descent fails. The least real field for either certificate is exactly $\mathbb{Q}(2^{1/\ell})$; every real number field carrying a certificate contains it and so has degree divisible by ℓ . In particular no finite tower of quadratic extensions suffices. The field degree grows only linearly with the number of variables here, and the field has a short radical description, so this is not an output-size lower bound; Section 9.3 provides one.

Corollary 9.5 (Rational lower bounds do not attain the minimum). *Let (f, A) be a certified quartic with minimum zero that is not a sum of squares of rational polynomials, for instance f_ℓ . Then $f + \eta$ has a rational positive definite Gram matrix for every rational $\eta > 0$, and*

$$\sup\{\gamma \in \mathbb{Q} : f - \gamma \text{ is a sum of squares in } \mathbb{Q}[X]\} = 0$$

is not attained.

Proof. Let $\mu = \mu_A$ and let p be the minimizer. The quantity $f(a) + \eta - \|\nabla f(a)\|^2 / (2\mu)$ tends to η as $a \rightarrow p$, so it is positive for some rational a . Lemma 8.1(c), applied to $f + \eta$ at this center, gives a rational positive definite Gram matrix, and Lemma 2.20 gives rational squares. Every negative rational γ is therefore feasible, $\gamma = 0$ is excluded by hypothesis, and positive γ are excluded by evaluating at p . $\square$

The real semidefinite program is not at fault: it attains the value zero, with an algebraic optimal Gram (Theorem 8.6). The failure is arithmetic. A procedure that rounds near-optimal Gram matrices to rational data obtains certified lower bounds $-\eta$ for every $\eta > 0$, and never the value zero. Corollary 9.5 makes no claim about the bit length of these certificates as $\eta \rightarrow 0$.

The construction behind Theorem 9.4 uses uniform constants and therefore large coefficients. A hand-sized example with the same mechanism is the following.

Example 9.6 (An explicit ternary quartic). In the variables (x, y, z) put

$$r_0 = 2 - 2xy, \quad r_1 = 2x^2 - 2yz, \quad r_2 = 2y^2 - 4z, \quad r_3 = 2z^2 - x, \quad r_4 = 2xz - y,$$

$A_0 = 4r_0 + 5r_1 + 3r_2 + 9r_3$, and

$$F_\star = A_0^2 + r_0^2 + r_1^2 + r_2^2 + r_3^2 - r_4^2. \quad (28)$$

This integer quartic has 31 monomials and coefficients of absolute value at most 448. All five r_i vanish at $p_\star = (a^4/2, a, a^2/2)$, where $a = 2^{1/5}$. Appendix G.3 prints a rational full Hessian Gram $M_\star$ of $F_\star$, specified exactly through rational quadratic data at the center $(3/4, 1, 1/2)$, and the twelve positive leading principal minors of the integer matrix $2(M_\star - I)$. They prove $\nabla^2 F_\star \succeq I$ by Sylvester's criterion. Hence $p_\star$ is the unique minimizer and $\min F_\star = 0$. With $\varphi_\star(P) = 2[z]P + 4[y^2]P$ one has

$$\varphi_\star(r_i r_j) = 4 \quad \text{if } i = j = 4, \quad \varphi_\star(r_i r_j) = 0 \quad \text{otherwise,}$$

so $\varphi_\star(F_\star) = -4$. For every real field E with $a \notin E$ the r_i form a basis of $\mathcal{I}_2^E(p_\star)$, so $\varphi_\star(g^2) \geq 0$ on that space, and Lemma 9.2 gives the same characterization as Theorem 9.4(d): a sum of squares or a positive semidefinite Gram over E exists exactly when $2^{1/5} \in E$. The fifteen products $r_i r_j$ are linearly independent, so $F_\star$ has the unique Gram matrix $(I_4 + gg^\top) \oplus (-1)$, $g = (4, 5, 3, 9)^\top$, on the basis $(r_0, \dots, r_4)$. Only the convexity certificate is a finite computation. Given it, the field statements are proved in Appendix G.3; sufficiency uses the rational Hessian Gram that the certificate provides.

The two examples differ from earlier failures of rational descent in the combination of hypotheses, not in the failure itself. Scheiderer proved that rational forms that are sums of squares over $\mathbb{R}$ need not be sums of squares over $\mathbb{Q}$, and that every prescribed real number field can be excluded as a field of coefficients by a rational nonnegative ternary quartic form [107, Theorem 2.1 and Corollary 2.11]. Descent does hold from totally real number fields [56, Theorem 1.4], [107, Theorem 1.2], and for rational polynomials with a positive definite real Gram matrix [56, Theorem 1.2]; exact rounding methods work in that interior case [27, 100]. A quaternary quartic form that is a sum of squares over $\mathbb{Q}(2^{1/3})$ but not over $\mathbb{Q}$ is attributed to Capco, Laplagne, and Scheiderer and proved in [75, Section 3.1]; that form is not convex. What Theorem 9.4 adds is the simultaneous presence of a strict rational full Hessian certificate, global strong convexity, an attained rational minimum, small integer data, and an exact description of all real fields carrying either certificate type, with an unbounded prime degree. The abstract mechanisms, kernel restriction of Gram matrices at real zeros and odd-degree field arithmetic, are classical.

Corollary 9.7 (Three variables are needed). *Every certified quartic with minimum zero in one or two variables is a sum of squares of rational polynomials. Consequently, among certified quartics with minimum zero, the ternary quartics of Theorem 9.4 with $\ell = 5$ and of Example 9.6 have the smallest possible number of variables for a failure of rational descent.*

Proof. A full positive definite Hessian Gram has a positive definite tensor block C , and comparing the terms of degree two in X gives $12f_4(X) = (X \otimes X)^\top C(X \otimes X) > 0$ for $X \neq 0$, where f_4 is the quartic part; so the quartic form is positive definite and the degree is four. The minimizer p is the only real zero, by strong convexity. In one variable the minimal polynomial of p divides both f and f' , hence its square divides f , so p has degree at most two; a real quadratic irrational zero would bring its real conjugate as a second zero. Thus p is rational, and Lemma 8.1(e) gives a sum of squares over $\mathbb{Q}$. In two variables f is nonnegative, has exactly one real zero, and has a positive definite quartic form, so Lemma 7.13 applies. $\square$

Lemma 7.13 rests on Scheiderer's classification of rational nonnegative ternary quartic forms that are not sums of squares of rational forms [107, Theorem 4.1]; such a form has at least two real projective zeros, whereas the homogenization of a certified bivariate quartic with minimum zero has exactly one. The corollary concerns the strict Hessian hypothesis; we make no minimality claim for weaker convexity hypotheses.

The obstruction in Theorem 9.4 and Example 9.6 is one of positivity: both quartics lie in the span $\mathcal{W}^{\mathbb{Q}}(p)$ of products of rational vanishing quadratics, but their unique or restricted Gram matrices are indefinite. A cruder obstruction is also possible: a certified quartic can lie outside that span altogether. This is the content of the next proposition, whose proof is in Appendix G.4.

Proposition 9.8 (A linear obstruction in four variables). *Let $\alpha = 2^{1/3}$, $\beta = 5^{1/3}$, and $p = (\alpha, \alpha^2, \beta, \beta^2)$ in the variables (x, y, z, w) . There is a rational quartic f with a rational positive definite full Hessian Gram, $\nabla^2 f \succeq I$, and minimum zero at p , such that $f \notin \mathcal{W}^{\mathbb{Q}}(p)$. In particular f is not a sum of squares of rational polynomials. It is a sum of squares over $\mathbb{Q}(\alpha, \beta)$, a field of degree nine.*

Here $\dim_{\mathbb{Q}} \mathcal{I}_2^{\mathbb{Q}}(p) = 6$, and no product of two rational vanishing quadratics contains the monomial yz^3 , whereas the stationary quartic $y(z^3 + w^3/5 - 3zw + 5)$ does. The proof adds a large multiple of a strictly convex rational sum of squares to this quartic. The general principle is that, whenever a sum of squares of rational quadratics vanishing at p has a positive definite full Hessian Gram, any rational stationary quartic outside $\mathcal{W}^{\mathbb{Q}}(p)$ can be made strictly convex in this way, so the quotient $\mathcal{J}_4^{\mathbb{Q}}(p)/\mathcal{W}^{\mathbb{Q}}(p)$ is itself an obstruction to rational descent. Theorem 9.14 below shows that, when a convex rational sum-of-squares baseline exists, these two obstructions, a large quadratic vanishing space and a stationary quartic outside the product span, are the only possible sources of a counterexample.

9.3 An exponentially large least field

The prime family needs dimension proportional to the field degree. A tower of fifth roots gives exponential degree in polynomial size. Fix $k \geq 1$, put $n = 3k$, and let

$$a_1 = 2^{1/5}, \quad a_i = a_{i-1}^{1/5} = 2^{1/5^i} \quad (2 \leq i \leq k), \quad p_k = (a_i, a_i^2, a_i^3)_{i=1}^k.$$

Use variables (x_i, y_i, z_i) for gate i , put $b_1 = 2$ and $b_i = x_{i-1}$ for $i \geq 2$, and define the five rational quadratics of gate i :

$$r_{i,1} = x_i^2 - y_i, \quad r_{i,2} = x_i y_i - z_i, \quad r_{i,3} = y_i^2 - x_i z_i, \quad r_{i,4} = y_i z_i - b_i, \quad r_{i,5} = z_i^2 - b_i x_i. \quad (29)$$

All of them vanish at p_k , since $a_i^5 = b_i(p_k)$. The three *chain residuals* $r_{i,1}, r_{i,2}, r_{i,4}$ of all gates have p_k as their only common real zero: they force $y_i = x_i^2$, $z_i = x_i^3$, and $x_i^5 = b_i$, and an odd root of a real number is unique. The relation $r_{k,3}$ of the last gate is the one that will be subtracted.

Theorem 9.9 (An exponentially large least field). *For every $k \geq 1$ one can construct, in time polynomial in k , a rational quartic $F_{k,0}$ that is an explicit sum of $3k+1$ squares of rational quadratics vanishing at p_k , a rational positive definite full Hessian Gram M_k of $F_{k,0}$, and a positive integer λ_k , all of total binary length polynomial in k , with the following properties for every rational $\lambda \geq \lambda_k$ and*

$$f_{k,\lambda} = \lambda F_{k,0} - r_{k,3}^2.$$

- (a) $f_{k,\lambda}$ has a rational full Hessian Gram at least I ; hence $\nabla^2 f_{k,\lambda} \succeq I$, the unique minimizer is p_k , and $\min f_{k,\lambda} = 0$.

(b) For every subfield $E \subseteq \mathbb{R}$ the following are equivalent: $2^{1/5^k} \in E$; $f_{k,\lambda}$ is a sum of squares over E ; $f_{k,\lambda}$ has a positive semidefinite Gram matrix over E .

In particular the least real field of both certificate types is $\mathbb{Q}(2^{1/5^k})$, of degree $5^k = 5^{n/3}$ over $\mathbb{Q}$.

The baseline has the form $F_{k,0} = \sigma_0 G_k^2 + \sigma_1 \sum_{i=1}^k \sum_{j \in \{1,2,4\}} r_{i,j}^2$ with positive rationals σ_0, σ_1 and a rational quadratic G_k vanishing at p_k , whose homogeneous quadratic part is positive definite and whose coefficient of z_k^2 is positive. Appendix G.5 constructs it from an exact exposing identity for one quintic gate, a lower bound for the Jacobian of the chain residuals, and the generic assembly of Lemma 7.6. We sketch the field argument, which is the heart of the theorem.

Suppose $a_k \notin E$ and put $L = E(a_k^5)$. By Lemma 9.2(d), $T^5 - a_k^5$ is irreducible over L , and all earlier coordinates $a_i = a_k^{5^{k-i}}$, $i < k$, lie in L . Substitute the earlier coordinates of p_k into a putative certificate over E . The result is a certificate over L for the three-variable slice $f^\flat$ of $f_{k,\lambda}$ through the last gate. Its five relations $r^\flat = (r_{k,1}^\flat, \dots, r_{k,5}^\flat)$ form a basis of the L -quadratics vanishing at (a_k, a_k^2, a_k^3) , and their fifteen products are linearly independent, because their leading forms $x^2, xy, y^2 - xz, yz, z^2$ multiply to a basis of all ternary quartic forms. All earlier residuals vanish on the slice, so

$$f^\flat = \lambda \left(\sigma_0 g^2 + \sigma_1 \sum_{j \in \{1,2,4\}} (r_{k,j}^\flat)^2 \right) - (r_{k,3}^\flat)^2,$$

where $g = \sum_j c_j r_{k,j}^\flat$ is the restriction of G_k . Product independence makes this Gram matrix on $r^\flat$ unique. The vector $(0, 0, c_5, 0, -c_3)$ is orthogonal to the coefficient vectors of g and of $r_{k,1}^\flat, r_{k,2}^\flat, r_{k,4}^\flat$, and the Gram value on it is $-c_3^2$. Since only $r_{k,5}$ contains z_k^2 , c_5 is the positive z_k^2 coefficient of G_k . A sum of squares or a positive semidefinite Gram over L would give a positive semidefinite Gram on $r^\flat$, by Lemma 9.2(b). This contradiction proves necessity; Lemma 8.1(e) proves sufficiency.

A joint field containing all coefficients does not by itself force any one coefficient to be complicated; a certificate could in principle spread the field over many coefficients of small degree. The tower excludes this.

Theorem 9.10 (One coefficient of exponential degree). *Let ℓ be an odd prime, $k \geq 1$, and $D = \ell^k$. If finitely many algebraic numbers $\beta_1, \dots, \beta_t$ generate a field containing $2^{1/D}$, then $[\mathbb{Q}(\beta_i) : \mathbb{Q}] \geq D$ for some i . Consequently, in every sum of squares of $f_{k,\lambda}$ with algebraic coefficients, and in every positive semidefinite Gram matrix of $f_{k,\lambda}$ with algebraic entries, some single coefficient or entry has degree at least 5^k over $\mathbb{Q}$. Writing its minimal polynomial densely requires at least $5^k + 1$ rational coefficients.*

The proof in Appendix G.6 combines the real-field lemma with a Galois argument. The splitting field of $T^D - 2$ has an automorphism of order D fixing the cyclotomic field, while every element of a product of symmetric groups of degrees less than D has order with ℓ -adic valuation below k . The input $f_{k,\lambda}$ and its Hessian certificate have length polynomial in k and the binary length of λ . For $\lambda = \lambda_k$, or any λ of binary length polynomial in k , the forced dense output is therefore superpolynomial in the input length and exponential in the number of variables. It is not claimed to be exponential in the full input bit length.

The obstruction concerns dense algebraic output of polynomial certificates. It does not extend to compact representations.

Theorem 9.11 (Compact exact certificates for the tower). *Let $k \geq 1$, let $\lambda \geq \lambda_k$ be rational, and let L_λ denote k plus the binary length of λ ; for $\lambda = \lambda_k$ this is polynomial in k . Use the signed decomposition $f_{k,\lambda} = \sum_{j=1}^{3k+2} \sigma_j g_j^2$ with rational weights σ_j and the rational quadratics g_j from Theorem 9.9: the exposing quadratic G_k , the $3k$ chain residuals, and $r_{k,3}$; the coefficient of $r_{k,3}^2$ is -1 .*

- (a) Shared root circuit. *In time polynomial in L_λ one can construct an arithmetic circuit of size polynomial in L_λ whose only nonrational operations are k real fifth-root gates computing $a_1, \dots, a_k$, and polynomially many quadratics whose coefficients are outputs of the circuit, such that $f_{k,\lambda}$ is the sum of their squares.*
- (b) Auxiliary root variables. *In time polynomial in L_λ one can construct a rational sum of squares $s(X, Y)$ of degree four in $6k$ variables and rational polynomials $H_j(X, Y)$ of degree at most two, of total size polynomial in L_λ , with*

$$f_{k,\lambda}(X) = s(X, Y) + \sum_{j=1}^{3k+2} H_j(X, Y) g_j(Y).$$

The equations $g_j(Y) = 0$ have the real solution $Y = p_k$, so the identity proves $f_{k,\lambda} \geq 0$. Eliminating the two redundant equations $G_k(Y) = 0$ and $r_{k,3}(Y) = 0$ gives an identity of degree at most five modulo the $3k$ chain residuals alone.

Part (a) is Taylor integration with symbolic coordinates (Lemma 8.1(e)); part (b) is the first-order convexity identity of SOS-convex polynomials [3, Theorem 3.1] written with the minimizer as a variable. Neither is a new proof system. The point of the theorem is the comparison on one family: the input and its convexity certificate are short; dense algebraic polynomial certificates of the optimal value are exponentially long; shared root circuits and rational certificates modulo the root equations are short. The existence of a real solution in (b) is essential, since an identity modulo an inconsistent system proves nothing. Rational-function certificates give another short alternative (Corollary 10.4). The theorem gives no lower bound for sparse polynomials, radicals, root towers, or arithmetic circuits; $T^{5^k} - 2$ itself is sparse.

9.4 Rational relations and stationary quartics at the tower point

The least-field proof uses one direction of one slice. We now describe all rational quadratic relations and all rational stationary quartics at the tower point; the recognition corollary then applies the required positive semidefinite test.

Theorem 9.12 (Rational relations and stationary quartics). *Let $k \geq 1$ and let r be the vector of the $5k$ quadratics (29).*

- (a) *The entries of r form a basis of $\mathcal{I}_2^{\mathbb{Q}}(p_k)$, and no nonzero rational affine polynomial vanishes at p_k . Thus $\dim_{\mathbb{Q}} \mathcal{I}_2^{\mathbb{Q}}(p_k) = 5k$.*
- (b) *The $\binom{5k+1}{2}$ products $r_{i,j} r_{i',j'}$ are linearly independent over $\mathbb{R}$.*
- (c) *$\mathcal{J}_3^{\mathbb{Q}}(p_k) = \{0\}$ and $\mathcal{J}_4^{\mathbb{Q}}(p_k) = \mathcal{W}^{\mathbb{Q}}(p_k) = \{r^{\top} S r : S \in \text{Sym}_{5k}(\mathbb{Q})\}$, with S unique.*

The proof is in Appendix G.8. Part (a) eliminates one pivot monomial per relation and identifies the remaining monomials with distinct base-five digit patterns of exponents of a_k below 5^k . Part (b) inducts over the gates, starting from the fifteen leading products of one gate. Part (c) uses a first-order interpolation lemma for one quintic gate, whose matrix splits into five blocks according to the weight of a monomial modulo five, and an induction over gates that removes the last gate's contribution. All arguments are uniform in k .

Corollary 9.13 (Exact rational recognition at the tower point). *Given k and a rational polynomial f of degree at most four, one can decide in time polynomial in k and the input length whether*

$f(p_k) = 0$ and, if so, whether f is a sum of squares of rational polynomials. The rational positive semidefinite Gram matrix of such an f on z is unique, and in the affirmative case a rational sum of squares of polynomial size is produced. If $f(p_k) = 0$ and $\nabla f(p_k) = 0$, for instance when p_k is an unconstrained minimizer with value zero, the linear step that writes $f = r^\top S r$ always succeeds, and f is then a sum of squares of rational polynomials exactly when $S \succeq 0$; the tower quartics $f_{k,\lambda}$ are stationary at p_k and have $S \not\succeq 0$.

Proof. Evaluate $f(p_k)$ by writing each monomial value as $2^t a_k^e$ with $0 \leq e < 5^k$; since $1, a_k, \dots, a_k^{5^k-1}$ are linearly independent over $\mathbb{Q}$, $f(p_k) = 0$ exactly when the coefficients collected in each residue class sum to zero. Exponents have $O(k)$ bits, and no dense field representation is formed. If f is a rational sum of squares, its factors lie in $\mathcal{I}_2^\mathbb{Q}(p_k)$ by Lemma 9.2(a,b), so $f = r^\top S r$ with $S \succeq 0$ rational. Conversely such a representation gives rational squares (Lemma 2.20). By Theorem 9.12(b) the matrix S is unique and is found by solving a rational linear system with $O(k^2)$ unknowns and $O(k^4)$ equations whose matrix has bounded integer entries; one rational symmetric elimination tests $S \succeq 0$. For a rational positive semidefinite Q with $f = z^\top Q z$, Lemma 9.2(b) gives $Q = B S B^\top$ with $r = B^\top z$, so Q is unique as well. The last sentence follows from Theorem 9.12(c), which places every rational quartic with zero value and gradient at p_k in the span of the products of relations, and from the slice argument for Theorem 9.9(b), which shows that the unique matrix of $f_{k,\lambda}$ is indefinite. $\square$

The recognition is exact and avoids both a semidefinite feasibility problem with a free matrix and any search over algebraic coefficient fields. Its scope is narrow: it requires the zero to be the supplied tower point, it decides rational, not real, sums of squares, and it does not locate an unknown zero. The theorem also explains the tower construction. Every rational stationary quartic at p_k is a quadratic form in the relations, so a strictly convex real sum of squares can fail rational descent only through an indefinite unique matrix S , which is exactly what Theorem 9.9 produces. The rational span of the relations is much smaller than the real space $\mathcal{I}_2^\mathbb{R}(p_k)$, which has dimension $\binom{3k+2}{2} - 1$; real square factors may use all of it.

9.5 A descent criterion

The constructions above subtract the square of a rational vanishing quadratic that the convex baseline does not use. This requires more rational vanishing quadratics than the dimension forces. The following criterion shows that this construction is impossible when the quadratic vanishing space has its minimum possible dimension. It uses the real sum-of-squares length bound of Theorem 7.9: a globally convex real quartic of degree exactly four with a zero at which its Hessian is positive definite is not a sum of fewer than $n + 1$ squares of real polynomials.

Theorem 9.14 (Descent from a small vanishing space). *Let $p \in \mathbb{R}^n$ satisfy*

$$\dim_{\mathbb{Q}} \mathcal{I}_2^\mathbb{Q}(p) = n + 1 \quad \text{and} \quad \mathcal{J}_4^\mathbb{Q}(p) = \mathcal{W}^\mathbb{Q}(p),$$

and suppose there is a globally convex rational polynomial G of degree exactly four that is a sum of squares of rational polynomials and satisfies $G(p) = 0$ and $\nabla^2 G(p) \succ 0$. Let g be a rational basis of $\mathcal{I}_2^\mathbb{Q}(p)$. Then every globally convex rational polynomial F of degree exactly four with

$$F(p) = 0, \quad \nabla F(p) = 0, \quad \nabla^2 F(p) \succ 0$$

has a unique symmetric matrix S with $F = g^\top S g$, and S is rational and positive definite. In particular F is a sum of squares of rational polynomials.

Proof. By Lemma 9.2(a,b), the rational square factors of G lie in $\mathcal{I}_2^{\mathbb{Q}}(p)$, so $G = g^{\top} S_0 g$ with $S_0 \succeq 0$ rational. If S_0 were singular, a real factorization of S_0 would write G as a sum of at most n real squares, contradicting Theorem 7.9. Thus $S_0 \succ 0$. Since $F \in \mathcal{J}_4^{\mathbb{Q}}(p) = \mathcal{W}^{\mathbb{Q}}(p)$, there is a rational symmetric S with $F = g^{\top} S g$. Suppose S is not positive definite. Along $S_t = (1-t)S_0 + tS$ the least eigenvalue is continuous, positive at $t = 0$ and nonpositive at $t = 1$, so there is a first $t_* \in (0, 1]$ at which S_{t_*} is positive semidefinite and singular. Then $F_{t_*} = (1-t_*)G + t_*F = g^{\top} S_{t_*} g$ is a sum of at most n real squares. On the other hand F_{t_*} is globally convex, vanishes at p , and has positive definite Hessian there. It also has degree exactly four. The quartic parts G_4 and F_4 are nonnegative, being limits of $t^{-4}G(tx)$ and $t^{-4}F(tx)$ and $F \geq F(p) = 0$ by convexity and stationarity; if $t_* < 1$ and $(1-t_*)G_4 + t_*F_4 = 0$ then $G_4 = 0$, which is false, and if $t_* = 1$ the degree of F is assumed. This contradicts Theorem 7.9, so $S \succ 0$, and Lemma 2.20 gives rational squares. $\square$

For uniqueness, let $K \neq 0$ be a real symmetric matrix with $g^{\top} K g = 0$; replacing K by $-K$ if necessary, K has a negative eigenvalue. The ray $S_0 + tK$, $t \geq 0$, reaches a first singular positive semidefinite matrix at a finite $t > 0$, which represents G by at most n real squares. This is again impossible. $\square$

The intermediate parameter t_* need not be rational; this is harmless, because Theorem 7.9 allows real square factors. The proof does not assume that the individual factors are convex.

Corollary 9.15 (Product independence). *If $\dim_{\mathbb{Q}} \mathcal{I}_2^{\mathbb{Q}}(p) = n+1$ and a baseline G as in Theorem 9.14 exists, then the products of a basis of $\mathcal{I}_2^{\mathbb{Q}}(p)$ are linearly independent over $\mathbb{R}$ and $\dim_{\mathbb{Q}} \mathcal{W}^{\mathbb{Q}}(p) = (n+1)(n+2)/2$. Under all hypotheses of Theorem 9.14, the quartic F has a unique rational positive semidefinite Gram matrix on z , of rank $n+1$, and is a sum of exactly $n+1$ real squares and of no fewer.*

Proof. The uniqueness part of the proof above used only G and the dimension. For the Gram statement, Lemma 9.2(b) gives $Q = BS'B^{\top}$ for any rational positive semidefinite Gram Q of F , where $g = B^{\top}z$, and uniqueness gives $S' = S$. A real factorization of $S \succ 0$ gives $n+1$ squares; Theorem 7.9 gives the lower bound. $\square$

The two hypotheses have different roles. The dimension bound turns a singular restricted Gram into too few real squares; the equality $\mathcal{J}_4^{\mathbb{Q}}(p) = \mathcal{W}^{\mathbb{Q}}(p)$ supplies a restricted Gram for every rational stationary quartic. Example 9.6 has $\dim \mathcal{I}_2^{\mathbb{Q}} = 5 > n+1 = 4$, Proposition 9.8 violates the second hypothesis, and the tower has $\dim \mathcal{I}_2^{\mathbb{Q}}(p_k) = 5k > 3k+1$. A counterexample to rational descent at a point carrying a rational convex sum-of-squares baseline must therefore use more than $n+1$ rational vanishing quadratics or a stationary quartic outside their product span. The criterion is an obstruction to a construction method; it gives no lower bound for other certificate systems.

The dimension hypothesis holds for the cyclic quartics of Theorem 7.8, whose zeros have field degree exponential in n .

Proposition 9.16 (Vanishing quadratics at the cyclic points). *Let $n \geq 4$, and let $m = n+1$, $\eta = (-1)^m$, $d_n = (2^m - \eta)/3$, $e_i = ((-2)^i - 1)/3$, and $p = (p_1, \dots, p_n)$ with $p_i = 2^{e_i/d_n}$, as in Theorem 7.8. With $x_0 = 1$ and indices modulo m , the m cyclic binomials $x_i^2 - 2^{\theta_i} x_{i+1} x_{i+2}$, where $\theta_i = 0$ except $\theta_{m-2} = \eta$ and $\theta_{m-1} = -\eta$, form a basis of $\mathcal{I}_2^{\mathbb{Q}}(p)$. Thus $\dim_{\mathbb{Q}} \mathcal{I}_2^{\mathbb{Q}}(p) = n+1$, and by Corollary 9.15 applied to the rational sum-of-squares quartic of Theorem 7.8, the products of these relations are linearly independent.*

The proof is in Appendix G.9. For $n = 3$ the space is larger, five-dimensional, and the hypothesis fails. Whether $\mathcal{J}_4^{\mathbb{Q}}(p) = \mathcal{W}^{\mathbb{Q}}(p)$ holds at all cyclic points is open. If the equality holds for a given $n \geq 4$, then Theorem 9.14 shows that no globally convex rational quartic with minimum zero at

that cyclic point and a positive definite Hessian there can fail rational descent, although the point has algebraic degree d_n .

10 Rational functions, multipliers, and block-separated certificates

The obstructions of Section 9 concern polynomial squares. Two changes of the certificate format are common in exact certification. One allows denominators: a nonnegative rational polynomial is a sum of squares of rational functions by Artin's theorem, and exact certificates with rational-function or multiplier terms are an established strategy [62]. The other imposes structure: sparse sum-of-squares methods split a certificate into blocks of variables [67]. This section measures both changes on the families of Section 9.

- (a) A quadratic in the span of the square factors whose homogeneous quadratic part is positive definite, together with an affine ideal membership of the subtracted quadratics, gives an explicit rational certificate with the multiplier $\omega = 1 + \|X\|^2$ (Theorem 10.2); higher-degree membership gives the multiplier ω^d (Theorem 10.3). After a polynomial-size increase of the scale, the tower quartics, whose polynomial certificates need a coefficient of degree 5^k , have short rational-function certificates with an everywhere-positive quadratic common denominator, and two is the minimum denominator degree (Corollary 10.4).
- (b) A prescribed multiplier behaves differently. In a scaling family of ternary quartics with a fixed Hessian at the minimizer, no fixed power of ω suffices (Theorem 10.7); every power below a bound of order $\log L / \log \log L$ in the input length L fails, although an adapted quadratic denominator gives a certificate of size $O(L)$ (Theorem 10.8).
- (c) Block-separated rational certificates can fail to exist, can require the field $\mathbb{Q}(2^{1/5^k})$, or can exist but require $\Omega(k2^k)$ bits, while an unrestricted rational certificate is short; this persists when both blocks are strongly SOS-convex (Theorems 10.10 and 10.11).

These are statements about specified certificate formats. None is a lower bound for unrestricted certificates or for the complexity of deciding nonnegativity.

10.1 Certificate formats

Let $f \in \mathbb{Q}[X]$. A *rational-function sum of squares with common denominator* $d \in \mathbb{Q}[X] \setminus \{0\}$ is an identity $f = \sum_j (u_j/d)^2$ of rational functions with $u_j \in \mathbb{Q}[X]$; equivalently $d^2 f = \sum_j u_j^2$. It is *regular* if d has no real zero. A *multiplier certificate* is an identity $uf = \sum_j g_j^2$ with $g_j \in \mathbb{Q}[X]$ and a fixed positive polynomial u . Put

$$\omega(X) = 1 + \|X\|^2,$$

and define the *radial order* $\nu(f) \in \mathbb{N} \cup \{\infty\}$ as the least $N \geq 0$ with $\omega^N f$ a sum of squares in $\mathbb{Q}[X]$. Since ω is a rational sum of squares, $\omega^N f$ is a rational sum of squares for every $N \geq \nu(f)$. A multiplier is not a common denominator: from $\omega f = \sum_j g_j^2$,

$$f = \sum_j \left(\frac{g_j}{\omega}\right)^2 + \sum_{i=1}^n \sum_j \left(\frac{X_i g_j}{\omega}\right)^2, \quad (30)$$

a regular certificate with denominator ω of degree two and numerators of degree $\max_j \deg g_j + 1$; the cleared identity is $\omega^2 f = \omega \sum_j g_j^2$.

For disjoint blocks of variables x and y and a polynomial $f(x) + g(y)$, a *block-separated sum of squares over E* is a sum of squares over E in which every square factor lies in $E[x]$ or in $E[y]$. A *separated Gram certificate over E* is a pair of positive semidefinite matrices Q_x, Q_y with entries in E , each on the monomial vector of its own block, including its own constant monomial, with $f(x) + g(y) = z(x)^\top Q_x z(x) + z(y)^\top Q_y z(y)$.

10.2 Quadratic denominators from ideal membership

Denominators of degree one never help.

Lemma 10.1 (Affine denominators cancel). *If $f \in \mathbb{Q}[X]$ has a rational-function sum of squares with a common denominator of degree at most one, then f is a sum of squares in $\mathbb{Q}[X]$.*

Proof. A constant denominator can be absorbed. Otherwise d is a nonconstant rational affine polynomial and $d^2 f = \sum_j u_j^2$. On the real hyperplane $d = 0$ every u_j vanishes. After a rational affine change of coordinates making d a multiple of X_1 , the polynomial $u_j(0, X_2, \dots, X_n)$ vanishes on $\mathbb{R}^{n-1}$, so it is zero and X_1 divides u_j in $\mathbb{Q}[X]$. Thus d divides every u_j , and cancelling d^2 gives a polynomial sum of squares. $\square$

Consequently every rational-function certificate of a polynomial that is not a rational sum of squares, such as f_ℓ or $f_{k,\lambda}$, has a common denominator of degree at least two. The next theorem shows how degree two is achieved, constructively and with a short certificate, whenever two pieces of algebraic data are available.

Theorem 10.2 (Quadratic multipliers from affine membership). *Let $g_1, \dots, g_m \in \mathbb{Q}[X_1, \dots, X_n]$ have degree at most two, and put $F_0 = \sum_i g_i^2$. Suppose the input also supplies*

- (i) *rational κ_i such that $G = \sum_i \kappa_i g_i = X^\top H X + \ell_G^\top X + c_G$ has $H \succ 0$;*
- (ii) *rational quadratics $R_1, \dots, R_s$ with affine rational multipliers, $R_a = \sum_i (\alpha_{ai} + \beta_{ai}^\top X) g_i$;*
- (iii) *a rational symmetric $s \times s$ matrix J , of any signature.*

One can compute, in time polynomial in the input length, a positive integer λ_ with the following property. For every rational $\lambda \geq \lambda_*$, the polynomial $\omega(\lambda F_0 - R^\top J R)$ has a rational positive definite Gram matrix on an explicit vector of polynomials of degree at most three, and hence is a sum of squares of rational polynomials of degree at most three, with a certificate of size polynomial in the input and the length of λ . If a rational positive definite full Hessian Gram of F_0 is supplied as well, λ_* can also be chosen so that $\lambda F_0 - R^\top J R$ has a rational full Hessian Gram at least I for all $\lambda \geq \lambda_*$. If the g_i have a common real zero p , then every R_a vanishes at p , and so do the value and the gradient of $\lambda F_0 - R^\top J R$.*

The proof in Appendix H.1 rests on an exact identity that represents zero by a matrix with a positive definite block. On the list $w = (X_j g_i)_{0 \leq j \leq n, 1 \leq i \leq m}$, with $X_0 = 1$, the identity matrix represents ωF_0 . The new polynomials $X_j R_a$ carry the block $I_s \otimes H$ in a Gram representation of 0, whose correction terms are fixed by the identity $GR_a^2 - GR_a(X^\top T_a X + r_a^\top X + \sigma_a) = 0$, where $R_a = X^\top T_a X + r_a^\top X + \sigma_a$. A small positive multiple of this zero representation added to the identity matrix gives a positive definite Gram matrix of ωF_0 on the enlarged list, which then dominates the Gram matrix of $\omega R^\top J R$ after scaling. No convexity of F_0 and no common zero are needed.

Affine membership is a genuine restriction. In $\mathbb{Q}[x, y, z]$ the quadratics $g_1 = x^2 + y^2 + z^2 - 1$, $g_2 = x^2$, $g_3 = xy - z$, $g_4 = yz - 1$ have no common complex zero, g_1 has positive definite quadratic part, and

$$1 = xg_1 + (-x + y^2 - 1)g_2 + (-xy + xz - y)g_3 + (-x^2 - x - 1)g_4,$$

yet 1 is not in the span of the g_i and $X_j g_i$ (Appendix H.2). The following extension handles membership of higher degree.

Theorem 10.3 (Multipliers from higher-degree membership). *Let g_i , F_0 , G , and J be as in Theorem 10.2, let $d \geq 1$, and let rational quadratics R_a be supplied with representations $R_a = \sum_i u_{ai} g_i$, $u_{ai} \in \mathbb{Q}[X]$, $\deg u_{ai} \leq d$. One can compute a positive integer λ_* such that for every rational $\lambda \geq \lambda_*$ the polynomial $\omega^d(\lambda F_0 - R^\top J R)$ is a sum of squares of rational polynomials of degree at most $d + 2$. Time and output size are polynomial in the input length, n , m , s , d , and $\binom{n+d}{d}$. The real zero set of $\lambda F_0 - R^\top J R$ is the common real zero set of the g_i . In particular this polynomial has a regular rational-function sum of squares with common denominator $\omega^{\lceil d/2 \rceil}$.*

The size bound counts the expanded basis of monomials of degree at most d ; it is not polynomial in a binary encoding of d . The targets must remain quadratic: a negative square of higher degree could make $\lambda F_0 - R^\top J R$ negative at infinity for every λ . The proof (Appendix H.3) attaches one zero identity to each monomial multiple $X^\alpha R_a$ and weights the identities of level $|\alpha| + 1$ by $\rho^{2(|\alpha|+1)}$ for a small rational ρ ; a single diagonal congruence then shows that every correction term carries at least one factor ρ . The example above shows that the class with $d = 2$ is strictly larger than the affine class; it does not show that the multiplier ω^2 is necessary there.

Corollary 10.4 (Quadratic denominators for large least fields).

- (a) *For every $k \geq 1$ one can compute, in time polynomial in k , an integer $\lambda'_k \geq \lambda_k$ such that for every rational $\lambda \geq \lambda'_k$ the tower quartic $f_{k,\lambda}$ of Theorem 9.9 satisfies $\omega f_{k,\lambda} \in \Sigma\mathbb{Q}[X]^2$, with square factors of degree at most three and a certificate of size polynomial in k and the length of λ . Hence $f_{k,\lambda}$ has a regular rational-function sum of squares with common denominator ω , and the minimum degree of a common denominator of a rational-function sum of squares of $f_{k,\lambda}$ is two. All conclusions of Theorems 9.9 and 9.10 hold for $f_{k,\lambda}$.*
- (b) *For every prime $\ell \geq 5$ one can compute, in time polynomial in ℓ , an integer $c_\ell \geq 1$ such that for every integer $c \geq c_\ell$ the integer quartic $f_\ell^{(c)} = c(f_\ell + r^2) - r^2$ of Remark G.1 satisfies $\omega f_\ell^{(c)} \in \Sigma\mathbb{Q}[X]^2$. The monomial count in Theorem 9.4(a) and its conclusions (b)–(d) hold for $f_\ell^{(c)}$. Given c , construction takes time polynomial in ℓ and the binary length of c ; the coefficient and Hessian-Gram entry lengths are $O(\log \ell + \log c)$.*

Proof. (a) By construction (Appendix G.5), $F_{k,0} = (G_k/(t\nu))^2 + \sum_{i,j} (r_{i,j}/\nu)^2$, $j \in \{1, 2, 4\}$, for positive rationals t, ν . Apply Theorem 10.2 to these factors, so that its F_0 is exactly $F_{k,0}$, with κ selecting the factor $G_k/(t\nu)$, whose quadratic part is positive definite, and the single target $R_1 = r_{k,3}$ with $J = (1)$. The exact identity

$$r_{k,3} = x_k r_{k,2} - y_k r_{k,1} = \nu x_k (r_{k,2}/\nu) - \nu y_k (r_{k,1}/\nu)$$

is an affine membership in these factors. The theorem gives λ_* with $\omega(\lambda F_{k,0} - r_{k,3}^2) \in \Sigma\mathbb{Q}[X]^2$ for every rational $\lambda \geq \lambda_*$; take $\lambda'_k = \max\{\lambda_k, \lambda_*\}$. The field proof of Theorem 9.9 holds for every $\lambda \geq \lambda_k$, and $f_{k,\lambda}$ is not a rational sum of squares, so Lemma 10.1 excludes denominators of degree at most one; (30) gives degree two. (b) Apply the theorem to the factors A and $(N_1/N_0)C_i$ of $f_\ell + r^2 = A^2 + (N_1/N_0)^2 \sum_i C_i^2$ (Appendix G.2), with κ selecting A , whose quadratic part is positive definite, and $J = (1)$. The identity

$$r = x_0 C_1 - x_1 C_0 = \frac{N_0}{N_1} \left(x_0 \frac{N_1}{N_0} C_1 - x_1 \frac{N_1}{N_0} C_0 \right)$$

is an affine membership with rational multipliers, and $c(f_\ell + r^2) - r^2 = f_\ell^{(c)}$; take $c_\ell = \lambda_*$. $\square$

For $\lambda \geq \lambda'_k$, every polynomial sum of squares of $f_{k,\lambda}$ with algebraic coefficients needs one coefficient of degree at least 5^k , while a rational-function sum of squares with the fixed denominator ω has polynomial size. Without any size bound, some power of ω works for each such quartic, with an exponent that depends on the quartic.

Proposition 10.5 (Finite radial order). *Let $f \in \mathbb{Q}[X_1, \dots, X_n]$ be nonnegative on $\mathbb{R}^n$, of even degree $2e \geq 2$, with positive definite homogeneous part of degree $2e$, finitely many real zeros, and positive definite Hessian at each real zero. Then $\omega^N f$ is a sum of squares in $\mathbb{Q}[X]$ for some integer $N \geq 0$, that is, $\nu(f) < \infty$. In particular every certified quartic with minimum zero has finite radial order.*

Nonnegativity cannot be omitted: $x^2 - 1$ satisfies the other conditions, but $\omega^N(x^2 - 1)$ is negative at the origin. A certified quartic with minimum zero satisfies all the hypotheses: its quartic form is positive definite (proof of Corollary 9.7), its only real zero is the minimizer, and its Hessian is positive definite everywhere. The proof (Appendix H.6) works in the coordinate ring of the unit sphere, where sums of squares form an archimedean preordering, and applies the pure-state and order-unit theory of Burgdorf, Scheiderer, and Schweighofer [21] to the square of the rational vanishing ideal of the zeros of the homogenized polynomial. Rational coefficients come from working in a ring over $\mathbb{Q}$ throughout; a statement about real coefficients alone would not give them. The proposition gives no bound on N and no certificate size, and the lower bounds of Section 10.3 do not use it.

Example 10.6 (The explicit ternary quartic). For the quartic $F_\star$ of Example 9.6, which is not a rational sum of squares, Appendix H.4 prints a vector b of fifteen rational polynomials of degree at most three, which is a basis of the rational cubics vanishing at $p_\star$, and an integer symmetric matrix N with entries of absolute value at most 4240, such that

$$b^\top N b = 8\omega F_\star, \quad N - 8I \succ 0,$$

the latter certified by fifteen positive leading principal minors. Thus $\omega F_\star$ is a rational sum of squares of cubics, $F_\star$ has a regular rational-function sum of squares with denominator ω , and two is the minimum common-denominator degree. Here the scale of $F_\star$ is not increased; the certificate is a finite exact identity.

10.3 Prescribed radial multipliers

The denominator in Corollary 10.4 is the fixed polynomial ω , but the scale of the quartic was chosen to fit it. If instead we prescribe the radial hierarchy ω^N for every quartic in a scaling family, no fixed order N suffices. Let $f^\circ = f_5^{(c_5)}$ be the ternary integer quartic of Corollary 10.4(b) in the variables (x_0, x_1, x_2) . It has a rational positive definite full Hessian Gram, $\nabla^2 f^\circ \succeq I$, minimum zero at $p = (a^2, a^3, a^4)$ with $a = 2^{1/5}$, $\varphi_5(f^\circ) = -4$ for the functional $\varphi_5 = [x_2] + [x_0^2]$ of (27), and $\omega f^\circ \in \Sigma\mathbb{Q}[X]^2$. For integers $t \geq 1$ put

$$f_t(X) = t^{-2} f^\circ(tX).$$

Theorem 10.7 (Unbounded radial order).

- (a) *Each f_t has a rational positive definite full Hessian Gram, $\nabla^2 f_t \succeq I$, unique minimizer p/t with value zero, and $\nabla^2 f_t(p/t) = \nabla^2 f^\circ(p)$. Its binary input length satisfies $L(t) = \Theta(\log t)$.*

- (b) $(1 + t^2 \|X\|^2)f_t$ is a sum of squares of rational cubics with a certificate of size $O(L(t))$. Hence f_t has a regular rational-function sum of squares with a quadratic common denominator, and two is the minimum degree of a common denominator.
- (c) For every $N \geq 0$ one can compute an integer τ_N such that $\omega^N f_t$ is not a sum of squares in $\mathbb{Q}[X]$ for every integer $t \geq \tau_N$. More generally, for each fixed $u \in \mathbb{Q}[X]$ with $u(0) > 0$, the polynomial uf_t is not a sum of squares in $\mathbb{Q}[X]$ for all sufficiently large integers t .

Theorem 10.8 (Quantitative radial order). *There are constants $c > 0$ and t_0 , depending only on f° , such that*

$$\nu(f_t) \geq c \frac{\log \log t}{\log \log \log t} \quad \text{for every integer } t \geq t_0.$$

Equivalently $\nu(f_t) = \Omega(\log L / \log \log L)$ with $L = L(t)$, while f_t has an adapted rational certificate of size $O(L)$. Each $\nu(f_t)$ is finite by Proposition 10.5.

The proofs are in Appendix H.5. The mechanism is a sequence of rational linear functionals ψ_d on polynomials of degree at most $2d$ that are positive on squares of real combinations of rational polynomials of degree at most d vanishing at p , and negative on f° . The first is φ_5 plus a small multiple of the evaluation sum over the grid $\{0, 1, 2\}^3$; each further one adds a large integer multiple of a grid functional of top degree, chosen through an exact Schur complement. A rational sum of squares of $\omega^N f_t$ would, after the substitution $X = x/t$, have factors in the domain of ψ_{N+2} , and ψ_{N+2} is negative on it once t is large. Tracking the heights of the functionals gives $\log \log \tau_N = O(N \log N)$, which inverts to Theorem 10.8.

The scaling-and-closedness method is Reznick's [103, Theorem 1 and Corollary 2], who used it to show that no single multiplier, and no finite menu of multipliers, works for all nonnegative forms of a given degree and number of variables outside the cases of Hilbert's theorem. Here every f_t is a real sum of squares already at order zero and is strongly SOS-convex with a strict rational certificate; the obstruction is purely arithmetic and comes from the rational vanishing space of the irrational minimizer. The Hessian at the minimizer is the same for all t , so the growth of the order is not explained by local ill-conditioning. The theorem concerns one prescribed hierarchy. It does not say that real sum-of-squares optimization needs high order, and it does not say that finding an adapted denominator is hard; here the scale makes it explicit.

10.4 Block-separated certificates

Sparse certificates restrict the variables that may appear together in a square. For a separable polynomial $f(x) + g(y)$, the strongest such restriction asks for squares in x alone or y alone. Real sparse decompositions of this kind are studied in [67], and for the separable quartics below real separated certificates exist (Theorem 10.10(b) with $E = \mathbb{R}$). Over $\mathbb{Q}$ the answer differs. The constructions use a lift of a certified quartic to a quadratic graph.

Lemma 10.9 (A rational quadratic lift). *Let (f, A) be a certified quartic in n variables with minimum zero at p ; the point p is not part of the input. Introduce variables $y \in \mathbb{R}^n$ and $z = (z_{ij})_{1 \leq i \leq j \leq n}$. In time polynomial in the size of (f, A) one can construct a rational quartic $g(y, z)$ and a rational sum of squares of $f(x) + g(y, z)$ with quadratic factors, of polynomial size, such that $g \geq 0$ and $g(p, (p_i p_j)_{i \leq j}) = 0$.*

The construction (Appendix H.7) replaces a quadratic pair in each cubic monomial of $\nabla f(y)$ by a variable z_{ij} to obtain a quadratic map $a(y, z)$, and sets $g = \|a\|^2 / (2\mu) - f(y) + \lambda \sum_{i \leq j} (y_i y_j - z_{ij})^2$ with $\mu = \mu_A / 2$ and an explicit integer λ . The joint certificate combines the Taylor Gram of Lemma 8.1 in the difference $x - y$ with a completed square. Substituting $x = p$ shows $g \geq 0$.

Theorem 10.10 (Rational block separation fails). *Let $k \geq 1$, $f = f_{k,\lambda_k}$, and let g be its lift from Lemma 10.9, in the variables $w = (y, z)$. Let $B_k(w)$ be the rational quartic obtained by applying Theorem 7.15 to f and its Hessian Gram, with full Hessian Gram $A_{B_k} \succ 0$, and let C be a rational symmetric full Hessian Gram of g . Put*

$$T_k = \left\lceil \frac{1 + \sum_{i,j} |C_{ij}|}{\mu_{A_{B_k}}} \right\rceil, \quad \tilde{g} = g + T_k B_k.$$

- (a) *The quartic $f(x) + \tilde{g}(w)$ has a rational sum of squares with quadratic factors, of size polynomial in k . Both f and $\tilde{g}$ have rational full Hessian Grams at least I , $\tilde{g} \geq 0$, and the unique zero of $\tilde{g}$ is $w_* = (p_k, (p_i p_j)_{i \leq j})$.*
- (b) *For every subfield $E \subseteq \mathbb{R}$ the following are equivalent: $2^{1/5^k} \in E$; $f(x) + \tilde{g}(w)$ has a block-separated sum of squares over E ; it has a separated Gram certificate over E . In particular, for $k = 1$ no rational block-separated certificate exists, although the joint rational sum of squares has fixed size and real separated certificates exist.*
- (c) *Every separated certificate with algebraic coefficients has one coefficient of degree at least 5^k . The total number of variables is $3k + 3k + \frac{3k(3k+1)}{2} = \Theta(k^2)$, so this degree is $5^{\Theta(\sqrt{n})}$ in the total dimension n .*
- (d) *Every positive semidefinite full Hessian Gram matrix of $f(x) + \tilde{g}(w)$ is singular.*

Proof. (a) The rational full Hessian Gram $C + T_k A_{B_k}$ of $\tilde{g}$ is at least $(T_k \mu_{A_{B_k}} - \sum_{i,j} |C_{ij}|)I \succeq I$. Since $g \geq 0$, $B_k \geq 0$, $g(w_*) = 0$, and w_* is the only zero of B_k , the only zero of $\tilde{g}$ is w_* . Adding T_k times the rational sum of squares of B_k to the certificate of Lemma 10.9 gives the joint certificate. (b) Let a block-separated certificate over E be given, and group its terms: $f(x) + \tilde{g}(w) = S_x(x) + S_w(w)$, where S_x and S_w are sums of squares, or Gram forms with positive semidefinite matrices, over E . Then $S_x - f = \tilde{g} - S_w$ depends on neither block, so it is a constant $c \in E$. Evaluating $S_x = f + c$ at p_k gives $c \geq 0$, and evaluating $S_w = \tilde{g} - c$ at w_* gives $c \leq 0$. Thus f itself has a certificate of the same type over E , and Theorem 9.9(b) gives $2^{1/5^k} \in E$. Conversely, if $2^{1/5^k} \in E$, then f is a sum of squares over E , and specializing the joint rational certificate at $x = p_k \in E^{3k}$ shows that $\tilde{g}$ is too. For $k = 1$ and $E = \mathbb{R}$ this gives real separated certificates. (c) follows from (b) and Theorem 9.10. (d) Let x_j be a variable of the first block and v_i a direction coordinate of the second. The Hessian of $f(x) + \tilde{g}(w)$ is block diagonal, and the second block depends only on w , so the Hessian biform has no monomial $x_j^2 v_i^2$. In the full joint Hessian basis this monomial arises only as the square of the coordinate $x_j v_i$. Hence every Gram matrix has a zero diagonal entry there, and a positive semidefinite one has a zero row. $\square$

The joint polynomial is strongly convex with a short rational Hessian sum of squares, obtained by adding the two block certificates, but by (d) it has no positive definite full Hessian Gram; strong convexity on each block does not couple the blocks. The obstruction is about rational coefficients: each block is a certified quartic with minimum zero, hence a real sum of squares by Lemma 8.1(e), so real block-separated certificates exist. When the minimum is positive, rational separated certificates exist, but they can be long.

Theorem 10.11 (Exponentially long rational block certificates). *Let $f = f_{1,\lambda_1}$ and $\tilde{g}$ be the quartics of Theorem 10.10 with $k = 1$. For $k \geq 1$ let F_k° and A_k° denote the quartic F_k and its rational positive definite full Hessian Gram A_k from the construction preceding Theorem 8.3, with $M_k = 1000^{k+3}$,*

and let u_k be the rational affine function defined there. Put $h_k = \lceil 3/\mu_{A_k^\circ} \rceil F_k^\circ + u_k^2$, a quartic whose $2k$ variables we rename ξ . Consider

$$P_k(x, w, \xi) = f(x) + \tilde{g}(w) + h_k(\xi)$$

with the two blocks x and (w, ξ) .

- (a) P_k is a rational quartic in $12 + 2k$ variables with a rational sum of squares and a rational sum-of-squares Hessian certificate, both of size polynomial in k , and $\nabla^2 P_k \succeq I$. Its minimum $m_k = \min h_k$ satisfies $0 < m_k \leq M_k^{-2^{k+1}}$.
- (b) Rational block-separated sums of squares of P_k exist.
- (c) In every separated Gram certificate $(Q_x, Q_{w,\xi})$ over $\mathbb{Q}$, the constant entry of Q_x has reduced denominator at least $M_k^{2^{k+1}}/b_0$, where b_0 is the denominator of $f(0)$. In every rational block-separated sum of squares, the constant terms of the square factors in x have reduced denominators whose binary lengths sum to at least $2^k \log_2 M_k - \frac{1}{2} \log_2 b_0$.

In particular every rational separated certificate has $\Omega(k2^k)$ bits, while the unrestricted rational certificate has size polynomial in k .

The proof is in Appendix H.8. Its core is the same grouping argument: a separated certificate determines a constant c with $S_x = f + c$, and now $0 < c \leq m_k$, because $c = 0$ would certify f over $\mathbb{Q}$ and c cannot exceed the minimum of the other block. A positive rational below $M_k^{-2^{k+1}}$ has a long denominator, and the constant entry of the x -block Gram is $f(0) + c$. The bound concerns two separately stored local Gram matrices, each with its own constant monomial, or an unweighted separated sum of squares; a single joint matrix with one shared constant coordinate can hide the split. The lower bound is exponential in the dimension and superpolynomial in the polynomially bounded input length; it is not claimed to be $2^{\Omega(L)}$. Real separation questions for sparse certificates are classical [67]; the point here is that a restriction harmless for real feasibility can remove all rational certificates, or make them exponentially long, even under certified strong convexity.

11 Core noise: values, selected cores, and full recourse points

Many structured problems have a few complicating *core* variables v and many *residual* variables z , and the residual problem is convex once v is fixed. Two-stage models, where z is the recourse of a first-stage decision v , and decomposition methods that branch only on v have this form. At a rational core point the residual problem is a convex program, so its value can be computed to any certified accuracy in polynomial time (Lemma 3.1). The projected value $V(v) = \min_z f(v, z)$ is nevertheless nonconvex, and the earlier sections show that approximating values, approximating optimizer coordinates, and deciding exact relations are different problems.

This section studies these outputs after one random linear perturbation of the core coefficients. The topic belongs to the output account of this paper for three reasons. First, the perturbation is drawn once from a finite rational law that is fixed before any accuracy is requested. The perturbed instance is therefore an ordinary rational instance, and the contracts of Section 2 apply to it without change. Second, every draw is answered correctly. Probability only bounds the expected cost; rare draws are solved by exact real-algebraic computation on the same instance and with the same selector. Third, the outputs separate sharply. On product boxes, values and one selected core are available under residual convexity alone; on coupled polytopes they need a supplied convexifier. Full optimizer points need more structure: a supplied convexifier together with a cubic objective or a

convexified objective that is convex on all of $\mathbb{R}^n$, or, for cubics on product boxes, exact certificates that we construct. For residual quartics, core noise alone does not help unless Square Root Sum and PosSLP have Las Vegas algorithms with expected polynomial running time (Remark 11.14).

The main results are Theorem 11.2 (values and selected cores), Corollaries 11.3 and 11.4 (joint convexity and exact quadratic output), Theorem 11.5 (full points under a supplied convexifier), Proposition 11.7 together with Proposition 11.8 (the structure of cubic residual fibers and two obstructions), and Theorem 11.9 (full points for residual-convex cubics on product boxes, with no convexifier). All proofs are in Appendix I.

11.1 Model, noise law, and selector

Definition 11.1 (Core-noise instances). Fix a degree bound $D \geq 1$. A *core-noise instance* consists of integers $k \geq 1$ and $m \geq 0$, a nonempty compact set $X \subseteq [0, 1]^k \times \mathbb{R}^m$, an explicitly encoded polynomial $f \in \mathbb{Q}[v, z]$ of total degree at most D in the core variables $v \in \mathbb{R}^k$ and residual variables $z \in \mathbb{R}^m$, a rational curvature parameter $\kappa \geq 0$, and a rational noise width $\sigma > 0$. The instance has one of two types.

- (B) *Product box*. $X = [0, 1]^k \times [0, 1]^m$; the function $z \mapsto f(v, z)$ is convex on $[0, 1]^m$ for every $v \in [0, 1]^k$; and $\kappa = \Lambda$ satisfies $\partial^2 f / \partial v_i^2 \leq \Lambda$ on X for $i = 1, \dots, k$.
- (P) *Coupled polytope*. $X = \{x : Ex \leq e\}$ is a bounded rational polytope contained in $[0, 1]^k \times \prod_{j=1}^m [l_j, u_j]$ for a supplied rational box, and $\kappa = \alpha$ makes $f(v, z) + \frac{\alpha}{2} \|v\|^2$ convex on X .

We write $n = k + m$ and let L be the total binary length of all these data, including any supplied certificate for the convexity and curvature promises.

Core coordinates with other rational ranges can be mapped affinely onto $[0, 1]^k$. This rescales Λ or α and changes the physical noise directions, and the rescaled data are part of the input; no invariance under such rescaling is claimed. Fixed coordinates are substituted first, and empty or single-point feasible sets are detected by linear programming.

Convexity and curvature are promises. The algorithms never test them; when a certificate is supplied, the stated bounds assume that it can be verified in time polynomial in L , and otherwise its verification cost must be added. In type (B) a valid Λ can always be computed from coefficient sums, but the numerical factor in our bounds depends on the supplied value, which may be much smaller. Type (P) asks for more than convex residual fibers: it supplies a convex problem on every cell of the core. For example, $f(v, z) = vz^2$ on the unit square has convex fibers, but $f + \frac{\alpha}{2}v^2$ is convex for no α (Proposition 11.8).

For a power of two $M \geq 2$ put

$$G_M = \left\{ -\sigma + \frac{2\sigma j}{M-1} : j = 0, 1, \dots, M-1 \right\},$$

the endpoint-inclusive uniform grid in $[-\sigma, \sigma]$. A *draw* is a vector γ with independent coordinates, each uniform on G_M , and the perturbed objective is

$$f_\gamma(v, z) = f(v, z) + \gamma^\top v, \quad f_\gamma^* = \min_X f_\gamma.$$

Only core coefficients are perturbed. The *selected core* $a = a(\gamma)$ is the lexicographically least core among the global minimizers of f_γ on X . The *optimal fiber*

$$S(\gamma) = \{z : (a, z) \in X, f_\gamma(a, z) = f_\gamma^*\}$$

is nonempty, compact, and convex in both types (Lemma 1.6). Let $p = p(\gamma)$ be its point of least Euclidean norm. The *selected point* is $x^\sharp(\gamma) = (a, p)$. It is a global minimizer of f_γ , and it does not depend on any requested accuracy.

An *evaluator* is a deterministic algorithm that receives the instance, the draw γ , and an integer $q \geq 0$, and returns rational numbers. We count its bit operations and the bit length of its output. A quantity is *base-computable* if it is computed from the instance before sampling, in time polynomial in L ; in particular the grid size M below has $\log M$ polynomial in L , so each sampled coefficient has polynomial bit length. The notation P_D stands for a polynomial whose degree depends only on D ; it never depends on k, m, σ , the draw, or q .

The guarantees below have the same form. Correctness holds on every draw and at every precision. The work bound is a single random factor $\Omega(\gamma)$ times a polynomial in $L + q$: one factor controls all precision queries at once, so there is no union bound over q and no resampling when the accuracy grows. The outputs are Cauchy names for fixed numbers of the sampled instance: a value interval for f_γ^* , and approximations to $a(\gamma)$ or $x^\sharp(\gamma)$. They are not expanded algebraic representations, and they do not solve the unperturbed instance.

11.2 Values and selected cores

Theorem 11.2 (Values and selected cores under one finite law). *Fix D . For core-noise instances of type (B) or (P) there are a base-computable power of two M and an evaluator with the following properties. For every draw $\gamma \in G_M^k$ and every integer $q \geq 0$, the evaluator returns a rational point $x_q = (v_q, z_q) \in X$ and a rational number ℓ_q , both of bit length at most $P_D(L + q)$, with*

$$\ell_q \leq f_\gamma^* \leq f_\gamma(x_q) \leq \ell_q + 2^{-q}, \quad \|v_q - a(\gamma)\| \leq 2^{-q}. \quad (31)$$

There is a function $\Omega : G_M^k \rightarrow [1, \infty)$ such that, for every draw and every q , the bit work of the query is at most $\Omega(\gamma)P_D(L + q)$, and

$$\mathbb{E} \Omega \leq k^{O(k)} \left(1 + \frac{\kappa}{\sigma}\right)^k P_D(L),$$

with an absolute constant in $O(k)$.

The value part of (31) is an exact Cauchy name for f_γ^* together with feasible points of certified objective gap; the core part is a Cauchy name for the single vector $a(\gamma)$. The residual coordinates z_q need not approach any particular residual optimizer. Ties among optimal cores, flat residual fibers, and positive-dimensional optimal sets are allowed; no growth constant, residual strong convexity, or interiority is supplied.

How the proof works. The evaluator refines dyadic cells of the core cube. On each cell it computes a lower bound and a feasible witness by convex optimization: in type (B) at the cell corners, using the coordinate bound Λ to control rounding, and in type (P) by the secant underestimator $f_\gamma + \frac{\alpha + \sigma}{2} \sum_i (v_i - l_i)(v_i - u_i)$ on the cell. The latter is the classical α BB construction [7]. Pruning against the best witness gives a certified value interval on every draw, and the coordinate hull of the surviving cells contains every optimal core and the incumbent. Small hull diameter therefore certifies the core output without deciding uniqueness.

Three probabilistic estimates control the cost. (i) An *all-scale count*: the number of cells at every level of every query is at most $4^k W(\gamma)$, where W is defined from the largest simplices in padded near-optimal sets. A convex-conjugate pushforward measure and a covering argument give $\Pr\{W > s\} \leq a_k/s + C/M$ with $a_k = k^{O(k)}(1 + \kappa/\sigma)^k$, in every core dimension. (ii) A

projected-growth tail: the core quadratic growth constant g of the sampled objective satisfies $\Pr\{g < t\} \leq kt/\sigma + C/M$; when g is not small the hull shrinks at the expected rate. (iii) A *finite-law transfer*: each event in (i) and (ii) is defined by a formula with two quantified blocks, so a block quantifier-elimination theorem bounds the number of intervals in every one-dimensional section independently of coefficient heights. Replacing continuous marginals by grid marginals one at a time costs C/M , including atoms at ties and exact algebraic relations. Here C is the base-computable format bound from Appendix I, of size $2^{\text{poly}(L)}$; it is independent of the requested precision and sampled coefficient heights.

When a cell count exceeds a base cap or the hull test fails, the evaluator solves the same sampled instance exactly. Two-block singleton formulas define the coordinates of the selected point; block quantifier elimination and univariate real-root isolation give their values. The exponential factor of this fallback depends only on the format of the instance, and the sampled coefficient length and q enter polynomially. This separation is what allows the grid to be chosen after the fallback budget, with $\log M$ polynomial in L .

A simpler count based on growth alone gives at most $(c(1 + \sqrt{k\kappa/g}))^k$ cells per level, for an absolute constant c . Its truncated moment under the growth tail is bounded only for $k \leq 2$; for $k \geq 3$ it grows as a positive power of the cap, and that route fails. The all-scale weak first-moment bound in (i) removes the restriction on k . It is an instance of the weak-type estimate for a centered maximal function of a locally finite measure; the conjugate construction makes that measure available.

Corollary 11.3 (Joint convexity: no numerical smoothing factor). *Fix D . For instances of type (P) with $\alpha = 0$, that is, with f convex on X , there are a base-computable power of two M and an evaluator with the outputs (31) on every draw and at every precision, and a function Ω with $\mathbb{E}\Omega \leq 2$ such that every query costs at most $\Omega(\gamma)P_D(L + q)$ bit operations.*

The bound has no factor $k^{O(k)}$ and no numerical dependence on σ ; the noise width enters only through its bit length. The proof uses no cells. It encloses every optimal core and the incumbent's core by $2k$ binary searches whose probes are convex value queries on $X \cap \{v_i \geq \theta\}$ and $X \cap \{v_i \leq \theta\}$ (Appendix I.8). Taking every variable as a core variable ($m = 0$) yields the lexicographically least optimizer of a convex polynomial on a polytope inside the unit cube, in expected polynomial time, under independent noise on every coordinate. This is consistent with Theorem 3.12: there the hard coordinate is not perturbed, and full-coordinate noise changes the instance.

Corollary 11.4 (Exact rational output for quadratic objectives). *Let $D = 2$. For instances of type (B) or (P) , there are a base-computable M and an algorithm that, on every draw, returns an exact rational global minimizer of f_γ on X with core $a(\gamma)$, together with the exact value f_γ^* . The output has bit length polynomial in L on every draw, and the expected bit work is $k^{O(k)}(1 + \kappa/\sigma)^k \text{poly}(L)$; for type (P) with $\alpha = 0$ it is $\text{poly}(L)$.*

The proof shows that the core of the lexicographically first optimizer of a rational quadratic program has denominators bounded by a base-computable U_Q with $\log U_Q = \text{poly}(L)$, uniformly over all draws. One query of Theorem 11.2 at accuracy $1/(4U_Q^2)$ followed by continued-fraction reconstruction recovers that core exactly, and one exact convex quadratic program [71] completes it. The height argument is specific to quadratics; it does not give exact output for cubic or quartic objectives.

11.3 Full points under a supplied convexifier

Theorem 11.2 says nothing about residual coordinates. Substituting an approximate core into an exact residual solver does not repair this: the residual optimizer set can jump at the selected core

(Proposition 11.8). With a supplied convexifier the jump is avoided by optimizing a convex surrogate on the fixed domain instead.

Theorem 11.5 (Full points with a supplied convexifier). *Consider instances of type (P) in which either*

- (i) $D = 3$ and $f + \frac{\alpha}{2} \|v\|^2$ is convex on X ; or
- (ii) D is fixed and $f + \frac{\alpha}{2} \|v\|^2$ is convex on all of $\mathbb{R}^n$.

There are a base-computable power of two M and an evaluator that, for every draw γ and every $q \geq 0$, returns a rational $x_q \in X$ with $\|x_q - x^\#(\gamma)\| \leq 2^{-q}$. Every query costs at most $\Omega(\gamma)P_D(L+q)$ bit operations, where $\mathbb{E}\Omega \leq k^{O(k)}(1 + \alpha/\sigma)^k P_D(L)$. If $\alpha = 0$, then $\mathbb{E}\Omega \leq 2$.

With $\beta = \alpha + 1$ the surrogate is $\psi_a(x) = f_\gamma(x) + \frac{\beta}{2} \|v - a\|^2$. It is convex, its minimum value is f_γ^* , and its optimal set is exactly the selected fiber $\{a\} \times S(\gamma)$. Its linear coefficients involve the unknown, possibly irrational core a . The key estimate is an error bound $\text{dist}(x, \{a\} \times S(\gamma)) \leq \Gamma(\psi_a(x) - f_\gamma^*)^{1/D_e}$ whose constant Γ has polynomial bit length and is uniform in a ; here $D_e = 4$ in case (i) and $D_e = D$ in case (ii). It is obtained by describing the optimal set through a known rational matrix whose right-hand side alone depends on a , and applying a Hoffman bound. In case (i) the matrix comes from the affine Hessian of a cubic, as in Theorem 3.7; in case (ii) from gradient coefficient rows, as in Theorem 3.4. The completion then minimizes the rational convex function $\psi_b + \tau \|x\|^2$ for a short rational approximation b of a . The search for a uses α ; the larger β appears only in the polynomial-time completion and does not change the numerical factor.

Corollary 11.6 (Affine-power certificates). *Condition (ii) holds, and can be verified in time polynomial in L for fixed D , when $f + \frac{\alpha}{2} \|v\|^2$ is supplied in the form $c_0 + \ell^\top x + \frac{1}{2} x^\top A x + \sum_j w_j (a_j^\top x - b_j)^{p_j}$ with rational data, $A \succeq 0$, $w_j > 0$, and even $p_j \leq D$.*

This explicit class contains quartic and higher even-power terms with dense rational affine forms and arbitrary rational coupling constraints. It is a sufficient certificate, not a recognition algorithm for global convexity.

11.4 Residual-convex cubics without a convexifier

We now drop the convexifier and keep only type (B) with total degree three. Write $H(v, z) = \nabla_{zz}^2 f(v, z)$, which is affine in (v, z) and positive semidefinite on the product box. For a face A of the core cube $[0, 1]^k$ let $F \subseteq \{1, \dots, k\}$ be its set of free coordinates, c_A its relative center, and $c_z = (\frac{1}{2}, \dots, \frac{1}{2})$ the residual center. Put

$$H_A = H(c_A, c_z), \quad g_A(v) = \nabla_z f(v, c_z), \quad J_A(v) = \begin{bmatrix} H_A \\ g_A(v)^\top \\ I_m \end{bmatrix}.$$

For $v \in [0, 1]^k$ let $S_v = \arg \min_{z \in [0, 1]^m} f(v, z)$.

Proposition 11.7 (Fixed face kernels and a conditional error bound). *Let f be a cubic of type (B) and let A be a face of $[0, 1]^k$.*

- (a) $0 \preceq H(v, z) \preceq 2H_A$ on $A \times [0, 1]^m$. If the free coordinates of $v \in A$ lie in $[\delta, 1 - \delta]$ with $0 < \delta \leq \frac{1}{2}$, then $H(v, c_z) \succeq 2\delta H_A$. In particular $\ker H(v, c_z) = \ker H_A$ for every v in the relative interior of A .

(b) For v in the relative interior of A and any $y \in S_v$,

$$S_v = \{z \in [0, 1]^m : H_A(z - y) = 0, g_A(v)^\top(z - y) = 0\}.$$

Every square minor of $J_A(v)$, restricted to A , is a polynomial of degree at most two in the free coordinates.

(c) Let $\delta \in (0, \frac{1}{2}]$ and $\mu \in (0, 1]$ be supplied rational margins. Suppose the free coordinates of $v \in A$ lie in $[\delta, 1 - \delta]$ and every nonzero square minor of $J_A(v)$ has absolute value at least μ . Then

$$\text{dist}(z, S_v) \leq \Gamma(f(v, z) - \min f(v, \cdot))^{1/4} \quad \text{for } z \in [0, 1]^m \text{ with gap at most } 1,$$

for a rational $\Gamma \geq 1$ that is computed from f , A , δ , and μ in time polynomial in L and the bit lengths of δ and μ .

The kernel is rational and constant on the relative interior of each face, although it can grow on the boundary. Only one row of the equality description of S_v varies with the core. The family of square minors of J_A is exponentially large, but each member is a quadratic in the free core coordinates. Part (c) is conditional: the margins δ and μ at an unknown, possibly irrational core cannot be read from rational input heights, and computing a Hoffman constant at a nearby rational core is unsound, since a minor can vanish there while it is tiny but nonzero at the true core.

Proposition 11.8 (Two obstructions). (a) The cubic $f(v, z) = vz^2$ on $[0, 1]^2$ has convex residual fibers, but no constants $\Gamma, \theta > 0$ satisfy $\text{dist}(z, S_v) \leq \Gamma(f(v, z) - \min f(v, \cdot))^\theta$ for all $v \in (0, 1]$ and $z \in [0, 1]$. For no $\alpha \geq 0$ is $f + \frac{\alpha}{2}v^2$ convex on the square.

(b) For $\gamma > 1$, the cubic $f_\gamma(v, z) = \gamma v - v^2 z$ on $[0, 1]^2$ has affine residual fibers, optimal set $\{0\} \times [0, 1]$, and selected point $(0, 0)$. For every $v > 0$ the residual problem has the unique minimizer $z = 1$, so exact residual solutions at approximate cores $b_j \downarrow 0$ converge to $(0, 1)$. For no $\alpha \geq 0$ is $f_\gamma + \frac{\alpha}{2}v^2$ convex on the square. Under uniform core noise of width $\sigma > 1$ added to $f(v, z) = -v^2 z$, such draws have positive probability.

Example (b) defeats minimum-norm completion by approximate-core substitution. The completion below avoids it by first certifying the exact face of the selected core, here the vertex $v = 0$, and then regularizing the residual problem at that exact face.

Theorem 11.9 (Full selected optimizer for residual-convex cubic recourse). Let (X, f, Λ, σ) be a core-noise instance of type (B) with $D = 3$ and $k, m \geq 1$. There are a base-computable power of two M and an evaluator such that, for every draw $\gamma \in G_M^k$ and every integer $q \geq 0$, the evaluator returns a rational point $(v_q, z_q) \in [0, 1]^{k+m}$ with

$$\|(v_q, z_q) - x^\sharp(\gamma)\| \leq 2^{-q}.$$

There is a function Ω such that every query costs at most $\Omega(\gamma) \text{poly}(L + q)$ bit operations and output bits, and

$$\mathbb{E} \Omega \leq k^{O(k)} \left(1 + \frac{\Lambda}{\sigma}\right)^k \text{poly}(L).$$

The polynomials have absolute exponents. No joint convexifier, lower bound on residual curvature, or uniqueness of residual optimizers is assumed.

For $k = 0$ the selected point is the minimum-norm minimizer of a cubic convex on a box, and Theorem 3.7 gives it deterministically. For $m = 0$ Theorem 11.2 suffices. The examples of Proposition 11.8 are covered. The evaluator can also return a certified interval of width 2^{-q} for f_γ^* at the same order of cost (Remark 11.15).

The proof composes three certificates with Theorem 11.2, applied on the same draw to f and to the $2k$ baselines $f \pm tv_i$. Each certificate is sound on every draw; probability only bounds how often one of them fails.

Lemma 11.10 (Exact core endpoints from nearby tilts). *Let f be of type (B) , $\bar{\Lambda} = \max\{1, \Lambda\}$, $0 < t \leq \frac{1}{2}$, $\gamma \in \mathbb{R}^k$, and $1 \leq i \leq k$.*

- (a) *If some global minimizer of $f_{\gamma-te_i}$ has core b with $b_i < t/\bar{\Lambda}$, then every global minimizer of f_γ has $v_i = 0$. If some global minimizer of $f_{\gamma+te_i}$ has core b with $b_i > 1 - t/\bar{\Lambda}$, then every global minimizer of f_γ has $v_i = 1$.*
- (b) *There are numbers θ_i^0, θ_i^1 depending only on γ_{-i} such that the following holds. If some global minimizer of f_γ has $v_i = 0$ while some global minimizer of $f_{\gamma-te_i}$ has core b with $b_i > 0$, then $\gamma_i \in [\theta_i^0, \theta_i^0 + t]$. If some global minimizer of f_γ has $v_i = 1$ while some global minimizer of $f_{\gamma+te_i}$ has $b_i < 1$, then $\gamma_i \in [\theta_i^1 - t, \theta_i^1]$.*

Part (a) uses the one-dimensional projected value $V_i(s) = \min\{f_\gamma(v, z) : v_i = s\} - \gamma_i s$, which becomes concave after subtracting $\bar{\Lambda}s^2/2$. Optimal coordinates move monotonically with the tilt, and two interior contacts for tilts that differ by t must be at distance at least $t/\bar{\Lambda}$. Applied with certified approximations of the selected cores of $f_{\gamma \pm te_i}$ to accuracy $t/(8\bar{\Lambda})$, the rules in (a) fix a face A containing $a(\gamma)$. By part (b), the probability under the grid law that the selected core has an endpoint coordinate that the rules miss is at most $kt/\sigma + 2k/M$.

Lemma 11.11 (A lattice certificate for all bounded-height quadratic margins). *Let $a \in [0, 1]^r$, let $\chi(a) \in [0, 1]^s$, $s = \binom{r+2}{2}$, list the monomials of degree at most two in a , including 1, and let $B, T \geq 1$ be integers. Given a rational $\hat{a} \in [0, 1]^r$ with $\|\hat{a} - a\|_\infty \leq 1/(4T)$, put $\omega_j = \text{round}(T\chi_j(\hat{a}))$, and let $\mathbf{b}_1$ be the first vector of an LLL-reduced basis, with parameter $3/4$ [78], of the lattice in $\mathbb{Z}^{s+1}$ generated by the rows of $[I_s \mid \omega]$.*

- (a) *If $\|\mathbf{b}_1\|^2 > 2^{s-1}(4sB)^2$, then $|h^\top \chi(a)| > sB/T$ for every $h \in \mathbb{Z}^s$ with $0 < \|h\|_\infty \leq B$.*
- (b) *Let $R_s = 2^s 4sB$. If $|h^\top \chi(a)| \geq \eta_0$ for every $h \in \mathbb{Z}^s$ with $0 < \|h\|_\infty \leq R_s$, and $T\eta_0 > (s+1)R_s$, then the test in (a) accepts.*

The test costs time polynomial in $s + \log T + \log B + \langle \hat{a} \rangle$.

Apply the lattice lemma to $a_F(\gamma)$, where F is the set of core coordinates left provisionally free by the endpoint rules defining A , after substituting the certified endpoint values into the minors. The tested family contains v_i and $1 - v_i$ for $i \in F$, so acceptance proves that these coordinates are truly interior. After clearing a common denominator Δ , every square minor of J_A on every face becomes an integer quadratic of height at most a base-computable B . Acceptance therefore certifies $\delta = \min\{\frac{1}{2}, sB/T\}$ and $\mu = \min\{1, sB/(\Delta T)\}$ in Proposition 11.7(c) for all exponentially many minors at once, with no enumeration. Part (b) needs the larger height R_s , because a short lattice vector can have coefficients larger than B .

Proposition 11.12 (Small margins are rare). *Let f be a cubic of type (B) and let $U \geq 1$ be rational with $\nabla_{vv}^2 f \preceq UI_k$ on X . Let A be a face of $[0, 1]^k$ with $r \geq 1$ free coordinates, let ψ be a nonzero*

integer polynomial of degree at most two in those coordinates, and let $0 < \eta < 1$. Under continuous uniform noise on $[-\sigma, \sigma]^k$, the probability that some global minimizer of f_γ has core a in the relative interior of A with $|\psi(a_F)| \leq \eta$ is at most $8(U/(2\sigma))^r \sqrt{\eta}$. Under the grid law it is at most this bound plus C/M , where $C = 2^{\text{poly}(L)}$ is base-computable and does not depend on ψ or η .

The proof does not use any continuity of residual optimizers. On a face, the projected value is semiconcave; at a minimizer under a linear tilt it has an affine lower support, and the slope of that support is a locally U -Lipschitz function of the contact point. Hence the noise must fall in the image of a small set under a Lipschitz map. An elementary slicing argument bounds the volume of $\{|\psi| \leq \eta\}$ by $8\sqrt{\eta}$ for every nonzero integer quadratic. The constant U can be much larger than Λ ; it enters only through bit lengths.

Proposition 11.13 (Certified completion). *Let f be a cubic of type (B) , and suppose that evaluators with the guarantees of Theorem 11.2 are available for the selected cores of f_γ and $f_{\gamma \pm te_i}$, $1 \leq i \leq k$, on a fixed draw. For base-computable parameters t and T , a base stage makes $2k + 1$ core queries at accuracy $\text{poly}(L)$ bits, applies the rules of Lemma 11.10, and runs the test of Lemma 11.11 once; it either rejects or accepts. Acceptance certifies a face A whose relative interior contains $a(\gamma)$ and a rational Γ of polynomial bit length with $\text{dist}(z, S(\gamma)) \leq \Gamma(f(a, z) - \min f(a, \cdot))^{1/4}$ for gaps at most one. After acceptance, a query q makes one core query at accuracy $\text{poly}(L) + O(q)$ bits and one convex minimization of the rational cubic $z \mapsto f(b, z) + \tau \|z\|^2$, and returns a point within 2^{-q} of $x^\sharp(\gamma)$. Here b is the rational core approximation lying in A , with the certified endpoint coordinates imposed exactly, and $\tau > 0$ is the regularization parameter specified below. Every acceptance is correct, whatever the draw.*

Proof of Theorem 11.9 in outline. Choose the fallback budget $\Xi = 2^{\text{poly}(L)}$ first, then $t = \min\{\frac{1}{2}, \sigma/(16k\Xi)\}$, a margin threshold η with $A_*\sqrt{\eta} \leq 1/(16\Xi)$ for the union-bound constant A_* over all faces and all quadratics of height at most R_{s_*} , $s_* = \binom{k+2}{2}$, and a lattice scale T with $T\eta > (s_* + 1)R_{s_*}$. Finally choose one grid size M that meets the requirements of Theorem 11.2 for all $2k + 1$ baselines and makes the finite-law error terms at most $1/(8\Xi)$. All choices precede sampling and have polynomial bit length. If no endpoint is missed and no tested quadratic has margin at most η , the lattice test accepts by Lemma 11.11(b), so rejection has probability at most $1/(4\Xi)$ by Lemma 11.10(b) and Proposition 11.12. On rejection the evaluator computes $x^\sharp(\gamma)$ exactly from two-block singleton formulas that encode the same selector: global optimality, lexicographically least core, and least residual norm within that core. The random factor is the sum of the $2k + 1$ factors of Theorem 11.2 plus Ξ times the indicator of rejection. The completion step is proved in Appendix L.16: regularization with $\tau = e^6/(1024R_z^4\Gamma^4)$, where $e = 2^{-q-1}$ and $R_z \geq \max\{1, \sqrt{m}\}$, controls the distance to the least-norm point of $S(\gamma)$, and a uniform objective perturbation bound controls the error from the approximate core.

11.5 Scope and relation to other work

Remark 11.14 (Degree four). Let F be a quartic from Theorem 3.12, convex on a unit box, with a unique minimizer whose designated coordinate is 0 or 1 according to an instance of Square Root Sum or PosSLP. Put $f(v, y) = v^2/2 + F(y)$ with one core coordinate $v \in [0, 1]$. Its residual fibers are convex and $\Lambda = 1$. For every draw the residual part of the selected point is the minimizer of F . If the conclusion of Theorem 11.9 held for this degree-four family, its evaluator would decide the source problem by a Las Vegas algorithm in expected polynomial time. Core noise cannot by itself make full points of residual-convex quartics tractable; the cubic structure of Proposition 11.7 is essential. This is a conditional statement, not a hardness theorem.

Remark 11.15 (Certified values for the sampled objective). To add a value interval to Theorem 11.9 or Theorem 11.5, query the point at accuracy $2^{-q-1}/G$, where $G \geq 1$ bounds $\|\nabla f_\gamma\|$ on X uniformly over the noise support, and pair its objective value with the lower bound ℓ_{q+1} of Theorem 11.2. This certifies the sampled objective only; the unperturbed objective is not solved at the same cost.

The results have clear limits. All guarantees concern one sampled rational instance. Theorem 11.9 uses the product box in the nearby-tilt argument, in the fixed face kernels, and in the conditional error bound; it does not extend to arbitrary coupled domains without a convexifier, and Remark 11.14 explains why degree four needs more than core noise. On accepted draws the algorithm internally certifies the exact core face and the margins needed for completion. Its output contract gives point and value approximations; it does not include all active-constraint labels or expanded algebraic coordinates, in line with Proposition 3.10. The factor $k^{O(k)}$ is not optimized, and the dependence on κ/σ is numerical, not through its bit length, except in Corollary 11.3.

The individual tools are classical: branch-and-bound on core cells with secant underestimators [7]; weak convex optimization [49]; the almost-everywhere differentiability of convex conjugates, which underlies the growth tail [104]; covering arguments for maximal functions [40]; block quantifier elimination [15, 102]; Hoffman error bounds [57]; and lattice reduction for small integer relations [78]. Smoothed analysis [112] also perturbs instances randomly to bound expected cost. What is shown here is a set of output guarantees for one fixed finite rational law: certified values and one selected core for nonconvex cores in every core dimension, with convex residuals on product boxes or a supplied convexifier on coupled polytopes; ordinary polynomial expected work under joint convexity; exact rational output for quadratics; full selected points under a supplied convexifier, for cubic objectives or when the convexified objective is convex on all of $\mathbb{R}^n$; and, for residual-convex cubics on product boxes, full selected points with no convexifier, obtained from exact face and lattice-margin certificates. Kannan and Rademacher optimize a convex objective plus a low-dimensional polynomial perturbation [63] to a prescribed objective accuracy; that guarantee is an objective approximation, whereas the guarantees here are Cauchy names for the value and for a fixed selected point of one sampled instance.

12 Discussion and open questions

12.1 What the results establish

The four outputs of Section 1 have different complexities. The results locate boundaries for convex polynomial problems with explicit rational data and give related arithmetic interfaces for nonconvex quadratic constraints and cone problems with explicit algebraic data.

Values become points when an effective growth modulus is available. Global convexity supplies one at every degree, with exponent $1/D$ and a constant of polynomial bit length, so a fixed optimizer can be approximated in time $\text{poly}(L, D, q)$ (Theorem 3.4). Convexity on the feasible set supplies one for cubics (Theorem 3.7) but not for quartics, where a constant-accuracy point already decides Square Root Sum or PosSLP (Section 3). With a nonconvex core of fixed dimension and convex cubic recourse on product boxes, one random tilt of the core gives full point output with the stated expected-work bound, correct for every draw and every precision (Section 11).

Points become exact decisions when a separation bound can be reached. For explicit globally strongly convex objectives of unary degree, every exact predicate at the minimizer reduces to one PosSLP instance (Theorem 4.4). Order predicates are PosSLP-hard already for quartics with a checked strict convexity certificate, even when the minimum is promised to be nonzero, and coordinate order predicates remain hard when the minimizer is rational and the minimum value is

known (Theorem 5.1 and Section 5). Linear constraints add the search for the active face, which our methods resolve under structural assumptions, a small nonlinear dimension, or randomization; integer variables add a discrete search that needs only ordinary precision (Section 6).

Exact descriptions have costs that are independent of the cost of exact decisions. The coordinates of points isolated by rational convex polynomial inequalities are characterized (Theorem 7.4). Their degree, the height of rational optimizers and witnesses, the size of interior Gram matrices, and the degree of the least coefficient field of a sum-of-squares certificate can all grow exponentially with the dimension for inputs of polynomial size (Sections 7–10). For the families that prove these lower bounds, the paper also gives polynomial-size descriptions of the same objects in compact formats: repeated-squaring circuits for the long rational minimizers, shared rational circuits for strictly feasible points and for interior Gram matrices, short radical expressions for the coordinates of the cyclic family, and root circuits or auxiliary root variables for the certificates of the tower family. Lower bounds on output size therefore depend on the format and should be stated for a specific one.

Common-field representations also let exact outputs serve as inputs to a later computation. Appendix L bounds the degree and total length of feasible witnesses, finite infima, and attained minimum-norm optimizers for rational quadratic constraints whose Hessians span a space of dimension h , without assuming convexity. It separately gives exact cone feasibility and witness recovery over one explicitly represented real number field, with bounded integer dimension and few independent continuous Hessians of the squared cone residuals. These are specified algebraic input and output contracts; they do not turn the nonconvex problems into ordinary polynomial-time optimization in unrestricted dimension.

12.2 Consequences for exact and verified optimization

The results bear on the design of exact and proof-producing solvers in the following ways. None of them is a running-time claim about existing software.

Bounds and pruning. For globally convex relaxations of fixed or unary degree, certified value enclosures cost polynomial time, so bound-based pruning with a tolerance is efficient. Exact pruning decisions, such as whether a relaxation bound is strictly below an incumbent value, are PosSLP-hard in the worst case even for unconstrained quartics with a checked strict convexity certificate. The hardness is a statement about how close to zero a nonzero optimal value can be. If there were a polynomial s such that every nonzero minimum in this class satisfied $|f_*| \geq 2^{-s(L)}$, then approximating f_* to $s(L) + 1$ bits would decide its sign in polynomial time, and PosSLP would be in P. Exact methods that refine numerical approximations must therefore expect to need precision that is not bounded by any fixed polynomial in L on some inputs, unless PosSLP is in P. In this class, Theorem 4.4 shows that $2^{e(L)}$ bits of precision always suffice, and its Newton circuits reach that precision with polynomially many gates.

Points and active sets. Branching rules and rounding heuristics use points. For globally convex relaxations of fixed or unary degree, a fixed approximate optimizer is available in polynomial time. For objectives that are convex only on the current box, a class that contains convex underestimators built on a box, no comparable guarantee holds for the class unless Square Root Sum or PosSLP is easy. Whether a bound is active at the relaxation optimizer is an exact decision: it is Square Root Sum-hard for strongly convex cubics on a box, although their optimizers are easy to approximate. If an explicit lower bound on the nonzero slacks is supplied, ordinary approximation recovers the active set (Proposition 4.9); without such a bound, an active set read off from an approximate

solution carries no guarantee. These statements concern the active set of the relaxation optimizer. Using it to fix variables of the original mixed-integer problem needs an additional optimization argument, which this paper does not provide.

Integer search and exact selection. For strongly convex mixed-integer problems, candidate lists that contain every optimal integer block can be built with ordinary arithmetic in fixed-parameter time, and exact arithmetic is needed only to select among the candidates. This separation suggests performing the discrete search numerically and reserving exact or algebraic arithmetic for the final comparisons. The binary extraction result shows that the final comparisons cannot be avoided in general, even when the optimal value is known.

Certificates for independent checking. A rational strict convexity certificate does not imply a rational sum-of-squares certificate of optimality: the ternary quartic of Section 9 has neither a rational sum-of-squares decomposition of $f - f_*$ nor a rational positive semidefinite Gram matrix for it. The alternatives have different costs. For this minimum-zero example a rational certificate for $f + \eta$ exists for every rational $\eta > 0$, but our results give no bound on its size as η tends to zero. Common denominators of degree two give short exact rational certificates for the ternary example and, after a polynomial-size increase of a scale parameter, for the tower family, whose polynomial certificates need coefficients of degree 5^k (Section 10). Auxiliary variables for the irrational coordinates, together with their defining equations, also give polynomial-size rational certificates for the tower family. When the minimum is positive, Theorem 8.12 gives a polynomial-size shared rational circuit for an interior Gram matrix, but checking that the matrix is positive definite is as hard as deciding the sign of the minimum. Imposing additional structure on a certificate, such as separation by blocks of variables or a fixed power of $1 + \|x\|^2$ as multiplier, can make it fail to exist or force it to be long, while the unrestricted or adapted certificate remains short.

Reporting exact optimizers. For the constrained strongly convex problems of Section 6, an active face and a polynomial system with a unique real solution on that face provide an implicit exact description. Expanded rational coordinates and coefficient lists of minimal polynomials can be exponentially long in the dimension even for well-conditioned instances with small coefficients.

Passing algebraic data between problems. A cone problem over rational data can acquire algebraic coefficients when an exact value from a preceding computation is used as a threshold. Appendix L handles known algebraic thresholds by placing all data in one specified real number field. With fixed integer dimension t and a fixed dimension h for the span of the continuous Hessians of the squared cone residuals, its feasibility and witness algorithms run in ordinary polynomial time and require neither a bounding box nor a Slater condition. The continuous witness is represented in the joint field generated by the input field and its coordinates. This charges the cost of selecting the real embedding and writing a common-field output, which a coordinatewise degree estimate alone would not control.

12.3 Open questions

The following questions are not settled by the results above. No theorem of this paper depends on an answer to any of them.

1. *Arbitrary polyhedra.* Is exact comparison for globally strongly convex quartics on arbitrary rational polyhedra in $\mathsf{P}^{\mathsf{PosSLP}}$, equivalently reducible to one PosSLP instance? It lies in $\mathsf{UP}^{\mathsf{PosSLP}} \cap$

$\text{coUP}^{\text{PosSLP}}$, and our methods remove the search under structure, small nonlinear dimension, or randomization. The constrained Newton recurrence converges without knowledge of the active set. What our approach lacks is an exact algorithm for convex quadratic programs whose number of arithmetic operations is polynomial independently of the encoding length of the Hessian, because the Hessians of the Newton subproblems are given by circuits. Bit-polynomial quadratic programming algorithms, smooth penalties, and barrier methods with ordinary accuracy bounds do not supply this interface (Section 6.7).

2. *Equality.* Is deciding $h(p) = 0$ PosSLP-hard in the class of Theorem 5.1, or is it easier? Our lower bounds concern order relations only. Deciding whether an integer circuit computes zero lies in coRP [6], but $h(p)$ is the limit of the Newton circuits rather than the value of one, so this fact does not apply directly.
3. *Degenerate convex quartics.* Is there an upper bound relative to PosSLP for exact comparisons at the minimum-norm optimizer of a globally convex quartic that is not strongly convex? If the Hessian vanishes entirely at an optimizer, the optimizer set is a rational affine space computable by linear algebra. For partially degenerate optimizers, Newton's method and its simple accelerations fail to converge quadratically, and the Hessian kernel at the optimizer need not contain a nonzero rational vector (Appendix K). These examples rule out particular proof routes, not an upper bound.
4. *Rational sums of squares at minimum zero.* In the class of Theorem 5.1, membership in the cone of rational sums of squares is PosSLP-hard. Is it in P^{PosSLP} ? At a supplied zero of the tower family it reduces to rational linear algebra and one semidefiniteness test (Section 9), but in general the zero is unknown and the answer depends on its coordinate field.
5. *Size of all Gram matrices.* Is there a family of short certified quartics with positive minimum for which every rational positive semidefinite polynomial Gram matrix has superpolynomial bit length? Our lower bounds concern interior, maximal-rank, exposing, or block-separated certificates. The families in this paper have short singular rational Gram matrices, and the determinant argument for interior Gram matrices does not apply to singular ones.
6. *Degree maxima.* For globally convex rational quartics that are rational sums of squares and have a unique zero with positive definite Hessian, the largest possible degree of the coordinate field is known for $n \leq 5$. For $n \geq 6$ it lies between $(2^{n+1} - (-1)^{n+1})/3$ and $2^n - 5$; for $n = 6$, between 43 and 59. Without the sum-of-squares hypothesis Section 7 proves the upper bound $2 \cdot 3^{n-1} - 1$, and the exact maxima are not known for $n \geq 3$.
7. *Domain-convex cubics.* The optimal error exponent for cubics that are convex on a polytope lies between $1/4$, proved here, and $1/3$, forced by t^3 on $[0, 1]$.
8. *Recourse beyond product boxes.* Can full point output under core noise be obtained for residual-convex cubics on polytopes that couple core and residual variables, without additional structure such as a supplied convexifier for the core? For residual quartics, a guarantee of the form of Section 11 would give a Las Vegas expected polynomial-time algorithm for PosSLP (Remark 11.14), so further structure is needed.
9. *Input models.* Our point theorem and exact upper bound are polynomial in the numerical degree. For sparse polynomials with exponents written in binary, or for objectives given by arithmetic circuits, the value, point, and exact problems need new arguments.

Finally, every reduction to one PosSLP instance in this paper would become an ordinary polynomial-time algorithm if PosSLP were in P, whereas the results stated with $\text{UP}^{\text{PosSLP}} \cap \text{coUP}^{\text{PosSLP}}$ or with Las Vegas oracle algorithms would yield the corresponding classes without the oracle. The size lower bounds for expanded outputs and the field lower bounds for certificates do not depend on the complexity of PosSLP.

A Proofs for Section 4

This appendix proves the results of Section 4. We keep the conventions of Section 4.1. All norms of vectors are Euclidean, and matrix and multilinear norms are the induced operator norms. $\bar{B}(c, \varrho)$ denotes the closed Euclidean ball with center c and radius ϱ . For a polynomial F , $\|F\|_1$ is the sum of the absolute values of its coefficients.

A.1 Derivative bounds

Lemma A.1. *Let $L' \geq 2$ be an integer and put $\Lambda = 2^{10L'^2}$. Let $F \in \mathbb{Q}[X_1, \dots, X_n]$ have $n \leq L'$, total degree at most L' , at most L' terms, and coefficients of absolute value at most $2^{L'}$. Then for all $x, y \in \bar{B}(0, 2^{5L'})$,*

$$|F(x)| \leq \Lambda, \quad \|\nabla F(x)\| \leq \Lambda, \quad \|\nabla^2 F(x)\| \leq \Lambda, \quad \|\nabla^2 F(x) - \nabla^2 F(y)\| \leq \Lambda \|x - y\|.$$

If $T = (T_1, \dots, T_n)$ is a polynomial map whose components satisfy the same assumptions, with at most L' terms in total, then for all $x, y \in \bar{B}(0, 2^{5L'})$,

$$\|T(x)\| \leq \Lambda, \quad \|J_T(x)\| \leq \Lambda, \quad \|J_T(x) - J_T(y)\| \leq \Lambda \|x - y\|.$$

Proof. Put $\varrho_1 = 2^{5L'}$, so $|x_i| \leq \varrho_1$ for every coordinate of a point of the ball. Let X^α be a monomial with $|\alpha| \leq L'$ and let β be a multi-index with $j = |\beta| \leq 3$. The derivative $\partial^\beta X^\alpha$ is a monomial of degree at most L' multiplied by a product of at most j factors from $\{1, \dots, L'\}$, so $|\partial^\beta X^\alpha(x)| \leq L'^j \varrho_1^{L'} = L'^j 2^{5L'^2}$ on the ball. Since $\|F\|_1 \leq L' 2^{L'} \leq 2^{2L'}$, every partial derivative of F of order $j \leq 3$ is at most $2^{2L'} L'^j 2^{5L'^2}$ there. A symmetric j -linear form on $\mathbb{R}^n$ whose coefficients are bounded by c has operator norm at most its Frobenius norm, hence at most $n^{j/2}c$. With $n \leq L'$ and $\log_2 L' \leq L'$, the derivative of order $j \leq 3$ has norm at most

$$2^{5L'^2+2L'+(9/2)\log_2 L'} \leq 2^{5L'^2+(13/2)L'} \leq 2^{10L'^2},$$

because $(13/2)L' \leq 5L'^2$ for $L' \geq 2$. The Lipschitz bound for $\nabla^2 F$ follows by integrating the third derivative along the segment from y to x , which lies in the ball.

For the map, let $C_i = \|T_i\|_1$, so $\sum_i C_i \leq 2^{2L'}$. On the ball, $\|T(x)\| \leq \sum_i |T_i(x)| \leq 2^{2L'} 2^{5L'^2}$. Next, $\|J_T(x)\| \leq \sum_{i,j} |\partial_j T_i(x)| \leq n \sum_i C_i L' 2^{5L'^2} \leq 2^{5L'^2+2L'+2\log_2 L'}$. Finally, $\|\nabla \partial_j T_i(z)\| \leq \sqrt{n} C_i L'^2 2^{5L'^2}$ on the ball, so

$$\|J_T(x) - J_T(y)\| \leq \sum_{i,j} |\partial_j T_i(x) - \partial_j T_i(y)| \leq n^{3/2} \left(\sum_i C_i \right) L'^2 2^{5L'^2} \|x - y\|.$$

All three constants are at most Λ by the estimate above. $\square$

A.2 Algebraic separation

We use the following special case of the one-block quantifier-elimination theorem in the form stated by Basu [12, Theorem 2.27]: one block of s existentially quantified variables, one free variable, and integer coefficients. In the notation of that theorem this is the case $\omega = 1$, $k_1 = s$, $\ell = 1$ and coefficient ring $\mathbb{Z}$.

Imported theorem. Let $\mathcal{P} \subset \mathbb{Z}[y_1, \dots, y_s, z]$ be a finite set of polynomials of total degree at most d whose coefficients have bit size at most τ , and let $\Psi(y, z)$ be a quantifier-free Boolean formula whose atoms are sign conditions on elements of $\mathcal{P}$. Then the formula $\exists y \Psi(y, z)$ is equivalent over $\mathbb{R}$ to a formula

$$\Theta(z) = \bigvee_i \bigwedge_j \left(\bigvee_l \text{sign}(P_{ijl}(z)) = \sigma_{ijl} \right), \quad \sigma_{ijl} \in \{-1, 0, 1\},$$

with polynomials $P_{ijl} \in \mathbb{Z}[z]$ of degree at most $d^{O(s)}$ whose coefficients have bit size at most $\tau d^{O(s)}$.

The implied constants are absolute. Fix an integer $c_0 \geq 1$ such that, for all $s \geq 1$, $d \geq 2$ and $\tau \geq 1$, the degrees are at most $d^{c_0 s}$ and the coefficient bit sizes at most $\tau d^{c_0 s}$. Every algorithm below uses this one fixed integer; nothing depends on the instance. The constant can be read off from the complexity analysis of the elimination algorithm cited in Basu [12], but its numerical value plays no role.

Proof of Lemma 4.1. Let $\Theta(z)$ be the formula given by the imported theorem for $\exists y \Phi(y, z)$, and let $\mathcal{Q}$ be the finite set of polynomials P_{ijl} that occur in it. The set defined by Θ is $\{\alpha\}$.

First, some nonzero $P \in \mathcal{Q}$ vanishes at α . Otherwise there is an open interval $U \ni \alpha$ on which every nonzero element of $\mathcal{Q}$ has constant nonzero sign; the zero polynomials have sign 0 everywhere. Then every atom of Θ has the same truth value at each $z \in U$ as at α , so $\Theta(z)$ holds on all of U . This contradicts the fact that Θ defines the single point α . In particular α is algebraic.

Write $P(z) = \sum_{i=0}^e p_i z^i$ with $e \leq d^{c_0 s}$ and integer coefficients satisfying $|p_i| < 2^\gamma$, where $\gamma = \tau d^{c_0 s}$. Suppose $\alpha \neq 0$, and let m be the least index with $p_m \neq 0$. The polynomial $Q(z) = P(z)/z^m$ has $Q(\alpha) = 0$ and nonzero integer constant term p_m ; it is not constant, because it vanishes at α . If $|\alpha| < 1$, then

$$1 \leq |p_m| = \left| \sum_{i=m+1}^e p_i \alpha^{i-m} \right| \leq |\alpha| \sum_{i=m+1}^e |p_i| \leq |\alpha| e 2^\gamma.$$

Since $e \leq d^{c_0 s} \leq \gamma \leq 2^\gamma$, this gives $|\alpha| \geq 2^{-2^\gamma}$. If $|\alpha| \geq 1$, the same bound also holds. This proves (b).

For (a), let $s, d, \tau \leq L^c$. Using $\log_2 L \leq L$,

$$\log_2(2\tau d^{c_0 s}) \leq 1 + c \log_2 L + c_0 L^c \cdot c \log_2 L \leq 1 + cL + cc_0 L^{c+1} \leq c(c_0 + 2)L^{c+1},$$

so (b) gives $|\alpha| \geq 2^{-2^e}$ with $e = c(c_0 + 2)L^{c+1}$. □

A.3 Division elimination and shifted comparisons

Proof of Lemma 4.2. An integer a with binary digits $a_k \dots a_0$ is built from 0 and 1 by Horner's rule: start with 0, and for each digit replace the current node v by $v + v$ and then add 1 if the digit is one. A negative integer is $0 - |a|$. This uses $O(k)$ nodes. A rational constant a/b with $b > 0$ is

represented by the pair (a, b) . Every other node v of C , with operands represented by (N_a, Q_a) and (N_b, Q_b) , receives the pair

$$\begin{aligned} a + b &: (N_a Q_b + N_b Q_a, Q_a Q_b), \\ a - b &: (N_a Q_b - N_b Q_a, Q_a Q_b), \\ a \times b &: (N_a N_b, Q_a Q_b), \\ a/b &: (N_a Q_b N_b, Q_a N_b N_b). \end{aligned}$$

Each rule adds at most four nodes and refers to existing nodes, so sharing is preserved and the size is $O(\sigma + b)$. The construction inspects no values.

Suppose every division of C is by a nonzero value. We show by induction along a topological order that $Q_v > 0$ and that the value of v is N_v/Q_v . Constants satisfy this. For sums, differences and products the claim follows from $Q_a, Q_b > 0$. For a division, the divisor has value $N_b/Q_b \neq 0$, so $N_b \neq 0$ and $Q_a N_b^2 > 0$; moreover $N_a Q_b N_b / (Q_a N_b^2) = (N_a/Q_a)(Q_b/N_b)$. Since $Q_v > 0$, the value of v and N_v have the same sign. $\square$

When the hypothesis fails, the integer circuit is still defined, but a denominator can be zero; every application below verifies the hypothesis.

Lemma A.2 (Shifted comparisons). *Let $g > 0$ and let $\alpha, \hat{\alpha}$ be real numbers such that $\alpha = 0$ or $|\alpha| \geq g$, and $|\hat{\alpha} - \alpha| \leq g/8$. Then*

$$\begin{aligned} \alpha > 0 &\iff \hat{\alpha} - g/2 > 0, & \alpha \geq 0 &\iff \hat{\alpha} + g/2 > 0, & \alpha = 0 &\iff g^2/4 - \hat{\alpha}^2 > 0, \\ \alpha < 0 &\iff -\hat{\alpha} - g/2 > 0, & \alpha \leq 0 &\iff -\hat{\alpha} + g/2 > 0, & \alpha \neq 0 &\iff \hat{\alpha}^2 - g^2/4 > 0. \end{aligned}$$

Each of the six expressions has absolute value at least $3g/8$ in the first two columns and at least $15g^2/64$ in the last column.

Proof. If $\alpha > 0$, then $\alpha \geq g$ and $\hat{\alpha} \geq 7g/8$, so $\hat{\alpha} - g/2 \geq 3g/8$. If $\alpha \leq 0$, then $\hat{\alpha} \leq g/8$ and $\hat{\alpha} - g/2 \leq -3g/8$. If $\alpha \geq 0$, then $\hat{\alpha} \geq -g/8$ and $\hat{\alpha} + g/2 \geq 3g/8$; if $\alpha < 0$, then $\alpha \leq -g$ and $\hat{\alpha} + g/2 \leq -3g/8$. The rows for $<$ and $\leq$ follow by replacing $\alpha, \hat{\alpha}$ with $-\alpha, -\hat{\alpha}$. If $\alpha = 0$, then $|\hat{\alpha}| \leq g/8$ and $g^2/4 - \hat{\alpha}^2 \geq 15g^2/64$. If $\alpha \neq 0$, then $|\hat{\alpha}| \geq 7g/8$ and $\hat{\alpha}^2 - g^2/4 \geq -33g^2/64$. The last expression is the negative of the equality expression. $\square$

A.4 Local Newton convergence and elimination without pivoting

Lemma A.3. *Let $A \in \mathbb{R}^{m \times m}$ satisfy $v^\top A v > 0$ for every $v \neq 0$; A need not be symmetric. Then every leading principal minor of A is positive. Consequently Gaussian elimination without row or column exchanges applies to A , all of its pivots are positive, and a system $Ax = c$ is solved with $O(m^3)$ arithmetic operations in which every division is by a pivot.*

Proof. Let A_k be the leading $k \times k$ submatrix. It satisfies $v^\top A_k v > 0$ for $v \in \mathbb{R}^k \setminus \{0\}$ (pad v with zeros). For $\theta \in [0, 1]$ the matrix $(1 - \theta)I + \theta A_k$ has the same property and is therefore nonsingular: a null vector v would give $v^\top((1 - \theta)I + \theta A_k)v = 0$. Its determinant is a continuous nonvanishing function of θ that equals one at $\theta = 0$, so $\det A_k > 0$. Elimination without exchanges does not change leading principal minors, and after $k - 1$ steps the leading $k \times k$ block is upper triangular with the first k pivots on its diagonal. Hence the k th pivot equals $\det A_k / \det A_{k-1} > 0$, with $\det A_0 = 1$. Forward elimination and back substitution divide only by pivots. $\square$

Lemma A.4 (Local Newton convergence). *Let $T: \mathbb{R}^m \rightarrow \mathbb{R}^m$ be continuously differentiable with Jacobian J , and let $T(p) = 0$. Let $0 < \mu \leq \Lambda$ and $\rho = \mu/(4\Lambda)$. Assume that for all $x, y \in \bar{B}(p, \rho)$ and $v \in \mathbb{R}^m$,*

$$v^\top J(x)v \geq \mu \|v\|^2, \quad \|J(x) - J(y)\| \leq \Lambda \|x - y\|.$$

Let $x_0 \in \bar{B}(p, \rho)$ and $x_{t+1} = x_t - J(x_t)^{-1}T(x_t)$. Then every x_t lies in $\bar{B}(p, \rho)$, every $J(x_t)$ has positive leading principal minors, and

$$\|x_t - p\| \leq \frac{2\mu}{\Lambda} 2^{-3 \cdot 2^t} \quad (t \geq 0).$$

Proof. For $x \in \bar{B}(p, \rho)$ and $v \in \mathbb{R}^m$, $\|J(x)v\| \|v\| \geq v^\top J(x)v \geq \mu \|v\|^2$, so $J(x)$ is invertible with $\|J(x)^{-1}\| \leq 1/\mu$, and Lemma A.3 applies to it. Suppose $x_t \in \bar{B}(p, \rho)$ and put $e_t = \|x_t - p\|$. The segment from p to x_t lies in the ball, and $T(x_t) = \int_0^1 J(p + \theta(x_t - p))(x_t - p) d\theta$. Hence

$$x_{t+1} - p = J(x_t)^{-1} \int_0^1 (J(x_t) - J(p + \theta(x_t - p)))(x_t - p) d\theta,$$

and since $\|J(x_t) - J(p + \theta(x_t - p))\| \leq \Lambda(1 - \theta)e_t$,

$$e_{t+1} \leq \frac{1}{\mu} \int_0^1 \Lambda(1 - \theta)e_t^2 d\theta = \frac{\Lambda}{2\mu} e_t^2.$$

Put $q_t = \Lambda e_t / (2\mu)$. Then $q_0 \leq \Lambda\rho / (2\mu) = 1/8$ and $q_{t+1} \leq q_t^2$, so $q_t \leq 2^{-3 \cdot 2^t}$ and $e_{t+1} \leq q_t e_t \leq e_t / 8$. Thus all iterates stay in $\bar{B}(p, \rho)$, and $e_t = (2\mu/\Lambda)q_t$ gives the bound. $\square$

A.5 Starting points

For gradients we use convex value optimization in the following form, which is the content of Lemma 3.1 that we need: given an explicit sparse convex polynomial f , an explicit rational polyhedron P contained in a box with rational bounds, and a rational $\eta > 0$, a deterministic algorithm computes, in time polynomial in the encoding length and $\log(1/\eta)$, a rational point $x \in P$ with $f(x) \leq \min_P f + \eta$.

For maps that are not gradients there is no objective. We use the central-cut ellipsoid method, in the following form.

Lemma A.5 (Rounded-cut warm start). *Let $n \geq 1$, and let $R_0 \geq 1$ and $0 < \delta_0 < \rho_0$ be rational. Let $p \in \mathbb{R}^n$ be unknown with $\bar{B}(p, \delta_0) \subseteq Q_0 = [-R_0, R_0]^n$. Suppose a deterministic procedure, given a rational $x \in Q_0$, either accepts x , in which case $\|x - p\| \leq \rho_0$, or returns a nonzero rational vector a such that $a^\top(y - x) \leq 0$ for every $y \in \bar{B}(p, \delta_0)$. Then a deterministic algorithm finds an accepted point, provided that every returned vector has encoding length polynomial in the query length, n , and the encoding lengths of R_0, ρ_0, δ_0 . It calls the procedure only on points of encoding length polynomial in n and the encoding lengths of R_0, ρ_0, δ_0 , and it performs polynomially many further bit operations in these quantities.*

Proof. Extend the procedure to all rational points: if $x \notin Q_0$, choose i with $|x_i| > R_0$ and return $a = \text{sign}(x_i)e_i$. For $y \in \bar{B}(p, \delta_0) \subseteq Q_0$ this gives $a^\top(y - x) \leq R_0 - |x_i| < 0$. In all cases divide a returned vector by its largest absolute entry, which preserves the cut and gives $\|a\|_\infty = 1$.

Let $n = 1$. Start with $I_0 = [-R_0, R_0]$ and query its midpoint x . If x is not accepted, the cut shows that $\bar{B}(p, \delta_0)$ lies on one side of x ; keep that half of the interval. Each interval contains $\bar{B}(p, \delta_0)$, whose length is $2\delta_0$, and the j th interval has length $2^{1-j}R_0$. Hence a query is accepted after at most $\lceil \log_2(R_0/\delta_0) \rceil + 1$ steps, and all midpoints are rational numbers of polynomial length.

Let $n \geq 2$. Put $K = \bar{B}(p, \rho_0)$ and $K_1 = \bar{B}(p, \delta_0) \subseteq K$. Since $\|p\| \leq \sqrt{n}R_0$, K is contained in the ball of radius $R = nR_0 + \rho_0$ around 0. The extended procedure is an oracle of the kind allowed in Grötschel et al. [49, Remark 3.2.33]: for every rational query y and every rational tolerance, it either asserts that y lies in K , which is true for accepted points, or returns c with $\|c\|_\infty = 1$ and $c^\top x \leq c^\top y$.

for all $x \in K_1$. Run the central-cut ellipsoid method of Grötschel et al. [49, Theorem 3.2.1] with this oracle, initial radius R and volume bound $\varepsilon = (\delta_0/n)^n$. By that theorem, it returns the queried center when the oracle asserts membership. The remark retains this acceptance alternative and allows cuts valid only on K_1 . Thus the method stops either at an accepted point or with an ellipsoid of volume less than ε that contains K_1 . The second alternative is impossible: the cube $p + [-\delta_0/n, \delta_0/n]^n$ lies in K_1 , because each of its points is within $\sqrt{n} \delta_0/n \leq \delta_0$ of p , and its volume $(2\delta_0/n)^n$ exceeds ε . The method is oracle-polynomial in n , $\langle R \rangle$ and $\langle \varepsilon \rangle$, and all its centers have encoding length polynomial in these quantities [49, Lemma 3.2.8]. $\square$

The rounding and enlargement steps of the cited method are what keep the centers short while preserving containment of K_1 ; rounding an exact ellipsoid update arbitrarily would not have this property. The procedure need not decide membership in the unknown ball K_1 . It only has to accept correctly and to return cuts that keep K_1 .

A.6 Proof of Lemma 4.3

Existence, uniqueness and size of the zero. In case (a), strong convexity makes f coercive and strictly convex, so it has a unique minimizer p , which is the unique zero of $T = \nabla f$; and ∇f satisfies the monotonicity inequality of case (b) with the same μ . In case (b), let $\varrho > \|T(0)\|/\mu$ and let Π be the Euclidean projection onto $\bar{B}(0, \varrho)$. The continuous map $x \mapsto \Pi(x - T(x))$ sends this ball to itself and has a fixed point x^* by Brouwer's theorem. The projection inequality at x^* reads $T(x^*)^\top(y - x^*) \geq 0$ for all $y \in \bar{B}(0, \varrho)$. If $\|x^*\| = \varrho$, the choice $y = 0$ contradicts

$$T(x^*)^\top x^* = \langle T(x^*) - T(0), x^* \rangle + T(0)^\top x^* \geq \mu \varrho^2 - \|T(0)\| \varrho > 0.$$

Hence x^* is interior, and choosing $y = x^* \pm \theta u$ for small $\theta > 0$ and every direction u gives $T(x^*) = 0$. If $T(p) = T(p') = 0$, monotonicity gives $0 \geq \mu \|p - p'\|^2$, so the zero p is unique. Applying the inequality to p and 0 gives $\mu \|p\|^2 \leq -T(0)^\top p$, so

$$\|p\| \leq \|T(0)\|/\mu \leq 2^{2L} \cdot 2^L = 2^{3L},$$

because $T(0)$ is the vector of constant terms of T (in case (a), of linear coefficients of f) and $\mu \geq 2^{-L}$. Finally, dividing $\langle T(x + \theta v) - T(x), \theta v \rangle \geq \mu \theta^2 \|v\|^2$ by θ^2 and letting $\theta \rightarrow 0$ gives $v^\top J_T(x) v \geq \mu \|v\|^2$ for all x, v .

Constants. Put $\Lambda = 2^{10L^2}$ and $\rho = \mu/(4\Lambda)$; note that $\mu \leq 2^L \leq \Lambda$ and $\rho < 1$. Since $\|p\| + 1 \leq 2^{5L'}$, the ball $\bar{B}(p, \rho)$ lies in $\bar{B}(0, 2^{5L'})$. On that larger ball, Lemma A.1 applies to f (case (a)) or T (case (b)) and to every observable h of encoding length at most L' : J_T is Λ -Lipschitz, $\|J_T\| \leq \Lambda$, $\|T\| \leq \Lambda$, and $\|\nabla h\| \leq \Lambda$. In case (a), $J_T = \nabla^2 f$ and its Lipschitz bound comes from the third derivatives of f .

Starting point. In case (a), apply Lemma 3.1 to f on the box $[-2^{3L}, 2^{3L}]^n$, which contains p , with $\eta = \mu \rho^2/2 = \mu^3/(32\Lambda^2)$. The output x_0 satisfies $f(x_0) - f(p) \leq \eta$, and Lemma 2.10 gives $\frac{\mu}{2} \|x_0 - p\|^2 \leq f(x_0) - f(p)$, so $\|x_0 - p\| \leq \rho$.

In case (b), apply Lemma A.5 with $R_0 = 2^{4L}$, $\rho_0 = \rho$ and

$$\delta_0 = \frac{\mu^3 \rho^2}{2\Lambda^3} = \frac{\mu^5}{32\Lambda^5} < \rho.$$

The procedure evaluates $T(x)$ exactly at the rational query $x \in Q_0$; it accepts if $\|T(x)\|^2 \leq (\mu\rho)^2$ and otherwise returns $a = T(x)$. Evaluation takes polynomial time because the degree is at

most L . Since $\|p\|_\infty \leq 2^{3L}$ and $\delta_0 < 1$, the ball $\bar{B}(p, \delta_0)$ lies in Q_0 . If x is accepted, then $\mu \|x - p\|^2 \leq \langle T(x) - T(p), x - p \rangle \leq \|T(x)\| \|x - p\|$ gives $\|x - p\| \leq \|T(x)\| / \mu \leq \rho$. If x is rejected, then $T(x) \neq 0$. Every point of Q_0 lies in $\bar{B}(0, \sqrt{n}2^{4L}) \subseteq \bar{B}(0, 2^{5L'})$, so $\|T(x)\| = \|T(x) - T(p)\| \leq \Lambda \|x - p\|$ and $\|T(x)\| \leq \Lambda$. Rejection gives $\|x - p\| > \mu\rho/\Lambda$, and for $y \in \bar{B}(p, \delta_0)$,

$$T(x)^\top(y - x) = -\langle T(x) - T(p), x - p \rangle + T(x)^\top(y - p) < -\mu\left(\frac{\mu\rho}{\Lambda}\right)^2 + \Lambda\delta_0 = -\Lambda\delta_0 < 0.$$

So every rejection returns a cut that keeps $\bar{B}(p, \delta_0)$, and the lemma yields a rational x_0 with $\|x_0 - p\| \leq \rho$ in time polynomial in L' . In both cases x_0 is printed in binary and has polynomially many bits.

Newton circuit. Lemma A.4 applies on $\bar{B}(p, \rho)$ with $m = n$ and the constants μ, Λ . Put $t_* = e(L') + 1$ and $\hat{x} = x_{t_*}$. The circuit takes the coordinates of x_0 as rational constants. One Newton step evaluates T and J_T at the current nodes and solves $J_T(x_t)d = T(x_t)$ by Gaussian elimination without exchanges (Lemma A.3); then $x_{t+1} = x_t - d$. For the evaluation, note that every entry of T and J_T (in case (a), of ∇f and $\nabla^2 f$) is a rational combination of monomials X^β obtained from the at most L monomials of the input by removing at most two variables. There are at most $L(1 + n + n^2)$ such monomials, each of degree at most L , and each is computed from the coordinate nodes by at most L multiplications. Thus one step adds $O(L^4 + n^3)$ nodes and rational constants of polynomial bit length, and the whole circuit has $O(L^2(L^4 + n^3))$ nodes. All divisions are by elimination pivots, which are positive, or occur inside rational constants. Earlier steps are referenced, never copied, so the expanded fractions are never formed.

Accuracy. By Lemma A.4, $\|\hat{x} - p\| \leq (2\mu/\Lambda)2^{-3 \cdot 2^{t_*}}$, and all iterates lie in $\bar{B}(p, \rho)$. For an observable h of encoding length at most L' , the segment from p to $\hat{x}$ lies in this ball, where $\|\nabla h\| \leq \Lambda$. Writing $e = e(L')$,

$$|h(\hat{x}) - h(p)| \leq \Lambda \|\hat{x} - p\| \leq 2\mu 2^{-6 \cdot 2^e} \leq 2^{L+1-6 \cdot 2^e} \leq 2^{-2^e-3} = g_{L'}/8,$$

where the last inequality uses $5 \cdot 2^e \geq L + 4$.

Separation. Let δ_T be the product of the denominators of the coefficients of f (case (a)) or of each component T_i (case (b)), and δ_h the product of the denominators of the coefficients of h . Consider the formula with quantified variables $X \in \mathbb{R}^n$ and free variable z ,

$$\delta_T T_1(X) = 0, \dots, \delta_T T_n(X) = 0, \quad \delta_h z - \delta_h h(X) = 0,$$

with δ_T chosen per component in case (b). Its polynomials have integer coefficients: in case (a) they are $\delta_T c_\alpha \alpha_i$ with $|\delta_T c_\alpha \alpha_i| \leq L2^{2L} < 2^{3L}$, in case (b) they are at most 2^{2L} , and those of the last equation are at most $2^{2L'}$. Hence their bit sizes are at most $\tau = 4L'$, their degrees at most $d = \max\{2, \deg T, \deg h\} \leq L'$, and there are $s = n \leq L'$ quantified variables. Since p is the only zero of T , the formula defines the single value $z = h(p)$. Lemma 4.1(b) gives $h(p) = 0$ or

$$|h(p)| \geq 2^{-8L' \cdot L'^{c_0 L'}} \geq 2^{-2^{e(L')}} ,$$

because $\log_2(8L') + c_0 L' \log_2 L' \leq 3L' + c_0 L'^2 \leq (c_0 + 3)L'^2$ for $L' \geq 2$. This proves Lemma 4.3. $\square$

A.7 Proofs of Theorems 4.4 and 4.7

Proof of Theorem 4.4. Case (a). On input $(f, h, \mu, \bowtie)$ of length L , construct the circuit $\hat{x}$ of Lemma 4.3(a) with $L' = L$. Append the nodes of $\hat{\alpha} = h(\hat{x})$, computed monomial by monomial, and the node $g = g_L$, obtained from the constant $1/2$ by $e(L)$ squarings. Append the expression of Lemma A.2 for the relation $\bowtie$. Apply Lemma 4.2 and output the integer circuit whose output is the numerator of the final node.

On a promised input, Lemma 4.3 gives $h(p) = 0$ or $|h(p)| \geq g$, and $|\hat{\alpha} - h(p)| \leq g/8$, so the final rational node is positive exactly when $h(p) \bowtie 0$ by Lemma A.2. All divisions in the rational circuit are by nonzero values: elimination pivots, the constant 2 in $1/2$, and denominators of rational constants. Hence Lemma 4.2 shows that the output integer has the same sign. The construction makes no oracle calls and uses polynomial time. The warm start of Lemma 3.1 is run with a clock equal to its polynomial time bound, so that the algorithm also halts in polynomial time on inputs that violate the promise; its output is then irrelevant.

Case (b). First check the certificate. The list w must consist of distinct monomials $X^{\beta}v_i$ and contain $v_1, \dots, v_n$, and M must be a symmetric rational $m \times m$ matrix. Expand $v^T \nabla^2 f(X) v = \sum_{i,j} v_i v_j \partial_i \partial_j f(X)$, a sparse polynomial with at most n^2 times as many terms as f , and $w^T M w = \sum_{k,l} M_{kl} w_k w_l$, a list of m^2 monomials; collect terms and compare coefficients. Test $M \succ 0$ by symmetric Gaussian elimination in exact rational arithmetic: $M \succ 0$ exactly when all pivots are positive, and the pivots are ratios of leading principal minors, so all intermediate numbers have polynomial length. If a check fails, output the circuit 0. Otherwise put $\mu = \det M / (\text{tr } M)^{m-1}$. By Lemma 2.7, $0 < \mu \leq \lambda_{\min}(M)$. For real X and v ,

$$v^T \nabla^2 f(X) v = w^T M w \geq \mu \|w(X, v)\|^2 \geq \mu \|v\|^2,$$

because every v_i is a coordinate of $w(X, v)$. Thus (f, h, μ) satisfies the promise of case (a), and its encoding length is polynomial in L ; apply case (a). $\square$

Proof of Theorem 4.7. Case (a) is the same as in Theorem 4.4, using Lemma 4.3(b); existence and uniqueness of p are part of that lemma. The ellipsoid procedure is run with a clock as well; if it ends without an accepted point, which cannot happen on promised inputs, the algorithm uses $x_0 = 0$. In case (b), the certificate is checked as before, with $v^T J_T(X) v = \sum_{i,j} v_i v_j \partial_i \partial_j T_i(X)$. A valid certificate gives $v^T J_T(x) v \geq \mu \|v\|^2$ for all x, v , and then

$$\langle T(x) - T(y), x - y \rangle = \int_0^1 (x - y)^T J_T(y + \theta(x - y)) (x - y) d\theta \geq \mu \|x - y\|^2,$$

which is the promise of case (a). $\square$

The Newton step in the monotone case solves the nonsymmetric system $J_T(x_t)d = T(x_t)$ directly. One can instead solve the symmetric positive definite system $J_T^T J_T d = J_T^T T$ by symmetric elimination; both have the same solution. Symmetric elimination must not be applied to J_T itself.

A.8 Proof of Proposition 4.9

Let L be the input length, $D = \max\{2, \deg f, \deg h\} \leq L$, and write the rows of A as a_i^T . All steps before the final circuit are ordinary rational computations of polynomial bit complexity; every algorithmic step is run with a clock, so that inputs violating the promises also finish in polynomial time.

Step 1: feasibility and a containing box. Decide whether $P = \emptyset$ by linear programming; if so, output the circuit 0. Otherwise compute a rational $\bar{x} \in P$ of polynomial encoding length. Lemma 2.10 with $K = P$ gives $\|p_P - \bar{x}\| \leq 2\|\nabla f(\bar{x})\|/\mu < R_P$ with $R_P = 1 + 2\|\nabla f(\bar{x})\|_1/\mu$. Let $B_P = \{x : \|x - \bar{x}\|_\infty \leq R_P\}$ and $B'_P = \{x : \|x - \bar{x}\|_\infty \leq R_P + 1\}$; then p_P lies in the interior of B_P , and every point within distance one of p_P lies in B'_P .

Step 2: constants. Put $R_2 = \|\bar{x}\|_\infty + R_P + 1$. As in the proof of Lemma A.1, the rationals

$$\Lambda_f = n^2 D^3 \|f\|_1 R_2^D, \quad \Lambda_h = nD \|h\|_1 R_2^D$$

bound $\|\nabla^3 f\|$ and $\|\nabla h\|$ on B'_P ; they have polynomial encoding length because $D \leq L$. Multiply each row of (A, b) by the product of its denominators to obtain an integer matrix $(\bar{A}, \bar{b})$ with the same rows up to positive factors. Each integer row of $\bar{A}$ has Euclidean norm at most 2^{ℓ_i} , where ℓ_i is the encoding length of row i , so by Hadamard's inequality every minor of $\bar{A}$ has absolute value at most 2^L . Consequently, for every set I of independent rows and every invertible column block $\bar{A}_{I,B}$, Cramer's rule bounds every entry of $\bar{A}_{I,B}^{-1} \bar{A}_{I,N}$ by 2^L . Row scaling by positive factors cancels in this product, so $A_{I,B}^{-1} A_{I,N} = \bar{A}_{I,B}^{-1} \bar{A}_{I,N}$, and the chart matrix Z of Step 5 below satisfies $\|Z\| \leq \|Z\|_F \leq \zeta := 2^{2L+1}$. Put

$$\Lambda_\varphi = \zeta^3 \Lambda_f + \zeta \Lambda_h + \mu \zeta, \quad \rho_\varphi = \frac{\mu}{4\Lambda_\varphi}, \quad \sigma = \max\{1, \max_i \|a_i\|_1\},$$

$$\varepsilon = \min\left\{1, \frac{\mu \delta^2}{128\sigma^2}, \frac{\mu \rho_\varphi^2}{2}\right\}.$$

Then $\zeta \rho_\varphi \leq 1/4$, and all these numbers have polynomial encoding length.

Step 3: an approximate minimizer. Apply Lemma 3.1 to f on the polyhedron $P \cap B_P$, whose minimizer is p_P , with accuracy ε . The output $\tilde{x} \in P$ satisfies $f(\tilde{x}) \leq f(p_P) + \varepsilon$. Lemma 2.10 with $K = P$ gives $\frac{\mu}{2} \|\tilde{x} - p_P\|^2 \leq f(\tilde{x}) - f(p_P)$, hence

$$\|\tilde{x} - p_P\| \leq \sqrt{2\varepsilon/\mu} \leq \min\{\delta/(8\sigma), \rho_\varphi\}.$$

Step 4: the active set. For every row, $|a_i^\top(\tilde{x} - p_P)| \leq \|a_i\|_1 \|\tilde{x} - p_P\| \leq \delta/8$. A row that is active at p_P therefore has slack at most $\delta/8$ at $\tilde{x}$, and by the promise an inactive row has slack at least δ at p_P and at least $7\delta/8$ at $\tilde{x}$. Hence the active set is exactly $S = \{i : b_i - a_i^\top \tilde{x} < \delta/2\}$, computed by exact rational comparisons.

Step 5: restriction to the active affine space. Let $E = \{x : A_S x = b_S\}$. If $d \in \ker A_S$, then $p_P \pm \theta d \in P$ for all small $\theta > 0$, because the rows in S stay tight and the other rows have positive slack. First-order optimality gives $\pm \theta \nabla f(p_P)^\top d \geq 0$, so $\nabla f(p_P) \perp \ker A_S$. Hence p_P is a stationary point of the strictly convex restriction of f to E , and it is the unique minimizer of f on E .

Choose a maximal independent set $I \subseteq S$ of rows by rational elimination; let $r = |I|$. Every row in S is a linear combination of rows in I , and p_P satisfies all of them, so $E = \{x : A_I x = b_I\}$. If $r = n$, then $p_P = A_I^{-1} b_I$ is rational; evaluate $h(p_P)$ exactly and output the circuit 1 or 0 according to whether $h(p_P) \bowtie 0$. If $r < n$, choose columns B with $A_{I,B}$ invertible and let N be the other columns. The chart $x_N = y$, $x_B = A_{I,B}^{-1}(b_I - A_{I,N}y)$ writes $E = \{\bar{x}_E + Zy\}$, where the rows of Z indexed by N form an identity matrix. Hence $Z^\top Z \succeq I$. The function $\varphi(y) = f(\bar{x}_E + Zy)$ satisfies $\nabla^2 \varphi(y) = Z^\top \nabla^2 f(\bar{x}_E + Zy) Z \succeq \mu I$, and its unique minimizer is $y^* = (p_P)_N$. The case $S = \emptyset$ is covered with $r = 0$, $Z = I$ and $\bar{x}_E = 0$.

Step 6: Newton steps in the chart. Apply Lemma A.4 to $T_\varphi(y) = Z^\top \nabla f(\bar{x}_E + Zy)$ with $J_\varphi(y) = Z^\top \nabla^2 f(\bar{x}_E + Zy) Z$, the moduli μ and Λ_φ , and $x_0 = y_0 := \tilde{x}_N$. Then $\|y_0 - y^*\| \leq$

$\|\tilde{x} - p_P\| \leq \rho_\varphi$. For y in $\bar{B}(y^*, \rho_\varphi)$, the point $\bar{x}_E + Zy$ is within $\zeta\rho_\varphi \leq 1/4$ of p_P and so lies in B'_P . Therefore $\|J_\varphi(y) - J_\varphi(y')\| \leq \zeta^3\Lambda_f\|y - y'\|$ on that ball, and the observable $\eta(y) = h(\bar{x}_E + Zy)$ has $\|\nabla\eta\| \leq \zeta\Lambda_h \leq \Lambda_\varphi$ there. The chart is not expanded into monomials: one step computes $x = \bar{x}_E + Zy_t$, the sparse derivatives $\nabla f(x)$ and $\nabla^2 f(x)$ as in Appendix A.6, the products with Z and $Z^\top$, and one elimination without exchanges, whose pivots are positive by Lemma A.3.

Step 7: separation. Consider the formula with quantified variables $(x, \lambda) \in \mathbb{R}^{n+r}$ and free variable z ,

$$\bar{A}_I x - \bar{b}_I = 0, \quad \delta_f \nabla f(x) - \bar{A}_I^\top \lambda = 0, \quad \delta_h z - \delta_h h(x) = 0,$$

where δ_f, δ_h clear the denominators of f and h . A pair (x, λ) satisfies the first two groups exactly when $x \in E$ and $\nabla f(x)$ lies in the row space of A_I , that is, when x is the stationary point p_P of f on E ; a multiplier exists at p_P . So the formula defines the single value $h(p_P)$. It has $s = n + r \leq 2L$ quantified variables, degree at most $D \leq L$, and coefficient bit sizes at most $4L$. Lemma 4.1(b) gives $h(p_P) = 0$ or $|h(p_P)| \geq 2^{-8L \cdot L^{2c_0L}} \geq g := 2^{-2^{e_2(L)}}$ with $e_2(L) = (2c_0 + 3)L^2$.

Step 8: the sign. Run $t_* = e_2(L) + 1$ Newton steps and put $\hat{\alpha} = \eta(y_{t_*})$. By Lemma A.4, $|\hat{\alpha} - h(p_P)| \leq \Lambda_\varphi \|y_{t_*} - y^*\| \leq 2\mu 2^{-6 \cdot 2^{e_2(L)}} \leq g/8$. Lemmas A.2 and 4.2 give the integer circuit, as in the proof of Theorem 4.4. All divisions in the rational circuit are by positive pivots or inside constants, and its size is polynomial in L . The set S computed in Step 4 is the active set. $\square$

A.9 Proof of Theorem 4.10

Magnitudes. Order the nodes of C topologically. On a Boolean input all node values are integers. Inputs and constants have absolute value at most $2 = 2^0$. If the first k nodes are bounded by $\beta_k \geq 2$ in absolute value, a sum or difference is bounded by $2\beta_k \leq \beta_k^2$, a product by β_k^2 , and a threshold value by 1. Hence every value is bounded by

$$B = 2^{2^S}.$$

This uses binary gates and counts every node of C , including inputs, constants, and the Boolean control logic of part (b).

Range compression. For $\sigma \geq 1$ let $F_\sigma(x) = 2\sigma x/(\sigma + x^2)$. If $1 \leq |x| \leq \sigma$, then $F_\sigma(x)$ has the sign of x and $1 \leq |F_\sigma(x)| \leq \sqrt{\sigma}$. Indeed, F_σ is odd with positive denominator; for $t \in [1, \sigma]$,

$$2\sigma t - (\sigma + t^2) = (t - 1)(\sigma - t) + (\sigma - 1)t \geq 0,$$

which gives $F_\sigma(t) \geq 1$, and $\sigma + t^2 \geq 2\sqrt{\sigma}t$ gives $F_\sigma(t) \leq \sqrt{\sigma}$. Let $\sigma_j = 2^{2^j}$, so $\sqrt{\sigma_j} = \sigma_{j-1}$, and these constants are shared by repeated squaring from 2. If $1 \leq |x| \leq \sigma_J$, applying $F_{\sigma_J}, F_{\sigma_{J-1}}, \dots, F_{\sigma_1}$ yields a number of the same sign with absolute value in $[1, 2]$.

Sign refinement. Let $G(x) = 2x/(1 + x^2)$. For $t \in [1, 2]$, $G(t) \in [4/5, 1]$. For $t \in [0, 1]$, $G(t) \geq t$ and

$$1 - G(t) = \frac{(1 - t)^2}{1 + t^2} \leq (1 - t)^2.$$

So if $|x| \in [1, 2]$ and G is applied $k \geq 1$ times, the result y has the sign of x and $1 - |y| \leq (1/5)^{2^{k-1}} \leq 2^{-2^{k-1}}$.

Threshold replacement. Put $\varepsilon = 2^{-2^{2S+4}}$. Let v be the exact integer input of a threshold gate and $\tilde{v}$ a rational number with $|\tilde{v} - v| \leq 1/4$. Form $x = 4\tilde{v} - 2$. If $v \geq 1$ then $x \geq 1$, and if $v \leq 0$ then $x \leq -1$; also $|x| \leq 4B + 3 \leq B^4 = \sigma_{S+2}$, using $B \geq 4$. Apply $F_{\sigma_{S+2}}, \dots, F_{\sigma_1}$, then G exactly $2S + 5$ times, and return $\tilde{H} = (1 + y)/2$. By the two previous paragraphs, $|\tilde{H} - H(v)| \leq \frac{1}{2} 2^{-2^{2S+4}} \leq \varepsilon$.

Error induction. Replace every arithmetic gate of C by the same operation on the approximate values, and every threshold gate by the replacement above, applied to the approximate value of its input. Inputs and constants are exact. Let Δ_k be the largest error among the first k nodes and put $\kappa = 3B$. We show $\Delta_k \leq \varepsilon \kappa^k$; it holds for input and constant nodes. Note first that

$$\varepsilon \kappa^S < \frac{1}{4},$$

because $\log_2(\varepsilon \kappa^S) \leq -2^{2S+4} + S(2S + 2) < -2$ for $S \geq 1$. At an addition or subtraction the error is at most $2\Delta_{k-1}$. At a multiplication with exact inputs a, b and errors at most $\Delta_{k-1} \leq 1$,

$$|\tilde{a}\tilde{b} - ab| \leq |a| |\tilde{b} - b| + |\tilde{b}| |\tilde{a} - a| \leq (2B + 1)\Delta_{k-1} \leq \kappa \Delta_{k-1}.$$

At a threshold gate the input error is at most $\Delta_{k-1} < 1/4$, so the output error is at most ε . In all cases $\Delta_k \leq \varepsilon \kappa^k$. Errors are absolute, so cancellation and reuse of a node twice are covered. The approximate output therefore lies below $1/4$ when $C(u) = 0$ and above $3/4$ when $C(u) = 1$.

Positive denominators. Represent every approximate value by integer circuits (P, Q) with $Q > 0$, starting from $(u_i, 1)$ and the constants $(0, 1), (1, 1)$. Arithmetic gates use the rules of Appendix A.3, without division. The threshold replacement uses

$$\begin{aligned} 4\tilde{v} - 2 : (4P - 2Q, Q), & \quad \frac{1+y}{2} : (Q + P, 2Q), \\ F_\sigma(P/Q) = \frac{2\sigma PQ}{\sigma Q^2 + P^2}, & \quad G(P/Q) = \frac{2PQ}{Q^2 + P^2}, \end{aligned}$$

and every new denominator is positive because $Q > 0$ and $\sigma > 0$. If the final pair is (P, Q) , output the integer circuit $C' = 2P - Q$. Then $C'(u) > 0$ exactly when $P/Q > 1/2$, that is, when $C(u) = 1$. Each threshold gate costs $O(S)$ nodes, the constants σ_j cost $O(S)$ shared nodes, and each arithmetic gate costs $O(1)$ nodes. So C' has $O(S^2)$ nodes and is built in time polynomial in S , without evaluating any value. This proves (a).

Part (b): unrolling an oracle machine. Fix an encoding of integer circuits as binary strings, for instance a list of instructions, each of which is a constant 0 or 1 or an operation $(\circ, i, j)$ with $\circ \in \{+, -, \times\}$ and earlier indices i, j , the last instruction being the output; a string that encodes no circuit is not in PosSLP. Let the machine use a query state in which it moves to a yes or no state according to whether its oracle tape holds a string of PosSLP. Let $t = t(L)$ be a polynomial bound on its running time on inputs of length L , valid for every sequence of oracle answers; if only a bound for the true oracle is known, add a clock that rejects after t steps, which does not change the decided problem.

Encode the configuration after $\theta \leq t$ steps by Boolean values: for each tape, cell position at most t and symbol, whether the cell holds the symbol; for each tape and position, whether the head is there; and for each state, whether it is the current state. The symbol under each head is the disjunction over positions of “the head is here and the cell holds this symbol”, which takes $O(t)$ gates per tape. Every other value of the next configuration is a Boolean function of the current

state, the symbols under the heads, a bounded number of neighboring cell and head values, and, in the query state, one answer bit. Halting states are kept unchanged. Implement Boolean operations by $\neg a = 1 - a$, $a \wedge b = ab$ and $a \vee b = a + b - ab$, which are exact on bits. This gives $O(t)$ nodes per tape and step, plus the query modules below. The input nodes are the L input bits.

At every step θ attach a query module that computes the bit $a_\theta = [\text{oracle tape content} \in \text{PosSLP}]$; the transition uses a_θ only when the state is the query state. The module works as follows.

- (i) A string of length at most t encodes at most t instructions. A deterministic polynomial-time algorithm maps the tape content to a validity bit, and for each slot $k \leq t$ to an opcode in $\{\text{absent}, 0, 1, +, -, \times\}$, two operand indices below k in binary, and the index of the output instruction. Unrolling this algorithm in the same way as the machine gives a Boolean circuit of size polynomial in t .
- (ii) Equality tests on the binary indices give bits $\epsilon_{k,i}$ and $\epsilon'_{k,i}$ ($i < k$) for the first and second operand of slot k , bits o_k for the output slot, and one-hot opcode bits w_k (constant one), s_k^+ , s_k^- , $s_k^\times$. All of them are 0 for absent or invalid slots.
- (iii) Define integer nodes

$$Y_k = \sum_{i < k} \epsilon_{k,i} V_i, \quad Z_k = \sum_{i < k} \epsilon'_{k,i} V_i, \quad V_k = w_k + s_k^+(Y_k + Z_k) + s_k^-(Y_k - Z_k) + s_k^\times Y_k Z_k,$$

$$\text{and } a_\theta = \text{valid} \cdot H(\sum_k o_k V_k).$$

If the string encodes a circuit, induction on k shows that V_k is the value of instruction k , so a_θ is the correct answer. If it encodes none, $a_\theta = 0$, which is also correct. Every node value is an integer, and the module has $O(t^2)$ nodes besides the parser. The module enumerates addresses, not sequences of earlier answers, so the total number of nodes is polynomial in t although the queries depend on earlier answers. Products $Y_k Z_k$ are formed even when they are not selected; they are nodes of the circuit and are covered by the bound B .

The output of the unrolled circuit is the acceptance bit after t steps, a Boolean value. Part (a) gives an integer circuit C' of polynomial size with $C'(u) > 0$ exactly when the machine accepts u . Replacing the input nodes by the constants given by the actual input yields one integer circuit, which is the required PosSLP instance. On inputs satisfying the promise, the machine answers correctly, and so does the instance. Conversely, a many-one reduction to PosSLP is a P^{PosSLP} algorithm with one query.

Part (c). Build one circuit containing $C_1, \dots, C_k$ with their shared nodes, a threshold gate on each output, and the gates of β implemented arithmetically as above. Its values are integers and its output is a bit, so part (a) applies. $\square$

B Proofs for the lower reductions

This appendix proves the results of Section 5. All algorithms use exact rational arithmetic. Unless stated otherwise, norms of vectors are Euclidean, norms of matrices are operator norms, and $\|\cdot\|_F$ is the Frobenius norm. For a rational polynomial g of degree at most two, $\|g\|_1$ denotes the sum of the absolute values of its coefficients in the monomial basis. Two elementary facts are used repeatedly. First, for a real $n \times n$ matrix J the product of the singular values is $|\det J|$ and each singular value is at most $\|J\|_F$, so

$$\sigma_{\min}(J) \geq |\det J| / \|J\|_F^{n-1}. \quad (32)$$

Second, if g is a quadratic with $g(p) = 0$, write $g(p+u) = \ell^\top u + u^\top H u$ with H symmetric. Then

$$\|H\| \leq \|g\|_1, \quad \|\ell\| \leq 2 \max\{1, \|p\|_\infty\} \|g\|_1. \quad (33)$$

Indeed, a monomial $x_a x_b$ contributes a symmetric matrix of norm at most one and a gradient $x_b e_a + x_a e_b$ of norm at most $2\|p\|_\infty$ at p , while a monomial x_a contributes no quadratic part and a gradient of norm one.

B.1 Certified odd-root circuits

Lemma B.1 (Normalization). *Let a certified odd-root circuit be given, let σ_i be the sign of the box of gate i , and let $\kappa = \max\{1, \max_i 1/\min(|\underline{\xi}_i|, |\bar{\xi}_i|)\}$. Then $\alpha_i = \sigma_i \kappa \xi_i$ satisfy*

$$\alpha_i^{d_i} = c'_i + \sum_{j < i} \sum_{e=1}^{n_j} a'_{ije} \alpha_j^e, \quad c'_i = \sigma_i \kappa^{d_i} c_i, \quad a'_{ije} = \sigma_i \sigma_j^e \kappa^{d_i - e} a_{ije}, \quad (34)$$

and the circuit with these coefficients and the boxes $[\underline{\alpha}_i, \bar{\alpha}_i] = \sigma_i \kappa [\underline{\xi}_i, \bar{\xi}_i]$ is a certified odd-root circuit with $\underline{\alpha}_i \geq 1$ for every i . Its encoding length is polynomial in that of the original circuit.

Proof. Since d_i is odd, $\sigma_i^{d_i} = \sigma_i$, and $\xi_j^{d_i} = \sigma_j^e \kappa^{-e} \alpha_j^e$. Multiplying (5) by $\sigma_i \kappa^{d_i}$ gives (34). For the interval test, the map $s \mapsto \sigma_j \kappa s$ sends $R_{j,e}$ to $R'_{j,e} = \sigma_j^e \kappa^e R_{j,e}$, so $a'_{ije} R'_{j,e} = \sigma_i \kappa^{d_i} a_{ije} R_{j,e}$ as sets. Hence the left side of the new test is $\sigma_i \kappa^{d_i}$ times the left side of the old one, and the new target $[\underline{\alpha}_i^{d_i}, \bar{\alpha}_i^{d_i}]$ is $\sigma_i \kappa^{d_i} [\underline{\xi}_i^{d_i}, \bar{\xi}_i^{d_i}]$, because odd powers are increasing. Inclusion is preserved, and $\underline{\alpha}_i = \kappa \min(|\underline{\xi}_i|, |\bar{\xi}_i|) \geq 1$. The exponents $|d_i - e|$ are at most $2 \sum_i n_i$, and the degrees are written in unary, so all new rationals have polynomial bit length. $\square$

From now on in this subsection the boxes satisfy $\underline{\xi}_i \geq 1$. Fix a rational $A \geq 3$ with

$$A \geq \sum_i \bar{\xi}_i + \sum_i |c_i| + \sum_{i,j,e} |a_{ije}|, \quad (35)$$

for example $A = 3$ plus the right side. Then $1 \leq \xi_i \leq A$, every retained power satisfies $1 \leq \xi_i^e \leq A^N$ with $N = \sum_i n_i$, and $\log_2 A$ is polynomial in the input length.

Lemma B.2 (Certified root approximation). *For a certified odd-root circuit with $\underline{\xi}_i \geq 1$ and a positive integer P , rationals $\hat{\xi}_i$ with $|\hat{\xi}_i - \xi_i| \leq 2^{-P}$ for all i can be computed in time polynomial in the input length and P .*

Proof. Put $C = NA^N$ and $\eta = 2^{-P}/(2(C+2)^k)$. For $i = 1, \dots, k$ we compute rational intervals $J_i = [l_i, u_i] \subseteq [\underline{\xi}_i, \bar{\xi}_i]$ containing ξ_i . Given J_j for $j < i$, form the rational interval $I_i = c_i + \sum_{j < i, e} a_{ije} [l_j^e, u_j^e] = [y_-, y_+]$. It contains $\rho_i(\xi)$, and it is contained in the left side of (4), hence in $[\underline{\xi}_i^{d_i}, \bar{\xi}_i^{d_i}] \subseteq [1, \infty)$. Its width is at most $\sum_{j,e} |a_{ije}| e A^{e-1} (u_j - l_j) \leq C \max_{j < i} (u_j - l_j)$. Bisection of $[\underline{\xi}_i^{d_i}, \bar{\xi}_i^{d_i}]$, with exact comparisons of s^{d_i} against $y_\pm$, yields rationals $l_i \leq y_-^{1/d_i}$ and $u_i \geq y_+^{1/d_i}$ in the box, each within η of the respective root. Since the real d_i -th root has derivative at most $1/d_i < 1$ on $[1, \infty)$,

$$u_i - l_i \leq (y_+ - y_-) + 2\eta \leq C \max_{j < i} (u_j - l_j) + 2\eta,$$

and $\xi_i = \rho_i(\xi)^{1/d_i} \in J_i$. By induction $\max_{j < i} (u_j - l_j) \leq 2\eta(C+2)^i$: this holds for $i = 1$, where I_1 is a point, and the step is $(C+1) \cdot 2\eta(C+2)^{i-1} + 2\eta \leq 2\eta(C+2)^i$. Hence every width is at most 2^{-P} , and $\hat{\xi}_i = l_i$ works. Each bisection takes $O(\log_2(A/\eta))$ steps on rationals whose denominators are those of the box endpoints times powers of two, and the exponents e, d_i are unary; all numbers have polynomially many bits. $\square$

B.2 Near-one boxes and small signals

Lemma B.3 (Near-one boxes and small signals). *Let $Q \geq 1$, $\delta_0 = 1000^{-(Q+3)}$ and $w_i = 1000^i \delta_0$ for $1 \leq i \leq Q$, so that $w_i \leq 10^{-9}$. Consider a cube-root circuit with at most Q gates in which every radicand ρ_i has one of the following forms:*

- (a) *a rational constant in $[1 - 2w_i, 1 + 3w_i]$, for example 1 or $1 + 3\delta_0^2$;*
- (b) *$1 + \sum_{j < i} \lambda_{ij}(\xi_j - 1)$ with rational λ_{ij} and $\sum_j |\lambda_{ij}| \leq 6$;*
- (c) *$3\xi_j^2 - 6\xi_j + 4$ for some $j < i$.*

Then the boxes $[1 - w_i, 1 + w_i]$ pass the direct interval test, so every gate value lies in its box. Moreover, $\text{sq}(z) = (1 + 3z^2)^{1/3} - 1$ satisfies $0 < \text{sq}(z) \leq z^2$ for $z > 0$. Hence a chain consisting of a gate with radicand $1 + 3\delta_0^2$ followed by $r - 1$ gates of form (c), each applied to its predecessor, has values $\xi_i = 1 + \delta_i$ with $0 < \delta_i \leq \delta_{i-1}^2$, and in particular $0 < \delta_r \leq \delta_0^{2^r}$.

Proof. Let $w = w_{i-1}$ for $i \geq 2$; all earlier boxes have half-width at most w . For form (b), the direct interval evaluation is $1 + \sum_j \lambda_{ij}[-w_j, w_j] \subseteq [1 - 6w, 1 + 6w]$; combining repeated occurrences of a predecessor into one coefficient can only decrease the coefficient sum. For form (c), the ranges $[(1 - w)^2, (1 + w)^2]$ of ξ_j^2 and $[1 - w, 1 + w]$ of ξ_j give the interval $[1 - 12w + 3w^2, 1 + 12w + 3w^2] \subseteq [1 - 13w, 1 + 13w]$. Since $w_i = 1000w$ and $w_i \leq 1/3$,

$$(1 - w_i)^3 \leq 1 - 3w_i + 3w_i^2 \leq 1 - 2w_i \leq 1 - 13w, \quad (1 + w_i)^3 \geq 1 + 3w_i \geq 1 + 13w,$$

so the test passes. For form (a) the same two displayed inequalities show $(1 - w_i)^3 \leq 1 - 2w_i$ and $1 + 3w_i \leq (1 + w_i)^3$; this also covers $i = 1$.

For $z > 0$, $(1 + z^2)^3 > 1 + 3z^2$ gives $\text{sq}(z) < z^2$, and $\text{sq}(z) > 0$ is clear. The first chain gate has $\xi = (1 + 3\delta_0^2)^{1/3} = 1 + \text{sq}(\delta_0)$. A gate of form (c) applied to $\xi_j = 1 + \delta_j$ has radicand $1 + 3\delta_j^2$, so its value is $1 + \text{sq}(\delta_j)$. Induction gives the chain bounds. $\square$

B.3 The analytic gadgets

For complex z with $|z| < 1/3$ let $(1 + 3z)^{1/3}$ be the principal branch; it is analytic there and equals the real cube root for real z . Define ϕ , sq , θ and mul as in Section 5.3.

Lemma B.4 (Analytic gadgets).

- (i) *For complex $|z| \leq 1/12$, $|\phi(z)| \leq 2|z|$. For real $|z| \leq 1/12$, $|\phi(z) - z| \leq 2z^2$.*
- (ii) *The function mul is analytic on a neighborhood of the closed polydisk $\{|x| \leq \frac{1}{100}, |y| \leq \frac{1}{100}\}$ and bounded by one there. It satisfies $\text{mul}(x, 0) = \text{mul}(0, y) = 0$, and its Taylor coefficient of xy is one.*
- (iii) *For complex $|x|, |y| \leq \frac{1}{400}$,*

$$|\text{mul}(x, y) - xy| \leq 2^{24} |xy| (|x| + |y|).$$

Proof. (i) For $|\zeta| \leq 1/12$ we have $|1 + 3\zeta| \geq 3/4$, so $|\phi'(\zeta)| = |1 + 3\zeta|^{-2/3} \leq (4/3)^{2/3} < 2$ and $|\phi''(\zeta)| = 2|1 + 3\zeta|^{-5/3} \leq 2(4/3)^{5/3} < 4$. Integrating ϕ' along the segment from 0 to z gives the first bound. For real z , Taylor's theorem with $\phi(0) = 0$ and $\phi'(0) = 1$ gives the second.

(ii) If $|x|, |y| \leq \frac{1}{100}$, then $|\pm(x \pm y)| \leq \frac{1}{50}$, so $|\phi(\pm(x \pm y))| \leq \frac{1}{25}$ by (i); its square has modulus at most 0.0016, so sq of it has modulus at most 0.0032, and hence $|\theta(x \pm y)| \leq 0.0032$. The

argument of the outer ϕ has modulus at most 0.0016, and $|\text{mul}| \leq 0.0032$. All arguments stay strictly inside the disks where the bounds of (i) hold, so the same holds on a slightly larger polydisk, where every composition is analytic. The function θ is even, so $\theta(x+0) - \theta(x-0) = 0$ and $\theta(0+y) - \theta(0-y) = 0$; since $\phi(0) = 0$, mul vanishes on both axes. From $\phi(z) = z - z^2 + O(z^3)$ we get $\text{sq}(\phi(z)) = \phi(z)^2 + O(z^4) = z^2 - 2z^3 + O(z^4)$, so $\theta(z) = z^2 + O(z^4)$ and $\frac{1}{4}[\theta(x+y) - \theta(x-y)] = xy + O(|(x,y)|^4)$. Applying ϕ gives $\text{mul}(x,y) = xy + O(|(x,y)|^4)$.

(iii) Write $\text{mul}(x,y) = \sum_{i,j \geq 0} m_{ij} x^i y^j$. By (ii) and Cauchy's inequality on the polydisk of radius $r = \frac{1}{100}$, $|m_{ij}| \leq r^{-i-j}$. By (ii), $m_{i0} = m_{0j} = 0$ for all i, j and $m_{11} = 1$. With $s = |x|/r \leq \frac{1}{4}$ and $t = |y|/r \leq \frac{1}{4}$,

$$|\text{mul}(x,y) - xy| \leq \sum_{\substack{i,j \geq 1 \\ (i,j) \neq (1,1)}} s^i t^j = st \frac{s+t-st}{(1-s)(1-t)} \leq \frac{16}{9} st(s+t) = \frac{16}{9r^3} |xy| (|x| + |y|).$$

Finally $\frac{16}{9} \cdot 10^6 < 2^{24}$. □

The bound in (iii) is proportional to $|xy|$. A bound of the form $O((|x| + |y|)^3)$ would not suffice: when one input is much smaller than the other, it can exceed the product itself. The exact vanishing on the axes is what removes the terms x^i and y^j .

B.4 Proof of Theorem 5.5

Signals. Fix $0 < \delta < 1$. For an integer $d \geq 1$, an integer C and a real $\Theta \geq 2$, a real number s is a (d, C, Θ) -signal if $|C| \leq \Theta$ and $|s - C\delta^d| \leq \Theta\delta^{d+1}$. The integer C may be zero. A signal s is stored as the root $1 + s$.

Lemma B.5 (Signal arithmetic). *Suppose $\Theta\delta \leq 2^{-30}$, and let $\Theta' = 2^{30}\Theta^3$.*

- (i) *Every (d, C, Θ) -signal satisfies $|s| \leq 2\Theta\delta^d \leq 2^{-29}$.*
- (ii) *If s, s' are (d, C, Θ) - and (d, C', Θ) -signals, then $\phi(s \pm s')$ is a $(d, C \pm C', \Theta')$ -signal.*
- (iii) *If s, s' are (a, C, Θ) - and (b, C', Θ) -signals, then $\text{mul}(s, s')$ is an $(a+b, CC', \Theta')$ -signal.*

Proof. (i) $|s| \leq \Theta\delta^d + \Theta\delta^{d+1} \leq 2\Theta\delta^d \leq 2\Theta\delta$. (ii) Put $z = s \pm s'$. Then $|z - (C \pm C')\delta^d| \leq 2\Theta\delta^{d+1}$ and $|z| \leq 4\Theta\delta^d \leq 2^{-28}$, so Lemma B.4(i) gives $|\phi(z) - z| \leq 2z^2 \leq 32\Theta^2\delta^{2d} \leq 32\Theta^2\delta^{d+1}$. The total error is at most $34\Theta^2\delta^{d+1} \leq \Theta'\delta^{d+1}$, and $|C \pm C'| \leq 2\Theta \leq \Theta'$. (iii) Write $s = C\delta^a + \epsilon$ and $s' = C'\delta^b + \epsilon'$. Then $ss' - CC'\delta^{a+b} = C\delta^a\epsilon' + C'\delta^b\epsilon + \epsilon\epsilon'$ has modulus at most $3\Theta^2\delta^{a+b+1}$. By (i), $|s|, |s'| \leq 2^{-29}$, so Lemma B.4(iii) applies and gives $|\text{mul}(s, s') - ss'| \leq 2^{24} \cdot 4\Theta^2\delta^{a+b} \cdot 4\Theta\delta = 2^{28}\Theta^3\delta^{a+b+1}$, using $|s| + |s'| \leq 4\Theta\delta$. The total is at most $\Theta'\delta^{a+b+1}$, and $|CC'| \leq \Theta^2 \leq \Theta'$. □

No step divides by a leading coefficient or uses a lower bound on $|C|$. Hence integer values equal to zero, exact cancellation, and repeated use of one signal (for example $\text{mul}(s, s)$ or $\phi(s - s) = 0$) need no separate treatment.

The schedule. Build binary constants of the program from 1 by doubling and addition, and append two gates computing $W = 2V - 1$. Every node ν of the resulting program receives a pair of signals (f_ν, g_ν) , a *prescribed* pair of integer coefficients $(V_\nu, 1)$, where V_ν is the integer value of the node, and an order $d_\nu \geq 1$:

- the constant 1 receives (δ, δ) with order one, where δ is the parameter signal defined below; the constant 0 receives (z, δ) , where z is the signal of a gate with constant radicand 1, so $z = 0$;
- a product $\nu = \nu_1 \nu_2$ receives $f_\nu = \text{mul}(f_{\nu_1}, f_{\nu_2})$ and $g_\nu = \text{mul}(g_{\nu_1}, g_{\nu_2})$;
- a sum or difference $\nu = \nu_1 \pm \nu_2$ receives $f_\nu = \phi(\text{mul}(f_{\nu_1}, g_{\nu_2}) \pm \text{mul}(f_{\nu_2}, g_{\nu_1}))$ and $g_\nu = \text{mul}(g_{\nu_1}, g_{\nu_2})$;

and in the last two cases $d_\nu = d_{\nu_1} + d_{\nu_2}$. The prescribed coefficients are consistent: the two products inside the sum have prescribed coefficients V_{ν_1} and V_{ν_2} and the common order $d_{\nu_1} + d_{\nu_2}$, so the sum has prescribed coefficient $V_{\nu_1} \pm V_{\nu_2}$, and the denominator coefficient is $1 \cdot 1$. Thus ϕ is only applied to signals of equal order. Each arithmetic node uses at most four macros, each an application of ϕ or mul ; let T be the number of macros. The orders are proof annotations and are never printed; since an order at most doubles at each arithmetic node, $d_\nu \leq 2^T$. The integers V_ν are never computed.

Raw gates. Each ϕ macro is one root gate with radicand $1 + 3(\xi_s - 1) \pm 3(\xi_{s'} - 1)$, where $\xi_s = 1 + s$. Each $\text{mul}(x, y)$ macro is nine root gates: four gates with radicands $1 \pm 3(\xi_x - 1) \pm 3(\xi_y - 1)$ computing $\phi(\pm(x + y))$ and $\phi(\pm(x - y))$; four square gates computing sq of these four signals; and one gate with radicand $1 + \frac{3}{8}(\zeta_1 + \zeta_2 - \zeta_3 - \zeta_4)$, where ζ_1, ζ_2 are the roots holding $\text{sq}(\phi(x + y))$ and $\text{sq}(\phi(-x - y))$ and ζ_3, ζ_4 those holding $\text{sq}(\phi(x - y))$ and $\text{sq}(\phi(y - x))$. Since $\frac{1}{4}[\theta(x + y) - \theta(x - y)] = \frac{1}{8} \sum_k \pm(\zeta_k - 1)$, the last gate holds exactly $1 + \text{mul}(x, y)$; every other gate likewise holds exactly 1 plus the indicated signal, because real cube roots are taken. Now put

$$r_0 = 2T + 5, \quad r_1 = T + 1, \quad Q = r_0 + r_1 + 9T + 1, \quad \delta_0 = 1000^{-(Q+3)}.$$

The circuit consists of: a parameter chain of r_0 gates, the first with radicand $1 + 3\delta_0^2$ and the others square gates of their predecessors; an auxiliary chain of r_1 square gates continuing from the last parameter gate; one gate with radicand 1; and the raw gates of the T macros, in topological order. There are at most Q gates. Gate o holds f_ν for the node ν computing W , and gate e is the last auxiliary gate. Give gate i the box $[1 - w_i, 1 + w_i]$ with $w_i = 1000^i \delta_0$.

Correctness. Every radicand has form (a), (b) or (c) of Lemma B.3: the coefficient sums of the affine radicands are 6 and $\frac{3}{2}$. Hence the boxes pass the direct interval test. The boxes and the first parameter radicand $1 + 3\delta_0^2$ have $O(Q)$ bits each, and the remaining coefficients have constant size. Since $Q = 12T + 7$ and there are at most Q gates, the whole circuit has size $O(T^2)$ bits and is computed in polynomial time.

Let $\delta = \xi_{r_0} - 1$ be the parameter signal. By Lemma B.3, $0 < \delta \leq \delta_0^{2^{r_0}} < 2^{-4 \cdot 2^{r_0}}$. Let $\Theta_0 = 2$ and $\Theta_t = 2^{30} \Theta_{t-1}^3$; then $\log_2 \Theta_t = 16 \cdot 3^t - 15$, because $\log_2 \Theta_t + 15 = 3(\log_2 \Theta_{t-1} + 15)$. Since $4 \cdot 2^{r_0} = 2^{2T+7} \geq 16 \cdot 3^T + 15 = 30 + \log_2 \Theta_T$, we have $\Theta_T \delta \leq 2^{-30}$.

We claim that after the t -th macro every available signal s with prescribed coefficient C and order d is a (d, C, Θ_t) -signal. Before the first macro the available signals are δ , a $(1, 1, \Theta_0)$ -signal with zero error, and $z = 0$, a $(1, 0, \Theta_0)$ -signal. Each macro applies Lemma B.5 with $\Theta = \Theta_{t-1}$, whose hypothesis holds because $\Theta_{t-1} \delta \leq \Theta_T \delta \leq 2^{-30}$, and earlier signals remain signals for the larger bound Θ_t . This proves the claim; it holds for arbitrary sharing of nodes and signals.

The numerator signal $s_o = \xi_o - 1$ of the node computing W is therefore a (d, W, Θ_T) -signal for some $1 \leq d \leq 2^T$:

$$|s_o - W\delta^d| \leq \Theta_T \delta^{d+1} \leq 2^{-30} \delta^d.$$

As W is a nonzero integer, $s_o \neq 0$, $\text{sign } s_o = \text{sign } W$, and $|s_o| \geq \frac{1}{2}\delta^d$. By Lemma B.5(i), $|s_o| \leq 2^{-29}$. The auxiliary signal $s_e = \xi_e - 1$ continues the chain for $r_1 = T+1$ more square gates, so $0 < s_e \leq \delta^{2^{T+1}}$ by Lemma B.3. Since $d \leq 2^T$, $\delta < 1$ and $\delta \leq 2^{-30}$,

$$s_e^2 \leq \delta^{2^{T+2}} \leq \delta^d \delta^3 \leq 2|s_o| \delta^3 \leq \frac{1}{8}|s_o|.$$

The gates o and e are distinct, since o belongs to a macro and e to the auxiliary chain. This proves (6). $\square$

B.5 Proof of Theorem 5.6

The lower bounds were derived in Section 5.3. For the upper bound, first check the format and the direct interval test, and map invalid inputs to a fixed program with output -1 . Let the valid input be $\xi_i^3 = c_i + \sum_{j < i} (a_{ij}\xi_j + b_{ij}\xi_j^2)$, $1 \leq i \leq k$, with positive boxes, a gate o and a threshold r . Let $L \geq 2$ be its total binary length. Then $k \leq L$, every printed numerator and denominator has absolute value at most 2^L , and the product D of the denominators of all c_i, a_{ij}, b_{ij} and r satisfies $D \leq 2^L$. A positive box endpoint is at least 2^{-L} , so $2^{-L} \leq \xi_i \leq 2^L$.

Separation. The numbers $\beta_i = D\xi_i$ satisfy $\beta_i^3 = D^3c_i + \sum_{j < i} (D^2a_{ij}\beta_j + Db_{ij}\beta_j^2)$, whose coefficients are integers. By induction each β_i is integral over the ring generated by $\beta_1, \dots, \beta_{i-1}$, hence an algebraic integer. The field $K = \mathbb{Q}(\xi_1, \dots, \xi_k)$ has degree at most $3^k \leq 3^L$. For every field embedding $\tau: K \rightarrow \mathbb{C}$, the numbers $\tau(\xi_i)$ satisfy the same gate equations, and $|\tau(\xi_i)| \leq \Xi := (2L+1)2^L$ by induction: if the earlier conjugates are bounded by $\Xi \geq 1$, the radicand has modulus at most $2^L(1 + L\Xi + L\Xi^2) \leq (2L+1)2^L\Xi^2 = \Xi^3$. If $\xi_o \neq r$, then $D(\xi_o - r) = \beta_o - Dr$ is a nonzero algebraic integer in K . Its norm is a nonzero integer, and each of its conjugates has modulus at most $2^L(\Xi + 2^L) \leq 2^{4L+1}$. Isolating the factor of the identity embedding,

$$|\xi_o - r| \geq 2^{-L} 2^{-(4L+1)(3^L-1)} \geq g := 2^{-2^{6L}},$$

because $L + (4L+1)(3^L-1) \leq (4L+2)3^L \leq 2^{2L} \cdot 2^{2L} \leq 2^{6L}$ for $L \geq 2$. The rational g is computed by $6L$ squarings of $\frac{1}{2}$.

Newton steps. Let $z > 0$ have cube root $\eta \in [2^{-L-1}, 2^{L+1}]$, and iterate $x_{t+1} = (2x_t + z/x_t^2)/3$ from $x_0 = 2^{L+1}$. By the inequality of arithmetic and geometric means $x_t \geq \eta$ for all t . For the relative error $\epsilon_t = x_t/\eta - 1 \geq 0$,

$$\epsilon_{t+1} = \frac{\epsilon_t^2(3 + 2\epsilon_t)}{3(1 + \epsilon_t)^2} \leq \min\{\epsilon_t^2, \frac{2}{3}\epsilon_t\}.$$

Since $\epsilon_0 \leq 2^{2L+2}$ and $(\frac{2}{3})^2 < \frac{1}{2}$, after $4L+6$ steps $\epsilon_t \leq \frac{1}{2}$; after $10L$ further steps $|x_t - \eta| \leq \epsilon_N := 2^{L+1-2^{10L}}$.

Propagation. Compute $\hat{\xi}_1, \dots, \hat{\xi}_k$ in order: form $\hat{z}_i = c_i + \sum_{j < i} (a_{ij}\hat{\xi}_j + b_{ij}\hat{\xi}_j^2)$ and apply $14L+6$ Newton steps to $z = \hat{z}_i$. Let $E_i = \max_{j \leq i} |\hat{\xi}_j - \xi_j|$ and $E_0 = 0$. Suppose $E_{i-1} \leq 2^{-7L-1}$. Then $|\hat{\xi}_j| \leq 2^{L+1}$ for $j < i$ and

$$|\hat{z}_i - \xi_i^3| \leq L 2^L (1 + 2^{L+1} + 2^L) E_{i-1} \leq 2^{4L} E_{i-1} \leq 2^{-3L-1}.$$

Since $\xi_i^3 \geq 2^{-3L}$, this gives $\hat{z}_i \geq 2^{-3L-1} > 0$ and $\hat{z}_i \leq 2^{3L+1}$, so its cube root η_i lies in $[2^{-L-1}, 2^{L+1}]$, and the Newton bound applies. The cube root has derivative at most $\frac{1}{3}(2^{-3L-1})^{-2/3} \leq 2^{2L+2}$ between $\hat{z}_i$ and ξ_i^3 , so $|\eta_i - \xi_i| \leq 2^{6L+2}E_{i-1}$ and $E_i \leq 2^{6L+2}E_{i-1} + \varepsilon_N$. By induction $E_i \leq 2^{(6L+3)i}\varepsilon_N \leq 2^{6L^2+4L+1-2^{10L}}$. This is at most 2^{-7L-1} , which justifies the inductive hypothesis at every gate, and also at most $2^{-2^{6L}-3} = g/8$, because $2^{10L} - 2^{6L} \geq 2^{6L} \geq 6L^2 + 11L + 4$ for $L \geq 2$.

The comparison. Put $\hat{\Delta} = \hat{\xi}_o - r$, so $|\hat{\Delta} - (\xi_o - r)| \leq g/8$, and $\xi_o \neq r$ implies $|\xi_o - r| \geq g$. The rational numbers $\hat{\Delta} - g/2$, $\hat{\Delta} + g/2$, $-\hat{\Delta} - g/2$, $-\hat{\Delta} + g/2$ and $g^2/4 - \hat{\Delta}^2$ are positive exactly when, respectively, $\xi_o > r$, $\xi_o \geq r$, $\xi_o < r$, $\xi_o \leq r$ and $\xi_o = r$; in the zero case $|\hat{\Delta}| \leq g/8$, and otherwise $|\hat{\Delta}| \geq 7g/8$. Each is the output of a rational circuit with $O(kL)$ Newton steps, divisions only by the positive numbers x_i^2 , and shared gates. The positive-denominator representation of rational circuits (Definition 2.16) converts it into one integer straight-line program of polynomial size with the same sign. $\square$

B.6 Proof of Theorem 5.7

By Lemma B.1 we may replace the circuit by its normalization. We keep the notation c_i, a_{ije}, ξ_i for the normalized data, write α_i for the normalized gate values, and fix A by (35); thus $1 \leq \alpha_i \leq A$ and the absolute coefficient sum is at most A . The variables are $X_{i,e}$ for $1 \leq e \leq n_i$, and $X_{i,0} = 1$ is a constant. Put $b_i(X) = c_i + \sum_{j < i, e} a_{ije} X_{j,e}$ and let $p \in \mathbb{R}^N$ have $p_{i,e} = \alpha_i^e$. Centered coordinates are $u = X - p$, with gate blocks $u_i = (u_{i,1}, \dots, u_{i,n_i})$.

Lemma B.6 (Residuals). *The N rational quadratics*

$$r_{i,e} = X_{i,1}X_{i,e} - X_{i,e+1} \quad (1 \leq e < n_i), \quad r_{i,n_i} = X_{i,n_i-1}X_{i,n_i} - b_i(X)$$

have p as their unique common real zero. Each has a homogeneous quadratic part of operator norm at most one and a gradient of norm at most $3A^N$ at p . Their Jacobian J at p satisfies $|\det J| = \prod_i d_i \alpha_i^{d_i-1} \geq 1$ and $\|J\|_F \leq 3NA^N$. Consequently, with $\beta = 4NA^N$ and $\nu = \beta^{-(N-1)}$, $J^\top J \succeq \nu^2 I$.

Proof. If all residuals vanish, the chain equations give $X_{i,e} = X_{i,1}^e$, and then $X_{i,1}^{2n_i-1} = b_i(X)$, whose right side depends only on earlier gates. By induction over i , $b_i(X) = \alpha_i^{d_i}$, and the unique real root of odd degree gives $X_{i,1} = \alpha_i$. Conversely p is a zero.

The quadratic parts are $X_{i,1}^2$, $X_{i,1}X_{i,e}$ or $X_{i,n_i-1}X_{i,n_i}$, with symmetric matrices of norm one or one half. Ordering variables and residuals by gates, J is block lower triangular, since $r_{i,e}$ involves only gate i and, through b_i , earlier gates. Fix a gate and write $\alpha = \alpha_i$, $n = n_i$, $d = d_i$. Along the curve $t \mapsto (t, t^2, \dots, t^n)$ in the gate's own coordinates, with earlier gates fixed at p , every chain residual vanishes identically and the terminal one equals $t^d - \alpha^d$. Hence $J_i w = (0, \dots, 0, d\alpha^{d-1})^\top$ for the diagonal block J_i and the tangent $w = (1, 2\alpha, \dots, n\alpha^{n-1})^\top$. Replacing the first basis vector by w is a change of basis of determinant one. In the new basis the first column of J_i is $d\alpha^{d-1}$ times the last unit vector, and the minor formed by the first $n-1$ rows and the last $n-1$ columns is triangular with diagonal entries -1 . Expanding along the first column gives $|\det J_i| = d\alpha^{d-1} \geq 1$.

At p , the chain row $e = 1$ has entries 2α and -1 ; a chain row $e \geq 2$ has entries α^e , α and -1 ; the terminal row has entries α^n, α^{n-1} and the coefficients $-a_{ije}$. Each row therefore has ℓ_1 norm at most $3A^N$, which bounds the gradient norms, and $\|J\|_F \leq \sqrt{N} 3A^N$. The bound on $J^\top J$ is (32), since $\|J\|_F \leq \beta$. $\square$

The local exposing form. For one gate write $\alpha = \alpha_i \in [1, A]$, $n = n_i \geq 2$, $d = 2n - 1$, and $X_j = X_{i,j}$ with $X_0 = 1$. Put $\ell_j = X_{j+1} - \alpha X_j$ for $0 \leq j < n$, and let T_α be the symmetric tridiagonal matrix indexed by $0, \dots, n-1$ with

$$(T_\alpha)_{jj} = \alpha^{d-1-2j}, \quad (T_\alpha)_{j,j+1} = (T_\alpha)_{j+1,j} = \frac{1}{2}\alpha^{d-2-2j}. \quad (36)$$

Set $P_\alpha = \ell^\top T_\alpha \ell$, a quadratic in the gate's own variables, and

$$E_i^* = P_{\alpha_i}(X_i) + (\alpha_i - X_{i,1})(b_i(X) - \alpha_i^{d_i}).$$

Lemma B.7 (Local exposing form).

- (i) $T_\alpha \succeq (n+1)^{-2}I$ and $\|T_\alpha\| \leq 2A^{d-1}$.
- (ii) On the power curve, $P_\alpha(t, t^2, \dots, t^n) = (t - \alpha)(t^d - \alpha^d)$.
- (iii) $E_i^*(p+u) = u_i^\top H_i u_i - u_{i,1} \sum_{j < i, e} a_{ije} u_{j,e}$, where H_i is symmetric with $h_0 I \preceq H_i \preceq 8A^{2N} I$ and $h_0 = [(N+1)^2 N^2 A^{2N}]^{-1}$.

Proof. (i) $T_\alpha = \Delta T_0 \Delta$ with $\Delta = \text{diag}(\alpha^{n-1}, \dots, \alpha, 1)$ and T_0 the tridiagonal matrix with diagonal 1 and neighboring entries $\frac{1}{2}$; indeed $\alpha^{n-1-j} \alpha^{n-1-j} = \alpha^{d-1-2j}$ and $\alpha^{n-1-j} \alpha^{n-2-j} = \alpha^{d-2-2j}$. The eigenvalues of T_0 are $1 + \cos(\pi s/(n+1))$, $1 \leq s \leq n$, so $\lambda_{\min}(T_0) = 1 - \cos \frac{\pi}{n+1} = 2 \sin^2 \frac{\pi}{2(n+1)} \geq \frac{2}{(n+1)^2}$ by $\sin x \geq 2x/\pi$ on $[0, \pi/2]$. As $\Delta \succeq I$, $y^\top \Delta T_0 \Delta y \geq \lambda_{\min}(T_0) \|\Delta y\|^2 \geq \lambda_{\min}(T_0) \|y\|^2$. The absolute row sums of T_α are at most $\alpha^{d-1} + \frac{1}{2}\alpha^{d-2}$ in row zero and at most $2\alpha^{d-2j}$ in row $j \geq 1$, hence at most $2A^{d-1}$.

(ii) On the curve, $\ell_j = t^j(t - \alpha)$. The diagonal terms give $(t - \alpha)^2 \sum_{j=0}^{n-1} \alpha^{d-1-2j} t^{2j}$, and each pair of off-diagonal terms gives $(t - \alpha)^2 \alpha^{d-2-2j} t^{2j+1}$. Together they give $(t - \alpha)^2 \sum_{h=0}^{d-1} \alpha^{d-1-h} t^h = (t - \alpha)(t^d - \alpha^d)$.

(iii) At $X = p+u$ we have $\ell_j = u_{j+1} - \alpha u_j$ with $u_0 = 0$, so $\ell = D_\alpha u_i$ for the lower bidiagonal matrix $D_\alpha = I - \alpha Z$, where Z is the lower shift matrix. Hence $P_\alpha(p_i + u_i) = u_i^\top H_i u_i$ with $H_i = D_\alpha^\top T_\alpha D_\alpha$. The inverse $D_\alpha^{-1} = \sum_{t < n} \alpha^t Z^t$ has entries at most A^{n-1} , so $\|D_\alpha^{-1}\| \leq \|D_\alpha^{-1}\|_F \leq nA^{n-1}$ and $H_i \succeq (n+1)^{-2} n^{-2} A^{-2(n-1)} I \succeq h_0 I$. Also $\|D_\alpha\| \leq 1 + \alpha \leq 2A$, so $\|H_i\| \leq 2A^{d-1} \cdot 4A^2 \leq 8A^{2N}$. For the second term, $b_i(p+u) = \alpha_i^{d_i} + \sum_{j < i, e} a_{ije} u_{j,e}$ and $\alpha_i - X_{i,1} = -u_{i,1}$. $\square$

In particular $E_i^*(p) = 0$ and $\nabla E_i^*(p) = 0$, including in the directions of earlier gates: the affine dependence on predecessors contributes only the quadratic cross term in (iii). Repeated or signed predecessor terms, and terms of every retained power, are all part of this cross term.

Lemma B.8 (Coupled exposing form). *Let $\rho = (h_0/(4A))^2$ and $E^* = \sum_i \rho^{i-1} E_i^*$. Then $E^*(p+u) = u^\top H^* u$ with*

$$H^* \succeq \frac{3}{2} \gamma I, \quad \|H^*\| \leq \Lambda_*, \quad \gamma = \frac{1}{2} h_0 \rho^{k-1}, \quad \Lambda_* = 8kA^{2N} + A.$$

Proof. Let Ω be the diagonal matrix equal to $\rho^{(i-1)/2}$ on the coordinates of gate i , and set $y = \Omega u$. Then $u^\top H^* u = y^\top \tilde{H} y$, where $\tilde{H}$ has diagonal blocks H_i and off-diagonal entries coming from the terms $-\rho^{i-1} u_{i,1} a_{ije} u_{j,e} = -\rho^{(i-j)/2} a_{ije} y_{i,1} y_{j,e}$ with $j < i$. The corresponding symmetric entries have modulus at most $\frac{1}{2} \rho^{1/2} |a_{ije}|$. In the row of a coordinate $y_{i,1}$ these entries come from its own radicand and from later gates that use it, and in the row of $y_{j,e}$, $e \geq 2$, only from later gates. Each family has absolute coefficient sum at most A , so every absolute row sum of the off-diagonal part is at most $\rho^{1/2} A = h_0/4$. Hence $\tilde{H} \succeq \frac{3}{4} h_0 I$ by Lemma B.7(iii), and $H^* = \Omega \tilde{H} \Omega \succeq \frac{3}{4} h_0 \rho^{k-1} I$. Without rescaling, the diagonal blocks contribute at most $\sum_i \rho^{i-1} \|H_i\| \leq 8kA^{2N}$ and the off-diagonal part, with entries at most $\frac{1}{2} |a_{ije}|$, at most A . $\square$

The bound holds in every direction, for arbitrary signs of the coefficients, for gates used by many later gates, and for radicands using several powers of one predecessor.

A rational vanishing representation. For a monomial $\mathbf{m}$ of degree at most two in the gate's own variables $X_1, \dots, X_n$, including $\mathbf{m} = 1$, let $w(\mathbf{m})$ be its weighted degree, with $w(X_a) = a$, $w(X_a X_b) = a + b$ and $w(1) = 0$, so that $\mathbf{m}(t, \dots, t^n) = t^{w(\mathbf{m})}$. For $0 \leq h \leq 2n$ put $m_0 = 1$, $m_h = X_h$ for $1 \leq h \leq n$, and $m_h = X_n X_{h-n}$ for $n < h \leq 2n$. Let $R_{\mathbf{m}} = \mathbf{m} - m_{w(\mathbf{m})}$. These rational quadratics vanish on the power curve, so the gate- i versions $R_{i,\mathbf{m}}$ vanish at p . Expanding (36) writes $P_\alpha = \sum_{\mathbf{m}} \chi_{\mathbf{m}}(\alpha) \mathbf{m}$ with polynomials $\chi_{\mathbf{m}}$ having integer coefficients, degrees at most $2n$, and total absolute coefficient sum at most $8n$: each diagonal term $\alpha^{d-1-2j} \ell_j^2$ and each neighboring pair $\alpha^{d-2-2j} \ell_j \ell_{j+1}$ contributes at most four.

Lemma B.9 (Exact vanishing representation). *With $S_i = X_{i,n_i}^2 - b_i(X) X_{i,1}$ and $T_i = X_{i,n_i-1} X_{i,n_i} - b_i(X) = r_{i,n_i}$,*

$$E_i^* = S_i - \alpha_i T_i + \sum_{\mathbf{m}} \chi_{\mathbf{m}}(\alpha_i) R_{i,\mathbf{m}}.$$

All of S_i , T_i and $R_{i,\mathbf{m}}$ are rational quadratics vanishing at p .

Proof. Write $\alpha = \alpha_i$, $d = d_i$, $n = n_i$. Since $\sum_{w(\mathbf{m})=h} \chi_{\mathbf{m}}(\alpha)$ is the coefficient of t^h in $P_\alpha(t, \dots, t^n) = t^{d+1} - \alpha t^d - \alpha^d t + \alpha^{d+1}$ and $d+1 = 2n$,

$$P_\alpha = \sum_{\mathbf{m}} \chi_{\mathbf{m}}(\alpha) R_{\mathbf{m}} + X_n^2 - \alpha X_{n-1} X_n - \alpha^d X_1 + \alpha^{d+1}.$$

Adding $(\alpha - X_1)(b_i - \alpha^d) = \alpha b_i - \alpha^{d+1} - X_1 b_i + \alpha^d X_1$ gives the identity. At p , $T_i = \alpha^d - \alpha^d = 0$ and $S_i = \alpha^{2n} - \alpha^d \alpha = 0$. $\square$

Assembly. Put

$$\mu = \gamma, \quad \Lambda = \Lambda_* + 1, \quad \varepsilon_{\max} = \min\left\{1, \frac{\mu^2}{2N}, \frac{\nu^2 \mu^2}{36N(\Lambda + N\beta)^2}\right\},$$

with β, ν from Lemma B.6, and let $t = 2^{-s}$ with s the least nonnegative integer such that $\varepsilon = t^2 \leq \varepsilon_{\max}$. Let

$$B_0 = A + 1 + 32N^2(2A)^{2N}, \quad \tau = \min\left\{1, \frac{\gamma}{2kB_0}, \frac{\varepsilon}{2A^N k B_0}\right\},$$

compute rationals $\hat{\alpha}_i$ with $|\hat{\alpha}_i - \alpha_i| \leq \tau$ by Lemma B.2, and define

$$G = \sum_{i=1}^k \rho^{i-1} \left[S_i - \hat{\alpha}_i T_i + \sum_{\mathbf{m}} \chi_{\mathbf{m}}(\hat{\alpha}_i) R_{i,\mathbf{m}} \right], \quad F = \left(\frac{G}{t\nu} \right)^2 + \sum_{i,e} \left(\frac{r_{i,e}}{\nu} \right)^2.$$

By Lemma B.9, G is a rational quadratic with $G(p) = 0$, whatever the approximations. The difference $G - E^*$ has $\|G - E^*\|_1 \leq k\tau B_0$. Indeed, for each gate, $|\hat{\alpha}_i - \alpha_i| \|T_i\|_1 \leq \tau(A+1)$; and on $[-2A, 2A]$ the derivative of $\chi_{\mathbf{m}}$ is at most its absolute coefficient sum times $2n(2A)^{2n-1}$, so $\sum_{\mathbf{m}} |\chi_{\mathbf{m}}(\hat{\alpha}_i) - \chi_{\mathbf{m}}(\alpha_i)| \leq 16N^2(2A)^{2N}\tau$, while $\|R_{i,\mathbf{m}}\|_1 \leq 2$. Since $G - E^*$ vanishes at p and $\|p\|_\infty \leq A^N$, (33) gives $G(p+u) = \ell^\top u + u^\top H u$ with

$$\|H - H^*\| \leq k\tau B_0 \leq \frac{1}{2}\gamma, \quad \|\ell\| \leq 2A^N k\tau B_0 \leq \varepsilon.$$

With Lemma B.8, $\mu I \preceq H \preceq \Lambda I$.

Thus the hypotheses of Lemma 7.6 hold with $n = N$ variables, the $m = N$ residuals $r_{i,e}$, the quadratic $g = G$, the constants μ, Λ, β, ν , and $\varepsilon = t^2$: the residual quadratic parts have norm at most one and their gradients at p have norm at most β (Lemma B.6); $\sum_{i,e} \nabla r_{i,e}(p) \nabla r_{i,e}(p)^\top = J^\top J \succeq \nu^2 I$; $G(p) = 0$ with quadratic part between μI and ΛI and $\|\nabla G(p)\| \leq \varepsilon$; and $\varepsilon \leq \varepsilon_{\max}$. The lemma shows that F is a sum of $N + 1$ squares of rational quadratics with $\nabla^2 F \succeq \frac{3}{2} I$ and unique zero p , and that the canonical Hessian Gram of this representation is a rational positive definite Hessian Gram of F , computed by polynomially many rational operations.

Complexity. The numbers $A, h_0, \rho, \gamma = \mu, \Lambda, \beta, \nu, B_0, \varepsilon_{\max}$ and τ are explicit rationals. Their bit lengths are polynomial in the input length, because they are built from A and $N \leq L$ by products and powers with exponents polynomial in k and N ; for example $\log_2(1/\nu) = (N - 1) \log_2 \beta$ and $\log_2(1/\gamma) = O(kN \log_2 A + k \log_2 N)$. Lemma B.2 with $P = \lceil \log_2(1/\tau) \rceil$ runs in polynomial time. The polynomials χ_m are explicit, there are $O(N^2)$ monomials per gate, and F has $O(N^4)$ monomials. Hence the whole construction runs in polynomial time. The normalization, the residuals, and the rational vanishing representation never use a minimal polynomial of any gate value or of the joint field. $\square$

B.7 Proof of Theorem 5.12

We use the notation of Section 5.6, write $\mathbf{j} = (0, 0, 1, 0)$ and $\mathbf{k} = (0, 0, 0, 1)$, and let e_1, e_2, e_3 be the standard basis of the vector part. For unit quaternions put $[u, w] = uw\bar{u}\bar{w}$, $\iota(u) = \mathbf{i}u\bar{\mathbf{i}}$, $\text{rot}(u) = \mathbf{c}u\bar{\mathbf{c}}$ with $\mathbf{c} = \frac{1}{2}(1, 1, 1, 1)$, $\text{pr}(u) = u \iota(u)$, $\text{add}(u, w) = \text{pr}(uw)$ and $\text{prod}(u, w) = \text{pr}(\text{rot}([u, \text{rot}(w)]))$.

Lemma B.10 (Quaternion identities). *For unit quaternions u, w :*

(i) *The three maps are*

$$\begin{aligned} \iota(u) &= (u_0, u_1, -u_2, -u_3), & \text{rot}(u) &= (u_0, u_3, u_1, u_2), \\ \text{pr}(u) &= (1 - 2u_1^2, 2u_0u_1, 2u_1u_3, -2u_1u_2). \end{aligned}$$

(ii) $[u, w] - \mathbf{1} = 2(0, \vec{u} \times \vec{w}) \bar{u}\bar{w}$, hence $|[u, w] - \mathbf{1}| \leq 2|\vec{u}||\vec{w}|$.

(iii) *If $u_0, w_0 \geq 0$, then*

$$|[u, w] - \mathbf{1} - 2(0, \vec{u} \times \vec{w})| \leq 4|\vec{u}||\vec{w}|(|\vec{u}| + |\vec{w}|).$$

(iv) *If $u_0 \geq 0$ and $|\vec{u}| \leq \frac{1}{2}$, then $|\overrightarrow{\text{pr}(u)} - 2u_1e_1| \leq 4|\vec{u}|^2$ and $\text{pr}(u)_0 \geq \frac{1}{2}$.*

Proof. (i) Conjugation by a unit quaternion fixes the scalar part and acts linearly on the vector part. Direct multiplication gives $\mathbf{i}\bar{\mathbf{j}} = -\mathbf{j}$ and $\mathbf{i}\bar{\mathbf{k}} = -\mathbf{k}$, which is the formula for ι . For rot , write $\mathbf{c} = \frac{1}{2}(\mathbf{1} + \mathbf{i} + \mathbf{j} + \mathbf{k})$; expanding the sixteen products in $(\mathbf{c}\mathbf{i})\bar{\mathbf{c}} = \frac{1}{4}(-\mathbf{1} + \mathbf{i} + \mathbf{j} - \mathbf{k})(\mathbf{1} - \mathbf{i} - \mathbf{j} - \mathbf{k})$ gives $\mathbf{j}$. The cyclic permutation $\mathbf{i} \mapsto \mathbf{j} \mapsto \mathbf{k} \mapsto \mathbf{i}$ is an automorphism of the quaternions fixing $\mathbf{c}$, so also $\mathbf{c}\bar{\mathbf{j}} = \mathbf{k}$ and $\mathbf{c}\bar{\mathbf{k}} = \mathbf{i}$. Thus rot sends $u_1e_1 + u_2e_2 + u_3e_3$ to $u_1e_2 + u_2e_3 + u_3e_1$. Finally, for $\vec{u}' = (u_1, -u_2, -u_3)$, the product $u \iota(u)$ has scalar part $u_0^2 - \vec{u} \cdot \vec{u}' = u_0^2 - u_1^2 + u_2^2 + u_3^2 = 1 - 2u_1^2$, using $|u| = 1$, and vector part $u_0(\vec{u} + \vec{u}') + \vec{u} \times \vec{u}' = (2u_0u_1, 0, 0) + (0, 2u_1u_3, -2u_1u_2)$.

(ii) Since $wu\bar{u}\bar{w} = \mathbf{1}$, we have $[u, w] - \mathbf{1} = (uw - wu)\bar{u}\bar{w}$, and $uw - wu = (0, 2\vec{u} \times \vec{w})$ by the product formula. The norm is multiplicative and $|\bar{u}| = |\bar{w}| = 1$.

(iii) The left side equals $|2(0, \vec{u} \times \vec{w})| |\bar{u}\bar{w} - \mathbf{1}|$, and $|\bar{u}\bar{w} - \mathbf{1}| = |u(\bar{u}\bar{w} - \mathbf{1})| = |\bar{w} - u| \leq |w - \mathbf{1}| + |u - \mathbf{1}|$. For a unit quaternion with $u_0 \geq 0$, $u_0 = (1 - |\vec{u}|^2)^{1/2} \geq 1 - |\vec{u}|^2$, so $|u - \mathbf{1}|^2 = 2 - 2u_0 \leq 2|\vec{u}|^2$ and $|u - \mathbf{1}| \leq 2|\vec{u}|$.

(iv) By (i), $\overrightarrow{\text{pr}(u)} - 2u_1e_1 = 2u_1(u_0 - 1, u_3, -u_2)$. Here $0 \leq 1 - u_0 \leq |\vec{u}|^2 \leq \frac{1}{2}|\vec{u}|$, so the vector $(u_0 - 1, u_3, -u_2)$ has norm at most $\frac{\sqrt{5}}{2}|\vec{u}| \leq 2|\vec{u}|$. Also $\text{pr}(u)_0 = 1 - 2u_1^2 \geq \frac{1}{2}$. $\square$

Lemma B.11 (Small rational signal). *Let g be a unit quaternion with $g_0 > 0$, $0 < g_1 \leq 2^{-16}$ and $|(g_2, g_3)| \leq 64g_1^2$. Then $g' = \text{prod}(g, g)$ satisfies the same three conditions and $g_1^2 \leq g'_1 \leq 6g_1^2$. Consequently the iterates $g^{(0)} = \mathbf{w}_\vartheta = (\frac{1-\vartheta^2}{1+\vartheta^2}, \frac{2\vartheta}{1+\vartheta^2}, 0, 0)$, $\vartheta = 2^{-20}$, and $g^{(j+1)} = \text{prod}(g^{(j)}, g^{(j)})$ are rational unit quaternions with $0 < g_1^{(j)} < 2^{-16 \cdot 2^j}$ and $|(g_2^{(j)}, g_3^{(j)})| \leq 64(g_1^{(j)})^2$.*

Proof. Write $x = g_1$. Then $|\vec{g}| \leq x + 64x^2 \leq 2x$. Let $U = \text{rot}([g, \text{rot}(g)])$. Both g and $\text{rot}(g)$ have positive scalar parts and vector parts of norm $|\vec{g}|$. By Lemma B.10(iii), and since rot is an isometry fixing $\mathbf{1}$, $|U - \mathbf{1} - 2(0, \text{rot}(\vec{g} \times \text{rot}(\vec{g})))| \leq 8|\vec{g}|^3 \leq 64x^3$, where rot acts on vectors by $(a_1, a_2, a_3) \mapsto (a_3, a_1, a_2)$. The first coordinate of $\text{rot}(\vec{g} \times \text{rot}(\vec{g}))$ is $g_1^2 - g_2g_3$, and $2|g_2g_3| \leq g_2^2 + g_3^2 \leq 4096x^4 \leq x^3$. Hence $|U_1 - 2x^2| \leq 65x^3 \leq x^2$, so $x^2 \leq U_1 \leq 3x^2$. By Lemma B.10(ii), $|U - \mathbf{1}| \leq 2|\vec{g}|^2 \leq 8x^2$, so $|\vec{U}| \leq 8x^2$ and $U_0 \geq 1 - 8x^2 > \frac{1}{2}$. By part (i), $g' = \text{pr}(U)$ has $g'_1 = 2U_0U_1 \in [x^2, 6x^2]$, scalar part $1 - 2U_1^2 > 0$, and $|(g'_2, g'_3)| = 2U_1|(U_2, U_3)| \leq 48x^4 \leq 64(g'_1)^2$. Finally $g'_1 \leq 6x^2 \leq x \leq 2^{-16}$.

The constant $\mathbf{w}_\vartheta$ is a rational unit quaternion with positive scalar part, $0 < (\mathbf{w}_\vartheta)_1 < 2\vartheta = 2^{-19}$ and zero transverse part. Products and conjugates of rational unit quaternions are rational unit quaternions. Iterating, $6g_1^{(j+1)} \leq (6g_1^{(j)})^2$, so $6g_1^{(j)} \leq (6g_1^{(0)})^{2^j} < (2^{-16})^{2^j}$, and $g_1^{(j+1)} \geq (g_1^{(j)})^2 > 0$. $\square$

Quaternion signals. Fix $0 < \delta < 1$. A (d, C, Θ) -signal is now a unit quaternion u with $u_0 > 0$, $|C| \leq \Theta$ and $|\vec{u} - C\delta^d e_1| \leq \Theta\delta^{d+1}$, where $d \geq 1$, $C \in \mathbb{Z}$ and $\Theta \geq 1$. Then $|\vec{u}| \leq 2\Theta\delta^d$.

Lemma B.12 (Quaternion signal arithmetic). *Suppose $\Theta\delta \leq 2^{-30}$ and let $\Theta' = 2^{20}\Theta^4$.*

- (i) *If u is a (d, C, Θ) -signal, then $\bar{u}$ is a $(d, -C, \Theta)$ -signal and $\text{pr}(u)$ is a $(d, 2C, \Theta')$ -signal.*
- (ii) *If u, w are (d, C, Θ) - and (d, C', Θ) -signals, then $\text{add}(u, w)$ is a $(d, 2(C + C'), \Theta')$ -signal.*
- (iii) *If u, w are (a, C, Θ) - and (b, C', Θ) -signals, then $\text{prod}(u, w)$ is an $(a + b, 4CC', \Theta')$ -signal.*

Proof. Let $h = 2\Theta\delta^d$ in (i) and (ii); then $h \leq 2^{-29}$. All scalar parts below are positive, and all vectors to which Lemma B.10(iv) is applied have norm at most $\frac{1}{2}$.

(i) Conjugation negates the vector part. For $\text{pr}(u)$, part (iv) of Lemma B.10 gives

$$|\overrightarrow{\text{pr}(u)} - 2C\delta^d e_1| \leq 4h^2 + 2\Theta\delta^{d+1} \leq 18\Theta^2\delta^{d+1}.$$

(ii) Write $\vec{u} = C\delta^d e_1 + \epsilon_u$ and $\vec{w} = C'\delta^d e_1 + \epsilon_w$. Then

$$\vec{u}\vec{w} - (C + C')\delta^d e_1 = \epsilon_w + \epsilon_u + (u_0 - 1)\vec{w} + (w_0 - 1)\vec{u} + \vec{u} \times \vec{w}.$$

The first two terms have norm at most $2\Theta\delta^{d+1}$; $0 \leq 1 - u_0 \leq h^2$, so the next two have norm at most $2h^3 = 16\Theta^3\delta^{3d}$; and since $e_1 \times e_1 = 0$, $|\vec{u} \times \vec{w}| \leq 3\Theta^2\delta^{2d+1}$. As $d \geq 1$, the total is at most $21\Theta^3\delta^{d+1}$. Also $(uw)_0 \geq u_0w_0 - h^2 > 0$ and $|\vec{u}\vec{w}| \leq 2h + h^2 \leq 5\Theta\delta^d$. Applying Lemma B.10(iv) to uw , $|\overrightarrow{\text{pr}(uw)} - 2(C + C')\delta^d e_1| \leq 4 \cdot 25\Theta^2\delta^{2d} + 42\Theta^3\delta^{d+1} \leq 142\Theta^3\delta^{d+1}$, and $|2(C + C')| \leq 4\Theta$.

(iii) Now $|\vec{u}| \leq 2\Theta\delta^a$ and $|\vec{w}| \leq 2\Theta\delta^b$, and $\text{rot}(w)$ has positive scalar part and vector part $\text{rot}(\vec{w}) = C'\delta^b e_2 + \text{rot}(\epsilon_w)$. Since $e_1 \times e_2 = e_3$, $|\vec{u} \times \text{rot}(\vec{w}) - CC'\delta^{a+b} e_3| \leq 3\Theta^2\delta^{a+b+1}$. By Lemma B.10(iii), $|[u, \text{rot}(w)] - \mathbf{1} - 2(0, \vec{u} \times \text{rot}(\vec{w}))| \leq 4 \cdot 4\Theta^2\delta^{a+b} \cdot 4\Theta\delta = 64\Theta^3\delta^{a+b+1}$. The rotation

rot maps e_3 to e_1 , so $U = \text{rot}([u, \text{rot}(w)])$ satisfies $|\vec{U} - 2CC'\delta^{a+b}e_1| \leq (6\Theta^2 + 64\Theta^3)\delta^{a+b+1}$, and by part (ii) of that lemma $|U - \mathbf{1}| \leq 8\Theta^2\delta^{a+b}$, so $U_0 > 0$ and $|\vec{U}| \leq 8\Theta^2\delta^{a+b}$. Applying part (iv), $|\overrightarrow{\text{pr}(U)} - 4CC'\delta^{a+b}e_1| \leq 256\Theta^4\delta^{2(a+b)} + (12\Theta^2 + 128\Theta^3)\delta^{a+b+1} \leq 396\Theta^4\delta^{a+b+1}$, and $|4CC'| \leq 4\Theta^2$. All bounds are at most Θ' times the relevant power of δ . $\square$

As in the cube-root compiler, no bound divides by a coefficient, so zero values, cancellation, and repeated arguments such as $\text{prod}(u, u)$ or $\text{add}(u, \bar{u})$ are covered.

The compiler. Prepare the program as in Appendix B.4, so that its output is $W = 2V - 1$. Each node ν receives a pair of quaternion signals (f_ν, g_ν) of a common order d_ν and prescribed coefficients (C_ν, D_ν) with $D_\nu > 0$ and $C_\nu = D_\nu V_\nu$. The constant 1 receives (g, g) and the constant 0 receives $(\mathbf{1}, g)$, both of order one, where g is the generator quaternion fixed below, with coefficients $(1, 1)$ and $(0, 1)$. A product $\nu = \nu_1 \nu_2$ receives $f_\nu = \text{prod}(f_{\nu_1}, f_{\nu_2})$ and $g_\nu = \text{prod}(g_{\nu_1}, g_{\nu_2})$, with coefficients $(4C_{\nu_1}C_{\nu_2}, 4D_{\nu_1}D_{\nu_2})$. A sum or difference receives

$$f_\nu = \text{add}(\text{prod}(f_{\nu_1}, g_{\nu_2}), \text{prod}(f_{\nu_2}, g_{\nu_1})^{\pm 1}), \quad g_\nu = \text{pr}(\text{prod}(g_{\nu_1}, g_{\nu_2})),$$

where $u^{+1} = u$ and $u^{-1} = \bar{u}$, with coefficients $(8(C_{\nu_1}D_{\nu_2} \pm C_{\nu_2}D_{\nu_1}), 8D_{\nu_1}D_{\nu_2})$. In both cases $d_\nu = d_{\nu_1} + d_{\nu_2}$, the two arguments of add have equal order, and $C_\nu = D_\nu V_\nu$ with $D_\nu > 0$. Let T be the number of signal operations prod, add, pr and conjugation used.

Let $r_0 = 2T + 2$ and let $g = g^{(r_0)}$ from Lemma B.11, with $\delta = g_1$. Then g is a $(1, 1, 128)$ -signal and $\mathbf{1}$ a $(1, 0, 128)$ -signal. Put $\Theta_0 = 128$ and $\Theta_t = 2^{20}\Theta_{t-1}^4$, so that $\log_2 \Theta_t = (41 \cdot 4^t - 20)/3 < 14 \cdot 4^t$. Since $\delta < 2^{-16 \cdot 2^{2T+2}} = 2^{-64 \cdot 4^T}$, we get $\Theta_T \delta < 2^{-50 \cdot 4^T} \leq 2^{-30}$. Induction over the operations, using Lemma B.12, shows that every signal is a (d, C, Θ_T) -signal for its order and prescribed coefficient, for any sharing pattern. For the output node, $C = DW$ is a nonzero integer, so the selected coordinate x of f_ν satisfies $|x - C\delta^d| \leq \Theta_T \delta^{d+1} < \frac{1}{2}\delta^d \leq \frac{1}{2}|C|\delta^d$. Hence $x \neq 0$ and $\text{sign } x = \text{sign } W$.

Each of ι , rot, pr, add, prod and the commutator is a fixed composition of at most twelve products and conjugations with the constants $\mathbf{i}, \mathbf{c}$ and their conjugates, so the circuit has $O(T)$ gates and is computed in polynomial time. Every gate value is a rational unit quaternion, and nothing is expanded.

For the upper bound, first check by rational arithmetic that every given constant has norm one, and map malformed inputs to a fixed program with output -1 . Then represent each gate value u by four integer straight-line programs $N_0, \dots, N_3$ and one program $D > 0$ with $u_s = N_s/D$; a constant with binary numerators and denominators has a representation of linear size. A product multiplies the denominators and forms the four bilinear numerators of the product formula; a conjugate negates three numerators. The sign of a coordinate u_s is the sign of N_s , so $u_s > 0$ is one PosSLP instance, and $u_s \geq 0$ is the instance $2N_s + 1 > 0$. $\square$

B.8 Proof of Proposition 5.13

Let the circuit have gates $1, \dots, k$ with values p_i , and variables $X_i \in \mathbb{R}^4$. For any quaternions, $|ab| = |a||b|$, $\langle ab, c \rangle = \langle a, \bar{c}\bar{b} \rangle$, and left or right multiplication by a unit quaternion is an orthogonal map of $\mathbb{R}^4$.

Residuals. The vector residual of gate i is $r_i = X_i - p_i$ for a constant gate, whose value p_i is given, $r_i = X_i - X_a X_b$ for a product, and $r_i = X_i - \bar{X}_a$ for a conjugate. Its four components are rational quadratics. Their common real zero is exactly p , by induction over the gates. At p the

Jacobian J is block lower triangular with identity diagonal blocks, so $\det J = 1$. The block row of a product gate is $(I, -\mathcal{R}_{p_b}, -\mathcal{L}_{p_a})$, with $\mathcal{R}_{p_b}: x \mapsto xp_b$ and $\mathcal{L}_{p_a}: x \mapsto p_ax$ orthogonal; if $a = b$ the parent block is $-(\mathcal{R}_{p_a} + \mathcal{L}_{p_a})$. The block row of a conjugate gate is $(I, -\text{diag}(1, -1, -1, -1))$. Hence every block row has operator norm at most three, every scalar residual gradient has norm at most three, and $\|J\|_F^2 \leq \sum_i 4 \cdot 9 = 9N$. With $\beta = 4N$ and $\nu = \beta^{-(N-1)}$, (32) gives $J^\top J \succeq \nu^2 I$. A component of a product $X_a X_b$ with $a \neq b$ is a bilinear form $X_a^\top K X_b$ with K a signed permutation matrix, whose symmetric matrix has norm $\frac{1}{2}$. If $a = b$ and $X_a = (x_0, x_1, x_2, x_3)$, the components are $x_0^2 - x_1^2 - x_2^2 - x_3^2$, $2x_0x_1$, $2x_0x_2$ and $2x_0x_3$, whose matrices have norm one. Constant and conjugate residuals are affine. So every residual quadratic part has norm at most one.

Exposing form. Let $q_i = |X_i|^2 - 1$ and define

$$E_i = q_i - 2\langle p_i, r_i \rangle - \begin{cases} 0 & \text{constant gate,} \\ q_a + q_b & \text{product gate,} \\ q_a & \text{conjugate gate,} \end{cases}$$

where for $a = b$ the product term is $2q_a$. For a product gate, expanding and using $\langle X_a X_b, p_i \rangle = \langle X_a, p_i \bar{X}_b \rangle$ and $|p_i \bar{X}_b| = |X_b|$ gives

$$E_i = |X_i - p_i|^2 - |X_a - p_i \bar{X}_b|^2.$$

Since $p_i = p_a p_b$ and $|p_b| = 1$, we have $p_i \bar{p}_b = p_a$, so with $U = X - p$,

$$E_i(p + U) = |U_i|^2 - |U_a - p_i \bar{U}_b|^2. \quad (37)$$

The order of the factors matters because multiplication is not commutative, and (37) holds verbatim when $a = b$. For a conjugate gate, $p_i = \bar{p}_a$ gives $E_i(p + U) = |U_i|^2 - |U_a|^2$; for a constant gate, $E_i(p + U) = |U_i|^2$. In each case E_i has zero value and zero gradient at p , and its negative part is at most $2|U_a|^2 + 2|U_b|^2$, or $4|U_a|^2$ if $a = b$, or $|U_a|^2$.

Let $\omega_i = 16^{-(i-1)}$, $\omega_* = \omega_k$ and $E^* = \sum_i \omega_i E_i$, and write $E^*(p + U) = U^\top H_* U$. Every block U_j receives $\omega_j |U_j|^2$ from its own gate and loses at most $4\omega_i |U_j|^2$ to each later gate i , whatever the sharing pattern. As $\sum_{i>j} \omega_i \leq \omega_j/15$,

$$\frac{11}{15} \omega_* I \preceq H_* \preceq I.$$

Rounding. Fix $\varepsilon = t^2$ below and let $\eta = \min\{1, \omega_*/(32k), \varepsilon/(16k)\}$. Choose a dyadic mesh $h \leq \eta/(2 \cdot 3^k)$ and compute $\hat{p}_i$ in topological order: $\hat{p}_i$ rounds each coordinate of the given constant p_i , of $\hat{p}_a \hat{p}_b$, or of $\bar{\hat{p}}_a$ to the nearest multiple of h . Let $e_i = |\hat{p}_i - p_i|$. Rounding changes a vector by at most h , and if $e_a, e_b \leq 1$ then $|\hat{p}_a \hat{p}_b - p_a p_b| \leq e_a(1 + e_b) + e_b \leq 3 \max\{e_a, e_b\}$. Induction gives $e_i \leq h(3^i - 1) \leq \eta/2$; in particular every coordinate error is at most η , and the assumption $e \leq 1$ is maintained. The mesh has $O(k + \log_2(1/\eta))$ bits, and the gate fractions are never expanded. Define G as E^* with each coefficient vector p_i in the term $-2\langle p_i, r_i \rangle$ replaced by $\hat{p}_i$, and with every residual and every q_i kept exact. Then $G(p) = 0$. Writing $G(p + U) = \ell^\top U + U^\top H U$ and $\delta_i = \hat{p}_i - p_i$, the difference $G - E^* = -2 \sum_i \omega_i \langle \delta_i, r_i \rangle$ gives $\|\ell\| \leq 2 \sum_i 3 |\delta_i| \leq 12k\eta$ and $\|H - H_*\| \leq 2 \sum_i \|\delta_i\|_1 \leq 8k\eta$. With $\mu = \omega_*/4$ and $\Lambda = 2$, the choice of η gives

$$\mu I \preceq H \preceq \Lambda I, \quad \|\nabla G(p)\| = \|\ell\| \leq \varepsilon.$$

Assembly. Let $t = 2^{-s}$ with s the least nonnegative integer such that

$$\varepsilon = t^2 \leq \min\left\{1, \frac{\mu^2}{2N}, \frac{\nu^2 \mu^2}{36N(\Lambda + N\beta)^2}\right\},$$

and let

$$F = \left(\frac{G}{t\nu}\right)^2 + \frac{1}{\nu^2} \sum_{i=1}^k |r_i|^2,$$

where $|r_i|^2$ is the sum of the squares of the four components of r_i . The hypotheses of Lemma 7.6 hold with $n = N$ variables, the $m = N$ scalar residuals, $g = G$, and the constants $\mu, \Lambda, \beta, \nu, \varepsilon$. Hence F is a sum of $N + 1$ squares of rational quadratics with $\nabla^2 F \succeq \frac{3}{2}I$ and unique zero p , and the canonical Hessian Gram of this representation is a rational positive definite Hessian Gram of F , computed in polynomial time. The derived scale and rounding parameters $\omega_i, \mu, \Lambda, \beta, \nu, \varepsilon, t, \eta$, and h , and the rounded coefficients $\hat{p}_i$, have $O(k + N \log N)$ bits. The given rational generators may have coordinates of any bit length; they enter the constant-gate residuals exactly. Every coefficient of G , of the square factors of F , and of its Hessian Gram therefore has bit length polynomial in the total input length. The zero is the exact list of gate values, so $p \in \mathbb{Q}^N \cap [-1, 1]^N$. $\square$

B.9 Proof of Proposition 5.16

Every y_t has a triangular bound $y_t \leq g_t(y_{<t})$: for a root variable, $g_t = \rho_i(Y)^{1/d_i}$, where the radicand is affine in the earlier power variables and is at least $\xi_i^{d_i} \geq 1$ on the boxes by the direct interval test; for a power variable, $g_t = -X_i^e$. The feasible set is convex, because powers are convex on the positive boxes, and compact. The circuit point y^* is feasible with every bound tight. On the boxes, which are convex, the partial derivatives of g_t are bounded by $|a_{ije}| \leq A$ for a root and by $eX^{e-1} \leq NA^{N-1}$ for a power, so $|g_t(y) - g_t(y')| \leq C \sum_{s < t} |y_s - y'_s|$ with $C = NA^N$.

Let y be feasible, $\delta_t = g_t(y_{<t}) - y_t \geq 0$ and $\epsilon_t = y_t^* - y_t$. Then $\epsilon_t = \delta_t + g_t(y_{<t}^*) - g_t(y_{<t})$, so $|\epsilon_t| \leq \delta_t + C \sum_{s < t} |\epsilon_s|$. Hence $\Sigma_t \leq (1 + C)\Sigma_{t-1} + \delta_t$ and $\Sigma_{t-1} \leq \sum_{s < t} (1 + C)^{t-1-s} \delta_s$. With $B = 2(C + 1)$ and $W_t = B^{N-t}$,

$$\sum_t W_t \epsilon_t \geq \sum_t W_t (\delta_t - C \Sigma_{t-1}) \geq \sum_s \delta_s \left[W_s - C \sum_{t > s} W_t (1 + C)^{t-1-s} \right] \geq \frac{1}{1 + C} \sum_s W_s \delta_s,$$

because $\sum_{t > s} W_t (1 + C)^{t-1-s} \leq W_s \sum_{h \geq 1} B^{-h} (1 + C)^{h-1} = W_s / (B - 1 - C) = W_s / (1 + C)$. Thus $\sum_t W_t y_t^* \geq \sum_t W_t y_t$, with equality only if every $\delta_s = 0$, in which case triangular recursion gives $y = y^*$. Finally, $\log_2 W_t \leq N \log_2 (2C + 2)$ is polynomial, and all other data are given. $\square$

C Proofs for point output

This appendix proves the results of Section 3. Section C.1 collects the tools: explicit sparse evaluation, interpolation inequalities, integer eigenvalue and Hoffman bounds, relative coordinates, the convex value interface, and the selector lemma. Sections C.2 and C.3 prove Theorems 3.4 and 3.7. Section C.4 contains the constructions behind the boundary results of Sections 3.4 and 3.5.

Throughout, $\langle r \rangle$ denotes the binary encoding length of a rational number, vector, or matrix, written with integer numerators and positive integer denominators. The norm $\|\cdot\|$ is Euclidean, and for a matrix it is the spectral norm. For a matrix T we write $\|T\|_\Delta = \sum_{i,j} |T_{ij}|$; it bounds the spectral norm of T and of $T^\top$. Running times count bit operations. A bound “polynomial in $a, b, \dots$ ” is a polynomial in $a + b + \dots$ with an absolute exponent.

C.1 Tools

Classical rational subroutines. Three classical facts are used without proof. First, Gaussian elimination over $\mathbb{Q}$ computes ranks, bases of kernels and ranges, inverses, and solutions of linear systems in polynomial time, and all its outputs have polynomial encoding length [49, Section 1.4]. Second, rational linear programs are solvable in polynomial time: given rational A, b, c , one can decide whether $\{x : Ax \leq b\}$ is empty and whether $c^\top x$ is unbounded above on it, and otherwise compute an optimal solution of polynomial encoding length; systems that also contain linear equations are handled in the same way [49, Theorem (6.4.12) and Section 6.5]. Third, Farkas' lemma: for real matrices G_1, G_2 and a real vector w , the system $G_1 y_1 + G_2 y_2 = w$, $y_1 \geq 0$, has no solution if and only if some d satisfies $G_1^\top d \leq 0$, $G_2^\top d = 0$, and $w^\top d > 0$. With rational data such a d with $w^\top d = 1$ is an optimal solution of the linear program $\max\{w^\top d : G_1^\top d \leq 0, G_2^\top d = 0, w^\top d \leq 1\}$.

Explicit sparse evaluation. An explicit sparse polynomial is a list of pairs (f_α, α) with distinct exponent vectors α and nonzero rational coefficients. Its numerical degree is the largest $|\alpha| = \sum_i \alpha_i$ in the list. Exponents may be written in binary; the bounds below are polynomial in the numerical degree, which is polynomial in the input length exactly when the degree is.

Lemma C.1 (Explicit sparse evaluation). *Let $f = \sum_{\alpha \in A} f_\alpha y^\alpha \in \mathbb{Q}[y_1, \dots, y_N]$ be an explicit sparse polynomial of encoding length L_f and numerical degree at most $D \geq 1$. Let $T \in \mathbb{Q}^{N \times n}$, $t \in \mathbb{Q}^N$, and $\tau \in \mathbb{Q}$ with $\tau \geq 0$, and put*

$$\varphi(x) = f(Tx + t) + \tau \|x\|^2.$$

- (a) *For every rational x , the exact rational values $\varphi(x)$ and $\nabla \varphi(x)$ can be computed in time polynomial in $L_f, D, \langle T, t, \tau \rangle, \langle x \rangle$. The computation does not form the expanded polynomial $f(Tx + t)$.*
- (b) *Let $\varrho \geq 1$ be rational, put $\varrho_1 = \max\{1, \varrho \max_i \sum_j |T_{ij}| + \max_i |t_i|\}$, and*

$$V_f = \sum_{\alpha} |f_\alpha| \varrho_1^{|\alpha|}, \quad G_f = \sum_{|\alpha| \geq 1} |f_\alpha| |\alpha| \varrho_1^{|\alpha|-1}, \quad H_f = \sum_{|\alpha| \geq 2} |f_\alpha| |\alpha|^2 \varrho_1^{|\alpha|-2}.$$

Then on the box $[-\varrho, \varrho]^n$,

$$|\varphi| \leq V_f + \tau n \varrho^2, \quad \|\nabla \varphi\| \leq \|T\|_\Delta G_f + 2\tau n \varrho, \quad \|\nabla^2 \varphi\| \leq \|T\|_\Delta^2 H_f + 2\tau.$$

These bounds are rationals of encoding length polynomial in $L_f, D, \langle T, t, \tau, \varrho \rangle$.

Proof. (a) Compute $y = Tx + t$. If every coordinate of y has encoding length at most σ , then each power $y_i^{\alpha_i}$, and hence each monomial y^α , is a rational whose numerator and denominator have at most $D\sigma$ bits, computed by repeated multiplication. Summing the at most L_f terms $f_\alpha y^\alpha$ gives $f(y)$ with polynomially many bits. The partial derivatives satisfy $\partial_i f(y) = \sum_{\alpha_i > 0} \alpha_i f_\alpha y^{\alpha - e_i}$ and are computed in the same way. Finally $\nabla \varphi(x) = T^\top \nabla f(y) + 2\tau x$.

(b) For $\|x\|_\infty \leq \varrho$ we have $\|Tx + t\|_\infty \leq \varrho_1$. For $\|y\|_\infty \leq \varrho_1$ and $\varrho_1 \geq 1$, a monomial satisfies $|y^\alpha| \leq \varrho_1^{|\alpha|}$, the entries of its gradient satisfy $\sum_i |\alpha_i y^{\alpha - e_i}| \leq |\alpha| \varrho_1^{|\alpha|-1}$, and the entries of its Hessian satisfy $\sum_{i,j} |\alpha_i (\alpha_j - \delta_{ij}) y^{\alpha - e_i - e_j}| \leq |\alpha|^2 \varrho_1^{|\alpha|-2}$. The Euclidean norm of a vector and the spectral norm of a matrix are bounded by the sums of the absolute values of their entries. The chain rule gives $\nabla_x f(Tx + t) = T^\top \nabla f(y)$ and $\nabla_x^2 f(Tx + t) = T^\top \nabla^2 f(y) T$. The regularizer contributes at most $\tau n \varrho^2$, $2\tau \|x\| \leq 2\tau n \varrho$, and 2τ . Each displayed bound is a sum of at most $L_f + 1$ rationals whose lengths are polynomial in the stated quantities. $\square$

Interpolation inequalities. For an integer $D \geq 1$ put

$$c_0(D) = (D+1)D^D, \quad c_1(D) = (D+1)D^{D+1}, \quad c_2(D) = (D+1)D^{D+2},$$

and for an integer $k \geq 1$ put $\beta_k = (k+1)2^k k^k$. These integers have $O(D \log(D+1))$ and $O(k \log(k+1))$ bits.

Lemma C.2 (Interpolation). *Let r be a real univariate polynomial of degree at most $D \geq 1$.*

- (a) $|r(t)| \leq c_0(D)(1+|t|)^D \max_{0 \leq j \leq D} |r(j/D)|$ for every real t .
- (b) If $\varrho > 0$ and $|r(t)| \leq W$ for $0 \leq t \leq \varrho$, then $|r'(0)| \leq c_1(D)W/\varrho$.
- (c) $|r''(0)| \leq c_2(D) \max_{0 \leq j \leq D} |r(j/D)|$.
- (d) Let $k \geq 1$ and let $p \in \mathbb{R}[z_1, \dots, z_n]$ have degree at most k in each variable. If $|p(z)| \leq V$ for every $z \in [0, 1]^n$, then every coefficient of p has absolute value at most $\beta_k^n V$.

Proof. For $j = 0, \dots, D$ let $\lambda_j(t) = \prod_{i \neq j} (t - i/D) / \prod_{i \neq j} (j/D - i/D)$ be the Lagrange basis for the nodes j/D , so that $r = \sum_j r(j/D) \lambda_j$. The denominator of λ_j has absolute value $j!(D-j)!/D^D \geq D^{-D}$.

(a) Every numerator factor satisfies $|t - i/D| \leq 1 + |t|$, so $|\lambda_j(t)| \leq D^D(1+|t|)^D$. Sum over the $D+1$ nodes.

(b) The derivative at zero of $\prod_{i \neq j} (t - i/D)$ is a sum of D products of $D-1$ numbers in $[-1, 0]$, so it has absolute value at most D , and $|\lambda_j'(0)| \leq D^{D+1}$. Apply this to $s \mapsto r(\varrho s)$, whose values at the nodes lie in $[-W, W]$ and whose derivative at zero is $\varrho r'(0)$.

(c) The second derivative at zero of the numerator is a sum over the $D(D-1)$ ordered pairs of distinct indices of products of numbers in $[-1, 0]$. Hence $|\lambda_j''(0)| \leq D(D-1)D^D \leq D^{D+2}$.

(d) Use the nodes j/k in $[0, 1]$ and the corresponding univariate Lagrange polynomials $\mu_0, \dots, \mu_k$. The sum of the absolute values of the coefficients is submultiplicative, and for a factor $z = i/k$ it equals $1 + i/k \leq 2$; with the denominator bound, each μ_j has coefficient sum at most $2^k k^k$. Tensor interpolation

$$p(z) = \sum_{J \in \{0, \dots, k\}^n} p(J/k) \prod_{i=1}^n \mu_{J_i}(z_i)$$

is exact for polynomials of degree at most k in each variable. The coefficient sum of each product is at most $(2^k k^k)^n$, and there are $(k+1)^n$ terms. Hence the coefficient sum of p is at most $\beta_k^n V$. The interpolation grid enters only this proof; no algorithm below evaluates p on it. $\square$

Integer matrices.

Lemma C.3 (Positive eigenvalues of integer matrices). *Let N be a symmetric positive semidefinite integer matrix of order n , rank $r \geq 1$, and entries bounded in absolute value by $C \geq 1$. Every nonzero eigenvalue of N is at least $(nC)^{-(r-1)}$. Consequently, if $M \succeq 0$ is rational, Q_M is a positive integer with $Q_M M$ integral, $C \geq 1$ bounds the entries of $Q_M M$, and Π is the orthogonal projection onto the range of M , then $x^\top M x \geq \|\Pi x\|^2 / (Q_M (nC)^{n-1})$ for every x .*

Proof. The product of the nonzero eigenvalues of N equals, up to sign, a coefficient of its characteristic polynomial; it is the sum of the principal minors of order r , an integer. It is positive, so it is at least one. Each eigenvalue is at most the largest absolute row sum, which is at most nC . Hence the smallest nonzero eigenvalue is at least $(nC)^{-(r-1)}$. For the consequence, $Q_M M$ and M have the same range, $x^\top M x = (\Pi x)^\top M (\Pi x)$, and Πx lies in the span of the eigenvectors of $Q_M M$ with nonzero eigenvalues. If $M = 0$, both sides vanish. $\square$

Proof of Lemma 3.2. Let $x \in Q$ and let y be the Euclidean projection of x onto the nonempty closed convex set Z . If $x = y$ there is nothing to prove. Let I be the set of inequality rows active at y . The vector $x - y$ lies in the normal cone of the polyhedron Z at y : the linear function $(x - y)^\top z$ is maximized over Z at $z = y$. By linear programming duality this normal cone is $\{A_I^\top \beta + T^\top \alpha : \beta \geq 0\}$, so $x - y = A_I^\top \beta + T^\top \alpha$ for some $\beta \geq 0$ and some α .

We pass to a linearly independent representation. Choose rows of T that form a basis of its row space; the representation can be rewritten with equality coefficients supported on these rows. If the chosen equality rows together with the inequality rows of positive coefficient are linearly dependent, some nontrivial relation among them gives a positive coefficient to at least one inequality row, because the chosen equality rows are independent. Subtracting the largest multiple of the relation that keeps all inequality coefficients nonnegative removes at least one inequality row. After finitely many steps $x - y = R^\top \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix}$, where R stacks linearly independent rows T' of T and A' of A_I , with $\beta' \geq 0$. Let k' be the number of rows of R .

By the Cauchy–Binet formula, $\det(RR^\top)$ is the sum of the squares of the $k' \times k'$ minors of R , so it is at least μ^2 . The largest eigenvalue of $RR^\top$ is at most $\|R\|_1 \|R\|_\infty \leq (k'C)(sC) \leq (sC)^2$. Therefore the smallest eigenvalue is at least $\mu^2 (sC)^{-2(k'-1)} \geq \mu^2 (sC)^{-2(s-1)}$, and $\sigma_{\min}(R) \geq \mu (sC)^{-(s-1)}$. Since R has full row rank,

$$\left\| \begin{pmatrix} \alpha' \\ \beta' \end{pmatrix} \right\| \leq \frac{\|x - y\|}{\sigma_{\min}(R)}.$$

Now

$$\|x - y\|^2 = \alpha'^\top T'(x - y) + \beta'^\top A'(x - y).$$

Each row of A' is active at y and satisfied at $x \in Q$, so $A'(x - y) \leq 0$ and the second term is nonpositive. Since $T'y = t'$, the first term equals $\alpha'^\top (T'x - t')$, which is at most $\|\alpha'\| \|Tx - t\|$. Combining, $\|x - y\|^2 \leq \|x - y\| \|Tx - t\| (sC)^{s-1}/\mu$. Divide by $\|x - y\|$.

For integer matrices, a linearly independent family of k' rows has rank k' , so some $k' \times k'$ minor is a nonzero integer and has absolute value at least one. $\square$

The proof uses neither boundedness of Q nor rationality of b, t . When the equality rows are absent, $Z = Q$ and the bound reads $\text{dist}(x, Z) = 0$.

Relative coordinates.

Lemma C.4 (Relative coordinates and an inner ball). *Let $P = \{x \in \mathbb{R}^n : Ax \leq b\}$ be nonempty, with rational A, b , and let $\varrho \geq 1$ be rational with $P \subseteq [-\varrho, \varrho]^n$. In polynomial time one can compute an integer m with $0 \leq m \leq n$, a point $x_0 \in P \cap \mathbb{Q}^n$, a matrix $V \in \mathbb{Q}^{n \times m}$ of rank m that contains the $m \times m$ identity matrix as a submatrix, and, when $m \geq 1$, a rational system $Gu \leq h$ with nonzero rows $g_i^\top$ and $h > 0$ such that, with $P' = \{u \in \mathbb{R}^m : Gu \leq h\}$,*

(i) $P = \{x_0 + Vu : u \in P'\}$ (for $m = 0$, $P = \{x_0\}$);

(ii) the ball of radius $r_0 = \min_i h_i / \|g_i\|_1$ about 0 lies in P' , and $\|u\| \leq R_0 = 2m\varrho$ for every $u \in P'$.

All outputs have encoding length polynomial in $\langle A, b, \varrho \rangle$. In particular $m = \dim P$.

Proof. For each row $a_i^\top x \leq b_i$ solve $\max\{b_i - a_i^\top x : x \in P\}$. Call the row an implicit equality if the optimum is zero; otherwise record an optimal point $x^{(i)}$, which has positive slack in row i . Let x_0 be the average of the recorded points, or any rational point of P if none was recorded. Then $x_0 \in P$ and x_0 has positive slack in every row that is not an implicit equality.

Let $Ex = e$ be the implicit equalities. Gaussian elimination brings this system to reduced row echelon form, with free coordinates $F \subseteq \{1, \dots, n\}$, $|F| = m$. Its solution set is $\{x_0 + Vu : u \in \mathbb{R}^m\}$, where V has the identity matrix in the rows indexed by F and $u = (x - x_0)_F$. Every point of P satisfies $Ex = e$, so (i) holds with $P' = \{u : x_0 + Vu \in P\}$.

Substitute $x = x_0 + Vu$ into $Ax \leq b$. An implicit equality row becomes $0 \leq 0$: it is constant on the solution set of $Ex = e$ and vanishes at x_0 . Any other row with $V^\top a_i = 0$ is constant on that set and has positive slack at x_0 , so it is redundant. Delete these rows. The remaining rows read $g_i^\top u \leq h_i$ with $g_i = V^\top a_i \neq 0$ and $h_i = b_i - a_i^\top x_0 > 0$. If $\|u\| \leq r_0$, then $g_i^\top u \leq \|g_i\| \|u\| \leq \|g_i\|_1 r_0 \leq h_i$. For $u \in P'$ we have $u = (x - x_0)_F$ with $x \in P$, so $\|u\| \leq \|u\|_1 \leq 2m\rho$. If $m \geq 1$, boundedness of P' forces at least one remaining row. All steps are linear programs and Gaussian elimination. The ball in P' shows that P' is full dimensional, hence $\dim P = m$. $\square$

The convex value interface. We use the following theorem of Grötschel, Lovász, and Schrijver. For a convex body $\mathcal{K} \subseteq \mathbb{R}^d$ (a compact convex set with nonempty interior) and $\varepsilon > 0$, let $S(\mathcal{K}, \varepsilon)$ be the set of points within distance ε of $\mathcal{K}$, and let $S(\mathcal{K}, -\varepsilon)$ be the set of points whose closed ε -ball lies in $\mathcal{K}$. A *weak separation oracle* for $\mathcal{K}$ receives a rational y and a rational $\delta > 0$ and either asserts $y \in S(\mathcal{K}, \delta)$ or returns a rational c with $\|c\|_\infty = 1$ and $c^\top z \leq c^\top y + \delta$ for all $z \in S(\mathcal{K}, -\delta)$.

Lemma C.5 (Weak optimization from weak separation; 49). *There is an oracle algorithm that, given d , a rational $R > 0$ with $\mathcal{K} \subseteq \{z : \|z\| \leq R\}$, a rational vector c , a rational $\varepsilon > 0$, and a weak separation oracle for $\mathcal{K}$, either returns a rational $y \in S(\mathcal{K}, \varepsilon)$ with $c^\top z \leq c^\top y + \varepsilon$ for all $z \in S(\mathcal{K}, -\varepsilon)$, or asserts that $S(\mathcal{K}, -\varepsilon)$ is empty. If the oracle is realized by an algorithm whose running time is polynomial in the encoding lengths of its query and of a fixed description of $\mathcal{K}$, the total running time is polynomial in d , $\langle R \rangle$, $\langle c \rangle$, $\langle \varepsilon \rangle$, and the length of that description.*

This is the weak optimization problem [49, (2.1.10)] for the circumscribed convex body $(\mathcal{K}; d, R)$ given by a weak separation oracle [49, (2.1.13), (2.1.16)], solved in oracle-polynomial time by Grötschel et al. [49, Corollary (4.2.7)]. The running-time consequence is their composition rule for oracle-polynomial algorithms with polynomially bounded oracle answers [49, (1.2.1) and Section 4.1]; we include the description of $\mathcal{K}$ in every oracle query.

Proof of Lemma 3.1. (a) Decide by linear programming whether P is empty. If it is not, apply Lemma C.4. If $m = 0$, return $\hat{x} = x_0$ and $\ell = \varphi(x_0)$. Assume $m \geq 1$ and let $\psi(u) = \varphi(x_0 + Vu)$ on P' ; it is continuous and convex. For $u \in P'$ the point $x_0 + Vu$ lies in P , so $\psi(u)$ and the vector $s_\psi(u) = V^\top s(x_0 + Vu)$ are computable in time polynomial in $\Delta + \langle u \rangle$. The vector $s_\psi(u)$ is a subgradient of ψ on P' , that is, $\psi(u') \geq \psi(u) + s_\psi(u)^\top (u' - u)$ for $u' \in P'$, and on P'

$$|\psi| \leq W, \quad \|s_\psi\| \leq G_V = \|V\|_\Delta G.$$

Consider the capped epigraph

$$\mathcal{K} = \{(u, s) \in \mathbb{R}^m \times \mathbb{R} : u \in P', \psi(u) \leq s \leq W + 2\}.$$

It is compact and convex. Let $a_0 = (0, W + 1)$ and $r_{\mathcal{K}} = \min\{r_0, 1/2\}$. A point (u, s) with $\|(u, s) - a_0\| \leq r_{\mathcal{K}}$ has $\|u\| \leq r_0$, hence $u \in P'$, and $W + \frac{1}{2} \leq s \leq W + \frac{3}{2}$, hence $\psi(u) \leq W < s \leq W + 2$. Thus the ball of radius $r_{\mathcal{K}}$ about a_0 lies in $\mathcal{K}$, and $\mathcal{K}$ is a convex body. Since $\|u\| \leq R_0$ and $|s| \leq W + 2$ on $\mathcal{K}$, it lies in the ball of radius $R_{\mathcal{K}} = R_0 + W + 2$ about the origin.

Separation. Given rational (u, s) and $\delta > 0$, proceed as follows. If $g_i^\top u > h_i$ for some i , return $(g_i, 0)/\|g_i\|_\infty$; every point of $\mathcal{K}$ satisfies $g_i^\top u' \leq h_i < g_i^\top u$. Otherwise, if $s > W + 2$, return $(0, 1)$.

Otherwise $u \in P'$; compute $\psi(u)$ and $s_\psi(u)$. If $s < \psi(u)$, return $(s_\psi(u), -1)$ divided by its maximum norm, which is at least one. This is valid because for $(u', s') \in \mathcal{K}$ the subgradient inequality gives

$$s' \geq \psi(u') \geq \psi(u) + s_\psi(u)^\top(u' - u) > s + s_\psi(u)^\top(u' - u).$$

Otherwise $(u, s) \in \mathcal{K}$, and we assert $(u, s) \in S(\mathcal{K}, \delta)$. Every returned vector separates (u, s) from all of $\mathcal{K}$, so the answers meet the weak separation requirement for every $\delta > 0$. The running time is polynomial in $\Delta + \langle u, s \rangle$.

Optimization. Put

$$A_0 = 1 + \frac{2W + 1}{r_{\mathcal{K}}}, \quad C_0 = A_0 + 1 + G_V \left(1 + \frac{R_0}{r_0}\right), \quad \varepsilon = \min\left\{\frac{r_{\mathcal{K}}}{2}, \frac{\eta}{C_0}\right\}.$$

Apply Lemma C.5 with objective $c = (0, \dots, 0, -1)$ and tolerance ε . The set $S(\mathcal{K}, -\varepsilon)$ contains the ball of radius $r_{\mathcal{K}} - \varepsilon > 0$ about a_0 , so the algorithm returns a rational $(u, s) \in S(\mathcal{K}, \varepsilon)$ with $s \leq s' + \varepsilon$ for every $(u', s') \in S(\mathcal{K}, -\varepsilon)$.

Lower bound. Let u_* minimize ψ over P' , $\psi_* = \psi(u_*)$, $z_* = (u_*, \psi_*) \in \mathcal{K}$, and $\lambda = \varepsilon/r_{\mathcal{K}} \leq \frac{1}{2}$. For $\|e\| \leq \varepsilon$,

$$(1 - \lambda)z_* + \lambda a_0 + e = (1 - \lambda)z_* + \lambda(a_0 + e/\lambda),$$

and $\|e/\lambda\| \leq r_{\mathcal{K}}$, so this point lies in $\mathcal{K}$. Hence $(1 - \lambda)z_* + \lambda a_0 \in S(\mathcal{K}, -\varepsilon)$. Its last coordinate is $\psi_* + \lambda(W + 1 - \psi_*) \leq \psi_* + \lambda(2W + 1)$, because $\psi_* \geq -W$. Therefore $s \leq \psi_* + A_0\varepsilon$, and $\ell = s - A_0\varepsilon$ satisfies $\ell \leq \psi_* = \min_P \varphi$.

Feasible repair. Some $(\bar{u}, \bar{s}) \in \mathcal{K}$ lies within ε of (u, s) . Put $\hat{u} = r_0 u / (r_0 + \varepsilon)$. For each row, $g_i^\top u \leq g_i^\top \bar{u} + \|g_i\| \varepsilon \leq h_i + \|g_i\| \varepsilon$, and $r_0 \|g_i\| \leq r_0 \|g_i\|_1 \leq h_i$, so

$$g_i^\top \hat{u} \leq \frac{r_0 h_i + r_0 \|g_i\| \varepsilon}{r_0 + \varepsilon} \leq h_i.$$

Thus $\hat{u} \in P'$. Moreover $\hat{u} - \bar{u} = [r_0(u - \bar{u}) - \varepsilon \bar{u}] / (r_0 + \varepsilon)$, and $\|\bar{u}\| \leq R_0$, so $\|\hat{u} - \bar{u}\| \leq \varepsilon(1 + R_0/r_0)$. Both points lie in P' , and the subgradient inequality at $\hat{u}$ gives $\psi(\hat{u}) \leq \psi(\bar{u}) + s_\psi(\bar{u})^\top(\hat{u} - \bar{u}) \leq \psi(\bar{u}) + G_V \|\hat{u} - \bar{u}\|$. Hence

$$\psi(\hat{u}) \leq \psi(\bar{u}) + G_V \varepsilon(1 + R_0/r_0) \leq \bar{s} + G_V \varepsilon(1 + R_0/r_0) \leq s + \varepsilon + G_V \varepsilon(1 + R_0/r_0).$$

Return $\hat{x} = x_0 + V\hat{u} \in P$ and ℓ . Then $\ell \leq \min_P \varphi \leq \varphi(\hat{x})$ and $\varphi(\hat{x}) - \ell \leq C_0 \varepsilon \leq \eta$.

Complexity. The data of Lemma C.4, and $A_0, C_0, \varepsilon, R_{\mathcal{K}}$, have encoding length polynomial in $\Delta + \langle \eta \rangle$. The separation routine runs in polynomial time in its query length and Δ . Lemma C.5 and one final evaluation give the stated bound.

(b) Lemma C.1 evaluates φ and $\nabla \varphi$ exactly and supplies $W = \max\{1, V_f + \tau n \varrho^2\}$ and $G = \max\{1, \|T\|_\Delta G_f + 2\tau n \varrho\}$. For a function that is continuously differentiable near P and convex on P , the gradient is a subgradient on P : for $x, x' \in P$ the convex function $\theta \mapsto \varphi(x + \theta(x' - x))$ on $[0, 1]$ lies above its tangent at $\theta = 0$. Part (a) applies with Δ polynomial in D and the stated encoding lengths. $\square$

Remark C.6 (An independently checkable lower bound). If φ is twice continuously differentiable with $\|\nabla^2 \varphi\| \leq H$ on P , a lower bound can also be certified by linear programming duality. Let $\hat{x} \in P$ have $\varphi(\hat{x}) - \min_P \varphi \leq \delta$, let $\gamma = \nabla \varphi(\hat{x})$, and let $\hat{\ell} = \varphi(\hat{x}) + \min_{x \in P} \gamma^\top(x - \hat{x})$. Convexity gives $\hat{\ell} \leq \min_P \varphi$, and an optimal dual solution of the tangent linear program certifies $\hat{\ell}$. If x' attains the tangent minimum and $\Delta_P \geq \max_{x, x' \in P} \|x - x'\|$, the points $\hat{x} + t(x' - \hat{x})$, $0 \leq t \leq 1$, give $t(\varphi(\hat{x}) - \hat{\ell}) \leq \delta + H t^2 \Delta_P^2 / 2$. With $K_0 = \max\{1, H \Delta_P^2\}$, $\delta = \min\{\eta/2, \eta^2/(8K_0)\}$, and $t = \min\{1, \eta/(2K_0)\}$ this yields $\varphi(\hat{x}) - \hat{\ell} \leq \eta$. A verifier checks feasibility of $\hat{x}$, the exact values $\varphi(\hat{x})$ and γ , and the dual certificate; it does not repeat the ellipsoid computation. This certificate is optional and is not used in the theorems.

The selector lemma.

Proof of Lemma 3.3. Write $\Phi = \varphi + \tau \|\cdot\|^2$. Since $\varphi \geq \varphi_*$ on P , every sublevel set of Φ in P is bounded; it is closed by continuity. Hence Φ attains its minimum on P , and the minimizer x_τ is unique because $\tau \|\cdot\|^2$ is strictly convex. Comparison with p gives

$$\varphi(x_\tau) + \tau \|x_\tau\|^2 \leq \varphi_* + \tau \|p\|^2,$$

so $\|x_\tau\| \leq \|p\| \leq R$ and $\varphi(x_\tau) - \varphi_* \leq \tau R^2 \leq 1$; the last inequality holds because $\varepsilon \leq 1 \leq R, \Gamma$ and $d \geq 2$.

Let s be the projection of x_τ onto the closed convex set S , and $e = \|s - x_\tau\|$. Comparison of Φ at x_τ and s gives

$$\varphi(x_\tau) - \varphi_* \leq \tau(\|s\|^2 - \|x_\tau\|^2) \leq \tau e(\|s\| + \|x_\tau\|) \leq \tau e(2\|x_\tau\| + e).$$

Since $p \in S$, $e \leq \|p - x_\tau\| \leq 2R$, and the right side is at most $4R\tau e$. If every point of S has norm at most R , then $\|s\| + \|x_\tau\| \leq 2R$ and the bound is $2R\tau e$. The error bound applies to x_τ , so $e^d \leq \Gamma^d \cdot 4R\tau e$. If $e > 0$, the choice of τ gives

$$e^{d-1} \leq 4R\tau\Gamma^d = \left(\frac{\varepsilon^2}{8R}\right)^{d-1},$$

so $e \leq \varepsilon^2/(8R)$; this also holds if $e = 0$, and in the bounded variant with the factor 2.

The point p is the projection of the origin onto S , so $p^\top(z - p) \geq 0$ for all $z \in S$; in particular $p^\top s \geq \|p\|^2$. Using $\|x_\tau\| \leq \|p\|$,

$$\|x_\tau - p\|^2 \leq 2\|p\|^2 - 2p^\top x_\tau \leq 2p^\top(s - x_\tau) \leq 2Re \leq \frac{\varepsilon^2}{4}.$$

Finally, for $x \in P$ and $0 < \theta \leq 1$, the point $x_\theta = x_\tau + \theta(x - x_\tau)$ lies in P , and convexity of φ and the identity $\|x_\theta\|^2 = (1 - \theta)\|x_\tau\|^2 + \theta\|x\|^2 - \theta(1 - \theta)\|x - x_\tau\|^2$ give

$$\Phi(x_\tau) \leq \Phi(x_\theta) \leq (1 - \theta)\Phi(x_\tau) + \theta\Phi(x) - \tau\theta(1 - \theta)\|x - x_\tau\|^2.$$

Dividing by θ and letting $\theta \rightarrow 0$ yields $\Phi(x) - \Phi(x_\tau) \geq \tau\|x - x_\tau\|^2$. If the left side is at most $\tau\varepsilon^2/4$, then $\|x - x_\tau\| \leq \varepsilon/2$ and $\|x - p\| \leq \varepsilon$. $\square$

No differentiability of φ is needed. The lemma does not assume that S is bounded; the factor 4 pays for a projection s that may lie far from the origin.

C.2 Proof of Theorem 3.4

Fix the data of Theorem 3.4: an explicit sparse $f = \sum_\alpha f_\alpha x^\alpha$, convex on $\mathbb{R}^n$, and $P = \{x : Ax \leq b\}$. If $n = 0$, the problem is the evaluation of a constant and a check of the constraints, so assume $n \geq 1$. Put $D = \max\{2, \deg f\}$ and $\kappa_D = c_2(D) = (D + 1)D^{D+2}$. Let $\int \cdot da$ denote integration over $[0, 1]^n$ with respect to Lebesgue measure.

Lemma C.7 (Integrated quadratic minorant). *Let*

$$\begin{aligned} M &= \int \nabla^2 f(a) da, & g &= \int \left[\nabla f(a) - \frac{2}{\kappa_D} \nabla^2 f(a) a \right] da, \\ m_0 &= \int \left[f(a) - \nabla f(a)^\top a + \frac{1}{\kappa_D} a^\top \nabla^2 f(a) a \right] da. \end{aligned}$$

- (a) M , g , and m_0 are rational, computable in time polynomial in L, D , and $f(x) \geq m_0 + g^\top x + x^\top Mx/\kappa_D$ for all $x \in \mathbb{R}^n$.
- (b) $M \succeq 0$, and $\ker M$ is the set of d with $\nabla^2 f(a)d = 0$ for every $a \in \mathbb{R}^n$. For such d , $\nabla f(x)^\top d = \nabla f(0)^\top d$ for all x .
- (c) Let Π be the orthogonal projection onto the range of M ($\Pi = 0$ if $M = 0$) and $w = -(I - \Pi)\nabla f(0)$. Then Π and w are rational and computable in polynomial time, and

$$f(x) = f(\Pi x) - w^\top x \quad (x \in \mathbb{R}^n), \quad g = \Pi g - w.$$

- (d) Let Q_M be a positive integer with $Q_M M$ integral, let $C_M \geq 1$ bound the entries of $Q_M M$, and put $\mu_0 = 1/(\kappa_D Q_M (nC_M)^{n-1})$. Then $x^\top Mx/\kappa_D \geq \mu_0 \|\Pi x\|^2$ for all x .

Proof. (a) Fix a, x and let $h(t) = f(a + t(x - a)) - f(a) - t\nabla f(a)^\top (x - a)$. It has degree at most D . By global convexity $h \geq 0$ on $\mathbb{R}$, $h(0) = 0$, and h is convex, so $0 \leq h(t) \leq th(1) \leq h(1)$ on $[0, 1]$. Since $h''(0) = (x - a)^\top \nabla^2 f(a)(x - a)$, Lemma C.2(c) gives

$$f(x) \geq f(a) + \nabla f(a)^\top (x - a) + \frac{1}{\kappa_D} (x - a)^\top \nabla^2 f(a)(x - a).$$

Expanding the right side in powers of x and integrating over a gives the minorant. For computation, use the identities

$$(\nabla^2 f(a)a)_i = \sum_{\alpha} f_{\alpha} \alpha_i (|\alpha| - 1) a^{\alpha - e_i}, \quad a^\top \nabla^2 f(a)a = \sum_{\alpha} f_{\alpha} |\alpha| (|\alpha| - 1) a^{\alpha},$$

$\nabla f(a)^\top a = \sum_{\alpha} f_{\alpha} |\alpha| a^{\alpha}$, and $\int a^{\gamma} da = \prod_i (\gamma_i + 1)^{-1}$. Thus

$$M_{ij} = \sum_{\alpha} f_{\alpha} \alpha_i (\alpha_j - \delta_{ij}) \int a^{\alpha - e_i - e_j} da, \quad g_i = \sum_{\alpha} f_{\alpha} \alpha_i \left(1 - \frac{2(|\alpha| - 1)}{\kappa_D}\right) \int a^{\alpha - e_i} da,$$

and $m_0 = \sum_{\alpha} f_{\alpha} (1 - |\alpha| + |\alpha|(|\alpha| - 1)/\kappa_D) \int a^{\alpha} da$, where terms with a negative exponent are omitted. Each entry is a sum of at most one term per input monomial, with denominators of $O(n \log(D + 1) + D \log D)$ bits.

(b) M is an integral of positive semidefinite matrices. If $d^\top M d = 0$, the continuous nonnegative polynomial $a \mapsto d^\top \nabla^2 f(a) d$ has zero integral over the cube, so it vanishes on the cube and hence identically. Positive semidefiniteness gives $\nabla^2 f(a)d = 0$ for all a . The converse is immediate. For such d , the gradient of $x \mapsto \nabla f(x)^\top d$ is $\nabla^2 f(x)d = 0$.

(c) A rational basis E of the range of M gives $\Pi = E(E^\top E)^{-1}E^\top$. Since $(I - \Pi)x \in \ker M$, part (b) gives

$$f(x) - f(\Pi x) = \int_0^1 \nabla f(\Pi x + \theta(I - \Pi)x)^\top (I - \Pi)x d\theta = \nabla f(0)^\top (I - \Pi)x = -w^\top x.$$

Every $\nabla^2 f(a)$ is symmetric and annihilates $\ker M$, so its range lies in the range of M and $(I - \Pi)\nabla^2 f(a) = 0$. By (b), $(I - \Pi)\nabla f(a) = (I - \Pi)\nabla f(0)$. Applying $I - \Pi$ to the definition of g gives $(I - \Pi)g = -w$.

(d) This is Lemma C.3. □

The vector w is the slope of f along the common Hessian kernel. Kernel directions with $w^\top d \neq 0$ are directions of linear change, not of invariance; part (c) records both.

Lemma C.8 (Boundedness, attainment, and a radius). *Let $P \neq \emptyset$, let $a \in P$ be rational, and put $f_a = f(a)$. Consider the linear system*

$$w = A^\top \lambda + Mz, \quad \lambda \geq 0. \quad (38)$$

- (a) *If (38) is infeasible, there is a rational d with $Ad \leq 0$, $Md = 0$, and $w^\top d = 1$, and $f(a + td) = f_a - t$ for all $t \geq 0$.*
- (b) *If it is feasible, let (λ, z) be a rational solution and put $v = Mz$, $\nu = \|\Pi g - v\|_1$, and $\beta_0 = m_0 - \lambda^\top b$. Then*

$$f(x) \geq \beta_0 - \nu \|\Pi x\| + \mu_0 \|\Pi x\|^2 \quad (x \in P). \quad (39)$$

With $r_ = 1 + (\nu + |f_a - \beta_0| + 1)/\mu_0$ and $W_0 = 1 + |\lambda^\top b| + \|v\|_1 r_* + |f(0)| + \|\nabla f(0)\|_1 r_* + |f_a|$, every $x \in P$ with $f(x) \leq f_a$ satisfies $\|\Pi x\| \leq r_*$ and $|w^\top x| \leq W_0$.*

- (c) *In case (b), let $J = \begin{pmatrix} M \\ w^\top \end{pmatrix}$, let Q_J be a positive integer with $Q_J J$ integral, let A_Z be obtained from A by multiplying each row by the least positive integer that makes it integral, let $C \geq 1$ bound the entries of $Q_J J$ and A_Z , let $m_1 = \max_{ij} |M_{ij}|$, and put*

$$R = 1 + \|a\|_1 + (nC)^{n-1} Q_J (\|Ja\|_1 + n^2 m_1 r_* + W_0).$$

Every $y \in P$ with $f(y) \leq f_a$ has a point $y' \in P$ with $f(y') = f(y)$ and $\|y'\| < R$. Consequently f attains its infimum on P , the optimal set S is nonempty, closed, and convex, and its minimum-norm point p satisfies $\|p\| < R$.

- (d) *All quantities above are rational and computable in time polynomial in L, D .*

Proof. (a) By Farkas' lemma, infeasibility gives d with $Ad \leq 0$, $Md = 0$, and $w^\top d > 0$; a linear program finds one with $w^\top d = 1$. Then $a + td \in P$ for $t \geq 0$, $\Pi d = 0$, and Lemma C.7(c) gives $f(a + td) = f(\Pi a) - w^\top a - t = f_a - t$.

(b) For $x \in P$, $\lambda \geq 0$ and $Ax \leq b$ give $w^\top x = \lambda^\top Ax + z^\top Mx \leq \lambda^\top b + v^\top \Pi x$, because v lies in the range of M . By Lemma C.7,

$$f(x) \geq m_0 + (\Pi g)^\top \Pi x - w^\top x + \mu_0 \|\Pi x\|^2 \geq \beta_0 + (\Pi g - v)^\top \Pi x + \mu_0 \|\Pi x\|^2,$$

and $|(\Pi g - v)^\top \Pi x| \leq \nu \|\Pi x\|$. This proves (39). If $f(x) \leq f_a$ and $\rho = \|\Pi x\| > r_*$, then $\rho > 1$ and $\mu_0 \rho - \nu > |f_a - \beta_0| + 1$, so $\mu_0 \rho^2 - \nu \rho > |f_a - \beta_0| + 1 > f_a - \beta_0$, contradicting (39). Hence $\|\Pi x\| \leq r_*$. The upper bound $w^\top x \leq \lambda^\top b + \|v\|_1 r_*$ was shown above. For the lower bound, global convexity at 0 and Lemma C.7(c) give

$$w^\top x = f(\Pi x) - f(x) \geq f(0) + \nabla f(0)^\top \Pi x - f_a \geq f(0) - \|\nabla f(0)\|_1 r_* - f_a.$$

The constraints $Ax \leq b$ alone give only the upper bound.

(c) Let $y \in P$ with $f(y) \leq f_a$. Since $My = M\Pi y$, $\|My\|_1 \leq nm_1 \|\Pi y\|_1 \leq n^2 m_1 r_*$, and $\|Q_J J y\|_1 \leq Q_J (n^2 m_1 r_* + W_0)$. The slice $Z_y = \{x \in P : Q_J J x = Q_J J y\}$ contains y . Apply Lemma 3.2 with inequality matrix A_Z and equality matrix $Q_J J$, both integral, to the point $a \in P$. The projection y' of a onto Z_y satisfies

$$\|y'\| \leq \|a\| + (nC)^{n-1} \|Q_J J(a - y)\| < R.$$

From $My' = My$ we get $y' - y \in \ker M$ and $\Pi y' = \Pi y$; with $w^\top y' = w^\top y$, Lemma C.7(c) gives $f(y') = f(y)$.

By (b), f is bounded below on P . If a is optimal, its own representative is a , of norm at most $\|a\|_1 < R$. Otherwise a minimizing sequence eventually lies in $\{f \leq f_a\}$; replace it by representatives in the compact set $P \cap \{\|x\| \leq R\}$, extract a convergent subsequence, and use continuity. Either way f attains its infimum at a point of norm less than R . The optimal set is a sublevel set of a convex continuous function on a closed convex set, so its minimum-norm point exists and has norm less than R . The argument does not assume attainment in order to construct R .

(d) The point a , the system (38), and its solution or Farkas certificate come from linear programs of polynomial size. Lemma C.7 supplies M, g, m_0, Π, w, μ_0 . The values $f_a, f(0), \nabla f(0)$ and all products in r_*, W_0, R have polynomial encoding length; the power $(nC)^{n-1}$ has $(n-1)\log_2(nC)$ bits. $\square$

Lemma C.9 (Global error modulus). *Assume that $S \neq \emptyset$ and that its minimum-norm point p satisfies $\|p\| \leq R$ for a rational $R \geq 1$. Let $\mathcal{E} = \{\alpha - e_i : f_\alpha \neq 0, \alpha_i > 0\}$ and define $N \in \mathbb{Q}^{\mathcal{E} \times n}$ by*

$$N_{\gamma, i} = (\gamma_i + 1)f_{\gamma + e_i},$$

where $f_{\gamma + e_i} = 0$ if $\gamma + e_i$ is not an exponent of f . Let Q_N be a positive integer with $Q_N N$ integral, let $C_N \geq 1$ bound the entries of $Q_N N$ and A_Z , let $k_N = \max\{1, |\mathcal{E}|\}$, and put

$$V_f = \sum_{\alpha} |f_\alpha| (R+2)^{|\alpha|}, \quad G_f = \sum_{|\alpha| \geq 1} |f_\alpha| |\alpha| (R+2)^{|\alpha|-1}, \quad h_f = \max\{1, 2V_f + 2G_f(R+1)\},$$

$$C_* = 1 + c_1(D)(h_f + 3^D c_0(D)), \quad \Gamma = \max\{1, (nC_N)^{n-1} Q_N k_N \beta_{D-1}^n C_*\}.$$

Then $S = \{x \in P : Nx = Np\}$ and

$$\text{dist}(x, S) \leq \Gamma (f(x) - f_*)^{1/D} \quad (x \in P, f(x) - f_* \leq 1).$$

The number Γ is computable in time polynomial in $L, D, \langle R \rangle$.

Proof. Coefficient rows. For every d and z ,

$$\nabla f(z)^\top d = \sum_i d_i \sum_{\alpha_i > 0} \alpha_i f_\alpha z^{\alpha - e_i} = \sum_{\gamma \in \mathcal{E}} (Nd)_\gamma z^\gamma,$$

and distinct γ give distinct monomials. Hence $Nd = 0$ if and only if $\nabla f(z)^\top d = 0$ for all z , that is, if and only if $f(z + \theta d) = f(z)$ for all z and all real θ .

A Bregman estimate. Fix $x \in P$ with $E = f(x) - f_* \in [0, 1]$, put $d = x - p$, and let

$$h(u) = f(p + u) - f(p) - \nabla f(p)^\top u.$$

Global convexity makes h convex and nonnegative on $\mathbb{R}^n$. Optimality of p over P gives $\nabla f(p)^\top d \geq 0$, and convexity gives $\nabla f(p)^\top d \leq E$. For $0 \leq \theta \leq 1$,

$$0 \leq h(\theta d) \leq f(p + \theta d) - f_* \leq \theta E \leq E.$$

The polynomial $\theta \mapsto h(\theta d)$ has degree at most D , so Lemma C.2(a) gives $h(2\theta d) \leq c_0(D)(1 + 2|\theta|)^D E$ for every real θ .

Let $c \in [0, 1]^n$ and $\pi_c(\theta) = h(c - p + \theta d)$. Since $c - p + \theta d$ is the midpoint of $2(c - p)$ and $2\theta d$, convexity of h gives

$$0 \leq \pi_c(\theta) \leq \frac{1}{2}h(2(c - p)) + \frac{1}{2}h(2\theta d). \quad (40)$$

The points p and $2c - p$ lie in the box of radius $R + 2$, and $\|c - p\|_\infty \leq R + 1$. For $\|y\|_\infty \leq R + 2$ every monomial satisfies $|y^\alpha| \leq (R + 2)^{|\alpha|}$ and $\sum_i |\partial_i y^\alpha| \leq |\alpha|(R + 2)^{|\alpha|-1}$, so $|f(y)| \leq V_f$ and $\|\nabla f(y)\|_1 \leq G_f$. Hence

$$h(2(c - p)) \leq |f(2c - p)| + |f(p)| + 2 \|\nabla f(p)\|_1 \|c - p\|_\infty \leq h_f.$$

Only the norm of p enters here; the point x may be arbitrarily far away.

If $E > 0$, then on $0 \leq \theta \leq E^{-1/D}$ we have $E(1 + 2\theta)^D \leq E(3E^{-1/D})^D = 3^D$, so $0 \leq \pi_c \leq h_f + 3^D c_0(D)$ there. Lemma C.2(b) with $\varrho = E^{-1/D}$ gives $|\pi'_c(0)| \leq c_1(D)(h_f + 3^D c_0(D))E^{1/D}$. Since $\pi'_c(0) = (\nabla f(c) - \nabla f(p))^\top d$ and $E \leq E^{1/D}$,

$$|\nabla f(c)^\top d| \leq C_* E^{1/D} \quad (c \in [0, 1]^n). \quad (41)$$

If $E = 0$, then $h(\theta d) = 0$ on $[0, 1]$, so this polynomial vanishes identically, and (40) bounds π_c on the whole real line. A bounded polynomial is constant, so $\pi'_c(0) = 0$; also $\nabla f(p)^\top d = 0$. Thus (41) holds with right side zero.

From residuals to coefficients. The polynomial $z \mapsto \nabla f(z)^\top d = \sum_\gamma (Nd)_\gamma z^\gamma$ has degree at most $D - 1 \geq 1$ in each variable, and (41) bounds it on $[0, 1]^n$. Lemma C.2(d) gives

$$|(N(x - p))_\gamma| \leq \beta_{D-1}^n C_* E^{1/D} \quad (\gamma \in \mathcal{E}).$$

The optimal slice. If $x \in S$, then $E = 0$ and $N(x - p) = 0$. Conversely, if $x \in P$ and $N(x - p) = 0$, then $f(x) = f(p)$ by invariance, so $x \in S$. Hence $S = \{x \in P : Q_N N x = Q_N N p\}$. Apply Lemma 3.2 with the integer matrices A_Z and $Q_N N$:

$$\text{dist}(x, S) \leq (nC_N)^{n-1} \|Q_N N(x - p)\| \leq (nC_N)^{n-1} Q_N \sqrt{k_N} \beta_{D-1}^n C_* E^{1/D} \leq \Gamma E^{1/D}.$$

If $\mathcal{E}$ is empty, f is constant, $S = P$, and the bound is trivial.

Complexity. The matrix N has at most n times as many rows as f has monomials, and its entries are products of input coefficients and exponents. The numbers V_f, G_f, h_f are sums of at most one term per monomial, each of $O(\langle f_\alpha \rangle + D \langle R \rangle)$ bits. The constants $c_0(D), c_1(D), 3^D$ have $O(D \log(D + 1))$ bits and β_{D-1}^n has $O(nD \log(D + 1))$ bits. $\square$

Proof of Theorem 3.4. Part (a). Decide emptiness of P by linear programming. If $P \neq \emptyset$, compute a rational $a \in P$ and the data of Lemma C.7, and solve (38). By Lemma C.8, the system is infeasible exactly when f is unbounded below on P , and in that case the lemma supplies the certificate (a, d) .

Part (b). If (38) is feasible, Lemma C.8(c) gives attainment and a rational R with $\|p\| < R$, and Lemma C.9 with this R gives Γ . Both are computed in time polynomial in L, D , so $\log R$ and $\log \Gamma$ are polynomial in L, D .

Part (c). Let $\varepsilon = 2^{-q}$ and define τ by Lemma 3.3 with $d = D$ and the computed R, Γ . Then

$$\log_2(1/\tau) \leq (2D - 2)q + (3D - 1) + D \log_2 R + D \log_2 \Gamma,$$

which is polynomial in L, D, q . By Lemma 3.3, $\|x_\tau\| \leq \|p\| < R$, so x_τ lies in the bounded polytope $Q = P \cap [-R, R]^n$ and minimizes $f + \tau \|\cdot\|^2$ over Q as well. Apply Lemma 3.1 to Q , $\varrho = R$, the function $\varphi = f + \tau \|\cdot\|^2$, and $\eta = \tau \varepsilon^2/4$. The evaluation and the bounds W, G come from Lemma C.1 with $T = I$, $t = 0$; the composition $f(x_0 + Vu)$ inside Lemma 3.1 is evaluated, not expanded. The returned rational $x_q \in Q \subseteq P$ satisfies $\varphi(x_q) - \varphi(x_\tau) \leq \eta$, hence $\|x_q - p\| \leq \varepsilon$.

For a value enclosure of width $\delta = 2^{-q_f}$, let $G_Q = \max\{1, \sum_{|\alpha| \geq 1} |f_\alpha| |\alpha| R^{|\alpha|-1}\}$, which bounds $\|\nabla f\|$ on $[-R, R]^n$. Run the preceding procedure with accuracy $\varepsilon' = \min\{2^{-q}, \delta/(2G_Q)\}$. The returned point x_q is within 2^{-q} of p , and since the segment $[p, x_q]$ lies in Q , $f(x_q) - f_* \leq G_Q \varepsilon' \leq \delta/2$. Apply Lemma 3.1 to f on Q with $\eta = \delta/2$ to get $\ell \leq \min_Q f$. Because $p \in Q$, $\min_Q f = f_*$. Thus $\ell \leq f_* \leq f(x_q) \leq \ell + \delta$. The encoding lengths involved are polynomial in $L, D, q + q_f$. $\square$

Proof of Corollary 3.5. For bounded $P \subseteq [-\varrho, \varrho]^n$, the function f attains its minimum by compactness, and $R_x = \max\{1, n\varrho\}$ bounds the norm of every point of P . Lemma C.9 applies with $R = R_x$, without Lemmas C.7 and C.8. Every optimal point has norm at most R_x , so Lemma 3.3 holds with the factor 2, and the proof of part (c) above applies with $Q = P$. For fixed D , every bound polynomial in L, D, q is polynomial in $L + q$. $\square$

Proof of the claim in Remark 3.6. Let $n = 2r \geq 4$, $f(y) = \sum_i y_i^{2r}$, and $U = I - \frac{2}{n} \mathbf{1}\mathbf{1}^\top$. Then $U^\top U = I - \frac{4}{n} \mathbf{1}\mathbf{1}^\top + \frac{4}{n^2} n \mathbf{1}\mathbf{1}^\top = I$, and every entry of U is nonzero because $n \geq 4$. For an exponent β with $|\beta| = r$, the coefficient of $y^{2\beta}$ in $f(Uy) = \sum_i (\sum_j U_{ij} y_j)^{2r}$ is

$$\frac{(2r)!}{\prod_j (2\beta_j)!} \sum_i \prod_j U_{ij}^{2\beta_j} > 0,$$

a sum of positive terms. There are $\binom{3r-1}{r} \geq 2^r$ such exponents, while f has n monomials of degree $2r$. $\square$

C.3 Proof of Theorem 3.7

Lemma C.10 (Reflection). *Let $H : \mathbb{R}^m \rightarrow \mathbb{R}^{m \times m}$ be an affine map into symmetric matrices, let $\Omega \subseteq \mathbb{R}^m$ be convex with $H(u) \succeq 0$ for $u \in \Omega$, and let $c \in \Omega$ and $\theta \in (0, 1]$ be such that $c - \theta(u - c) \in \Omega$ for every $u \in \Omega$. Then $0 \preceq H(u) \preceq (1 + \theta^{-1})H(c)$ for $u \in \Omega$; in particular $\ker H(c) \subseteq \ker H(u)$. If Ω contains the ball of radius r about c and lies in the ball of radius $R \geq r$ about c , one can take $\theta = r/R$. If Ω is a box and c is its center, one can take $\theta = 1$.*

Proof. With $u' = c - \theta(u - c) \in \Omega$ we have $c = \frac{\theta}{1+\theta}u + \frac{1}{1+\theta}u'$, so affinity gives $H(c) = \frac{\theta}{1+\theta}H(u) + \frac{1}{1+\theta}H(u') \succeq \frac{\theta}{1+\theta}H(u)$. If $H(c)d = 0$, then $0 \leq d^\top H(u)d \leq 0$ and positive semidefiniteness gives $H(u)d = 0$. In the ball case, $\|u' - c\| = \theta \|u - c\| \leq r$; in the box case u' is the reflection of u through the center. $\square$

For a polynomial φ of degree at most three write $H(u) = \nabla^2 \varphi(u)$; it is affine in u . Let $D^3 \varphi$ be the constant, fully symmetric third derivative, so that $H(u) - H(v) = D^3 \varphi[u - v]$ and Taylor's formula is exact:

$$\varphi(v + d) = \varphi(v) + \nabla \varphi(v)^\top d + \frac{1}{2} d^\top H(v) d + \frac{1}{6} D^3 \varphi[d, d, d]. \quad (42)$$

Proof of Lemma 3.8. Put $a = d^\top H(v)d$ and $b = d^\top H(u)d$, both nonnegative. Since $b = a + D^3 \varphi[d, d, d]$, (42) gives $E = \nabla \varphi(v)^\top d + (2a + b)/6$. First-order optimality of v over the convex set Ω gives $\nabla \varphi(v)^\top d \geq 0$, so $\sigma = a + b \leq 6E$. Symmetry of $D^3 \varphi$ gives

$$d^\top H(c)d = a + D^3 \varphi[c - v, d, d] = a + (c - v)^\top (H(u) - H(v))d.$$

For a positive semidefinite Q , $\|Qd\|^2 = d^\top Q^2 d \leq \|Q\| d^\top Q d$. Hence $\|H(u)d\| + \|H(v)d\| \leq \sqrt{\Lambda b} + \sqrt{\Lambda a} \leq \sqrt{2\Lambda \sigma}$, and $|d^\top H(c)d| \leq \sigma + \rho_1 \sqrt{2\Lambda \sigma}$. Also $\sigma \leq 2\Lambda \rho_2^2$, so $\sigma \leq \rho_2 \sqrt{2\Lambda \sigma}$. Therefore $|d^\top H(c)d| \leq (\rho_1 + \rho_2) \sqrt{2\Lambda \sigma}$, and squaring with $\sigma \leq 6E$ proves the lemma. $\square$

Proof of Lemma 3.9. Let $\psi(u) = \varphi(u) - g_c^\top u$. For $u \in \Omega$ the segment from c to u lies in Ω , and $\nabla \psi(u) = \int_0^1 H(c + \theta(u - c))(u - c) d\theta$. Each H on Ω is symmetric and annihilates K , so $\nabla \psi(u) \in K^\perp$.

Let w minimize φ over Ω , put $d = w - v$, $a = d^\top H(v)d$, and $b = d^\top H(w)d$. Since $\varphi(w) = \varphi(v)$, the computation in the previous proof gives $0 = \nabla \varphi(v)^\top d + (2a + b)/6$ with three nonnegative terms, so $a = b = 0$ and $H(v)d = H(w)d = 0$. The identity in the previous proof gives $d^\top H(c)d = 0$, hence

$H(c)d = 0$ and $d \in K$. Then $\psi(w) - \psi(v) = \int_0^1 \nabla \psi(v + \theta d)^\top d d\theta = 0$ and $0 = \varphi(w) - \varphi(v) = g_c^\top d$. Conversely, if $w \in \Omega$, $H(c)(w - v) = 0$, and $g_c^\top(w - v) = 0$, then $w - v \in K$, $\psi(w) = \psi(v)$, and $\varphi(w) = \varphi(v)$.

For the estimate, let $u \in \Omega$ and $d = u - v$. Since $\nabla \psi \in K^\perp$, $\nabla \psi^\top d = \nabla \psi^\top \Pi_K d$, and the mean value bound on $[c, z] \subseteq \Omega$ gives $\|\nabla \psi(z)\| = \|\nabla \varphi(z) - \nabla \varphi(c)\| \leq \Lambda R_\Omega$. Hence

$$|g_c^\top d| = \left| \varphi(u) - \varphi(v) - \int_0^1 \nabla \psi(v + \theta d)^\top \Pi_K d d\theta \right| \leq \varphi(u) - \varphi(v) + \Lambda R_\Omega \|\Pi_K d\|. \quad \square$$

Proof of Theorem 3.7. Reduction. Compute $\varrho = \max\{1, \max_j \max_{x \in P} |x_j|\}$ by $2n$ linear programs and apply Lemma C.4. If $m = 0$, then $P = S = \{x_0\}$; take $\Gamma_P = 1$ and output x_0 . Otherwise let $\varphi(u) = f(x_0 + Vu)$, a cubic in u , and $P' = \{Gu \leq h\}$ with inner radius r_0 and outer bound R_0 about 0. Convexity of f on P makes φ convex on the full-dimensional set P' , so $H(u) = \nabla^2 \varphi(u) = V^\top \nabla^2 f(x_0 + Vu) V \succeq 0$ on the interior of P' and, by continuity, on P' . This step needs the reduction: convexity on a lower-dimensional polytope does not make the ambient Hessian positive semidefinite. We never expand φ ; only $H_c = H(0) = V^\top \nabla^2 f(x_0) V$, $g_c = \nabla \varphi(0) = V^\top \nabla f(x_0)$, and evaluations of f are used.

Center Hessian. Lemma C.10 with $c = 0$ and $\theta = r_0/R_0$ gives $0 \preceq H(u) \preceq \omega H_c$ on P' , where $\omega = 1 + R_0/r_0$, and $K = \ker H_c$ lies in every $\ker H(u)$, $u \in P'$. Put $\Lambda = \max\{1, \omega \max_i \sum_j |(H_c)_{ij}|\}$; then $\|H(u)\| \leq \Lambda$ on P' . Let Δ be a positive integer that clears the denominators of H_c , g_c , and G , let $C_* \geq 1$ bound the entries of $T_Z = \Delta \binom{H_c}{g_c^\top}$ and ΔG , and let S' be the optimal set of φ on P' , so that $S = x_0 + VS'$ and $f(x_0 + Vu) - f_* = \varphi(u) - \varphi_*$.

Affine branch. If $H_c = 0$, then H vanishes on the open set $\text{int } P'$, hence everywhere, and φ is affine with gradient g_c . If $g_c = 0$, then $S' = P'$ and $\Gamma_P = 1$ works. Otherwise $S' = \{u \in P' : g_c^\top u = g_c^\top v\}$ for any $v \in S'$, and $\Delta g_c^\top(u - v) = \Delta E$ with $E = \varphi(u) - \varphi_*$. Lemma 3.2 gives $\text{dist}(u, S') \leq (mC_*)^{m-1} \Delta E$, which is at most $(mC_*)^{m-1} \Delta E^{1/4}$ for $E \leq 1$.

Curved branch. Let $H_c \neq 0$ have rank r . The product of its nonzero eigenvalues is the sum of its principal minors of order r , a positive rational with denominator dividing Δ^r ; so it is at least Δ^{-r} . Each eigenvalue is at most $\|H_c\| \leq \Lambda$. Hence the least positive eigenvalue λ satisfies $\lambda \geq \lambda_0 = \Delta^{-m} \Lambda^{-(m-1)}$. Let $v \in S'$, $u \in P'$, $d = u - v$, and $E = \varphi(u) - \varphi(v) \leq 1$. With $\rho_1 = R_0 \geq \|v\|$ and $\rho_2 = 2R_0 \geq \|d\|$, Lemma 3.8 gives

$$\lambda_0^2 \|\Pi_K d\|^4 \leq (d^\top H_c d)^2 \leq 108 \Lambda R_0^2 E,$$

so $\|\Pi_K d\| \leq \rho_T E^{1/4}$ with $\rho_T = \max\{1, 108 \Lambda R_0^2 / \lambda_0^2\}$. By Lemma 3.9 with $\Omega = P'$, $c = 0$, and $R_\Omega = R_0$, the optimal set is $S' = \{u \in P' : T_Z u = T_Z v\}$, and $|g_c^\top d| \leq E + \Lambda R_0 \|\Pi_K d\|$. Since $H_c d = H_c \Pi_K d$,

$$\|T_Z d\| \leq \Delta(\Lambda \|\Pi_K d\| + E + \Lambda R_0 \|\Pi_K d\|) \leq \Delta(1 + \Lambda(1 + R_0)\rho_T) E^{1/4}.$$

Lemma 3.2, applied to P' with the integer matrices ΔG and T_Z , gives $\text{dist}(u, S') \leq \Gamma_u E^{1/4}$ with

$$\Gamma_u = (mC_*)^{m-1} \Delta(1 + \Lambda(1 + R_0)\rho_T).$$

Original coordinates. For $x = x_0 + Vu \in P$, $\text{dist}(x, S) \leq \|V\| \text{dist}(u, S')$. Hence $\Gamma_P = \max\{1, \|V\|_\Delta\} \Gamma_u$ (or the affine-branch constant multiplied by the same factor) satisfies the error bound. All quantities are rational, computed in polynomial time, and have polynomial encoding length.

Output. For a set approximation, apply Lemma 3.1 to f on P with $\eta = (2^{-q}/\Gamma_P)^4 \leq 1$; the returned $\hat{x}$ has $\text{dist}(\hat{x}, S) \leq \Gamma_P \eta^{1/4} = 2^{-q}$. For the fixed selector, let $R_x = \max\{1, n\varrho\}$, which

bounds the norm of every point of P , and apply Lemma 3.3 with $d = 4$, $\Gamma = \Gamma_P$, $R = R_x$, and the factor 2:

$$\tau = \frac{\varepsilon^6}{1024R_x^4\Gamma_P^4}, \quad \eta = \frac{\tau\varepsilon^2}{4}, \quad \varepsilon = 2^{-q}.$$

Lemma 3.1, applied to $f + \tau \|x\|^2$ on P in the original coordinates, returns $\hat{x} \in P$ with regularized gap at most η , hence $\|\hat{x} - p\| \leq \varepsilon$. The penalty is the original norm $\|x\|^2 = \|x_0 + Vu\|^2$, not $\|u\|^2$; the latter would select a different optimizer in general. The encoding lengths of τ and η are $O(q)$ plus a polynomial in L . $\square$

C.4 Boundary constructions

Treewidth. Every objective below for which a treewidth bound is claimed is a sum of terms in at most three variables each. Every monomial of the expanded sum occurs in some term, so every edge of the interaction graph joins two variables of one term. Hence a tree of bags in which the variables of each term lie together in some bag, and each variable occurs in a connected set of bags, is a tree decomposition of the interaction graph; bags of at most three variables give treewidth at most two.

Two generic devices. The reductions combine a strongly convex base problem with a one-sided sign test and a quartic amplifier.

Lemma C.11 (Sign test). *Let $Y \subseteq \mathbb{R}^k$ be a box and Φ a polynomial with $\nabla^2\Phi \succeq \kappa I$ on Y for some $\kappa \geq 1/8$, and let y^0 be its minimizer over Y . Let $\zeta(y) = \sigma^\top y + \sigma_0$ be affine with $\|\sigma\| \leq 1$ and $|\zeta| \leq 2$ on Y . Put $\epsilon_0 = 1/32$ and*

$$H(y, \theta) = \Phi(y) + \theta^2 + \epsilon_0\theta\zeta(y) \quad \text{on } Y \times [0, 1].$$

Then $\nabla^2 H \succeq \frac{1}{16}I$ on $Y \times [0, 1]$, so H has a unique minimizer (y^, θ^*) . If $\zeta(y^0) \geq 0$, then $(y^*, \theta^*) = (y^0, 0)$; if $\zeta(y^0) < 0$, then $0 < \theta^* < 1$.*

Proof. The Hessian of $\Phi(y) + \theta^2$ is at least $\min\{\kappa, 2\}I \succeq \frac{1}{8}I$, and the Hessian of $\epsilon_0\theta\zeta(y)$ has spectral norm $\epsilon_0 \|\sigma\| \leq \frac{1}{32}$. On the face $\theta = 0$, $H = \Phi$, with the unique minimizer y^0 . At $(y^0, 0)$ the partial derivatives in y equal $\nabla\Phi(y^0)$, which satisfies the first-order condition for Φ on Y , and $\partial_\theta H = \epsilon_0\zeta(y^0)$. If $\zeta(y^0) \geq 0$, the point satisfies the first-order condition on $Y \times [0, 1]$ and is the minimizer by convexity. If $\zeta(y^0) < 0$, the direction $+e_\theta$ is a feasible descent direction at $(y^0, 0)$, and no other point of the face $\theta = 0$ can be optimal, since its restriction has only the minimizer y^0 ; so $\theta^* > 0$. At $\theta = 1$, $\partial_\theta H \geq 2 - 2\epsilon_0 > 0$, which excludes $\theta^* = 1$. $\square$

Lemma C.12 (Quartic amplifiers). *Let $\eta_0 = 1/192$.*

- (a) *Let H be a polynomial on a box Z with $\nabla^2 H \succeq \frac{1}{16}I$ on Z , unique minimizer z^* , and a distinguished coordinate $\theta \in [0, 1]$. Then $F'(z, y) = H(z) + \eta_0\theta^2(y-1)^2$ is convex on $Z \times [0, 1]$, and its optimal set is $\{z^*\} \times [0, 1]$ if $\theta^* = 0$ and $\{(z^*, 1)\}$ if $\theta^* > 0$.*
- (b) *Let $H_\pm$ be polynomials on boxes $Z_\pm$ with Hessians at least $\frac{1}{16}I$, unique minimizers $z_\pm^*$, and distinguished coordinates $\theta_\pm \in [0, 1]$, such that exactly one of θ_+^*, θ_-^* is positive. Then*

$$F(z_+, z_-, y) = H_+(z_+) + H_-(z_-) + \eta_0[\theta_+^2(y-1)^2 + \theta_-^2 y^2]$$

is convex on $Z_+ \times Z_- \times [0, 1]$ and has a unique minimizer, whose y -coordinate is 1 if $\theta_+^ > 0$ and 0 if $\theta_-^* > 0$.*

Proof. For real θ, y, γ and a direction (π, ς) in (θ, y) ,

$$\frac{d^2}{dt^2} \left[(\theta + t\pi)^2 (y + t\varsigma - \gamma)^2 \right]_{t=0} = 2(\theta\varsigma + 2(y - \gamma)\pi)^2 - 6(y - \gamma)^2\pi^2, \quad (43)$$

as direct expansion shows. For $y \in [0, 1]$ and $\gamma \in \{0, 1\}$ the negative term is at least $-6\pi^2$. Multiplied by η_0 , it is at least $-\frac{1}{32}\pi^2$, which the base Hessian absorbs; in (b) the two losses affect the distinct coordinates θ_+ and θ_- . Hence the Hessian quadratic form is at least $\frac{1}{32}$ times the squared norm of the base part of the direction, with no division by θ . This proves convexity, including where the amplitudes vanish.

(a) Since $F' \geq H \geq H(z^*)$, a point is optimal exactly when $z = z^*$ and $\theta^{*2}(y - 1)^2 = 0$, and $y = 1$ always attains the bound.

(b) F is at least the sum of the two base minima. If $\theta_+^* > 0$ and $\theta_-^* = 0$, the point $(z_+^*, z_-^*, 1)$ attains this sum; if $\theta_-^* > 0$ and $\theta_+^* = 0$, the point $(z_+^*, z_-^*, 0)$ does. Every optimal point must attain both base minima, hence equal $z_\pm^*$ in those coordinates, and must make both added terms vanish, which fixes y . $\square$

At the minimizer in (b) the curvature in the y direction is $2\eta_0((\theta_+^*)^2 + (\theta_-^*)^2)$. It is positive but can be extremely small; the amplifier transfers a tiny amplitude into an order-one coordinate without a usable growth constant.

Square Root Sum. An instance consists of positive integers $a_1, \dots, a_n$ and an integer B ; the question is whether $\sum_i \sqrt{a_i} \leq B$. If $B < 0$ the answer is no.

Lemma C.13 (Averaging tree). *For a Square Root Sum instance with $B \geq 0$, compute powers of two R_i with $R_i^2 \leq a_i < 4R_i^2$, and put $c_i = a_i/R_i^2 \in [1, 4)$, $M_0 = \max_i R_i$, $w_i = R_i/M_0 \in (0, 1]$, and let K be the least power of two with $K \geq \max\{2, n\}$. Take a complete binary tree with K leaf slots. Each real leaf $i \leq n$ has a variable $u_i \in [1, 2]$ and signal $w_i u_i$; padded leaves have signal 0. Each internal node ν has a variable $v_\nu \in [0, 2]$, its own signal v_ν , and residual $\varsigma_\nu = v_\nu - \frac{1}{2}(\text{left signal} + \text{right signal})$. Let s be the root variable and*

$$F_0(u, v) = \sum_{i=1}^n (u_i^3/3 - c_i u_i) + \sum_{\nu} \varsigma_\nu^2$$

on the box Y of these variables. Then $\nabla^2 F_0 \succeq \frac{1}{8}I$ on Y , and the unique minimizer has root value $s^0 = \sum_i \sqrt{a_i}/(KM_0) \in [0, 2]$. The construction takes polynomial time; all coefficients and box widths are bounded by absolute constants; and the bags $\{v_\nu\} \cup \{\text{variable children of } \nu\}$, joined along the tree, form a tree decomposition with bags of size at most three.

Proof. The leaf term has derivative $u_i^2 - c_i$ and is minimized on $[1, 2]$ at $u_i = \sqrt{c_i} \in [1, 2)$. Setting internal variables to the averages of their children's signals makes every residual zero and keeps them in $[0, 2]$. This point minimizes every summand, and its root is the average of the K leaf signals, $\frac{1}{K} \sum_i w_i \sqrt{c_i} = \sum_i \sqrt{a_i}/(KM_0)$.

For the Hessian, order the variables and write the residuals as the rows of $(I - P)$ applied to a direction d : the row of a leaf variable is the identity row, and the row of an internal variable subtracts half of each variable child's coefficient (w_i for a leaf, 1 for an internal node). Every variable is a child of at most one parent, so distinct rows of P have disjoint supports and $PP^\top$ is diagonal with entries at most $2 \cdot \frac{1}{4}$. Thus $\|P\| \leq 1/\sqrt{2} < \frac{3}{4}$. Since $2u_i \geq 2$,

$$d^\top \nabla^2 F_0 d \geq 2 \sum_i d_{u_i}^2 + 2 \sum_{\nu} (\text{residual row of } d)^2 = 2 \|(I - P)d\|^2 \geq 2 \cdot \frac{1}{16} \|d\|^2.$$

For the decomposition, a leaf variable occurs only in its parent's bag, which also covers its unary term; adjacent bags share the child internal variable; every residual lies in its own bag. The normalization uses integer square roots of bounded size and binary arithmetic on numbers of polynomial length, and the tree has fewer than $4n + 4$ nodes. Coefficients are bounded because $c_i < 4$, $w_i \leq 1$, and every residual has coefficients in $[-1, 1]$. $\square$

Lemma C.14 (Integral radical sums). *If positive integers $a_1, \dots, a_n$ satisfy $\sum_i \sqrt{a_i} \in \mathbb{Z}$, then every a_i is a perfect square.*

Proof. Let $\mathcal{F} = \mathbb{Q}(\sqrt{a_1}, \dots, \sqrt{a_n})$, a number field, and let $\text{Tr} = \text{Tr}_{\mathcal{F}/\mathbb{Q}}$. If a_i is not a square, $\mathbb{Q}(\sqrt{a_i})$ has degree two and $\text{Tr}_{\mathbb{Q}(\sqrt{a_i})/\mathbb{Q}}(\sqrt{a_i}) = 0$, so transitivity of the trace gives $\text{Tr}(\sqrt{a_i}) = [\mathcal{F} : \mathbb{Q}(\sqrt{a_i})] \cdot 0 = 0$. A rational number z has $\text{Tr}(z) = [\mathcal{F} : \mathbb{Q}]z$. Applying the trace to $\sum_i \sqrt{a_i} = B' \in \mathbb{Z}$ gives $\sum_{a_i \text{ square}} \sqrt{a_i} = B'$, and subtracting, the sum of the positive numbers $\sqrt{a_i}$ over the nonsquares is zero. So there are no nonsquares. $\square$

The field $\mathcal{F}$ appears only in the proof. The reduction tests squareness by integer square roots.

Proof of Proposition 3.10. If $B < 0$, answer no. If $B \geq 2KM_0$, answer yes, since $s^0 \leq 2$ gives $\sum_i \sqrt{a_i} = KM_0 s^0 \leq B$. Otherwise $b = B/(KM_0) \in [0, 2)$. Apply Lemma C.11 to $\Phi = F_0$ of Lemma C.13 with $\zeta = b - s$, and let $F = H$. Then $\nabla^2 F \succeq \frac{1}{16}I$, and the distinguished coordinate θ satisfies $\theta^* = 0$ exactly when $b \geq s^0$, that is, when $\sum_i \sqrt{a_i} \leq B$, including equality; otherwise $0 < \theta^* < 1$. The polynomial F is a cubic without constant term, with coefficients bounded by an absolute constant. Its diagonal Hessian entries are at most $4 + \frac{1}{2}$ for leaf variables, $2 + \frac{1}{2}$ for internal variables, and 2 for θ . Adding the bag $\{s, \theta\}$ at the root keeps bags of size at most three, so the interaction graph has treewidth at most two. Point approximation follows from strong convexity: Lemma 3.1 with $\eta = 2^{-2q}/32$ returns a feasible point of gap at most η , and $F(z) - F_* \geq \frac{1}{32} \|z - z^*\|^2$ gives distance at most 2^{-q} . $\square$

Proof of Proposition 3.11. Apply Lemma C.12(a) to the cubic $H = F$ of Proposition 3.10, with distinguished coordinate θ , and a new coordinate $y \in [0, 1]$. If $\sum_i \sqrt{a_i} \leq B$, then $\theta^* = 0$ and $S = \{z^*\} \times [0, 1]$, whose minimum-norm point has $y = 0$; otherwise $S = \{(z^*, 1)\}$. The point $(z^*, 1)$ is optimal in both cases, and a rational point within 2^{-q} of it is obtained by approximating z^* as in the proof of Proposition 3.10 and appending $y = 1$. The new term has scope $\{\theta, y\}$; attaching the bag $\{\theta, y\}$ to the bag $\{s, \theta\}$ keeps the treewidth at most two. Its coefficients are bounded. $\square$

Proof of Theorem 3.12(a). Handle $B < 0$ and $B \geq 2KM_0$ as above, and decide the instance by integer arithmetic if every a_i is a perfect square. Otherwise Lemma C.14 shows $\sum_i \sqrt{a_i} \neq B$, so $s^0 \neq b$. Build two independent copies of F_0 with root variables $s_{\pm}$, and apply Lemma C.11 with $\zeta_+ = b - s_+$ and $\zeta_- = s_- - b$. The resulting $H_{\pm}$ have Hessians at least $\frac{1}{16}I$, and

$$\theta_+^* > 0 \iff s^0 > b, \quad \theta_-^* > 0 \iff s^0 < b.$$

Exactly one amplitude is positive. Lemma C.12(b) gives a quartic F , convex on its box, with a unique minimizer whose y -coordinate is 1 if $\sum_i \sqrt{a_i} > B$ and 0 if $\sum_i \sqrt{a_i} < B$.

For the structural claims, take the two tree decompositions with their root bags $\{s_{\pm}, \theta_{\pm}\}$, add the bags $\{\theta_+, y\}$ and $\{\theta_-, y\}$ attached to $\{s_+, \theta_+\}$ and $\{s_-, \theta_-\}$, and join the two new bags. This is a tree decomposition with bags of size at most three; y occurs in two adjacent bags. Before rescaling, the diagonal Hessian entries are at most 5: the amplifier adds at most $2\eta_0$ to the $\theta_{\pm}$ entries and contributes $2\eta_0(\theta_+^2 + \theta_-^2)$ to the y entry. The coordinate intervals have width at most two; the affine maps onto $[0, 1]$ preserve degrees and factor scopes and multiply diagonal Hessian entries by at

most four, giving the bound 20. Each factor (leaf term, residual square, θ^2 , sign test, amplifier) has degree at most four, at most three variables, and bounded coefficients after the substitution, and each variable lies in at most three factors. Hence every nonconstant monomial of the expanded polynomial has a coefficient bounded by an absolute constant. The constant term is a sum over all factors and need not be bounded; deleting it does not change the minimizer. $\square$

PosSLP. A straight-line program uses the constants 0 and 1 and the operations $+$, $-$, $\times$ on earlier results; its output is an integer A , and PosSLP asks whether $A > 0$ [6].

Lemma C.15 (A strongly convex gate objective). *From a straight-line program with output A one computes in polynomial time a rational quartic $\mathcal{G}$ on the box $[-\frac{1}{4}, \frac{1}{4}]^N$ with $\nabla^2 \mathcal{G} \succeq I$ there, whose unique minimizer x^* has last coordinate t^* with $t^* > 0$ if $A > 0$ and $t^* < 0$ if $A \leq 0$. The number N is linear in the program length, and the exact values of the gates are never written.*

Proof. Represent each gate value α by a pair (u, v) with $v > 0$ and $u/v = \alpha$: the constants 0 and 1 by $(0, \frac{1}{4})$ and $(\frac{1}{4}, \frac{1}{4})$, a sum or difference of (u_1, v_1) and (u_2, v_2) by $((u_1 v_2 \pm u_2 v_1)/4, v_1 v_2/4)$, and a product by $(u_1 u_2/4, v_1 v_2/4)$. These preserve the ratios and the positivity of v , and all entries stay in $[-\frac{1}{4}, \frac{1}{4}]$. For the output pair (u, v) add $t = (2u - v)/4 = \frac{v}{4}(2A - 1)$, which is nonzero and positive exactly when $A > 0$.

List the pair coordinates and t in topological order as $x_1, \dots, x_N$, with $x_i = p_i(x_1, \dots, x_{i-1})$, where each p_i is a constant, an affine function, or a quadratic with at most two monomials. On the whole box, repeated operands included,

$$|p_i| \leq \frac{1}{4}, \quad \|\nabla p_i\|_1 \leq 1, \quad \|\nabla^2 p_i\| \leq 1.$$

Let $r_i = x_i - p_i$, $a_i = 64^{N-i}$, and $\mathcal{G} = \sum_i a_i r_i^2$. The triangular system $r = 0$ has the unique solution x^* given by the exact gate values, so $\mathcal{G} \geq 0 = \mathcal{G}(x^*)$.

Let $D_a = \text{diag}(8^{N-i})$, let L_p be the strictly lower triangular matrix with entries $\partial_j p_i$, and $\mathcal{E} = D_a L_p D_a^{-1}$, with entries $8^{j-i} \partial_j p_i$ for $j < i$. Row sums of $|\mathcal{E}|$ are at most $\frac{1}{8}$ and column sums at most $\sum_{k \geq 1} 8^{-k} = \frac{1}{7}$, so $\|\mathcal{E}\| \leq (1/56)^{1/2} < \frac{1}{4}$. For a direction h ,

$$2 \|D_a(I - L_p)h\|^2 = 2 \|(I - \mathcal{E})D_a h\|^2 \geq \frac{9}{8} \|D_a h\|^2.$$

The remaining Hessian term is $-2 \sum_i a_i r_i h^\top \nabla^2 p_i h$. Since $|r_i| \leq \frac{1}{2}$ and p_i depends only on earlier coordinates, it is at least

$$-\sum_i a_i \|h_{<i}\|^2 = -\sum_j h_j^2 \sum_{i>j} a_i \geq -\frac{1}{63} \|D_a h\|^2.$$

Thus $\nabla^2 \mathcal{G} \succeq (\frac{9}{8} - \frac{1}{63}) D_a^2 \succeq I$. The polynomial has $O(N)$ monomials and weights of $O(N)$ bits. $\square$

Proof of Theorem 3.12(b). Take two copies of $\mathcal{G}$ with outputs $t_\pm$ and apply Lemma C.11 with $\kappa = 1$, $\zeta_+ = -t_+$, and $\zeta_- = t_-$; here $|\zeta_\pm| \leq \frac{1}{4}$. Since $t^* \neq 0$, exactly one of $\theta_+^* > 0$ (when $A > 0$) and $\theta_-^* > 0$ (when $A \leq 0$) holds. Lemma C.12(b) gives a quartic, convex on its box, with a unique minimizer whose y -coordinate is 1 if $A > 0$ and 0 if $A \leq 0$.

Multiply the whole objective by 64^{-N} . This preserves the minimizer and convexity. The gate weights become 64^{-i} , and every squared residual has coefficients bounded by an absolute constant even when a monomial occurs in many residuals, because $\sum_i 64^{-i} < \frac{1}{63}$; the remaining terms are bounded as well. The affine map from $[-\frac{1}{4}, \frac{1}{4}]$ to $[0, 1]$ keeps these bounds, including that of the constant term. The encoding length is polynomial; the weights have $O(N)$ bits, so bounded magnitude does not mean bounded bit length. No treewidth bound is claimed. $\square$

Proof of Theorem 3.12(c). Given a source instance, run the reduction. If it answers directly, stop. Otherwise query the point algorithm for a point $\hat{z}$ with $\|\hat{z} - z^*\| \leq \frac{1}{4}$ and answer according to whether the designated coordinate of $\hat{z}$ exceeds $\frac{1}{2}$; its true value is 0 or 1. Feasibility of $\hat{z}$ is not used. The running time is polynomial if the point algorithm is. If the point algorithm is always correct and runs in expected polynomial time, so does the decision procedure. $\square$

Regularization.

Proof of Proposition 3.13. Write $x_0 = \frac{1}{2}$ (a constant) and $G_n(x) = 8 \sum_{i=1}^n x_i^2 - \sum_{i=1}^n x_{i-1}^2 x_i$, so that the $i = 1$ term is $-x_1/4$ and $F_n = G_n + x_n^2(y-1)^2$.

(a) On $[0, 1]^n$ the Hessian of G_n is tridiagonal with diagonal entries $16 - 2x_{i+1} \geq 14$ ($i < n$) and 16 ($i = n$), and off-diagonal entries $-2x_i$ in positions $(i, i+1)$. Each absolute off-diagonal row sum is at most 4, so $\nabla^2 G_n \succeq 10I$. For a direction (π, ς) in (x, y) , (43) with $\gamma = 1$ gives a Hessian form of at least $(10 - 6) \|\pi\|^2 \geq 0$, so F_n is convex on the box.

Let x^* minimize G_n on $[0, 1]^n$. At a lower bound $x_i = 0$ the partial derivative is $-x_{i-1}^2$, negative once $x_{i-1} > 0$; since $x_0 = \frac{1}{2}$, induction excludes all lower bounds. At an upper bound the derivative is at least $16 - 1 - 2 > 0$. Hence x^* is interior and

$$x_i^* = \frac{(x_{i-1}^*)^2}{16 - 2x_{i+1}^*} \quad (i < n), \quad x_n^* = \frac{(x_{n-1}^*)^2}{16}.$$

So $0 < x_i^* \leq (x_{i-1}^*)^2/14$, and with $\omega_i = x_i^*/14$ we get $\omega_i \leq \omega_{i-1}^2$, $\omega_0 = \frac{1}{28}$, hence $x_n^* \leq 14 \cdot 28^{-2^n} \leq 2^{-2^{n+1}}$ for $n \geq 1$. Since $F_n \geq G_n \geq G_n(x^*)$ with equality at $(x^*, 1)$ and $x_n^* > 0$, the point $(x^*, 1)$ is the unique minimizer.

(b) The point $(x^*, 0)$ has gap $(x_n^*)^2 \leq 2^{-2^{n+2}}$ and distance one. An error bound $\text{dist} \leq \Gamma \text{gap}^\vartheta$ at this point gives $1 \leq \Gamma(x_n^*)^{2\vartheta}$, so $\log_2 \Gamma \geq \vartheta 2^{n+2}$.

(c) For $\lambda > 0$, the same boundary arguments show that the regularized minimizer has interior x -coordinates. Minimizing over y for fixed x gives $y_\lambda = x_{n,\lambda}^2/(x_{n,\lambda}^2 + \lambda)$, and stationarity in x gives

$$x_{i,\lambda} = \frac{x_{i-1,\lambda}^2}{16 + 2\lambda - 2x_{i+1,\lambda}} \quad (i < n), \quad x_{n,\lambda} = \frac{x_{n-1,\lambda}^2}{16 + 2\lambda + 2(1 - y_\lambda)^2}.$$

All denominators are at least 14, so the bound of (a) applies: $x_{n,\lambda} \leq 2^{-2^{n+1}}$. If $y_\lambda \geq \frac{1}{2}$ then $\lambda \leq x_{n,\lambda}^2 \leq 2^{-2^{n+2}}$. A positive rational $\lambda = p_\lambda/q_\lambda$ with this property has $q_\lambda \geq 2^{2^{n+2}}$.

(d) Optimality of z_λ and $\|z\|^2 \leq n+1$ on the box give $F_n(z_\lambda) - F_{n,*} \leq (n+1)\lambda$. For the point problem, $F_n(x, 1) = G_n(x)$, so $(x^*, 1)$ is approximated by applying Lemma 3.1 to the 10-strongly convex G_n and appending $y = 1$. $\square$

Algebraic output on a conditioned path.

Proof of Proposition 3.14. (a) The chain $a_n = 1$, $a_i = \sqrt{a_{i+1} + 2}$ stays in $[1, 2)$ and makes every square in Ψ_n vanish; conversely $\Psi_n = 0$ forces $x_n = 1$ and then the chain. For $x \in [1, 2]^n$ and $e = x - a$ the residuals are

$$r_i = x_i^2 - x_{i+1} - 2 = (x_i + a_i)e_i - e_{i+1} \quad (i < n), \quad r_n = e_n.$$

Thus $r = D_r(I - P_r)e$ with $D_r = \text{diag}(x_1 + a_1, \dots, x_{n-1} + a_{n-1}, 1)$ and $P_r = D_r^{-1}S_n$, where S_n is the upper shift. Since $x_i + a_i \geq 2$, P_r is a nilpotent weighted shift with $\|P_r\| \leq \frac{1}{2}$, so $\|(I - P_r)^{-1}\| \leq 2$;

also $\|D_r^{-1}\| \leq 1$. Hence $\|e\| \leq 2\|r\|$ and $\Psi_n(x) = \|r\|^2 \geq \frac{1}{4}\|x - a\|^2$. The outgoing factor contributes $12x_i^2 - 4x_{i+1} - 8 \leq 36$ to $\partial_{ii}\Psi_n$ and the incoming factor contributes 2, so $\partial_{ii}\Psi_n \leq 38$.

(b) Let $\phi(t) = t^2 - 2$ and $P_k = \phi^{\circ k} - 1$, $k = n - 1$. Since $\phi(a_i) = a_{i+1}$, $P_k(a_1) = 0$. Modulo 2, $\phi(t) \equiv t^2$, so $P_k(t + 1) \equiv (t + 1)^{2^k} - 1 \equiv t^{2^k}$. Moreover $\phi(\pm 1) = -1$ gives $P_k(1) = -2$. Hence $P_k(t + 1)$ is monic, all its other coefficients are even, and its constant term is -2 ; Eisenstein's criterion at 2 shows that it is irreducible, and so is P_k . Therefore P_k is the minimal polynomial of a_1 and $[\mathbb{Q}(a_1) : \mathbb{Q}] = 2^{n-1}$.

For the support, we show by induction that $\phi^{\circ k}$ ($k \geq 1$) has a nonzero coefficient at every even power t^{2j} , $0 \leq j \leq 2^{k-1}$, with sign $(-1)^{2^{k-1}-j}$. This holds for $t^2 - 2$. If $\chi = \phi^{\circ k}$ has this property, every product contributing to the coefficient of t^{2J} in χ^2 has sign $(-1)^{2^k-J}$, so these coefficients are nonzero with that sign. The constant coefficient of χ is $\phi^{\circ k}(0) \in \{-2, 2\}$, so the constant coefficient of $\chi^2 - 2$ is 2, which has the required sign. Finally the constant coefficient of P_k is -3 for $k = 1$ and 1 for $k \geq 2$. Hence P_k has exactly $2^{k-1} + 1 = 2^{n-2} + 1$ nonzero coefficients.

(c) Let $\alpha_n = 1$ and, for $i = n - 1, \dots, 1$, let $\alpha_i = \lfloor 2^8 \sqrt{\alpha_{i+1} + 2} \rfloor / 2^8$, computed by an integer square root. With $e_i = |\alpha_i - a_i|$, we have $e_n = 0$ and

$$e_i \leq 2^{-8} + \frac{e_{i+1}}{\sqrt{\alpha_{i+1} + 2} + \sqrt{a_{i+1} + 2}} \leq 2^{-8} + \frac{1}{2}e_{i+1},$$

so $e_i \leq 2^{-7}$ by induction. Put $\ell_i = \max\{1, \alpha_i - 2^{-6}\}$ and $u_i = \min\{2, \alpha_i + 2^{-6}\}$. Then $a \in B = \prod_i [\ell_i, u_i] \subseteq [1, 2]^n$, the endpoints have $O(1)$ bits, and $|x_i - a_i| \leq 3 \cdot 2^{-7}$ for $x \in B$. Consequently $|r_i| \leq 4 \cdot 3 \cdot 2^{-7} + 3 \cdot 2^{-7} = \frac{15}{128}$. The residual Jacobian is $J = \text{diag}(2x_1, \dots, 2x_{n-1}, 1) - S_n$; as in (a), $\|J^{-1}\| \leq 2$ because $x_i \geq 1$. The exact Hessian is

$$\nabla^2 \Psi_n = 2J^T J + 4 \text{diag}(r_1, \dots, r_{n-1}, 0) \succeq \left(\frac{1}{2} - \frac{60}{128}\right)I = \frac{1}{32}I$$

on B . □

Rectangular certificates.

Proof of Proposition 3.15. (a) The residuals are $\varrho_0 = x_0 - \frac{1}{4}$ and $\varrho_i = x_i - \frac{1}{4}x_{i-1}^2$. They vanish exactly at $a_0 = \frac{1}{4}$, $a_i = \frac{1}{4}a_{i-1}^2$, which gives $a_i = 2^{-(4 \cdot 2^i - 2)}$, a point strictly inside $X = [0, \frac{1}{2}]^{n+1}$; so a is the unique minimizer, with value 0. With $e = x - a$ the residuals are $\varrho = (I - P_\varrho)e$, where P_ϱ is a lower shift with weights $(x_{i-1} + a_{i-1})/4 \leq \frac{1}{4}$. Hence $\|e\| \leq \frac{4}{3}\|\varrho\|$ and $\Xi_n \geq \frac{9}{16}\|x - a\|^2$. Direct differentiation gives $\partial_{ii}\Xi_n = 2 + \frac{3}{4}x_i^2 - x_{i+1} \leq \frac{35}{16}$ for $i < n$ and $\partial_{nn}\Xi_n = 2$. With the residual Jacobian J , which has unit diagonal and subdiagonal entries $-x_{i-1}/2 \in [-\frac{1}{4}, 0]$, the exact Hessian is $2J^T J - \text{diag}(\varrho_1, \dots, \varrho_n, 0)$. Since $\|I - J\| \leq \frac{1}{4}$ and $\varrho_i \leq \frac{1}{2}$ on X , $\nabla^2 \Xi_n \succeq (2 \cdot \frac{9}{16} - \frac{1}{2})I = \frac{5}{8}I$.

If a rectangle with rational endpoints contains a and lies in $(0, \frac{1}{2})^{n+1}$, its lower endpoint in coordinate n satisfies $0 < \ell_n \leq a_n = 2^{-m}$ with $m = 4 \cdot 2^n - 2$. Writing $\ell_n = p_n/q_n$ with integers $p_n \geq 1$, $q_n \geq 1$ gives $q_n \geq 2^m$.

(b) Let $h(x) = x_n - \frac{1}{8}x_{n-1}^2 = \frac{1}{2}(x_n + \varrho_n)$ and $\tilde{\Xi}_n = \Xi_n + y^2 + \frac{1}{4}yh$ on $X \times [0, \frac{1}{2}]$. Since $x_n, y \geq 0$, $\frac{1}{4}yh \geq \frac{1}{8}y\varrho_n \geq -\frac{1}{16}(y^2 + \varrho_n^2)$, and $\varrho_n^2 \leq \Xi_n$, so

$$\tilde{\Xi}_n \geq \frac{15}{16}(\Xi_n + y^2) \geq \frac{135}{256}(\|x - a\|^2 + y^2).$$

Thus $(a, 0)$ is the unique minimizer, with value 0, and $\partial_y \tilde{\Xi}_n(a, 0) = \frac{1}{4}h(a) = \frac{1}{8}a_n > 0$ because $a_{n-1}^2 = 4a_n$. The added term changes the diagonal only by $-\frac{y}{16}$ in coordinate $n - 1$, and $\partial_{yy}\tilde{\Xi}_n = 2$. Its other Hessian entries are $-x_{n-1}/16$ in position $(n - 1, y)$ and $\frac{1}{4}$ in position (n, y) ; the absolute

row sums of this perturbation are at most $\frac{9}{32}$, so $\nabla^2 \tilde{\Xi}_n \succeq (\frac{5}{8} - \frac{9}{32})I = \frac{11}{32}I$. Treewidth two follows from the path bags $\{x_{i-1}, x_i\}$ and the extra bag $\{x_{n-1}, x_n, y\}$.

For a rectangle $\prod_i [\ell_i, u_i] \subseteq X$ containing a , h is increasing in x_n and decreasing in $x_{n-1} \geq 0$, so the exact minimum of $\partial_y \tilde{\Xi}_n(x, 0) = \frac{1}{4}h(x)$ over the rectangle is $\frac{1}{4}(\ell_n - u_{n-1}^2/8)$, attained at a corner. If it is nonnegative, then $\ell_n \geq u_{n-1}^2/8 \geq a_{n-1}^2/8 = a_n/2 > 0$, while containment gives $\ell_n \leq a_n = 2^{-m}$. The denominator bound follows as in (a).

(c) The recurrence $a_i = \frac{1}{4}a_{i-1}^2$ is an arithmetic circuit of size $O(n)$. The map $T(x) = (\frac{1}{4}, \frac{1}{4}x_0^2, \dots, \frac{1}{4}x_{n-1}^2)$ sends the closed box X into itself and has Lipschitz constant at most $\frac{1}{4}$ there, so the contraction principle on X certifies the unique fixed point a , and $\Xi_n = \|x - T(x)\|^2 \geq 0$ certifies optimality. In the second family, $\varrho_n(a) = 0$ gives $h(a) = a_n/2 > 0$ symbolically, and the displayed lower bound certifies global optimality. Neither certificate requires a rectangle in the open box or a sign on a whole rectangle. $\square$

D Proofs for Section 6

This appendix proves the results of Section 6. We keep the notation of Section 6.1. Rational numbers, vectors, and polynomials are encoded in binary, and $\text{bits}(\cdot)$ denotes encoding length. A *rational circuit* is a pair of integer circuits (N, Q) over $\{0, 1, +, -, \times\}$ with $Q > 0$, denoting N/Q ; sums, differences, and products add a constant number of gates, division by a nonzero N_2/Q_2 uses

$$\frac{N_1/Q_1}{N_2/Q_2} = \frac{N_1 Q_2 N_2}{Q_1 N_2^2},$$

and the sign of N/Q is the sign of N (Definition 2.16 and Lemma 4.2). Comparing two rational circuits costs at most two PosSLP queries. For their difference numerator N , equality holds exactly when both strict-sign tests $N > 0$ and $-N > 0$ return false.

External results are used only through the following contracts. Each contract is stated where it is used.

- (E1) Convex value optimization (Lemma 3.1): for an explicit rational convex polynomial of fixed degree, a nonempty explicit rational polyhedron contained in a supplied rational box, and a rational $\varepsilon \in (0, 1]$, an exactly feasible rational point with objective value at most the minimum plus ε is computed in ordinary time polynomial in the input length and $\log(1/\varepsilon)$.
- (E2) Separation (Lemma 4.1) and the exact unconstrained comparisons of Theorems 4.4 and 4.7, including the uniform Newton circuit of Lemma 4.3 and the compilation Theorem 4.10.
- (E3) Linear programming with an arithmetic count independent of objective and right-hand-side lengths [49, Theorem 6.6.3], exact convex quadratic programming [71], exact separable quadratic flows [117, Theorem 20], Clarkson-type sampling for violator spaces [44], integer feasibility with integer-point separation oracles [8, 11, 54], and fixed-dimensional integer and mixed-integer linear feasibility [31, 55].

D.1 Geometry, radii, and supports

Two elementary consequences of strong convexity are used repeatedly. For completeness we derive them in the form needed; they are instances of Lemma 2.10.

Lemma D.1 (Radius and distance). *Let φ be a twice differentiable function on $\mathbb{R}^N$ with $\nabla^2 \varphi \succeq \mu I$.*

(a) Let $x_0 \in \mathbb{R}^N$, put $A_0 = \|\nabla\varphi(0)\|_1 + |\varphi(x_0) - \varphi(0)| + 1$ and $R_0 = 2 + 2A_0/\mu$. Then $\varphi(x) > \varphi(x_0)$ whenever $\|x\| \geq R_0$. In particular, every minimizer of φ over a closed set containing x_0 has norm less than R_0 .

(b) If φ is minimized over a closed convex set K at p , then $\varphi(x) - \varphi(p) \geq \frac{\mu}{2}\|x - p\|^2$ for every $x \in K$.

If φ is an explicit rational polynomial of fixed degree and x_0 is rational, then R_0 is rational with bit length polynomial in $\text{bits}(\varphi) + \text{bits}(\mu) + \text{bits}(x_0)$.

Proof. (a) Write $\rho = \|x\|$ and $\Delta = \varphi(x_0) - \varphi(0)$. Strong convexity and $|\nabla\varphi(0)^\top x| \leq \|\nabla\varphi(0)\|_1 \rho$ give $\varphi(x) \geq \varphi(0) - \|\nabla\varphi(0)\|_1 \rho + \frac{\mu}{2}\rho^2$. If $\rho \geq R_0$, then $\frac{\mu}{2}\rho - \|\nabla\varphi(0)\|_1 \geq A_0 - \|\nabla\varphi(0)\|_1 = |\Delta| + 1$, so $\varphi(x) - \varphi(x_0) \geq \rho(|\Delta| + 1) - \Delta > 0$ because $\rho \geq 2$. (b) $\varphi(x) \geq \varphi(p) + \nabla\varphi(p)^\top(x - p) + \frac{\mu}{2}\|x - p\|^2$, and $\nabla\varphi(p)^\top(x - p) \geq 0$ by optimality of p on K . The size statement holds because $\varphi(0)$, $\varphi(x_0)$, and $\nabla\varphi(0)$ are fixed-degree evaluations. $\square$

Proof of Lemma 6.1. (a) Fix $x_0 \in P$ and $R_1 > \|T(x_0)\|/\mu$, and let $K = P \cap \{x : \|x - x_0\| \leq R_1\}$. The map $x \mapsto \Pi_K(x - T(x))$, where Π_K is Euclidean projection, is a continuous self-map of the nonempty compact convex set K , so Brouwer's theorem gives a fixed point $\bar{p}$. The projection inequality at $\bar{p}$ reads $T(\bar{p})^\top(x - \bar{p}) \geq 0$ for $x \in K$. If $\|\bar{p} - x_0\| = R_1$, then (9) gives

$$T(\bar{p})^\top(\bar{p} - x_0) \geq T(x_0)^\top(\bar{p} - x_0) + \mu R_1^2 \geq R_1(\mu R_1 - \|T(x_0)\|) > 0,$$

contradicting the inequality at $x = x_0$. Hence $\|\bar{p} - x_0\| < R_1$. For any $x \in P$ and small $\varepsilon > 0$, the point $\bar{p} + \varepsilon(x - \bar{p})$ lies in K , so $T(\bar{p})^\top(x - \bar{p}) \geq 0$; thus $\bar{p}$ solves VI(T, P). If p, p' are two solutions, adding $T(p)^\top(p' - p) \geq 0$ and $T(p')^\top(p - p') \geq 0$ gives $(T(p) - T(p'))^\top(p - p') \leq 0$, and (9) forces $p = p'$. For $T = \nabla f$, integrating (8) along segments gives (9), and convexity shows that the solutions of VI($\nabla f, P$) are exactly the minimizers of f on P .

(b) If $-T(x) = \sum_{i \in S(x)} \lambda_i a_i$ with $\lambda \geq 0$, then for $y \in P$ we get $T(x)^\top(y - x) = -\sum_i \lambda_i (a_i^\top y - b_i) \geq 0$. Conversely, let x solve the variational inequality and let d satisfy $a_i^\top d \leq 0$ for $i \in S(x)$. The rows outside $S(x)$ have positive slack at x , so $x + \varepsilon d \in P$ for small $\varepsilon > 0$, and $T(x)^\top d \geq 0$. By Farkas' lemma, $-T(x)$ is a nonnegative combination of $\{a_i : i \in S(x)\}$. Choose such a combination with the smallest number of positive coefficients, supported on J . If the rows a_i , $i \in J$, were dependent, a nonzero β with $\sum_{i \in J} \beta_i a_i = 0$ could be chosen with some $\beta_i > 0$; subtracting $\vartheta\beta$ with $\vartheta = \min\{\lambda_i/\beta_i : \beta_i > 0\}$ would remove a coefficient while keeping the others nonnegative. Hence the rows in J are independent and $|J| \leq \text{rank } A$; if $T(x) = 0$, then $J = \emptyset$.

(c) Let $|J| = s$. Gaussian elimination with a fixed pivoting rule selects s columns $\mathcal{B}$ for which $A_{J,\mathcal{B}}$ is invertible; let $\mathcal{N}$ be the other columns. Setting $x_{\mathcal{N}} = y$ and $x_{\mathcal{B}} = A_{J,\mathcal{B}}^{-1}(b_J - A_{J,\mathcal{N}}y)$ parametrizes $\{x : A_J x = b_J\}$ as $\bar{x} + Zy$, with $Z_{\mathcal{N}} = I$ and $Z_{\mathcal{B}} = -A_{J,\mathcal{B}}^{-1}A_{J,\mathcal{N}}$. Thus $Z^\top Z = I + Z_{\mathcal{B}}^\top Z_{\mathcal{B}} \succeq I$. After the rows of A are scaled to integers, every entry of $\bar{x}$ and Z is a ratio of minors of the scaled matrix $[A \ b]$, so Hadamard's inequality bounds all bit lengths by one polynomial in L , for every J . For $y, y' \in \mathbb{R}^{n-s}$,

$$(U_J(y) - U_J(y'))^\top(y - y') = (T(\bar{x} + Zy) - T(\bar{x} + Zy'))^\top Z(y - y') \geq \mu\|Z(y - y')\|^2 \geq \mu\|y - y'\|^2.$$

Substituting an affine map into a polynomial of fixed degree creates polynomially many monomials with polynomially bounded coefficients. For $T = \nabla f$, the chain rule gives $\nabla g_J(y) = Z^\top \nabla f(\bar{x} + Zy)$ and $\nabla^2 g_J = Z^\top \nabla^2 f Z \succeq \mu Z^\top Z \succeq \mu I$.

(d) Since $A_J p = b_J$ and Z has full column rank, $p = \bar{x} + Zy_p$ for a unique y_p . Because $A_J(\bar{x} + Zy) = b_J$ for all y , we have $A_J Z = 0$, so $Z^\top a_i = 0$ for $i \in J$. As $T(p)$ lies in the span of

these rows, $U_J(y_p) = Z^T T(p) = 0$. By (a) applied to U_J on all of $\mathbb{R}^{n-s}$, the zero of U_J is unique, so $y_p = y_J$. A support J from (b) satisfies $A_J p = b_J$ and $T(p) = -A_J^T \lambda$. A row basis J of $A_{S(p)}$ satisfies $A_J p = b_J$, and $T(p) \in -\text{cone}\{a_i : i \in S(p)\} \subseteq \text{span}\{a_i : i \in J\}$. $\square$

The next lemma is the deterministic test behind the support certificate of Section 6.2 and behind the violation primitive of Theorem 6.16. For a set G of rows write $P_G = \{x : a_i^T x \leq b_i \ (i \in G)\}$.

Lemma D.2 (Support test). *Let G be a set of rows with $P_G \neq \emptyset$, let p_G be the solution of $\text{VI}(T, P_G)$, and let $J \subseteq G$ be a set of independent rows with chart $(\bar{x}, Z)$, zero y_J of U_J , and $p_J = \bar{x} + Z y_J$. With λ_J as in (10), consider the tests*

$$b_i - a_i^T p_J \geq 0 \ (i \in G), \quad \lambda_J(y_J) \geq 0, \quad T(p_J) + A_J^T \lambda_J(y_J) = 0. \quad (44)$$

If they hold, then $p_J = p_G$. Conversely, every support $J \subseteq G$ of p_G given by Lemma 6.1(b) passes them. Each test is a sign condition on an explicit polynomial of degree at most three in y , with coefficient length polynomial in L , at the zero of U_J . Hence the tests are decided in polynomial time with a PosSLP oracle, by Theorem 4.7, or by rational arithmetic if $|J| = n$.

Proof. Suppose (44) holds. Then $p_J \in P_G$, $a_i^T p_J = b_i$ for $i \in J$, and $-T(p_J) = A_J^T \lambda$ with $\lambda = \lambda_J(y_J) \geq 0$. For $x \in P_G$, $T(p_J)^T(x - p_J) = -\lambda^T(A_J x - b_J) \geq 0$. Thus p_J solves $\text{VI}(T, P_G)$, and $p_J = p_G$ by uniqueness. Conversely, if J is a support of p_G with $T(p_G) = -A_J^T \lambda$ and $\lambda \geq 0$, then $p_J = p_G$ by Lemma 6.1(d), and $\lambda_J(y_J) = (A_J A_J^T)^{-1} A_J A_J^T \lambda = \lambda$, so all tests hold. The matrix $(A_J A_J^T)^{-1}$ is rational of polynomial size because the rows of J are independent; the tested expressions are affine or cubic in y after substitution. $\square$

Guessing J and running these tests with $G = \{1, \dots, m\}$, followed by a comparison of $h(p_J)$, proves the $\text{NP}^{\text{PosSLP}} \cap \text{coNP}^{\text{PosSLP}}$ bound mentioned in Remark 6.3.

D.2 Unambiguous certificates

A linear program with a circuit objective. We use the following linear-programming theorem, whose arithmetic count is independent of the objective and right-hand-side lengths.

Contract (E3a) [49, Theorem 6.6.3]. There is an algorithm that, given rational D_0 , e_0 , and c such that $\{v : D_0 v \leq e_0\}$ is nonempty and bounded, returns a vertex minimizing $c^T v$ over this polytope. Its operations on numbers are additions, subtractions, multiplications, divisions by nonzero numbers, and comparisons, and their number is bounded by a polynomial in $\text{bits}(D_0)$, independently of c and e_0 . The integer roundings in its preprocessing of the objective are carried out by comparisons of numbers whose range is bounded in advance.

Lemma D.3 (Linear programming with a circuit objective). *Let $Q_0 = \{v \in \mathbb{R}^n : D_0 v \leq e_0\}$ be a nonempty bounded polytope with explicit rational data, and let c be a vector of rational circuits. A vertex of Q_0 minimizing $c^T v$ can be computed, with its coordinates printed as rationals in binary, in deterministic time polynomial in $\text{bits}(D_0, e_0)$ and the circuit size, with a PosSLP oracle.*

Proof. Run the algorithm of contract (E3a) with every number held as a rational circuit. Binary rational constants are built from 0 and 1 by doubling and addition. Each arithmetic operation adds a constant number of shared gates, and each comparison is answered exactly by PosSLP. The simulation therefore follows the execution of the algorithm on the rational input c , all divisions are

by nonzero numbers, and the output is an optimal vertex v_* represented by circuits. The number of gates and queries is polynomial.

To print v_* , decide for each row i whether $(D_0)_i v_* = (e_0)_i$, by two strict-sign queries. The rows that are tight at a vertex span $\mathbb{R}^n$: otherwise a nonzero d orthogonal to all of them gives $v_* \pm \varepsilon d \in Q_0$ for small $\varepsilon > 0$, because the other rows have positive slack, and v_* would not be extreme. Choose n independent tight rows by rational elimination on the explicit matrix D_0 and solve the corresponding square system. Its unique solution is v_* , and Cramer's rule bounds its bit length polynomially in $\text{bits}(D_0, e_0)$. No circuit is expanded. $\square$

The normal-cone test. We use Lemma 4.3 in the following form. Let U be an explicit cubic map on $\mathbb{R}^d$, $d \geq 1$, satisfying (9) with modulus μ , with unique zero y_* , and let M be an integer, written in unary, that is at least the encoding length of (U, μ) . Part (b) of that lemma, with $L' = M$, constructs in time polynomial in M and without oracle calls a shared rational circuit $\hat{y}$ such that every explicit polynomial η of encoding length at most M satisfies

$$\eta(y_*) = 0 \text{ or } |\eta(y_*)| \geq g_M, \quad \text{and} \quad |\eta(\hat{y}) - \eta(y_*)| \leq g_M/8, \quad (45)$$

where $g_M = 2^{-2^{e(M)}}$ has a circuit of $e(M)$ squarings of $1/2$. For gradient maps, part (a) of the lemma gives the same conclusion with a warm start from convex value optimization.

Lemma D.4 (Normal-cone test). *Let U and y_* be as above, let $C(y) \in \mathbb{Q}[y]^n$ be an explicit vector of polynomials of degree at most three, and let $Q_0 = \{v : D_0 v \leq e_0\}$ be a nonempty bounded rational polytope. Whether $C(y_*)^\top v \geq 0$ for all $v \in Q_0$ is decidable in deterministic polynomial time with a PosSLP oracle.*

Proof. Every vertex of Q_0 is the unique solution of n independent rows of $D_0 v = e_0$. After scaling rows to integers, Cramer's rule and Hadamard's inequality bound the coordinates of every vertex by a polynomial σ in the input length Λ_0 of (U, μ, C, D_0, e_0) . Hence there is a polynomial $M = M(\Lambda_0) \geq \Lambda_0$, computed without knowing any vertex, that bounds the encoding length of every vertex observable $\eta_v(y) = v^\top C(y)$. Construct $\hat{y}$ as in (45), form the rational circuit vector $\hat{c} = C(\hat{y})$, and use Lemma D.3 to print a vertex w minimizing $\hat{c}^\top v$ over Q_0 . Finally decide $\eta_w(y_*) \geq 0$ exactly, by Theorem 4.7 or by testing the sign of $\eta_w(\hat{y}) + g_M/2$; answer accordingly.

If $C(y_*)^\top v \geq 0$ on Q_0 , then in particular $\eta_w(y_*) \geq 0$. Otherwise the linear function $v \mapsto C(y_*)^\top v$ is negative somewhere on the polytope, hence at some vertex v_1 . Then $\eta_{v_1}(y_*) \leq -g_M$ and $\hat{c}^\top v_1 \leq -7g_M/8$, so $\hat{c}^\top w \leq -7g_M/8$. If $\eta_w(y_*) \geq 0$, then $\hat{c}^\top w \geq -g_M/8$, which is impossible. Thus $\eta_w(y_*) < 0$ and the test answers correctly. Ties among vertex values cause no difficulty. $\square$

The linear program has a rational objective given by a circuit; it is never run with the algebraic vector $C(y_*)$. The only algebraic quantity whose sign is decided is the explicit cubic η_w at y_* , so the observable theorems are applied only to explicit polynomials.

Proof of Theorem 6.2. We describe two nondeterministic oracle machines, $\mathbf{M}$ for the promise problem and $\overline{\mathbf{M}}$ for its complement. Both first perform the following deterministic steps. If a certificate is supplied and is invalid, $\mathbf{M}$ rejects and $\overline{\mathbf{M}}$ accepts. If $P = \emptyset$, decided by linear programming, $\mathbf{M}$ rejects and $\overline{\mathbf{M}}$ accepts. If $n = 0$, everything is a rational computation. Otherwise both machines guess $S \subseteq \{1, \dots, m\}$, one bit per row, and then deterministically:

- (1) choose the row basis J of A_S obtained by scanning the rows of S in increasing order, and its chart $(\bar{x}, Z)$; let y_J be the zero of U_J and $p_S = \bar{x} + Zy_J$;

- (2) reject unless $b_i - a_i^\top p_S = 0$ for all $i \in S$ and $b_i - a_i^\top p_S > 0$ for all $i \notin S$;
- (3) reject unless (11) holds, decided by Lemma D.4 with $C(y) = T(\bar{x} + Zy)$ and $Q_0 = Q_S$ (if $|J| = n$, by ordinary linear programming with the rational objective $T(\bar{x})$);
- (4) M accepts if $h(p_S) \bowtie 0$, and $\bar{M}$ accepts otherwise.

Each slack in step (2) and the observable in step (4) is an explicit polynomial in y at the zero of U_J , which satisfies (9) by Lemma 6.1(c), so Theorem 4.7 decides it. All steps take polynomial time with a PosSLP oracle.

Soundness. Suppose S passes steps (2) and (3). Then $p_S \in P$ and $A_S p_S = b_S$. For $x \in P$, the vector $x - p_S$ satisfies $A_S(x - p_S) \leq 0$, and $v = (x - p_S) / \max\{1, \|x - p_S\|_\infty\}$ lies in Q_S ; hence $T(p_S)^\top(x - p_S) \geq 0$. Thus p_S solves $\text{VI}(T, P)$ and equals p . Step (2) now says $S = S(p)$.

Completeness. For $S = S(p)$, Lemma 6.1(d) gives $p_S = p$, so step (2) holds by the definition of $S(p)$. By Lemma 6.1(b), $-T(p) = \sum_{i \in S(p)} \lambda_i a_i$ with $\lambda \geq 0$, so $T(p)^\top v = -\sum_i \lambda_i a_i^\top v \geq 0$ for $v \in Q_S$, and step (3) holds.

Consequently exactly one guess reaches step (4), and it uses $p_S = p$. On a yes-instance M has exactly one accepting computation and $\bar{M}$ has none; on a no-instance the roles are exchanged. Inconsistent guessed equalities, redundant rows, zero rows, zero multipliers, and lower-dimensional polyhedra are all handled by the exact mask in step (2). $\square$

D.3 Few nonlinear directions

Proof of Lemma 6.5. (a) For $v \in \mathbb{R}^n$, $D_v(f_3 + f_4) = \sum_i v_i \partial_i(f_3 + f_4) = \sum_\beta (Wv)_\beta x^\beta$. Since $D_v f_3$ and $D_v f_4$ are homogeneous of degrees two and three, both vanish identically exactly when $Wv = 0$. Each monomial of $f_3 + f_4$ contributes at most four monomials to the partial derivatives, so W has at most $4L$ nonzero rows, and its rank, a row basis U of its row space, and $V = U^\top(UU^\top)^{-1}$ are computed by rational elimination in polynomial time.

(b) For $x \in \mathbb{R}^n$, the vector $e = x - VUx$ lies in $\ker U = K_f$. The derivative of $\vartheta \mapsto (f_3 + f_4)(VUx + \vartheta e)$ is $D_e(f_3 + f_4)(VUx + \vartheta e) = 0$, so $(f_3 + f_4)(x) = (f_3 + f_4)(VUx) = \theta(Ux)$.

(c) Let $x = Mx' + a$ with M invertible and put $f'(x') = f(Mx' + a)$. Expanding $f_4(y + a) = f_4(y) + D_a f_4(y) + (\text{terms of degree } \leq 2)$ and $f_3(y + a) = f_3(y) + (\text{terms of degree } \leq 2)$ shows that the homogeneous parts of f' of degrees four and three are $f_4(Mx')$ and $f_3(Mx') + (D_a f_4)(Mx')$. For $v' \in \mathbb{R}^n$ and $v = Mv'$, the derivatives in direction v' are $(D_v f_4)(Mx')$ and $(D_v f_3 + D_v D_a f_4)(Mx')$. If $D_v f_4 \equiv 0$, then $D_v D_a f_4 = D_a D_v f_4 \equiv 0$. Hence $v' \in K_{f'}$ if and only if $D_v f_4 \equiv 0$ and $D_v f_3 \equiv 0$, that is, $K_{f'} = M^{-1}K_f$. $\square$

The next lemma is the elimination on an arbitrary affine face. Its point is the uniform size bound: the same polynomial T works for every set of rows, so the unknown active face of the optimizer can be used in proofs without being found.

Lemma D.5 (Elimination of the quadratic directions). *Let f satisfy (8) with nonlinear dimension k and U, V, θ as in Lemma 6.5. Let J be a set of independent rows with chart $(\bar{x}, Z)$, $Z \in \mathbb{Q}^{n \times d}$, and let y_J be the unconstrained minimizer of $y \mapsto f(\bar{x} + Zy)$. Rational linear algebra computes $s = \text{rank}(UZ) \leq k$, $\bar{c} \in \mathbb{Q}^n$, and $D \in \mathbb{Q}^{n \times s}$ of full column rank such that, for $\varphi(u) = f(\bar{c} + Du)$:*

- (i) $\nabla^2 \varphi(u) \succeq \mu D^\top D \succ 0$ for all $u \in \mathbb{R}^s$; for $s \geq 1$, the rational number

$$\nu = \frac{\mu \det(D^\top D)}{\text{tr}(D^\top D)^{s-1}}$$

satisfies $\mu D^\top D \succeq \nu I$ (Lemma 2.7);

- (ii) $\bar{x} + Zy_J = \bar{c} + Du_*$, where u_* is the unique zero of $\nabla\varphi$;
- (iii) for every observable h , the coefficients of $\bar{c}$, D , φ , and $\eta(u) = h(\bar{c} + Du)$, after clearing denominators, have bit length at most $T(L)$, where T is a polynomial with an absolute exponent that depends on neither J nor k .

Proof. Let $Y = UZ \in \mathbb{Q}^{k \times d}$. From the reduced row echelon form of Y , let $W_0 \in \mathbb{Q}^{d \times (d-s)}$ be the standard basis of $\ker Y$ (one vector per nonpivot column) and let $T_0 \in \mathbb{Q}^{d \times s}$ consist of the unit vectors of the pivot columns. In the coordinates ordered as pivot and nonpivot columns, $[T_0 \ W_0]$ is block upper triangular with identity diagonal blocks, hence invertible, and YT_0 has rank s because $Y[T_0 \ W_0] = [YT_0 \ 0]$.

Write $y = T_0u + W_0w$ and $F(u, w) = f(\bar{x} + ZT_0u + ZW_0w)$. Since $UZW_0 = 0$, Lemma 6.5(b) gives

$$F(u, w) = f_{\leq 2}(\bar{x} + ZT_0u + ZW_0w) + \theta(U\bar{x} + YT_0u).$$

The second term does not depend on w , and the first has degree at most two. With $Q_2 = \nabla^2 f_{\leq 2}$, a constant rational matrix,

$$F(u, w) = \frac{1}{2}w^\top Hw + w^\top(Gu + a) + F(u, 0),$$

where

$$H = (ZW_0)^\top Q_2 ZW_0, \quad G = (ZW_0)^\top Q_2 ZT_0, \quad a = (ZW_0)^\top \nabla f_{\leq 2}(\bar{x}).$$

The Hessian of F in w equals $(ZW_0)^\top \nabla^2 f(x) ZW_0$, because the term $\theta(\cdot)$ does not depend on w ; by (8) it is at least $\mu(ZW_0)^\top ZW_0 \succ 0$, since ZW_0 has full column rank. Thus $H \succ 0$, and for fixed u the unique minimizer in w is $w(u) = -H^{-1}(Gu + a)$. Put $\bar{c} = \bar{x} - ZW_0H^{-1}a$ and $D = ZT_0 - ZW_0H^{-1}G$, so that $\bar{x} + ZT_0u + ZW_0w(u) = \bar{c} + Du$ and $\varphi(u) = \min_w F(u, w)$.

Since $UZW_0 = 0$, we have $UD = YT_0$, which has rank s ; so D has full column rank, and (i) follows from $\nabla^2\varphi = D^\top \nabla^2 f(\bar{c} + Du)D$. For (ii), write $y_J = T_0u_J + W_0w_J$. Minimizing first over w shows $w_J = w(u_J)$ and that u_J minimizes φ ; by (i), u_J is the unique zero u_* of $\nabla\varphi$. For (iii), the construction is a fixed sequence of rational operations: the chart, the product UZ , an echelon form, the matrices H , G , a and H^{-1} , and the substitution of the affine map $\bar{c} + Du$ into f and h . Each step multiplies, inverts, or row-reduces matrices of dimension at most $n \leq L$ whose entries have polynomially bounded bit length; by Cramer's rule and Hadamard's inequality each such step preserves a polynomial bound, and the number of steps is fixed, so one polynomial bound holds for every J . Substituting an affine map in $s \leq n$ variables into a monomial of degree at most four gives at most $(n+1)^4$ terms, with additive growth of coefficient bit length. Clearing denominators by the product of all denominators keeps the bit length polynomial. None of these bounds depends on k beyond $k \leq n$. $\square$

Proof of Theorem 6.6. A uniform gap. Assume $P \neq \emptyset$ and let J be a row basis of $A_{S(p)}$. By Lemma 6.1(d), $p = \bar{x} + Zy_J$ with y_J the unconstrained minimizer of $f(\bar{x} + Zy)$, so Lemma D.5 gives $p = \bar{c} + Du_*$ and $h(p) = \eta(u_*)$. The formula

$$\exists u \in \mathbb{R}^s : \quad \partial_1\varphi(u) = 0 \wedge \cdots \wedge \partial_s\varphi(u) = 0 \wedge z - \eta(u) = 0$$

defines the singleton $\{h(p)\}$, because u_* is the only real zero of $\nabla\varphi$. After clearing denominators it has $s \leq k$ quantified variables and $s+1$ integer polynomials of degree at most four with coefficients of bit size at most $T(L)$. Lemma 4.1, with s quantified variables, degree $d = 4$, and $\tau = T(L)$, gives $|h(p)| \geq 2^{-2T(L)4^{c_0s}}$ when $h(p) \neq 0$. With $G(k) = 2 \cdot 4^{c_0k}$ this yields

$$h(p) \neq 0 \implies |h(p)| \geq \gamma := 2^{-\omega}, \quad \omega = G(k)T(L). \quad (46)$$

The bound is linear in $T(L)$, with a factor that depends only on k . If $s = 0$, then $h(p) = \eta()$ is rational of bit length at most $T(L)$, and the same bound holds. The bound (46) holds for every observable of encoding length at most L , in particular for the slack observables $b_i - a_i^\top x$. The set $S(p)$ is used only to prove it.

The algorithm. Decide $P = \emptyset$ by linear programming; otherwise obtain a rational $x_0 \in P$ of polynomial bit length. Compute k, U, V (Lemma 6.5), the radius R_0 of Lemma D.1(a) for f and x_0 , so $\|p\| < R_0$, and $\omega = G(k)T(L)$. Let C_h be the sum of the absolute values of the coefficients of h , and put

$$B = 1 + \max\left\{4nC_h(R_0 + 1)^3, \max_i \|a_i\|_1\right\}, \quad \delta = \min\left\{1, \frac{\gamma}{8B}\right\}, \quad \varepsilon = \min\left\{1, \frac{\mu\delta^2}{2}\right\}.$$

On the ball $\|x\| \leq R_0 + 1$, each partial derivative of a monomial cx^α of degree at most four is bounded by $4|c|(R_0 + 1)^3$, so B bounds $\|\nabla h\|$ and the gradients of all slack observables there. Contract (E1), applied to f on $P \cap \{x : \|x\|_\infty \leq R_0\}$, returns a rational $\hat{x} \in P$ with $f(\hat{x}) \leq f(p) + \varepsilon$. Its running time is polynomial in L and $\log(1/\varepsilon) = O(\omega + \text{bits}(B) + \text{bits}(\mu))$, hence at most $F_1(k)L^{C_1}$. By Lemma D.1(b), $\|\hat{x} - p\| \leq \delta \leq 1$, so the segment from p to $\hat{x}$ lies in the ball and $|h(\hat{x}) - h(p)| \leq B\delta \leq \gamma/8$. Return 0 if $|h(\hat{x})| < \gamma/2$ and the sign of $h(\hat{x})$ otherwise. If $h(p) = 0$, then $|h(\hat{x})| \leq \gamma/8$; if $h(p) \neq 0$, then $|h(\hat{x})| \geq 7\gamma/8$ and the signs agree. The rational evaluations take time polynomial in L and ω .

The same point $\hat{x}$ decides every slack, because B also bounds their gradients; this yields $S(p)$. Finally compute a row basis J of $A_{S(p)}$ and apply Lemma D.5 to output $\bar{c}, D$, and φ . All steps take time $F(k)L^C$ with an absolute C : the only superpolynomial quantity is the number ω of accuracy bits, which enters polynomially. $\square$

The algorithm prints $\omega = 2 \cdot 4^{c_0 k} T(L)$ bits. Since $\log(1/\varepsilon) = O(\omega + \text{poly}(L))$ and every step is polynomial in L and ω , one can take $F(k) = 2^{O(k)}$.

D.4 Exact constrained Newton steps

Proof of Theorem 6.12. Let $(f, P) \in \mathcal{C}$, let h be an observable of degree at most D , and let L be the total input length. Decide $P \neq \emptyset$ by linear programming and assume $n \geq 1$.

Constants. By linear programming compute a rational point $x_{00} \in P$, and let R_0 be the radius of Lemma D.1(a) for f and x_{00} , so that $\|p\| < R_0$. Put $\varrho = R_0 + 1$, $\Omega = \{x : \|x\|_\infty \leq \varrho\}$, and let C_f and C_h be the sums of the absolute values of the coefficients of f and h , increased to at least one. Define

$$\Lambda = n^2 D^3 C_f \varrho^D, \quad \kappa = \max\{1, \Lambda/(2\mu)\}, \quad B_h = 1 + n D C_h \varrho^D.$$

On Ω , every third partial derivative of f is bounded by $D^3 C_f \varrho^D$. Hence each Hessian entry is $\sqrt{n} D^3 C_f \varrho^D$ -Lipschitz there, and the Frobenius norm shows that Λ is a Lipschitz constant of $\nabla^2 f$ on Ω in operator norm. Similarly $\|\nabla h\| \leq B_h$ on Ω . All these numbers are rational of polynomial bit length.

Warm start and convergence. Contract (E1), applied to f on $P \cap \{x : \|x\|_\infty \leq R_0\}$ with accuracy $\varepsilon_0 = \min\{1, \mu/(8\kappa^2)\}$, returns a rational $x_0 \in P$ with $f(x_0) \leq f(p) + \varepsilon_0$, so $\|x_0 - p\| \leq 1/(2\kappa) \leq 1/2$ by Lemma D.1(b). Let $x_{j+1} = N(x_j)$. Every iterate lies in P . If $\|x_j - p\| \leq 1/2$, then x_j and p lie in the convex set Ω , and Lemma 6.11 gives $\|x_{j+1} - p\| \leq \kappa \|x_j - p\|^2$. With $e_j = \kappa \|x_j - p\|$ we get $e_0 \leq 1/2$ and $e_{j+1} \leq e_j^2$, so by induction

$$\|x_j - p\| \leq \kappa^{-1} 2^{-2^j} \leq \frac{1}{2} \quad (j \geq 0). \quad (47)$$

Separation. The formula in the free variable z

$$\exists x \in \mathbb{R}^n \exists \lambda \in \mathbb{R}^m : Ax \leq b, \lambda \geq 0, \nabla f(x) + A^\top \lambda = 0, \lambda_i(b_i - a_i^\top x) = 0 \ (i \leq m), \ z = h(x)$$

defines $\{h(p)\}$. Indeed, Lemma 6.1(b) at p gives a multiplier supported on $S(p)$. Conversely, a solution (x, λ) has $x \in P$ and $-\nabla f(x) \in \text{cone}\{a_i : i \in S(x)\}$ by complementarity, so x solves $\text{VI}(\nabla f, P)$ and equals p . The formula has $n + m$ quantified variables and polynomially many integer polynomials of degree at most D with coefficients of polynomial bit length. Lemma 4.1, with $s = n + m$ quantified variables, degree at most $\max\{D, 2\}$, and coefficient bit size τ polynomial in L , gives $h(p) = 0$ or $|h(p)| \geq 2^{-2\tau \max\{D, 2\} c_0(n+m)}$. Hence there is a polynomial e_D , depending only on D , with $h(p) = 0$ or $|h(p)| \geq \gamma := 2^{-2e_D(L)}$; enlarge e_D so that also $2^{e_D(L)} \geq \log_2 B_h + 3$. Then (47) with $j_0 = e_D(L) + 1$ gives

$$|h(x_{j_0}) - h(p)| \leq B_h 2^{-2 \cdot 2^{e_D(L)}} \leq 2^{-2^{e_D(L)} - 3} = \gamma/8.$$

Circuit implementation. The point x_0 is printed in binary and turned into rational circuits. For $j < j_0$, the entries of $H_j = \nabla^2 f(x_j)$ and $c_j = \nabla f(x_j) - H_j x_j$ are computed from the circuits of x_j with polynomially many gates, since the degree is fixed. The solver is then run on (H_j, c_j) with every number held as a rational circuit and every comparison answered exactly by PosSLP. Because the answers are exact, the simulated execution is the execution of the solver on the rational data (H_j, c_j) : every division is by a nonzero number and the output is $x_{j+1} = N(x_j)$. The number of operations per step is polynomial in L and independent of the bit lengths of (H_j, c_j) , so after j_0 steps all circuits have polynomial size. Finally build γ by $e_D(L)$ squarings of $1/2$, let $\hat{\alpha} = h(x_{j_0})$, and ask for the sign of the numerator of

$$\hat{\alpha} - \gamma/2, \quad \hat{\alpha} + \gamma/2, \quad \text{or} \quad \gamma^2/4 - \hat{\alpha}^2,$$

which is positive exactly when $h(p) > 0$, $h(p) \geq 0$, or $h(p) = 0$, respectively; negations give the other relations. A zero value gives $|\hat{\alpha}| \leq \gamma/8$, and a nonzero value gives $|\hat{\alpha}| \geq 7\gamma/8$ with the same sign.

All work is polynomial in L , so the predicate is in P^{PosSLP} . Applying the procedure to the slacks $b_i - a_i^\top x$ gives $S(p)$. If $f(p) < \tau$, then for $h = f - \tau$ we have $h(p) \leq -\gamma$, hence $f(x_{j_0}) - \tau \leq -7\gamma/8 < 0$, and $x_{j_0} \in P$ is the claimed circuit point. The comparisons inside the solver depend on earlier answers, so the procedure is a Turing reduction. $\square$

D.4.1 Boxes

Lemma D.6 (Inactive truncation). *Let f satisfy (8), let $P \neq \emptyset$ have minimizer p , and let $q \in P$ be rational. Put $R_1 = 1 + 2\|\nabla f(q)\|_1/\mu$ and $P' = P \cap \{x : |x_i - q_i| \leq R_1 \ (i \leq n)\}$. Then $|p_i - q_i| < R_1$ for all i , so p is the minimizer of f on P' and none of the added bounds is active at p . The bit length of R_1 is polynomial in $L + \text{bits}(q)$.*

Proof. Since $f(p) \leq f(q)$ and $f(p) \geq f(q) - \|\nabla f(q)\| \|p - q\| + \frac{\mu}{2} \|p - q\|^2$, we get $\|p - q\| \leq 2\|\nabla f(q)\|/\mu \leq 2\|\nabla f(q)\|_1/\mu < R_1$. $\square$

For a box, q is obtained by clipping 0 to the finite bounds; for a flow polyhedron, by linear programming. In both cases P' is again a box or a flow polyhedron, now with finite bounds, and the active bounds of P' at p are the active original bounds.

Lemma D.7 (Admissible vector from the comparison matrix). *Let H be symmetric positive definite with $M(H) \succ 0$. Put $d = M(H)^{-1}\mathbf{1}$ and $v = \frac{1}{2}(H + M(H))d$. Then $d > 0$, $v \geq \mathbf{1}$, and $H_{SS}^{-1}v_S > 0$ entrywise for every nonempty $S \subseteq \{1, \dots, n\}$. If all off-diagonal entries of H are nonpositive, then $v = \mathbf{1}$.*

Proof. Nonnegative inverse. Let B_0 be positive definite with nonpositive off-diagonal entries, and let $B_0 u = w \geq 0$. Write $u = u_+ - u_-$ with $u_{\pm} \geq 0$ of disjoint supports. Then $0 \leq w^\top u_- = u_-^\top B_0 u_+ - u_-^\top B_0 u_-$. The first term involves only off-diagonal entries, so it is nonpositive, and the second is positive unless $u_- = 0$. Hence $u \geq 0$. Applied to $B_0 = M(H)$ and $w = \mathbf{1}$, this gives $d \geq 0$; if $d_i = 0$, then $(M(H)d)_i = \sum_{j \neq i} M(H)_{ij} d_j \leq 0$, contradicting $(M(H)d)_i = 1$. So $d > 0$.

Lower bound on v . The identity $M(H)d = \mathbf{1}$ reads $H_{ii}d_i = 1 + \sum_{j \neq i} |H_{ij}|d_j$. Therefore

$$v_i = H_{ii}d_i + \sum_{j \neq i, H_{ij} < 0} H_{ij}d_j = 1 + \sum_{j \neq i, H_{ij} > 0} H_{ij}d_j \geq 1.$$

If H has nonpositive off-diagonal entries, the last sum is empty and $v = \mathbf{1}$.

Principal systems. Fix a nonempty S and consider the Jacobi map $\mathcal{J}$ on $\mathbb{R}^S$ given by $(\mathcal{J}x)_i = (v_i - \sum_{j \in S \setminus \{i\}} H_{ij}x_j)/H_{ii}$. Its fixed points are the solutions of $H_{SS}x = v_S$. For $x \in [0, d_S]$, the numerator is smallest when $x_j = d_j$ for $H_{ij} > 0$ and $x_j = 0$ otherwise, and then equals $1 + \sum_{j \notin S, H_{ij} > 0} H_{ij}d_j \geq 1$. It is largest when $x_j = d_j$ for $H_{ij} < 0$ and $x_j = 0$ otherwise, and then equals $H_{ii}d_i + \sum_{j \notin S, H_{ij} < 0} H_{ij}d_j \leq H_{ii}d_i$. Hence $\mathcal{J}$ maps $[0, d_S]$ into $[0, d_S]$, with every coordinate at least $1/H_{ii} > 0$. For $x, x' \in \mathbb{R}^S$, with $\|x\|_d = \max_i |x_i|/d_i$,

$$\frac{|(\mathcal{J}x - \mathcal{J}x')_i|}{d_i} \leq \frac{\sum_{j \neq i} |H_{ij}|d_j}{H_{ii}d_i} \|x - x'\|_d = \left(1 - \frac{1}{H_{ii}d_i}\right) \|x - x'\|_d,$$

so $\mathcal{J}$ is a contraction. Its unique fixed point lies in $[0, d_S]$ and has positive coordinates; since H_{SS} is invertible, the fixed point is $H_{SS}^{-1}v_S$. $\square$

The iteration $\mathcal{J}$ appears only in the proof. An algorithm computes $M(H)$ by sign tests, d by a positive definite solve, and v by matrix arithmetic.

Lemma D.8 (Parametric box quadratic programming). *Let $H \in \mathbb{Q}^{n \times n}$ be symmetric positive definite, $u \in \mathbb{Q}^n$ with $u > 0$, $c \in \mathbb{Q}^n$, and let $v \in \mathbb{Q}^n$, $v > 0$, satisfy $H_{SS}^{-1}v_S \geq 0$ for every nonempty S . The procedure below returns the minimizer of $\frac{1}{2}x^\top Hx + c^\top x$ over $[0, u]$ after at most $2n$ pivots. It uses $O(n^4)$ arithmetic operations and $O(n^2)$ comparisons, and it divides only by pivots of positive definite matrices and by numbers it has tested to be positive.*

Procedure. Maintain a partition $\{1, \dots, n\} = \mathcal{L} \cup \mathcal{F} \cup \mathcal{U}$ and a parameter $\theta \geq 0$; initially $\mathcal{L}$ is everything and $\theta = \max\{0, \max_i(-c_i/v_i)\}$. In each round compute, using $LDL^\top$ factorization of $H_{\mathcal{F}\mathcal{F}}$,

$$\alpha_{\mathcal{F}} = H_{\mathcal{F}\mathcal{F}}^{-1}(c_{\mathcal{F}} + H_{\mathcal{F}\mathcal{U}}u_{\mathcal{U}}), \quad \beta_{\mathcal{F}} = H_{\mathcal{F}\mathcal{F}}^{-1}v_{\mathcal{F}},$$

and for $i \notin \mathcal{F}$, $\alpha_i = c_i + H_{i\mathcal{U}}u_{\mathcal{U}} - H_{i\mathcal{F}}\alpha_{\mathcal{F}}$ and $\beta_i = v_i - H_{i\mathcal{F}}\beta_{\mathcal{F}}$. The event values are $-\alpha_i/\beta_i$ for $i \in \mathcal{L}$ with $\beta_i > 0$ and $-(u_i + \alpha_i)/\beta_i$ for $i \in \mathcal{F}$ with $\beta_i > 0$. Let θ' be the maximum of 0 and all event values. If $\theta' = 0$, return the point x with $x_{\mathcal{L}} = 0$, $x_{\mathcal{U}} = u_{\mathcal{U}}$, and $x_{\mathcal{F}} = -\alpha_{\mathcal{F}}$. Otherwise move the smallest index whose event value equals θ' from $\mathcal{L}$ to $\mathcal{F}$ or from $\mathcal{F}$ to $\mathcal{U}$, set $\theta = \theta'$, and start a new round.

Proof. For a partition and a parameter θ , let $x(\theta)$ be the point with $x_{\mathcal{L}} = 0$, $x_{\mathcal{U}} = u_{\mathcal{U}}$, and $x_{\mathcal{F}} = -\alpha_{\mathcal{F}} - \theta\beta_{\mathcal{F}}$. It solves the free equations $(Hx + c + \theta v)_{\mathcal{F}} = 0$, and for $i \notin \mathcal{F}$ the gradient of $q_{\theta}(x) = \frac{1}{2}x^\top Hx + (c + \theta v)^\top x$ is $\partial_i(\theta) = \alpha_i + \theta\beta_i$. We prove that each round starts with the following invariant: $0 \leq x(\theta) \leq u$, $\partial_i(\theta) \geq 0$ for $i \in \mathcal{L}$, and $\partial_i(\theta) \leq 0$ for $i \in \mathcal{U}$. These are the Karush–Kuhn–Tucker conditions for minimizing q_{θ} over $[0, u]$. Initially $x = 0$ and $\partial_i = c_i + \theta v_i \geq 0$ by the choice of θ .

Signs of the slopes. Admissibility gives $\beta_{\mathcal{F}} \geq 0$. For $i \notin \mathcal{F}$ let $S = \mathcal{F} \cup \{i\}$ and $w = H_{SS}^{-1}v_S \geq 0$. Eliminating $w_{\mathcal{F}}$ from $H_{SS}w = v_S$ gives $(H_{ii} - H_{i\mathcal{F}}H_{\mathcal{F}\mathcal{F}}^{-1}H_{\mathcal{F}i})w_i = \beta_i$. The Schur complement on the left is positive, so $\beta_i \geq 0$; for $\mathcal{F} = \emptyset$, $\beta_i = v_i > 0$.

One round. As θ decreases with the partition fixed, the free coordinates $-\alpha_i - \theta\beta_i$ do not decrease, and the gradients $\alpha_i + \theta\beta_i$ outside $\mathcal{F}$ do not increase. Hence the invariant persists until a gradient in $\mathcal{L}$ reaches zero or a free coordinate reaches its upper bound. These are exactly the event values, and each of them is at most the current θ , because the invariant holds at the current θ ; coordinates with zero slope never produce events. So the invariant holds on $[\theta', \theta]$. If $\theta' = 0$, it holds at $\theta = 0$, where q_0 is the original objective; the conditions are sufficient for optimality because the problem is convex, and the minimizer is unique because $H \succ 0$.

A pivot. If $\theta' > 0$ and the selected index i is in $\mathcal{L}$, then $\partial_i(\theta') = 0$ and $x_i = 0$, so the current point satisfies the free equations for $\mathcal{F} \cup \{i\}$. These equations have a unique solution because $H_{\mathcal{F} \cup \{i\}, \mathcal{F} \cup \{i\}} \succ 0$, so the new partition represents the same point at θ' , and the invariant holds. If $i \in \mathcal{F}$ with $x_i(\theta') = u_i$, the current point satisfies the free equations for $\mathcal{F} \setminus \{i\}$ with $x_i = u_i$, and $\partial_i(\theta') = 0 \leq 0$; again the invariant holds. Ties are processed one index at a time.

An index moves only from $\mathcal{L}$ to $\mathcal{F}$ and from $\mathcal{F}$ to $\mathcal{U}$, so there are at most $2n$ pivots and at most $2n + 1$ rounds. Each round uses $O(n^3)$ arithmetic operations; the LDL^T factorization of a positive definite matrix needs no comparisons and has positive pivots. Each round needs $O(n)$ comparisons to test $\beta_i > 0$ and to find the maximum and the smallest maximizing index. $\square$

Proof of Corollary 6.13. If some $\ell_i > u_i$, the box is empty. Substitute every fixed coordinate $\ell_i = u_i$. If every coordinate is fixed, then $P = \{\ell\}$ and $p = \ell$ is rational: evaluate $h(p)$ exactly, report every bound as active, and stop; neither the transfer theorem nor the procedure of Lemma D.8 is called on an empty set of variables. Otherwise the Hessian of the reduced objective is a principal submatrix of $\nabla^2 f$, so each of (i)–(iii) is inherited: principal submatrices of a comparison matrix are comparison matrices, and forests and sign patterns restrict. Apply Lemma D.6 to obtain a box with finite bounds $\ell < u$ and the same minimizer; its added bounds are inactive. Observables are substituted in the same way, and the active original bounds are decided by the slack observables.

Conditions (i) and (ii) imply (iii). Under (ii), $M(H) = H$. Under (i), root each tree of the forest, set $\sigma_{\text{root}} = 1$, and for an edge from a parent i to a child j set $\sigma_j = -\text{sgn}(H_{ij})\sigma_i$ if $H_{ij} \neq 0$ and $\sigma_j = \sigma_i$ otherwise. With $\Sigma = \text{diag}(\sigma)$, every off-diagonal entry of $\Sigma H \Sigma$ is $-|H_{ij}|$, so $M(H) = \Sigma H \Sigma \succ 0$.

The Taylor-QP solver for the box translates $x = \ell + x'$, which replaces c by $c + H\ell$ and the box by $[0, u - \ell]$, computes $M(H)$ with n^2 sign tests, d and v as in Lemma D.7 with $O(n^3)$ operations, and runs the procedure of Lemma D.8. Under (iii) at the current point, Lemma D.7 makes v admissible, so the solver is exact. It uses $O(n^4)$ operations and comparisons, independently of the bit lengths of H and c , with a control computable in polynomial time. It is therefore an arithmetic Taylor-QP solver, and Theorem 6.12 applies. Condition (i) can be checked from the coefficients after the substitution: the reduced box is full-dimensional, and a polynomial that vanishes on a full-dimensional box vanishes identically. $\square$

D.4.2 Separable flows

We use the quadratic case of Véggh's strongly polynomial algorithm through the following contract.

Contract (E3c) [117, Theorem 20 and Section 6.1]. There is an algorithm that, given a directed graph with $m' \geq 1$ arcs and no isolated nodes, rational node demands, finite rational lower and upper arc capacities, and arc costs $a_e s^2 + c_e s$ with rational $a_e \geq 0$ and c_e , decides whether a feasible flow exists and, if so, returns an optimal flow. Its operations on numbers are additions, subtractions, multiplications, divisions, and comparisons;

their number is $O(m'^4 \log(m' + 2))$ and does not depend on the numerical data. On rational data the returned flow is the exact optimum.

An isolated node carries no arc, so its demand must be zero in every feasible flow. Isolated nodes are therefore deleted beforehand, with $O(|V|)$ comparisons, and a nonzero demand at one of them shows that no feasible flow exists.

Implementations of minimum-cost flow algorithms often make the network strongly connected by artificial arcs of large linear cost. In the arithmetic model, any such cost is computed from the data by the algorithm's own operations. The following lemma gives, in addition, an explicit valid cost that is a rational function of the data, so that this step never inspects the bit length of a circuit.

Lemma D.9 (Explicit artificial costs). *Consider a network with N_0 nodes, demands summing to zero, arcs e with capacity intervals $[0, U_e]$ (possibly $U_e = \infty$), and convex differentiable costs C_e , and suppose that it has an optimal flow s^* . Let $C_0 \geq 1$ satisfy $|C'_e(s_e^*)| \leq C_0$ for all arcs. Add a node ω with zero demand and, for each node v , uncapacitated arcs (v, ω) and (ω, v) with linear cost $M_0 s$, where $M_0 > (N_0 - 1)C_0$. Then the optimal flows of the augmented network are exactly the optimal flows of the original network, extended by zero on the new arcs.*

Proof. With node potentials π , the reduced cost of an arc $e = (i, j)$ at a flow s is $r_e = C'_e(s_e) - \pi_i + \pi_j$. A feasible flow is optimal if and only if some potential satisfies, for every arc with $U_e > 0$, $r_e \geq 0$ when $s_e = 0$, $r_e = 0$ when $0 < s_e < U_e$, and $r_e \leq 0$ when $s_e = U_e$; an arc with $U_e = 0$ imposes no condition, because the multipliers of its two bounds absorb any reduced cost. This is the Karush–Kuhn–Tucker system of a convex problem with linear constraints. Such potentials exist at s^* . In the residual graph of s^* , which has an arc $i \rightarrow j$ of length $C'_e(s_e^*)$ when $s_e^* < U_e$ and an arc $j \rightarrow i$ of length $-C'_e(s_e^*)$ when $s_e^* > 0$, these conditions say that $-\pi$ satisfies the triangle inequality, so there is no negative cycle. Hence shortest-path distances $\delta(v)$ from an extra source joined to every node by an arc of length zero are well defined, simple shortest paths have at most $N_0 - 1$ residual arcs, and $-(N_0 - 1)C_0 \leq \delta(v) \leq 0$. The potential $\pi = -\delta$ satisfies the optimality conditions and takes values in $[0, (N_0 - 1)C_0]$. Set $\pi_\omega = 0$. The new arcs then have reduced costs $M_0 - \pi_v > 0$ and $M_0 + \pi_v > 0$ at zero flow, so s^* extended by zero is optimal in the augmented network.

Let s' be any feasible augmented flow and let r be the reduced costs at s^* extended by zero. Convexity gives $\text{cost}(s') \geq \text{cost}(s^*) + \sum_a c'_a(s'_a - s_a^*)$, where c'_a is $C'_a(s_a^*)$ or M_0 . Since s' and s^* have the same demands, $\sum_a (\pi_i - \pi_j)(s'_a - s_a^*) = 0$, so the sum equals $\sum_a r_a(s'_a - s_a^*)$. Each term is nonnegative: on an arc with $U_a > 0$ by the sign conditions, and on an arc with $U_a = 0$ because $s'_a = s_a^* = 0$. Each new arc contributes $r_a s'_a$ with $r_a > 0$. Hence an augmented optimum carries no flow on the new arcs, and its restriction is optimal for the original network. $\square$

For quadratic costs on finite capacity intervals, a valid C_0 is $1 + \sum_e (|C'_e(\text{lower bound})| + |C'_e(\text{upper bound})|)$, because C'_e is monotone. It is computed from circuit data with two sign tests per arc.

Proof of Corollary 6.14. Decide feasibility by linear programming and obtain a rational feasible flow q . Substitute every fixed arc ($\ell_e = u_e$) and adjust the demands. If no arc remains, the feasible flow is the rational point p given by the fixed values: evaluate $h(p)$ exactly, report both bounds of every arc as active, and stop. Otherwise delete the isolated nodes; their adjusted demands are zero because $P \neq \emptyset$. Apply Lemma D.6 with q to obtain finite capacities with the same minimizer and inactive added bounds; below, ℓ and u denote these finite bounds. At a feasible x , the Taylor model is, up to a constant, the separable quadratic flow problem in the flow variable y with arc costs $a_e y_e^2 + c_e y_e$, where $a_e = \frac{1}{2} f''_e(x_e) > 0$ and $c_e = f'_e(x_e) - f''_e(x_e)x_e$ are rational circuits computed from

those of x with a constant number of gates per arc. The shift $s = y - \ell$ turns the capacity intervals into $[0, u_e - \ell_e]$, replaces the demands b by $b - A_G \ell$, computed once by rational arithmetic, and replaces c_e by $c_e + 2a_e \ell_e$, with a constant number of gates per arc; it puts every subproblem in the form of Lemma D.9. The algorithm of contract (E3c), run on the reduced graph, which has $m' \geq 1$ arcs and no isolated nodes, simulated on rational circuits with PosSLP comparisons and, if needed, with the artificial cost of Lemma D.9, is an arithmetic Taylor-QP solver, and Theorem 6.12 applies. For $D \leq 4$, the promise $f_e'' \geq \mu$ is checked as follows: a quadratic $\alpha s^2 + \beta s + \gamma_0$ is nonnegative on $\mathbb{R}$ exactly when $\alpha > 0$ and $4\alpha\gamma_0 \geq \beta^2$, or $\alpha = \beta = 0 \leq \gamma_0$. $\square$

D.4.3 One sign instance

We use Theorem 4.10(b): if a deterministic oracle Turing machine M , whose oracle queries are encodings of integer circuits over $\{0, 1, +, -, \times\}$, halts within $\pi_0(L)$ steps on every input of length L and for all oracle answers, for a polynomial π_0 , then there is a polynomial-time map $x \mapsto \Phi_x$ to integer circuits with $\Phi_x > 0$ if and only if M with a PosSLP oracle accepts x .

Proof of Corollary 6.15. Let M_0 be the decision procedure of the proof of Theorem 6.12 for $(h, \bowtie)$, with the preprocessing of Corollary 6.13 or Corollary 6.14 when it applies. Its queries are numerators of rational circuits that it has written, hence encodings of integer circuits. On every promised input of length L its running time is at most $\pi_1(L)$ for a fixed polynomial π_1 : the ordinary steps (linear programming, the warm start, the constants) are polynomial in bit time, and the number of Newton steps, of solver operations per step, and of gates per operation are bounded by polynomials in L that do not depend on the expanded values of any circuit. Each query has length at most the running time.

Let M check that its input is a well-formed instance, run M_0 for at most $\pi_1(L)$ steps, and reject if M_0 has not halted. Then M halts within a polynomial number of steps on every input and agrees with M_0 on promised inputs. On inputs that violate a promise, a division by a number that happens to be zero only produces a circuit with a zero denominator, which is still a syntactically valid query; the clock bounds the computation. Invalid supplied certificates and empty feasible sets are decided by ordinary steps before any query. The adaptive form of Theorem 4.10(b) now gives the required circuit Φ . For a polynomial-size Boolean combination of predicates, let one machine run the corresponding procedures one after the other and evaluate the combination; it satisfies the same hypotheses. $\square$

The proof uses only that the decision procedure is deterministic and has a worst-case polynomial bound on time and query length. A procedure with these properties and a time bound $F(\kappa)L^C$ compiles into one circuit in time polynomial in $F(\kappa)L^C$, which is an FPT many-one reduction. The Las Vegas algorithm of Theorem 6.16 has only an expected time bound, and fixing its random bits does not yield a correct deterministic procedure; the guessed certificate of Theorem 6.2 is likewise not removed by compilation.

D.5 Constraint rank

A *violator space* [44] is a pair (H, V) of a finite set H and a map $V : 2^H \rightarrow 2^H$ such that $G \cap V(G) = \emptyset$ for all $G \subseteq H$ (consistency) and $V(G) = V(F)$ whenever $F \subseteq G \subseteq H$ and $G \cap V(F) = \emptyset$ (locality). A *basis* of G is an inclusion-minimal $B \subseteq G$ with $V(B) = V(G)$, and the *combinatorial dimension* is the largest size of a basis of any $G \subseteq H$.

Lemma D.10 (A violator space of rank r). *Let $P \neq \emptyset$, let $H = \{1, \dots, m\}$ index the rows of A , and for $G \subseteq H$ let p_G be the solution of $\text{VI}(T, P_G)$. Then $V(G) = \{i \in H : a_i^\top p_G > b_i\}$ defines a*

violator space. Every basis consists of linearly independent rows, so the combinatorial dimension is at most $r = \text{rank } A$. If B is a basis of H , then $p_B = p$.

Proof. Every P_G contains P , so p_G exists and is unique by Lemma 6.1(a). Consistency holds because $p_G \in P_G$. For locality, $G \cap V(F) = \emptyset$ says that $p_F \in P_G$. Since $P_G \subseteq P_F$, the inequality $T(p_F)^\top(x - p_F) \geq 0$ holds for all $x \in P_G$, so $p_F = p_G$ and $V(F) = V(G)$. No potential function or ordering of values is used.

Let B be a basis of some G . By Lemma 6.1(b), applied to $\text{VI}(T, P_B)$, there is a set $J \subseteq B$ of independent rows with $a_i^\top p_B = b_i$ for $i \in J$ and $-T(p_B) = A_J^\top \lambda$, $\lambda \geq 0$. For $x \in P_J$, $T(p_B)^\top(x - p_B) = -\lambda^\top(A_J x - b_J) \geq 0$, and $p_B \in P_B \subseteq P_J$. Hence $p_J = p_B$, so $V(J) = V(B) = V(G)$, and minimality of B forces $J = B$. Finally, $V(H) = \emptyset$ because $p_H = p \in P$; so a basis B of H has $V(B) = \emptyset$, that is, $p_B \in P$. Its variational inequality holds on $P_B \supseteq P$, so $p_B = p$. $\square$

The bound concerns every basis of every subset. A small active support at the final solution alone would not bound the combinatorial dimension.

Contract (E3d) [44, Theorem 27 and Sections 4.2–4.5]. Let (H, V) be a violator space with $|H| = m$ and combinatorial dimension at most a given $\delta \geq 1$. Clarkson’s two randomized algorithms, as formulated for violator spaces, return a basis of H . They access V only through the *violation test*, which decides $i \in V(G)$ for given $G \subseteq H$ and $i \in H \setminus G$, and they perform an expected $O(\delta m + \delta^{O(\delta)})$ violation tests. In every test $|G| \leq \delta$: the tests ask whether an element violates a basis returned by a recursive call, or, inside the routine that finds a basis of a sample of $O(\delta^2)$ elements by examining its subsets B with $|B| \leq \delta$, whether an element violates B or whether $b \in B$ violates $B \setminus \{b\}$. The remaining operations are random sampling of a prescribed number of elements from a set or from a multiset given by integer multiplicities, doubling of multiplicities, and set bookkeeping. In each call of the reweighting algorithm the multiplicities are doubled $O(\delta \log m)$ times.

Lemma D.11 (Violation primitive). *In the setting of Lemma D.10, given G with $|G| \leq r$ and $i \notin G$, one decides $i \in V(G)$ in deterministic time $2^r L^{C_0}$ with a PosSLP oracle, where C_0 is an absolute constant.*

Proof. Enumerate the subsets $J \subseteq G$ with independent rows in a fixed order; there are at most 2^r . For each, compute the chart and U_J and run the tests (44) with this G . By Lemma D.2, a passing J gives $p_J = p_G$, and some J passes, namely a support of p_G , which lies in G . At the first passing J , decide $a_i^\top p_J - b_i > 0$, an affine observable at the zero of U_J . Each J costs polynomially many tests, each a PosSLP instance of polynomial length (Theorem 4.7). $\square$

Proof of Theorem 6.16. Decide $P = \emptyset$ by linear programming and compute r . If $r = 0$, then all rows are zero and feasibility gives $P = \mathbb{R}^n$; the solution is the zero of T , and Theorem 4.7 decides the predicate. Otherwise run Clarkson’s algorithms of contract (E3d) with $\delta = r$ on the violator space of Lemma D.10, implementing every violation test by Lemma D.11. By contract (E3d), every test has a first argument of size at most r , so the primitive applies. The returned basis B satisfies $p_B = p$ by Lemma D.10. Enumerating the independent subsets of B and running the support test with $G = B$ yields J and $(\bar{x}, Z, U_J)$ with $p = \bar{x} + Zy_J$. The predicate $h(p) \bowtie 0$ is then decided by Theorem 4.7 applied to U_J and $h(\bar{x} + Zy)$, or by rational arithmetic if $|J| = n$.

Bit model of the sampling. Each element of H is a row index, so repeated inequalities are distinct elements. Multiplicities are stored as binary integers. Since each call of the reweighting algorithm doubles multiplicities $O(r \log m)$ times, every multiplicity, and their sum, has $O(r \log m)$

bits. To draw one labeled copy from a multiset with total weight W , draw a uniform integer in $\{1, \dots, W\}$ by rejection from $\lceil \log_2 W \rceil$ random bits, which needs an expected constant number of trials, and locate it by prefix sums; for sampling without replacement, decrement the copy count temporarily. Before the basis routine, duplicate copies are identified with their row index, since multiplicity affects sampling probabilities but not the polyhedron P_G . Between consecutive violation tests the algorithms perform a constant number of sampling rounds and set operations on at most m elements. Hence the expected bit time outside the tests is polynomial in L times one plus the expected number of tests.

Expected time and correctness. The expected number of tests is $O(rm + r^{O(r)})$ and each costs $2^r L^{C_0}$, with $m \leq L$. The total expected time is therefore $(r + 1)^{O(r)} L^C$ with an absolute C . Every query has length polynomial in L , independently of the random choices. The algorithms stop only with a basis of H , and every test is answered exactly, so every returned answer is correct; randomness affects only the running time. $\square$

D.6 Candidate lists with unconstrained fibers

We use the following integer feasibility theorem, which queries its oracle only at integer points.

Theorem D.12 (Integer feasibility with integer-point separation; [8, 11, 54]). *There is a deterministic algorithm with the following property. Its input is an integer $t \geq 1$ and an integer $R \geq 2$, and it has access to an oracle for a closed convex set $S \subseteq [-R, R]^t$ that is queried only at integer points. On a query $z \in \mathbb{Z}^t$ the oracle either asserts $z \in S$, or returns $(\pi, \beta) \in \mathbb{Q}^t \times \mathbb{Q}$ with $\pi^\top x \leq \beta$ for all $x \in S$ and $\pi^\top z > \beta$. The algorithm either returns a point of $S \cap \mathbb{Z}^t$ or correctly reports $S \cap \mathbb{Z}^t = \emptyset$. If the oracle's answers on queries of bit length ℓ have length and computation time at most $\Phi(\ell)$, its running time is*

$$2^{O(t \log(t+1))} \text{poly}(t, 1 + \text{bits}(R), \Phi(c_0 t(1 + \text{bits}(R))))$$

for an absolute constant c_0 and a polynomial of absolute degree. Queries outside $[-R, R]^t$ may be answered by a violated box inequality.

This is Ari and Hildebrand [8, Definition 3.4 and Theorem 3.5], who attribute it to Hildebrand and Göß [54, Theorem 9 and Appendix B] and Basu [11, Theorem 5.7 and Remark 5.11]. No inradius or full-dimensionality assumption on S is required, empty and one-point sets are allowed, and the queries are made in the original integer coordinates. The algorithm is deterministic. Basu's remarks also contain the idea of optimizing without bisection on the objective by keeping visited candidates, which our lists use in a form that never needs objective values.

For the initial integer point we use deterministic integer linear feasibility in fixed dimension: for an explicit rational polyhedron $Q \subseteq \mathbb{R}^t$, one can decide $Q \cap \mathbb{Z}^t = \emptyset$ or return a point of $Q \cap \mathbb{Z}^t$ in time $2^{O(t \log(t+1))} L^{O(1)}$, with output of bit length $2^{O(t)} L^{O(1)}$ and absolute exponents [55, Theorem 1.1, specialized to linear constraints and a constant objective].

Lemma D.13 (Gradient cuts for unconstrained fibers). *Let $P = Q \times \mathbb{R}^n$ and $g(z) = \min_y f(z, y)$ for $z \in \mathbb{R}^t$. Then g is differentiable and μ -strongly convex on $\mathbb{R}^t$, with $\nabla g(z) = \nabla_z f(z, y(z))$, where $y(z)$ is the unique fiber minimizer. For an integer $R \geq 1$ and $z \in \mathbb{Z}^t \cap [-R, R]^t$, a rational c_z with $\|c_z - \nabla g(z)\| \leq \mu/4$, and hence satisfying (13), is computed in ordinary time $(L + \text{bits}(R) + \text{bits}(z))^{C_1}$ for an absolute constant C_1 .*

Proof. Each fiber $y \mapsto f(z, y)$ has Hessian at least μI , so $y(z)$ is unique, and the implicit function theorem applied to $\nabla_y f(z, y) = 0$ makes $y(\cdot)$ smooth. Then $g(z) = f(z, y(z))$ and $\nabla g(z) = \nabla_z f(z, y(z)) + Dy(z)^\top \nabla_y f(z, y(z)) = \nabla_z f(z, y(z))$. For z_1, z_2 with fiber minimizers y_1, y_2 and

$\vartheta \in [0, 1]$, strong convexity of f gives $g(\vartheta z_1 + (1 - \vartheta)z_2) \leq \vartheta g(z_1) + (1 - \vartheta)g(z_2) - \frac{\mu}{2}\vartheta(1 - \vartheta)\|z_1 - z_2\|^2$, after dropping the y -part of the squared distance. If c_z satisfies the stated accuracy and $w \neq z$ is an integer point, then

$$g(w) - g(z) \geq c_z^\top(w - z) - \frac{\mu}{4}\|w - z\| + \frac{\mu}{2}\|w - z\|^2 \geq c_z^\top(w - z) + \frac{\mu}{4}\|w - z\|^2,$$

because $\|w - z\| \geq 1$.

To compute c_z , let $m_0 = t + n$, $C_f = \max\{1, \sum_\alpha |f_\alpha|\}$, and

$$D_1 = 4m_0 C_f (1 + R)^3, \quad S_1 = 1 + R + D_1/\mu, \quad K_1 = 1 + 12m_0 C_f S_1^2, \quad \eta = \min\left\{1, \frac{\mu}{4tK_1}\right\}.$$

Each partial derivative of f at $(z, 0)$ is at most $4C_f(1 + R)^3$ in absolute value, so $\|\nabla_y f(z, 0)\| \leq D_1$, and strong monotonicity of $\nabla_y f(z, \cdot)$ gives $\|y(z)\| \leq D_1/\mu$. On the set of points (z, y) with $\|y - y(z)\| \leq 1$, every coordinate is at most S_1 in absolute value, every second partial derivative of f is at most $12C_f S_1^2$, and the block $\nabla_{zy}^2 f$ has operator norm less than K_1 . If $n = 0$, put $c_z = \nabla f(z)$. Otherwise contract (E1), applied to $y \mapsto f(z, y)$ on the box $\|y\|_\infty \leq D_1/\mu + 1$, returns a rational $\hat{y}$ with $f(z, \hat{y}) \leq f(z, y(z)) + \min\{1, \mu\eta^2/2\}$, hence $\|\hat{y} - y(z)\| \leq \eta$. Put $c_z = \nabla_z f(z, \hat{y})$. For each $i \leq t$, the mean value theorem along the segment from $y(z)$ to $\hat{y}$ gives $|(c_z)_i - \partial_i g(z)| \leq K_1 \eta \leq \mu/(4t)$, so $\|c_z - \nabla g(z)\| \leq \sqrt{t} \mu/(4t) \leq \mu/4$. All numbers involved have bit length polynomial in $L + \text{bits}(R) + \text{bits}(z)$, and the accuracy requested in (E1) has polynomially many bits. $\square$

Proof of Theorem 6.18. If $t = 0$, the only possible block is the empty vector, which is feasible exactly when $Q = \mathbb{R}^0$ satisfies its constraints. Let $t \geq 1$. Use integer linear feasibility to certify $Q \cap \mathbb{Z}^t = \emptyset$ or to find $z_0 \in Q \cap \mathbb{Z}^t$. Apply Lemma D.1(a) to f on $\mathbb{R}^{t+n}$ and the point $(z_0, 0)$; every optimal solution has $f \leq f(z_0, 0)$, so its norm is less than R_0 . Let $R = \lceil R_0 \rceil + 1$ and $P_0 = Q \cap [-R, R]^t$, which contains z_0 and every optimal integer block. Since $\text{bits}(z_0) \leq 2^{O(t)} L^{O(1)}$, also $\text{bits}(R) \leq 2^{O(t)} L^{O(1)}$.

Define a deterministic answer rule $\mathcal{O}$ for integer queries z . If $z \notin P_0$, return the first violated inequality of P_0 in a fixed order. Otherwise compute c_z by Lemma D.13. If $c_z \neq 0$, return $c_z^\top x \leq c_z^\top z - \mu/4$. The answer depends only on the instance and on z . The list algorithm starts with $\mathcal{Z} = \{z_0\}$ and runs the algorithm of Theorem D.12 with radius R , answering queries by $\mathcal{O}$ and adding every queried point of P_0 to $\mathcal{Z}$. If some query $z \in P_0$ has $c_z = 0$, it stops at once and outputs $\mathcal{Z}$. Otherwise it outputs $\mathcal{Z}$ when the feasibility algorithm halts.

Running time. Let $\mathcal{O}_\emptyset$ agree with $\mathcal{O}$, except that at a query $z \in P_0$ with $c_z = 0$ it returns $x_1 \leq z_1 - 1$. Every answer of $\mathcal{O}_\emptyset$ is an inequality violated by its query, and every inequality is valid on the empty set. So $\mathcal{O}_\emptyset$ is a valid oracle for $S = \emptyset$ that never asserts membership, and its answers on queries of length ℓ cost at most $\Phi(\ell) = (L + \text{bits}(R) + \ell)^{C_1}$. Up to the possible early stop, the list algorithm performs the run of Theorem D.12 with $\mathcal{O}_\emptyset$. That run halts, reporting emptiness, within $2^{O(t \log(t+1))} \text{poly}(t, 1 + \text{bits}(R), \Phi(c_0 t(1 + \text{bits}(R)))) = 2^{O(t \log(t+1))} L^C$ steps, with C absolute. The list and its total length are bounded by the same quantity.

Every optimal block is listed. If the algorithm stops early at z with $c_z = 0$, then z is the unique optimal block by Lemma 6.17, and it is listed. Otherwise suppose an optimal block w is not in $\mathcal{Z}$. Define the oracle $\mathcal{O}_w$ that asserts membership exactly at w and answers every other query as $\mathcal{O}$ does. It is a valid oracle for $S = \{w\} \subseteq [-R, R]^t$: inequalities of P_0 are valid at $w \in P_0$, and at a query $z \in P_0$ with $z \neq w$ we have $g(w) \leq g(z)$, so $c_z \neq 0$ and the cut is valid at w by Lemma 6.17. Its answer cost exceeds Φ only by a comparison with the stored w . The actual run never queried w , and the feasibility algorithm is deterministic, so the run with $\mathcal{O}_w$ asks the same queries, receives the same answers, and also reports $S \cap \mathbb{Z}^t = \emptyset$. This contradicts the correctness of the algorithm, since $S \cap \mathbb{Z}^t = \{w\}$.

At most 2^t optimal blocks. If $w_1 \neq w_2$ are optimal blocks with $w_1 \equiv w_2 \pmod{2}$, then $(w_1 + w_2)/2 \in Q \cap \mathbb{Z}^t$ and strict convexity of g gives $g((w_1 + w_2)/2) < \min g$. Hence distinct optimal blocks have distinct parity vectors. For $f(z, y) = \sum_{i \leq t} (z_i - \frac{1}{2})^2 + \|y\|^2$ and $Q = \mathbb{R}^t$, with $\mu = 2$, the optimal blocks are exactly $\{0, 1\}^t$. $\square$

Proof of Corollary 6.19. If $Q \cap \mathbb{Z}^t = \emptyset$, the predicates have fixed truth values under the convention $\min \emptyset = +\infty$. Otherwise the list $\mathcal{Z}$ contains every optimal block, so $\min = \min_{z \in \mathcal{Z}} g(z)$. Thus $\min \leq \tau$ and $\min < \tau$ are disjunctions of the predicates $g(z) \leq \tau$ and $g(z) < \tau$, and the other relations are Boolean combinations of these two. For $n \geq 1$, the predicate $g(z) \bowtie \tau$ is the observable $f(z, y) - \tau$ at the minimizer of the strongly convex quartic $y \mapsto f(z, y)$, so Theorem 4.4 maps it to one PosSLP instance; for $n = 0$ it is a rational comparison and maps to a fixed circuit. All instances are built from $\mathcal{Z}$ without oracle calls, and the Boolean form, Theorem 4.10(c), combines them into one instance. The total construction time is polynomial in the length of $\mathcal{Z}$ and L , hence $2^{O(t \log(t+1))} L^C$.

For (b), compare every pair $z, z' \in \mathcal{Z}$ by Corollary 4.5: the observable $f(z, y) - f(z', y')$ at the minimizer of the strongly convex quartic $f(z, y) + f(z', y')$ on $\mathbb{R}^{2n}$, whose Hessian is block diagonal and at least μI , equals $g(z) - g(z')$. A listed block is optimal exactly when its value is at most every other listed value, because the list contains an optimal block. All queries are fixed before any answer is received. The continuous part of the optimum for an optimal block z_* is $y(z_*)$, the unique real solution of $\nabla_y f(z_*, y) = 0$. $\square$

Lemma D.14 (Padding that keeps a full Hessian Gram). *Let $f \in \mathbb{Q}[x_1, \dots, x_n]$ be a quartic with a positive definite rational full Hessian Gram M on $(v, x \otimes v)$, of size $m_M = n + n^2$, and put $\mu_0 = \det M / (\text{tr } M)^{m_M-1}$. For $t \geq 1$ and $\tau = 1 + 8nt/\mu_0$, the quartic*

$$P(x, z) = f(x) + \|z\|^2 + \tau\|z\|^4 + \|z\|^2\|x\|^2, \quad z \in \mathbb{R}^t,$$

has $\min\{P(x, z) : x \in \mathbb{R}^n, z \in \mathbb{Z}^t\} = \min f$, attained exactly at $z = 0$, and a positive definite rational full Hessian Gram on the joint basis $(v_x, v_z, x \otimes v_x, x \otimes v_z, z \otimes v_x, z \otimes v_z)$, computable in polynomial time.

Proof. The added terms are nonnegative and vanish exactly when $z = 0$, so the mixed-integer minimum is $\min f$, attained only with $z = 0$. In the direction (v_x, v_z) , the second derivatives of the added terms are $2\|v_z\|^2$, $\tau(4\|z \otimes v_z\|^2 + 8(z^\top v_z)^2)$, and $2\|x \otimes v_z\|^2 + 2\|z \otimes v_x\|^2 + 8(x^\top v_x)(z^\top v_z)$. Keep M on $(v_x, x \otimes v_x)$; put $2I$ on $v_z, x \otimes v_z$ and $z \otimes v_x$; put $4\tau I + 8\tau \iota_t \iota_t^\top$ on $z \otimes v_z$, where ι_t is the vectorized identity, so that $\iota_t^\top(z \otimes v_z) = z^\top v_z$; and represent the remaining term by the off-diagonal block $4\iota_n \iota_t^\top$ between $x \otimes v_x$ and $z \otimes v_z$. For formal vectors a in the $x \otimes v_x$ block and c in the $z \otimes v_z$ block, the cross term is at most $8\sqrt{nt} \|a\| \|c\| \leq \frac{\mu_0}{2} \|a\|^2 + \frac{32nt}{\mu_0} \|c\|^2$ in absolute value. Since $M \succeq \mu_0 I$ (Lemma 2.7) and $4\tau - 32nt/\mu_0 = 4$, the assembled matrix is at least $\min\{\mu_0/2, 2\}I$ on the whole basis, not only on vectors of tensor form. $\square$

D.7 Mixed linear constraints

Two further external results enter.

Contract (E3b) [71]. A convex quadratic function with rational coefficients that attains its minimum on a nonempty explicit rational polyhedron has a rational minimizer that is computed exactly in polynomial time.

Contract (E3e) [31, Theorem 3, applied with the zero objective]. Given explicit rational A, B, d with t columns of A , one decides in deterministic time $F_0(t)L^{C_0}$, with F_0

computable and C_0 absolute, whether $\{(z, y) : z \in \mathbb{Z}^t, Az + By \leq d\}$ is empty, and otherwise returns a rational point of it.

Lemma D.15 (Primal–dual cuts for constrained fibers). *Let $P = \{(z, y) : Az + By \leq d\}$, let $z \in \mathbb{Z}^t$ have a nonempty fiber, let $\hat{y}$ be a rational point of the fiber and $\lambda \geq 0$ a rational vector, and put*

$$\sigma = d - Az - B\hat{y} \geq 0, \quad \rho = \nabla_y f(z, \hat{y}) + B^\top \lambda, \quad E = \lambda^\top \sigma + \frac{\|\rho\|^2}{2\mu}, \quad c_z = \nabla_z f(z, \hat{y}) + A^\top \lambda.$$

(a) *For every feasible w , $g(w) - g(z) \geq c_z^\top(w - z) + \frac{\mu}{2}\|w - z\|^2 - E$. In particular, if $E \leq \mu/4$, then c_z satisfies (13).*

(b) *If $n \geq 1$ and $z \in [-R, R]^t$ for an integer $R \geq 1$, then $\hat{y}$ and λ with $E \leq \mu/4$ are computed in ordinary time polynomial in $L + \text{bits}(R) + \text{bits}(z)$.*

Proof. (a) Let $(w, v) \in P$, and put $\Delta_z = w - z$ and $\Delta_y = v - \hat{y}$. Joint strong convexity gives

$$f(w, v) \geq f(z, \hat{y}) + \nabla_z f^\top \Delta_z + \nabla_y f^\top \Delta_y + \frac{\mu}{2}(\|\Delta_z\|^2 + \|\Delta_y\|^2),$$

with derivatives at $(z, \hat{y})$. Feasibility of (w, v) gives $A\Delta_z + B\Delta_y \leq \sigma$, so $\lambda^\top \sigma - \lambda^\top A\Delta_z - \lambda^\top B\Delta_y \geq 0$. Subtracting this quantity from the right side yields

$$f(w, v) \geq f(z, \hat{y}) + c_z^\top \Delta_z + \frac{\mu}{2}\|\Delta_z\|^2 - \lambda^\top \sigma + \rho^\top \Delta_y + \frac{\mu}{2}\|\Delta_y\|^2,$$

and $\rho^\top \Delta_y + \frac{\mu}{2}\|\Delta_y\|^2 \geq -\|\rho\|^2/(2\mu)$. Minimize over feasible v and use $f(z, \hat{y}) \geq g(z)$. For integer $w \neq z$, $\frac{\mu}{2}\|w - z\|^2 - \frac{\mu}{4} \geq \frac{\mu}{4}\|w - z\|^2$.

(b) Write $h_z(y) = f(z, y)$ and let y_z be its minimizer on the fiber $F_z = \{y : By \leq d - Az\}$. Linear programming gives a rational $\bar{y} \in F_z$, and Lemma D.1(a), applied to h_z and $\bar{y}$, gives an integer Y with $\|y_z\| < Y$. With $m_0 = t + n$ and C_f as before, put

$$S = 2 + R + Y, \quad \beta_1 = 1 + 4m_0 C_f S^3, \quad \beta_2 = 1 + 12m_0 C_f S^2, \quad \delta = \min\left\{1, \frac{\mu}{16\beta_1}, \frac{\mu}{4\beta_2}\right\}.$$

Then $\|\nabla h_z(y_z)\| \leq \beta_1$, and $\|\nabla^2 h_z(y)\| \leq \beta_2$ when $\|y - y_z\| \leq 1$, because all coordinates involved are at most S in absolute value. Contract (E1), applied to h_z on $F_z \cap \{y : \|y\|_\infty \leq Y\}$, gives a rational $\hat{y} \in F_z$ with $h_z(\hat{y}) \leq h_z(y_z) + \min\{1, \mu\delta^2/2\}$, so $\|\hat{y} - y_z\| \leq \delta$ by Lemma D.1(b).

By Lemma 6.1(b), applied to ∇h_z on F_z , there is a real $\lambda_* \geq 0$, supported on the rows active at y_z , with $\nabla h_z(y_z) + B^\top \lambda_* = 0$. It need not be rational or unique, and the proof uses no bound on its norm. Complementary slackness and stationarity give the identity

$$\lambda_*^\top \sigma = \lambda_*^\top (d - Az - By_z) + (B^\top \lambda_*)^\top (y_z - \hat{y}) = \nabla h_z(y_z)^\top (\hat{y} - y_z),$$

whose value lies in $[0, \beta_1 \delta]$; it is nonnegative because $\hat{y}$ is feasible and y_z is optimal on F_z . Also $\|\nabla h_z(\hat{y}) + B^\top \lambda_*\| = \|\nabla h_z(\hat{y}) - \nabla h_z(y_z)\| \leq \beta_2 \delta$. Hence the convex quadratic function

$$\mathcal{E}(\lambda) = \sigma^\top \lambda + \frac{\|\nabla h_z(\hat{y}) + B^\top \lambda\|^2}{2\mu}$$

satisfies $\inf_{\lambda \geq 0} \mathcal{E} \leq \mathcal{E}(\lambda_*) \leq \beta_1 \delta + \beta_2^2 \delta^2 / (2\mu) \leq \frac{\mu}{16} + \frac{\mu}{32}$.

The infimum is attained. The set $\mathcal{K} = \{(B^\top \lambda, \sigma^\top \lambda) : \lambda \geq 0\}$ is a finitely generated cone, hence closed, and its second coordinate is nonnegative. In these coordinates the objective is $\tau + \|\nabla h_z(\hat{y}) + u\|^2 / (2\mu)$, whose sublevel set at the value of $\lambda = 0$ is nonempty and compact within $\mathcal{K}$. A minimizer $(u, \tau) \in \mathcal{K}$ lifts to some $\lambda \geq 0$. Contract (E3b) therefore returns an exact rational minimizer $\hat{\lambda} \geq 0$, and $E = \mathcal{E}(\hat{\lambda}) \leq 3\mu/32 < \mu/4$. All data of the two convex problems are rational of polynomial bit length, and the accuracy requested in (E1) has polynomially many bits; no bound on $\|\lambda_*\|$ enters. $\square$

Proof of Theorem 6.20. Use contract (E3e) to certify infeasibility or to find a rational feasible (z_0, y_0) . If $t = 0$, output the empty block. Otherwise apply Lemma D.1(a) to f on $\mathbb{R}^{t+n}$ and (z_0, y_0) to get R with every optimal block in $[-R, R]^t$; $\text{bits}(R) \leq F_1(t)L^{C_1}$. Only the integer block is truncated; the fibers keep all their constraints. Let $\mathcal{D} = \{z \in [-R, R]^t : \exists y, Az + By \leq d\}$, a compact convex set containing z_0 and every optimal block, on which g is μ -strongly convex by the argument of Lemma D.13.

Answer an integer query z as follows. If $z \notin [-R, R]^t$, return a violated box inequality. If the fiber of z is empty, linear programming returns a rational $\alpha \geq 0$ with $B^\top \alpha = 0$ and $\alpha^\top(d - Az) < 0$; return $\alpha^\top Ax \leq \alpha^\top d$, which is violated by z and valid on $\mathcal{D}$ because $\alpha^\top(Az' + By') \leq \alpha^\top d$ and $\alpha^\top By' = 0$. Otherwise record z , compute c_z by Lemma D.15 (or $c_z = \nabla f(z)$ if $n = 0$), stop if $c_z = 0$, and return $c_z^\top x \leq c_z^\top z - \mu/4$. These answers depend only on the instance and the query, and cost polynomial time in $L + \text{bits}(R) + \text{bits}(z)$.

The running-time argument (target $\emptyset$), the completeness argument (target $\{w\}$ for an omitted optimal block w), and the early stop are exactly as in the proof of Theorem 6.18: domain and Farkas cuts are valid on $\mathcal{D} \ni w$, and residual cuts at $z \neq w$ are valid at w by Lemma 6.17, because $g(w) \leq g(z)$. The total time is $F_0(t)L^{C_0}$ plus $2^{O(t \log(t+1))} \text{poly}(\text{bits}(R), L)$, which is $F(t)L^C$. For the parity bound, if (w_1, y_1) and (w_2, y_2) are optimal with $w_1 \neq w_2$ and $w_1 \equiv w_2 \pmod{2}$, their midpoint is feasible, has an integer block, and has a strictly smaller objective value. $\square$

Proof of Corollary 6.21. Let $\mathcal{Z}$ be the list of Theorem 6.20, of total length at most $Q_L = F_1(t)L^{C_1}$. For $z \in \mathcal{Z}$, the fiber instance $(f(z, \cdot), \{y : By \leq d - Az\}, \mu)$ has length at most $(Q_L + L)^{C_2}$ with absolute C_2 , because the degree is fixed. Its curvature is at least μ , since $\nabla_{yy}^2 f$ is a principal block of $\nabla^2 f$. Write $f_3 + f_4 = \theta(U_z z + U_y y)$ with $U = [U_z \ U_y]$ from Lemma 6.5. For fixed z , the homogeneous parts of degrees three and four in y of $\theta(U_z z + U_y y)$ are polynomials in $U_y y$, and $f_{\leq 2}(z, y)$ has degree at most two in y . Hence the nonlinear kernel of $f(z, \cdot)$ contains $\ker U_y$, and its nonlinear dimension is at most k . For $z, z' \in \mathcal{Z}$, the product instance with objective $f(z, y) + f(z', y')$, feasible set $F_z \times F_{z'}$, and observable $f(z, y) - f(z', y')$ has curvature at least μ and nonlinear kernel containing $\ker U_y \times \ker U_y$, hence nonlinear dimension at most $2k$. Theorem 6.6 compares $g(z)$ and $g(z')$ in time $F(2k)(Q_L + L)^{C_2 C}$, and compares $g(z)$ with τ using $h = f(z, y) - \tau$. At most $|\mathcal{Z}|^2 + |\mathcal{Z}|$ comparisons select every optimal block and decide the predicate, and Theorem 6.6(iii) represents the continuous part. The total time is $F'(k + t)L^{C'}$ with $F'(\kappa)$ the maximum of the parameter factors over $k + t = \kappa$. $\square$

Proof of Theorem 6.22. Compute the list $\mathcal{Z}$ of Theorem 6.20, remove duplicates, and sort it lexicographically. Each fiber has constraint matrix B , of rank r , and curvature at least μ . For $z, z' \in \mathcal{Z}$, the product instance of the previous proof has constraint matrix $\text{diag}(B, B)$, of rank $2r$, so Theorem 6.16, with $T = \nabla(f(z, y) + f(z', y'))$, decides $g(z) < g(z')$ in expected time $(2r + 1)^{O(r)}(Q_L + L)^{C_3}$. Scan the sorted list, keeping the first block, and replace the current block only by a block of strictly smaller value. The result z_* is the lexicographically smallest optimal block, because the list contains every optimal block and ties keep the earlier one. One more call of Theorem 6.16 on the fiber of z_* returns a support J with chart $(\bar{y}, Z)$ and the explicit strongly monotone cubic map U_J whose unique zero u_* gives $y_* = \bar{y} + Zu_*$; the same call with $h = f(z_*, y) - \tau$ decides $\min \bowtie \tau$. Comparing every listed block with z_* returns all optimal blocks.

The number of calls is at most $2|\mathcal{Z}| + 2$, a deterministic bound. Each call uses fresh random bits, and its expected running time, conditioned on everything computed before it, is at most $(2r + 1)^{O(r)}(Q_L + L)^{C_3}$, because this bound holds for every input of that length. Linearity of expectation bounds the total expected time by $F(t, r)L^C$. Every call returns a correct answer, so every output is correct. Different runs may return different charts, but all represent the same

point. $\square$

Proof of Corollary 6.23. Both machines first compute, deterministically, feasibility and the list $\mathcal{Z} = \{z_1, \dots, z_s\}$; infeasible instances are decided by the convention. They then guess sets $S_1, \dots, S_s$ of rows, one bit per row and fiber, and verify each S_i for the fiber of z_i by steps (1)–(3) of the proof of Theorem 6.2, with $T = \nabla_y f(z_i, \cdot)$. By that proof, exactly one guess passes for every fiber, namely its active set, and it yields a chart $(\bar{y}_i, Z_i)$ such that the fiber minimizer is $\bar{y}_i + Z_i u_i$ with u_i the unconstrained minimizer of $q_i(u) = f(z_i, \bar{y}_i + Z_i u)$, whose Hessian is at least μI . The predicate $\min \bowtie \tau$ is a Boolean combination of the predicates $q_i(u_i) \leq \tau$ and $q_i(u_i) < \tau$, each decided by Theorem 4.4 (or by rational arithmetic for a zero-dimensional chart). One machine accepts when the combination is true, the other when it is false. Each machine has at most one accepting computation, and the guesses and verifications take time polynomial in the length of the list. $\square$

D.8 A hard binary decision

Proof of Corollary 6.24. By Theorem 5.2, an integer straight-line program with output V is mapped in polynomial time to a rational quartic F on $\mathbb{R}^N$, an index j , $N + 1$ rational quadratic square factors whose sum is F , and a positive definite rational full Hessian Gram Q , such that $F \geq 0$ has a unique zero $p \in \mathbb{Q}^N \cap [-1, 1]^N$, $\nabla^2 F \succeq \frac{3}{2}I$, $p_j \neq 0$, and $p_j > 0$ exactly when $V > 0$.

In (14), the integer z is 0 or 1, and then $H = F(x) + \frac{1}{4} \geq \frac{1}{4}$, with equality only at $x = p$. The constraint $z - 1 \leq x_j \leq z$ allows $x_j \in [-1, 0]$ for $z = 0$ and $x_j \in [0, 1]$ for $z = 1$. Both fibers are nonempty ($x = 0$), and since $0 < |p_j| \leq 1$, exactly the fiber $z = \mathbf{1}[p_j > 0]$ contains p . Hence the optimal value is $1/4$, the unique optimal solution is $(p, \mathbf{1}[p_j > 0])$, and $z_* = 1$ exactly when $V > 0$. The Hessian of H is $\text{diag}(\nabla^2 F, 2) \succeq \frac{3}{2}I$. The constraints $0 \leq z \leq 1$ have zero continuous part, and $z - 1 \leq x_j \leq z$ have continuous normals $\pm e_j$, so the continuous constraint rank is one. The other coordinates of x are unconstrained; the optimizer is bounded, although the feasible set is not.

For the full Gram, let $\varrho = \det(Q)/\text{tr}(Q)^{d_Q-1}$, where $d_Q = N + N^2$ is the size of Q , so $Q \succeq \varrho I$ (Lemma 2.7), and let $\epsilon = \min\{\varrho/4, 1/(100N)\}$. Put

$$\hat{H} = H + \epsilon z(z - 1)\|x\|^2 + \frac{1}{4}z^2(z - 1)^2.$$

Both added terms vanish at $z \in \{0, 1\}$, so every feasible objective value, the optimum, the optimizer, and the constraints are unchanged. Write $\vartheta = z - \frac{1}{2}$, let (v, s) be a direction in (x, z) , and let $w = (v, x \otimes v)$. Direct differentiation of $\hat{H} = F + \vartheta^2 + \epsilon(\vartheta^2 - \frac{1}{4})\|x\|^2 + \frac{1}{4}(\vartheta^2 - \frac{1}{4})^2$ gives

$$D^2 \hat{H}[(v, s)]^2 = w^\top Q w - \frac{\epsilon}{2}\|v\|^2 + \frac{7}{4}s^2 + 2\epsilon\|\vartheta v\|^2 + 2\epsilon\|xs\|^2 + 8\epsilon(x^\top v)(\vartheta s) + 3(\vartheta s)^2.$$

In the basis $(w, s, \vartheta v, xs, \vartheta s)$, which contains every direction coordinate and every product of a coordinate of (x, z) with a direction coordinate exactly once after the invertible rational change $\vartheta v = zv - \frac{1}{2}v$, $\vartheta s = zs - \frac{1}{2}s$, this is the quadratic form of the block matrix with diagonal blocks $Q - \frac{\epsilon}{2}E$, $\frac{7}{4}$, $2\epsilon I$, $2\epsilon I$, 3 , and one off-diagonal block $4\epsilon d_0$ between w and ϑs . Here E is the identity on the v -part of w and zero elsewhere, and d_0 is the vector with $d_0^\top w = x^\top v$, so $\|d_0\|^2 = N$. Since $Q - \frac{\epsilon}{2}E \succeq \frac{\varrho}{2}I$, the Schur complement of the last entry is at least $3 - 32\epsilon^2 N/\varrho \geq 3 - 8\epsilon N \geq 3 - \frac{2}{25} > 0$. Hence the matrix is positive definite, and so is its rational congruence in the original basis. All entries have polynomial bit length. $\square$

The base objective H has no positive definite full joint Hessian Gram: at $x = 0$ in the direction s , its Hessian form is the constant $2s^2$, while a positive definite Gram on a basis containing zs would force growth like $z^2 s^2$. The correction restores the full Gram at the cost of the specific modulus $3/2$ and of the supplied objective square factors. The value gap between the two fibers is at least

$\frac{3}{4}p_j^2$, by strong convexity and the distance from p to the wrong half-space, and at most $\frac{\Lambda_F}{2}p_j^2$ with $\Lambda_F = \max_{[-1,1]^N} \|\nabla^2 F\|$, by evaluating at $p - p_j e_j$.

E Proofs for Section 7

This appendix proves every result of Section 7. Norms of vectors are Euclidean and norms of matrices are operator norms unless the Frobenius norm $\|\cdot\|_F$ is indicated. For a rational number we use its reduced fraction. A *quadratic* is a polynomial of degree at most two. Imported results are stated with their exact contracts where they are used; the algebraic-geometric ones are collected in Appendix E.9.

E.1 Canonical Hessian Grams

Proof of Lemma 7.1. (a) First let $f = g^2$ with $g(x) = g_0 + b^\top x + x^\top T x$. Then $\nabla g(x) = b + 2Tx$ and $\nabla^2 g = 2T$, so

$$v^\top \nabla^2(g^2)(x)v = 2(v^\top \nabla g(x))^2 + 2g(x) v^\top \nabla^2 g v = 2(b^\top v + 2x^\top T v)^2 + 4(g_0 + b^\top x + x^\top T x)v^\top T v.$$

On the other hand, with $w = w(x, v)$, the three blocks of $\Gamma(g)$ contribute

$$\begin{aligned} v^\top (2bb^\top + 4g_0 T)v &= 2(b^\top v)^2 + 4g_0 v^\top T v, \\ 2v^\top \Delta(b, T)(x \otimes v) &= 2 \sum_{i,k,j} v_i (2b_k T_{ij} + 4b_i T_{kj}) x_k v_j \\ &= 4(b^\top x)(v^\top T v) + 8(b^\top v)(x^\top T v), \\ (x \otimes v)^\top \Xi(T)(x \otimes v) &= 8 \left(\sum_{k,j} x_k v_j T_{kj} \right)^2 + 4 \sum_{k,j,k',j'} x_k v_j T_{kk'} T_{jj'} x_{k'} v_{j'} \\ &= 8(x^\top T v)^2 + 4(x^\top T x)(v^\top T v). \end{aligned}$$

Their sum equals the previous display. Thus $v^\top \nabla^2(g^2)(x)v = w^\top \Gamma(g)w$ for every quadratic g . Both sides are quadratic functions of the coefficient vector of g . Polarizing, $v^\top \nabla^2(g_k g_l)(x)v = w^\top \hat{\Gamma}(g_k, g_l)w$, because $2g_k g_l = (g_k + g_l)^2 - g_k^2 - g_l^2$. Summing with the weights S_{kl} proves (a).

(b) Again it suffices to treat one quadratic: both sides of the claimed identity are quadratic in the coefficients of g , the recentring $g \mapsto g(\cdot + c)$ is linear in those coefficients, and the polarized and weighted identities follow as in (a). Put $A_c = c \otimes I_n$, so that $(A_c v)_{(k,j)} = c_k v_j$ and $(A_c^\top y)_i = \sum_k c_k y_{(k,i)}$. For every symmetric T and every vector b , direct contraction of indices gives

$$A_c^\top \Xi(T) = \Delta(2Tc, T), \quad (48)$$

$$A_c^\top \Xi(T) A_c = 8(Tc)(Tc)^\top + 4(c^\top T c) T, \quad (49)$$

$$\Delta(b, T) A_c + A_c^\top \Delta(b, T)^\top = 4(b^\top c) T + 4b(Tc)^\top + 4(Tc)b^\top. \quad (50)$$

For (48), the $(i, (k, j))$ entry of the left side is $\sum_{k'} c_{k'} \Xi_{(k', i), (k, j)} = \sum_{k'} c_{k'} (8T_{k' i} T_{kj} + 4T_{k' k} T_{ij}) = 8(Tc)_i T_{kj} + 4(Tc)_k T_{ij}$, which is the corresponding entry of $\Delta(2Tc, T)$. Multiplying this entry by $c_k \delta_{ji'}$ and summing over (k, j) gives $8(Tc)_i (Tc)_{i'} + 4(c^\top T c) T_{ii'}$, which is (49). Finally $(\Delta(b, T) A_c)_{ii'} = \sum_k (2b_k T_{ii'} + 4b_i T_{ki'}) c_k = 2(b^\top c) T_{ii'} + 4b_i (Tc)_{i'}$; adding the transpose gives (50).

Let $h = g(\cdot + c)$. Then $h(x) = h_0 + b'^\top x + x^\top T x$ with $b' = b + 2Tc$ and $h_0 = g(c) = g_0 + b^\top c + c^\top T c$. Write $\Gamma(h) = \begin{pmatrix} C' & \Delta' \\ \Delta'^\top & \Xi \end{pmatrix}$ with $C' = 2b'b'^\top + 4h_0 T$, $\Delta' = \Delta(b', T)$, and $\Xi = \Xi(T)$. Block multiplication gives

$$\Psi_c^\top \Gamma(h) \Psi_c = \begin{pmatrix} C' - \Delta' A_c - A_c^\top \Delta'^\top + A_c^\top \Xi A_c & \Delta' - A_c^\top \Xi \\ (\Delta' - A_c^\top \Xi)^\top & \Xi \end{pmatrix}.$$

By (48) and linearity of Δ in its first argument, $\Delta' - A_c^\top \Xi = \Delta(b + 2Tc, T) - \Delta(2Tc, T) = \Delta(b, T)$. By (49) and (50) with b' in place of b , the upper left block equals

$$2b'b'^\top - 4b'(Tc)^\top - 4(Tc)b'^\top + 8(Tc)(Tc)^\top + 4(h_0 - b'^\top c + c^\top Tc)T.$$

Substituting $b' = b + 2Tc$, the first four terms equal $2(b' - 2Tc)(b' - 2Tc)^\top = 2bb^\top$, and $h_0 - b'^\top c + c^\top Tc = g_0 + b^\top c + c^\top Tc - b^\top c - 2c^\top Tc + c^\top Tc = g_0$. Hence $\Psi_c^\top \Gamma(h) \Psi_c = \Gamma(g)$.

(c) The inverse of Ψ_c is Ψ_{-c} , and $\|\Psi_{-c}\| \leq 1 + \|c \otimes I_n\| = 1 + \|c\|$, since $\|c \otimes v\| = \|c\| \|v\|$. If $\Gamma_S(g(\cdot + c)) \succeq \lambda I$ then, by (b), $y^\top \Gamma_S(g)y \geq \lambda \|\Psi_c y\|^2 \geq \lambda \|y\|^2 / \|\Psi_c^{-1}\|^2$ for every vector y . $\square$

The following elementary bounds are used repeatedly.

Lemma E.1 (Block bounds). *For a vector $b \in \mathbb{R}^n$ and symmetric $T \in \mathbb{R}^{n \times n}$, $\|\Delta(b, T)\| \leq \|\Delta(b, T)\|_F \leq 6 \|b\| \|T\|_F \leq 6\sqrt{n} \|b\| \|T\|$ and $\Xi(T) \succeq -4 \|T\|^2 I$. If $T \succeq \mu I$ with $\mu \geq 0$, then $\Xi(T) \succeq 4\mu^2 I$; if $T \succeq 0$, then $\Xi(T) \succeq 0$.*

Proof. The two summands of $\Delta(b, T)$ have squared Frobenius norms $\sum_{i,k,j} 4b_k^2 T_{ij}^2 = 4 \|b\|^2 \|T\|_F^2$ and $16 \|b\|^2 \|T\|_F^2$, and $\|T\|_F \leq \sqrt{n} \|T\|$. The term $8 \text{vec}(T) \text{vec}(T)^\top$ is positive semidefinite, and the eigenvalues of $T \otimes T$ are the products $\lambda_i \lambda_j$ of eigenvalues of T . $\square$

E.2 The bivariate quartic

Proof of Proposition 7.2. Put $r = 2^{1/3}$, $p = (r, r^2)$, and $q_3 = (x - y)^2 - 2x - y + 4$. Using $r^3 = 2$ and $r^4 = 2r$ one checks $q_1(p) = q_2(p) = q_3(p) = 0$. Expanding the right side coefficient by coefficient proves the identities

$$A = 7599q_1 + 2937q_2 + 5000q_3, \quad (2r - 1)q_1 + (r^2 - 1)q_2 + q_3 = (x - r, y - r^2) \begin{pmatrix} 2r & -1 \\ -1 & r^2 \end{pmatrix} (x - r, y - r^2)^\top; \quad (51)$$

in the second, the constant terms agree because $r^6 = 4$, and the linear terms because $r^4 = 2r$. In particular $A(p) = 0$, so $F(p) = 0$.

Zero set (b). Since F is a sum of squares, $F \geq 0$ and $F^{-1}(0) = \{F \leq 0\}$ is the common zero set of A, q_1, q_2 . Now $q_1 = 0$ gives $y = x^2$, and then $q_2 = x(x^3 - 2)$; the choice $x = 0$ forces $y = 0$ and $A(0, 0) = 20000 \neq 0$. Hence the only common zero is p . The polynomial $T^3 - 2$ is irreducible by Eisenstein's criterion at 2, so r is irrational and $\{F \leq 0\}$ contains no rational point. All coefficients of F are integers; the largest in absolute value is the coefficient 755983876 of x^2 , which is below 2^{30} .

Local data. Put $G = A/5000$, $\varepsilon = 1/2500$, and $f = F/5000^2 = G^2 + \varepsilon(q_1^2 + q_2^2)$. For $u \in \mathbb{R}^2$,

$$q_1(p + u) = j_1^\top u + u_1^2, \quad q_2(p + u) = j_2^\top u + u_2^2, \quad G(p + u) = \ell^\top u + u^\top H u,$$

where $j_1 = (2r, -1)$, $j_2 = (-2, 2r^2)$, $H = \begin{pmatrix} 12599/5000 & -1 \\ -1 & 7937/5000 \end{pmatrix}$, and, by (51) (whose right side has zero gradient at p), $\ell = -d_1 j_1 - d_2 j_2$ with $d_1 = 2r - 1 - \frac{7599}{5000}$ and $d_2 = r^2 - 1 - \frac{2937}{5000}$. We claim

$$H \succeq \frac{2937}{5000} I \succ \frac{1}{2} I, \quad \|H\| < 5, \quad \|j_1\|, \|j_2\| < 4, \quad J^\top J \succeq \frac{36}{23} I \succ I, \quad \|\ell\| < \frac{1}{5000}, \quad (52)$$

where J has rows $j_1^\top, j_2^\top$. Gershgorin's theorem gives the first bound, and $\text{tr } H < 5$ the second. Cubing shows $1.259921 < r < 1.259922$; hence $r < 13/10$, $\|j_1\|^2 = 4r^2 + 1 < 8$, and $\|j_2\|^2 = 4 + 8r < 15$. Also $\det J = 4r^3 - 2 = 6$ and $\text{tr}(J^\top J) = 5 + 4r^2 + 8r < 23$, so the smaller eigenvalue of $J^\top J$

exceeds $36/23$. Finally the interval for r gives $0 < d_1 < 4.4 \cdot 10^{-5}$ and $0 < d_2 < 3.5 \cdot 10^{-6}$, so $\|\ell\| \leq 4(d_1 + d_2) < 4 \cdot 4.75 \cdot 10^{-5} < 1/5000$.

Global Hessian bound (a). Write $f(p+u) = f_2 + f_3 + f_4$ with homogeneous parts

$$f_2 = (\ell^\top u)^2 + \varepsilon \|Ju\|^2, \quad f_3 = 2(\ell^\top u)h(u) + 2\varepsilon((j_1^\top u)u_1^2 + (j_2^\top u)u_2^2), \quad f_4 = h(u)^2 + \varepsilon(u_1^4 + u_2^4),$$

where $h(u) = u^\top Hu$. Then $\nabla^2 f_2 \succeq 2\varepsilon J^\top J \succeq 2\varepsilon I$. Next, $\nabla^2(h^2) = 8(Hu)(Hu)^\top + 4h(u)H \succeq 4 \cdot \frac{1}{2} \|u\|^2 \cdot \frac{1}{2} I = \|u\|^2 I$, and u_1^4, u_2^4 are convex, so $\nabla^2 f_4 \succeq \|u\|^2 I$. For a vector c and a symmetric S ,

$$\nabla^2[2(c^\top u)(u^\top Su)] = 4[(c^\top u)S + c(Su)^\top + (Su)c^\top],$$

whose norm is at most $12\|c\|\|S\|\|u\|$. Applying this to the three terms of f_3 , with $\|e_i e_i^\top\| = 1$ and (52), gives $\|\nabla^2 f_3\| \leq (60/5000 + 96\varepsilon)\|u\| = (63/1250)\|u\|$. With $s = \|u\|$,

$$\nabla^2 f(p+u) \succeq \left(s^2 - \frac{63}{1250}s + \frac{1}{1250}\right)I = \left(s - \frac{63}{2500}\right)^2 I + \frac{1031}{6250000}I \succeq \frac{1031}{6250000}I.$$

Multiplying by 5000^2 gives $\nabla^2 F \succeq 4124I$ on $\mathbb{R}^2$.

The canonical Gram (c). Order the coordinates of $w(x, v)$ as $(v_1, v_2, x_1 v_1, x_1 v_2, x_2 v_1, x_2 v_2)$. Applying (15) to the three factors A , $100q_1 = 100x^2 - 100y$, and $100q_2 = 100y^2 - 200x$, and adding, gives the integer matrix

$$\Gamma_F = \begin{pmatrix} 1511967752 & -6948 & -1199979156 & 476220000 & -9602 & -378057752 \\ -6948 & 952449602 & -476239204 & -43876 & 377970000 & -599989578 \\ -1199979156 & -476239204 & 1904937612 & -755940000 & -755940000 & 899986104 \\ 476220000 & -43876 & -755940000 & 599993052 & 300000000 & -476220000 \\ -9602 & 377970000 & -755940000 & 300000000 & 599993052 & -476220000 \\ -378057752 & -599989578 & 899986104 & -476220000 & -476220000 & 756071628 \end{pmatrix}. \quad (53)$$

For instance, its first diagonal entry is $2 \cdot 15874^2 + 4 \cdot 20000 \cdot 12599 + 2 \cdot 200^2$. By Lemma 7.1(a), $v^\top \nabla^2 F(x)v = w(x, v)^\top \Gamma_F w(x, v)$.

To prove $\Gamma_F \succ 0$, recentre the three factors at p . Their recentred canonical Gram is $5000^2 M$ with $M = \begin{pmatrix} C & \Delta \\ \Delta^\top & \Xi \end{pmatrix}$, where, because the recentred factors vanish at the origin,

$$C = 2\ell\ell^\top + 2\varepsilon J^\top J, \quad \Delta = \Delta(\ell, H) + \varepsilon(\Delta(j_1, e_1 e_1^\top) + \Delta(j_2, e_2 e_2^\top)), \quad \Xi = \Xi(H) + \varepsilon(\Xi(e_1 e_1^\top) + \Xi(e_2 e_2^\top)).$$

By (52) and Lemma E.1, $C \succeq \frac{72}{23}\varepsilon I \succeq 0.00125 I$ and $\Xi \succeq 4(2937/5000)^2 I \succeq 1.38 I$. Since $\|H\|_F^2 < 11$, $\|\Delta\| \leq 6(\|\ell\| \|H\|_F + \varepsilon(\|j_1\| + \|j_2\|)) < 6(3.32/5000 + 8/2500) < 0.0232$. The Schur complement therefore satisfies $C - \Delta\Xi^{-1}\Delta^\top \succeq (0.00125 - 0.0232^2/1.38)I \succ 0$, so $M \succ 0$. Lemma 7.1(b) gives $\Gamma_F = 5000^2 \Psi_p^\top M \Psi_p \succ 0$.

A direct rational certificate (d). For $x = (x_1, x_2)$ and $v = (a, b)$ put $X = 100x_1 - 126$, $Y = 100x_2 - \frac{1587}{10}$, and

$$\omega = (2a - 2b, 2b, 2Xa - Xb - Ya + Yb, 2Xb + Ya - Yb, 2Ya - Yb, 2Yb)^\top.$$

Let N_F be the symmetric integer matrix

$$N_F = \begin{pmatrix} 812297303152 & 102427849072 & 22618204800 & -13093175680 & 5654551200 & -130053040 \\ 102427849072 & 262472489120 & 22618204800 & -5643994880 & 1929960800 & -3911932400 \\ 22618204800 & 22618204800 & 761975044800 & 78611522400 & 39305761200 & -2034439000 \\ -13093175680 & -5643994880 & 78611522400 & 128114982000 & -15939729800 & -13708914300 \\ 5654551200 & 1929960800 & 39305761200 & -15939729800 & 112025966300 & 13144848050 \\ -130053040 & -3911932400 & -2034439000 & -13708914300 & 13144848050 & 120298532375 \end{pmatrix}.$$

Expanding both sides, using $v^\top \nabla^2 F(x) v = w^\top \Gamma_F w$ and (53), proves the polynomial identity

$$16 \cdot 10^6 (v^\top \nabla^2 F(x) v - 4096 \|v\|^2) = \omega^\top N_F \omega. \quad (54)$$

Every row of N_F is strictly diagonally dominant; the margins $\delta_i = (N_F)_{ii} - \sum_{j \neq i} |(N_F)_{ij}|$ are

$$\begin{aligned} (\delta_0, \delta_1, \delta_2) &= (668373469360, 125940547168, 596786912600), \\ (\delta_3, \delta_4, \delta_5) &= (1117644940, 36051115250, 87368345585). \end{aligned}$$

Hence $\omega^\top N_F \omega = \sum_i \delta_i \omega_i^2 + \sum_{i < j} |(N_F)_{ij}| (\omega_i + \text{sign}((N_F)_{ij}) \omega_j)^2$, a positive integer combination of 21 squares of polynomials with rational coefficients. This proves $\nabla^2 F \succeq 4096I$ by a rational certificate; equivalently, $F - 2048 \|x\|^2$ is SOS-convex with an explicit rational Hessian sum of squares. $\square$

E.3 One real conjugate

We use two standard facts. First, the field $\mathbb{R}_{\text{alg}}$ of real algebraic numbers is real closed, and by the Tarski–Seidenberg transfer principle a system of polynomial equations and inequalities with rational coefficients that has a solution in $\mathbb{R}^n$ has a solution in $\mathbb{R}_{\text{alg}}^n$ [15, Theorem 2.80 and Corollary 2.83]. Second, a number field K of degree $[K : \mathbb{Q}]$ has r_1 real embeddings and $2r_2$ nonreal embeddings occurring in conjugate pairs, with $[K : \mathbb{Q}] = r_1 + 2r_2$.

Proof of Lemma 7.3. Algebraicity. In both cases the singleton is the solution set of a finite system of rational polynomial equations or weak inequalities. The system has a real solution, hence a solution in $\mathbb{R}_{\text{alg}}^n$, which is also a real solution and therefore equals p .

Reduction to equalities. In the zero-set case let E be the given rational polynomials. In the convex case let $\mathcal{A} = \{i : f_i(p) = 0\}$ and $E = \{f_i : i \in \mathcal{A}\}$. We claim that $\{x : f_i(x) \leq 0, i \in \mathcal{A}\} = \{p\}$. Suppose $z \neq p$ satisfies these inequalities. For $0 \leq s \leq 1$ and $i \in \mathcal{A}$, convexity gives $f_i(p + s(z - p)) \leq (1 - s)f_i(p) + sf_i(z) \leq 0$. For $i \notin \mathcal{A}$ we have $f_i(p) < 0$, so by continuity and finiteness there is $s_0 > 0$ with $f_i(p + s(z - p)) < 0$ for all $i \notin \mathcal{A}$ and $0 \leq s \leq s_0$. Then $p + s_0(z - p) \neq p$ satisfies all original inequalities, a contradiction.

One real embedding. Put $K = \mathbb{Q}(p)$ and let $\sigma : K \rightarrow \mathbb{R}$ be a field embedding. Apply σ coordinatewise to p . For every $e \in E$, rational coefficients give $e(\sigma(p)) = \sigma(e(p)) = 0$. In the zero-set case $\sigma(p)$ is therefore a real common zero, so $\sigma(p) = p$. In the convex case $\sigma(p)$ satisfies $f_i(\sigma(p)) = 0 \leq 0$ for $i \in \mathcal{A}$, so $\sigma(p) = p$ by the claim. As K is generated by the coordinates of p , σ is the inclusion. Hence $r_1 = 1$, and $[K : \mathbb{Q}] = 1 + 2r_2$ is odd.

Coordinates. Let $K_j = \mathbb{Q}(p_j) \subseteq K$ and let $\tau : K_j \rightarrow \mathbb{R}$ be a real embedding. Since $[K : \mathbb{Q}]$ is odd, so is $[K : K_j]$. Choose a primitive element ϑ of K/K_j with minimal polynomial $m_\vartheta \in K_j[T]$ of odd degree $[K : K_j]$. The real polynomial $\tau(m_\vartheta)$ has odd degree, hence a real root ϑ' , and $\vartheta \mapsto \vartheta'$ extends τ to a real embedding of $K = K_j(\vartheta)$. Distinct real embeddings of K_j would give distinct real embeddings of K . So K_j has exactly one real embedding, which means that the minimal polynomial of p_j has exactly one real root. $\square$

Proof of Theorem 7.4. (v) $\Rightarrow$ (ii): the real zero set of f is the given singleton. (v) $\Rightarrow$ (iii): f is a sum of squares, so $f \geq 0$ and $\{f \leq 0\} = f^{-1}(0)$; take the single convex inequality $f \leq 0$. (v) $\Rightarrow$ (iv): the unique zero of the nonnegative strongly convex f is its unique minimizer.

(ii) $\Rightarrow$ (i) and (iii) $\Rightarrow$ (i) are Lemma 7.3. For (iv) $\Rightarrow$ (i), let p be the unique minimizer of a rational convex polynomial h . For a convex differentiable function the critical points are exactly the global minimizers, so the real zero set of the rational polynomial $\sum_i (\partial_i h)^2$ is $\{p\}$, and Lemma 7.3 applies.

(i) $\Rightarrow$ (v): if α is rational, $f = (x - \alpha)^2 + (x - \alpha)^4$ is a sum of two rational squares with $f'' = 2 + 12(x - \alpha)^2 \geq 2$. Writing f as $(x - \alpha)^2 + (x^2 - 2\alpha x + \alpha^2)^2$, its canonical Hessian Gram on $w = (v, xv)$ is $\begin{pmatrix} 2+12\alpha^2 & -12\alpha \\ -12\alpha & 12 \end{pmatrix}$, which has determinant 24 and is positive definite. If α is irrational, its minimal polynomial has degree $d > 1$ and exactly one real root, so the nonreal roots pair up and $d \geq 3$ is odd. Theorem 7.7 with $n = (d + 1)/2$ gives (v), with $\nabla^2 f \succeq \frac{3}{2}I \succeq I$ and $n + 1$ rational squares. Its algorithm accepts any rational multiple of the minimal polynomial, because the first step normalizes the input to a primitive integer polynomial. $\square$

E.4 The generic realization lemma

Proof of Lemma 7.6. Put $\Phi = g^2 + \varepsilon \sum_j r_j^2 = \varepsilon \nu^2 f$. The representation of f uses the $m + 1$ rational squares $g/(t\nu)$ and r_j/ν , and every factor vanishes at p .

Centred Gram. Let M be the canonical Hessian Gram of the recentred representation $\Phi(p + \cdot) = g(p + \cdot)^2 + \varepsilon \sum_j r_j(p + \cdot)^2$. The recentred factors have no constant terms, so by (15)

$$M = \begin{pmatrix} C & \Delta \\ \Delta^\top & \Xi \end{pmatrix}, \quad C = 2\ell\ell^\top + 2\varepsilon \sum_j b_j b_j^\top, \\ \Delta = \Delta(\ell, H) + \varepsilon \sum_j \Delta(b_j, T_j), \quad \Xi = \Xi(H) + \varepsilon \sum_j \Xi(T_j).$$

By (16) and Lemma E.1, $C \succeq 2\varepsilon\nu^2 I$, $\Xi \succeq (4\mu^2 - 4\varepsilon m)I \succeq 2\mu^2 I$ (using $\varepsilon \leq \mu^2/(2m)$), and

$$\|\Delta\| \leq 6\sqrt{n}(\|\ell\| \Lambda + \varepsilon m\beta) \leq 6\sqrt{n} \varepsilon(\Lambda + m\beta) =: B_\Delta.$$

The individual matrices $\Xi(T_j)$ may be indefinite; the bound $\Xi(T_j) \succeq -4I$ is what the estimate uses. The Schur complement satisfies

$$S := C - \Delta\Xi^{-1}\Delta^\top \succeq \left(2\varepsilon\nu^2 - \frac{B_\Delta^2}{2\mu^2}\right)I = \left(2\varepsilon\nu^2 - \frac{18n\varepsilon^2(\Lambda + m\beta)^2}{\mu^2}\right)I \succeq \frac{3}{2}\varepsilon\nu^2 I,$$

where the last step uses $\varepsilon \leq \nu^2\mu^2/(36n(\Lambda + m\beta)^2)$. For $y = (y_1, y_2)$ conformal with the blocks, put $y'_2 = y_2 + \Xi^{-1}\Delta^\top y_1$. Completing the square,

$$y^\top M y = y_1^\top S y_1 + y_2'^\top \Xi y_2'. \quad (55)$$

(b). Lemma 7.1(a), applied to the recentred representation, gives

$$v^\top \nabla^2 \Phi(p + u) v = w(u, v)^\top M w(u, v) \quad (u, v \in \mathbb{R}^n).$$

The first block of $w(u, v)$ is v , so (55) gives $v^\top \nabla^2 \Phi(p + u) v \geq \frac{3}{2}\varepsilon\nu^2 \|v\|^2$ for all u, v . Dividing by $\varepsilon\nu^2$ gives $\nabla^2 f \succeq \frac{3}{2}I$. A nonnegative strongly convex function vanishing at p has p as its unique zero and unique minimizer.

(a). The quartic part of Φ is $(x^\top H x)^2 + \varepsilon \sum_j (x^\top T_j x)^2 \geq \mu^2 \|x\|^4$, so f has degree exactly four and positive definite quartic form.

(c). With $\theta' = \|\Xi^{-1}\Delta^\top\| \leq B_\Delta/(2\mu^2) = \theta$, we have $\|y\|^2 \leq ((1 + \theta)\|y_1\| + \|y'_2\|)^2 \leq ((1 + \theta)^2 + 1)(\|y_1\|^2 + \|y'_2\|^2)$. Hence (55) yields $M \succeq \min\{\frac{3}{2}\varepsilon\nu^2, 2\mu^2\}((1 + \theta)^2 + 1)^{-1}I$. The canonical Gram of the representation of Φ is $\varepsilon\nu^2\Gamma_f$; by Lemma 7.1(b) and (c) with $c = p$, $\varepsilon\nu^2\Gamma_f = \Psi_p^\top M \Psi_p \succeq \lambda_{\min}(M)(1 + \|p\|)^{-2}I$. Dividing by $\varepsilon\nu^2$ gives $\Gamma_f \succeq \gamma I$. By Lemma 7.1(a), Γ_f is a full Hessian Gram of f , and it is rational because it is computed by (15) from the rational factors. Each of the $m + 1$ matrices $\Gamma(\cdot)$ has $(n + n^2)^2$ entries, each a fixed polynomial of degree two in the coefficients. $\square$

Remark E.2 (Curvature without the factor n). Part (b) of Lemma 7.6 remains true when the last bound in (17) is replaced by $\varepsilon \leq \nu^2 \mu^2 / (36(\Lambda + m\beta)^2)$, the other conditions being kept. In the proof above the factor n enters only through the operator-norm bound $\|\Delta(b, T)\| \leq 6\sqrt{n} \|b\| \|T\|$, while part (b) needs the Gram only at vectors of the form $w(u, v)$. A direct estimate at these vectors avoids it. For a quadratic $q(p + u) = c^\top u + u^\top S u$,

$$v^\top \nabla^2(q^2)(p + u) v = 2(c^\top v + 2u^\top S v)^2 + 4(c^\top u + u^\top S u) v^\top S v.$$

Fix u, v and put $s = \|u\|$. The terms of this expression of degree zero in u are $2(c^\top v)^2$; those of degree one, $8(c^\top v)(u^\top S v) + 4(c^\top u)(v^\top S v)$, have absolute value at most $12\|c\| \|S\| s \|v\|^2$; and those of degree two are $8(u^\top S v)^2 + 4(u^\top S u)(v^\top S v)$. Apply this to g ($c = \ell$, $S = H$) and to each r_j ($c = b_j$, $S = T_j$), and add with the weights of $\Phi = g^2 + \varepsilon \sum_j r_j^2$. By (16), the terms of degree zero sum to at least $2\varepsilon \nu^2 \|v\|^2$. The terms of degree two are at least $4\mu^2 s^2 \|v\|^2$ for g and at least $-4s^2 \|v\|^2$ for each r_j , so their weighted sum is at least $(4\mu^2 - 4\varepsilon m)s^2 \|v\|^2 \geq 2\mu^2 s^2 \|v\|^2$. The terms of degree one have weighted sum of absolute value at most $12(\|\ell\| \Lambda + \varepsilon m\beta)s \|v\|^2 \leq 12\varepsilon(\Lambda + m\beta)s \|v\|^2$. Hence

$$v^\top \nabla^2 \Phi(p + u) v \geq (2\varepsilon \nu^2 - 12\varepsilon(\Lambda + m\beta)s + 2\mu^2 s^2) \|v\|^2.$$

The minimum of the bracket over $s \geq 0$ is $2\varepsilon \nu^2 - 18\varepsilon^2(\Lambda + m\beta)^2/\mu^2$, which is at least $\frac{3}{2}\varepsilon \nu^2$ under the stated bound on ε . Dividing by $\varepsilon \nu^2$ gives $\nabla^2 f \succeq \frac{3}{2}I$. The proof of part (c) keeps the factor n , because its matrix inequality concerns all vectors of $\mathbb{R}^{n+n^2}$, not only those of the form $w(u, v)$.

A rational positive semidefinite Gram matrix also yields an explicit rational sum of squares of the Hessian biform. Rational symmetric elimination writes $\Gamma_f = \sum_k c_k \lambda_k \lambda_k^\top$ with rational vectors λ_k and rational $c_k > 0$. A positive rational u/v , with u, v positive integers, equals uv/v^2 ; writing $uv = \sum_h \epsilon_h 2^h$ in binary and using $2^{2h}/v^2 = (2^h/v)^2$ and $2^{2h+1}/v^2 = 2(2^h/v)^2$ expresses it as a sum of at most $2\lceil \log_2(uv + 1) \rceil$ rational squares, without factoring integers. This conversion is not needed for any result of Section 7.

E.5 Power coordinates

The algorithm uses one imported numerical subroutine.

Lemma E.3 (Certified root approximation; imported). *There is a deterministic algorithm that, given a square-free $P \in \mathbb{Z}[T]$ of degree d with coefficients of absolute value at most 2^τ and an integer $N_0 \geq 1$, outputs Gaussian dyadic numbers $z_1, \dots, z_d$ and the guarantee that the complex roots of P can be enumerated as $\beta_1, \dots, \beta_d$ with $|z_i - \beta_i| \leq 2^{-N_0}$ for every i . Its bit complexity is polynomial in d , τ , and N_0 .*

This is a classical result; one primary source is [87, Theorem 5], which isolates all complex roots and refines the isolating disks to any requested radius in $\tilde{O}(d^3 + d^2\tau + dN_0)$ bit operations; see also [97, 108]. Only polynomiality is used. An uncertified numerical root finder cannot replace this lemma.

Proof of Theorem 7.7. Normalization and root bounds. Clearing denominators and dividing by the content and the sign of the leading coefficient gives a primitive $P_\mathbb{Z} = \sum_{i=0}^d a_i T^i \in \mathbb{Z}[T]$ with $a_d > 0$ and $|a_i| \leq 2^\tau$, $\tau \geq 1$, in polynomial time; τ is polynomial in L . Let $P_1 = P_\mathbb{Z}/a_d = \sum_i c_i T^i$ be the monic minimal polynomial of α , so $|c_i| \leq 2^\tau$. Put

$$R = 2^{\tau+1}, \quad \sigma = 2^{-\tau(d-1)}(2R)^{-d(d-1)/2}.$$

Every complex root β satisfies $|\beta| \leq 1 + \max_i |c_i| < R$. The discriminant of $P_{\mathbb{Z}}$ is a nonzero integer, since an irreducible polynomial over $\mathbb{Q}$ is separable, and equals $a_d^{2d-2} \prod_{i < j} (\beta_i - \beta_j)^2$. Bounding all but one difference by $2R$ shows that distinct roots satisfy $|\beta - \beta'| \geq a_d^{-(d-1)} (2R)^{-(d(d-1)/2-1)} > \sigma$. Hence every nonreal root has $|\operatorname{Im} \beta| > \sigma/2$, and $|P'_1(\alpha)| = \prod_{\beta \neq \alpha} |\alpha - \beta| > \sigma^{d-1}$. The nonreal roots form $s = (d-1)/2$ conjugate pairs; let $\beta_j = u_j + iv_j$, $v_j > \sigma/2$, $1 \leq j \leq s$, be those in the upper half-plane. Write $v_k(T) = (1, T, \dots, T^k)^\top$.

A positive definite Gram of $P_1(T)(1+T^2)^e/(T-\alpha)$. Put $e = n - s - 1 \geq 0$. For $1 \leq j \leq s$ let $B_j = \begin{pmatrix} u_j^2 + v_j^2 & -u_j \\ -u_j & 1 \end{pmatrix}$, so that $v_1^\top B_j v_1 = (T - \beta_j)(T - \bar{\beta}_j)$, and for $s < j \leq n-1$ let $B_j = I_2$, so that $v_1^\top B_j v_1 = 1 + T^2$. Since $\det B_j = v_j^2 > \sigma^2/4$ and $\operatorname{tr} B_j < 1 + R^2$,

$$b_0 I \preceq B_j, \quad \|B_j\| \leq 1 + R^2, \quad b_0 = \frac{\sigma^2}{4(1+R^2)}.$$

Let S_j be the 0/1 matrix with $S_j v_j(T) = (v_{j-1}(T)^\top, T v_{j-1}(T)^\top)^\top$; then $S_j^\top S_j$ is diagonal with entries 1 and 2, so $I \preceq S_j^\top S_j \preceq 2I$. Define $G_0 = [1]$ and $G_j = S_j^\top (B_j \otimes G_{j-1}) S_j$. Because $S_j v_j = v_1 \otimes v_{j-1}$, the mixed-product rule gives $v_j^\top G_j v_j = (v_1^\top B_j v_1)(v_{j-1}^\top G_{j-1} v_{j-1})$. Hence $G' = G_{n-1}$ satisfies

$$v_{n-1}^\top G' v_{n-1} = \frac{P_1(T)}{T-\alpha} (1+T^2)^e, \quad G' \succeq b_0^{n-1} I, \quad \|G'\| \leq (2(1+R^2))^{n-1}.$$

The exposing matrix. Let D_c be the $n \times (n+1)$ matrix with $(D_c z)_k = z_{k+1} - cz_k$, $0 \leq k < n$, so that $D_c v_n(T) = (T - c)v_{n-1}(T)$, and put $Q_* = D_\alpha^\top G' D_\alpha$. Then

$$v_n(T)^\top Q_* v_n(T) = (T - \alpha) P_1(T) (1+T^2)^e, \quad Q_* \succeq 0, \quad \ker Q_* = \ker D_\alpha = \mathbb{R} v_n(\alpha). \quad (56)$$

For $y \in \mathbb{R}^n$, the equation $D_\alpha(0, y) = \omega$ is solved by $y_k = \sum_{i < k} \alpha^{k-1-i} \omega_i$, so $\|y\| \leq n R^{n-1} \|\omega\|$, and therefore

$$(0, y)^\top Q_*(0, y) \geq \gamma_0 \|y\|^2, \quad \gamma_0 = \frac{b_0^{n-1}}{n^2 R^{2n-2}}, \quad \|Q_*\| \leq \Lambda_* = (1+R)^2 (2(1+R^2))^{n-1}. \quad (57)$$

The rational vanishing space. Let $\mathcal{L}_n$ be the set of symmetric $(n+1) \times (n+1)$ matrices Q with $v_n(T)^\top Q v_n(T) \equiv 0 \pmod{P_1}$. For $0 \leq k < d$ let E_k be the rational symmetric matrix whose (i, j) entry is the coefficient of T^k in the remainder of T^{i+j} modulo P_1 . With $\langle X, Y \rangle = \operatorname{tr}(XY)$, $v_n^\top Q v_n \equiv \sum_k \langle E_k, Q \rangle T^k$, so $\mathcal{L}_n$ is the common kernel of the forms $\langle E_k, \cdot \rangle$, and $Q_* \in \mathcal{L}_n$. The E_k are linearly independent: since $d-1 \leq 2n$, every $k < d$ equals $i+j$ for some $0 \leq i, j \leq n$, and at that entry $(E_l)_{ij} = \delta_{kl}$. Hence the Gram matrix $\Gamma_E = (\langle E_k, E_l \rangle)$ is invertible, and

$$\Pi(X) = X - \sum_{k,l} (\Gamma_E^{-1})_{kl} \langle E_l, X \rangle E_k$$

is the orthogonal projection onto $\mathcal{L}_n$. It maps rational matrices to rational matrices, and it does not increase Frobenius distances to points of $\mathcal{L}_n$. For $d \leq m \leq 2n$ write the remainder $T^m \bmod P_1$ as $\sum_i \rho_{m,i} T^i$. Then $\rho_{m+1,i} = \rho_{m,i-1} - \rho_{m,d-1} c_i$, so all $|\rho_{m,i}| \leq R^{m-d+1}$, and $a_d^{m-d+1} \rho_{m,i}$ is an integer. All entries of E_k , Γ_E , and, by Cramer's rule and Hadamard's inequality, of Γ_E^{-1} have bit length polynomial in d and τ .

Approximation. Let $0 < \zeta \leq \min\{1, \sigma/16\}$, and apply Lemma E.3 with $2^{-N_0} \leq \zeta$. An approximation z of the real root has $|\operatorname{Im} z| \leq \zeta < \sigma/4$, and an approximation of a nonreal root has $|\operatorname{Im} z| > \sigma/2 - \zeta > \sigma/4$; this identifies $\tilde{\alpha} = \operatorname{Re} z$ for the real root and approximations $\tilde{\beta}_j$ of the upper roots, each within ζ . Form $\tilde{B}_j$ from $\tilde{\beta}_j$ as above (and $\tilde{B}_j = I_2$ for $j > s$), the recursion

$\tilde{G}_j$, and the rational matrix $\tilde{X} = D_{\tilde{\alpha}}^T \tilde{G}_{n-1} D_{\tilde{\alpha}}$. With $\hat{B} = 1 + (R+1)^2$ and $C_B = 2R+3$ we have $\|\tilde{B}_j - B_j\| \leq (|\tilde{\beta}_j| + |\beta_j|)\zeta + \sqrt{2}\zeta \leq C_B\zeta$ and $\|\tilde{B}_j\|, \|B_j\| \leq \hat{B}$. By induction on j , using $\|S_j\|^2 \leq 2$,

$$\|\tilde{G}_j - G_j\| \leq 2(C_B\zeta(2\hat{B})^{j-1} + \hat{B}\|\tilde{G}_{j-1} - G_{j-1}\|) \leq 2jC_B\zeta(2\hat{B})^{j-1}.$$

Since $\|D_{\tilde{\alpha}} - D_{\alpha}\| \leq \zeta$, $\|D_{\tilde{\alpha}}\| \leq R+2$, and $\|D_{\alpha}\| \leq R+1$,

$$\|\tilde{X} - Q_*\|_F \leq (n+1)C_X\zeta, \quad C_X = (2\hat{B})^{n-1}(2R+3)((n-1)(R+2)^2 + 1).$$

For a rational $0 < \delta \leq 1$ choose $\zeta \leq \delta/(2(n+1)C_X)$ as well, and put $Q = \Pi(\tilde{X})$. Then $Q \in \mathcal{L}_n$ is rational and $\|Q - Q_*\|_F \leq \delta/2$.

The exposing quadratic. Let $g(x) = (1, x)^T Q(1, x)$ and $p = (\alpha, \dots, \alpha^n)$, so $(1, p) = v_n(\alpha)$. Since $Q \in \mathcal{L}_n$, $g(p) = v_n(\alpha)^T Q v_n(\alpha) = 0$. For $u \in \mathbb{R}^n$, $g(p+u) = \ell^T u + u^T H u$ with $\ell = 2[Qv_n(\alpha)]_{1:n} = 2[(Q - Q_*)v_n(\alpha)]_{1:n}$ and $H = Q_{1:n, 1:n}$. With $\kappa_v = (n+1)R^n \geq \|v_n(\alpha)\|$, (57) gives

$$\|\ell\| \leq \kappa_v \delta, \quad H \succeq (\gamma_0 - \delta/2)I, \quad \|H\| \leq \Lambda_* + \delta/2. \quad (58)$$

Residuals. Put $x_0 = 1$; for $0 \leq k \leq d$ let $\phi_k = x_k$ if $k \leq n$ and $\phi_k = x_n x_{k-n}$ if $n < k \leq d$ (then $1 \leq k-n \leq n-1$, because $d \leq 2n-1$). Let $\kappa = 1 + \sum_{k=n+1}^d |c_k|$ and

$$r_j = \frac{x_1 x_j - x_{j+1}}{\kappa} \quad (1 \leq j < n), \quad r_n = \frac{1}{\kappa} \sum_{k=0}^d c_k \phi_k.$$

On the moment curve $\gamma(T) = (T, T^2, \dots, T^n)$ we have $\phi_k(\gamma(T)) = T^k$, so $r_j(\gamma(T)) = 0$ for $j < n$ and $r_n(\gamma(T)) = P_1(T)/\kappa$. In particular every r_j vanishes at $p = \gamma(\alpha)$. Conversely, the first $n-1$ equations force $x_j = x_1^j$, and then $r_n = P_1(x_1)/\kappa$, so p is the only real common zero. Each r_j with $j < n$ has quadratic part $x_1 x_j / \kappa$, of matrix norm at most 1; the quadratic part of r_n is a combination of the $x_n x_{k-n}$ with coefficient sum at most $\kappa - 1$. Hence $\|T_j\| \leq 1$ for all j . The gradients satisfy $\|\nabla r_j(p)\| \leq 3R^n$ for $j < n$ and $\|\nabla r_n(p)\| \leq \sum_k |c_k| 2R^n \leq (d+1)R^{n+1}$, so the Jacobian J at p satisfies $\|J\| \leq \|J\|_F \leq \beta = (d+4)R^{n+1}$ and every row has norm at most β .

The determinant of J is $P'_1(\alpha)/\kappa^n$. Indeed, differentiating $r(\gamma(T)) = (0, \dots, 0, P_1(T)/\kappa)$ at $T = \alpha$ gives $J\omega = (0, \dots, 0, P'_1(\alpha)/\kappa)^T$ with $\omega = \gamma'(\alpha) = (1, 2\alpha, \dots, n\alpha^{n-1})$. Replace the first column of the identity by ω ; this change of basis has determinant $\omega_1 = 1$. Expanding along the new first column, $\det J = (-1)^{n+1} \kappa^{-1} P'_1(\alpha) \det J'$, where J' consists of rows $1, \dots, n-1$ and columns $2, \dots, n$ of J . Row j of J' has entry $-1/\kappa$ in column $j+1$ and α/κ in column j (for $j \geq 2$), so J' is triangular with diagonal $-1/\kappa$ and $\det J' = (-1/\kappa)^{n-1}$. The product of the singular values now gives

$$\sigma_{\min}(J) \geq \frac{|\det J|}{\|J\|^{n-1}} \geq \nu = \frac{\sigma^{d-1}}{\kappa^n \beta^{n-1}}, \quad \sum_j b_j b_j^T = J^T J \succeq \nu^2 I.$$

Parameters and output. Put $\mu = \gamma_0/2$ and $\Lambda = \Lambda_* + 1$. Let $t = 2^{-N_t}$ with N_t the least nonnegative integer such that $\varepsilon = 4^{-N_t} \leq \min\{1, \mu^2/(2n), \nu^2 \mu^2/(36n(\Lambda + n\beta)^2)\}$; then take $\delta = \min\{1, \gamma_0, \varepsilon/\kappa_v\}$ and the corresponding ζ . By (58), $H \succeq \mu I$, $\|H\| \leq \Lambda$, and $\|\ell\| \leq \varepsilon$. With $m = n$ residuals all hypotheses of Lemma 7.6 hold, which gives every stated property of F_P .

Complexity. Every bound above $(R, \sigma, b_0, \gamma_0, \Lambda_*, \kappa, \kappa_v, \beta, \nu, C_X)$ is a rational number obtained from d, n, τ and the c_i by a polynomial number of arithmetic operations, and the logarithms of these numbers and of their reciprocals are $O(d^3 \tau + d \log d)$. Thus N_0, N_t , and $\log(1/\delta)$ are polynomial in d and τ . Root approximation is polynomial by Lemma E.3. The matrices $\tilde{B}_j, \tilde{G}_j, D_{\tilde{\alpha}}, \tilde{X}$ and Q

have order at most $2n$ and are obtained by $O(n^4 + d^4)$ rational operations; the entries of $\tilde{G}_j$ have bit length $O(j(N_0 + \tau))$, and the projection uses the polynomial-size data of Γ_E^{-1} . The output g, r_j, t, ν and the canonical Gram therefore have polynomial bit length and are computed in polynomial time.

Converse. Let f be as in the converse statement and $h(T) = f(T, T^2, \dots, T^k) \in \mathbb{Q}[T]$. If $h = 0$, then f vanishes on the whole real moment curve, contradicting uniqueness of the zero. So h is a nonzero polynomial of degree at most $4k$, nonnegative on $\mathbb{R}$, with $h(\alpha) = 0$. A real zero of a nonnegative polynomial has even multiplicity, so α is at least a double root. Since P_1 is the minimal polynomial of α and is separable, $P_1 \mid h$; writing $h = P_1 h_1$ and using $P_1'(\alpha) \neq 0$, also $h_1(\alpha) = 0$ and $P_1 \mid h_1$. Hence $2d \leq \deg h \leq 4k$, so $k \geq d/2$, that is $k \geq (d+1)/2$. $\square$

Proof of Corollary 7.5. Use the preceding construction with $n = d-1$; write $v = v_{d-1}$ and $p' = (\alpha, \dots, \alpha^{d-1})$. Let C be the companion matrix of P_1 , with $(Cz)_k = z_{k+1}$ for $k \leq d-2$ and $(Cz)_{d-1} = -\sum_i c_i z_i$. Then $Cv(\beta) = \beta v(\beta)$ for every root β . Let $\ell_\alpha = (\ell_0, \dots, \ell_{d-1})$ be the coefficient vector of $P_1(T)/(T - \alpha) = \sum_i \ell_i T^i$; comparing coefficients in $(T - \alpha) \sum_i \ell_i T^i = P_1(T)$ gives $\ell_\alpha^\top C = \alpha \ell_\alpha^\top$. Since α is a simple eigenvalue, $U := \text{image}(C - \alpha I)$ has dimension $d-1$ and equals $\ker \ell_\alpha^\top$. Also $\ell_\alpha^\top v(\alpha) = P_1'(\alpha)$, and $|\ell_i| \leq \sum_{k>i} |c_k| R^{k-i-1} \leq dR^d$, so $\|\ell_\alpha\| \leq d^2 R^d$.

Margin of Q_ on U .* Let $w \in U$, write $w = w_0 v(\alpha) + w'$ with $w'_0 = 0$, and note $D_\alpha w' = D_\alpha w$, so $\|w'\| \leq (d-1)R^{d-2} \|D_\alpha w\|$ as in the proof above. From $0 = \ell_\alpha^\top w = w_0 P_1'(\alpha) + \ell_\alpha^\top w'$ we get $|w_0| \leq \|\ell_\alpha\| \|w'\| / \sigma^{d-1}$, hence $\|w\| \leq \theta_U \|D_\alpha w\|$ with $\theta_U = (d-1)R^{d-2}(1 + dR^{d-1}d^2 R^d \sigma^{-(d-1)})$. Therefore $w^\top Q_* w \geq b_0^{d-2} \|D_\alpha w\|^2 \geq \beta_* \|w\|^2$ on U , where $\beta_* = b_0^{d-2} / \theta_U^2$.

The pencil. Run the approximation with $\delta \leq \min\{1, \beta_*\}$ and let $B = Q$; then $B \in \mathcal{L}_{d-1}$ is rational, $\|B - Q_*\| \leq \beta_*/2$, so $B|_U \succeq (\beta_*/2)I$ and $\|B\| \leq \Lambda_* + 1$. For $\lambda = (\lambda_1, \lambda_2) \in \mathbb{R}^2$ put $Q_\lambda = C^\top B C - \lambda_1(C^\top B + B C) + \lambda_2 B$ and $q_\lambda(x) = (1, x)^\top Q_\lambda (1, x)$ for $x \in \mathbb{R}^{d-1}$. Since $Cv(\alpha) = \alpha v(\alpha)$ and $v(\alpha)^\top B v(\alpha) = 0$, every q_λ vanishes at p' . At $\lambda_* = (\alpha, \alpha^2)$, $Q_{\lambda_*} = (C - \alpha I)^\top B (C - \alpha I)$ is positive semidefinite with kernel $\mathbb{R}v(\alpha)$, because B is positive definite on $U = \text{image}(C - \alpha I)$. For z with $z_0 = 0$, the first $d-1$ coordinates of $(C - \alpha I)z$ are $D_\alpha z$, so $\|(C - \alpha I)z\| \geq \|z\| / ((d-1)R^{d-2})$ and

$$z^\top Q_{\lambda_*} z \geq \gamma_G \|z\|^2, \quad \gamma_G = \frac{\beta_*}{2(d-1)^2 R^{2d-4}}.$$

Since $\|C\| \leq \|C\|_F \leq dR$, every λ with $\max\{|\lambda_1 - \alpha|, |\lambda_2 - \alpha^2|\} \leq 3e$, where e is a dyadic number with $e \leq \gamma_G / (6(2dR + 1)(\Lambda_* + 1))$, satisfies $\|Q_\lambda - Q_{\lambda_*}\| \leq \gamma_G/2$, so q_λ has Hessian $2(Q_\lambda)_{1:d-1, 1:d-1} \succeq \gamma_G I$.

A rational triangle. Compute a dyadic $\hat{\alpha}$ with $|\hat{\alpha} - \alpha| \leq e/(16(2R + 1))$ (by Lemma E.3 or by bisection), and put $c = (\hat{\alpha}, \hat{\alpha}^2)$; then $\delta' := \lambda_* - c$ satisfies $\|\delta'\|_\infty \leq e/16$. Let $\mu_1 = c + (-e, -e)$, $\mu_2 = c + (2e, -e)$, $\mu_3 = c + (-e, 2e)$. Solving $\lambda_* = \sum_i \tau_i \mu_i$ with $\sum_i \tau_i = 1$ gives $\tau_2 = (1 + \delta'_1/e)/3$, $\tau_3 = (1 + \delta'_2/e)/3$, and $\tau_1 = 1 - \tau_2 - \tau_3$, all larger than $1/4$. Every vertex lies within $3e$ of λ_* in the maximum norm. Put $g_i = q_{\mu_i}$. These are rational quadratics with positive definite Hessians that vanish at p' , and since $\lambda \mapsto q_\lambda$ is affine,

$$\sum_i \tau_i g_i(x) = q_{\lambda_*}(x) = \omega^\top B \omega \geq \frac{\beta_*}{2} \|\omega\|^2, \quad \omega = (C - \alpha I)(1, x) \in U,$$

with equality to zero only if $\omega = 0$, that is, $(1, x) \in \mathbb{R}v(\alpha)$ and $x = p'$. If $g_i(x) \leq 0$ for all i , the left side is at most zero, so $x = p'$. Finally, each g_i is strictly convex with rational Hessian and gradient, so its unique minimizer is rational and differs from the irrational point p' ; its minimum is therefore negative, and $\{g_i \leq 0\}$ is a full-dimensional ellipsoid. All quantities have polynomial bit length, so the construction is polynomial. $\square$

E.6 The cyclic family

Throughout this subsection $n \geq 2$, $m = n + 1$, $\eta = (-1)^m$, indices are taken modulo m , and $x_0 = 1$. Recall $d_n = (2^m - \eta)/3$, $e_i = ((-2)^i - 1)/3$, the exponents θ_i , the point $p_i = 2^{e_i/d_n}$, and the binomials

$$r_i(x) = x_i^2 - 2^{\theta_i} x_{i+1} x_{i+2} \quad (0 \leq i \leq n).$$

Write $a = 2^{1/d_n}$, so $p_i = a^{e_i}$, and $w_i = p_i^{-2}$.

Lemma E.4 (Arithmetic of the cyclic point). *(a) d_n is a positive odd integer, every e_i is an integer, $e_0 = 0$, $e_1 = -1$, and $|e_i| < d_n$; hence $1/2 < p_i < 2$ and $1/4 < w_i < 4$.*

(b) $2e_i - e_{i+1} - e_{i+2} = d_n \theta_i$ for $0 \leq i \leq n$, with indices of e modulo m . Consequently $r_i(p) = 0$ for all i .

(c) $\mathbb{Q}(p) = \mathbb{Q}(p_1) = \mathbb{Q}(a)$ and $[\mathbb{Q}(a) : \mathbb{Q}] = d_n$.

(d) With $X_i = x_i/p_i$ (so $X_0 = 1$), $\sum_i w_i r_i(x) = \frac{1}{2} \sum_i (X_i - X_{i+1})^2$. In particular p is the only real common zero of the r_i .

Proof. (a) $2^m \equiv (-1)^m \pmod{3}$, so d_n is an integer; $2^m - \eta$ is odd, so d_n is odd. Likewise $(-2)^i \equiv 1 \pmod{3}$. For $i \leq n$, $|e_i| \leq (2^n + 1)/3 < (2^m - 1)/3 \leq d_n$ because $n \geq 2$. (b) The sequence $E_i = ((-2)^i - 1)/3$, $i \geq 0$, satisfies $2E_i = E_{i+1} + E_{i+2}$, $E_m = \eta d_n$, and $E_{m+1} = -2\eta d_n - 1$. For $i \leq m - 3$ no index wraps and $\theta_i = 0$. For $i = m - 2$, $2e_{m-2} - e_{m-1} - e_0 = E_m = \eta d_n$. For $i = m - 1$, $2e_{m-1} - e_0 - e_1 = 2E_{m-1} + 1 = E_m + E_{m+1} + 1 = -\eta d_n$. Since $a^{d_n} = 2$, $p_i^2 = a^{d_n \theta_i} p_{i+1} p_{i+2} = 2^{\theta_i} p_{i+1} p_{i+2}$. (c) Every p_i is an integer power of a , and $p_1 = a^{-1}$. The polynomial $T^{d_n} - 2$ is irreducible by Eisenstein's criterion at 2. (d) By (b), $w_i r_i = X_i^2 - X_{i+1} X_{i+2}$; summing over i modulo m gives $\sum_i X_i^2 - \sum_i X_i X_{i+1} = \frac{1}{2} \sum_i (X_i - X_{i+1})^2$. This vanishes only if all X_i equal $X_0 = 1$. $\square$

Lemma E.5 (Quantitative data). *Write $r_i(p + u) = b_i^\top u + u^\top T_i u$ and let J have rows $b_i^\top$. Then $\|T_i\| \leq 1$, $\|b_i\| < 7$, $\|J\| < 32$, $J^\top J \succeq (8n)^{-2} I$, and $G_* = \sum_i w_i r_i$ satisfies $G_*(p + u) = u^\top H_* u$ with $(8n^2)^{-1} I \preceq H_* \preceq 8I$.*

Proof. Because $m \geq 3$, the indices $i, i + 1, i + 2$ are distinct, so the quadratic part of r_i is a diagonal entry 1 (absent if $i = 0$) and a disjoint off-diagonal pair $-2^{\theta_i}/2$ (absent if an index is 0); its norm is at most 1. The gradient b_i has at most three nonzero entries, $2p_i$, $-2^{\theta_i} p_{i+2}$, and $-2^{\theta_i} p_{i+1}$, each of absolute value less than 4. Put $h = ED_p^{-1} u$, where $D_p = \text{diag}(p_1, \dots, p_n)$ and E inserts a zero coordinate with index 0. Using $2^{\theta_i} p_{i+1} p_{i+2} = p_i^2$,

$$b_i^\top u = p_i^2 (2h_i - h_{i+1} - h_{i+2}), \quad J = \text{diag}(p_0^2, \dots, p_n^2) (2I - \mathcal{S} - \mathcal{S}^2) ED_p^{-1},$$

where $(\mathcal{S}z)_i = z_{i+1}$ is the cyclic shift. The four factors have norms less than 4, at most 4, 1, and less than 2, so $\|J\| < 32$. For the lower bound, $2I - \mathcal{S} - \mathcal{S}^2 = (2I + \mathcal{S})(I - \mathcal{S})$ and $\|(2I + \mathcal{S})y\| \geq 2\|y\| - \|\mathcal{S}y\| = \|y\|$. Since $h_0 = 0$, each h_k is a sum of $k \leq n$ consecutive differences, so by the Cauchy–Schwarz inequality

$$\|h\|^2 \leq n^2 \sum_i (h_i - h_{i+1})^2 = n^2 \|(I - \mathcal{S})h\|^2. \quad (59)$$

With $\|h\| \geq \|u\|/2$ and $p_i^2 > 1/4$ this gives $\|Ju\| \geq \|u\|/(8n)$. Finally, Lemma E.4(d) with $X_i = 1 + h_i$ gives $G_*(p + u) = \frac{1}{2} \|(I - \mathcal{S})h\|^2$, which lies between $\|h\|^2/(2n^2) \geq \|u\|^2/(8n^2)$ and $2\|h\|^2 \leq 8\|u\|^2$. $\square$

Lemma E.6 (Rational rank-one square roots). *Let $\mathbf{g} \in \mathbb{Q}^m$ be nonzero, $R_{\mathbf{g}} = \mathbf{g}^\top \mathbf{g}$, and let $0 < \varrho < \sqrt{R_{\mathbf{g}}}$ be rational. Put $t = (R_{\mathbf{g}} - \varrho^2)/(2\varrho)$ and $L_\varrho = tI + (\varrho/R_{\mathbf{g}})\mathbf{g}\mathbf{g}^\top$. Then $t > 0$, L_ϱ is rational, symmetric and invertible, and $L_\varrho^2 = t^2I + \mathbf{g}\mathbf{g}^\top$. Hence for every vector $r = (r_0, \dots, r_{m-1})$ of polynomials, $(\mathbf{g}^\top r)^2 + t^2 \sum_i r_i^2 = \sum_i (L_\varrho r)_i^2$.*

Proof. Since $(\mathbf{g}\mathbf{g}^\top)^2 = R_{\mathbf{g}}\mathbf{g}\mathbf{g}^\top$, the coefficient of $\mathbf{g}\mathbf{g}^\top$ in L_ϱ^2 is $(2t\varrho + \varrho^2)/R_{\mathbf{g}} = 1$. The eigenvalues of L_ϱ are t and $t + \varrho$. $\square$

Proof of Theorem 7.8. Integer weights. Let $M_1 = 10^6 n^5$ and $Q_1 = 32mM_1^2$. We compute integers $z_0, \dots, z_n$ with $|z_i/Q_1 - w_i| \leq 1/Q_1$ in time polynomial in n , without ever forming a polynomial of degree d_n . Since $w_i = \exp(y_i)$ with $y_i = -(2e_i/d_n) \log 2 \in (-2 \log 2, 2 \log 2)$, use $\log 2 = 2 \sum_{j \geq 0} 3^{-(2j+1)}/(2j+1)$, whose tail after N terms is less than 9^{-N} . Replacing $\log 2$ by the N -term sum L_N changes y_i by less than $2 \cdot 9^{-N}$, and $\exp$ is 8-Lipschitz on $[-2, 2]$. The Taylor polynomial of degree $K \geq 2$ of $\exp$ approximates it on $[-2, 2]$ within $2^{K+2}/(K+1)!$. Choosing N and K of size $O(\log Q_1)$ gives a rational $\hat{w}_i$ with $|\hat{w}_i - w_i| \leq 1/(4Q_1)$; rounding $Q_1 \hat{w}_i$ to an integer gives z_i . The integers e_i and d_n have $O(n)$ bits, and all intermediate rationals have bit length polynomial in n .

The exposing quadratic. Let $\mathbf{g} = (z_i/Q_1)_{i=0}^n$ and $G = \mathbf{g}^\top r = \sum_i (z_i/Q_1)r_i$. Then $G(p) = 0$, and with $G(p+u) = \ell^\top u + u^\top H u$, the identity $\sum_i w_i b_i = \nabla G_*(p) = 0$ gives $\ell = \sum_i (z_i/Q_1 - w_i)b_i$ and $H - H_* = \sum_i (z_i/Q_1 - w_i)T_i$. By Lemma E.5,

$$\|\ell\| \leq \frac{8m}{Q_1} = \frac{1}{4M_1^2}, \quad \|H - H_*\| \leq \frac{m}{Q_1} = \frac{1}{32M_1^2}, \quad \frac{1}{16n^2}I \preceq H \preceq 9I.$$

Regularization with a rational square root. Since $1/5 < z_i/Q_1 < 5$, we have $m/25 < R_{\mathbf{g}} < 25m$. The function $\varrho \mapsto t(\varrho) = (R_{\mathbf{g}} - \varrho^2)/(2\varrho)$ decreases on $(0, \infty)$, and the set of ϱ with $1/M_1 \leq t(\varrho) \leq 2/M_1$ is an interval of length at least $M_1^{-1} \min_{1/M_1 \leq t \leq 2/M_1} R_{\mathbf{g}}/(\sqrt{R_{\mathbf{g}} + t^2}(\sqrt{R_{\mathbf{g}} + t^2} + t)) \geq 1/(1450M_1)$ contained in $(0, 5m)$. Rational bisection on $[0, 5m]$, testing $M_1(R_{\mathbf{g}} - \varrho^2) \geq 2\varrho$ and $M_1(R_{\mathbf{g}} - \varrho^2) \leq 4\varrho$ at dyadic midpoints, finds such a $\varrho = a_1/b_1$ after $O(\log n)$ steps, with a_1, b_1 of polynomial magnitude. Put $t = t(\varrho)$ and $\varepsilon = t^2$, so $M_1^{-2} \leq \varepsilon \leq 4M_1^{-2}$.

Curvature and Gram. Apply Lemma 7.6 to G and the $m = n+1$ residuals $r_0, \dots, r_n$ with $\mu = 1/(16n^2)$, $\Lambda = 9$, $\beta = 8$, and $\nu = 1/(8n)$. Here $\Lambda + m\beta = 8n + 17 \leq 17n$, and

$$\frac{\mu^2}{2m} \geq \frac{1}{768n^5}, \quad \frac{\nu^2 \mu^2}{36n(\Lambda + m\beta)^2} \geq \frac{1}{170459136n^9},$$

both larger than $4M_1^{-2} = 4 \cdot 10^{-12} n^{-10}$; also $\|\ell\| \leq \varepsilon/4$. Thus $\Phi = G^2 + \varepsilon \sum_i r_i^2$ satisfies $\nabla^2 \Phi \succeq \frac{3}{2} \varepsilon \nu^2 I$, and the canonical Gram of this representation is positive definite. By Lemma E.6, $\Phi = \sum_i (L_\varrho r)_i^2$. Both representations equal $r^\top (\mathbf{g}\mathbf{g}^\top + \varepsilon I) r$ with the same matrix, so they have the same canonical Gram.

Integer squares. Let $S_0 = \sum_i z_i^2$ and $D_0 = 2a_1 b_1 Q_1^2 S_0$. Then $L_\varrho = tI + \frac{a_1}{b_1 S_0} z z^\top$ with $t = (S_0 b_1^2 - a_1^2 Q_1^2)/(2a_1 b_1 Q_1^2)$, so $D_0 L_\varrho$ is an integer matrix; each $2r_i$ has integer coefficients. Put $s_i = 32M_1 n D_0 (L_\varrho r)_i = 16M_1 n \sum_j (D_0 L_\varrho)_{ij} (2r_j) \in \mathbb{Z}[x]$ and $F_n = \sum_i s_i^2 = (32M_1 n D_0)^2 \Phi$. The integers used in this denominator-clearing and scaling step, namely z_i , Q_1 , M_1 , a_1 , b_1 , S_0 , D_0 , and the coefficients of the $2r_j$, have magnitude polynomial in n , so the coefficients of the s_i have $O(\log(n+1))$ bits. The integers d_n and e_i , which have $O(n)$ bits, enter only through the weights w_i and are never coefficients. Since $D_0 \geq 1$ and $\varepsilon \geq M_1^{-2}$, $\nabla^2 F_n \succeq (32M_1 n)^2 \cdot \frac{3}{2} M_1^{-2} (8n)^{-2} I = 24I$.

The canonical Gram of $\sum_i s_i^2$ is $(32M_1 n D_0)^2$ times that of Φ , hence positive definite. It is an integer matrix: for an integer quadratic $s = s_0 + b^\top x + x^\top T x$ the matrices b , $2T$ are integral, and

every entry of (15) is an integer combination of products of entries of b , $2T$ and s_0 . This proves (a). Part (b) is Lemma E.4, since F_n vanishes exactly at the common zeros of the r_i (L_ρ is invertible).

(c) G has $O(n)$ monomials, so G^2 and $\sum_i r_i^2$ have $O(n^2)$, and every coefficient of $F_n = \sum_i s_i^2$ is a sum of $O(n^2)$ products of polynomial-magnitude integers.

(d) is Theorem 7.9, which applies because F_n is globally convex of degree four with $\nabla^2 F_n(p) \succ 0$.

(e) Since all factors vanish at p , $\nabla^2 \Phi(p) = 2\ell\ell^\top + 2\varepsilon J^\top J$. Its least eigenvalue is at least $2\varepsilon\nu^2 = \varepsilon/(32n^2)$ and its largest at most $2\|\ell\|^2 + 2\varepsilon\|J\|^2 \leq \varepsilon(\varepsilon/8 + 2048)$. The ratio is at most $32n^2(1/8 + 2048) < 65600n^2$, and scaling by a positive constant does not change it.

(f) Substituting $x_1 = y_1 - 1$ maps each s_i to an integer quadratic with $O(\log(n+1))$ -bit coefficients; the curvature bound, the integer canonical Gram (Lemma 7.1(b)), and the number of squares are unchanged. The zero has first coordinate $1 + a^{-1}$. The number a^{-1} is a root of $2T^{d_n} - 1$, which is irreducible because its reversal $T^{d_n} - 2$ is; so $1 + a^{-1}$ has the irreducible primitive minimal polynomial $2(T - 1)^{d_n} - 1$, primitive because its leading coefficient is 2 and its constant coefficient is $2(-1)^{d_n} - 1 = -3$. The coefficient of T^j for $j \geq 1$ is $\pm 2\binom{d_n}{j} \neq 0$. Each coefficient has $O(d_n)$ bits, and for the $\Theta(d_n)$ indices $d_n/3 \leq j \leq 2d_n/3$ one has $\binom{d_n}{j} \geq 2^{\min(j, d_n-j)} \geq 2^{d_n/3-1}$; hence the total binary length is $\Theta(d_n^2)$. $\square$

Remark E.7 (Rational ellipsoids for the cyclic point). With $\tau_e = 1/(32n^2)$, $c_i = z_i/Q_1$, $C_z = \sum_i c_i$ and $W = \sum_i w_i$, the $n+1$ rational quadratics $G + \tau_e r_i$ have quadratic parts at least $(\mu - \tau_e)I = I/(32n^2)$ and vanish at p . The real weights $\lambda_i = \tau_e^{-1}(w_i - Wc_i/(\tau_e + C_z))$ satisfy $\sum_i \lambda_i(G + \tau_e r_i) = G_*$, and $\tau_e(\tau_e + C_z)\lambda_i = \tau_e w_i + w_i \sum_j (c_j - w_j) - W(c_i - w_i) \geq \tau_e/4 - 8m/Q_1 > 0$. Since $G_* \geq 0$ vanishes only at p , the intersection of the ellipsoids $\{G + \tau_e r_i \leq 0\}$ is $\{p\}$. Each ellipsoid is full-dimensional because its rational minimizer differs from p .

E.7 The minimum number of squares

We use the Brouwer degree of continuous maps $S^r \rightarrow S^r$, $r \geq 1$: it is a homotopy invariant, and if a continuous map is smooth near the finite preimage of a point y at which it is a local diffeomorphism, its degree is the sum over that preimage of the signs of the Jacobian determinants [52, Proposition 2.30], [88, Sections 4–5]. A continuous map $q : \mathbb{R}^r \rightarrow \mathbb{R}^r$ with $\|q(w)\| \rightarrow \infty$ as $\|w\| \rightarrow \infty$ extends continuously to the one-point compactification S^r by $\infty \mapsto \infty$; its degree is that of the extension. A homotopy q_s , $0 \leq s \leq 1$, extends to a homotopy of the extensions when $\|q_s(w)\| \geq \psi(\|w\|)$ for a function ψ with $\psi(\rho) \rightarrow \infty$, uniformly in s . Sard's theorem gives regular values in every nonempty open set [88, Section 3].

Proof of Theorem 7.9. Step 1: degrees and the count $m \geq n$. If some g_i had degree $e > 2$, the sum of the squares of the degree- e parts of the factors of maximal degree would be a nonzero form of degree $2e > 4$ (a nonzero real form squared is nonzero, and squares cannot cancel), contradicting $\deg f = 4$. So every g_i is a quadratic. Write $q = (g_1, \dots, g_m)$. All g_i vanish at p , so $\nabla^2 f(p) = 2Dq(p)^\top Dq(p)$; positive definiteness forces $\text{rank } Dq(p) = n$ and $m \geq n$. The zero p is unique: if $f(p') = 0$ with $p' \neq p$, then by convexity and $f \geq 0$, f vanishes on the segment $[p, p']$, so the second derivative of f at p in the direction $p' - p$ is zero. Suppose now, for a contradiction, that $m = n$, so $q : \mathbb{R}^n \rightarrow \mathbb{R}^n$.

Step 2: the quartic-flat subspace. Write $q = q_2 + q_1 + q_0$ by homogeneous degree and $H(x) = \|q_2(x)\|^2$, the quartic form of f . The function H is convex, as the pointwise limit of $s^{-4}f(sx)$ when $s \rightarrow \infty$, and it is nonnegative and even. Hence $K = \{x : H(x) = 0\}$ is convex and closed under real scalar multiples, so it is a linear subspace. For $v \in K$, $x \in \mathbb{R}^n$, $s \in \mathbb{R}$, and $0 < \eta < 1$,

$$H(x + sv) \leq (1 - \eta)H\left(\frac{x}{1 - \eta}\right) + \eta H\left(\frac{sv}{\eta}\right) = (1 - \eta)^{-3}H(x).$$

Letting $\eta \rightarrow 0$ gives $H(x + sv) \leq H(x)$, and applying this to $x + sv$ and $-s$ gives equality. Let B_2 be the symmetric bilinear map with $q_2(x) = B_2(x, x)$. Since $q_2(v) = 0$, $q_2(x + sv) = q_2(x) + 2sB_2(x, v)$, and the coefficient of s^2 in the constant function $\|q_2(x + sv)\|^2$ is $4\|B_2(x, v)\|^2$. Hence $B_2(x, v) = 0$, and $q_2(x + v) = q_2(x)$ for all $v \in K$.

Step 3: coordinates. Choose orthonormal coordinates $x = (w, v)$ with $v \in K \cong \mathbb{R}^k$ and $w \in K^\perp \cong \mathbb{R}^r$, $r = n - k$. Then

$$q(w, v) = q_2(w) + Aw + Bv + c$$

with matrices A, B and a vector c . The matrix $Dq(p) = (Dq_2(p_w) + A \ B)$ is nonsingular, so B has full column rank k . Because $\deg f = 4$, $q_2 \neq 0$ and $r \geq 1$. For $w \neq 0$ in $K^\perp$, $w \notin K$, so $\|q_2(w)\| > 0$.

Step 4: convexity removes the coupling. The ww block of the Hessian of $f = \sum_i q_i^2$ is

$$\nabla_{ww}^2 f(w, v) = C(w) + 2 \sum_i (Bv)_i \nabla^2 q_{2,i},$$

where $C(w) = \sum_i [2\nabla_w q_i \nabla_w q_i^\top + 2(q_{2,i}(w) + (Aw)_i + c_i) \nabla^2 q_{2,i}]$ does not depend on v , because $\nabla_w q_i = \nabla q_{2,i}(w) + A_i^\top$. Global convexity makes this block positive semidefinite for all v . Taking $v = se_j$ with $s \in \mathbb{R}$ arbitrary shows that the symmetric matrix $\sum_i (Be_j)_i \nabla^2 q_{2,i}$ vanishes, since otherwise $y^\top (C(w) + 2s \sum_i (Be_j)_i \nabla^2 q_{2,i}) y < 0$ for a suitable y and s . So the homogeneous quadratic $(Be_j)^\top q_2(w)$ has zero Hessian and is identically zero:

$$B^\top q_2(w) = 0 \quad (w \in \mathbb{R}^r). \quad (60)$$

Step 5: elimination of v . Let $T_\perp$ be an $r \times n$ matrix whose rows form an orthonormal basis of $(\operatorname{im} B)^\perp$, and put

$$v_0(w) = -(B^\top B)^{-1} B^\top (Aw + c), \quad \bar{q}(w) = T_\perp (q_2(w) + Aw + c).$$

If $k = 0$, take $T_\perp = I$ and omit v . By (60), $q_2(w) \in (\operatorname{im} B)^\perp$, so orthogonal decomposition of $q(w, v)$ along $\operatorname{im} B$ and its complement gives

$$f(w, v) = \|B(v - v_0(w))\|^2 + \|\bar{q}(w)\|^2.$$

The function $\bar{f}(w) = f(w, v_0(w)) = \|\bar{q}(w)\|^2$ is convex, being the restriction of f to the graph of the affine map v_0 . Its zeros lift to zeros of f , so its only zero is $w_* = p_w$, and $p_v = v_0(p_w)$. With $E = \begin{pmatrix} I_r \\ D_{v_0} \end{pmatrix}$ of full column rank, $\nabla^2 \bar{f}(w_*) = E^\top \nabla^2 f(p) E \succ 0$; as $\nabla^2 \bar{f}(w_*) = 2D\bar{q}(w_*)^\top D\bar{q}(w_*)$, the square map $\bar{q} : \mathbb{R}^r \rightarrow \mathbb{R}^r$ has the unique zero w_* , and it is regular. Its quadratic part $\bar{q}_2 = T_\perp q_2$ satisfies $\|\bar{q}_2(w)\| = \|q_2(w)\|$, which is positive on the unit sphere; by compactness and homogeneity there is $a_0 > 0$ with $\|\bar{q}_2(w)\| \geq a_0 \|w\|^2$.

Step 6: parity. For $0 \leq s \leq 1$ put $\bar{q}^{(s)}(w) = \bar{q}_2(w) + s(T_\perp Aw + T_\perp c)$. Then $\|\bar{q}^{(s)}(w)\| \geq a_0 \|w\|^2 - \|T_\perp A\| \|w\| - \|T_\perp c\|$ uniformly in s , so $\bar{q} = \bar{q}^{(1)}$ and $\bar{q}_2 = \bar{q}^{(0)}$ have the same degree. The point 0 has the single preimage w_* under $\bar{q}$, where $\bar{q}$ is a local diffeomorphism, so $\deg \bar{q} = \pm 1$. On the other hand, choose a regular value $y \neq 0$ of $\bar{q}_2$. Its preimage is compact, by the growth bound, and discrete, so it is finite; it does not contain 0, and it is invariant under $w \mapsto -w$ because $\bar{q}_2$ is even. Since $D\bar{q}_2$ is linear in w , $D\bar{q}_2(-w) = -D\bar{q}_2(w)$, and the two points of each pair $\{w, -w\}$ contribute ϵ and $(-1)^r \epsilon$ for some $\epsilon = \pm 1$. Each pair contributes an even number, so $\deg \bar{q}_2$ is even. This contradiction proves $m \geq n + 1$.

Sharpness. For $f = (\sum_i x_i^2)^2 + \sum_i x_i^2$, $\nabla^2 f = 8xx^\top + (4\|x\|^2 + 2)I \succeq 2I$, $f(0) = 0$, and f is a sum of $n + 1$ squares. Lemma 7.6 with $m = n$ and Theorem 7.8 give further examples with exactly $n + 1$ squares. The examples after Theorem 7.9 are verified directly; for the sextic, $\frac{d^2}{dx_1^2}(x_1 + x_1^3)^2 = 2 + 24x_1^2 + 30x_1^4$. $\square$

E.8 Binomial networks

We use the following classical facts about a directed multigraph $\mathcal{D}$ on $\{0, \dots, n\}$ with out-degree two everywhere, A its adjacency matrix (counting multiplicities and loops), and $L = 2I - A$. Parallel arcs are distinguished.

- (N1) (Directed matrix-tree theorem [114, Theorem 3.6].) The principal minor of L obtained by deleting row and column i equals the number τ_i of spanning arborescences in which every vertex has a directed path to i . Loops cannot enter a tree, and their contributions to the degree and adjacency terms of L cancel; hence the theorem includes loops here.
- (N2) (BEST theorem [115].) If $\mathcal{D}$ is Eulerian (every in-degree is also two), the number of Euler circuits, counted as cyclic arc sequences, equals $\tau_i \prod_v (\deg^+ v - 1)! = \tau_i$ for every i .
- (N3) (Interlace polynomials [9].) If $\mathcal{D}$ is Eulerian with Euler circuit $\mathcal{C}$, the interlace graph H of $\mathcal{C}$ on the $m = n + 1$ vertices has an interlace polynomial $q_H(x) = \sum_{j \geq 1} \alpha_j x^j$ with nonnegative integer coefficients and no constant term, $q_H(1)$ equals the number of Euler circuits of $\mathcal{D}$, and $q_H(2) = 2^m$.
- (N4) ([10, Theorem 1].) For a graph H on m vertices with adjacency matrix A_H , $q_H(-1) = (-1)^{\text{rk}} 2^{m - \text{rk}}$, where $\text{rk} = \text{rank}_{\mathbb{F}_2}(I + A_H)$.

Proof of Proposition 7.10. Lattice count. Let B be L with column 0 deleted; its rows are the exponent vectors of the Laurent monomials $y_i^2 / (y_{a(i)} y_{b(i)})$ in $y_1, \dots, y_n$ (with $y_0 = 1$). Strong connectivity gives $\tau_i > 0$, so by (N1) $\text{rank } L = m - 1$. As $L\mathbf{1} = 0$ and $L \text{adj}(L) = \text{adj}(L)L = \det(L)I = 0$, the columns of $\text{adj}(L)$ lie in $\mathbb{R}\mathbf{1}$ and its rows in the one-dimensional left kernel; its diagonal is (τ_i) , so $\text{adj}(L) = \mathbf{1}\tau^\top$ and $\tau^\top L = 0$. The maximal minor of B that omits row i is $\pm \text{adj}(L)_{0i} = \pm \tau_i$. By the Smith normal form, the index of the row lattice Λ_B of B in $\mathbb{Z}^n$ equals the gcd of the maximal minors, $g = \gcd(\tau_0, \dots, \tau_n)$.

Solutions. Every complex solution has nonzero coordinates: $x_0 = 1$, and if $x_i \neq 0$ then $c_i x_{a(i)} x_{b(i)} = x_i^2 \neq 0$; strong connectivity propagates this to all vertices. Writing $x_i = p_i y_i$, the equations become $y^{B_i} = 1$ for every row B_i , because $p^{B_i} = c_i$. Thus normalized solutions correspond to characters of the finite abelian group $\Gamma_B = \mathbb{Z}^n / \Lambda_B$, of which there are exactly g . If ϵ is the exponent of Γ_B , then $\epsilon e_j = \sum_i k_i B_i$ with integers k_i , so $p_j^\epsilon = \prod_i c_i^{k_i} \in \mathbb{Q}$; hence p is algebraic, every embedding of $\mathbb{Q}(p)$ gives a distinct complex solution, and $[\mathbb{Q}(p) : \mathbb{Q}] \leq g$. Real normalized solutions are characters with values in $\{\pm 1\}$, the only real roots of unity; there is a nontrivial one exactly when g is even. Since p is the only real solution, g is odd.

Non-Eulerian graphs. The primitive positive integer vector $\omega = \tau/g$ spans the left kernel of L . It equals $\mathbf{1}$ exactly when $\mathbf{1}^\top L = 0$, that is, when every in-degree is two. Otherwise some $\omega_i \geq 2$. An arborescence toward i chooses one of the two outgoing arcs at each of the n other vertices, so $\tau_i \leq 2^n$. Hence $g \leq \tau_i/2 \leq 2^{n-1}$, and since g is odd, $g \leq 2^{n-1} - 1 < d_n$.

Eulerian graphs. Now all τ_i equal g , which by (N2) and (N3) equals $q_H(1)$ for the interlace graph of any Euler circuit. Since $q_H(1) - q_H(-1) = 2 \sum_j \text{odd } \alpha_j$ is even and $q_H(1) = g$ is odd, $q_H(-1)$ is odd, so (N4) forces $\text{rk} = m$ and $q_H(-1) = (-1)^m$. For $j \geq 1$, $2^j - (-1)^j \geq 3$, so

$$3g \leq \sum_j \alpha_j (2^j - (-1)^j) = q_H(2) - q_H(-1) = 2^m - (-1)^m,$$

that is, $g \leq d_n$.

The exposing normal form. Let $\mathcal{Q} = \sum_j \lambda_j r_j$ be the positive semidefinite combination with kernel $\mathbb{R}p$, $p = (1, p_1, \dots, p_n)$. Every diagonal entry of $\mathcal{Q}$ is positive: a zero diagonal entry would put

a coordinate vector into the kernel, but p has no zero coordinate and $m \geq 2$. A nonzero binomial vanishing at p has two distinct monomials with nonzero coefficients (a single monomial cannot vanish at p), and it cannot have two square monomials with positive coefficients. So each summand $\lambda_j r_j$ contributes a positive coefficient to at most one square X_i^2 , while all m diagonal entries must receive one. Hence each summand supplies exactly one positive square, at distinct indices. Assigning to the binomial its square X_i^2 and rescaling rationally gives $X_i^2 - c_i X_{a(i)} X_{b(i)}$ with $c_i \in \mathbb{Q} \setminus \{0\}$, which defines a two-out multigraph. The same count shows that fewer than m binomials cannot supply all diagonal entries.

In the coordinates $X_i = p_i Y_i$, the i -th summand becomes $h_i(Y_i^2 - Y_{a(i)} Y_{b(i)})$ with $h_i > 0$. The kernel contains $\mathbf{1}$, which gives $2h = A^\top h$: every vertex has inflow equal to outflow when arc $i \rightarrow j$ carries flow h_i . If $\mathcal{D}$ were not strongly connected, take a strongly connected component with no incoming arcs from other components; summing the flow balance over it shows that its outgoing arcs carry zero flow, so it has none, and the argument repeats on the rest. Thus the components are mutually disconnected. By stationarity,

$$\sum_i h_i (Y_i^2 - Y_{a(i)} Y_{b(i)}) = \frac{1}{2} \sum_i h_i (Y_{a(i)} - Y_{b(i)})^2,$$

because the diagonal coefficients agree, so the indicator vector of each component lies in the kernel. Corank one forces a single component. Finally, the common real zero is unique: if all binomials vanish at $(1, x)$, then $\mathcal{Q}(1, x) = 0$ and $(1, x) \in \mathbb{R}p$, so $x = (p_1, \dots, p_n)$. The first part now applies. $\square$

E.9 Imported results from algebraic geometry

All varieties are over $\mathbb{C}$ and projective unless stated otherwise; a variety is reduced and irreducible, and its degree is the number of points in its intersection with a general linear space of complementary dimension [50]. $S = \mathbb{C}[x_0, \dots, x_N]$, and for a zero-dimensional subscheme $R \subset \mathbb{P}^N$ and $k \geq 0$, $H_R(k)$ is the rank of the restriction map $S_k \rightarrow H^0(\mathcal{O}_R(k))$.

- (AG1) [51, Propositions I.7.1, I.7.2 and Theorem I.7.7] If $X \subseteq \mathbb{P}^N$ is a variety of dimension $r \geq 1$ and V is a hypersurface of degree e not containing X , then $X \cap V$ is nonempty, every irreducible component of $X \cap V$ has dimension $r - 1$, and the sum of their degrees is at most $e \deg X$.
- (AG2) [43, Proposition 8.4 and Example 8.4.6] If forms $F_1, \dots, F_N$ of degrees $e_1, \dots, e_N$ have finitely many common zeros in $\mathbb{P}^N$, then they form a regular sequence, the scheme $Z = V(F_1, \dots, F_N)$ (a *complete intersection*) has length $e_1 \cdots e_N$, $S/(F_1, \dots, F_N)$ is Cohen–Macaulay of dimension one, and every linear form that vanishes at no point of Z is a nonzerodivisor on $S/(F_1, \dots, F_N)$.
- (AG3) (Cayley–Bacharach [36, Theorem CB7].) Let $N \geq 3$, let $Z \subset \mathbb{P}^N$ be a zero-dimensional complete intersection of hypersurfaces of degrees $e_1, \dots, e_N$, and put $s = \sum_i e_i - N - 1$. If Z is reduced, every form of degree s vanishing at all but one point of Z vanishes on Z . In general, if $\Gamma, R \subseteq Z$ are closed subschemes residual to each other, then for every integer t with $0 \leq t \leq s$, $h^0(\mathcal{I}_\Gamma(t)) - h^0(\mathcal{I}_Z(t)) = h^1(\mathcal{I}_R(s - t))$. Moreover, for every finite subscheme $R \subset \mathbb{P}^N$ and every $k \geq 0$, $h^1(\mathcal{I}_R(k)) = \text{length}(R) - H_R(k)$, by the cohomology sequence of $0 \rightarrow \mathcal{I}_R(k) \rightarrow \mathcal{O}(k) \rightarrow \mathcal{O}_R(k) \rightarrow 0$ and $H^1(\mathbb{P}^N, \mathcal{O}(k)) = 0$. We use the reduced statement for $N = n \geq 3$ and the residual statement only for $N = n \geq 4$, with $t = 2$ and $s - t = n - 3 \geq 1$.
- (AG4) (Gorenstein duality [33, Chapter 21].) If $\bar{q}_1, \dots, \bar{q}_N \in \mathbb{C}[y_1, \dots, y_N]$ is a regular sequence of quadratic forms, the graded algebra $\mathcal{G} = \mathbb{C}[y]/(\bar{q})$ has Hilbert series $(1 + t)^N$, $\mathcal{G}_N$ is one-dimensional, and multiplication $\mathcal{G}_i \times \mathcal{G}_{N-i} \rightarrow \mathcal{G}_N$ is a perfect pairing. The same holds over $\mathbb{Q}$.

- (AG5) (Residual intersections [37, Theorem 3.2 and Remark 3.3]; Segre classes of local complete intersections [43, Proposition 4.1].) Let $Y \subset \mathbb{P}^5$ be a nonempty reduced pure-dimensional local complete intersection, not necessarily smooth (for example a reduced curve that is a complete intersection of type $(1, 1, 2, 2)$ with singular points), such that $\mathcal{I}_Y(2)$ is generated by global sections. Then for general quadrics $h_1, \dots, h_5$ vanishing on Y , the set $V(h_1, \dots, h_5) \setminus Y$ is finite, and the sum of the local lengths of $V(h_1, \dots, h_5)$ at the points of $V(h_1, \dots, h_5) \setminus Y$ equals $32 - \int_Y (1 + 2H)^5 \cap s(Y, \mathbb{P}^5)$. For a local complete intersection, singular or not, the Segre class is $s(Y, \mathbb{P}^5) = c(N_{Y/\mathbb{P}^5})^{-1} \cap [Y]$, with $N_{Y/\mathbb{P}^5}$ the normal bundle, so the count is $32 - e(Y)$ with $e(Y) = \int_Y \{(1 + 2H)^5 c(N_{Y/\mathbb{P}^5})^{-1}\}_{\dim Y}$. The cited residual count holds for an arbitrary closed subscheme cut out by forms of degree at most two; the local complete intersection property is used only for this formula for the Segre class.
- (AG6) (Minimal degree [34, Proposition 0, Theorem 1 and Lemma 2.1].) An irreducible variety of dimension k that spans $\mathbb{P}^r$ has degree at least $r - k + 1$. An irreducible curve of degree r spanning $\mathbb{P}^r$ is a rational normal curve, which is smooth of genus zero and whose ideal is generated by quadrics.
- (AG7) A smooth curve $C \subset \mathbb{P}^5$ of degree δ and genus g has $\deg N_{C/\mathbb{P}^5} = 6\delta + 2g - 2$, by the Euler sequence and adjunction; consequently $e(C) = 4\delta - 2g + 2$.

We also use two elementary consequences. First, a variety X that is invariant under complex conjugation and has odd degree has a real point: intersecting with general real linear subspaces of complementary dimension gives $\deg X$ distinct points, permuted by conjugation, and nonreal points pair up. In particular a conjugation-invariant linear space has real points. Second, a finite subscheme of $\mathbb{P}^N$ defined over $\mathbb{R}$ and of odd length has a real point, because conjugation pairs nonreal points with equal local lengths.

Lemma E.8 (Weighted Bézout count). *Let $X_1, \dots, X_s \subseteq \mathbb{P}^N$ be varieties, $e \geq 1$, and let $h_1, \dots, h_t$ be forms of degree at most e . Put $Z = (\bigcup_k X_k) \cap V(h_1, \dots, h_t)$, let $\mathcal{I}(Z)$ be its set of isolated points and $\mathcal{P}(Z)$ its set of positive-dimensional irreducible components. Then*

$$\#\mathcal{I}(Z) + \sum_{Y \in \mathcal{P}(Z)} e^{\dim Y} \deg Y \leq \sum_k e^{\dim X_k} \deg X_k.$$

Proof. Assign to a variety X the weight $e^{\dim X} \deg X$ and start with the list $X_1, \dots, X_s$. Process $h_1, \dots, h_t$ in turn: keep every listed variety contained in $V(h_i)$, and replace every other listed variety X by the irreducible components of $X \cap V(h_i)$; a point not in $V(h_i)$ is simply removed. By (AG1), when $\dim X \geq 1$ the new components have dimension $\dim X - 1$ and total degree at most $e \deg X$, so the total weight does not increase. Keep repeated entries. After the last step the union of the listed varieties is Z . Each isolated point of Z and each $Y \in \mathcal{P}(Z)$ is a listed variety: it is contained in some listed irreducible closed subset of Z , which must coincide with it by maximality. These entries are distinct, so their weights are bounded by the final, hence by the initial, total weight. $\square$

E.10 Degree bounds

Lemma E.9 (Flat directions). *Let $f \in \mathbb{Q}[x_1, \dots, x_n]$ be globally convex of degree at most four, with $f \geq 0$, a zero p , and $\nabla^2 f(p) \succ 0$, and let f_4, f_3 be its homogeneous parts of degrees four and three. Then $K_4 = \{d : f_4(d) = 0\}$ is a linear subspace defined over $\mathbb{Q}$, and f_4 and f_3 are invariant under translations by K_4 . In rational coordinates $x = U(y, \vartheta)$, $U \in GL_n(\mathbb{Q})$, with $K_4 = \{y = 0\}$, there are*

a rational matrix $A_\vartheta \succ 0$, a rational affine map ϑ_* , and a rational globally convex $\hat{f}$ of degree at most four whose quartic form is positive definite, such that

$$f(U(y, \vartheta)) = \hat{f}(y) + (\vartheta - \vartheta_*(y))^\top A_\vartheta (\vartheta - \vartheta_*(y)).$$

The unique zero $\hat{p}$ of $\hat{f}$ is the y -part of $U^{-1}p$, $\nabla^2 \hat{f}(\hat{p}) \succ 0$, and $\mathbb{Q}(\hat{p}) = \mathbb{Q}(p)$. If $f = \sum_j r_j^2$ with rational quadratics r_j , then $\hat{f} = \sum_j r_j(U(y, \vartheta_*(y)))^2$ is a sum of squares of rational quadratics.

Proof. As in Step 2 of the proof of Theorem 7.9, f_4 is convex, nonnegative and even, K_4 is a linear subspace, and $f_4(x + d) = f_4(x)$ for $d \in K_4$. Hence $K_4 = \{d : D_d f_4 \equiv 0\}$, the kernel of a rational linear map, and it has a rational basis. For $d \in K_4$, $\nabla^2 f(x + sd) = \nabla^2 f(x) + s \nabla^2 (D_d f_3)$, since $\nabla^2 f_4$ is invariant and $\nabla^2 f_3$ is linear in the point. Positive semidefiniteness for all real s forces $\nabla^2 (D_d f_3) = 0$; the quadratic form $D_d f_3$ therefore vanishes, and f_3 is invariant along K_4 . In the new coordinates the parts f_4, f_3 depend only on y , so $f(U(y, \vartheta)) = G(y) + \vartheta^\top A_\vartheta \vartheta + \vartheta^\top (B_\vartheta y + c_\vartheta)$ with rational data, and $A_\vartheta \succ 0$ because it is half the $\vartheta\vartheta$ block of the Hessian at p . Completing the square with $\vartheta_*(y) = -\frac{1}{2} A_\vartheta^{-1} (B_\vartheta y + c_\vartheta)$ gives the displayed identity with $\hat{f}(y) = f(U(y, \vartheta_*(y)))$, which is convex as the restriction of f to an affine graph. Its quartic form is $f_4(U(y, 0))$, positive for $y \neq 0$. Zeros of $\hat{f}$ correspond to zeros of f , and the coordinates of p are rational affine functions of those of $\hat{p}$ and conversely. The Hessian of $\hat{f}$ at $\hat{p}$ is a Schur complement of the positive definite Hessian of $f \circ U$. The last statement follows by substitution. $\square$

Proof of Theorem 7.11. By Lemma 7.3 (zero-set case), p is algebraic and D is odd. Write $R_j(x_0, x) = x_0^2 r_j(x/x_0)$ for the quadratic forms homogenizing the r_j , and identify $x \in \mathbb{C}^n$ with $[1 : x] \in \mathbb{P}^n$. Let Γ be the set of the D conjugates $\sigma(p)$ of p .

(a) All r_j vanish at p , and $\nabla^2 f(p) = 2D r(p)^\top D r(p)$, so some n of them, say $h_1, \dots, h_n$, have nonsingular Jacobian at p . Applying σ to the rational determinant shows that their Jacobian is nonsingular at every point of Γ . So every point of Γ is a simple, hence isolated, zero of $h_1, \dots, h_n$, and an isolated point of $Z = V(H_1, \dots, H_n) \subseteq \mathbb{P}^n$, where H_i homogenizes h_i . Lemma E.8 with $X_1 = \mathbb{P}^n$ and $e = 2$ gives $D \leq 2^n$, and oddness gives $D \leq 2^n - 1$.

Reduction for (b) and (c). Assume f globally convex. Apply Lemma E.9. If $K_4 \neq 0$, then $\hat{f}$ is a rational sum of quadratic squares in at most $n - 1$ variables with the same field degree, so part (a) gives $D \leq 2^{n-1} - 1$, which is smaller than every bound in (b) and (c); if no variable remains, p is rational. So we may assume that f_4 is positive definite. Then the forms R_j have no common real zero at infinity: $R_j(0, d) = r_{j,2}(d)$ for the quadratic parts, and $f_4(d) = \sum_j r_{j,2}(d)^2 > 0$ for real $d \neq 0$. Their only common real affine zero is p .

(b), *first bound.* Let $n \geq 3$ and suppose $D = 2^n - 1$. With Z as in (a), Lemma E.8 gives $D + 2\#\mathcal{P}(Z) \leq 2^n$, so Z is finite. By (AG2) it is a complete intersection of length 2^n . Points of Γ have local length one, so the residual has length one: it is a single reduced point b , and Z is reduced. Since Z and Γ are defined over $\mathbb{Q}$, b is fixed by $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, so $b \in \mathbb{P}^n(\mathbb{Q})$. Choose a rational linear form Λ_b with $\Lambda_b(b) \neq 0$. For every j , the form $R_j \Lambda_b^{n-3}$ has degree $n - 1 = \sum_i 2 - n - 1$ and vanishes on $\Gamma = Z \setminus \{b\}$, so by (AG3) it vanishes at b , hence $R_j(b) = 0$. If $b = [1 : b']$, then $b' \in \mathbb{Q}^n$ is a real zero of f different from p (as $b \notin \Gamma$). If $b = [0 : d]$, then $f_4(d) = 0$ with $d \neq 0$ real. Both are impossible. So $D \leq 2^n - 3$.

(b), *second bound.* Let $n \geq 4$ and suppose $D = 2^n - 3$. Let $\mathcal{W} = V(R_1, \dots, R_m) \subseteq \mathbb{P}^n$. Its only real point is $[1 : p]$, which is isolated because the Jacobian of the r_j has rank n . Hence no positive-dimensional component of $\mathcal{W}$ has a real point.

Generic combinations. For a matrix $A \in \mathbb{C}^{n \times m}$ put $h^A = Ar$ and $Z^A = V(AR)$. On $U_{\mathcal{W}} = \mathbb{P}^n \setminus \mathcal{W}$ the vector $R(x)$ is nonzero, so $\{(x, A) \in U_{\mathcal{W}} \times \mathbb{C}^{n \times m} : AR(x) = 0\}$ is a vector bundle of rank $nm - n$ over the n -dimensional $U_{\mathcal{W}}$, of dimension nm . By the theorem on the dimension of fibres, for A in a

nonempty Zariski open set the fibre $Z^A \cap U_{\mathcal{W}}$ is finite, so every positive-dimensional component of Z^A lies in $\mathcal{W}$. The condition that $\mathbf{A} \operatorname{Dr}(\sigma(p))$ be nonsingular for every σ is another nonempty Zariski open condition. A nonempty Zariski open subset of $\mathbb{C}^{n \times m}$ contains rational points, so we may choose $\mathbf{A}$ rational with both properties. Put $Z = Z^A$; it is defined over $\mathbb{Q}$, and Γ consists of simple isolated points of Z .

Finiteness. The set $\mathcal{P}(Z)$ is permuted by complex conjugation, and its members lie in $\mathcal{W}$, so they have no real points. If $\mathcal{P}(Z) \neq \emptyset$, its total weight $\sum_{Y \in \mathcal{P}(Z)} 2^{\dim Y} \deg Y$ is at least four: weight below four would mean a single member, a line, which would then be conjugation-invariant and have real points. Lemma E.8 would give $D + 4 \leq 2^n$, a contradiction. So Z is finite, a complete intersection of length 2^n by (AG2), and $Z = \Gamma \sqcup R$ with a residual subscheme R of length three, defined over $\mathbb{Q}$; R is disjoint from Γ because points of Γ have local length one.

The residual imposes independent conditions. If $H_R(1) \leq 2$, the linear forms vanishing on R cut out a linear space of dimension at most one containing R . A reduced point cannot contain a scheme of length three, so R lies on a line ℓ . Each $\mathbf{A}R$ restricts to a binary quadratic form on ℓ vanishing on a scheme of length three, hence vanishes on ℓ ; then $\ell \subseteq Z$, contradicting finiteness. So $H_R(1) = 3$. A linear form Λ that does not vanish at the support of R is a unit on R ; multiplication by it maps the image of S_k into that of S_{k+1} injectively, so $H_R(k) = 3$ for all $k \geq 1$.

Conclusion. Apply (AG3) with $s = n - 1$ and $t = 2$: since $n - 3 \geq 1$, $h^0(\mathcal{I}_\Gamma(2)) - h^0(\mathcal{I}_Z(2)) = 3 - H_R(n - 3) = 0$. Every R_j vanishes on Γ , hence on $Z \supseteq R$. Since R is defined over $\mathbb{R}$ and has odd length, it has a real point, which is a common zero of all R_j outside Γ . As shown in the reduction, this is impossible. With oddness, $D \leq 2^n - 5$.

(c) is proved in Appendix E.11.

Maxima. For $n = 1$, (a) gives $D = 1$. For $n = 2, 3, 4, 5$ the bounds are 3, 5, 11, 21, and the quartics of Theorem 7.8 are globally convex rational sums of quadratic squares with field degrees $d_2, \dots, d_5 = 3, 5, 11, 21$. $\square$

Proof of Proposition 7.12. The real zero set of f is $\{p\}$, so Lemma 7.3 shows that p is algebraic and $D = [\mathbb{Q}(p) : \mathbb{Q}]$ is odd. Since $f \geq 0$ attains its minimum at p , $\nabla f(p) = 0$, and $\nabla^2 f(p) \succ 0$. For every conjugate $b = \sigma(p)$: $f(b) = 0$, $\nabla f(b) = 0$, and the complex symmetric matrix $\mathcal{H}_b = \nabla^2 f(b) = \sigma(\nabla^2 f(p))$ is nonsingular.

Generic gradient combinations. For each b , the set of vectors $\varpi \in \mathbb{C}^n$ with $\varpi^\top \mathcal{H}_b^{-1} \varpi = 0$ is a proper quadric cone. Choose $\varpi \in \mathbb{Q}^n$ outside these finitely many cones, and a rational $(n - 1) \times n$ matrix $\mathbf{A}$ of rank $n - 1$ with $\ker \mathbf{A} = \mathbb{R}\varpi$. At each b , the polynomial map $\mathbf{A}\nabla f$ has derivative $\mathbf{A}\mathcal{H}_b$ of rank $n - 1$, so near b its zero set is a smooth complex curve $\gamma_b(t)$ with $\gamma_b(0) = b$ and tangent $v_b = \mathcal{H}_b^{-1} \varpi$ (because $\mathbf{A}\mathcal{H}_b v_b = \mathbf{A}\varpi = 0$). The function $\varphi_b(t) = f(\gamma_b(t))$ satisfies $\varphi_b(0) = \varphi'_b(0) = 0$ and $\varphi''_b(0) = v_b^\top \mathcal{H}_b v_b = \varpi^\top \mathcal{H}_b^{-1} \varpi \neq 0$.

Splitting the double points. For a sufficiently small $c \neq 0$, Rouché's theorem gives two distinct zeros of $\varphi_b(t) - c$ near $t = 0$, at which $\varphi'_b \neq 0$. At such a point $x = \gamma_b(t)$ the Jacobian of the system $(f - c, \mathbf{A}\nabla f)$ is nonsingular: the rows of $\mathbf{A}\nabla^2 f(x)$ span the conormal space of the curve, and $\nabla f(x)^\top \gamma'_b(t) = \varphi'_b(t) \neq 0$. Taking c small for all D conjugates at once, the system has at least $2D$ distinct simple, hence isolated, solutions.

Bézout. Homogenize $f - c$ to a form of degree $\deg f \leq 4$ and the entries of $\mathbf{A}\nabla f$ to forms of degree at most three. The irreducible components of the quartic hypersurface have dimension $n - 1$ and total degree at most four, so their total weight with $e = 3$ is at most $4 \cdot 3^{n-1}$. Lemma E.8 bounds the number of isolated points of the full intersection, which include the $2D$ affine isolated solutions, by $4 \cdot 3^{n-1}$. Hence $D \leq 2 \cdot 3^{n-1}$, and oddness gives $D \leq 2 \cdot 3^{n-1} - 1$.

Two variables with convexity. Let $n = 2$ and f globally convex. A convex polynomial of odd degree at least three does not exist, because its leading form would be convex and odd, hence linear.

If $\deg f \leq 2$, p is the minimizer of a rational strictly convex quadratic, so it is rational. If f_4 has a nonzero real zero, Lemma E.9 reduces to one variable, where the bound just proved gives $D \leq 1$. Otherwise $\deg f = 4$ and f_4 is positive definite; Lemma 7.13 shows that f is a sum of squares of rational polynomials, necessarily quadratics, and Theorem 7.11(a) gives $D \leq 3$. $\square$

Lemma 7.13 uses the following classification.

(Sch) [107] Let $F \in \mathbb{Q}[x_0, x_1, x_2]$ be a quartic form that is nonnegative on $\mathbb{R}^3$ but is not a sum of squares of quadratic forms with rational coefficients. Then $F = \lambda \ell_1 \ell_2 \ell_3 \ell_4$ with $\lambda \in \mathbb{C}$ and linear forms $\ell_i \in \mathbb{C}[x_0, x_1, x_2]$ such that no three of the lines $V(\ell_i)$ pass through a common point.

Proof of Lemma 7.13. Let $F(x_0, x) = x_0^4 f(x/x_0)$. Then $F \geq 0$ on $\mathbb{R}^3$: for $x_0 \neq 0$ by homogeneity, and for $x_0 = 0$ because $F(0, x) = f_4(x)$. Its real projective zeros are the real zero of f only, since $f_4(x) > 0$ for real $x \neq 0$. Suppose F is not a sum of squares of rational quadratic forms, and write $F = \lambda \ell_1 \ell_2 \ell_3 \ell_4$ as in (Sch). The four lines are distinct. No ℓ_i is proportional to a real form: otherwise $F = \ell_i K$ with a real cubic form K not vanishing identically on the real line $V(\ell_i)$, and near a real point of that line where $K \neq 0$, F would change sign. Since F is real and its factorization into lines is unique, conjugation permutes the four lines without fixed points; they form two conjugate pairs $\{\ell_1, \bar{\ell}_1\}$ and $\{\ell_2, \bar{\ell}_2\}$. The point $V(\ell_1) \cap V(\bar{\ell}_1)$ is conjugation invariant, hence real, and it is a zero of F ; so is $V(\ell_2) \cap V(\bar{\ell}_2)$, and the two points differ because no three lines are concurrent. This gives two real projective zeros, a contradiction. Hence $F = \sum_i G_i^2$ with rational quadratic forms G_i , and $f(x) = \sum_i G_i(1, x)^2$.

For a certified quartic (f, A) in two variables with minimum zero, the full Hessian Gram gives $12f_4(x) = (x \otimes x)^\top A_{22}(x \otimes x) > 0$ for $x \neq 0$, where A_{22} is the lower right block of A , and strong convexity gives a unique zero; so the hypotheses hold. $\square$

Example E.10 (Global convexity is needed in Theorem 7.11(b)). Let $r_1 = z - x^2$, $r_2 = yz - y^2 - 1$, $r_3 = xz + y^2$ and $f = r_1^2 + r_2^2 + r_3^2$ on $\mathbb{R}^3$. A common zero has $z = x^2$ and $y^2 = -x^3$, and then $r_2 = 0$ reads $yx^2 + x^3 - 1 = 0$; $x = 0$ is impossible, so $y = (1 - x^3)/x^2$ and $P_7(x) = x^7 + x^6 - 2x^3 + 1 = 0$. Conversely every root of P_7 gives a common zero. Modulo two, $\bar{P}_7 = x^7 + x^6 + 1$ has no root in $\mathbb{F}_2$, and repeated squaring in $\mathbb{F}_2[x]/(\bar{P}_7)$, using $x^8 \equiv x^6 + x + 1$, $x^{10} \equiv x^6 + x^3 + x^2 + x + 1$, and $x^{12} \equiv x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$, gives

$$x^{16} \equiv x^6 + x^5 + x^4 + x^3 + x, \quad x^{32} \equiv x^5 + x^4 + x^2 + x + 1, \quad x^{64} \equiv x^4 + x^3 + 1, \quad x^{128} \equiv x.$$

Since 7 is prime, $\bar{P}_7$ is irreducible, hence so is P_7 over $\mathbb{Q}$. For $x \geq 0$, $P_7(x) = x^7 + (x^3 - 1)^2 > 0$; $P_7(-t) = -t^7 + t^6 + 2t^3 + 1$ has one sign change, so by Descartes' rule P_7 has at most one negative root, and $P_7(-2) < 0 < P_7(-1)$. Thus f has exactly one real zero p , with $\mathbb{Q}(p) = \mathbb{Q}(x)$ of degree $7 > 2^3 - 3$. At every common zero the Jacobian determinant of (r_1, r_2, r_3) equals $-P_7'(x)/x^2 \neq 0$, so $\nabla^2 f(p) = 2Dr^\top Dr \succ 0$. However $\partial_x^2 f(0, 0, 1) = -2$, and the homogenized quadrics share the real point $[x : y : z : x_0] = [0 : 0 : 1 : 0]$ at infinity.

E.11 Five variables

Proposition E.11 (Finite odd residuals in $\mathbb{P}^5$). *Let $Z \subset \mathbb{P}^5$ be a zero-dimensional complete intersection of five quadrics with rational coefficients, and let $Z = \Gamma \sqcup R$ be a decomposition into closed subschemes defined over $\mathbb{Q}$, with Γ reduced and $\text{length}(R) \in \{1, 3, 5, 7, 9\}$. Then some real point of R is a common zero of all quadratic forms that vanish on Γ .*

Proof. Choose a rational linear form Λ that does not vanish at any point of Z , and work in the affine chart $\Lambda = 1$. Let $\mathcal{A}$ be the coordinate ring of Z over $\mathbb{Q}$ in this chart, a $\mathbb{Q}$ -algebra of dimension 32, filtered by the images $F_i\mathcal{A}$ of polynomials of degree at most i . By (AG2), Λ is a nonzerodivisor on the homogeneous coordinate ring $S_{\mathbb{Q}}/I_Z$, so $F_i\mathcal{A} \cong (S_{\mathbb{Q}}/I_Z)_i$ and the associated graded algebra is $S_{\mathbb{Q}}/(I_Z + (\Lambda))$, a complete intersection of five quadrics in five variables. By (AG4) it has Hilbert series $(1+t)^5$ and perfect pairings $\text{gr}_i \times \text{gr}_{5-i} \rightarrow \text{gr}_5$.

A nondegenerate functional. Let $\lambda : \mathcal{A} \rightarrow \mathbb{Q}$ vanish on $F_4\mathcal{A}$ and be nonzero on the one-dimensional space $\mathcal{A}/F_4\mathcal{A} = \text{gr}_5$. If $a \neq 0$ has leading degree i , choose $b' \in \text{gr}_{5-i}$ pairing nontrivially with its leading class and lift it to $b \in F_{5-i}\mathcal{A}$; then $ab \in F_5\mathcal{A}$ has nonzero image in gr_5 , so $\lambda(ab) \neq 0$. Hence $(a, b) \mapsto \lambda(ab)$ is nondegenerate on $\mathcal{A}$. The decomposition $Z = \Gamma \sqcup R$ gives $\mathcal{A} = \mathcal{A}_{\Gamma} \times \mathcal{A}_R$, and the restriction λ_R of λ to $\mathcal{A}_R$ is again nondegenerate.

The quotient by an annihilator. Let $\mathbf{q}$ be a rational quadric vanishing on Γ , dehomogenized in the chart; its image in $\mathcal{A}$ is $(0, \mathbf{q}_R)$. Put $\mathcal{B} = \mathcal{A}_R / \text{Ann}(\mathbf{q}_R)$, $\ell = \dim \mathcal{B}$, and $\beta(a, b) = \lambda_R(\mathbf{q}_R ab)$. This is a well-defined nondegenerate symmetric bilinear form on $\mathcal{B}$: an element pairs to zero with everything exactly when its product with $\mathbf{q}_R$ vanishes. Let $\mathcal{V} \subseteq \mathcal{B}$ be the image of the polynomials of degree at most one and $v = \dim \mathcal{V}$. For a, b of degree at most one, $\mathbf{q}ab$ has degree at most four and vanishes on Γ , so $\beta(a, b) = \lambda(\mathbf{q}ab) = 0$. Thus $\mathcal{V}$ is totally isotropic and $2v \leq \ell$.

A Bézout bound. The subscheme $T = \text{Spec } \mathcal{B}$ of R lies in the projective linear space $\Pi \cong \mathbb{P}^{v-1}$ cut out by the linear forms vanishing on it. The five quadrics restrict to quadrics on Π whose common zero set $Z \cap \Pi$ is finite; choosing successively general combinations that do not contain any positive-dimensional component of the current intersection gives $v - 1$ quadrics on Π with finite common zero scheme containing T . By (AG2) its length is 2^{v-1} , so $\ell \leq 2^{v-1}$ (for $v = 1$, T is contained in a reduced point and $\ell \leq 1$). If $1 \leq \ell \leq 9$, the inequalities $2v \leq \ell \leq 2^{v-1}$ force $v = 4$ and $\ell = 8$.

Real parity. Suppose that no real point of R is a common zero of all quadrics through Γ . Then a suitable rational combination $\mathbf{q}$ of a basis of these quadrics is nonzero at every one of the finitely many real points of R . Over $\mathbb{R}$, $\mathcal{A}_R \otimes \mathbb{R}$ is a product of local algebras with residue fields $\mathbb{R}$ (at real points) or $\mathbb{C}$ (at pairs of conjugate points). On a factor with residue field $\mathbb{R}$, $\mathbf{q}$ is a unit, so the factor survives unchanged in $\mathcal{B} \otimes \mathbb{R}$. A factor with residue field $\mathbb{C}$ is a $\mathbb{C}$ -algebra, so it and its quotients have even real dimension. Hence $\ell \equiv \text{length}(R) \pmod{2}$ is odd. Since $\text{length}(R)$ is odd, R has a real point, $\mathbf{q}$ is nonzero there, and $1 \leq \ell \leq 9$. This contradicts $\ell = 8$. $\square$

Proposition E.12 (Positive-dimensional base loci in $\mathbb{P}^5$). *Let $\mathcal{Q}$ be a complex vector space of quadratic forms on $\mathbb{P}^5$ spanned by real forms, with base locus $\mathcal{W}$. Suppose no positive-dimensional component of $\mathcal{W}$ has a real point, and that $\Gamma \subset \mathcal{W}$ is a set of $D \geq 23$ points at each of which the differentials of the forms in $\mathcal{Q}$ span a space of dimension five (in an affine chart). Then $\mathcal{W}$ is finite.*

Proof. Suppose $\mathcal{W}$ has positive-dimensional components. Each point of Γ is an isolated point of $\mathcal{W}$, so it lies on none of them.

Weight. As in the proof of Theorem 7.11, five general forms $h_1, \dots, h_5 \in \mathcal{Q}$ have simple common zeros at Γ and no positive-dimensional common component outside $\mathcal{W}$; every positive-dimensional component of $\mathcal{W}$ is then a component of $V(h_1, \dots, h_5)$. Lemma E.8 gives

$$\sum_Y 2^{\dim Y} \deg Y \leq 32 - D \leq 9, \quad (61)$$

the sum running over the positive-dimensional components of $\mathcal{W}$. This set is permuted by conjugation.

The possible configurations. Components of dimension at least four have weight at least 16. A three-dimensional component has weight at least 8, so it would be the only positive-dimensional

component, conjugation invariant, of degree one, hence a real linear space, which has real points. If there is a surface component, the surface components have total degree at most two: either a single plane, which would be invariant and real; two planes, which must be conjugate (invariant planes are real); or one integral quadric surface, which is invariant. In the last two cases no curve component fits in (61). Otherwise all components are curves of total degree at most four. An invariant curve has even degree (odd degree would give a real point), so it is an integral conic or an integral quartic. A non-invariant curve comes with its conjugate, and both have degree one or two. In each case we choose a subvariety $Y_0 \subseteq \mathcal{W}$ in the list below. It is a nonempty reduced pure-dimensional local complete intersection with $\mathcal{I}_{Y_0}(2)$ globally generated, and $e(Y_0) \geq 10$.

Two conjugate planes. They are disjoint, since their intersection would be an invariant linear space with real points. In coordinates with the planes $\{x_3 = x_4 = x_5 = 0\}$ and $\{x_0 = x_1 = x_2 = 0\}$, the nine products $x_i x_j$, $i \leq 2 < j$, generate the ideal. For a plane, $N = \mathcal{O}(1)^3$ and $e = [H^2](1 + 2H)^5(1 + H)^{-3} = 40 - 30 + 6 = 16$; so $e(Y_0) = 32$.

A real integral quadric surface. By (AG6) an irreducible surface of degree two spans a linear space of dimension at most three; it cannot span a plane, since then it would be the plane, of degree one. So it spans a $\mathbb{P}^3$, which is real because the surface is conjugation invariant, and it is a hypersurface of degree two there. An irreducible and reduced quadric in $\mathbb{P}^3$ has rank three or four; if it had rank three, its unique singular point would be conjugation invariant, hence a real point of $\mathcal{W}$ on a positive-dimensional component. So it has rank four and is smooth. Its ideal is generated by two linear forms and one quadric, $N = \mathcal{O}(1)^2 \oplus \mathcal{O}(2)$, and $e = 2[H^2](1 + 2H)^4(1 + H)^{-2} = 2(24 - 16 + 3) = 22$.

A real integral conic. By (AG6) it spans a plane, in which it is a smooth conic, cut out by three linear forms and a quadric; (AG7) gives $e = 4 \cdot 2 + 2 = 10$.

A real integral quartic curve C . By (AG6) its span has dimension at most four. If the span is a plane, every quadric of $\mathcal{Q}$ restricts to a conic containing the integral plane quartic, hence vanishes on the plane; the plane would lie in $\mathcal{W}$, contradicting the configuration. If the span is a $\mathbb{P}^3$ (necessarily real) and the quadrics of $\mathcal{Q}$ restrict there to a space of dimension at most one, $\mathcal{W}$ contains the span or a quadric surface, again a contradiction. Otherwise two independent restricted quadrics contain C ; they have no common component, since a common plane would contain the nondegenerate curve or leave it on a line. Their complete intersection has degree four and contains the reduced curve C of degree four; being unmixed and generically reduced, it equals C . So C is a complete intersection of type $(1, 1, 2, 2)$ in $\mathbb{P}^5$, with $N = \mathcal{O}(1)^2 \oplus \mathcal{O}(2)^2$ and $e = 4[H](1 + 2H)^3(1 + H)^{-2} = 4(6 - 2) = 16$, smooth or not. If the span is $\mathbb{P}^4$, C is a rational normal quartic by (AG6), with ideal generated by quadrics and one linear form, and (AG7) gives $e = 4 \cdot 4 + 2 = 18$.

Two conjugate lines. They are disjoint, span a real $\mathbb{P}^3$, and in coordinates there with lines $\{x_0 = x_1 = 0\}$, $\{x_2 = x_3 = 0\}$ the four products $x_i x_j$, $i \leq 1 < j$, together with two linear forms generate the ideal. Each line has $e = 6$, so $e(Y_0) = 12$.

Two conjugate conics $C, \bar{C}$ in planes $\Pi, \bar{\Pi}$. If $\Pi = \bar{\Pi}$, the plane is real, and every quadric of $\mathcal{Q}$ restricts to a conic vanishing on two distinct integral conics, hence vanishes on the plane; this contradicts the configuration. If the planes are disjoint, the products between the two coordinate blocks together with the two conic equations generate the ideal sheaf of $C \sqcup \bar{C}$; then $e = 10 + 10 = 20$. If the planes meet in one point a , it is real and lies on neither conic (else it would be a real point of $\mathcal{W}$ on a curve component). In coordinates on the span $\mathbb{P}^4$ with $\Pi = V(x_3, x_4)$, $\bar{\Pi} = V(x_1, x_2)$, $a = [1 : 0 : 0 : 0 : 0]$, and conic equations $c(x_0, x_1, x_2)$, $c'(x_0, x_3, x_4)$ normalized to have coefficient one on x_0^2 , the four products $x_i x_j$ ($i \in \{1, 2\}$, $j \in \{3, 4\}$), the quadric $c + c' - x_0^2$, and the linear form of the span generate the ideal sheaf of $C \sqcup \bar{C}$: away from a the products cut out one plane locally and the quadric restricts to that plane's conic, while at a the quadric does not vanish. Again $e = 20$. If the planes meet along a line, their span is a real $\mathbb{P}^3$ and $\Pi \cup \bar{\Pi} = V(Q_0)$ for a real quadric Q_0 there. Some real $q \in \mathcal{Q}$ does not vanish on Π (otherwise, by conjugation, $\mathcal{W}$ would contain both

planes and their real common line). Its restrictions to Π and $\bar{\Pi}$ are nonzero multiples of the two conic equations, so $V(\mathbf{Q}_0, \mathbf{q})$ in that $\mathbb{P}^3$ is a complete intersection of degree four containing $C \cup \bar{C}$; as before it equals $C \cup \bar{C}$, of type $(1, 1, 2, 2)$, and $e = 16$.

Conclusion. In every case $Y_0 \subseteq \mathcal{W}$, so the five forms h_i vanish on Y_0 and have the D simple zeros Γ outside Y_0 . By the implicit function theorem, every quintuple of quadrics vanishing on Y_0 that is close enough to $(h_1, \dots, h_5)$ has at least D distinct simple zeros outside Y_0 , one near each point of Γ . General quintuples in $H^0(\mathcal{I}_{Y_0}(2))^5$ are dense, so by (AG5) $D \leq 32 - e(Y_0) \leq 22$, a contradiction. $\square$

Proof of Theorem 7.11(c). Let $n = 5$, f globally convex, and suppose $D \geq 23$; by (a) and oddness, $D \in \{23, 25, 27, 29, 31\}$. As in the reduction of the proof of (b), we may assume that f_4 is positive definite, since otherwise $D \leq 15$. Let $\mathcal{Q}$ be the complex span of the forms R_j , $\mathcal{W}$ its base locus, and Γ the orbit of p . The only real point of $\mathcal{W}$ is $[1 : p]$, and it is isolated, so positive-dimensional components of $\mathcal{W}$ have no real points; the differentials of $\mathcal{Q}$ span a five-dimensional space at each point of Γ . Proposition E.12 shows that $\mathcal{W}$ is finite. Choose a rational $5 \times m$ matrix A as in the proof of (b); then $Z = V(AR)$ has no positive-dimensional component (those would lie in the finite $\mathcal{W}$), so it is a zero-dimensional complete intersection of length 32, defined over $\mathbb{Q}$, with the simple points Γ . Its residual R has length $32 - D \in \{1, 3, 5, 7, 9\}$ and is defined over $\mathbb{Q}$. Proposition E.11 gives a real point of R at which every R_j vanishes; it is not in Γ . As in (b), this is a second real zero of f or a real zero of f_4 at infinity, a contradiction. Hence $D \leq 21$. $\square$

E.12 Univariate realizations

For a univariate f and a centre c , put $z = x - c$ and $w = (1 + z^2)^N$. Direct differentiation of $g = fw$ gives the identity

$$\frac{g''}{w} = f'' + \frac{4Nz}{1 + z^2}f' + \left[\frac{2N}{1 + z^2} + \frac{4N(N-1)z^2}{(1 + z^2)^2} \right]f. \quad (62)$$

Proof of Theorem 7.14(b). Write $f = P^2$, of degree $2d$, and define the rational constants

$$\begin{aligned} R_0 &= 2^{\tau+1}, \quad \sigma_1 = 2^{-4d^2(\tau+1)}, \quad m_1 = \sigma_1^{2(d-1)}, \quad C_0 = (d+1)2^{2\tau}, \quad Q_0 = (2d+1)^2C_0, \\ R_1 &= 4(R_0 + Q_0 + 1), \quad H_1 = (2d+1)^4C_0R_1^{2d}, \quad r_1 = \frac{m_1}{2H_1}, \quad \phi_1 = r_1^2(\sigma_1/2)^{2(d-1)}, \\ \psi_1 &= \frac{\phi_1 r_1^2}{(1 + 4R_1^2)^2}, \quad N = 2 + \left\lceil \frac{3H_1 + 1}{\psi_1} \right\rceil, \quad \kappa = \frac{2}{m_1}. \end{aligned}$$

Bounds on f . All roots satisfy $|\gamma| < R_0$. As in the proof of Theorem 7.7, the nonzero integer discriminant gives $|\gamma - \gamma'| \geq 2^{-\tau(d-1)}(2R_0)^{-d(d-1)/2+1} > \sigma_1$ for distinct roots when $d \geq 2$. Hence $|P'(\alpha)| \geq \sigma_1^{d-1}$ (the leading coefficient is a positive integer) and $f''(\alpha) = 2P'(\alpha)^2 \geq 2m_1$. The coefficients of f are bounded by C_0 and those of f', f'' by Q_0 , so $|f^{(j)}(x)| \leq H_1$ for $|x| \leq R_1$ and $j \leq 3$. We use three regions.

- (I) If $|x - \alpha| \leq r_1$, then $|x| < R_1$, $f''(x) \geq 2m_1 - H_1r_1 \geq m_1$, $f'' \leq H_1$, and f' has the sign of $x - \alpha$ with $|f'(x)| \leq H_1|x - \alpha|$.
- (II) If $|x| \leq R_1$ and $|x - \alpha| \geq r_1$, then $f(x) = a_d^2 \prod_\gamma |x - \gamma|^2 \geq r_1^2(\sigma_1/2)^{2(d-1)} = \phi_1$, because $|x - \gamma| \geq |\operatorname{Im} \gamma| > \sigma_1/2$ for the nonreal roots.
- (III) If $|x| \geq R_1$, then $f''(x) \geq 1$ and $xf'(x) > 0$: the leading coefficients of f' and f'' are at least two, and since $R_1 > 4(Q_0 + 1)$ the lower terms are less than a quarter of the leading term in absolute value. (For $d = 1$, f'' is a constant at least two.)

The centre. Because P has a single, simple real root and a positive leading coefficient, d is odd and $P(-R_0) < 0 < P(R_0)$. Rational bisection with exact sign evaluation by Horner's rule produces a dyadic c with $\delta = |c - \alpha| \leq r_1/2$ and $\delta^2 \leq m_1/(8NH_1)$ after a number of steps polynomial in $d + \tau$; if a midpoint is a root, take it.

Curvature. In region (I), the last term of (62) is nonnegative, and the middle term is negative only when x lies between c and α , where $|z| \leq \delta$ and $|f'| \leq H_1\delta$; so $g''/w \geq m_1 - 4NH_1\delta^2 \geq m_1/2$. In region (II), $r_1/2 \leq |z| \leq 2R_1$, so the last term is at least $N(N-1)\phi_1 r_1^2/(1+4R_1^2)^2 = N(N-1)\psi_1$, the middle term is at least $-2NH_1$ (as $|z|/(1+z^2) \leq 1/2$), and $g''/w \geq N(N-1)\psi_1 - 2NH_1 - H_1 \geq N(H_1 + 1) - H_1 \geq 1$ because $(N-1)\psi_1 \geq 3H_1 + 1$. In region (III), $|c| < R_0 + 1 < R_1 \leq |x|$, so z and x have the same sign, $zf' > 0$, and $g''/w \geq f'' \geq 1$. Since $w \geq 1$ and $m_1 \leq 1$, $g'' \geq m_1/2$ everywhere, and $G = \kappa g$ satisfies $G'' \geq 1$. The factor w is positive, so $G^{-1}(0) = \{\alpha\}$.

Size. The logarithms of R_0, R_1, H_1 are $O(d(\tau + \log(d+1)))$, and those of $1/\sigma_1, 1/m_1, 1/r_1, 1/\phi_1, 1/\psi_1, N$ and κ are $O(d^3(\tau + \log(d+1)))$. A circuit computes P by Horner's rule, squares it, forms $1 + (T - c)^2$, raises it to the power N by repeated squaring with shared intermediate results ($O(\log N)$ gates), and multiplies by κ . For the integer $x_0 = 2^{\tau+2}$, $x_0 - c \geq 2^{\tau+1} \geq 4$ and $P(x_0)$ is a nonzero integer, so $G(x_0) \geq \kappa 2^N \geq 2^N$; its reduced numerator has at least $N + 1$ bits. $\square$

Proof of Theorem 7.14(a). If α is the unique real zero of a rational univariate polynomial, Lemma 7.3 (zero-set case, $n = 1$) shows that it is algebraic with one real conjugate. Conversely, apply part (b) to the primitive integer multiple of the minimal polynomial of α . $\square$

The order of choices in part (b) matters: the exponent N is fixed first and the centre is then chosen close to α . A fixed centre does not suffice in general. For $f = (x-1)^2$, $c = 0$ and $x_N = 1 - 2/N$, the right side of (62) tends to $2 - 8 + 4 = -2$ as $N \rightarrow \infty$, so $f(x)(1+x^2)^N$ is not convex for large N .

Proof of Theorem 7.14(c). The cubic. $P'_k(x) = 3(x^2 - 1)$, $P_k(-1) = 4 + 2^{-2k} > 0$ and $P_k(1) = 2^{-2k} > 0$, so P_k has exactly one real root, and $P_k(-2) = 2^{-2k} > 0$ shows that it is less than -2 . Write $\alpha_k = -2 - t$ with $t > 0$; then $P_k(\alpha_k) = 0$ reads $t(3+t)^2 = 2^{-2k}$. If $t = a/b$ in lowest terms with positive integers, then $t(3+t)^2 = a(a+3b)^2/b^3$ in lowest terms (both a and $a+3b$ are coprime to b), with numerator at least 16; but the reduced numerator of 2^{-2k} is 1. So P_k has no rational root and, being a cubic, is irreducible. Comparing coefficients, the other roots are $\beta \pm i\eta$ with

$$\beta = 1 + \frac{t}{2}, \quad \eta^2 = 3t + \frac{3}{4}t^2, \quad \beta - \alpha_k = 3 + \frac{3}{2}t > 3, \quad 2^{-2k} - \eta^2 = t(6 + \frac{21}{4}t + t^2) > 0,$$

so $0 < \eta < 2^{-k}$. Clearing denominators shows that P_k has binary length $O(k)$.

An interval estimate. We use Markov's inequality: a real polynomial $\mathcal{P}$ of degree at most D_1 satisfies $\max_{[a,b]} |\mathcal{P}'| \leq \frac{2D_1^2}{b-a} \max_{[a,b]} |\mathcal{P}|$ [20, Theorem 5.1.8]. Iterating it with the same D_1 bounds the j -th derivative by $(2D_1^2/(b-a))^j \max |\mathcal{P}|$. Suppose $\mathcal{P}(b) = 1$, $0 \leq \mathcal{P} \leq 1$ on $[a, b]$, and $\mathcal{P}(b + i\eta) = 0$. Taylor expansion at b gives $1 \leq \sum_{j=1}^{D_1} q_M^j/j!$ with $q_M = 2D_1^2 |\eta|/(b-a)$. If $q_M < 1/2$ the right side is less than $q_M/(1 - q_M) < 1$; hence

$$D_1^2 \geq \frac{b-a}{4|\eta|}. \quad (63)$$

Convex realizations. Let F be rational, globally convex, with only real zero α_k . Since $F(\alpha_k) = 0$ and P_k is irreducible, $P_k \mid F$, so $F(\beta + i\eta) = 0$; in particular F is not affine. A nonaffine convex polynomial tends to $+\infty$ at both ends, so a negative value would produce two real zeros; hence $F \geq 0$ and $F(\beta) > 0$. On $[\alpha_k, \beta]$, $\mathcal{P} = F/F(\beta)$ lies between 0 and the chord from 0 to 1, so (63) with $a = \alpha_k$, $b = \beta$ gives $(\deg F)^2 \geq (\beta - \alpha_k)/(4\eta) > \frac{3}{4}2^k$. The normalizing constant may be irrational; rationality is used only to force the complex zero.

Systems. Remove identically zero rows from a finite system of rational convex inequalities $F_i \leq 0$ with solution set $\{\alpha_k\}$. Some active row F_i has $F'_i(\alpha_k) \geq 0$: otherwise all active rows would be negative on a short interval to the right of α_k , where the inactive rows are also negative. Its tangent line gives $F_i \geq 0$ on $[\alpha_k, \infty)$, and $F_i(\beta) > 0$, since otherwise convexity would make F_i vanish on $[\alpha_k, \beta]$. As before $P_k \mid F_i$, and the same argument applies to $F_i/F_i(\beta)$.

Objectives. If a nonconstant rational convex J has a minimizer at α_k , then $J'(\alpha_k) = 0$, so $P_k \mid J'$. Convexity makes J' nondecreasing, so $0 \leq J' \leq J'(\beta)$ on $[\alpha_k, \beta]$ with $J'(\beta) > 0$ (otherwise J' vanishes on an interval and J is constant). Apply (63) to $J'/J'(\beta)$, of degree $\deg J - 1$. $\square$

Proof of Theorem 7.14(d). Theorem 7.7 with $d = 3$ and $n = 2$ applies to P_k . For J_k , $J'_k = P_k$ is negative left of α_k and positive right of it, so J_k decreases and then increases: its sublevel sets are intervals, and α_k is its unique minimizer. But $J''_k(0) = -3$. $\square$

E.13 The quadratic graph

Proof of Theorem 7.15. Let $h = n + n^2$, $U_A = \text{tr } A$, and $\varrho_A = \det A/U_A^{h-1}$. Every eigenvalue of A is at most U_A , so $A \succeq \varrho_A I$, and therefore $\nabla^2 f \succeq \varrho_A I$ on $\mathbb{R}^n$. Strong monotonicity of ∇f gives $\varrho_A \|p\|^2 \leq -\nabla f(0)^\top p$, so $\|p\| \leq P_A := 1 + \|\nabla f(0)\|_1 / \varrho_A$. All these rationals have polynomial bit length.

Approximate minimizers. For a rational $0 < \eta \leq 1$, a rational $\hat{p}$ with $\|\hat{p} - p\| \leq \eta$ is computed in time polynomial in L and $\log(1/\eta)$: Lemma 3.1, applied to f on the box $[-P_A - 1, P_A + 1]^n$, which contains p , returns a rational $\hat{p}$ in the box with $f(\hat{p}) \leq f(p) + \varrho_A \eta^2/2$, and strong convexity gives $\frac{\varrho_A}{2} \|\hat{p} - p\|^2 \leq f(\hat{p}) - f(p)$. The promise $\min f = 0$ is not used here.

A quadratic lift of f . Let C_{dup} map $(s_{ij})_{i \leq j}$ to the symmetric $n \times n$ matrix, viewed as a vector in $\mathbb{R}^{n^2}$; its singular values lie in $[1, \sqrt{2}]$. For rational $\hat{p}$ put $d = y - \hat{p}$ and $s_{ij} = z_{ij} - \hat{p}_i y_j - \hat{p}_j y_i + \hat{p}_i \hat{p}_j$, and define

$$g_{\hat{p}}(y, z) = f(\hat{p}) + \nabla f(\hat{p})^\top d + \int_0^1 (1 - \varsigma) \left(\hat{p} \otimes d + \varsigma C_{\text{dup}} s \right)^\top A \left(\hat{p} \otimes d + \varsigma C_{\text{dup}} s \right) d\varsigma.$$

The integrand is a quadratic polynomial in (y, z) whose coefficients are polynomials of degree at most two in ς , and $\int_0^1 (1 - \varsigma) \varsigma^k d\varsigma = \frac{1}{2}, \frac{1}{6}, \frac{1}{12}$ for $k = 0, 1, 2$; so $g_{\hat{p}}$ is a rational quadratic. On the graph $z_{ij} = y_i y_j$ we have $s_{ij} = d_i d_j$, so $C_{\text{dup}} s = d \otimes d$ and $\hat{p} \otimes d + \varsigma d \otimes d = (\hat{p} + \varsigma d) \otimes d$. Taylor's formula with integral remainder and the full Hessian Gram give

$$g_{\hat{p}}(y, (y_i y_j)) = f(\hat{p}) + \nabla f(\hat{p})^\top d + \int_0^1 (1 - \varsigma) d^\top \nabla^2 f(\hat{p} + \varsigma d) d d\varsigma = f(y).$$

In particular $g_{\hat{p}}(w_*) = f(p) = 0$ for every rational $\hat{p}$.

Uniform bounds. Assume $\|\hat{p} - p\| \leq 1$, so $\|\hat{p}\| \leq P_A + 1$, and put $J_A = 4(P_A + 2)$. The quadratic part of $g_{\hat{p}}$ is $\int_0^1 (1 - \varsigma) (y, \hat{p} \otimes y + \varsigma C_{\text{dup}} s^{\text{lin}})^\top A(\cdots) d\varsigma$, where $s_{ij}^{\text{lin}} = z_{ij} - \hat{p}_i y_j - \hat{p}_j y_i$. The scalar integrals turn $\|a\|^2 + \|b + \varsigma c\|^2$ into the quadratic form of $\frac{1}{2} I \oplus \begin{pmatrix} 1/2 & 1/6 \\ 1/6 & 1/12 \end{pmatrix} \otimes I$ in (a, b, c) ; the 2×2 matrix has determinant $1/72$ and trace $7/12$, hence least eigenvalue at least $1/42$. With $A \succeq \varrho_A I$ and $\|C_{\text{dup}} s^{\text{lin}}\| \geq \|s^{\text{lin}}\|$, the quadratic part is at least $\frac{\varrho_A}{42} (\|y\|^2 + \|s^{\text{lin}}\|^2)$. The linear map $(y, z) \mapsto (y, s^{\text{lin}})$ and its inverse have norm at most $1 + 2\|\hat{p}\| \leq J_A$. Hence the quadratic-part matrix $H_{\hat{p}}$ satisfies

$$\mu_0 I \preceq H_{\hat{p}} \preceq \Lambda_0 I, \quad \mu_0 = \frac{\varrho_A}{48 J_A^2}, \quad \Lambda_0 = 10 U_A (P_A + 2)^2 J_A^2,$$

the upper bound following from $\|A\| \leq U_A$, $\|\hat{p} \otimes y\| \leq (P_A + 1)\|y\|$, $\|C_{\text{dup}}\|^2 \leq 2$, and $\|b + \varsigma c\|^2 \leq 2\|b\|^2 + 2\|c\|^2$. The point w_* has norm at most $K_A = 2P_A^2$.

The gradient at w_ .* The vector $\Phi_A(\hat{p}, y) = \nabla_{(y,z)} g_{\hat{p}}(y, (y_i y_j))$ is a polynomial in $(\hat{p}, y)$ of degree bounded by an absolute constant, with rational coefficients computed from f and A . Let $\deg \Phi_A$ be its degree, S_A the sum of the absolute values of its coefficients, and $D_A = 1 + \deg \Phi_A S_A (P_A + 2)^{\deg \Phi_A}$. At $\hat{p} = p$, the definition is a homogeneous quadratic in (d, s) , because $f(p) = 0$ and $\nabla f(p) = 0$, and d and s vanish at w_* ; so $\Phi_A(p, p) = 0$. On the box where all coordinates of $\hat{p}$ have absolute value at most $P_A + 1$, the Jacobian of Φ_A in $\hat{p}$ has norm at most D_A , so $\|\nabla g_{\hat{p}}(w_*)\| = \|\Phi_A(\hat{p}, p)\| \leq D_A \|\hat{p} - p\|$.

Residuals. Let $r_{ij}^{\text{gr}} = y_i y_j - z_{ij}$ ($i \leq j$), and let $\mathbf{a}(y, z)$ be obtained from $\nabla f(y)$ by replacing, in every cubic monomial $y_i y_j y_k$ with $i \leq j \leq k$, the factor $y_j y_k$ by z_{jk} . Then $\mathbf{a}(y, (y_i y_j)) = \nabla f(y)$, and the N quadratics $\mathbf{R} = (\mathbf{a}, r^{\text{gr}})$ have the single real common zero w_* , because $r^{\text{gr}} = 0$ restricts to the graph and ∇f has the single zero p . In the coordinates (y, r^{gr}) , related to (y, z) by a map of Jacobian determinant ± 1 , one has $\mathbf{a} = \nabla f(y) - E(y) r^{\text{gr}}$ with E linear, so the Jacobian of $\mathbf{R}$ at w_* is block triangular with diagonal blocks $\nabla^2 f(p)$ and I , and $|\det D\mathbf{R}(w_*)| = \det \nabla^2 f(p) \geq \varrho_A^n$. Let $W_0 = 1 + \sum_j \|\mathbf{R}_j\|_1$ (sums of absolute values of coefficients) and $r_j = \mathbf{R}_j / (2W_0)$. The quadratic-part matrices of the r_j have norm at most $1/2$, their gradients at w_* have norm at most $\beta_A = 1 + K_A$, the Jacobian of (r_j) at w_* has norm at most $N\beta_A$, and its determinant is at least $\varrho_A^n / (2W_0)^N$ in absolute value. Hence $\sum_j b_j b_j^\top \succeq \nu_A^2 I$ with $\nu_A = \min\{1, \varrho_A^n (2W_0)^{-N} (N\beta_A)^{-(N-1)}\}$.

Assembly. Choose a dyadic t with $\varepsilon = t^2 \leq \min\{1, \mu_0^2 / (2N), \nu_A^2 \mu_0^2 / (36N(\Lambda_0 + N\beta_A)^2)\}$, compute $\hat{p}$ with $\|\hat{p} - p\| \leq \min\{1, \varepsilon / D_A\}$, and put $g = g_{\hat{p}}$. Then g and $r_1, \dots, r_N$ satisfy the hypotheses of Lemma 7.6 at w_* in dimension N with N residuals, $\mu = \mu_0$, $\Lambda = \Lambda_0$, $\beta = \beta_A$ and $\nu = \nu_A$. The lemma gives the stated properties of B ; in particular the canonical Hessian Gram of B is computed directly from the rational factors, without a second approximation of p . Every quantity above has polynomial bit length, so the algorithm runs in polynomial time. By Theorem 7.9, B is not a sum of fewer than $N + 1$ real squares. $\square$

F Proofs for Section 8

This appendix proves the results of Section 8. We keep the notation of that section: $z(X)$ is the vector of monomials of degree at most two with the constant first, $N = \binom{n+2}{2}$, $w(X, v) = (v, X \otimes v)$, and a certified quartic is a rational quartic with a rational positive definite full Hessian Gram. For a polynomial P of degree at most two, $\text{coef}(P) \in \mathbb{R}^N$ denotes its coefficient vector, so that $P = \text{coef}(P)^\top z$. For a rational a/b in lowest terms we use $\text{bits}(a/b) = \lceil \log_2(|a| + 1) \rceil + \lceil \log_2(b + 1) \rceil$, which is at least $\log_2 \max\{1, |a|\} + \log_2 b$.

F.1 The Taylor Gram

Proof of Lemma 8.1. (a) For fixed X the polynomial $\varphi(t) = f(a + t(X - a))$ satisfies $\varphi(1) = \varphi(0) + \varphi'(0) + \int_0^1 (1 - t) \varphi''(t) dt$. With $d = X - a$ this reads

$$f(X) = f(a) + b^\top d + \int_0^1 (1 - t) d^\top \nabla^2 f(a + td) d dt.$$

By (18) with the point $a + td$ and the direction d , the integrand is $(1 - t) w(a + td, d)^\top A w(a + td, d)$, and $w(a + td, d) = (d, (a + td) \otimes d) = U + tV$. Since $\int_0^1 (1 - t) dt = \frac{1}{2}$, $\int_0^1 (1 - t)t dt = \frac{1}{6}$, and $\int_0^1 (1 - t)t^2 dt = \frac{1}{12}$, the integral equals

$$\frac{1}{2} U^\top A U + \frac{1}{3} U^\top A V + \frac{1}{12} V^\top A V = \frac{1}{2} (U + V/3)^\top A (U + V/3) + \frac{1}{36} V^\top A V,$$

where symmetry of A was used. Substituting $U = C_U(a)z(X)$ and $V = C_V(a)z(X)$ gives $z(X)^\top \mathcal{T}_a(A)z(X)$. The same computation shows, for every symmetric B of order $n + n^2$,

$$z(X)^\top \mathcal{T}_a(B)z(X) = \int_0^1 (1-t) (U + tV)^\top B (U + tV) dt. \quad (64)$$

(b) The matrix $\mathcal{T}_a(A)$ is a sum of congruences of A with positive scalar factors, so it is positive semidefinite when A is. Suppose $A \succ 0$. Then $\xi^\top \mathcal{T}_a(A)\xi = 0$ holds exactly when $(C_U(a) + \frac{1}{3}C_V(a))\xi = 0$ and $C_V(a)\xi = 0$, that is, when $C_U(a)\xi = 0$ and $C_V(a)\xi = 0$. The rows of $C_U(a)$ and $C_V(a)$ are the coefficient vectors of the polynomials d_i , $a_i d_j$, and $d_i d_j$ for $i, j \in \{1, \dots, n\}$. Each of them vanishes at a , and the d_i together with the $d_i d_j$, $i \leq j$, form a basis of the hyperplane $\mathcal{H}_a = \{P : \deg P \leq 2, P(a) = 0\}$, since they are the nonconstant monomials in the translated variables. Because $P(a) = \text{coef}(P)^\top z(a)$, the coefficient vectors of $\mathcal{H}_a$ form the hyperplane $z(a)^\perp$. Hence ξ lies in the kernel exactly when $\xi \perp z(a)^\perp$, that is, when $\xi \in \mathbb{R}z(a)$. For the translated form, let S be the invertible matrix with $Sz(X) = (1, (d^\alpha)_{1 \leq |\alpha| \leq 2})$. Then $Sz(a)$ is the first unit vector, so the symmetric matrix $S^{-\top} \mathcal{T}_a(A) S^{-1}$ has first row and column zero and a positive definite complementary block.

(c) Since $A \succeq \mu I$, the matrix $\bar{A} = A - \mu \text{diag}(I_n, 0)$ satisfies $\bar{A} \succeq \mu \text{diag}(0, I_{n^2}) \succeq 0$. The map $B \mapsto \mathcal{T}_a(B)$ is linear, and (64) with $B = \text{diag}(I_n, 0)$ gives $z^\top \mathcal{T}_a(\text{diag}(I_n, 0))z = \int_0^1 (1-t) \|d\|^2 dt = \frac{1}{2} \|d\|^2$, because the first block of $U + tV$ is d . Therefore part (a) gives

$$f = z^\top \mathcal{T}_a(\bar{A})z + f(a) + b^\top d + \frac{\mu}{2} \|d\|^2 = z^\top \mathcal{T}_a(\bar{A})z + \frac{\mu}{2} \|d + b/\mu\|^2 + c_a.$$

As $z_0 = 1$, this is $f = z^\top \mathcal{Q}_a z$. All three summands of $\mathcal{Q}_a$ are positive semidefinite when $c_a \geq 0$. Suppose $c_a > 0$. The first n rows of $C_V(a)$ vanish, so $\mathcal{T}_a(\bar{A}) \succeq \frac{1}{36} C_V(a)^\top \bar{A} C_V(a) \succeq \frac{\mu}{36} C_V(a)^\top C_V(a)$, and

$$\mathcal{Q}_a \succeq \frac{\mu}{36} C_V(a)^\top C_V(a) + \frac{\mu}{2} C_{\text{aff}}(a)^\top C_{\text{aff}}(a) + c_a e_0 e_0^\top.$$

A vector ξ in the kernel of the right side is orthogonal to the coefficient vectors of every $d_i d_j$, every $d_i + b_i/\mu$, and the constant polynomial 1. These polynomials span all polynomials of degree at most two, because $d_i = (d_i + b_i/\mu) - (b_i/\mu) \cdot 1$ and the constant, the d_i , and the $d_i d_j$ form a basis. Hence $\xi = 0$ and $\mathcal{Q}_a \succ 0$.

(d) The entries of $C_U(a)$ and $C_V(a)$ are integer polynomials in the coordinates of a . The vector $b = \nabla f(a)$ has entries in E , so $C_{\text{aff}}(a)$ and c_a have entries in E when $\mu \in E$. The matrices $\mathcal{T}_a(A)$ and $\mathcal{Q}_a$ are obtained from these and from A by ring operations and rational constants. At a critical point $b = 0$, and part (a) reads $f - f(a) = z^\top \mathcal{T}_a(A)z$.

(e) Substituting $A = \sum_k s_k \ell_k \ell_k^\top$ into the integral of part (a) gives $\sum_k s_k \int_0^1 (1-t)(u_k + tv_k)^2 dt$ with $u_k = \ell_k^\top U$ and $v_k = \ell_k^\top V$. The scalar identity $\int_0^1 (1-t)(u + tv)^2 dt = \frac{1}{2}(u + v/3)^2 + (v/6)^2$ and $\frac{1}{2}y^2 = 2(y/2)^2$ give the displayed sum. If $A \succeq 0$ is rational, rational symmetric elimination gives $A = \sum_k s_k \ell_k \ell_k^\top$ with rational $s_k \geq 0$ and rational ℓ_k , and each positive s_k is a sum of rational squares (Lemma 2.20): if $s_k = \sum_l \sigma_{kl}^2$ with rational σ_{kl} , then $s_k y^2 = \sum_l (\sigma_{kl} y)^2$, and $2y^2 = y^2 + y^2$. Every resulting square factor is a rational multiple of $\ell_k^\top (U + V/3)$ or of $\ell_k^\top V$. The entries of U and V are d_i , $a_i d_j$, and $d_i d_j$, whose coefficients are integer polynomials of degree at most two in the coordinates of a , and ℓ_k is rational; this proves the statement about coefficients. At a critical point $b = 0$, so $f - f(a)$ is a sum of squares of polynomials with coefficients in $\mathbb{Q}(a_1, \dots, a_n)$. $\square$

F.2 The realization interface and the cube-root chain

All quartics in Section 8 are assembled by Lemma 7.6, applied with as many residuals as variables. We restate its hypotheses in the form used below; our lower curvature bound m is the parameter

denoted μ in that lemma. Let $n \geq 1$ and $p \in \mathbb{R}^n$, and let $G, r_1, \dots, r_n$ be rational quadratic polynomials vanishing at p , with exact translated forms

$$G(p+u) = \ell^\top u + u^\top H u, \quad r_j(p+u) = b_j^\top u + u^\top T_j u.$$

Suppose positive rationals m, Λ, β, ν and a rational $t > 0$ satisfy

$$mI \preceq H, \quad \|H\| \leq \Lambda, \quad \|T_j\| \leq 1, \quad \|b_j\| \leq \beta, \quad \sum_j b_j b_j^\top \succeq \nu^2 I, \quad \|\ell\| \leq \varepsilon := t^2, \quad (65)$$

$$\varepsilon \leq \min\left\{1, \frac{m^2}{2n}, \frac{\nu^2 m^2}{36n(\Lambda + n\beta)^2}\right\}. \quad (66)$$

Then $F = (G/(t\nu))^2 + \sum_{j=1}^n (r_j/\nu)^2$ is a rational quartic of degree exactly four, a sum of $n+1$ squares of rational quadratics, with $\nabla^2 F \succeq \frac{3}{2}I$ on $\mathbb{R}^n$; the point p is its unique zero and unique minimizer; and the canonical Hessian Gram of the displayed squares is a rational positive definite full Hessian Gram of F , computed from the rational factors by polynomially many rational operations on numbers of the input bit length. No further approximation of p is needed for the Gram.

Lemma F.1 (The cube-root chain). *Let $k \geq 1$, $n = 2k$, $M_k = 1000^{k+3}$, and let δ_i , ξ_i , and p be as in (21) and (22). Then $0 < \delta_i \leq \delta_{i-1}^2$ for $1 \leq i \leq k$; in particular $0 < \delta_k \leq M_k^{-2^k}$ and $1 < \xi_i \leq 1 + M_k^{-2}$. The number ξ_1 is irrational. In time polynomial in k one can construct rational quadratics $F_{k,1}, \dots, F_{k,n+1}$ and a rational matrix A_k , all of binary length polynomial in k , such that $F_k = \sum_j F_{k,j}^2$ satisfies $\nabla^2 F_k \succeq \frac{3}{2}I$, $F_k(p) = 0$, and $F_k(X) > 0$ for $X \neq p$, and A_k is a positive definite full Hessian Gram of F_k .*

Proof. The chain. Since $(1 + \delta_i)^3 = 1 + 3\delta_{i-1}^2 > 1$, every δ_i is positive. As $(1+t)^3 \geq 1+3t$ for $t \geq 0$, $1 + 3\delta_i \leq 1 + 3\delta_{i-1}^2$. Hence $\delta_i \leq \delta_{i-1}^2 = M_k^{-2^i}$, which is at most $M_k^{-2} \leq 10^{-24}$ for $i \geq 1$. The gate equations (22) follow from $1 + 3\delta_{i-1}^2 = 1 + 3(\xi_{i-1} - 1)^2$. For irrationality, note that $M_k^2 = K_0^3$ with $K_0 = 10^{2(k+3)}$, so $\xi_1 = (K_0^3 + 3)^{1/3}/K_0$. A rational cube root of an integer is an integer, and $K_0^3 < K_0^3 + 3 < (K_0 + 1)^3$, so ξ_1 is irrational.

Residuals. Put $c_1 = 1 + 3M_k^{-2}$ and $c_i(X) = 3y_{i-1} - 6x_{i-1} + 4$ for $2 \leq i \leq k$, and define the n rational quadratics

$$r_i = x_i^2 - y_i, \quad s_i = x_i y_i - c_i \quad (1 \leq i \leq k).$$

On their common real zero set $y_i = x_i^2$ and $x_i^3 = c_i$. By induction on i and uniqueness of real cube roots, $x_i = \xi_i$; hence p is the unique common real zero. With $u = X - p$ and coordinates u_{x_i}, u_{y_i} ,

$$\begin{aligned} r_i(p+u) &= 2\xi_i u_{x_i} - u_{y_i} + u_{x_i}^2, \\ s_i(p+u) &= \xi_i^2 u_{x_i} + \xi_i u_{y_i} + [i \geq 2] (6u_{x_{i-1}} - 3u_{y_{i-1}}) + u_{x_i} u_{y_i}. \end{aligned}$$

The quadratic parts are $e_{x_i} e_{x_i}^\top$ and $\frac{1}{2}(e_{x_i} e_{y_i}^\top + e_{y_i} e_{x_i}^\top)$, of norms one and one half. Since $1 < \xi_i < 1.001$, the gradients at p satisfy $\|\nabla r_i(p)\|^2 = 4\xi_i^2 + 1 < 6$ and $\|\nabla s_i(p)\|^2 \leq \xi_i^4 + \xi_i^2 + 45 < 48$, so $\beta = 7$ bounds all of them. The Jacobian J , with rows $\nabla r_1(p)^\top, \nabla s_1(p)^\top, \dots$ and columns ordered as X , is block lower triangular with diagonal blocks $\begin{pmatrix} 2\xi_i & -1 \\ \xi_i^2 & \xi_i \end{pmatrix}$ of determinant $3\xi_i^2 > 3$. Hence $|\det J| > 1$, and $\|J\|^2 \leq \|J\|_F^2 < 54k \leq (8k)^2$. Since $|\det J|$ is the product of the singular values, $\sigma_{\min}(J) \geq |\det J| / \|J\|^{n-1} > (8k)^{-(n-1)}$. Thus $\sum(\nabla r_i \nabla r_i^\top + \nabla s_i \nabla s_i^\top) = J^\top J \succeq \nu^2 I$ with $\nu = (8k)^{-(2k-1)}$.

An exact exposing quadratic. For $1 \leq i \leq k$ let

$$E_i = y_i^2 - c_i x_i - \xi_i s_i + \xi_i^2 r_i.$$

Each of $y_i^2 - c_i x_i$, s_i , and r_i vanishes at p , hence so does E_i . Expanding, $E_i = y_i^2 - \xi_i x_i y_i + \xi_i^2 x_i^2 - \xi_i^2 y_i - c_i(x_i - \xi_i)$. For a real α let $P_\alpha(x, y) = \alpha^2(x - \alpha)^2 + \alpha(x - \alpha)(y - \alpha x) + (y - \alpha x)^2$, which expands to $y^2 - \alpha xy + \alpha^2 x^2 - \alpha^2 y - \alpha^3 x + \alpha^4$. Therefore

$$E_i = P_{\xi_i}(x_i, y_i) + (\xi_i - x_i)(c_i - \xi_i^3).$$

At $p + u$ we have $x_i - \xi_i = u_{x_i}$, $y_i - \xi_i x_i = u_{y_i} - \xi_i u_{x_i}$, and $c_i(p + u) - \xi_i^3 = c_i(p + u) - c_i(p)$, which is $-6u_{x_{i-1}} + 3u_{y_{i-1}}$ for $i \geq 2$ and zero for $i = 1$. Hence

$$E_i(p + u) = \xi_i^2 u_{x_i}^2 - \xi_i u_{x_i} u_{y_i} + u_{y_i}^2 + [i \geq 2] (6u_{x_i} u_{x_{i-1}} - 3u_{x_i} u_{y_{i-1}}). \quad (67)$$

In particular E_i has neither a constant nor a linear term at p . The local matrix $H_i = \begin{pmatrix} \xi_i^2 & -\xi_i/2 \\ -\xi_i/2 & 1 \end{pmatrix}$ satisfies $\frac{1}{2}I \preceq H_i \preceq 2I$: the matrix $H_i - \frac{1}{2}I$ has positive diagonal and determinant $(\xi_i^2 - 1)/4 \geq 0$, and $\text{tr } H_i < 2.01$.

Let $\omega_i = 2^{-10(i-1)}$ and $E_* = \sum_i \omega_i E_i$, so that $E_*(p + u) = u^\top H_* u$. Write $U_i = (u_{x_i}, u_{y_i})$ and $\tilde{U}_i = \sqrt{\omega_i} U_i$. The local terms contribute $\sum_i \tilde{U}_i^\top H_i \tilde{U}_i$, which lies between $\frac{1}{2} \|\tilde{U}\|^2$ and $2 \|\tilde{U}\|^2$. The cross term of index $i \geq 2$ has absolute value at most

$$\sqrt{45} \omega_i |U_i| |U_{i-1}| = \sqrt{45} 2^{-5} |\tilde{U}_i| |\tilde{U}_{i-1}| \leq \frac{\sqrt{45}}{64} (|\tilde{U}_i|^2 + |\tilde{U}_{i-1}|^2),$$

since $\omega_i/\omega_{i-1} = 2^{-10}$. Summing over i , all cross terms together have absolute value at most $\frac{\sqrt{45}}{32} \|\tilde{U}\|^2 < 0.21 \|\tilde{U}\|^2$. Hence $\frac{1}{4} \|\tilde{U}\|^2 \leq u^\top H_* u \leq \frac{5}{2} \|\tilde{U}\|^2$, and $\omega_k \|u\|^2 \leq \|\tilde{U}\|^2 \leq \|u\|^2$ gives

$$\frac{\omega_k}{4} I \preceq H_* \preceq \frac{5}{2} I. \quad (68)$$

Rational coefficients. Let $0 < \eta \leq 10^{-3}$, and let $\hat{\xi}_i$ be rationals with $|\hat{\xi}_i - \xi_i| \leq \eta$. Define the rational quadratic

$$\hat{G} = \sum_{i=1}^k \omega_i (y_i^2 - c_i x_i - \hat{\xi}_i s_i + \hat{\xi}_i^2 r_i).$$

The difference $\hat{G} - E_* = \sum_i \omega_i (-(\hat{\xi}_i - \xi_i)s_i + (\hat{\xi}_i^2 - \xi_i^2)r_i)$ is a combination of residuals, so $\hat{G}(p) = 0$ exactly. Write $\hat{G}(p + u) = \ell^\top u + u^\top H u$. Since $|\hat{\xi}_i^2 - \xi_i^2| \leq \eta(2\xi_i + \eta) \leq 3\eta$ and $\sum_i \omega_i < 1.001$,

$$\|\ell\| \leq \sum_i \omega_i (\eta \|\nabla s_i(p)\| + 3\eta \|\nabla r_i(p)\|) < 15\eta.$$

The matrix $H - H_*$ is block diagonal, and its block i is $\omega_i (-(\hat{\xi}_i - \xi_i)T_{s_i} + (\hat{\xi}_i^2 - \xi_i^2)T_{r_i})$, where T_{r_i}, T_{s_i} are the quadratic parts above. Hence $\|H - H_*\| \leq \frac{7}{2}\eta$.

Put $m = \omega_k/8$, $\Lambda = 3$, $\beta = 7$, and $\nu = (8k)^{-(2k-1)}$. Let $t = 2^{-\tau}$ for the least integer $\tau \geq 0$ such that $\varepsilon = 4^{-\tau}$ satisfies (66), and let η be the largest power of two with $\eta \leq \min\{10^{-3}, \omega_k/32, \varepsilon/15\}$. Then (68) and the perturbation bounds give $H \succeq (\frac{1}{4} - \frac{7}{64})\omega_k I \succeq mI$, $H \preceq (\frac{5}{2} + \frac{7}{64})I \preceq \Lambda I$, and $\|\ell\| < 15\eta \leq \varepsilon$. Hence (65) and (66) hold for $\hat{G}$ and the residuals $r_1, s_1, \dots, r_k, s_k$.

Computing $\hat{\xi}_i$. Let $\phi(s) = (1 + 3s^2)^{1/3} - 1$, so that $\delta_i = \phi(\delta_{i-1})$ and $|\phi'(s)| = 2|s|(1 + 3s^2)^{-2/3} \leq 2|s|$. Put $\hat{\delta}_0 = \delta_0$. Given $\hat{\delta}_{i-1}$, compute by bisection on $[1, 2]$, with exact rational comparisons of ρ^3 against $1 + 3\hat{\delta}_{i-1}^2$, a dyadic ρ_i within $\eta/2$ of $(1 + 3\hat{\delta}_{i-1}^2)^{1/3}$, and put $\hat{\delta}_i = \rho_i - 1$. If $|\hat{\delta}_{i-1} - \delta_{i-1}| \leq \eta$, then $|\hat{\delta}_{i-1}|, |\delta_{i-1}| \leq M_k^{-1} + \eta < \frac{1}{500}$, so the mean value theorem gives

$$|\hat{\delta}_i - \delta_i| \leq \frac{\eta}{2} + |\phi(\hat{\delta}_{i-1}) - \phi(\delta_{i-1})| \leq \frac{\eta}{2} + \frac{2}{500}\eta \leq \eta.$$

By induction $\hat{\xi}_i = 1 + \hat{\delta}_i$ satisfies $|\hat{\xi}_i - \xi_i| \leq \eta$ for all i . The numbers $\log_2(1/\nu)$, $\log_2(1/m)$, $\log_2(1/\varepsilon)$, and $\log_2(1/\eta)$ are $O(k \log k)$. Every $\hat{\delta}_i$ is a dyadic rational with $O(k \log k)$ bits, each bisection uses $O(k \log k)$ comparisons of rationals of that length, and so all $\hat{\xi}_i$ are computed in time polynomial in k .

Assembly. The realization interface yields $F_k = (\hat{G}/(t\nu))^2 + \sum_i ((r_i/\nu)^2 + (s_i/\nu)^2)$, a sum of $n+1$ squares with $\nabla^2 F_k \succeq \frac{3}{2}I$ and unique zero p , and the rational positive definite canonical Hessian Gram A_k of these factors. Each factor has $O(k)$ monomials with coefficients of $O(k \log k)$ bits, so F_k , its square factors, and A_k have binary length polynomial in k . Since F_k is a sum of squares that vanish simultaneously only where all residuals vanish, $F_k(X) > 0$ for $X \neq p$. $\square$

F.3 Proof of Theorem 8.3

Proof of Theorem 8.3. Lemma 2.8(a) and $B_k \succeq I$ give $\nabla^2 g_k \succeq I$. Since $F_k \geq 0$ vanishes at p , we have $\nabla F_k(p) = 0$, so

$$g_k(p) = -\delta_k^2 < 0, \quad \nabla g_k(p) = -2\delta_k e_{x_k}, \quad \|\nabla g_k(p)\| = 2\delta_k.$$

For any X with $r = \|X - p\|$, strong convexity gives $g_k(X) \geq -\delta_k^2 - 2\delta_k r + \frac{1}{2}r^2$. If $g_k(X) \leq 0$, the quadratic inequality forces $r \leq (2 + \sqrt{6})\delta_k$. Hence every point of S_k , including every boundary point, satisfies

$$\|X - p\| \leq (2 + \sqrt{6})\delta_k < 5M_k^{-2^k}. \quad (69)$$

Thus S_k is bounded; it is closed and convex, and it contains the open set $\{g_k < 0\} \ni p$. For distinct $X, Y \in S_k$ and $0 < t < 1$, strong convexity gives $g_k(tX + (1-t)Y) < 0$, so S_k is strictly convex. The coordinates of p lie in $(1, 1 + 3M_k^{-2}]$, and $5M_k^{-2^k} \leq 5 \cdot 10^{-24}$, which proves the box inclusion.

For (b), let $X \in S_k$ have rational first coordinate a/b . The first coordinate of p is ξ_1 , which is irrational (Lemma F.1; M_k^2 is a perfect cube and $M_k^2 + 3$ is not). Hence $(a/b)^3 \neq \xi_1^3 = 1 + 3M_k^{-2}$, and $M_k^2 a^3 - (M_k^2 + 3)b^3$ is a nonzero integer. Both a/b and ξ_1 lie in $(0.99, 1.01)$, so (69) gives

$$\begin{aligned} 1 &\leq |M_k^2 a^3 - (M_k^2 + 3)b^3| = M_k^2 b^3 \left| \frac{a}{b} - \xi_1 \right| \left| \frac{a^2}{b^2} + \frac{a}{b} \xi_1 + \xi_1^2 \right| \\ &< M_k^2 b^3 \cdot 5M_k^{-2^k} \cdot 3.1 < 16b^3 M_k^{2-2^k}. \end{aligned}$$

Taking logarithms gives the displayed bound, since $\log_2 M_k = (k+3)\log_2 1000$. For $k \geq 2$ we have $2^k - 2 \geq 2^{k-1}$ and $\log_2 1000 > 9.96$, so the right side exceeds $2^{k-1}(3.3k + 9.9) - 2 > k2^k$. Part (c) is Proposition 4.6, applied to g_k with the curvature bound $\mu = 1$ certified by $B_k \succeq I$, since $\min g_k \leq g_k(p) < 0$. The size of (g_k, B_k) is polynomial in k by Lemma F.1 and Lemma 2.7. $\square$

F.4 Circle chains and the proof of Theorem 8.5

Both quartics of Theorem 8.5 come from one family of circle chains. Fix $k \geq 0$, $n = 2(k+1)$, and a ratio $\rho \in \{1, \frac{1}{2}\}$. Put $R_j = \rho^j$ and $c_j = R_j/R_{j-1}^2$ for $j \geq 1$; thus $c_j = 1$ when $\rho = 1$ and $c_j = 2^{j-2}$ when $\rho = \frac{1}{2}$. Let $z_j = ((3+4i)/5)^{2^j} = a_j + ib_j$, and define the rational residuals

$$r_0 = x_0 - \frac{3}{5}, \quad s_0 = y_0 - \frac{4}{5}, \quad r_j = x_j - c_j(x_{j-1}^2 - y_{j-1}^2), \quad s_j = y_j - 2c_j x_{j-1} y_{j-1} \quad (1 \leq j \leq k), \quad (70)$$

which reduce to (24) when $\rho = 1$. Put $P = (R_j a_j, R_j b_j)_{j=0}^k$, $q_j = x_j^2 + y_j^2 - R_j^2$, $\kappa_j = 2c_j^2 R_{j-1}^2$, and $t_{j-1} = (-b_{j-1}, a_{j-1})$. Write $u = X - P$, with $u_j \in \mathbb{R}^2$ the displacement of the pair (x_j, y_j) .

Lemma F.2 (Circle chains). *In this setting the following hold.*

(a) P is the unique common real zero of the residuals, and every q_j vanishes at P .

(b) For $1 \leq j \leq k$ the rational-coefficient quadratic $E_j = q_j - 2R_j a_j r_j - 2R_j b_j s_j - \kappa_j q_{j-1}$ satisfies

$$E_j(P + u) = \|u_j\|^2 - 2\kappa_j (t_{j-1}^\top u_{j-1})^2.$$

Here $\kappa_j = 2$ when $\rho = 1$ and $\kappa_j = \frac{1}{2}$ when $\rho = \frac{1}{2}$.

(c) The Jacobian of $(r_0, s_0, \dots, r_k, s_k)$ at P is $J = I - T$, where T is block strictly lower triangular, its only nonzero blocks are $T_{j,j-1} = 2c_j R_{j-1} \begin{pmatrix} a_{j-1} & -b_{j-1} \\ b_{j-1} & a_{j-1} \end{pmatrix}$, and $2c_j R_{j-1}$ equals 2 when $\rho = 1$ and 1 when $\rho = \frac{1}{2}$. In particular $T^{k+1} = 0$.

Proof. (a) The residuals force $(x_0, y_0) = (3/5, 4/5)$ and then, inductively, $x_j + iy_j = c_j(x_{j-1} + iy_{j-1})^2$. Since $c_j(R_{j-1}z_{j-1})^2 = c_j R_{j-1}^2 z_j = R_j z_j$, the unique solution is P . Also $|R_j z_j| = R_j$, so $q_j(P) = 0$.

(b) Fix $j \geq 1$ and write $(a, b) = (a_{j-1}, b_{j-1})$, $(\alpha', \beta') = R_{j-1}(a, b)$ for the predecessor point, $(A', B') = R_j(a_j, b_j) = c_j(\alpha'^2 - \beta'^2, 2\alpha'\beta')$ for the successor point, $c = c_j$, $u_{j-1} = (u_1, u_2)$, and $u_j = (w_1, w_2)$. Then

$$\begin{aligned} q_j(P + u) &= 2A'w_1 + 2B'w_2 + \|u_j\|^2, \\ r_j(P + u) &= w_1 - c(2\alpha'u_1 - 2\beta'u_2 + u_1^2 - u_2^2), \\ s_j(P + u) &= w_2 - 2c(\beta'u_1 + \alpha'u_2 + u_1u_2), \\ q_{j-1}(P + u) &= 2\alpha'u_1 + 2\beta'u_2 + \|u_{j-1}\|^2. \end{aligned}$$

In $E_j = q_j - 2A'r_j - 2B's_j - \kappa_j q_{j-1}$ the terms linear in u_j cancel, and there are no products of u_j with u_{j-1} . The identities $A'\alpha' + B'\beta' = c\alpha'R_{j-1}^2$ and $-A'\beta' + B'\alpha' = c\beta'R_{j-1}^2$ show that the terms linear in u_{j-1} are $(4c^2R_{j-1}^2 - 2\kappa_j)(\alpha'u_1 + \beta'u_2) = 0$. The quadratic terms in u_{j-1} have matrix $2c \begin{pmatrix} A' & B' \\ B' & -A' \end{pmatrix} - \kappa_j I$. Since $2cR_j = 2c^2R_{j-1}^2 = \kappa_j$, this matrix equals $\kappa_j \begin{pmatrix} a_{j-1} & b_{j-1} \\ b_{j-1} & -a_{j-1} \end{pmatrix}$, and $a_j = a^2 - b^2$, $b_j = 2ab$, $a^2 + b^2 = 1$ turn it into $-2\kappa_j \begin{pmatrix} b^2 & -ab \\ -ab & a^2 \end{pmatrix} = -2\kappa_j t_{j-1} t_{j-1}^\top$. The constant term vanishes by (a). The values of κ_j follow from $c_j = 1$, $R_{j-1} = 1$ when $\rho = 1$, and from $c_j R_{j-1} = 2^{j-2} \cdot 2^{-(j-1)} = \frac{1}{2}$ when $\rho = \frac{1}{2}$.

(c) The derivatives of (r_j, s_j) with respect to (x_{j-1}, y_{j-1}) at P form the matrix $-2c_j \begin{pmatrix} \alpha' & -\beta' \\ \beta' & \alpha' \end{pmatrix}$, which is $-T_{j,j-1}$; the derivatives with respect to (x_j, y_j) form the identity, and no other derivative is nonzero. $\square$

Proof of Theorem 8.5. Both quartics are assembled by the realization interface of Appendix F.2.

Approximating the chain. Let $0 < \eta \leq 1$ be rational. Put $\hat{z}_0 = z_0$ and let $\hat{z}_j$ be the complex number obtained from $\hat{z}_{j-1}^2$ by rounding its real and imaginary parts to multiples of $h = 2^{-P'}$, where P' is the least integer with $3^k h \leq \eta$. Let $e_j = |\hat{z}_j - z_j|$. If $e_{j-1} \leq 1$, then $|\hat{z}_{j-1}^2 - z_{j-1}^2| \leq (|\hat{z}_{j-1}| + |z_{j-1}|)e_{j-1} \leq 3e_{j-1}$, and rounding adds at most $\sqrt{2}h < 2h$, so $e_j \leq 3e_{j-1} + 2h$. Since $e_0 = 0$, induction gives $e_j \leq (3^j - 1)h \leq \eta$ for all $j \leq k$. For $j \geq 1$ the real and imaginary parts of $\hat{z}_j$ are dyadic rationals of absolute value at most two with $O(k + \log(1/\eta))$ bits; the initial parts $\frac{3}{5}$ and $\frac{4}{5}$ of $\hat{z}_0$ have constant bit length. The computation takes time polynomial in $k + \log(1/\eta)$. Multiplying by R_j gives rational approximations of $R_j a_j$ and $R_j b_j$ with errors at most η .

The exposing quadratic. With weights $\omega_j > 0$ to be chosen, let $E_0 = r_0^2 + s_0^2$ and

$$\hat{G} = \omega_0 E_0 + \sum_{j=1}^k \omega_j (q_j - 2\hat{\alpha}_j r_j - 2\hat{\beta}_j s_j - \kappa_j q_{j-1}),$$

where $\hat{\alpha}_j, \hat{\beta}_j$ are the approximations of $R_j a_j, R_j b_j$. Only coefficients of residuals are approximated, so $\hat{G}(P) = 0$. Let H_* be the matrix of the exact sum $\omega_0 E_0 + \sum_j \omega_j E_j$; by Lemma F.2(b) it is block diagonal, with block $\omega_j I - 2\kappa_{j+1}\omega_{j+1}t_j t_j^\top$ for $j < k$ and $\omega_k I$ for $j = k$. Write $\hat{G}(P+u) = \ell^\top u + u^\top H u$, and let $\Delta_j = (\hat{\alpha}_j - R_j a_j, \hat{\beta}_j - R_j b_j)$, so $\|\Delta_j\| \leq \sqrt{2}\eta$. The quadratic parts of r_j and s_j are $-c_j \text{diag}(1, -1)$ and $-c_j \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ on the pair $j-1$. Hence $H - H_*$ is block diagonal with blocks of norm $2\omega_j c_j \|\Delta_j\| \leq 2\sqrt{2}\omega_j c_j \eta$, and $\|\ell\| \leq \sum_{j \geq 1} 2\omega_j \|\Delta_j\|_1 \max\{\|\nabla r_j(P)\|, \|\nabla s_j(P)\|\}$. By Lemma F.2(c), $\|\nabla r_j(P)\| = \|\nabla s_j(P)\| = (1 + (2c_j R_{j-1})^2)^{1/2}$ for $j \geq 1$, and $\|\nabla r_0\| = \|\nabla s_0\| = 1$.

The quartic f_k ($\rho = 1$). Take $\omega_j = 8^{-j}$. Each block of H_* with $j < k$ is at least $(\omega_j - 4\omega_{j+1})I = \frac{\omega_j}{2}I$ and at most $\omega_j I$, so $8^{-k}I \preceq H_* \preceq I$. The residual gradients have norms at most $\sqrt{5} < 3$, and $\|H - H_*\| \leq 2\sqrt{2}\eta/8 < \eta/2$, $\|\ell\| \leq 4\sqrt{5}\eta \sum_{j \geq 1} 8^{-j} < 2\eta$. By Lemma F.2(c), $\|T\| \leq 2$ and $J^{-1} = \sum_{i=0}^k T^i$, so $\|J^{-1}\| < 2^{k+1}$ and $J^\top J \succeq 4^{-(k+1)}I$. Put $m = 8^{-k}/2$, $\Lambda = 2$, $\beta = 3$, and $\nu = 2^{-(k+1)}$. Choose the dyadic t as in the cube-root construction, so that $\varepsilon = t^2$ satisfies (66), and take $\eta = \min\{8^{-k}, \varepsilon/2\}$. Then $H \succeq mI$, $H \preceq \Lambda I$, and $\|\ell\| \leq \varepsilon$, so the realization interface applies to $\hat{G}$ and the n residuals. It yields

$$f_k = \left(\frac{\hat{G}}{t\nu}\right)^2 + \sum_{j=0}^k \left[\left(\frac{r_j}{\nu}\right)^2 + \left(\frac{s_j}{\nu}\right)^2\right],$$

a sum of $n+1$ rational squares with $\nabla^2 f_k \succeq \frac{3}{2}I$, unique zero and minimizer $p = P$, and a rational positive definite Hessian Gram. Here $\log_2(1/m)$, $\log_2(1/\nu)$, $\log_2(1/\varepsilon)$, and $\log_2(1/\eta)$ are $O(k)$, so every coefficient has $O(k)$ bits and the construction takes time polynomial in k . Part (b) follows from Lemma 8.4 with $m = 2^k$, because the last two coordinates of p are a_k and b_k .

The quartic $\tilde{f}_k$ ($\rho = \frac{1}{2}$). Take $\omega_j = k+1-j$. Each block of H_* with $j < k$ is $\omega_j I - \omega_{j+1}t_j t_j^\top$, whose eigenvalues are ω_j and $\omega_j - \omega_{j+1} = 1$, and the last block is I ; hence $I \preceq H_* \preceq (k+1)I$. Let $C = \max\{1, 2^{k-2}\}$, so $c_j \leq C$. The residual gradients have norm $\sqrt{2}$ for $j \geq 1$, hence $\|H - H_*\| \leq 2\sqrt{2}kC\eta < 3kC\eta$ and $\|\ell\| \leq 4\sqrt{2}\eta \sum_{j \geq 1} \omega_j < 3k(k+1)\eta$. Divide every residual by C : the quadratic parts of r_j/C and s_j/C have norm $c_j/C \leq 1$, their gradients have norm at most $\beta = 2/C$, and, since $\|T\| \leq 1$ by Lemma F.2(c), $\|J^{-1}\| \leq k+1$, so $J^\top J/C^2 \succeq \nu^2 I$ with $\nu = 1/(C(k+1))$. Put $m = \frac{1}{2}$ and $\Lambda = k+2$. Choose t as before and $\eta = \min\{1, 1/(6kC), \varepsilon/(3k(k+1))\}$ (for $k=0$ no approximation is needed). Then $H \succeq \frac{1}{2}I$, $H \preceq (k+2)I$, and $\|\ell\| \leq \varepsilon$, and the interface yields

$$\tilde{f}_k = \left(\frac{\hat{G}}{t\nu}\right)^2 + \sum_{j=0}^k \left[\left(\frac{r_j}{C\nu}\right)^2 + \left(\frac{s_j}{C\nu}\right)^2\right] = \left(\frac{\hat{G}}{t\nu}\right)^2 + (k+1)^2 \sum_{j=0}^k (r_j^2 + s_j^2),$$

with $\nabla^2 \tilde{f}_k \succeq \frac{3}{2}I$, unique zero and minimizer $\tilde{p} = P$, and a rational positive definite Hessian Gram. All coefficients have $O(k \log k)$ bits. The coordinates of $\tilde{p}$ are $2^{-j}a_j$ and $2^{-j}b_j$; dividing by a power of two cannot remove the factor 5^{2^k} from the reduced denominators of a_k and b_k , and $\|\tilde{p}\|^2 = \sum_j 4^{-j} < \frac{4}{3}$.

The Hessian at $\tilde{p}$. All square factors vanish at $\tilde{p}$, so $\nabla^2(\phi^2)(\tilde{p}) = 2\nabla\phi(\tilde{p})\nabla\phi(\tilde{p})^\top$ for each factor ϕ , and $C\nu = 1/(k+1)$ gives

$$\nabla^2 \tilde{f}_k(\tilde{p}) = \frac{2\ell\ell^\top}{\varepsilon\nu^2} + 2(k+1)^2 J^\top J.$$

Since $\|T\| \leq 1$, we have $\|J\| \leq 2$ and $\|J^{-1}\| \leq k+1$, so the second term lies between $2I$ and $8(k+1)^2 I$. The first term is positive semidefinite with norm at most $2\|\ell\|^2/(\varepsilon\nu^2) \leq 2\varepsilon/\nu^2 \leq m^2/(18n(\Lambda+n\beta)^2) < 1$ by (66). This proves (c). Finally, every ordinary representation of p or $\tilde{p}$ contains a coordinate whose reduced denominator is at least 5^{2^k} , and $\log_2 5^{2^k} = 2^k \log_2 5$; the inputs have length polynomial in k . $\square$

F.5 Moments, optimal Grams, and exposing matrices

We use two elementary facts about positive semidefinite matrices. If $P, R \succeq 0$ and $\text{tr}(PR) = 0$, then $\text{tr}(PR) = \|R^{1/2}P^{1/2}\|_F^2$ forces $R^{1/2}P^{1/2} = 0$, hence $PR = 0$ and $\text{range } R \subseteq \ker P$. If $P \succeq 0$ and $v^\top P v = 0$, then $Pv = 0$.

Proof of Theorem 8.6. (a) Since $\nabla f(p) = 0$, Lemma 8.1 with $a = p$ gives $f - f_* = z^\top Q_* z$, where $Q_* = \mathcal{T}_p(A) \succeq 0$, $\ker Q_* = \mathbb{R}e$ by part (b) of that lemma, and the entries of Q_* lie in K by part (d), because f and A are rational. If Q is any positive semidefinite Gram of $f - f_*$, then $e^\top Q e = f(p) - f_* = 0$, so $Qe = 0$ and $\text{rank } Q \leq N - 1$.

(b) For y feasible in (P) and (γ, Q) feasible in (S),

$$\Lambda_y(f) - \gamma = \Lambda_y(z^\top Q z) = \sum_{\alpha, \beta} Q_{\alpha\beta} y_{\alpha+\beta} = \text{tr}(QM_2(y)) \geq 0.$$

The point sequence $y_\alpha^p = p^\alpha$ is feasible in (P), with $M_2(y^p) = ee^\top$ and value $f(p) = f_*$, and (f_*, Q_*) is feasible in (S). Hence both optimal values equal f_* and both are attained. If y is optimal in (P), then $\text{tr}(Q_* M_2(y)) = \Lambda_y(f) - f_* = 0$, so the range of $M_2(y)$ lies in $\ker Q_* = \mathbb{R}e$, and $M_2(y) = cee^\top$ for some $c \geq 0$. The constant entry gives $c = y_0 = 1$. Every multi-index γ with $|\gamma| \leq 4$ is a sum $\alpha + \beta$ with $|\alpha|, |\beta| \leq 2$, so $y_\gamma = M_2(y)_{\alpha\beta} = p^\gamma$, and $y = y^p$. The ranks of $ee^\top$ and Q_* add up to N , so the pair is strictly complementary. The moments of the standard Gaussian measure on $\mathbb{R}^n$ are strictly feasible in (P): for $c \neq 0$, $c^\top M_2(y)c$ is the Gaussian integral of the square of the nonzero polynomial $c^\top z$, which is positive. In (S), the pair $(f_* - 1, Q_* + e_0 e_0^\top)$ is feasible, and $Q_* + e_0 e_0^\top \succ 0$: a nonzero $\xi \notin \mathbb{R}e$ has $\xi^\top Q_* \xi > 0$, while $\xi = ce$ with $c \neq 0$ has $(e_0^\top \xi)^2 = c^2 > 0$ because the constant coordinate of e is one.

(c) If $K \subseteq E$, then Q_* is an optimal Gram of rank $N - 1$ with entries in E . Conversely, let Q be an optimal Gram of rank $N - 1$ with entries in E . Its rank is the same over E and over $\mathbb{R}$, so its kernel in E^N is spanned by one nonzero vector ζ . By (a), $\zeta \in \mathbb{R}e$, and since the constant coordinate of e is one, $\zeta_0 \neq 0$ and $e = \zeta/\zeta_0 \in E^N$. The coordinates of e include $p_1, \dots, p_n$, so $K \subseteq E$.

(d) Let $V = \text{span}_{\mathbb{Q}}\{p^\alpha : |\alpha| \leq 2\} \subset \mathbb{R}$, of dimension r over $\mathbb{Q}$. The $\mathbb{Q}$ -linear map $\mathbb{Q}^N \rightarrow V$, $\xi \mapsto \xi^\top e$, is onto, so its kernel has dimension $N - r$. If Q is a rational optimal Gram, then $Qe = 0$ by (a), so every row of Q lies in this kernel, and the rank of Q , which is its row rank over $\mathbb{Q}$, is at most $N - r$. If some p_i is irrational, then 1 and p_i are linearly independent over $\mathbb{Q}$, so $r \geq 2$ and $\text{rank } Q + \text{rank}(ee^\top) \leq N - 1$.

(e) The set $\mathcal{F}_p = \{Q \succeq 0 : Qe = 0\}$ is the face of the positive semidefinite cone exposed by $ee^\top$, and $\mathcal{G}_* \subseteq \mathcal{F}_p$ by (a). The minimal face of the positive semidefinite cone containing a matrix $Q_0 \succeq 0$ consists of all $Q \succeq 0$ with $\text{range } Q \subseteq \text{range } Q_0$. For $Q_0 = Q_*$ the range is $e^\perp$, so this minimal face is $\mathcal{F}_p$, and $\mathcal{F}_p$ is the smallest face containing $\mathcal{G}_*$; Q_* lies in its relative interior. If $Z \succeq 0$ exposes $\mathcal{G}_*$, then $\text{tr}(ZQ_*) = 0$, so $\text{range } Z \subseteq \mathbb{R}e$ and $Z = cee^\top$ with $c > 0$ because $Z \neq 0$. Conversely $\text{tr}(ee^\top Q) = e^\top Q e = 0$ for every $Q \in \mathcal{G}_*$. For $Z = cee^\top$, the entry in the row of the constant monomial and the column of X_i is cp_i , and the constant entry is c , so $p_i = Z_{0,X_i}/Z_{0,0}$ belongs to the field generated by the entries of Z . In particular no nonzero rational positive semidefinite matrix exposes $\mathcal{G}_*$ when $p \notin \mathbb{Q}^n$.

For facial reduction, let $\mathcal{A}$ map a symmetric matrix Q to the coefficient vector of $z^\top Q z$, indexed by the monomials of degree at most four. The optimal-level Gram system is $\mathcal{A}(Q) = \text{coef}(f - f_*)$, $Q \succeq 0$. For every vector y indexed by those monomials, $\langle y, \mathcal{A}(Q) \rangle = \sum_{\alpha, \beta} Q_{\alpha\beta} y_{\alpha+\beta} = \langle M_2(y), Q \rangle$, so $\mathcal{A}^*(y) = M_2(y)$. For $y = y^p$ we obtain $\mathcal{A}^*(y^p) = ee^\top$, which is positive semidefinite and nonzero, and $\langle y^p, \text{coef}(f - f_*) \rangle = f(p) - f_* = 0$. This is a facial-reduction certificate in the sense of Borwein

and Wolkowicz [19]: every feasible Q satisfies $\langle ee^\top, Q \rangle = \langle y^p, \mathcal{A}(Q) \rangle = 0$, so the system may be restricted to $\mathcal{F}_p$. After this one step the feasible point Q_* lies in the relative interior of $\mathcal{F}_p$, so no further step is needed. Zero steps do not suffice, because no feasible Gram is positive definite. Hence the real singularity degree, the least number of such steps after which the restricted system has a relatively interior feasible point, is exactly one. $\square$

Proof of Corollary 8.10. Let p be the optimizer of f_k . By Theorem 8.5(b) and Lemma 8.4, its last two coordinates are a_k and b_k , with reduced denominators exactly 5^{2^k} .

(a) By Theorem 8.6(b) the optimal moment matrix is $ee^\top$, whose entry in the row of the constant monomial and the column of x_k is a_k .

(b) The Taylor Gram Q_* of Theorem 8.6(a) is rational, since p and the Hessian Gram are rational, and has rank $N - 1$. Let Q be any rational optimal Gram of rank $N - 1$. Order z with the constant monomial first and write

$$Q = \begin{pmatrix} q_{00} & c^\top \\ c & Q' \end{pmatrix}, \quad e = \begin{pmatrix} 1 \\ w \end{pmatrix}.$$

The block Q' is positive definite: for $u \neq 0$ the vector $(0, u)$ is not a multiple of e , so $(0, u)^\top Q(0, u) = u^\top Q' u > 0$ by Theorem 8.6(a). The equation $Qe = 0$ gives $Q'w = -c$. For each row i of the $(N - 1) \times N$ matrix $[Q' \mid c]$, multiply by the product π_i of the reduced denominators of its N entries. This produces an integer system $Q''w = -c''$ with $Q'' = \text{diag}(\pi)Q'$ nonsingular. By Cramer's rule every coordinate of w is a ratio of an integer to $\det Q''$, so its reduced denominator divides $|\det Q''|$. One coordinate of w is a_k , hence

$$5^{2^k} \leq |\det Q''|. \quad (71)$$

If every entry of Q has height at most B , then $\pi_i \leq 2^{NB}$, every entry of Q'' has absolute value at most $2^{(N+1)B}$, and the Leibniz expansion gives $|\det Q''| \leq (N - 1)! 2^{(N+1)(N-1)B}$. With (71) this is the stated bound on B . For the total length, let s_i be the sum, over the entries a/b of row i of $[Q' \mid c]$, of $\log_2 b$, plus the largest $\log_2 \max\{1, |a|\}$ in that row. The entries of row i of Q'' have absolute value at most 2^{s_i} , so $|\det Q''| \leq (N - 1)! 2^{\sum_i s_i}$. Since s_i is at most the sum of the binary lengths of the entries in row i , (71) shows that the total binary length of the entries of Q is at least $H_k - \log_2((N - 1)!)$. As $\log_2((N - 1)!) = O(n^2 \log n)$ and $N^2 - 1 = \Theta(n^4)$, the entry bound is $\Omega(2^k/n^4)$.

(c) By Theorem 8.6(e), a nonzero positive semidefinite exposing matrix is $Z = cee^\top$ with $c > 0$, and $a_k = Z_{0,x_k}/Z_{0,0}$. If these two entries are α_1/β_1 and α_2/β_2 in lowest terms with heights at most B , then $a_k = \alpha_1\beta_2/(\beta_1\alpha_2)$, whose reduced denominator divides $\beta_1\alpha_2$ and is therefore at most 2^{2B} . Hence $5^{2^k} \leq 2^{2B}$.

(d) If $f_k = \sum_{j=1}^{n+1} F_j^2$ with rational quadratics F_j , then $\sum_j \text{coef}(F_j) \text{coef}(F_j)^\top$ is a rational positive semidefinite Gram of $f_k = f_k - 0$ of rank at most $n + 1$ and binary length polynomial in k . Since $n \geq 2$, $n + 1 < N - 1$.

For $\tilde{f}_k$ the same proofs apply, because the terminal coordinates of $\tilde{p}$ have reduced denominators divisible by 5^{2^k} : in (b) and (c) these denominators divide $|\det Q''|$ and $\beta_1\alpha_2$, respectively, so both quantities are still at least 5^{2^k} . $\square$

F.6 Interior Gram matrices

Lemma F.3 (A uniform trace bound). *Let $W_n = 2^{-n} \int_{[-1,1]^n} z(X)z(X)^\top dX$. Then $W_n \succeq I_N/(15(n + 1))$. Consequently every positive semidefinite Gram Q of a polynomial h of degree at most four satisfies $\text{tr } Q \leq 15(n + 1) \|h\|_1$.*

Proof. For $\xi \in \mathbb{R}^N$, $\xi^\top W_n \xi$ is the mean of P^2 over the cube, where $P = \xi^\top z = a_0 + \sum_i b_i X_i + \sum_i c_i X_i^2 + \sum_{i < j} d_{ij} X_i X_j$. Under the uniform probability measure on $[-1, 1]^n$, the functions 1, X_i , $X_i^2 - \frac{1}{3}$, and $X_i X_j$ ($i < j$) are pairwise orthogonal, with mean squares 1, $\frac{1}{3}$, $\frac{4}{45}$, and $\frac{1}{9}$. Writing $a' = a_0 + \frac{1}{3} \sum_i c_i$,

$$\xi^\top W_n \xi = a'^2 + \frac{1}{3} \sum_i b_i^2 + \frac{4}{45} \sum_i c_i^2 + \frac{1}{9} \sum_{i < j} d_{ij}^2.$$

By the Cauchy–Schwarz inequality, $a_0^2 \leq 2a'^2 + \frac{2}{9}(\sum_i c_i)^2 \leq 2a'^2 + \frac{2n}{9} \sum_i c_i^2$, so

$$\|\xi\|^2 \leq 2a'^2 + \sum_i b_i^2 + (1 + \frac{2n}{9}) \sum_i c_i^2 + \sum_{i < j} d_{ij}^2.$$

Each coefficient on the right is at most $15(n+1)$ times the corresponding coefficient of $\xi^\top W_n \xi$; for the squares this is $1 + \frac{2n}{9} \leq \frac{4}{3}(n+1)$. This proves $W_n \succeq I/(15(n+1))$. If $Q \succeq 0$ and $h = z^\top Q z$, then $\text{tr } Q/(15(n+1)) \leq \text{tr}(QW_n)$, which is the mean of h over the cube and hence at most $\max_{[-1,1]^n} |h| \leq \|h\|_1$. $\square$

Lemma F.4 (Denominators of a positive determinant). *Let Q be a rational symmetric matrix whose upper-triangular entries have reduced denominators b_{ij} , $i \leq j$. Then $(\prod_{i \leq j} b_{ij}^2) \det Q$ is an integer. If $Q \succ 0$, then $\det Q \geq \prod_{i \leq j} b_{ij}^{-2}$.*

Proof. In the Leibniz expansion of $\det Q$, a product $\prod_i Q_{i, \sigma(i)}$ uses each diagonal entry at most once and each off-diagonal entry $Q_{ij} = Q_{ji}$ at most twice, as Q_{ij} and as Q_{ji} . Hence every term, and therefore $\det Q$, becomes an integer after multiplication by $\prod_{i \leq j} b_{ij}^2$. For $Q \succ 0$ that integer is positive, hence at least one. $\square$

Proof of Theorem 8.11. (b) By Lemma F.1, $A_k \succ 0$ is rational, so $\mu_{A_k} > 0$, $\lambda_k A_k \succeq 3I$, and $\lambda_k A_k + 2\iota_k \iota_k^\top \succeq 3I$; this matrix is a full Hessian Gram of h_k because the Hessian biform of u_k^2 is $2v_{x_k}^2$. Hence $\nabla^2 h_k \succeq 3I$ (Lemma 2.8(a)). The binary length of λ_k is polynomial in k by Lemma 2.7, so the whole certified quartic has length polynomial in k . Since $F_k \geq 0$ vanishes only at p and $u_k(p) = \delta_k > 0$, $h_k > 0$ on $\mathbb{R}^n$, and its minimum m_k is attained by coercivity, so $m_k > 0$. As $\nabla h_k(p) = 2\delta_k e_{x_k} \neq 0$, the point p is not the minimizer, and $m_k < h_k(p) = \delta_k^2 \leq M_k^{-2^{k+1}}$. If $F_{k,j}$ are the square factors of F_k , the matrix $\lambda_k \sum_j \text{cof}(F_{k,j}) \text{cof}(F_{k,j})^\top + \text{cof}(u_k) \text{cof}(u_k)^\top$ is a rational positive semidefinite Gram of h_k of rank at most $n+2$, and $n+2 < N$ for $n \geq 2$. Writing $\lambda_k = \sum_{\epsilon \in S} 2^\epsilon$ in binary and $2^\epsilon F^2 = (2^{\epsilon/2} F)^2$ or $2^\epsilon F^2 = 2(2^{(\epsilon-1)/2} F)^2$ turns $\lambda_k \sum_j F_{k,j}^2 + u_k^2$ into an unweighted sum of $O((n+1) \log_2 \lambda_k)$ squares of rational quadratics. Finally, apply Lemma 8.1(c) to h_k with $\mu = 3$: the number $c_a = h_k(a) - \|\nabla h_k(a)\|^2/6$ depends continuously on a and equals $m_k > 0$ at the minimizer of h_k , so it is positive at some rational a , and Q_a is a rational positive definite Gram of h_k .

(c) Let $Q \succ 0$ be a rational Gram of h_k . Since the constant coordinate of $z(p)$ is one and $z(p)$ has further nonzero coordinates, $\|z(p)\| > 1$ and

$$\lambda_{\min}(Q) \leq \frac{z(p)^\top Q z(p)}{\|z(p)\|^2} < h_k(p) \leq M_k^{-2^{k+1}}.$$

By Lemma F.3, every eigenvalue of Q is at most $\text{tr } Q \leq T_k$, and $T_k \geq 1$. Therefore $0 < \det Q < M_k^{-2^{k+1}} T_k^{N-1}$. Lemma F.4 gives $\det Q \geq \prod_{i \leq j} b_{ij}^{-2}$, and taking logarithms proves the displayed inequality. The polynomial h_k has polynomially many coefficients of polynomial binary length in k , so $\log_2 T_k$ is polynomial in k , and $N = O(k^2)$. Since $\log_2 M_k = (k+3) \log_2 1000$, the right side

is $\Omega(k2^k)$. There are $N(N+1)/2 = O(k^4)$ upper-triangular entries, so one of them has $\Omega(2^k/k^3)$ denominator bits.

(a) Put $\mu = \mu_A$, which has binary length polynomial in L (Lemma 2.7), and $\varphi = f - \|\nabla f\|^2/(2\mu)$. This polynomial has degree at most six, $O(n^6)$ coefficients, and coefficients of binary length polynomial in L . At the minimizer p of f , $\varphi(p) = f(p) = \min f > 0$. Let Δ be the product of the denominators of the coefficients of φ ; then $\Delta\varphi$ has integer coefficients of binary length $\tau = \text{poly}(L)$, and $\{\Delta\varphi > 0\} = \{\varphi > 0\}$ is a nonempty open set. By the theorem of Basu, Pollack, and Roy on rational points in sets defined by strict inequalities [14, Theorem 4.1.2], it contains a point $a \in \mathbb{Q}^n$ whose coordinates have numerators and denominators of binary length at most $\tau 6^{O(n)} = \text{poly}(L) 2^{O(n)}$. Then $c_a = \varphi(a) > 0$, and $\mathcal{Q}_a$ of Lemma 8.1(c) is a rational positive definite Gram of f .

Each entry of $\mathcal{Q}_a$ is the value at a of a polynomial of degree at most six with $\text{poly}(n)$ terms and rational coefficients of binary length $\text{poly}(L)$; these polynomials do not depend on a , and they come from f , A , and μ through the fixed formulas (19) and (20). If the coordinates of a have binary length at most B_a , write them over a common denominator of length at most nB_a ; then every such value has binary length $\text{poly}(L) \cdot B_a$. Summing over the N^2 entries gives total length $\text{poly}(L) 2^{O(n)}$. A bound that only counted arithmetic operations would not suffice here; the bound uses the fixed degree of these expressions in a .

For the unweighted sum of squares, rational $LDL^\top$ elimination of $\mathcal{Q}_a \succ 0$ gives $f = \sum_i d_i (\ell_i^\top z)^2$ with rational $d_i > 0$, and the entries of L and the d_i are ratios of minors of $\mathcal{Q}_a$, hence of binary length polynomial in that of $\mathcal{Q}_a$. Each d_i is a sum of at most $2 \text{bits}(d_i)$ rational squares (Lemma 2.20), which gives the claimed expression. The argument proves existence and a size bound. It is not an algorithm polynomial in L , because the sampling theorem is applied to a set whose rational points may need exponentially long coordinates. $\square$

F.7 Shared-circuit Grams

Proof of Theorem 8.12. Validation of the input. Check that f has degree at most four and that A is symmetric, verify (18) by expanding $w^\top A w$ and comparing coefficients with $v^\top \nabla^2 f(X) v$, and verify $A \succ 0$ by rational $LDL^\top$ elimination. These are polynomial-time rational computations. Reject on failure.

The observable and the Newton point. Compute $\mu = \mu_A$, which has binary length polynomial in the input length (Lemma 2.7) and satisfies $\nabla^2 f \succeq \mu I$ (Lemma 2.8(a)), and compute the expanded polynomial $h = f - \|\nabla f\|^2/(2\mu)$. It has degree at most six, $O(n^6)$ coefficients, and binary length polynomial in the input length. Since $\nabla f(p) = 0$, we have $h(p) = f(p) = \min f$. Apply Lemma 4.3(a) to f and μ , with L' at least the encoding lengths of the input of that lemma and of h . It gives, in polynomial time and without oracle calls, a shared rational circuit for a point $\hat{x} \in \mathbb{Q}^n$, in which every division is by a nonzero value, and a number $g = g_{L'} > 0$ such that $h(p) = 0$ or $|h(p)| \geq g$, and $|h(\hat{x}) - h(p)| \leq g/8$.

The Gram. Let $Q = \mathcal{Q}_{\hat{x}}$ be the matrix (20) with center $\hat{x}$ and $\bar{A} = A - \mu \text{diag}(I_n, 0)$. Its ingredients are circuits: the entries of $C_U(\hat{x})$ and $C_V(\hat{x})$ are integer polynomials of degree at most two in $\hat{x}$, $C_{\text{aff}}(\hat{x})$ uses $\nabla f(\hat{x})/\mu$, and $c_{\hat{x}} = f(\hat{x}) - \|\nabla f(\hat{x})\|^2/(2\mu) = h(\hat{x})$. Evaluating ∇f at $\hat{x}$ takes $O(n^4)$ gates, and the matrix products in (19) have dimensions $O(n^2)$ and take $O(n^6)$ gates. All are appended to the shared circuit for $\hat{x}$, and the only new divisors are μ and the constants 2, 3, and 36. By Lemma 8.1(c), $f = z^\top Q z$ for every center; no approximation enters this identity, because $\hat{x}$ denotes an exact rational vector.

If $\min f > 0$, then $h(p) > 0$, so $h(p) \geq g$ and $c_{\hat{x}} = h(\hat{x}) \geq 7g/8 > 0$, and Lemma 8.1(c) gives $Q \succ 0$. Conversely, if $Q \succ 0$, then for every X , $f(X) = z(X)^\top Q z(X) \geq \lambda_{\min}(Q) \|z(X)\|^2 \geq \lambda_{\min}(Q) > 0$,

since the constant coordinate of $z(X)$ is one; as the minimum is attained, $\min f > 0$.

The remarks after the theorem. For the family h_k of Theorem 8.11, $\min h_k > 0$, so the value of Q is a rational positive definite Gram of h_k , whose expanded encoding has $\Omega(k2^k)$ denominator bits. For item (i), the map from a certified quartic to the circuit matrix Q is a polynomial-time reduction from the predicate $\min f > 0$ to positive definiteness of circuit matrices, and that predicate is PosSLP-hard by Theorem 5.1(i). For item (ii), $f(\hat{x}) \geq \min f$ for every point, and Proposition 4.6 gives $f(\hat{x}) < 0$ when $\min f < 0$. For item (iv), if $\min f > 0$ the leading principal minors of Q are positive, so symmetric elimination without pivoting divides only by positive quantities and yields circuits for $f = \sum_i d_i (\ell_i^\top z)^2$ with $d_i > 0$; when $\min f \leq 0$ some pivot may vanish, so this step is not part of the sign-free construction. $\square$

G Proofs for Section 9

Throughout this appendix, $\|\cdot\|$ is the Euclidean norm of a vector and the operator norm of a matrix, and $\|\cdot\|_F$ is the Frobenius norm. For a quadratic $g(X) = g_0 + b^\top X + X^\top T X$ with T symmetric, $\Gamma(g)$ is its canonical Hessian Gram (15), with blocks $2bb^\top + 4g_0T$, $\Delta(b, T)$, and $\Xi(T)$, and for a center $c \in \mathbb{R}^n$ we write $\Gamma_c(g) = \Gamma(g(\cdot + c))$ for the canonical Gram of the recentred quadratic; both are extended linearly to signed rational combinations $\sum_j \sigma_j g_j^2$. By Lemma 7.1, $v^\top \nabla^2(\sum_j \sigma_j g_j^2)(c+u)v = (v, u \otimes v)^\top \Gamma_c(\cdot)(v, u \otimes v)$, and $\Gamma = \Psi_c^\top \Gamma_c \Psi_c$ with the invertible matrix Ψ_c of that lemma. Two consequences are used repeatedly. If Γ_c of a combination is positive definite, then the matrix Γ computed from the same factors at the origin is a positive definite full Hessian Gram; it is rational when the factors and weights are rational, and an integer matrix when the factors are integer quadratics and the weights are integers, since g_0 , b , and $2T$ are then integral and every block of (15) is an integer combination of their entries. Second, $\|\Delta(b, T)\| \leq \|\Delta(b, T)\|_F \leq 6\|b\| \|T\|_F$, the two tensors in its definition having Frobenius norms $2\|b\| \|T\|_F$ and $4\|b\| \|T\|_F$; $\|\Xi(T)\| \leq 8\|T\|_F^2 + 4\|T\|^2$; and for a symmetric block matrix $\left\| \begin{pmatrix} C & D \\ D^\top & Q \end{pmatrix} \right\| \leq \max\{\|C\|, \|Q\|\} + \|D\|$.

Lemma 7.6 is used with $m = n$ residuals, in the notation of its statement: quadratics $g, r_1, \dots, r_n$ vanishing at p , positive rationals μ, Λ, β, ν satisfying (16), and a rational square $\varepsilon = t^2$ satisfying (17). Its conclusions (b) and (c) give, for $f = (g/(t\nu))^2 + \sum_j (r_j/\nu)^2$, the bound $\nabla^2 f \succeq \frac{3}{2}I$ and a rational canonical Gram $\Gamma_f \succeq \gamma I$ with $\gamma = \min\{3/2, 2\mu^2/(\varepsilon\nu^2)\}/(((1+\theta)^2+1)(1+\|p\|)^2)$ and $\theta = 3\sqrt{n}\varepsilon(\Lambda + n\beta)/\mu^2$.

G.1 Proof of Lemma 9.2

(a) Let δ be the largest degree of a factor and suppose $\delta \geq 3$. The homogeneous part of degree 2δ of $\sum_j g_j^2$ is $\sum_j (g_j^{(\delta)})^2$, where $g_j^{(\delta)}$ is the degree- δ part of g_j . It is positive at any real point where some $g_j^{(\delta)}$ is nonzero, so it is a nonzero polynomial of degree $2\delta > 4$, which is impossible.

(b) The real numbers $g_j(p)$ satisfy $\sum_j g_j(p)^2 = f(p) = 0$. For the Gram statement, $z(p)^\top Q z(p) = 0$ and $Q \succeq 0$ give $Q^{1/2}z(p) = 0$, hence $Qz(p) = 0$. The i th entry of $Qz(p)$ is the value at p of the quadratic whose coefficient vector is the i th column of Q . If Q has entries in E , every column of Q is therefore the coefficient vector of an element of $\mathcal{I}_2^E(p)$. Let B be the matrix with entries in E whose columns are the coefficient vectors of the basis g , so that $g = B^\top z$; it has full column rank, and we fix a left inverse C with entries in E . Every column of Q lies in the range of B , so $BCQ = Q$, and by symmetry $Q = QC^\top B^\top$. Thus $Q = B(CQC^\top)B^\top$, and $S = CQC^\top$ is positive semidefinite with entries in E . Finally $f = z^\top Q z = g^\top S g$.

(c) If $f = \sum_j g_j^2$ with $g_j \in E[X]$, parts (a) and (b) put every g_j in $\mathcal{I}_2^E(p)$, so $\varphi(f) = \sum_j \varphi(g_j^2) \geq 0$. If f has a positive semidefinite Gram over E , part (b) gives $f = g^\top S g$ with $S \succeq 0$. Let $\Phi_{ij} = \varphi(g_i g_j)$. For $c \in E^t$ we have $c^\top \Phi c = \varphi((c^\top g)^2) \geq 0$. Since $\mathbb{Q} \subseteq E$ is dense in $\mathbb{R}$ and Φ is a real matrix, $\Phi \succeq 0$. Therefore $\varphi(f) = \text{tr}(S\Phi) \geq 0$. Both cases contradict $\varphi(f) < 0$.

(d) Let P_a be the minimal polynomial of a over E and $r = \deg P_a$. It divides $T^D - 2$, whose roots are $a\zeta^j$ with $\zeta = e^{2\pi i/D}$. The constant term of P_a is $(-1)^r a^r \zeta^e$ for some integer e , and it is real. The only real D th roots of unity are ± 1 , and -1 is not one because D is odd, so $\zeta^e = 1$ and $a^r \in E$. Let $d' = \gcd(r, D) = ur + vD$ with integers u, v . Then $a^{d'} = (a^r)^u (a^D)^v = (a^r)^u 2^v \in E$, so a is a root of $T^{d'} - a^{d'} \in E[T]$, and minimality gives $r \leq d'$. Hence $r = d'$ divides D , and $T^r - a^r$ is the minimal polynomial. If $D = \ell$ is prime and $a \notin E$, then $r > 1$ forces $r = \ell$. If $D = 5^k$ and $a \notin E$, then $r = 5^j$ with $j \geq 1$. The element a^5 is a root of $T^{r/5} - a^r \in E[T]$, so $[E(a^5) : E] \leq r/5$, while $[E(a) : E(a^5)] \leq 5$ because a is a root of $T^5 - a^5$. Since $[E(a) : E] = r$, both inequalities are equalities.

G.2 The prime family

Fix a prime $\ell \geq 5$, $s = (\ell - 1)/2 \geq 2$, $n = s + 1 \geq 3$, $a = 2^{1/\ell}$, and $p = (a^s, \dots, a^{2s})$. All coordinates of p lie in $[1, 2)$, and $a < 2^{1/5} < 1.15$.

An exposing quadratic. Define the integer quadratics

$$P_0 = x_0^2 - x_s, \quad P_1 = x_s^2 - 2x_{s-1}, \quad P_2 = x_0 x_1 - 2, \quad P_{i+2} = x_i^2 - x_{i-1} x_{i+1} \quad (1 \leq i \leq s-1),$$

the real weights $w_0 = 2a$, $w_1 = a^2$, $w_2 = -2$, $w_{i+2} = a^{2s-2i+2}$ ($1 \leq i \leq s-1$), and $E_* = \sum_{j=0}^{s+1} w_j P_j$. Every P_j vanishes at p , because $a^{2s+1} = 2$. None of them has a linear term in x_0 ; for P_1 this uses $s \geq 2$. Let Θ be the symmetric tridiagonal matrix indexed by $0, \dots, s$ with $\Theta_{ii} = a^{2s-2i}$ and $\Theta_{i,i+1} = \Theta_{i+1,i} = \frac{1}{2} a^{2s-2i-1}$, and put

$$L(x) = (x_1 - ax_0, x_2 - ax_1, \dots, x_s - ax_{s-1}, 2 - ax_s)^\top.$$

We claim $E_* = L^\top \Theta L$. Write L_i for the entries. Using $a^{2s+1} = 2$, the coefficients of $\sum_i a^{2s-2i} L_i^2 + \sum_{i < s} a^{2s-2i-1} L_i L_{i+1}$ are as follows. The coefficient of x_0^2 is $a^{2s+2} = 2a$. For $1 \leq j \leq s-1$ the coefficient of x_j^2 is $a^{2s-2j+2}(1+1-1) = a^{2s-2j+2}$, from L_{j-1}^2 , L_j^2 , and $L_{j-1} L_j$; likewise the coefficient of x_s^2 is $a^2(1+1-1) = a^2$. The coefficient of $x_j x_{j+1}$ is $-2a^{2s-2j+1} + a^{2s-2j+1} + a^{2s-2j+1} = 0$ for $1 \leq j \leq s-1$, and $-2a^{2s+1} + a^{2s+1} = -2$ for $j = 0$. The coefficient of $x_j x_{j+2}$ is $-a^{2s-2j}$, from $L_j L_{j+1}$. The linear terms are $-2ax_s$ and $-2a^2 x_{s-1}$, and the constant is 4. These are exactly the coefficients of E_* , which proves the claim. In particular $E_*(p) = 0$ and $\nabla E_*(p) = 0$, since $L(p) = 0$. In centered coordinates $L(p+u) = \mathcal{B}u$ with $\mathcal{B} = -aI + \mathcal{U}$, where $\mathcal{U}$ has ones on the first superdiagonal. Hence $E_*(p+u) = u^\top H_* u$ with $H_* = \mathcal{B}^\top \Theta \mathcal{B}$.

We have $\Theta = D_a \mathcal{T} D_a$, where $D_a = \text{diag}(a^s, a^{s-1}, \dots, 1)$ has entries in $[1, \sqrt{2})$ and $\mathcal{T}$ has unit diagonal and entries $\frac{1}{2}$ next to it. The eigenvalues of $\mathcal{T}$ are $1 + \cos(j\pi/(n+1))$, $1 \leq j \leq n$, and $1 - \cos(\pi/(n+1)) = 2 \sin^2(\pi/(2n+2)) \geq 2/(n+1)^2 \geq 1/(2n^2)$. Thus $I/(2n^2) \preceq \Theta \preceq 4I$. Next $\|\mathcal{B}\| \leq 1 + a < 3$, and the finite series $\mathcal{B}^{-1} = -a^{-1} \sum_{j=0}^{n-1} (\mathcal{U}/a)^j$ gives $\|\mathcal{B}^{-1}\| \leq n$. Therefore

$$\frac{1}{2n^4} I \preceq H_* \preceq 36I. \quad (72)$$

Each P_j has quadratic part of norm at most one, and gradient at p of norm at most 5: for example $\nabla P_1(p)$ has entries -2 and $2a^{2s} < 4$.

Chain residuals and their Jacobian. Put

$$C_0 = x_0^2 - x_s, \quad C_1 = x_0 x_1 - 2, \quad C_i = x_0 x_i - x_1 x_{i-1} \quad (2 \leq i \leq s).$$

They vanish at p , have no linear term in x_0 , have quadratic parts of norm at most one, and have gradients at p of norm at most 5. For a direction u put $h_i = u_i/p_i$, and divide the differential of each C_i at p by the positive number p_s , $p_0 p_1$, or $p_0 p_i = p_1 p_{i-1}$, respectively, each in $[1, 4]$. The result is the linear map

$$\varrho_0 = 2h_0 - h_s, \quad \varrho_1 = h_0 + h_1, \quad \varrho_i = h_0 + h_i - h_1 - h_{i-1} \quad (2 \leq i \leq s).$$

With $\delta = h_1 - h_0$ the last equations give $h_i = h_0 + i\delta + \sum_{j=2}^i \varrho_j$, and the first two then give

$$h_0 = \frac{\varrho_0 + s\varrho_1 + \sum_{j=2}^s \varrho_j}{2s+1}, \quad \delta = \frac{\varrho_1 - 2\varrho_0 - 2\sum_{j=2}^s \varrho_j}{2s+1}.$$

By Cauchy–Schwarz, $|h_0| \leq \|\varrho\|$ and $|\delta| \leq \|\varrho\| \sqrt{4s+1}/(2s+1) \leq \|\varrho\|/\sqrt{s}$, so $|h_i| \leq (1+2\sqrt{s})\|\varrho\|$ and $\|h\| \leq 3n\|\varrho\|$. Since $\|u\| \leq 2\|h\|$ and the row scaling divides by numbers at least one, the Jacobian J of $(C_0, \dots, C_s)$ at p satisfies $\|Ju\| \geq \|u\|/(6n)$. Hence $J^\top J \succeq \nu^2 I$ with $\nu = 1/(8n)$.

The integer quartic. Put

$$N_0 = 2^{20}n^8, \quad t = N_0^{-1}, \quad \varepsilon = t^2, \quad N_1 = 64(n+1)N_0^2, \quad A = \sum_{j=0}^{s+1} \lfloor N_1 w_j \rfloor P_j,$$

and

$$f_\ell = A^2 + (N_1/N_0)^2 \sum_{i=0}^s C_i^2 - r^2, \quad r = x_1 x_s - 2x_0. \quad (73)$$

The weight -2 is kept exactly, and $N_1/N_0 = 64(n+1)N_0$ is an integer, so f_ℓ has integer coefficients. All of A , C_i , and r vanish at p , so $f_\ell(p) = 0$ and $\nabla f_\ell(p) = 0$.

Write $\bar{A} = A/N_1$. Each of the $n+1$ weights changes by at most $1/N_1$, so $\bar{A}(p+u) = \ell_A^\top u + u^\top H u$ with

$$\|\ell_A\| \leq \frac{5(n+1)}{N_1} < \frac{\varepsilon}{8}, \quad \|H - H_*\| \leq \frac{n+1}{N_1} = \frac{\varepsilon}{64}.$$

With (72) this gives $H \succeq \mu I$ for $\mu = 1/(4n^4)$ and $\|H\| \leq 40 =: \Lambda$. Take $\beta = 8$ and the ν above. The conditions (17) hold with $m = n$: $\mu^2/(2n) = 1/(32n^9)$, and since $\Lambda + n\beta = 40 + 8n \leq 24n$,

$$\frac{\nu^2 \mu^2}{36n(\Lambda + n\beta)^2} \geq \frac{1}{64n^2 \cdot 16n^8 \cdot 36n \cdot 576n^2} = \frac{1}{21233664 n^{13}},$$

while $\varepsilon = 2^{-40}n^{-16}$ is smaller than both and $\|\ell_A\| \leq \varepsilon$. Lemma 7.6 applies to $g = \bar{A}$ and the residuals $C_0, \dots, C_s$. Here $\theta \leq 3\sqrt{n}\varepsilon \cdot 24n \cdot 16n^8 < 1152n^{10}\varepsilon < 1$, the minimum in γ equals $3/2$, and $\|p\| \leq 2\sqrt{n}$ gives $(1 + \|p\|)^2 \leq 9n$. Hence the quartic $f_{\text{lem}} = (\bar{A}/(t\nu))^2 + \sum_i (C_i/\nu)^2$ has a rational canonical Gram $\Gamma_{f_{\text{lem}}} \succeq \gamma I$ with $\gamma \geq 1/(30n)$.

Since $f_\ell + r^2 = N_1^2(\bar{A}^2 + \varepsilon \sum_i C_i^2) = N_1^2 \varepsilon \nu^2 f_{\text{lem}}$, the canonical Gram of the representation (73) is $\Gamma_{f_\ell} = N_1^2 \varepsilon \nu^2 \Gamma_{f_{\text{lem}}} - \Gamma(r)$. At the origin, r has constant term zero, linear part $-2x_0$, and quadratic matrix T_r with $\|T_r\| = \frac{1}{2}$ and $\|T_r\|_F^2 = \frac{1}{2}$, so the blocks of $\Gamma(r)$ have norms at most 8, $6 \cdot 2 \cdot 2^{-1/2} < 9$, and 5, and $\|\Gamma(r)\| < 17$. As $N_1^2 \varepsilon \nu^2 \gamma \geq 64(n+1)^2 N_0^2 / (30n^3) \geq 2N_0^2/n$,

$$\Gamma_{f_\ell} \succeq (2N_0^2/n - 17)I \succeq I.$$

This is an integer matrix, because the factors in (73) are integer quadratics with integer weights. Since $\|w(x, v)\| \geq \|v\|$, it proves $\nabla^2 f_\ell \succeq I$ everywhere, and with $f_\ell(p) = 0$, $\nabla f_\ell(p) = 0$, it gives $f_\ell(x) \geq \frac{1}{2} \|x - p\|^2$. This proves Theorem 9.4(b).

Size and construction. The quadratic A has $O(n)$ monomials and coefficients bounded by $8N_1$, and N_1 and N_0 are bounded by a polynomial in n . Hence f_ℓ has $O(n^2)$ monomials with coefficients of magnitude polynomial in n , and the canonical Gram entries are polynomial in these coefficients. This proves Theorem 9.4(a). Each integer $\lfloor N_1 w_j \rfloor$ for $w_j = 2^{e/\ell}$, $e \in \{2, 4, \dots, 2s, \ell + 1\}$, is the largest integer y with $y^\ell \leq N_1^\ell 2^e$, found by binary search over $[0, 4N_1]$ with integers of $O(\ell \log n)$ bits. All other steps are arithmetic on $O(n^4)$ numbers of $O(\log n)$ bits, so the construction runs in time polynomial in ℓ . No number-field computation is needed.

The functional. For every quadratic $g = g_0 + \sum_i v_i x_i + \sum_{i < j} h_{ij} x_i x_j$ one has $\varphi_\ell(g^2) = v_0^2 + 2g_0(v_s + h_{00})$, as in the proof of Lemma 9.3. The quadratics A and C_i are rational and vanish at p , so $v_s + h_{00} = 0$ by that lemma with $E = \mathbb{Q}$, and they have $v_0 = 0$. Thus $\varphi_\ell(A^2) = \varphi_\ell(C_i^2) = 0$, while $r^2 = x_1^2 x_s^2 - 4x_0 x_1 x_s + 4x_0^2$ gives $\varphi_\ell(r^2) = 4$. Hence $\varphi_\ell(f_\ell) = -4$, which is Theorem 9.4(c).

Remark G.1 (Other scales). For every rational $\lambda \geq 1$ the quartic $f_\ell^{(\lambda)} = \lambda(f_\ell + r^2) - r^2$ has the canonical Gram $\Gamma_{f_\ell} + (\lambda - 1)N_1^2 \varepsilon \nu^2 \Gamma_{f_{\text{lem}}} \succeq I$ and $\varphi_\ell(f_\ell^{(\lambda)}) = -4$. The monomial count in Theorem 9.4(a) and its conclusions (b)–(d) therefore hold for $f_\ell^{(\lambda)}$, with the same proof. The quartic and this Hessian Gram are rational, and are integer when λ is integer. Given λ , construction takes time polynomial in ℓ and its binary length; coefficient and Gram-entry lengths are $O(\log \ell + \langle \lambda \rangle)$.

G.3 The explicit ternary quartic

The convexity certificate. Let $c = (3/4, 1, 1/2)$ and $u = X - c$, and let

$$M_\star = \Gamma_c(A_0) + \sum_{i=0}^3 \Gamma_c(r_i) - \Gamma_c(r_4),$$

computed from (15) with the coordinates ordered as v_1, v_2, v_3 followed by the products $u_k v_j$ in lexicographic order of (k, j) . By Lemma 7.1(a) it satisfies $v^\top \nabla^2 F_\star(c + u)v = (v, u \otimes v)^\top M_\star (v, u \otimes v)$. The matrix $2(M_\star - I)$ is an integer matrix, and its leading principal minors of orders $1, \dots, 12$ are

1	98	7	132268108059697184
2	8714	8	26367351626513255040
3	1060526	9	8176807399117693383936
4	1508095300	10	4716316942464353421986304
5	505804465336	11	2238789673795455159378824192
6	388208815855184	12	1231832682931718409670151748608

All are positive, so $M_\star \succ I$ by Sylvester's criterion. Since $\|(v, u \otimes v)\| \geq \|v\|$, this gives $\nabla^2 F_\star \succeq I$ on $\mathbb{R}^3$. By the covariance identity of Lemma 7.1(b), $\Psi_c^\top M_\star \Psi_c$ is a rational positive definite full Hessian Gram of $F_\star$ on $w(X, v)$. This finite certificate is the only computational ingredient of Example 9.6; the remaining statements are proved below.

The zero and the vanishing basis. With $a^5 = 2$ and $p_\star = (a^4/2, a, a^2/2)$ one checks $xy = a^5/2 = 1$, $x^2 = yz = a^3/2$, $y^2 = 2z = a^2$, $2z^2 = x = a^4/2$, and $2xz = y = a$ at $p_\star$, so all r_i vanish there. Let $E \subseteq \mathbb{R}$ be a field with $a \notin E$. The monomials $1, y, z, yz, x$ take the values $1, a, a^2/2, a^3/2, a^4/2$, which are linearly independent over E by Lemma 9.2(d). The evaluation map from the ten-dimensional space $E[X]_{\leq 2}$ to the five-dimensional E -space $E(a)$ therefore has rank five, and $\dim_E \mathcal{I}_2^E(p_\star) = 5$. The monomials xy, x^2, y^2, z^2, xz each occur in exactly one r_i , so the r_i are independent and form a basis.

The functional. Only r_0 has a constant term, namely 2; only r_2 has a z term, namely $-4z$, and a y^2 term, namely $2y^2$; only r_4 has a y term, namely $-y$. Hence $[z](r_i r_j) \neq 0$ only for $\{i, j\} = \{0, 2\}$, where it is -8 , and $[y^2](r_i r_j) \neq 0$ only for $\{i, j\} = \{0, 2\}$, where it is 4, and for $i = j = 4$, where it is 1. With $\varphi_* = 2[z] + 4[y^2]$ this gives $\varphi_*(r_0 r_2) = -16 + 16 = 0$, $\varphi_*(r_4^2) = 4$, and $\varphi_*(r_i r_j) = 0$ otherwise. For $g = \sum_i c_i r_i$ we obtain $\varphi_*(g^2) = 4c_4^2 \geq 0$, and $\varphi_*(F_*) = -4$ because A_0 has no r_4 component. Lemma 9.2(c) excludes both certificate types over every real field not containing a , and Lemma 8.1(e) gives a sum of squares over every field containing a .

Products. The leading forms of $r_0, \dots, r_4$ are nonzero multiples of $l_1 = xy$, $l_2 = x^2 - yz$, $l_3 = y^2$, $l_4 = z^2$, $l_5 = xz$. Ten of the fifteen products $l_i l_j$ are single monomials, namely

$$x^2 y^2, xy^3, xy^2 z, x^2 yz, y^4, y^2 z^2, xy^2 z, z^4, xz^3, x^2 z^2.$$

The remaining five quartic monomials are

$$x^3 y = l_1 l_2 + l_3 l_5, \quad x^4 = l_2^2 + 2l_1 l_5 - l_3 l_4, \quad y^3 z = l_1^2 - l_2 l_3, \quad yz^3 = l_5^2 - l_2 l_4, \quad x^3 z = l_2 l_5 + l_1 l_4.$$

Thus the fifteen products span the fifteen-dimensional space of ternary quartic forms and are independent. A linear relation among the products $r_i r_j$ would give one among their quartic parts, so the $r_i r_j$ are independent too, and the Gram matrix of F_* on $(r_0, \dots, r_4)$ is unique.

G.4 Proof of Proposition 9.8

The field. Let $K = \mathbb{Q}(\alpha)$, a real cubic field. If $\beta \in K$, write $\beta = b_0 + b_1 \alpha + b_2 \alpha^2$ with rational b_i . Then β has degree three and generates K , so its conjugates are the images of β under the three embeddings of K , and $\text{Tr}_{K/\mathbb{Q}}(\beta) = 0$ and $\text{Tr}_{K/\mathbb{Q}}(\beta^2) = 0$, because the minimal polynomial $T^3 - 5$ has zero coefficients of T^2 and T . Since $\text{Tr}(1) = 3$ and $\text{Tr}(\alpha) = \text{Tr}(\alpha^2) = 0$, the first equation gives $b_0 = 0$. Then $\beta^2 = b_1^2 \alpha^2 + 2b_1 b_2 \alpha^3 + b_2^2 \alpha^4 = 4b_1 b_2 + b_1^2 \alpha^2 + 2b_2^2 \alpha$, and the second equation gives $12b_1 b_2 = 0$. If $b_2 = 0$ then $b_1^3 = 5/2$, and if $b_1 = 0$ then $b_2^3 = 5/4$; neither is the cube of a rational. Hence $\beta \notin K$. The polynomial $T^3 - 5$ has no root in the real field K , since its only real root is β , so it is irreducible over K , and $[\mathbb{Q}(\alpha, \beta) : \mathbb{Q}] = 9$, with basis $\alpha^i \beta^j$, $0 \leq i, j \leq 2$.

Vanishing quadratics and their products. The fifteen monomials of degree at most two in (x, y, z, w) evaluate at p into $\mathbb{Q}(\alpha, \beta)$, and $1, x, y, z, w, xz, xw, yz, yw$ take the nine basis values. The evaluation map has rank nine, so $\dim_{\mathbb{Q}} \mathcal{I}_2^{\mathbb{Q}}(p) = 6$. The six quadratics

$$g_1 = x^2 - y, \quad g_2 = xy - 2, \quad g_3 = y^2 - 2x, \quad g_4 = z^2 - w, \quad g_5 = zw - 5, \quad g_6 = w^2 - 5z$$

vanish at p and are independent, their pivot monomials $x^2, xy, y^2, z^2, zw, w^2$ being distinct, so they form a basis. Products of two of them either involve only one of the blocks (x, y) and (z, w) or have degree at most two in each block. Neither kind contains the monomial yz^3 , so $[yz^3]P = 0$ for every $P \in \mathcal{W}^{\mathbb{Q}}(p)$.

The perturbation. Let $h = yC(z, w)$ with $C = z^3 + w^3/5 - 3zw + 5$. At $(z, w) = (\beta, \beta^2)$ we have $C = 5 + 5 - 15 + 5 = 0$, $C_z = 3z^2 - 3w = 0$, and $C_w = \frac{3}{5}w^2 - 3z = \frac{3}{5} \cdot 5\beta - 3\beta = 0$. Hence $h(p) = 0$ and $\nabla h(p) = 0$, while $[yz^3]h = 1$. The quartic h has a rational symmetric full Hessian Gram N_h : every monomial of its Hessian biform has degree two in v and at most two in X , and is therefore a product of two entries of $w(X, v)$, so coefficients can be distributed over the corresponding symmetric pairs of entries.

A strictly convex rational baseline. For $\rho > 0$ let $\mathcal{E}_\rho(X, Y) = \rho^2(X^2 - Y) - \rho(XY - \rho^3) + (Y^2 - \rho^3 X)$. Expanding at $X = \rho + u$, $Y = \rho^2 + t$ gives $\mathcal{E}_\rho = \rho^2 u^2 - \rho u t + t^2$, a positive definite form with determinant $3\rho^2/4$. Put $E_* = \mathcal{E}_\alpha(x, y) + \mathcal{E}_\beta(z, w) = \alpha^2 g_1 - \alpha g_2 + g_3 + \beta^2 g_4 - \beta g_5 + g_6$. It vanishes at p with zero gradient and positive definite quadratic part. For rationals $\tilde{c}_1, \dots, \tilde{c}_4$ put $E = \tilde{c}_1 g_1 - \tilde{c}_2 g_2 + g_3 + \tilde{c}_3 g_4 - \tilde{c}_4 g_5 + g_6$; it vanishes at p exactly, and as $(\tilde{c}_i) \rightarrow (\alpha^2, \alpha, \beta^2, \beta)$ its quadratic part tends to that of E_* and its gradient at p tends to zero. The residuals g_1, g_2, g_4, g_5 have Jacobian at p block diagonal with blocks $\begin{pmatrix} 2\rho & -1 \\ \rho^2 & \rho \end{pmatrix}$, $\rho \in \{\alpha, \beta\}$, of determinant $3\rho^2 \neq 0$, and their quadratic parts have norm at most one. Fix positive rationals $\mu, \Lambda, \beta_0, \nu$ with μ at most half the least eigenvalue of the quadratic part of E_* , Λ at least twice its norm, β_0 at least the norms of the residual gradients at p (this is the bound called β in (16)), and ν^2 at most the least eigenvalue of $J^\top J$ for their Jacobian J . Choose a rational t such that $\varepsilon = t^2$ satisfies the first part of (17) with $m = n = 4$, and then choose the $\tilde{c}_i$ so close that the quadratic part of E is at least μI and at most Λ in norm, and $\|\nabla E(p)\| \leq \varepsilon$. Lemma 7.6 applies to $g = E$ and the residuals g_1, g_2, g_4, g_5 : the rational sum of squares $G = (E/(t\nu))^2 + \sum_{j \in \{1,2,4,5\}} (g_j/\nu)^2$ has a rational positive definite canonical Gram M_G . Moreover $G \in \mathcal{W}^\mathbb{Q}(p)$.

Conclusion. Let λ be a rational number with $\lambda \mu_{M_G} \geq 1 + \sum_{i,j} |(N_h)_{ij}|$, where μ_{M_G} is the determinant–trace margin, and put $f = \lambda G + h$. The sum of absolute entries bounds $\|N_h\|$. Its full Hessian Gram $\lambda M_G + N_h \succeq I$ is rational, so $\nabla^2 f \succeq I$. Since $f(p) = 0$ and $\nabla f(p) = 0$, the point p is the unique minimizer and the minimum is zero. The coefficient of yz^3 in f is $\lambda \cdot 0 + 1 = 1$, so $f \notin \mathcal{W}^\mathbb{Q}(p)$. A sum of squares of rational polynomials would consist of elements of $\mathcal{I}_2^\mathbb{Q}(p)$ by Lemma 9.2(a,b) and would lie in $\mathcal{W}^\mathbb{Q}(p)$. Finally $p \in \mathbb{Q}(\alpha, \beta)^4$, and Lemma 8.1(e) gives a sum of squares over that field.

G.5 The tower baseline and the proof of Theorem 9.9

One quintic gate. For $\alpha > 0$ and $b = \alpha^5$ let $\bar{r}_1, \dots, \bar{r}_5$ be the five quadratics (29) of one gate with the constant b in place of b_i , and put

$$\mathcal{E}_\alpha = \bar{r}_5 - \alpha \bar{r}_4 + \alpha^4 \bar{r}_1 + \alpha^2 \bar{r}_3.$$

Let $\ell_\alpha = (x - \alpha, y - \alpha x, z - \alpha y)$ and let T_α be the symmetric tridiagonal matrix with diagonal $(\alpha^4, \alpha^2, 1)$ and off-diagonal entries $\alpha^3/2$ and $\alpha/2$. Expanding $\ell_\alpha^\top T_\alpha \ell_\alpha$ gives

$$\alpha^4 x^2 + \alpha^2 y^2 - \alpha^2 xz - \alpha yz + z^2 - \alpha^5 x - \alpha^4 y + \alpha^6,$$

because the coefficient of xy is $\alpha^3 - 2\alpha^3 + \alpha^3 = 0$; the same expansion of $\mathcal{E}_\alpha$ with $b = \alpha^5$ gives the identical polynomial. Hence

$$\mathcal{E}_\alpha = \ell_\alpha^\top T_\alpha \ell_\alpha, \quad \mathcal{E}_\alpha(\alpha + u_1, \alpha^2 + u_2, \alpha^3 + u_3) = u^\top H_\alpha u, \quad H_\alpha = \mathcal{B}_\alpha^\top T_\alpha \mathcal{B}_\alpha, \quad (74)$$

where $\mathcal{B}_\alpha$ is lower bidiagonal with unit diagonal and subdiagonal $-\alpha$. For $1 \leq \alpha \leq 2^{1/5}$ we have the following bounds. $T_\alpha = D_\alpha \mathcal{T}_3 D_\alpha$ with $D_\alpha = \text{diag}(\alpha^2, \alpha, 1) \succeq I$, and the tridiagonal matrix $\mathcal{T}_3$ has eigenvalues 1 and $1 \pm \sqrt{2}/2$, so $T_\alpha \succeq (1 - \sqrt{2}/2)I \succeq \frac{1}{4}I$ and $\|T_\alpha\| \leq (1 + \sqrt{2}/2)\alpha^4 \leq 3$. The inverse of $\mathcal{B}_\alpha$ is lower triangular with entries 1, α, α^2 , so $\|\mathcal{B}_\alpha^{-1}\|^2 \leq 3 + 2\alpha^2 + \alpha^4 < 9$, while $\|\mathcal{B}_\alpha\| \leq 1 + \alpha < 2.2$. Hence

$$\frac{1}{36}I \preceq H_\alpha \preceq 16I \quad (1 \leq \alpha \leq 2^{1/5}). \quad (75)$$

Coupling the gates. For the tower put $E_i = r_{i,5} - a_i r_{i,4} + a_i^4 r_{i,1} + a_i^2 r_{i,3}$, with $b_i = x_{i-1}$ for $i \geq 2$. Replacing the constant $a_i^5 = a_{i-1}$ by x_{i-1} in $\bar{r}_5 - a_i \bar{r}_4$ changes it by $-(x_{i-1} - a_{i-1})x_i + a_i(x_{i-1} - a_{i-1}) = -(x_{i-1} - a_{i-1})(x_i - a_i)$. Writing $u = X - p_k$ and $u_i = (u_{i,1}, u_{i,2}, u_{i,3})$, we obtain from (74)

$$E_i(p_k + u) = u_i^\top H_{a_i} u_i - u_{i-1,1} u_{i,1} \quad (i \geq 2), \quad E_1(p_k + u) = u_1^\top H_{a_1} u_1.$$

Let $\rho = 2^{-13}$ and $E_* = \sum_{i=1}^k \rho^{i-1} E_i$. By the arithmetic-geometric mean inequality, $\rho^{i-1} |u_{i-1,1} u_{i,1}| \leq \frac{\rho^{i-1}}{144} u_{i,1}^2 + 36 \rho^{i-1} u_{i-1,1}^2$, and $36 \rho^{i-1} \leq \rho^{i-2}/144$ because $\rho \leq 1/5184$. Summing and using (75),

$$E_*(p_k + u) \geq \sum_i \rho^{i-1} \left(\frac{1}{36} - \frac{2}{144} \right) \|u_i\|^2 \geq \frac{\rho^{k-1}}{72} \|u\|^2.$$

The quadratic part of E_* also has norm at most $16 + 1 = 17$, the coupling terms forming a matrix of norm at most one.

The rational exposing quadratic. Put $a_0 = 2$, $\vartheta = 2^{-32k-15}$, and compute dyadic rationals $\tilde{a}_1, \dots, \tilde{a}_k \in [1, 2]$ as follows: $\tilde{a}_0 = 2$, and $\tilde{a}_i$ is obtained by bisection on $[1, 2]$ for the increasing map $\tau \mapsto \tau^5$, compared exactly with $\tilde{a}_{i-1}$, so that $|\tilde{a}_i - \tilde{a}_{i-1}^{1/5}| \leq \vartheta/2$. Since $\tau \mapsto \tau^{1/5}$ is $\frac{1}{5}$ -Lipschitz on $[1, \infty)$, induction gives $|\tilde{a}_i - a_i| \leq \frac{\vartheta}{2} + \frac{1}{5} |\tilde{a}_{i-1} - a_{i-1}| \leq \vartheta$. Each bisection uses $O(k)$ steps on rationals of $O(k)$ bits. Define

$$G_k = \sum_{i=1}^k \rho^{i-1} (r_{i,5} - \tilde{a}_i r_{i,4} + \tilde{a}_i^4 r_{i,1} + \tilde{a}_i^2 r_{i,3}).$$

It is a rational quadratic, it vanishes at p_k exactly, and $[z_k^2]G_k = \rho^{k-1} > 0$, since z_k^2 occurs only in $r_{k,5}$. The difference $G_k - E_*$ is a combination of $r_{i,4}, r_{i,1}, r_{i,3}$ with coefficients ρ^{i-1} times $-(\tilde{a}_i - a_i)$, $\tilde{a}_i^4 - a_i^4$, and $\tilde{a}_i^2 - a_i^2$, of absolute values at most $\vartheta, 8\vartheta$, and 3ϑ . These relations have quadratic parts of norm at most one and gradients at p_k of norm at most 2.3, 2.6, and 3.3. As $\sum_i \rho^{i-1} < 2$, the quadratic part H of G_k and its gradient ℓ_G at p_k satisfy $\|H - H_{E_*}\| \leq 24\vartheta$ and $\|\ell_G\| \leq 70\vartheta$.

The chain Jacobian. Let J be the Jacobian at p_k of the $3k$ chain residuals, rows ordered by gates. The block of gate i with respect to (x_i, y_i, z_i) is

$$J_i = \begin{pmatrix} 2\alpha & -1 & 0 \\ \alpha^2 & \alpha & -1 \\ 0 & \alpha^3 & \alpha^2 \end{pmatrix}, \quad J_i^{-1} = \frac{1}{5\alpha^4} \begin{pmatrix} 2\alpha^3 & \alpha^2 & 1 \\ -\alpha^4 & 2\alpha^3 & 2\alpha \\ \alpha^5 & -2\alpha^4 & 3\alpha^2 \end{pmatrix}, \quad \alpha = a_i,$$

with $\det J_i = 5\alpha^4$; for $\alpha \in [1, 2^{1/5}]$ the Frobenius norm gives $\|J_i^{-1}\| < 2$. The only other entries of J are -1 in the row of $r_{i,4}$ and the column of x_{i-1} . Writing $J = \mathcal{D} + \mathcal{N}$ with $\mathcal{D} = \text{diag}(J_i)$, the matrix $\mathcal{D}^{-1}\mathcal{N}$ is block strictly lower triangular with norm less than two, so $J^{-1} = \sum_{j=0}^{k-1} (-\mathcal{D}^{-1}\mathcal{N})^j \mathcal{D}^{-1}$ has norm less than 2^{k+1} . Hence $J^\top J \succeq \nu^2 I$ with $\nu = 2^{-k-1}$. The chain residuals have quadratic parts of norm at most one and gradients at p_k of norm at most 3.

The baseline. Put $\mu = \rho^{k-1}/144$, $\Lambda = 18$, $\beta = 3$, $n = 3k$, $t = 2^{-16k-4}$, and $\varepsilon = t^2 = 2^{-32k-8}$. Then $24\vartheta \leq \mu$, so $H \succeq \mu I$ and $\|H\| \leq \Lambda$, and $\|\ell_G\| \leq 70\vartheta \leq \varepsilon$. The conditions (17) hold with $m = n$: since $\Lambda + n\beta \leq 27k$, $20736 \cdot 78732 < 2^{31}$, and $k^3 \leq 2^{3k}$,

$$\frac{\nu^2 \mu^2}{36n(\Lambda + n\beta)^2} \geq \frac{2^{-2k-2} 2^{-26(k-1)}}{20736 \cdot 78732 k^3} \geq 2^{-31k-7} \geq \varepsilon,$$

and $\mu^2/(2n)$ is larger. By Lemma 7.6, applied to $g = G_k$ and the $3k$ chain residuals,

$$F_{k,0} = \left(\frac{G_k}{t\nu}\right)^2 + \sum_{i=1}^k \sum_{j \in \{1,2,4\}} \left(\frac{r_{i,j}}{\nu}\right)^2,$$

an explicit sum of $3k + 1$ squares of rational quadratics, has a rational positive definite canonical Gram M_k , computed from these factors. In the notation of Section 9.3, $\sigma_0 = (t\nu)^{-2}$ and $\sigma_1 = \nu^{-2}$. Let $\lambda_k = \lceil 17/\mu_{M_k} \rceil$. All coefficients and the entries of M_k have $O(k)$ bits, the dimension of M_k is $3k + 9k^2$, and μ_{M_k} and λ_k are computed exactly in polynomial time with polynomial bit length.

Proof of (a). The quadratic $r_{k,3} = X^\top T X$ is homogeneous, with T supported on the last gate, $\|T\| = 1$, and $\|T\|_F^2 = \frac{3}{2}$. Its Gram $\Gamma_0(r_{k,3})$ has only the quartic block, of norm at most $8 \cdot \frac{3}{2} + 4 = 16$. For rational $\lambda \geq \lambda_k$ the rational full Hessian Gram $\lambda M_k - \Gamma_0(r_{k,3})$ of $f_{k,\lambda}$ is at least $(\lambda \mu_{M_k} - 16)I \succeq I$. All squares in $f_{k,\lambda}$ vanish at p_k , so $f_{k,\lambda}(p_k) = 0$ and $\nabla f_{k,\lambda}(p_k) = 0$, and strong convexity makes p_k the unique minimizer.

Proof of (b). If $a_k \in E$, then $p_k \in E^{3k}$ since $a_i = a_k^{5^{k-i}}$, and Lemma 8.1(e) gives a sum of squares over E , hence also a positive semidefinite Gram over E . Suppose $a_k \notin E$ and put $L = E(a_k^5)$. By Lemma 9.2(d), $[L(a_k) : L] = 5$, so $T^5 - a_k^5$ is irreducible over L ; for $k = 1$, $L = E$. The coordinates $a_i = (a_k^5)^{5^{k-1-i}}$, $i < k$, lie in L . For $P \in E[X]$ let $P^b(x, y, z) = P(p_{<k}, x, y, z) \in L[x, y, z]$, where $p_{<k}$ lists the coordinates of the first $k - 1$ gates. A sum of squares over E restricts to a sum of squares over L . If $f = z(X)^\top Q z(X)$ with $Q \succeq 0$ over E , then $z(p_{<k}, V) = \Pi z'(V)$ for the monomial vector z' in $V = (x, y, z)$ and a matrix Π over L , so $f^b = z'^\top (\Pi^\top Q \Pi) z'$ with $\Pi^\top Q \Pi \succeq 0$ over L .

On the slice, every relation of an earlier gate becomes the constant zero, and $r_{k,j}^b$ is the relation of one gate with $b = a_k^5$. Hence

$$f_{k,\lambda}^b = \lambda \left(\sigma_0 (G_k^b)^2 + \sigma_1 \sum_{j \in \{1,2,4\}} (r_{k,j}^b)^2 \right) - (r_{k,3}^b)^2.$$

Let $\bar{p} = (a_k, a_k^2, a_k^3)$. The monomials $1, x, y, z, xz$ take the values $1, a_k, \dots, a_k^4$, linearly independent over L , so $\dim_L \mathcal{I}_2^L(\bar{p}) = 5$; the relations $r_{k,j}^b$ vanish at $\bar{p}$ and have the distinct pivot monomials x^2, xy, y^2, yz, z^2 , so they form a basis. Their leading forms $x^2, xy, y^2 - xz, yz, z^2$ have products spanning all ternary quartic forms: ten products are monomials, and

$$x^3z = \hat{r}_2^2 - \hat{r}_1\hat{r}_3, \quad xy^3 = \hat{r}_2\hat{r}_3 + \hat{r}_1\hat{r}_4, \quad y^3z = \hat{r}_3\hat{r}_4 + \hat{r}_2\hat{r}_5, \quad xz^3 = \hat{r}_4^2 - \hat{r}_3\hat{r}_5, \quad y^4 = \hat{r}_3^2 + 2\hat{r}_2\hat{r}_4 - \hat{r}_1\hat{r}_5,$$

where $\hat{r}_j$ is the leading form of $r_{k,j}^b$. So the fifteen products $r_{k,j}^b r_{k,j'}^b$ are linearly independent over $\mathbb{R}$, and the map $S \mapsto (r^b)^\top S r^b$ on real symmetric 5×5 matrices is injective. Write $G_k^b = \sum_j c_j r_{k,j}^b$ with $c_j \in L$; then $c_5 = [z^2]G_k^b = [z_k^2]G_k > 0$. The slice is represented by

$$S_0 = \lambda(\sigma_0 c c^\top + \sigma_1(e_1 e_1^\top + e_2 e_2^\top + e_4 e_4^\top)) - e_3 e_3^\top, \quad y^\top S_0 y = -c_5^2 < 0 \text{ for } y = (0, 0, c_5, 0, -c_3).$$

A sum of squares of $f_{k,\lambda}$ over E restricts to $f^b = \sum_j (d_j^\top r^b)^2$ with $d_j \in L^5$, by Lemma 9.2(a,b), so injectivity gives $S_0 = \sum_j d_j d_j^\top \succeq 0$, a contradiction. A positive semidefinite Gram over E restricts to one over L , and Lemma 9.2(b) gives $f^b = (r^b)^\top S' r^b$ with $S' \succeq 0$, so $S' = S_0$, again a contradiction. Finally $T^{5^k} - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion at two, so $\mathbb{Q}(a_k)$ has degree 5^k .

G.6 Proof of Theorem 9.10

Let $D = \ell^k$, $a = 2^{1/D}$, $\zeta = e^{2\pi i/D}$, $Z = \mathbb{Q}(\zeta)$, and $Z^+ = \mathbb{Q}(\zeta + \zeta^{-1}) \subseteq \mathbb{R}$. The field Z^+ is a subfield of the abelian extension $Z/\mathbb{Q}$, hence Galois over $\mathbb{Q}$, and it is totally real: every embedding sends $\zeta + \zeta^{-1}$ to $2\cos(2\pi j/D)$.

Step 1: $[Z^+(a) : Z^+] = D$. By Lemma 9.2(d) the minimal polynomial of a over Z^+ is $T^r - a^r$ with $r \mid D$ and $a^r \in Z^+$. If $r < D$, then $a^r = 2^{1/(D/r)}$ has minimal polynomial $T^{D/r} - 2$ over $\mathbb{Q}$, by Eisenstein's criterion, and this polynomial has nonreal roots because $D/r \geq 3$ is odd. As Z^+ is normal over $\mathbb{Q}$, it would contain these nonreal conjugates, contradicting $Z^+ \subseteq \mathbb{R}$.

Step 2: $[Z(a) : Z] = D$. The real field $Z^+(a)$ does not contain ζ , which is a root of $T^2 - (\zeta + \zeta^{-1})T + 1 \in Z^+[T]$. Hence $[Z^+(a, \zeta) : Z^+(a)] = 2$ and likewise $[Z : Z^+] = 2$. Since $Z(a) = Z^+(a, \zeta)$, comparing $[Z(a) : Z^+] = 2D$ computed through $Z^+(a)$ and through Z gives the claim.

Step 3: an automorphism of order D . The field $S = Z(a)$ is the splitting field of $T^D - 2$ over $\mathbb{Q}$. By Step 2, $T^D - 2$ is irreducible over Z , so some $\sigma \in \text{Gal}(S/Z)$ maps a to ζa . Since σ fixes ζ , $\sigma^j(a) = \zeta^j a$, and σ has order exactly D .

Step 4: lifting. Let K' be the compositum of the normal closures of the fields $\mathbb{Q}(\beta_i)$; it is finite and Galois over $\mathbb{Q}$. It contains a , hence all conjugates of a , and therefore S . Restriction $\text{Gal}(K'/\mathbb{Q}) \rightarrow \text{Gal}(S/\mathbb{Q})$ is surjective, so some $\tau \in \text{Gal}(K'/\mathbb{Q})$ restricts to σ , and D divides the order of τ . On the other hand $\text{Gal}(K'/\mathbb{Q})$ acts faithfully on the union of the sets of conjugates of the β_i , since these generate K' , and preserves each set. This embeds the group into $\prod_i \mathfrak{S}_{d_i}$, $d_i = [\mathbb{Q}(\beta_i) : \mathbb{Q}]$. If every $d_i < D$, every cycle of every component permutation has length less than ℓ^k , so the ℓ -adic valuation of the order of every element of the product is at most $k - 1$. This contradicts $\ell^k \mid \text{ord}(\tau)$.

The certificate consequence. If the coefficients of the square factors of a sum of squares of $f_{k,\lambda}$ are algebraic, the real field E they generate admits a sum of squares, so $a_k \in E$ by Theorem 9.9, and the lemma applies to the list of these coefficients with $\ell = 5$. The same argument applies to the entries of a positive semidefinite Gram matrix with algebraic entries. A polynomial with a root of degree 5^k has degree at least 5^k , so a dense list of any annihilating polynomial of that coefficient has at least $5^k + 1$ entries.

G.7 Proof of Theorem 9.11

Let $M' = \lambda M_k - \Gamma_0(r_{k,3}) \succeq I$ be the rational full Hessian Gram of $f = f_{k,\lambda}$ from the proof of Theorem 9.9(a).

(a) Lemma 8.1(e) writes f as a sum of squares of quadratics whose coefficients are rational polynomials of degree at most two in the coordinates of p_k . Their number and the bit lengths of these rational polynomials are polynomial in L_λ , since the entries of M' have length polynomial in L_λ and rational elimination of M' has polynomial bit complexity, and a positive rational u/v is a sum of at most $2\log_2(2uv)$ rational squares, by the binary expansion of uv and the identity $u/v = uv/v^2$ (Lemma 2.20). The circuit computes a_i by fifth roots in sequence, a_i^2, a_i^3 by two multiplications per gate, and every coefficient by evaluating its rational polynomial.

(b) Let Y be a second vector of $3k$ variables, $u = X - Y$, $\mathcal{A} = (u, Y \otimes u)$, $\mathcal{B} = (0, u \otimes u)$, and

$$s(X, Y) = \frac{1}{2}(\mathcal{A} + \frac{1}{3}\mathcal{B})^\top M'(\mathcal{A} + \frac{1}{3}\mathcal{B}) + \frac{1}{36}\mathcal{B}^\top M'\mathcal{B}.$$

Writing $M' = \sum_j c_j c_j^\top$ with rational c_j makes s an explicit rational sum of squares of degree four. Since $w(Y + tu, u) = \mathcal{A} + t\mathcal{B}$, the integral identity of Lemma 8.1(a) and Taylor's formula with integral remainder, both identities between polynomials in (X, Y) , give

$$f(X) = f(Y) + \nabla f(Y)^\top (X - Y) + s(X, Y).$$

With $f = \sum_j \sigma_j g_j^2$ and $H_j = \sigma_j(g_j(Y) + 2\nabla g_j(Y)^\top(X - Y))$ we have $f(Y) + \nabla f(Y)^\top(X - Y) = \sum_j H_j g_j(Y)$, which is the identity. Each H_j has degree at most two. At $Y = p_k$ all $g_j(Y)$ vanish, so $f(X) = s(X, p_k) \geq 0$ for every real X . For the second identity, the exact relations

$$r_{i,3} = -y_i r_{i,1} + x_i r_{i,2}, \quad r_{i,5} = -z_i r_{i,2} + x_i r_{i,4}$$

express $r_{k,3}$ and G_k as combinations of chain residuals with affine rational multipliers. Substituting them multiplies H_j by affine polynomials, which gives multipliers of degree at most three and an identity of degree at most five.

G.8 Proof of Theorem 9.12

(a) The quadratic relations. Each relation $r_{i,j}$ has a pivot monomial, namely $x_i^2, x_i y_i, y_i^2, y_i z_i, z_i^2$, and its other monomials lie outside the set of pivots: they are $y_i, z_i, x_i z_i$, the constant or x_{i-1} , and x_1 or $x_{i-1} x_i$. The monomials of degree at most two that are not pivots are the constant, the four monomials $x_i, y_i, z_i, x_i z_i$ of each gate, and the nine products of a variable of gate i with a variable of gate $j, i < j$. Let $a = a_k$, so that $a_i = a^{5^{k-i}}$. These monomials evaluate at p_k to a^e with $0 \leq e < 5^k$, where the base-five expansion of e is, respectively, zero; a single digit in $\{1, 2, 3, 4\}$ at position $k - i$; or two digits in $\{1, 2, 3\}$ at positions $k - i$ and $k - j$. Distinct monomials give distinct expansions. By Eisenstein's criterion $T^{5^k} - 2$ is irreducible, so these values are linearly independent over $\mathbb{Q}$. Given $g \in \mathcal{I}_2^\mathbb{Q}(p_k)$, subtract multiples of the relations to remove all pivots. The remainder is a combination of non-pivot monomials vanishing at p_k , hence zero. So the relations span, and their distinct pivots make them independent. Since $1, x_i, y_i, z_i$ are non-pivot monomials, no nonzero rational affine polynomial vanishes at p_k . It follows that $\mathcal{J}_2^\mathbb{Q}(p_k) = \{0\}$: the partial derivatives of an element are rational affine polynomials vanishing at p_k , so the element is a constant equal to its value zero.

(b) Products. Induct on k . Let $V = (x_k, y_k, z_k)$ and grade polynomials by their degree in V . Relations of earlier gates have V -degree zero, and $r_{k,j} = \hat{r}_j + (\text{terms of } V\text{-degree at most one})$ with the leading forms $\hat{r} = (x_k^2, x_k y_k, y_k^2 - x_k z_k, y_k z_k, z_k^2)$. In a real linear relation among the products, the part of V -degree four involves only products of two last-gate relations and equals the same combination of the products $\hat{r}_j \hat{r}_{j'}$; these are independent (proof of Theorem 9.9(b)), so those coefficients vanish. The part of V -degree two then reads $\sum_j \hat{r}_j P_j = 0$, where P_j is the combination of earlier relations multiplying $r_{k,j}$. The forms $\hat{r}_j$ have linearly independent coefficient vectors, so they remain independent over the field of rational functions in the earlier variables, and every $P_j = 0$. By (a) for the earlier gates, the earlier relations are independent, so the coefficients of all mixed products vanish. The remaining relation involves earlier gates only and is trivial by induction; for $k = 1$ only the local products occur.

(c) A local interpolation lemma. Let $L \subseteq \mathbb{R}$ be a field, $b \in L \setminus \{0\}$, $T^5 - b$ irreducible over L , and α its real root. We claim that the L -linear map

$$\mathcal{H} : L[x, y, z]_{\leq 3} \rightarrow L(\alpha)^4, \quad P \mapsto (P, P_x, P_y, P_z)(\alpha, \alpha^2, \alpha^3)$$

is bijective. Both sides have dimension twenty over L . Compose with the invertible map multiplying the last three components by $\alpha, \alpha^2, \alpha^3$. The composite sends $x^i y^j z^h$ to $\alpha^w(1, i, j, h)$, where $w = i + 2j + 3h = 5t + \varrho$ with $0 \leq \varrho \leq 4$, and $\alpha^w = b^t \alpha^\varrho$. In the L -basis $\alpha^\varrho e_c$ of $L(\alpha)^4$, the composite is block diagonal over the residue ϱ . Each residue class contains four monomials:

ϱ	monomials	determinant of the columns $(1, i, j, h)$
0	$1, yz, x^2z, xy^2$	5
1	x, z^2, xyz, y^3	5
2	y, x^2, xz^2, y^2z	-5
3	z, xy, x^3, yz^2	-5
4	xz, y^2, x^2y, z^3	-5

For instance, for $\varrho = 0$ the columns are $(1, 0, 0, 0)$, $(1, 0, 1, 1)$, $(1, 2, 0, 1)$, $(1, 1, 2, 0)$, and the determinant equals that of $\begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & 2 \\ 1 & 1 & 0 \end{pmatrix}$, which is 5. The block determinants are these integers times nonzero powers of b , so the map is bijective.

(c) Stationary cubics. We show $\mathcal{J}_3^{\mathbb{Q}}(p_k) = \{0\}$ by induction. For $k = 1$ apply the local lemma with $L = \mathbb{Q}$ and $b = 2$. Let $k \geq 2$, $L = \mathbb{Q}(a_{k-1})$, $b = a_{k-1}$, and $\alpha = a_k$; then $[L(\alpha) : L] = 5$, so $T^5 - b$ is irreducible over L . Write U for the earlier variables and $F = \sum_{\mu} c_{\mu}(U)\mu(V)$ for $F \in \mathcal{J}_3^{\mathbb{Q}}(p_k)$, summing over monomials μ in V . The polynomial $F(p_{k-1}, V)$ has degree at most three in V , coefficients in L , and zero value and V -gradient at (a_k, a_k^2, a_k^3) , so it is zero. Hence every c_{μ} vanishes at p_{k-1} . For $\deg \mu \geq 2$ the coefficient c_{μ} is affine, hence zero by (a). Thus $F = h_0 + x_k h_1 + y_k h_2 + z_k h_3$ with $\deg h_0 \leq 3$, $\deg h_j \leq 2$, all vanishing at p_{k-1} . For an earlier variable ξ , $0 = \partial_{\xi} F(p_k) = \partial_{\xi} h_0 + a_k \partial_{\xi} h_1 + a_k^2 \partial_{\xi} h_2 + a_k^3 \partial_{\xi} h_3$ evaluated at p_{k-1} , with coefficients in L ; independence of $1, a_k, a_k^2, a_k^3$ over L makes each $\partial_{\xi} h_j(p_{k-1}) = 0$. Thus $h_1, h_2, h_3 \in \mathcal{J}_2^{\mathbb{Q}}(p_{k-1}) = \{0\}$ and $h_0 \in \mathcal{J}_3^{\mathbb{Q}}(p_{k-1}) = \{0\}$.

(c) Stationary quartics. The inclusion $\mathcal{W}^{\mathbb{Q}}(p_k) \subseteq \mathcal{J}_4^{\mathbb{Q}}(p_k)$ follows from the product rule; uniqueness of S is (b). For the reverse inclusion let $F \in \mathcal{J}_4^{\mathbb{Q}}(p_k)$. For $k = 1$, subtract a rational combination of the products $r_{1,j} r_{1,j'}$ with the same quartic part, which exists because the leading products span all ternary quartic forms; the remainder lies in $\mathcal{J}_3^{\mathbb{Q}}(p_1) = \{0\}$. For $k \geq 2$ proceed in four steps.

Step 1. The part of F of V -degree four has rational constant coefficients. Subtract the rational combination of products of last-gate relations with this V -degree-four part. The remainder F' lies in $\mathcal{J}_4^{\mathbb{Q}}(p_k)$ and has V -degree at most three.

Step 2. As in the cubic case, $F'(p_{k-1}, V) = 0$, so all coefficients c_{μ} of F' vanish at p_{k-1} . Coefficients of V -monomials of degree three are affine and hence zero. Coefficients of V -monomials of degree two have degree at most two and lie in $\mathcal{I}_2^{\mathbb{Q}}(p_{k-1})$.

Step 3. We claim $c_{y^2} + c_{xz} = 0$, where these are the coefficients of y_k^2 and $x_k z_k$. For an earlier variable ξ , the polynomial $\Phi_{\xi}(V) = \sum_{\mu} \partial_{\xi} c_{\mu}(p_{k-1}) \mu(V)$ has degree at most two, coefficients in L , and value $\partial_{\xi} F'(p_k) = 0$ at (a_k, a_k^2, a_k^3) . By the slice computation in the proof of Theorem 9.9(b), it is an L -combination of the five relations of one gate, and each of them has the same coefficient on y^2 as minus its coefficient on xz . Hence $\partial_{\xi}(c_{y^2} + c_{xz})(p_{k-1}) = 0$ for every earlier variable ξ . Also $(c_{y^2} + c_{xz})(p_{k-1}) = 0$, so $c_{y^2} + c_{xz} \in \mathcal{J}_2^{\mathbb{Q}}(p_{k-1}) = \{0\}$.

Step 4. By Step 3 the V -degree-two part of F' equals $\sum_j \hat{r}_j P_j$ with $P_1 = c_{x^2}$, $P_2 = c_{xy}$, $P_3 = c_{y^2}$, $P_4 = c_{yz}$, $P_5 = c_{z^2}$, all in $\mathcal{I}_2^{\mathbb{Q}}(p_{k-1})$, which by (a) is spanned by the earlier relations. Subtracting $\sum_j r_{k,j} P_j$, a combination of products of relations, leaves $F'' = h_0 + x_k h_1 + y_k h_2 + z_k h_3 \in \mathcal{J}_4^{\mathbb{Q}}(p_k)$ with $\deg h_0 \leq 4$ and $\deg h_j \leq 3$. The terms $-x_{k-1} P_4$ and $-x_{k-1} x_k P_5$ of the subtracted products only contribute to these coefficients. The value and V -gradient of F'' at p_k give $h_j(p_{k-1}) = 0$ for all j , and the earlier derivatives give $\nabla h_j(p_{k-1}) = 0$ as in the cubic case. Hence $h_1, h_2, h_3 \in \mathcal{J}_3^{\mathbb{Q}}(p_{k-1}) = \{0\}$, and $h_0 \in \mathcal{J}_4^{\mathbb{Q}}(p_{k-1})$ is a combination of products of earlier relations by induction. This proves $F \in \mathcal{W}^{\mathbb{Q}}(p_k)$.

G.9 Proof of Proposition 9.16

Put $B = -2$ and $a = 2^{1/d_n}$, so $e_i = (B^i - 1)/3$, $e_0 = 0$, $p_i = a^{e_i}$, and $3d_n = |B^m - 1|$. Since $2e_i - e_{i+1} - e_{i+2} = B^i(2 - B - B^2)/3 = 0$ for $0 \leq i \leq m-3$, and the two wrapped indices give $\pm d_n$, each cyclic binomial vanishes at p ; this is the vanishing identity of the cyclic construction. Homogenize with $X_0 = 1$ and index monomials of degree at most two as $X_i X_j$, $0 \leq i \leq j \leq m-1$. Such a monomial evaluates to $a^{e_i+e_j} = 2^{\lfloor (e_i+e_j)/d_n \rfloor} a^{(e_i+e_j) \bmod d_n}$, and $1, a, \dots, a^{d_n-1}$ are linearly independent over $\mathbb{Q}$. Group the monomials into classes with equal residue of $e_i + e_j$ modulo d_n . The evaluation map has rank equal to the number of classes, so $\dim \mathcal{I}_2^{\mathbb{Q}}(p)$ equals the number of monomials minus the number of classes.

Two monomials $X_i X_j$ and $X_u X_v$ lie in one class exactly when the integer $B^i + B^j - B^u - B^v = 3(e_i + e_j - e_u - e_v)$ is divisible by $3d_n$, that is, when $B^i + B^j \equiv B^u + B^v \pmod{B^m - 1}$. Suppose the two monomials are distinct. At most four indices occur, and $m \geq 5$, so some index is unused. Multiplication by a power of B , a unit modulo $B^m - 1$ with $B^m \equiv 1$, shifts all indices cyclically, and we may assume that the unused index is $m-1$. Then all four exponents are at most $m-2$. A term has modulus 2^{m-2} only if its exponent is $m-2$, and then it equals B^{m-2} ; the other terms have modulus at most 2^{m-3} . If n_+ of the two terms B^i, B^j and n_- of the two terms B^u, B^v have exponent $m-2$, then

$$|B^i + B^j - B^u - B^v| \leq |n_+ - n_-| 2^{m-2} + (4 - n_+ - n_-) 2^{m-3} \leq 3 \cdot 2^{m-2} < 2^m - 1 \leq |B^m - 1|.$$

So the congruence is an equality of integers. If $i \neq j$ and $u \neq v$, the 2-adic valuations give $\min\{i, j\} = \min\{u, v\}$, and cancelling that term gives $\max\{i, j\} = \max\{u, v\}$, so the monomials coincide. If $i = j$ and $u = v$, then $i = u$. In the remaining case $2B^e = B^u + B^v$ with $u < v$, the valuations give $u = e + 1$, and then $B^v = B^e(2 - B) = 4B^e = B^{e+2}$. Shifting back, the only nontrivial classes are the pairs $\{X_e^2, X_{e+1}X_{e+2}\}$, indices modulo m . These m pairs are distinct and pairwise disjoint, and a square has a unique partner, so every class has at most two elements and exactly m classes have two. Hence $\dim \mathcal{I}_2^{\mathbb{Q}}(p) = m = n + 1$, and the m cyclic binomials, which vanish at p and involve distinct squares, form a basis. The last assertion of the proposition is Corollary 9.15, applied with the globally convex rational sum of squares of Theorem 7.8, which has minimum zero at p and a positive definite Hessian there, and has degree exactly four because its positive definite canonical Hessian Gram makes its quartic form positive definite.

H Proofs for Section 10

For a rational symmetric matrix M we use the bound $\|M\| \leq \max_i \sum_j |M_{ij}| \leq \sum_{i,j} |M_{ij}|$, and for $M \succ 0$ of order h the rational margin $\mu_M = \det M / (\text{tr } M)^{h-1} \leq \lambda_{\min}(M)$ (Lemma 2.7). A positive definite rational Gram matrix on a vector of rational polynomials yields a sum of squares of rational polynomials of polynomial size (Lemma 2.20).

H.1 Proof of Theorem 10.2

Put $X_0 = 1$ and let $w = (X_j g_i)_{0 \leq j \leq n, 1 \leq i \leq m}$, a list of $m(n+1)$ polynomials of degree at most three, possibly with repetitions. Then $w^\top w = \omega F_0$. Let γ be the coordinate vector with $\gamma^\top w = G$ (entries κ_i at the positions $(0, i)$), let a_a be the coordinate vector with $a_a^\top w = R_a$ given by the affine multipliers, and let U be the $m(n+1) \times n$ matrix whose j th column represents $X_j G$ (entries κ_i at the positions (j, i)). Write $R_a = X^\top T_a X + r_a^\top X + \sigma_a$ with T_a symmetric, put $b_a = X R_a \in \mathbb{Q}[X]^n$,

and let $v = (w, b_1, \dots, b_s)$. Define

$$A_0 = c_G \sum_a a_a a_a^\top - \frac{1}{2} \sum_a \sigma_a (\gamma a_a^\top + a_a \gamma^\top), \quad C_a = \frac{1}{2} (a_a \ell_G^\top - U T_a - \gamma r_a^\top),$$

$C = (C_1 \ \cdots \ C_s)$, and $\mathcal{H} = I_s \otimes H$. Then

$$v^\top \begin{pmatrix} A_0 & C \\ C^\top & \mathcal{H} \end{pmatrix} v = 0. \quad (76)$$

Indeed, the contribution of target a is

$$\begin{aligned} & c_G R_a^2 - \sigma_a G R_a + R_a \ell_G^\top X R_a - (XG)^\top T_a X R_a - G r_a^\top X R_a + R_a^2 X^\top H X \\ & = R_a^2 G - G R_a (X^\top T_a X + r_a^\top X + \sigma_a) = 0, \end{aligned}$$

using $U^\top w = XG$. The terms with σ_a and r_a are needed because the targets need not be homogeneous.

Let $\eta = \mu_H$, $B_A = \sum_{i,j} |(A_0)_{ij}|$, and

$$\epsilon = \min \left\{ 1, \frac{1}{2(1+B_A)}, \frac{\eta}{4(1+\|C\|_F^2)} \right\}, \quad Q_\epsilon = \begin{pmatrix} I + \epsilon A_0 & \epsilon C \\ \epsilon C^\top & \epsilon \mathcal{H} \end{pmatrix}.$$

By (76), $v^\top Q_\epsilon v = \omega F_0$. The top block satisfies $I + \epsilon A_0 \succeq \frac{1}{2}I$, so its inverse has norm at most two, and the Schur complement is at least $(\epsilon\eta - 2\epsilon^2\|C\|_F^2)I \succeq \frac{\epsilon\eta}{2}I$. Hence $Q_\epsilon \succ 0$, whatever the redundancies in v .

Let $A_R = (a_1 \ \cdots \ a_s)$ and $D_J = \text{diag}(A_R J A_R^\top, J \otimes I_n)$. Then $v^\top D_J v = R^\top J R + \|X\|^2 R^\top J R = \omega R^\top J R$. With $B_J = \max_i \sum_j |(D_J)_{ij}|$ put $\lambda_* = \lceil (1+B_J)/\mu_{Q_\epsilon} \rceil$. For $\lambda \geq \lambda_*$,

$$\omega(\lambda F_0 - R^\top J R) = v^\top (\lambda Q_\epsilon - D_J) v, \quad \lambda Q_\epsilon - D_J \succeq I.$$

If a rational positive definite full Hessian Gram M_F of F_0 is supplied, let B_R be a rational symmetric full Hessian Gram of $R^\top J R$, obtained by distributing each coefficient of its Hessian biform over the pairs of entries of $w(X, v)$ whose product is the corresponding monomial, and enlarge λ_* to at least $(1 + \sum_{i,j} |(B_R)_{ij}|)/\mu_{M_F}$; then $\lambda M_F - B_R \succeq I$. If the g_i vanish at p , so do the R_a , and $\lambda F_0 - R^\top J R$ is a combination of products of two polynomials vanishing at p , so its value and gradient vanish there.

All matrices have dimension at most $m(n+1) + sn$ and rational entries obtained from the input by a bounded number of arithmetic operations; μ_H and μ_{Q_ϵ} are determinants and trace powers of such matrices. Hence λ_* and the Gram certificate have polynomial size, and rational elimination with the binary splitting of positive weights produces the squares in polynomial time. A prescribed λ contributes its own length.

H.2 Affine membership is a restriction

For $g_1 = x^2 + y^2 + z^2 - 1$, $g_2 = x^2$, $g_3 = xy - z$, $g_4 = yz - 1$, direct expansion confirms

$$xg_1 + (-x + y^2 - 1)g_2 + (-xy + xz - y)g_3 + (-x^2 - x - 1)g_4 = 1.$$

Define the linear functional on polynomials of degree at most three

$$\psi(P) = [1]P + [x]P + [yz]P + [z^2]P + [xy^2]P + [xyz]P.$$

It vanishes on all sixteen products mg_i , $m \in \{1, x, y, z\}$: for instance $\psi(g_1) = -1 + 1$, $\psi(xg_1) = \psi(x^3 + xy^2 + xz^2 - x) = 1 - 1$, $\psi(yg_3) = \psi(xy^2 - yz) = 1 - 1$, $\psi(zg_3) = \psi(xyz - z^2) = 1 - 1$, $\psi(g_4) = 1 - 1$, $\psi(xg_4) = \psi(xyz - x) = 1 - 1$, and the remaining products contain no monomial seen by ψ . Since $\psi(1) = 1$, the constant 1 is not in the span of the g_i and $X_j g_i$. The four quadratics have no common complex zero: $x^2 = 0$ forces $x = 0$, then $z = 0$, contradicting $yz = 1$.

H.3 Proof of Theorem 10.3

The lists. For $\alpha \in \mathbb{Z}_{\geq 0}^n$ with $|\alpha| \leq d$ let $\kappa_\alpha = d! / ((d - |\alpha|)! \alpha_1! \cdots \alpha_n!)$, a positive integer, so that $\omega^d = \sum_{|\alpha| \leq d} \kappa_\alpha X^{2\alpha}$. Let $w = (X^\alpha g_i)_{|\alpha| \leq d, 1 \leq i \leq m}$, of level zero, and $D_0 = \text{diag}(\kappa_\alpha)$; then $w^\top D_0 w = \omega^d F_0$ and $D_0 \succeq I$. For each target a and each α with $|\alpha| \leq d - 1$ add the block $b_{a,\alpha} = X^\alpha X R_a$ of n polynomials, of level $|\alpha| + 1$, and let v be the concatenation of w and all blocks. Its length is $t = m \binom{n+d}{d} + sn \binom{n+d-1}{d-1}$.

Every parent $X^\alpha R_a$, $|\alpha| \leq d - 1$, has an explicit representation: for $\alpha = 0$ by the membership coordinates in w , and for $|\alpha| = e \geq 1$ as the j th entry of $b_{a,\alpha - e_j}$, of level e , for any j with $\alpha_j > 0$. The polynomials $X^\alpha G$ and $X^\alpha X G$ are represented in w , because G is a constant combination of the g_i and $|\alpha| + 1 \leq d$.

Zero identities. Write $R_a = X^\top T_a X + r_a^\top X + \sigma_a$. For fixed (a, α) put $u = X^\alpha$, $\pi = u R_a$, $\chi = u G$, $\zeta = u X R_a$, and $\Upsilon = u X G$. Then

$$\zeta^\top H \zeta + \pi \ell_G^\top \zeta + c_G \pi^2 - \Upsilon^\top T_a \zeta - \chi r_a^\top \zeta - \sigma_a \chi \pi = 0, \quad (77)$$

because the first three terms equal $u^2 G R_a^2$ and the last three equal $-u^2 G R_a^2$. Let $Z_{a,\alpha}$ be the symmetric rational matrix on v that represents the left side through the representations above, cross terms split equally. Its block on $b_{a,\alpha} = \zeta$ is H ; let $E_{a,\alpha}$ be $Z_{a,\alpha}$ with that block removed. With $e = |\alpha|$, the entries of $E_{a,\alpha}$ join the following levels: $c_G \pi^2$ joins (e, e) ; $\sigma_a \chi \pi$ joins $(0, e)$; $\pi \ell_G^\top \zeta$ joins $(e, e + 1)$; and the two terms with Υ and χ join $(0, e + 1)$.

One perturbation. Let $K = \sum_{a,\alpha} \sum_{i,j} |(E_{a,\alpha})_{ij}|$, $\eta_0 = \min\{1, \mu_H\}$, and $\rho = \eta_0 / (2(1 + K)) \leq \frac{1}{2}$. Put

$$Q = (D_0 \oplus 0) + \sum_{a,\alpha} \rho^{2(|\alpha|+1)} Z_{a,\alpha},$$

so $v^\top Q v = \omega^d F_0$. Let S be diagonal with entry ρ^{-l} at every coordinate of level l . In SQS the blocks D_0 and one copy of H on every new block appear unchanged, because $\rho^{2(e+1)} \rho^{-2(e+1)} = 1$. An entry of $E_{a,\alpha}$ joining levels l, l' is multiplied by $\rho^{2(e+1)-l-l'}$, and the four level pairs give exponents 2, $e + 2$, 1, and $e + 1$, all at least one. Hence the corrections in SQS have total absolute entry sum at most $K\rho < \eta_0/2$, and $SQS \succeq \eta_0 I - \frac{\eta_0}{2} I$. Since levels are at most d ,

$$Q \succeq \frac{1}{2} \eta_0 \rho^{2d} I.$$

The target. For $|\beta| \leq d$ let V_β be the rational $s \times t$ matrix expressing the vector $X^\beta R$ in v , through the membership coordinates for $\beta = 0$ and through an entry of $b_{a,\beta - e_j}$ otherwise. Then $D_J = \sum_{|\beta| \leq d} \kappa_\beta V_\beta^\top J V_\beta$ represents $\omega^d R^\top J R$. With $B_J = \max_i \sum_j |(D_J)_{ij}|$ and $\lambda_* = \lceil 2(B_J + 1) / (\eta_0 \rho^{2d}) \rceil$, every rational $\lambda \geq \lambda_*$ gives $\omega^d (\lambda F_0 - R^\top J R) = v^\top (\lambda Q - D_J) v$ with $\lambda Q - D_J \succeq I$. The entries of v have degree at most $d + 2$. Because v contains every g_i , this identity gives $\omega^d (\lambda F_0 - R^\top J R) \geq \sum_i g_i^2$, so a real zero of $\lambda F_0 - R^\top J R$ is a common zero of the g_i ; conversely the membership representations make every R_a vanish there.

Size and denominators. The matrices have dimension t , polynomial in $\binom{n+d}{d}$, n , m , s ; the multinomials have $O(d \log(n+1))$ bits; K is a sum of polynomially many explicit rationals; and ρ^{2d} has polynomial length. This gives the complexity claim, with no successive determinant bounds. If d is even, the identity gives a rational-function sum of squares with denominator $\omega^{d/2}$; if d is odd, multiply once more by ω as in (30) and use $\omega^{(d+1)/2}$. For $d = 0$ the targets lie in the constant span of the g_i , and direct domination of their Gram matrix on (g_i) gives a polynomial certificate.

H.4 The explicit multiplier certificate of Example 10.6

Let

$$b = (z^2 - \frac{x}{2}, y^2 - 2z, xz - \frac{y}{2}, xy - 1, x^2 - yz, z^3 - \frac{y}{4}, yz^2 - \frac{1}{2}, y^2z - x, \\ y^3 - 2yz, xz^2 - \frac{yz}{2}, xyz - z, xy^2 - y, x^2z - \frac{1}{2}, x^2y - x, x^3 - z)^\top$$

and let N be the symmetric integer matrix whose rows are

4240	527	792	-124	992	-1728	696	-1112	216	-312	1212	-384	-1200	76	-8
527	556	376	-194	361	-16	136	-92	0	280	-340	-232	44	168	-128
792	376	3176	-175	568	-1000	584	-232	256	-1272	432	-912	-1112	990	-384
-124	-194	-175	980	-72	432	-1080	552	-200	104	-408	488	-320	-624	136
992	361	568	-72	1960	-216	480	400	-80	-912	184	-400	-576	440	-720
-1728	-16	-1000	432	-216	2624	-1440	700	-24	0	-912	8	1052	-256	-40
696	136	584	-1080	480	-1440	3784	-1896	440	-240	728	-680	-1344	1232	-352
-1112	-92	-232	552	400	700	-1896	2000	-480	-96	-832	400	560	-816	240
216	0	256	-200	-80	-24	440	-480	320	-24	240	-384	56	264	-80
-312	280	-1272	104	-912	0	-240	-96	-24	3368	-672	264	40	-488	460
1212	-340	432	-408	184	-912	728	-832	240	-672	2800	-552	-952	376	-304
-384	-232	-912	488	-400	8	-680	400	-384	264	-552	1296	24	-944	244
-1200	44	-1112	-320	-576	1052	-1344	560	56	40	-952	24	2760	-528	0
76	168	990	-624	440	-256	1232	-816	264	-488	376	-944	-528	1848	-640
-8	-128	-384	136	-720	-40	-352	240	-80	460	-304	244	0	-640	832

Expanding both sides shows $b^\top N b = 8(1 + x^2 + y^2 + z^2)F_\star$. The leading principal minors of $N - 8I$ of orders $1, \dots, 15$ are

$$\begin{aligned} &4232, \quad 2041407, \quad 5839006240, \quad 5263436936513, \quad 8309698919905512, \\ &11480475166599386752, \quad 24140997007816543732224, \quad 9072065142124268516803584, \\ &1221674797258437785568657408, \quad 1569452470480427736351156994048, \\ &1967046692952416690050593546305536, \quad 844438887950283618215158211684597760, \\ &196739716166438745148196850208516079616, \quad 21440610749731697832005735132127356256256, \\ &497197930466744687768853864373192977547264, \end{aligned}$$

all positive, so $N - 8I \succ 0$ by Sylvester's criterion and $N/8$ is a rational positive definite Gram matrix of $\omega F_\star$ on b . Each entry of b vanishes at $p_\star$, by the evaluation rule $x^i y^j z^h \mapsto 2^{\lfloor e/5 \rfloor - i - h} a^{e \bmod 5}$ with $e = 4i + j + 2h$; the fifteen entries have distinct leading monomials, none of which occurs in another entry, so they are independent; and the rational polynomials of degree at most three vanishing at $p_\star$ form a space of dimension $20 - 5 = 15$. Thus b is a basis of that space, which explains why a certificate of this shape can exist. The identity and the positivity are the only finite verifications in Example 10.6.

H.5 Proofs of Theorems 10.7 and 10.8

Throughout, $p = (a^2, a^3, a^4)$ with $a = 2^{1/5}$, the variables are $X = (x_0, x_1, x_2)$, and f° is as in Section 10.3.

Theorem 10.7(a) and (b). Let A° be a rational positive definite full Hessian Gram of f° . Since $\nabla^2 f_t(X) = \nabla^2 f^\circ(tX)$ and $w(tX, v) = \text{diag}(I_3, tI_9) w(X, v)$, the matrix $\text{diag}(I_3, tI_9) A^\circ \text{diag}(I_3, tI_9)$ is a rational positive definite full Hessian Gram of f_t , and $\nabla^2 f_t \succeq I$. The minimizer is p/t , with $f_t(p/t) = 0$ and $\nabla^2 f_t(p/t) = \nabla^2 f^\circ(p)$. Each coefficient of f_t is $t^{|\gamma|-2} [X^\gamma] f^\circ$ with $|\gamma| \leq 4$, so $L(t) = O(\log t)$; the quartic part of f° is nonzero, so some coefficient is a nonzero integer multiple of t^2 and $L(t) \geq \log_2 t$. For (b), write $\omega f^\circ = \sum_j g_j^2$ with fixed rational cubics. Then

$$(1 + t^2 \|X\|^2) f_t(X) = t^{-2} \omega(tX) f^\circ(tX) = \sum_j (g_j(tX)/t)^2,$$

a fixed number of rational cubics with coefficients of $O(\log t)$ bits. The identity (30), with ω replaced by $1 + t^2 \|X\|^2$, gives the rational-function certificate. If f_t were a rational sum of squares, the substitution $X = x/t$ would make f° one, contradicting Theorem 9.4 for f° ; Lemma 10.1 then excludes degree at most one.

Vanishing spaces. For $d \geq 2$ let $I_d = \mathcal{I}_d^\circ(p)$ and let V_d be its real span. Use the representatives $\varrho_0 = 1, \varrho_1 = x_0 x_2, \varrho_2 = x_0, \varrho_3 = x_1, \varrho_4 = x_2$, whose values at p are $1, 2a, a^2, a^3, a^4$. A monomial $m = x_0^i x_1^j x_2^h$ has $m(p) = a^e = 2^{\lfloor e/5 \rfloor} a^{e \bmod 5}$ with $e = 2i + 3j + 4h$, so $m(p) = c_m \varrho_{e \bmod 5}(p)$ with $c_m = 2^{\lfloor e/5 \rfloor - \lfloor e \equiv 1 \rfloor}$. For every monomial m of degree at most d other than the representatives, put $\pi_m = m - c_m \varrho_{e \bmod 5}$. These polynomials form a basis of I_d : each contains its own monomial m with coefficient one and no other non-representative monomial, and a rational combination of the representatives vanishing at p is zero because $1, a, \dots, a^4$ are linearly independent. The bases are nested in d . For $\deg m \leq d$ we have $e \leq 4d$, so $c_m \in \{2^{-1}, 1, \dots, 2^{d-1}\}$; every π_m has coefficients in $\frac{1}{2}\mathbb{Z}$ and ℓ_1 -norm at most 2^d . For $d \geq 3$ the new basis elements, those with $\deg m = d$, have leading form exactly m . Finally, $\varphi_5(g^2) \geq 0$ for every $g \in V_2$: every rational element of I_2 satisfies the linear condition $v_2 + h_{00} = 0$ of Lemma 9.3, hence so does every real combination, and then $\varphi_5(g^2) = v_0^2 + 2g_0(v_2 + h_{00}) = v_0^2$.

Grid functionals. A real polynomial in three variables of degree at most d in each variable that vanishes on $\{0, 1, \dots, d\}^3$ is zero: for fixed values of the last two variables in the grid the univariate polynomial in the first variable has $d+1$ roots, so its coefficients vanish on the grid in the remaining variables, and induction finishes. Let $E_2(P) = \sum_{u \in \{0,1,2\}^3} P(u)$ and, for $d \geq 3$, let $M_d(P) = \sum_{u \in \{0,\dots,d\}^3} P_{2d}(u)$, where P_{2d} is the homogeneous part of degree $2d$.

Separators. Let $K_0 = E_2(f^\circ) + 1$, a positive integer since $f^\circ \geq 0$ has integer coefficients, and $\psi_2 = \varphi_5 + E_2/K_0$ on $\mathbb{R}[X]_{\leq 4}$. Then $a_0 = -\psi_2(f^\circ) = 4 - E_2(f^\circ)/K_0 > 3$. For $0 \neq g \in V_2$, g has degree at most two in each variable, so $E_2(g^2) = \sum_u g(u)^2 > 0$, and $\psi_2(g^2) > 0$. All monomial values of ψ_2 lie in $K_0^{-1}\mathbb{Z}$.

Let $d \geq 3$ and suppose ψ_{d-1} is a linear functional on $\mathbb{R}[X]_{\leq 2d-2}$ with monomial values in $K_0^{-1}\mathbb{Z}$, positive on squares of nonzero elements of V_{d-1} , and with $\psi_{d-1}(f^\circ) = -a_0$. Extend it by zero to the monomials of degrees $2d-1$ and $2d$, obtaining $\hat{\psi}_{d-1}$. On the nested basis of I_d its Gram matrix is $\begin{pmatrix} A & B \\ B^\top & C \end{pmatrix}$, where $A \succ 0$ is the Gram matrix of ψ_{d-1} on the basis of I_{d-1} . The functional M_d vanishes on products of an old basis element with any basis element, which have degree at most $2d-1$; on two new elements with leading monomials m, m' it gives $D_{mm'} = \sum_u m(u)m'(u)$. The integer matrix D is positive definite: for a nonzero coefficient vector the form $\sum c_m m$ is nonzero, has degree at most d in each variable, and does not vanish on the grid. Put $S = C - B^\top A^{-1} B$, $m_d = \max_i \sum_j |S_{ij}|$, $\beta_d = \mu_D$, $T_d = \lceil (m_d + 1)/\beta_d \rceil$, and $\psi_d = \hat{\psi}_{d-1} + T_d M_d$. The Gram matrix of

ψ_d on the basis of I_d is $\begin{pmatrix} A & B \\ B^\top & C+T_d D \end{pmatrix}$, with Schur complement $S + T_d D \succeq I$, so it is positive definite. Its monomial values remain in $K_0^{-1}\mathbb{Z}$, and $\psi_d(f^\circ) = -a_0$ because $\deg f^\circ = 4 \leq 2d - 2$.

Theorem 10.7(c). Fix $N \geq 0$, put $d = N + 2$, $s = \|X\|^2$, and $A_N = \sum_{j=1}^N \binom{N}{j} |\psi_d(s^j f^\circ)|$, and let $\tau_N = \lceil \sqrt{\max\{1, 2(A_N + 1)/a_0\}} \rceil$. For an integer $t \geq \tau_N$ put $\epsilon = t^{-2} \leq \min\{1, a_0/(2(A_N + 1))\}$. Then

$$\psi_d((1 + \epsilon s)^N f^\circ) \leq -a_0 + \epsilon A_N < -a_0/2.$$

If $\omega^N f_t$ were a sum of squares in $\mathbb{Q}[X]$, the substitution $X = x/t$ and multiplication by t^2 would make $(1 + \epsilon s)^N f^\circ$ one. Its factors have degree at most d and vanish at p , so they lie in I_d , and ψ_d would be nonnegative on it. This contradiction proves the first claim; τ_N is computed by exact rational arithmetic. For a fixed $u \in \mathbb{Q}[X]$ with $u(0) > 0$, let $d = 2 + \lceil \deg u/2 \rceil$. As $t \rightarrow \infty$ the polynomial $u(x/t)f^\circ(x)$ converges coefficientwise to $u(0)f^\circ$, so $\psi_d(u(x/t)f^\circ) \rightarrow -u(0)a_0 < 0$, and the same argument excludes a rational sum of squares of $u f_t$ for large t .

Theorem 10.8. Let $H_d = \max\{1, \max_{\deg m \leq 2d} |K_0 \psi_d(m)|\}$, an integer. Fix $d \geq 3$, and let $n_d = \dim I_{d-1} \leq d^3$ and $h_d = \binom{d+2}{2} \leq 2d^2$ be the numbers of old and new basis elements. A product of two basis elements has coefficients in $\frac{1}{4}\mathbb{Z}$ and ℓ_1 -norm at most 2^{2d} , so every entry of the Gram matrix of $\hat{\psi}_{d-1}$ becomes an integer of absolute value at most $U_d = 2^{2d+2} H_{d-1}$ after multiplication by $Q_0 = 4K_0$. The integer matrix $Q_0 A$ is positive definite, so $\det(Q_0 A) \geq 1$, and its cofactors are at most $(n_d - 1)! U_d^{n_d-1}$. Hence $|(A^{-1})_{ij}| \leq Q_0(n_d U_d)^{n_d-1}$, and

$$|S_{ij}| \leq \frac{U_d}{Q_0} + n_d^2 \left(\frac{U_d}{Q_0}\right)^2 Q_0(n_d U_d)^{n_d-1} \leq 2(n_d U_d)^{n_d+1}, \quad m_d \leq 2h_d(n_d U_d)^{n_d+1}.$$

The entries of D are at most $J_d = (d+1)^3 d^{2d}$, $\det D \geq 1$, and $\text{tr } D \leq h_d J_d$, so $\beta_d^{-1} \leq (h_d J_d)^{h_d-1}$ and $T_d \leq (m_d + 2)(h_d J_d)^{h_d}$. The new monomial values satisfy $|K_0 T_d M_d(m)| \leq K_0 T_d J_d$, and old values are unchanged, so $H_d \leq \max\{H_{d-1}, K_0(m_d + 2)(h_d J_d)^{h_d+1}\}$. Taking logarithms, with $\log_2 U_d = 2d + 2 + \log_2 H_{d-1}$, $n_d + 1 \leq 2d^3$, and $\log_2 J_d \leq 5d \log_2(d+1)$, there is an absolute constant C_1 with

$$\log_2 H_d \leq C_1 d^3 (d + \log_2 H_{d-1}) + \log_2 K_0.$$

Put $\beta'_d = d + \log_2 H_d + \log_2 K_0 \geq 1$. Then $\beta'_d \leq C_2 d^3 \beta'_{d-1}$ for an absolute constant C_2 , so $\beta'_d \leq C_2^d (d!)^3 \beta'_2$ and $\log_2 \beta'_d \leq C_3 d \log_2 d$ for $d \geq 3$, with C_3 depending only on f° .

For the threshold, $|\psi_d(P)| \leq (H_d/K_0) \|P\|_1$, where $\|P\|_1$ is the sum of the absolute coefficients, and $\|s^j f^\circ\|_1 \leq 3^j \|f^\circ\|_1$. Hence $A_N \leq 6^N H_{N+2} \|f^\circ\|_1$, and since $a_0 > 3$, $\log_2 \tau_N \leq C_4 \beta'_{N+2}$ and $\log_2 \log_2 \tau_N \leq C_5 (N+2) \log_2 (N+2)$ for $N \geq 1$, with constants depending only on f° . The bound also holds for $\tau'_N = \max_{j \leq N} \tau_j$, since its right side increases with N .

For an integer t put $u = \log_2 \log_2 t$, and assume u is large enough that $N = \lfloor u/(2C_5 \log_2 u) \rfloor - 2 \geq 1$. Then $N + 2 \leq u$ and $C_5 (N+2) \log_2 (N+2) \leq u/2$, so $t \geq \tau'_N$. By part (c), $\omega^j f_t$ is not a rational sum of squares for any $j \leq N$, so $\nu(f_t) \geq N + 1$. This gives $\nu(f_t) \geq c u / \log_2 u$ for $t \geq t_0$, which is the claim, and $L(t) = \Theta(\log t)$ gives the form in L . The argument does not use the finiteness of $\nu(f_t)$, which is Proposition 10.5.

H.6 Proof of Proposition 10.5

Imported results. All rings are commutative and contain $\mathbb{Q}$. A *cone* in an additive group is closed under addition and contains zero. For a cone C in a group G , an element $u \in C$ is an *order unit* of (G, C) if for every $g \in G$ some positive integer r has $ru \pm g \in C$. A *state* of (G, C, u) is an

additive map $\psi : G \rightarrow \mathbb{R}$ that is nonnegative on C and has $\psi(u) = 1$; a *pure state* is an extreme point of the convex set of states. A *preordering* S of a ring A is a cone containing all squares and closed under multiplication; it is *archimedean* if for every $a \in A$ some positive integer R has $R - a \in S$. For an ideal I , an *S -pseudomodule* in I is a cone $C \subseteq I$ with $SC \subseteq C$. We use the following results of Burgdorf, Scheiderer, and Schweighofer.

- (I) [21, Theorem 2.5] If u is an order unit of (G, C) and $\psi(g) > 0$ for every pure state ψ of (G, C, u) , then $rg \in C$ for some positive integer r .
- (II) [21, Corollary 4.12] Let S be an archimedean preordering of A , I an ideal, $C \subseteq I$ an S -pseudomodule with order unit u , and ψ a pure state of (I, C, u) . If $\psi(u^2) \neq 0$, there is a ring homomorphism $\lambda : A \rightarrow \mathbb{R}$, nonnegative on S , with $\lambda(u) \neq 0$ and $\psi(h) = \lambda(h)/\lambda(u)$ for all $h \in I$. If $\psi(u^2) = 0$, there is a ring homomorphism $\lambda : A \rightarrow \mathbb{R}$, nonnegative on $S + I$, with $\lambda(I) = 0$ and $\psi(ah) = \lambda(a)\psi(h)$ for all $a \in A$ and $h \in I$.
- (III) [21, Theorem 6.2] If S is an archimedean preordering of A and $\lambda(h) > 0$ for every ring homomorphism $\lambda : A \rightarrow \mathbb{R}$ nonnegative on S , then $rh \in S$ for some positive integer r .

The order-unit statement of [21, Proposition 5.3(a) and Remark 5.5(1)] is also used; we prove the case needed below directly. Every positive rational is a sum of rational squares, so a preordering of sums of squares is closed under multiplication by positive rationals, and $rh \in S$ implies $h \in S$.

The sphere ring. Let $y = (y_0, \dots, y_n)$, $q = \sum_i y_i^2$, $A = \mathbb{Q}[y]/(q - 1)$, and let $S \subseteq A$ be the set of sums of squares, a preordering. Let B be the set of $a \in A$ with $N \pm a \in S$ for some positive integer N . It contains $\mathbb{Q}$, and it is closed under addition, negation, and multiplication by rationals. If $N \pm a \in S$, then

$$N^2 - a^2 = \frac{1}{2N}((N + a)^2(N - a) + (N - a)^2(N + a)) \in S$$

and $N^2 + a^2 \in S$, so $a^2 \in B$; hence $ab = ((a + b)^2 - (a - b)^2)/4 \in B$ for $a, b \in B$, and B is a subring. In A , $1 \pm y_i = \frac{1}{2}((1 \pm y_i)^2 + \sum_{j \neq i} y_j^2) \in S$, so every y_i lies in B , $B = A$, and S is archimedean. The ring homomorphisms $A \rightarrow \mathbb{R}$ are the evaluations at points of the real unit sphere S^n , and all of them are nonnegative on S .

Let $H(y) = y_0^{2e} f(y_1/y_0, \dots, y_n/y_0)$ be the homogenization of f , a form of degree $2e$. For $y_0 = 0$ and $y \neq 0$, $H(y)$ is the positive definite leading form of f at $(y_1, \dots, y_n)$, and for $y_0 \neq 0$, $H(y) = y_0^{2e} f(y'/y_0) \geq 0$. Hence $H \geq 0$ on $\mathbb{R}^{n+1}$, and its zeros on S^n form the finite set $Z = \{\pm(1, z)/\|(1, z)\| : f(z) = 0\}$. If Z is empty, (III) gives $H \in S$; continue with the last paragraph.

Algebraicity of the zeros. Let z be a real zero of f and $K = \mathbb{Q}(z_1, \dots, z_n)$. Since $f \geq 0$, $\nabla f(z) = 0$, and $\nabla^2 f(z)$ is invertible. Suppose K had positive transcendence degree over $\mathbb{Q}$. Choose a transcendence basis and the derivation of the rational function field that sends one basis element to one and the others to zero; in characteristic zero it extends uniquely to a derivation ∂ of the finite algebraic extension K . Applying ∂ to the rational identities $\nabla f(z) = 0$ gives $\nabla^2 f(z) \partial z = 0$, hence $\partial z = 0$, so ∂ vanishes on the generators of K and is zero, a contradiction. Thus z , and every point of Z , has algebraic coordinates.

The vanishing ideal. For $x \in Z$ the evaluation $a \mapsto a(x)$ maps A onto the number field $\mathbb{Q}(x)$, so its kernel $\mathfrak{m}_x$ is a maximal ideal. Let $\mathfrak{m}_1, \dots, \mathfrak{m}_t$ be the distinct ideals among them, and $J = \mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_t$. Distinct maximal ideals are comaximal, and so are their powers: if

$a + b = 1$ with $a \in \mathfrak{m}_i$, $b \in \mathfrak{m}_j$, expanding $(a + b)^3$ shows $1 \in \mathfrak{m}_i^2 + \mathfrak{m}_j^2$. Hence $J = \mathfrak{m}_1 \cdots \mathfrak{m}_t$ and $\mathfrak{m}_1^2 \cap \cdots \cap \mathfrak{m}_t^2 = \mathfrak{m}_1^2 \cdots \mathfrak{m}_t^2 = J^2$. A real point of S^n at which all of J vanishes gives a ring homomorphism $A/J \cong \prod_i A/\mathfrak{m}_i \rightarrow \mathbb{R}$, which factors through one residue field; since H lies in every $\mathfrak{m}_i$, it vanishes there, so the point is in Z . The real zero set of J on S^n is therefore exactly Z .

We show $H \in J^2$. Let $x \in Z$ and $\mathfrak{M} = \{g \in \mathbb{Q}[y] : g(x) = 0\}$, the preimage of $\mathfrak{m}_x$. Since $H \geq 0$ on $\mathbb{R}^{n+1}$ and $H(x) = 0$, the point x is a minimizer and $\nabla H(x) = 0$. Now a polynomial $g \in \mathfrak{M}$ with $\nabla g(x) = 0$ lies in $\mathfrak{M}^2$. Indeed, after an invertible rational linear change of coordinates, which transforms gradients linearly, the primitive element theorem lets us assume $\mathbb{Q}(x) = \mathbb{Q}(x_0)$. Let μ be the minimal polynomial of x_0 , so $\mu'(x_0) \neq 0$, and write $x_i = c_i(x_0)$ with $c_i \in \mathbb{Q}[T]$. The polynomials $g_0 = \mu(y_0)$ and $g_i = y_i - c_i(y_0)$, $1 \leq i \leq n$, vanish at x , and the quotient of $\mathbb{Q}[y]$ by the ideal they generate is $\mathbb{Q}[y_0]/(\mu)$, a field; so they generate $\mathfrak{M}$. Their gradients at x , namely $\mu'(x_0)e_0$ and $e_i - c'_i(x_0)e_0$, are linearly independent over $\mathbb{Q}(x)$. If $g = \sum_j a_j g_j$ and $\nabla g(x) = \sum_j a_j(x) \nabla g_j(x) = 0$, then every $a_j(x) = 0$, so $a_j \in \mathfrak{M}$ and $g \in \mathfrak{M}^2$. The same computation shows that $\mathfrak{M}/\mathfrak{M}^2$ has dimension $n + 1$ over $\mathbb{Q}(x)$, the class of g corresponding to $\nabla g(x)$. Applied to H , this gives $H \in \mathfrak{M}^2$, so the class of H lies in $\mathfrak{m}_x^2$ for every $x \in Z$, hence in J^2 .

An order unit. Let $b_1, \dots, b_r$ generate J , put $I = J^2$, $C = S \cap I$, and $u = \sum_i b_i^2$. The cone C is an S -pseudomodulo in I . Every $h \in I$ is a finite sum of terms $ab_i b_j$ with $a \in A$, and for an integer $R \geq 1$ with $R - a^2 \in S$,

$$R(b_i^2 + b_j^2) \pm 2ab_i b_j = (ab_i \pm b_j)^2 + (R - a^2)b_i^2 + (R - 1)b_j^2 \in C.$$

Summing over a representation of h , and using $u - b_i^2 \in C$, gives a positive integer N with $Nu \pm h \in C$. Since also $u \in C$, u is an order unit of (I, C) . The zeros of u on S^n are exactly Z .

Pure states. Let ψ be a pure state of (I, C, u) and λ the homomorphism given by (II), evaluation at a point $x \in S^n$.

If $\psi(u^2) \neq 0$, then $\lambda(u) \neq 0$; as u is a sum of squares, $\lambda(u) > 0$, so $x \notin Z$ and $\psi(H) = H(x)/u(x) > 0$.

If $\psi(u^2) = 0$, then $\lambda(I) = 0$. For $b \in J$ we have $b^2 \in I$, so $\lambda(b)^2 = 0$; thus $\lambda(J) = 0$ and $x \in Z$. Let $\mathfrak{m} = \mathfrak{m}_x$ and $K = A/\mathfrak{m} \cong \mathbb{Q}(x)$, embedded in $\mathbb{R}$ by λ . The relation $\psi(ah) = \lambda(a)\psi(h)$ shows that ψ vanishes on $\mathfrak{m}I$ and is K -semilinear. The K -vector spaces $I/\mathfrak{m}I$ and $\mathfrak{m}^2/\mathfrak{m}^3$ coincide: both are annihilated by $\mathfrak{m}$, hence unchanged by localization at $\mathfrak{m}$, and J and $\mathfrak{m}$ have the same localization because the other $\mathfrak{m}_i$ contain elements outside $\mathfrak{m}$. The space $\mathfrak{m}/\mathfrak{m}^2 = \mathfrak{M}/(\mathfrak{M}^2 + (q - 1))$ has dimension n over K , because the class of $q - 1$ corresponds to $\nabla q(x) = 2x \neq 0$; its elements correspond to the restrictions to $T_x S^n = x^\perp$ of the K -rational covectors $\nabla g(x)$. If $c_1, \dots, c_n \in \mathfrak{m}$ have classes forming a K -basis of $\mathfrak{m}/\mathfrak{m}^2$, every product of two elements of $\mathfrak{m}$ is congruent modulo $\mathfrak{m}^3$ to a combination of the $c_i c_j$ with coefficients in A , so $\dim_K \mathfrak{m}^2/\mathfrak{m}^3 \leq n(n + 1)/2$.

For $b \in \mathfrak{m}^2$ let $\mathcal{Q}_b$ be the Hessian at x of the restriction of b to S^n , a quadratic form on $T_x S^n$, well defined because b and its tangential differential vanish at x . The map $b \mapsto \mathcal{Q}_b$ vanishes on $\mathfrak{m}^3$, satisfies $\mathcal{Q}_{kb} = \lambda(k)\mathcal{Q}_b$, and sends cc' with $c, c' \in \mathfrak{m}$ to $dc \otimes dc' + dc' \otimes dc$, where dc is the tangential differential at x ; these differentials span the real cotangent space. So $b \mapsto \mathcal{Q}_b$ induces a surjective $\mathbb{R}$ -linear map from $(\mathfrak{m}^2/\mathfrak{m}^3) \otimes_{K, \lambda} \mathbb{R}$ onto the quadratic forms on $T_x S^n$, a space of dimension $n(n + 1)/2$, at least that of the source; it is therefore bijective. Hence there is a real symmetric form Φ on $T_x S^n$ with $\psi(h) = \langle \Phi, \mathcal{Q}_h \rangle$ for all $h \in I$.

The form Φ is positive semidefinite and nonzero. For $g \in J$ we have $g^2 \in C$, so $0 \leq \psi(g^2) = 2 \langle \Phi, dg \otimes dg \rangle$. Since J maps onto $\mathfrak{m}/\mathfrak{m}^2$, the differentials dg , $g \in J$, fill the K -rational points of the cotangent space, which are dense, so $\Phi \succeq 0$. Also $1 = \psi(u) = 2 \sum_i \langle \Phi, db_i \otimes db_i \rangle$, so $\Phi \neq 0$. Finally

$\mathcal{Q}_H \succ 0$. Write $x = (x_0, x')$ with $x_0 \neq 0$ and $z = x'/x_0$, so $f(z) = 0$, $\nabla f(z) = 0$, and $\nabla^2 f(z) \succ 0$. With $\pi(y) = (y_1/y_0, \dots, y_n/y_0)$ and $H = y_0^{2e} f \circ \pi$ near x , the vanishing value and gradient of $f \circ \pi$ at x give $\nabla^2 H(x) = x_0^{2e} D\pi(x)^\top \nabla^2 f(z) D\pi(x)$. The kernel of $D\pi(x)$ is $\mathbb{R}x$, which meets $x^\perp$ only in zero, and since $\nabla H(x) = 0$ the Hessian of $H|_{S^n}$ at x is the restriction of $\nabla^2 H(x)$ to $x^\perp$. Hence $\mathcal{Q}_H \succ 0$, and $\psi(H) = \langle \Phi, \mathcal{Q}_H \rangle > 0$, a nonzero positive semidefinite form paired with a positive definite one.

Every pure state is therefore positive at H . By (I), $rH \in C \subseteq S$ for some positive integer r , and so $H \in S$.

From the sphere to the radial multiplier. Write $H = \sum_i p_i^2$ in A , with polynomials $p_i \in \mathbb{Q}[y]$, so $H - \sum_i p_i^2 \in (q-1)$. Split $p_i = e_i + o_i$ into its parts that are even and odd under $y \mapsto -y$. Since H and $q-1$ are even, also $H - \sum_i p_i(-y)^2 \in (q-1)$, and averaging gives $H - \sum_i (e_i^2 + o_i^2) \in (q-1)$. Let $D \geq 2e$ be an even integer bounding all $\deg p_i$. Multiplying each monomial of e_i of degree $2j$ by $q^{(D-2j)/2}$, and each monomial of o_i of degree $2j+1$ by $q^{(D-2j-2)/2}$, gives forms E_i of degree D and O_i of degree $D-1$ with the same values on S^n . The forms $q^{D-e}H$ and $\sum_i E_i^2 + q \sum_i O_i^2$ of degree $2D$ agree on S^n , hence everywhere by homogeneity:

$$q^{D-e}H = \sum_i E_i^2 + \sum_i \sum_{j=0}^n (y_j O_i)^2.$$

Setting $y_0 = 1$ gives $q(1, X) = \omega(X)$ and $H(1, X) = f(X)$, so $\omega^{D-e}f$ is a sum of squares in $\mathbb{Q}[X]$. Multiplying by further powers of the rational sum of squares ω shows that every larger exponent also works.

H.7 Proof of Lemma 10.9

Let $\mu = \mu_A/2$, so that $A \succeq 2\mu I$, and $\nabla^2 f \succeq 2\mu I$. Put $r_{ij} = y_i y_j - z_{ij}$ for $i \leq j$. In each cubic monomial of the gradient $\nabla f(y)$ replace the product $y_i y_j$ of its two smallest indices, $i \leq j$, by z_{ij} , and keep the terms of degree at most two. This gives a rational quadratic map $a(y, z)$ with $a(y, (y_i y_j)_{i \leq j}) = \nabla f(y)$ and

$$\nabla f(y) - a(y, z) = E(y) r(y, z),$$

where the matrix $E(y)$ has homogeneous linear entries: a cubic term $c y_l y_i y_j$ is replaced by $c y_l z_{ij}$, and the difference is $c y_l r_{ij}$.

Put $d = x - y$ and let $\mathbf{w} = (d, y \otimes d, d \otimes d)$, of length $n + 2n^2$. Partitioning A into its blocks A_{00}, A_{01}, A_{11} for the coordinates v and $X \otimes v$, Lemma 8.1(a) with the polynomial center y , equivalently Taylor's formula with integral remainder, gives

$$f(x) - f(y) - \nabla f(y)^\top d = \mathbf{w}^\top \mathcal{M} \mathbf{w}, \quad \mathcal{M} = \begin{pmatrix} A_{00}/2 & A_{01}/2 & A_{01}/6 \\ A_{10}/2 & A_{11}/2 & A_{11}/6 \\ A_{10}/6 & A_{11}/6 & A_{11}/12 \end{pmatrix},$$

since $w(y + td, d) = (d, y \otimes d) + t(0, d \otimes d)$ and $\int_0^1 (1-t)t^j dt = \frac{1}{2}, \frac{1}{6}, \frac{1}{12}$ for $j = 0, 1, 2$. The same integration applied to $A \succeq 2\mu I$ gives $\mathcal{M} \succeq 2\mu [\frac{1}{2}I_n \oplus ((\frac{1}{6}I_n \oplus \frac{1}{12}I_{n^2})) \otimes I_n]$, and the 2×2 matrix has determinant $1/72 > 0$. Hence $\mathcal{N} = \mathcal{M} - \frac{\mu}{2} \text{diag}(I_n, 0, 0) \succ 0$. Let $\mathcal{D}$ be the rational matrix with $(\nabla f(y) - a)^\top d = r^\top \mathcal{D} \mathbf{w}$; only the columns of $y \otimes d$ are used. Let $\delta = \mu_{\mathcal{N}}$ and $\lambda = \lceil 1 + \|\mathcal{D}\|_F^2 / (4\delta) \rceil$. The matrix

$$\mathcal{K} = \begin{pmatrix} \mathcal{N} & \mathcal{D}^\top/2 \\ \mathcal{D}/2 & \lambda I \end{pmatrix}$$

is positive definite, its Schur complement being $\lambda I - \frac{1}{4}\mathcal{DN}^{-1}\mathcal{D}^\top \succeq I$. Put

$$g(y, z) = \frac{\|a(y, z)\|^2}{2\mu} - f(y) + \lambda \|r(y, z)\|^2.$$

We claim

$$f(x) + g(y, z) = (\mathbf{w}, r)^\top \mathcal{K}(\mathbf{w}, r) + \frac{\mu}{2} \left\| d + \frac{a(y, z)}{\mu} \right\|^2.$$

The first term equals $f(x) - f(y) - \nabla f(y)^\top d - \frac{\mu}{2} \|d\|^2 + (\nabla f(y) - a)^\top d + \lambda \|r\|^2$, and the second equals $\frac{\mu}{2} \|d\|^2 + a^\top d + \|a\|^2 / (2\mu)$; their sum is $f(x) - f(y) + \lambda \|r\|^2 + \|a\|^2 / (2\mu) = f(x) + g(y, z)$. Both matrices are rational and positive semidefinite, so this is a rational sum of squares with quadratic factors. Substituting $x = p$ gives $g \geq 0$, because $f(p) = 0$. At $(y, z) = (p, (p_i p_j))$ we have $r = 0$ and $a = \nabla f(p) = 0$, so $g = 0$. All quantities are obtained by rational linear algebra of polynomial size from (f, A) , without computing p .

H.8 Proof of Theorem 10.11

(a) The quartic F_k° is a sum of $2k + 1$ squares of rational quadratics with unique zero p_k° , and $\delta_k = u_k(p_k^\circ)$ satisfies $0 < \delta_k \leq M_k^{-2^k}$ (Section 8.2). With $\lambda = \lceil 3/\mu_{A_k^\circ} \rceil$, the rational matrix $\lambda A_k^\circ + 2ee^\top \succeq 3I$ is a full Hessian Gram of h_k , where e selects the direction coordinate of the variable in u_k . The quartic h_k is a rational sum of squares, and it is positive: a zero would be a zero of F_k° , hence p_k° , where $u_k^2 = \delta_k^2 > 0$. Strong convexity makes its minimum m_k attained, and $m_k \leq h_k(p_k^\circ) = \delta_k^2 \leq M_k^{-2^{k+1}}$. The blocks f and $\tilde{g}$ have minimum zero, attained at p_1 and w_* . The Hessian of P_k is block diagonal, so adding the three block Hessian certificates gives a rational sum-of-squares Hessian certificate and $\nabla^2 P_k \succeq I$; adding the rational sums of squares of $f + \tilde{g}$ and h_k gives a rational sum of squares of P_k . The variable count is $3 + 9 + 2k$.

(b) Choose rationals c and s with $0 < c$, $0 < s$, and $c + s < m_k$, and a rational point x' with $f(x') < s$, possible since $f(p_1) = 0$. By Lemma 8.1(c), applied at a rational center close to the minimizer, the certified quartics $f + c$ and $h_k - c - s$, whose minima are positive, have rational positive definite Gram matrices, hence rational sums of squares. Specializing the joint rational certificate of $f + \tilde{g}$ at $x = x'$ shows that $f(x') + \tilde{g}$ is a rational sum of squares, and so is $\tilde{g} + s = (f(x') + \tilde{g}) + (s - f(x'))$. Then $P_k = (f + c) + [(\tilde{g} + s) + (h_k - c - s)]$ is block separated. No size bound is claimed for these certificates.

(c) Group a separated certificate as $P_k = S_x(x) + S'(w, \xi)$ with S_x and S' nonnegative. As in the proof of Theorem 10.10, $S_x = f + c$ and $S' = \tilde{g} + h_k - c$ for a rational constant c . Evaluating S_x at p_1 gives $c \geq 0$, and evaluating S' at w_* and a minimizer of h_k gives $c \leq m_k$. If $c = 0$, then S_x would be a rational certificate of f , which Theorem 9.9(b) with $E = \mathbb{Q}$ excludes. Thus $0 < c \leq M_k^{-2^{k+1}}$, and writing c in lowest terms with a positive numerator shows $\text{den}(c) \geq M_k^{2^{k+1}}$. For a separated Gram certificate, the constant entry of Q_x equals $f(0) + c$, so $\text{den}(c) \leq b_0 \text{den}((Q_x)_{00})$. For a separated sum of squares with x -factors u_j , $c = \sum_j u_j(0)^2 - f(0)$, so $\text{den}(c) \leq b_0 \prod_j \text{den}(u_j(0))^2$. Taking logarithms gives both bounds, and $2^k \log_2 M_k = (k + 3)2^k \log_2 1000 = \Omega(k2^k)$.

I Proofs for core noise and recourse

This appendix proves every result of Section 11. The first six subsections build the common machinery: external tools, finite-law transfer, the exact fallback, certified cells, the all-scale count, and the projected-growth tail. The remaining subsections prove the theorems in the order of Section 11.

I.1 Conventions and external tools

Norms are Euclidean unless marked otherwise; $\|\cdot\|_\infty$ is the maximum norm. Leb is Lebesgue measure, Leb^* its outer measure, $B(c, r)$ the open Euclidean ball, and $v_k \leq 2^k$ the volume of the unit ball in $\mathbb{R}^k$. For a rational number, vector, or matrix r , $\langle r \rangle$ is its bit length (Definition 2.1). For a core-noise instance (Definition 11.1) we write $x = (v, z)$, $\kappa_+ = \kappa + \sigma > 0$, and

$$X_C = \{x \in X : v \in C\}$$

for a set $C \subseteq \mathbb{R}^k$. Every instance has a description of X by at most L explicit rational inequalities: the $2n$ box inequalities in type (B), and the rows of $Ex \leq e$ together with the supplied box in type (P). A bound is *base-computable* if it is computed from the instance before sampling, in time polynomial in L . The letter b denotes the largest bit length of a sampled coefficient. A point of the grid G_M is $\sigma(2j - M + 1)/(M - 1)$, so its numerator and denominator are those of σ multiplied by integers of absolute value less than M ; since M is a power of two, $b \leq L + 2 \log_2 M$. We use the following tools.

Convex values. Lemma 3.1 supplies the convex value interface. We use it in this form. Let Q be a polyhedron given by explicit rational inequalities and contained in a supplied rational box, and let ϕ be an explicitly encoded rational polynomial of degree at most D that is convex on Q . Given a rational $\eta > 0$, one decides whether Q is empty and, if not, computes a rational $x \in Q$ and a rational ℓ with

$$\ell \leq \min_Q \phi \leq \phi(x) \leq \ell + \eta, \quad (78)$$

in bit work $P_D(L_Q + \langle \eta \rangle)$, where L_Q is the bit length of Q , its box, and ϕ . The tolerance enters through its bit length $\langle \eta \rangle$, not only through $\log(1/\eta)$: a rational close to 1 can have long numerator and denominator. Every tolerance used below has bit length polynomial in L and the query precision. Rational linear programs, including feasibility problems, are solved exactly in polynomial time [49, Theorem (6.4.12) and Section 6.5], and quadratic programs on nonempty bounded rational polytopes with rational data and a positive semidefinite objective Hessian have rational optimal solutions computable in polynomial time [71].

Block quantifier elimination. We use formulas

$$\Phi(t) = Q_1 x^{(1)} Q_2 x^{(2)} \Psi(t, x^{(1)}, x^{(2)}), \quad Q_1, Q_2 \in \{\exists, \forall\}, \quad (79)$$

with one free variable t , blocks of n_1 and n_2 real variables, and a Boolean combination Ψ of at most s atoms $g \triangleright 0$, $\triangleright \in \{<, =, >\}$, whose polynomials have degree at most $d \geq 2$. Put

$$H(s, d, n_1, n_2) = (sd)^{c_{\text{QE}}(n_1+1)(n_2+1)}. \quad (80)$$

Theorem I.1 (Block quantifier elimination). *There are an algorithm and an explicit absolute constant c_{QE} with the following properties, where H is given by (80).*

- (a) *For arbitrary real coefficients, $\Phi(t)$ is equivalent to a quantifier-free formula in t that is a Boolean combination of at most H atoms $h(t) \triangleright 0$ with $\deg h \leq H$.*
- (b) *If all coefficients are integers of bit length at most $\tau \geq 2$, the algorithm computes such a formula, with integer coefficients of bit length at most $(\tau + 1)H$, in at most $\tau^{c_{\text{QE}}} H$ bit operations.*

This is the case of one free variable and two quantifier blocks of Renegar’s theorem [102, Theorem 1.1], whose bounds have the form $(sd)^{2^{O(\omega)\ell}} \prod_j n_j$ for ω blocks and ℓ free variables, with bit cost of order $\tau \log \tau \log \log \tau$ times the same factor and output coefficient length linear in τ times that factor; see also Basu et al. [15, Chapter 14]. Part (a) uses the statement of that theorem in the real-number model, whose coefficients are arbitrary reals; it is applied below with noise values and thresholds as coefficients. Part (b) uses the separate statement for integer coefficients in the bit model, and it is applied only after denominators are cleared. Two features matter below. The format factor H does not depend on the sizes of the coefficients, and the coefficient length τ enters the bit cost and output length only polynomially, with an absolute exponent. A general cylindrical-decomposition bound, doubly exponential in the number of variables, would not suffice.

Theorem I.2 (Univariate real algebraic numbers). *There is an absolute constant c_U with the following property. Given nonzero integer polynomials $P, P_1, \dots, P_N$ in one variable, of degree at most δ and coefficient bit length at most τ , and an integer $q \geq 0$, one computes the squarefree part of P , disjoint rational isolating intervals for all real roots of P , the sign of every P_j at every such root, and refinements of all isolating intervals to width at most 2^{-q} , in $(N + \delta + \tau + q)^{c_U}$ bit operations [15, Chapters 8 and 10].*

Theorem I.3 (Lattice reduction). *Given a basis of a lattice $\mathcal{L} \subseteq \mathbb{Z}^{s+1}$ of rank s with entries of bit length at most τ , the LLL algorithm with parameter $3/4$ computes, in time polynomial in $s + \tau$, a basis whose first vector $\mathbf{b}_1$ satisfies $\|\mathbf{b}_1\| \leq 2^{(s-1)/2} \lambda_1(\mathcal{L})$, where λ_1 is the length of a shortest nonzero lattice vector [78].*

Classical analytic facts. We use: (F1) a finite convex function on $\mathbb{R}^k$ is differentiable almost everywhere, and it is differentiable at every point where its subdifferential is a singleton; a finite concave function has a nonempty superdifferential at every interior point of its domain; and $y \in \partial\phi(c)$ if and only if $\phi(c) + \phi^*(y) = y^\top c$ [104, Theorems 23.4, 23.5, 25.1, 25.5]. (F2) If $S \subseteq \mathbb{R}^k$ and $\varphi : S \rightarrow \mathbb{R}^k$ has Lipschitz constant ℓ , then $\text{Leb}^*(\varphi(S)) \leq \ell^k \text{Leb}^*(S)$, since k -dimensional Hausdorff measure equals Lebesgue outer measure on $\mathbb{R}^k$ and does not increase by more than the factor ℓ^k under an ℓ -Lipschitz map; the covering proof applies to maps defined only on S [40, Section 2.4]. (F3) If ϕ is convex and differentiable on $\mathbb{R}^k$ with κ -Lipschitz gradient, then $\phi(y+u) \leq \phi(y) + \nabla\phi(y)^\top u + \frac{\kappa}{2} \|u\|^2$, and, if $\inf \phi > -\infty$, $\phi(y) - \inf \phi \geq \|\nabla\phi(y)\|^2 / (2\kappa)$; the second inequality follows from the first with $u = -\nabla\phi(y)/\kappa$.

I.2 Finite laws and one-dimensional sections

Lemma I.4 (Section counts). *Let $\Phi(t)$ be a formula (79) with arbitrary real coefficients, and let H be as in (80). Then $\{t \in \mathbb{R} : \Phi(t)\}$ and its complement are each a union of at most $2H^2 + 1$ intervals, some of which may be single points.*

Proof. By Theorem I.1(a), Φ is a Boolean combination of at most H atoms $h(t) \triangleright 0$ with $\deg h \leq H$. An identically zero polynomial has constant sign. The nonzero polynomials have at most H^2 real roots in total. Between consecutive roots, and at each root, every sign is constant, so the truth value of Φ is constant on each of the at most $H^2 + 1$ open intervals and H^2 points determined by the roots. Both sets are unions of some of these pieces. $\square$

Lemma I.5 (Replacing continuous marginals by grid marginals). *Let $E \subseteq \mathbb{R}^k$ be a Borel set such that, for every coordinate j and every fixed value of the other $k - 1$ coordinates, the section $\{t : (\dots, t, \dots) \in E\}$ is a union of at most C intervals or points. Let Pr_{cont} be the uniform law on $[-\sigma, \sigma]^k$ and Pr_{grid} the uniform law on G_M^k . Then $|\text{Pr}_{\text{grid}}(E) - \text{Pr}_{\text{cont}}(E)| \leq 2kC/M$.*

Proof. Let $F_c(t) = (t + \sigma)/(2\sigma)$ on $[-\sigma, \sigma]$ and let F_g be the distribution function of the uniform law on G_M . For the grid points $t_j = -\sigma + 2\sigma j/(M-1)$ and $t_j \leq t < t_{j+1}$ we have $F_g(t) = (j+1)/M$ and $j/(M-1) \leq F_c(t) < (j+1)/(M-1)$. Hence $-\frac{j+1}{M(M-1)} \leq F_g(t) - F_c(t) \leq \frac{M-1-j}{M(M-1)}$, so $|F_g - F_c| \leq 1/M$ everywhere; the same holds for left limits, since F_c is continuous. The probability of an interval or point, with any convention at its endpoints, is a difference of two such values, so the two laws differ by at most $2/M$ on it and by at most $2C/M$ on a union of C of them.

For $0 \leq j \leq k$ let ν_j be the product law whose first j coordinates are uniform on G_M and whose remaining coordinates are uniform on $[-\sigma, \sigma]$. Then $\nu_0 = \text{Pr}_{\text{cont}}$ and $\nu_k = \text{Pr}_{\text{grid}}$. The laws ν_{j-1} and ν_j differ only in coordinate j . By Fubini's theorem, $\nu_j(E) - \nu_{j-1}(E)$ is the average, over the other coordinates distributed according to their common marginals, of the difference of the two one-dimensional laws on the section of E . Each such difference is at most $2C/M$ in absolute value, whatever the values of the other coordinates. Summing over j proves the claim. $\square$

Every event to which we apply Lemma I.5 is defined by a formula (79) in which one noise coordinate is the free variable and all other noise coordinates and thresholds are real coefficients. It is semialgebraic, hence Borel, and Lemma I.4 gives the section bound. The bound covers ties and exact algebraic relations among the noise values; no almost-sure statement replaces it.

I.3 The selected point and the exact fallback

Lemma I.6 (The selected point). *Let (X, f, κ, σ) be a core-noise instance and $\gamma \in \mathbb{R}^k$.*

- (a) *The set $\arg \min_X f_\gamma$ is nonempty and compact, and the lexicographically least core a among its points exists.*
- (b) *The fiber $S(\gamma)$ is nonempty, compact, and convex. Its point p of least norm is unique, and $x^\# = (a, p)$ is a global minimizer.*
- (c) *Let $\text{LexGT}(v', v)$ be the formula $(v'_1 > v_1) \vee (v'_1 = v_1 \wedge ((v'_2 > v_2) \vee (v'_2 = v_2 \wedge (\dots \vee (v'_{k-1} = v_{k-1} \wedge v'_k > v_k) \dots))))$, with $2k-1$ atoms, and put*

$$\text{Sel}(x', x) = \text{LexGT}(v', v) \vee (v' = v \wedge \|z'\|^2 \geq \|z\|^2).$$

For each coordinate index i of $\mathbb{R}^n$, the formula

$$\exists x \forall x' : x \in X \wedge x_i = t \wedge [x' \notin X \vee f_\gamma(x') > f_\gamma(x) \vee (f_\gamma(x') = f_\gamma(x) \wedge \text{Sel}(x', x))] \quad (81)$$

holds exactly for $t = x_i^\#$.

Proof. (a) X is compact and f_γ is continuous. Minimizing v_1 over the compact optimal set, then v_2 over the compact set of minimizers, and so on, produces the lexicographically least core.

(b) In type (B), $S(\gamma) = \arg \min_{z \in [0,1]^m} f_\gamma(a, z)$, because a is an optimal core; this set is convex because $f(a, \cdot)$ is convex on $[0,1]^m$. In type (P), let $X_a = \{z : (a, z) \in X\}$. The function $z \mapsto f(a, z) + \frac{\alpha}{2} \|a\|^2$ is the restriction of the convex function $f + \frac{\alpha}{2} \|v\|^2$ to an affine slice of X , so $f_\gamma(a, \cdot)$ is convex on X_a and $S(\gamma) = \arg \min_{X_a} f_\gamma(a, \cdot)$ is convex. Compactness and nonemptiness are clear, and the least-norm point of a nonempty compact convex set is unique.

(c) Let x satisfy the matrix of (81). Every feasible x' has $f_\gamma(x') \geq f_\gamma(x)$, so x is a global minimizer. If its core v differed from a , then a would be lexicographically smaller, and a global minimizer x' with core a would violate the bracket. Hence $v = a$, $z \in S(\gamma)$, and the competitors (a, z') with $z' \in S(\gamma)$ force $\|z\| \leq \|z'\|$; thus $z = p$. Conversely $x^\#$ satisfies the matrix: a competitor of equal value is a global minimizer, so its core is lexicographically at least a , and if its core equals a its residual lies in $S(\gamma)$ and has norm at least $\|p\|$. $\square$

Lemma I.7 (Exact fallback with a base-only exponential factor). *Fix D . There are an absolute constant c_F and a polynomial p_D such that every core-noise instance has a base-computable integer $\Xi_0 \leq 2^{p_D(L)}$ with the following property. For every rational $\gamma \in [-\sigma, \sigma]^k$ whose entries have bit length at most b , and every integer $q \geq 0$, one computes a rational $\tilde{x} \in X$ with $\|\tilde{x} - x^\sharp(\gamma)\|_\infty \leq 2^{-q}$ and bit length at most $(L + b + q + 1)^{c_F}$, using at most $\Xi_0(L + b + q + 1)^{c_F}$ bit operations. Consequently, with a base-computable rational $G \geq \max\{1, \sup_X \|\nabla f_\gamma\|\}$ valid for all $\gamma \in [-\sigma, \sigma]^k$, one obtains within the same bound a rational $x_q \in X$ and ℓ_q satisfying (31).*

Proof. Clear the denominators of (81). Its polynomials have degree at most $d = \max\{D, 2\}$ and integer coefficients of bit length $\tau \leq c(L + kb)$ for an absolute c , because the products of all denominators of f , of the description of X , and of γ have total length at most $L + kb$. The formula has two blocks of n variables, one free variable, and $s \leq 4L + 4n + 2$ atoms. Apply Theorem I.1(b) to each of the n formulas. With $H = H(s, d, n, n)$, the cost is at most $n\tau^{c_{QE}}H$, and each output formula has at most H atoms of degree at most H with coefficients of bit length at most $(\tau + 1)H$.

Fix i and the output formula $\Phi_i(t)$. By Lemma I.6(c) its solution set is $\{x_i^\sharp\}$. Some atom polynomial of Φ_i is nonconstant and vanishes at $x_i^\sharp$: otherwise every atom would keep its sign on a neighborhood of $x_i^\sharp$, and Φ_i would hold on that neighborhood. Let P be the product of the nonconstant atom polynomials; its degree is at most H^2 and its coefficient length is at most $H^2((\tau + 1)H + \log(H + 1))$. Theorem I.2 isolates the real roots of the squarefree part of P and determines the signs of all atom polynomials at each root; evaluating the Boolean combination identifies the unique root satisfying Φ_i . Refining its isolating interval to width 2^{-q-2} and rounding the midpoint to a dyadic number with $q + 3$ fractional bits gives w_i with $|w_i - x_i^\sharp| \leq 2^{-q-1}$; since $|x_i^\sharp|$ is bounded by the supplied box, w_i has $O(q + L)$ bits. These steps cost $(H^3 + \tau H^3 + q)^{c_U}$ per coordinate.

In type (B), clip w to the box: the result $\tilde{x}$ is feasible and no farther from $x^\sharp$ in any coordinate. In type (P), solve the rational linear feasibility problem $\{x \in X : \|x - w\|_\infty \leq 2^{-q-1}\}$; it contains $x^\sharp$, and any solution $\tilde{x}$ satisfies $\|\tilde{x} - x^\sharp\|_\infty \leq 2^{-q}$ and has bit length polynomial in $L + q$. Since $\log H$ is polynomial in L for fixed D , all costs are at most $\Xi_0(L + b + q + 1)^{c_F}$ for a base-computable $\Xi_0 \leq 2^{p_D(L)}$ and an absolute c_F ; here Ξ_0 absorbs every power of H and the factor n .

For the last statement, take $G = \max\{1, G_f + k\sigma\}$, where G_f bounds $\|\nabla f\|$ on the supplied box by coefficient sums and $k\sigma \geq \|\gamma\|$. Apply the first part with precision $q + 1 + \lceil \log_2(nG) \rceil = q + O(L)$. Then $\|\tilde{x} - x^\sharp\| \leq \sqrt{n} 2^{-q-1}/(nG) \leq 2^{-q}/(2G)$; since X is convex, $|f_\gamma(\tilde{x}) - f_\gamma^*| \leq 2^{-q-1}$. Put $x_q = \tilde{x}$ and $\ell_q = f_\gamma(\tilde{x}) - 2^{-q-1}$. Then (31) holds, and $\|v_q - a\| \leq \|\tilde{x} - x^\sharp\| \leq 2^{-q}$. $\square$

The base factor Ξ_0 depends only on the format of the instance: its dimension, the number of inequalities describing X , and D . The sampled coefficient length b and the precision q enter only polynomially. This lets the grid size be chosen after Ξ_0 , with $\log M$ polynomial in L ; if the fallback cost were $2^{\text{poly}(L+b)}$, the grid choice would be circular.

I.4 Certified cells

For $j \geq 0$ and $h = 2^{-j}$, a *cell of level j* is a box $C = \prod_{i=1}^k [l_i, l_i + h]$ with $l_i \in h\mathbb{Z} \cap [0, 1 - h]$; its *children* are the 2^k cells of level $j + 1$ inside it. Put

$$e_h = \frac{\kappa_+ k h^2}{8}, \quad \varepsilon_h = 4e_h = \frac{\kappa_+ k h^2}{2}.$$

A *cell oracle* receives a draw γ and a cell C of side h . It returns $\perp$ only if $X_C = \emptyset$; otherwise it returns a rational number $\text{LB}(C)$ and a rational point $\omega_C \in X_C$ with

$$\text{LB}(C) \leq f_\gamma(x) \quad (x \in X_C), \quad f_\gamma(\omega_C) \leq \text{LB}(C) + 2e_h. \quad (82)$$

Lemma I.8 (Cell oracles). *Instances of both types have cell oracles. For a cell of level j the oracle uses at most $2^k P_D(L + b + j)$ bit operations, and its outputs have bit length at most $P_D(L + b + j)$.*

Proof. Type (B). For each corner w of C , apply (78) to the convex polynomial $z \mapsto f_\gamma(w, z)$ on $[0, 1]^m$ with accuracy e_h (if $m = 0$, evaluate $f_\gamma(w)$); it returns z_w and ℓ_w with $\ell_w \leq V_\gamma(w) \leq f_\gamma(w, z_w) \leq \ell_w + e_h$, where $V_\gamma(w) = \min_z f_\gamma(w, z)$. Put $\text{LB}(C) = \min_w \ell_w - e_h$ and $\omega_C = (w^*, z_{w^*})$ for a corner w^* attaining the minimum. The second inequality in (82) is immediate. For the first, let $(v, z) \in X_C$. Choose independent random corner coordinates $W_i \in \{l_i, l_i + h\}$ with $\Pr\{W_i = l_i + h\} = (v_i - l_i)/h$, so that $\mathbb{E}W_i = v_i$ and $\text{Var } W_i \leq h^2/4$. Replace the core coordinates one at a time. Conditionally on $W_1, \dots, W_{i-1}$, the function $\varphi(u) = f_\gamma(W_1, \dots, W_{i-1}, u, v_{i+1}, \dots, v_k, z)$ satisfies $\varphi'' \leq \Lambda$ on $[l_i, l_i + h]$, so $\varphi(u) \leq \varphi(v_i) + \varphi'(v_i)(u - v_i) + \frac{\Lambda}{2}(u - v_i)^2$ there, and $\mathbb{E}\varphi(W_i) \leq \varphi(v_i) + \Lambda h^2/8$. Summing over i , $\mathbb{E}f_\gamma(W, z) \leq f_\gamma(v, z) + e_h$, since $\Lambda \leq \kappa_+$. Some corner w therefore has $V_\gamma(w) \leq f_\gamma(w, z) \leq f_\gamma(v, z) + e_h$, and $\text{LB}(C) \leq \ell_w - e_h \leq f_\gamma(v, z)$.

Type (P). Put $\phi_C(x) = f_\gamma(x) + \frac{\kappa_+}{2} \sum_i (v_i - l_i)(v_i - l_i - h)$. Since $\kappa_+ = \alpha + \sigma$, ϕ_C equals $f + \frac{\alpha}{2} \|v\|^2$ plus the convex function $\frac{\sigma}{2} \|v\|^2$ plus an affine function, so it is convex on X . On X_C each product $(v_i - l_i)(v_i - l_i - h)$ lies in $[-h^2/4, 0]$, hence $f_\gamma - e_h \leq \phi_C \leq f_\gamma$ there. Apply (78) to ϕ_C on the polyhedron $X_C = X \cap \{l \leq v \leq l + h\}$ with accuracy e_h . If X_C is empty, return $\perp$. Otherwise it returns $\omega_C \in X_C$ and ℓ_C with $\ell_C \leq \min_{X_C} \phi_C \leq \phi_C(\omega_C) \leq \ell_C + e_h$; put $\text{LB}(C) = \ell_C$. Then $\text{LB}(C) \leq \min_{X_C} f_\gamma$ and $f_\gamma(\omega_C) \leq \phi_C(\omega_C) + e_h \leq \text{LB}(C) + 2e_h$.

In both types the data of each convex call have bit length $P_D(L + b + j)$, and the tolerance $e_h = \kappa_+ k 2^{-2j-3}$ is positive, because $\kappa_+ \geq \sigma > 0$, and has $\langle e_h \rangle \leq 2j + P_D(L)$. $\square$

The type (P) construction is the classical secant underestimator of α BB branch and bound [7]. The type (B) construction needs only the diagonal bound Λ and convex residual fibers. We add σ to the curvature parameter so that $e_h > 0$ also when $\kappa = 0$; this changes $1 + \kappa/\sigma$ by at most a factor two.

Refinement. Given a draw, a final level J , and a cap N , the *refinement* proceeds as follows. Level 0 consists of the single cell $[0, 1]^k$. For $j \geq 1$, if 2^k times the number of cells retained at level $j - 1$ exceeds N , stop with the outcome CAP; otherwise generate all children of the retained cells. At every level call the cell oracle on each generated cell, discard the cells that return $\perp$, set the incumbent x_{inc} to a witness of least value among all witnesses produced so far and $U = f_\gamma(x_{\text{inc}})$, and then, in a second pass using this final value of U , retain exactly the cells with $\text{LB}(C) \leq U$. The list work at a level is linear in the number of generated cells.

Lemma I.9 (Invariants of the refinement). *After level j has been completed, with $h = 2^{-j}$:*

- (a) $U - 2e_h \leq f_\gamma^* \leq U = f_\gamma(x_{\text{inc}})$ and $U \leq f_\gamma^* + 2e_h$;
- (b) for every global minimizer $x^* = (v^*, z^*)$ of f_γ , some retained cell contains v^* ;
- (c) some retained cell contains the core of x_{inc} ;
- (d) every retained cell C satisfies $f_\gamma(\omega_C) \leq f_\gamma^* + 4e_h$ and $C \subseteq v(\omega_C) + [-h, h]^k$, where $v(\omega_C)$ is the core of ω_C .

Proof. (b) At level 0 the single cell contains v^* and its lower bound is at most $f_\gamma(x^*) = f_\gamma^* \leq U$. If a retained cell of level $j - 1$ contains v^* , one of its children does; that child is generated, and (82) gives $\text{LB} \leq f_\gamma^* \leq U$, so it is retained.

(c) The incumbent was produced as a witness ω_{C_0} of a cell C_0 generated at some level $j_0 \leq j$, and it has remained the incumbent since. At level j_0 , (82) gives $\text{LB}(C_0) \leq f_\gamma(x_{\text{inc}}) = U$, so C_0 is retained. Induction over the levels after j_0 , exactly as in (b) with x_{inc} in place of x^* , proves (c).

(a) The cells retained at level j , together with all cells discarded at levels $0, \dots, j$, cover $[0, 1]^k$; this follows by induction, since the children of a retained cell cover it. A cell discarded at level $j' \leq j$ either has $X_C = \emptyset$ or has $\text{LB}(C) > U_{j'} \geq U$, where $U_{j'}$ is the incumbent value at level j' ; incumbent values never increase. A cell retained at level j has $\text{LB}(C) \geq f_\gamma(\omega_C) - 2e_h \geq U - 2e_h$, because ω_C was a candidate for the incumbent. Hence every feasible point has value at least $U - 2e_h$. Finally, the retained cell containing v^* from (b) has $f_\gamma(\omega_C) \leq \text{LB}(C) + 2e_h \leq f_\gamma^* + 2e_h$, so $U \leq f_\gamma^* + 2e_h$.

(d) A retained cell has $f_\gamma(\omega_C) \leq \text{LB}(C) + 2e_h \leq U + 2e_h \leq f_\gamma^* + 4e_h$ by (a). Since $v(\omega_C) \in C$ and C has side h , every point of C differs from $v(\omega_C)$ by at most h in each coordinate. $\square$

All four invariants hold for every draw and for every admissible answer of the convex value calls; no growth or probability assumption is used.

I.5 An all-scale count

For a draw $\gamma \in \mathbb{R}^k$ and $0 < h \leq 1$ define the projected near-optimal set, its padded hull, and the count proxy

$$\begin{aligned} S_h(\gamma) &= \{v : \exists z (v, z) \in X, f_\gamma(v, z) \leq f_\gamma^* + \varepsilon_h\}, \\ K_h(\gamma) &= \text{conv } S_h(\gamma) + [-h, h]^k, \\ \mathfrak{d}(K) &= \max\{|\det(y_1 - y_0, \dots, y_k - y_0)| : y_0, \dots, y_k \in K\}, \\ A(\gamma) &= \sup_{0 < h \leq 1} \mathfrak{d}(K_h(\gamma))/h^k, \quad W(\gamma) = \max\{1, A(\gamma)\}. \end{aligned}$$

The set $S_h(\gamma)$ is nonempty and compact, being the projection of a compact sublevel set, so $K_h(\gamma)$ is a compact convex body with nonempty interior. We allow $A(\gamma) = +\infty$.

Lemma I.10 (Simplex volume). *For every compact convex $K \subseteq \mathbb{R}^k$ with nonempty interior, $\mathfrak{d}(K)/k! \leq \text{vol}(K) \leq 2^k \mathfrak{d}(K)$.*

Proof. The simplex with vertices attaining $\mathfrak{d}(K)$ lies in K and has volume $\mathfrak{d}(K)/k!$. For the upper bound, fix such vertices $y_0, \dots, y_k$; since $\mathfrak{d}(K) > 0$, the edges $y_i - y_0$ form a basis. Write $y \in K$ as $y = y_0 + \sum_i \lambda_i (y_i - y_0)$. Replacing y_i by y multiplies the determinant by λ_i , so maximality gives $|\lambda_i| \leq 1$. Thus K lies in the parallelepiped $y_0 + \{\sum_i \lambda_i (y_i - y_0) : \lambda \in [-1, 1]^k\}$ of volume $2^k \mathfrak{d}(K)$. $\square$

Lemma I.11 (Cell count). *For every draw and every run of the refinement, at most $2^k W(\gamma)$ cells are retained at each completed level, and at most $4^k W(\gamma)$ cells are generated at each level.*

Proof. Let $h = 2^{-j}$. By Lemma I.9(d), the core of the witness of each retained cell lies in $S_h(\gamma)$, and the cell lies in $K_h(\gamma)$. Distinct cells of level j have disjoint interiors and volume h^k . By Lemma I.10 their number is at most $\text{vol}(K_h)/h^k \leq 2^k \mathfrak{d}(K_h)/h^k \leq 2^k A$. The next level has at most 2^k times as many cells, and level 0 has one. $\square$

We now bound the tail of W . Write $c = -\gamma$, so that $f_\gamma(x) = f(x) - c^\top v$; the uniform laws on $[-\sigma, \sigma]^k$ and on G_M^k are symmetric, so c and γ have the same law. Define, for $c \in \mathbb{R}^k$,

$$\Psi(c) = \max_{x \in X} (c^\top v - f(x)) = -\min_X f_{-c}, \quad \Upsilon(c) = \Psi(c) + \frac{\|c\|^2}{2\kappa_+}.$$

Lemma I.12 (A conjugate pushforward measure). *(a) Ψ is finite and convex, and every sub-gradient of Ψ lies in $[0, 1]^k$. The conjugate Υ^* is finite and differentiable on $\mathbb{R}^k$, $\nabla \Upsilon^*$ is κ_+ -Lipschitz, and $y \in \partial \Upsilon(c)$ if and only if $\nabla \Upsilon^*(y) = c$.*

(b) $\nu(E) = \text{Leb}\{y : \nabla \Upsilon^(y) \in E\}$ is a Borel measure, and $\nu(\prod_i [\alpha_i, \beta_i]) \leq \prod_i (1 + (\beta_i - \alpha_i)/\kappa_+)$.*

(c) For every $c \in \mathbb{R}^k$ and $0 < h \leq 1$, $\text{vol}(K_h(-c)) \leq \nu(B(c, 4\kappa_+kh))$.

Proof. (a) Ψ is a maximum of affine functions of c over a compact index set, hence finite and convex. If $s \in \partial \Psi(c)$ and $u \in \mathbb{R}^k$, then $s^\top u \leq \Psi(c+u) - \Psi(c) \leq \sum_i \max\{u_i, 0\}$, because every core lies in $[0, 1]^k$. Taking $u = \pm \epsilon e_i$ gives $0 \leq s_i \leq 1$. The function Υ is $(1/\kappa_+)$ -strongly convex. For fixed y , the function $c \mapsto y^\top c - \Upsilon(c)$ is strongly concave and continuous, so it has a unique maximizer $c(y)$ and $\Upsilon^*(y)$ is finite. By (F1), $\partial \Upsilon^*(y) = \{c : y \in \partial \Upsilon(c)\}$ is the set of these maximizers, namely $\{c(y)\}$, so Υ^* is differentiable with $\nabla \Upsilon^*(y) = c(y)$. If $y_j \in \partial \Upsilon(c_j)$ for $j = 1, 2$, then $y_j - c_j/\kappa_+ \in \partial \Psi(c_j)$, and monotonicity of $\partial \Psi$ gives $(y_1 - y_2)^\top (c_1 - c_2) \geq \|c_1 - c_2\|^2/\kappa_+$; hence $\|c_1 - c_2\| \leq \kappa_+ \|y_1 - y_2\|$.

(b) $\nabla \Upsilon^*$ is continuous, so ν is a Borel measure. If $\nabla \Upsilon^*(y) = c$, then $y \in \partial \Upsilon(c) = \partial \Psi(c) + c/\kappa_+ \subseteq [0, 1]^k + c/\kappa_+$. For a box E the preimage therefore lies in $[0, 1]^k + E/\kappa_+$, a box with the stated volume.

(c) Let $\gamma = -c$ and $v \in S_h(\gamma)$, with a residual z such that $(v, z) \in X$ and $f(v, z) - c^\top v \leq -\Psi(c) + \varepsilon_h$. Put $y_0 = v + c/\kappa_+$ and $r(y) = \Upsilon(c) + \Upsilon^*(y) - c^\top y \geq 0$. Since $\Psi(c') \geq c'^\top v - f(v, z)$ for all c' ,

$$\Upsilon^*(y_0) \leq \sup_{c'} (y_0^\top c' - c^\top v + f(v, z) - \frac{\|c'\|^2}{2\kappa_+}) = f(v, z) + \frac{\|c\|^2}{2\kappa_+},$$

and therefore $r(y_0) \leq \Psi(c) + f(v, z) - c^\top v \leq \varepsilon_h$. The function r is convex, has κ_+ -Lipschitz gradient $\nabla \Upsilon^* - c$, and has infimum zero. By (F3), $\|\nabla r(y_0)\| \leq \sqrt{2\kappa_+\varepsilon_h} = \kappa_+\sqrt{k}h$, and for $u \in [-h, h]^k$, where $\|u\| \leq \sqrt{k}h$,

$$r(y_0 + u) \leq \varepsilon_h + \kappa_+\sqrt{k}h\|u\| + \frac{\kappa_+}{2}\|u\|^2 \leq 2\kappa_+kh^2.$$

The sublevel set $\{r \leq 2\kappa_+kh^2\}$ is convex and contains $S_h(\gamma) + c/\kappa_+ + [-h, h]^k$, hence also $K_h(\gamma) + c/\kappa_+$. On it, (F3) gives $\|\nabla \Upsilon^*(y) - c\| = \|\nabla r(y)\| \leq 2\kappa_+\sqrt{k}h < 4\kappa_+kh$. Thus $K_h(\gamma) + c/\kappa_+ \subseteq \{y : \nabla \Upsilon^*(y) \in B(c, 4\kappa_+kh)\}$, and translation invariance of volume proves (c). $\square$

Lemma I.13 (Disjoint balls). *Let $\mathcal{F}$ be a family of open balls $B(c, r_c)$, one for each center c in a set $E \subseteq \mathbb{R}^k$, with $0 < r_c \leq R_*$, all contained in a bounded set. There is a countable disjoint subfamily $\{B(c_j, r_j)\}$ with $E \subseteq \bigcup_j B(c_j, 5r_j)$.*

Proof. Put $\mathcal{F}_0 = \mathcal{F}$. While $\mathcal{F}_i \neq \emptyset$, let ρ_i be the supremum of the radii in $\mathcal{F}_i$, choose $B(c_i, r_i) \in \mathcal{F}_i$ with $r_i > \rho_i/2$, and let $\mathcal{F}_{i+1}$ consist of the balls of $\mathcal{F}_i$ disjoint from it. The chosen balls are disjoint. A ball $B(c, r_c)$ removed at step i meets $B(c_i, r_i)$ and has $r_c \leq \rho_i < 2r_i$, so $\|c - c_i\| < r_i + r_c < 3r_i$. If some ball $B(c, r_c)$ were never chosen or removed, then $\rho_i \geq r_c$ for all i , and infinitely many disjoint chosen balls of radius greater than $r_c/2$ would lie in a bounded set, which is impossible. $\square$

Lemma I.14 (Weak first-moment tail). *For every $s > 0$,*

$$\Pr_{\text{cont}}\{A > s\} \leq \frac{a_k}{s}, \quad a_k = (180k^3)^k \left(1 + \frac{\kappa_+}{\sigma}\right)^k.$$

Proof. Let $R_* = 4\kappa_+k$, $Q = [-\sigma, \sigma]^k$, and $E_s = \{c \in Q : A(-c) > s\}$. For $c \in E_s$ choose $h \in (0, 1]$ with $\mathfrak{d}(K_h(-c)) > sh^k$. By Lemmas I.10 and I.12(c), the ball $B(c, r_c)$ with $r_c = R_*h \leq R_*$ satisfies

$$\nu(B(c, r_c)) \geq \text{vol}(K_h(-c)) \geq \frac{\mathfrak{d}(K_h(-c))}{k!} > \frac{s r_c^k}{k! R_*^k}.$$

All these balls lie in $Q_* = [-\sigma - R_*, \sigma + R_*]^k$. Lemma I.13 selects disjoint balls $B(c_j, r_j)$ whose fivefold enlargements cover E_s . Therefore

$$\text{Leb}^*(E_s) \leq \sum_j v_k 5^k r_j^k < \frac{5^k v_k k! R_*^k}{s} \sum_j \nu(B(c_j, r_j)) \leq \frac{5^k v_k k! R_*^k}{s} \nu(Q_*).$$

By Lemma I.12(b), $\nu(Q_*) \leq (1 + 2\sigma/\kappa_+ + 8k)^k$. Dividing by $(2\sigma)^k$ and using $v_k \leq 2^k$, $k! \leq k^k$, and $\kappa_+/\sigma + 2 + 8k\kappa_+/\sigma \leq 9k(1 + \kappa_+/\sigma)$ gives

$$\Pr_{\text{cont}}\{A > s\} \leq \frac{5^k 2^k k^k (2k)^k}{s} \left(\frac{\kappa_+}{\sigma} + 2 + \frac{8k\kappa_+}{\sigma} \right)^k \leq \frac{(180k^3)^k (1 + \kappa_+/\sigma)^k}{s}. \quad \square$$

The bound has the power s^{-1} in every core dimension. It is the weak-type estimate for the centered maximal function of the locally finite measure ν ; the conjugate construction converts near-optimal sets at every scale into balls of comparable ν -mass.

Lemma I.15 (The count event is a two-block formula). *There is a base-computable $C_W \leq 2^{P_D(L)}$ such that, for every power of two M and every $s \geq 1$, $\Pr_{\text{grid}}\{W > s\} \leq a_k/s + C_W/M$.*

Proof. For $s \geq 1$, $\{W > s\} = \{A > s\}$, and $A(\gamma) > s$ holds if and only if there are $h \in (0, 1]$ and $y_0, \dots, y_k \in K_h(\gamma)$ with $|\det(y_1 - y_0, \dots, y_k - y_0)| > sh^k$. By Carathéodory's theorem, $y_i \in K_h(\gamma)$ if and only if $y_i = \sum_{l=0}^k \lambda_{il} v_{il} + u_i$ with $\lambda_{il} \geq 0$, $\sum_l \lambda_{il} = 1$, $v_{il} \in S_h(\gamma)$, and $\|u_i\|_\infty \leq h$. Since minima over X are attained, $v_{il} \in S_h(\gamma)$ if and only if some z_{il} has $(v_{il}, z_{il}) \in X$ and $f_\gamma(v_{il}, z_{il}) \leq f_\gamma(x') + \kappa_+ kh^2/2$ for every $x' \in X$. One universal block x' serves all $(k+1)^2$ comparisons.

The determinant is not expanded into $k!$ monomials. Instead introduce existential matrices Q and R with $(y_1 - y_0, \dots, y_k - y_0) = QR$, $Q^\top Q = I$, R upper triangular, $R_{ii} > 0$, and product chains $\pi_1 = R_{11}$, $\pi_i = \pi_{i-1} R_{ii}$, $\epsilon_1 = h$, $\epsilon_i = \epsilon_{i-1} h$, and require $\pi_k > s \epsilon_k$. A nonsingular real matrix has such a factorization, and then π_k equals the absolute value of its determinant; conversely these equations force $|\det| = \pi_k$. The strict inequality forces nonsingularity.

The resulting formula has the form (79) with one existential block of $O(k^2(k+m))$ variables, one universal block of n variables, $O(k^2(k+L))$ atoms, and degree at most $\max\{D, 2\}$; the free variable is one noise coordinate, and the other noise coordinates and s are real coefficients. By Lemma I.4, every section of $\{A > s\}$ is a union of at most $C = 2H^2 + 1$ intervals or points, with H from (80), uniformly in s and in the fixed coordinates. Lemmas I.5 and I.14 give the claim with $C_W = 2kC$. Since $\log H$ is polynomial in L for fixed D , so is $\log C_W$. $\square$

The formula also shows that $\{A > s\}$ is semialgebraic, so A and W are measurable, including the event $\{W = \infty\}$, whose probability is at most C_W/M under the grid law.

I.6 Projected growth

For a draw γ let $g(\gamma)$ be the supremum of the numbers $t \geq 0$ for which some $x^* = (v^*, z^*) \in X$ satisfies

$$f_\gamma(x) - f_\gamma(x^*) \geq t \|v - v^*\|^2 \quad (x \in X). \quad (83)$$

Such an x^* is a global minimizer. Residual ties are allowed; if the core projection of X is a single point, $g = +\infty$.

Lemma I.16 (Projected-growth tail). *(a) For $t > 0$ the set G_t of draws admitting (83) is closed, and $g(\gamma) \geq t$ if and only if $\gamma \in G_t$. If $g(\gamma) > 0$, all global minimizers have the same core $a(\gamma)$, and $f_\gamma(x) - f_\gamma^* \geq t \|v - a(\gamma)\|^2$ on X for every finite $t \in (0, g(\gamma)]$.*

(b) $\Pr_{\text{cont}}\{g < t\} \leq kt/\sigma$ for every $t > 0$.

(c) There is a base-computable $C_g \leq 2^{P_D(L)}$ with $\Pr_{\text{grid}}\{g < t\} \leq kt/\sigma + C_g/M$ for every $t > 0$ and every power of two M .

Proof. (a) Let $\gamma_l \rightarrow \gamma$ with witnesses x_l^* . A subsequence of the witnesses converges in the compact set X , and (83) passes to the limit; so G_t is closed. If $g(\gamma) \geq t$, then $\gamma \in G_{t'}$ for all $t' < t$; letting $t' \uparrow t$ and passing to a convergent subsequence of witnesses shows $\gamma \in G_t$; the converse holds by definition. If (83) holds with $t > 0$ and witness x^* , then $f_\gamma(x^*) = f_\gamma^*$, and every global minimizer x has $0 \geq t \|v - v^*\|^2$, so its core is v^* . Hence $v^* = a(\gamma)$, and (83) is the displayed inequality. If $g(\gamma) > 0$, every finite $t \in (0, g(\gamma)]$ has $\gamma \in G_t$, which gives the last claim.

(b) Translate the core coordinates by $(\frac{1}{2}, \dots, \frac{1}{2})$; this changes f_γ by a constant and leaves (83) unchanged, and afterwards every core lies in $[-\frac{1}{2}, \frac{1}{2}]^k$. Write $c = -\gamma$ and define the finite convex function

$$\Psi_t(c) = \max_{x \in X} (c^\top v - f(x) + t \|v\|^2).$$

As in Lemma I.12(a), every subgradient lies in $[-\frac{1}{2}, \frac{1}{2}]^k$, and the core of every maximizer at c is a subgradient. Let c be a point where Ψ_t is differentiable, with gradient a . Then every maximizer $\bar{x}$ has core a , and for $y = c + 2ta$ and every $x \in X$,

$$f(x) - f(\bar{x}) \geq c^\top (v - a) + t(\|v\|^2 - \|a\|^2) = y^\top (v - a) + t \|v - a\|^2.$$

Hence the draw $-y$ satisfies (83) with witness $\bar{x}$. At differentiability points put $\hat{c} = c + 2t\nabla\Psi_t(c)$; then $\|\hat{c} - c\|_\infty \leq t$ and $-\hat{c} \in G_t$.

Let $P(y)$ be the unique minimizer of $\Psi_t(c) + \|c - y\|^2/(4t)$. Its optimality condition is $y - P(y) \in 2t \partial\Psi_t(P(y))$. If y, y' are two points, adding the two monotonicity inequalities gives $\|P(y) - P(y')\|^2 \leq (y - y')^\top (P(y) - P(y'))$, so P is 1-Lipschitz. If c is a differentiability point, then c satisfies the optimality condition for $y = \hat{c}$, so $P(\hat{c}) = c$.

Let $Q = [-\sigma, \sigma]^k$, $t < \sigma$, $Q' = [-\sigma + t, \sigma - t]^k$, and let $\tilde{G} = \{c \in Q : -c \in G_t\}$, a closed set. For every differentiability point $c \in Q'$ we have $\hat{c} \in \tilde{G}$ and $P(\hat{c}) = c$. By (F1), the differentiability points have full measure, so $P(\tilde{G})$ contains Q' up to a null set. By (F2), $\text{Leb}(\tilde{G}) \geq \text{Leb}^*(P(\tilde{G})) \geq (2\sigma - 2t)^k$. The uniform law of c equals that of γ , so $\Pr_{\text{cont}}\{g < t\} \leq 1 - (1 - t/\sigma)^k \leq kt/\sigma$. For $t \geq \sigma$ the bound is trivial.

(c) G_t is defined by $\exists x^* \forall x : x^* \in X \wedge [x \notin X \vee f_\gamma(x) - f_\gamma(x^*) - t \|v - v^*\|^2 \geq 0]$, a formula (79) with two blocks of n variables, at most $2L + 2$ atoms, and degree at most $\max\{D, 2\}$. Its complement is $\{g < t\}$. Lemmas I.4 and I.5 and part (b) give the claim with $C_g = 2k(2H^2 + 1)$. $\square$

The proof refines the classical observation that a linear tilt in general position has a unique maximizer because convex conjugates are differentiable almost everywhere. Here the conjugate includes the growth term $t \|v\|^2$, and the proximal map turns differentiability into a quantitative bound.

I.7 Proof of Theorem 11.2

Base choices. Let Ξ_0 be the fallback factor of Lemma I.7 and let C_W, C_g be the constants of Lemmas I.15 and I.16. Choose, before sampling,

$$\Xi \geq \max\{2, \Xi_0\} \text{ with } \log \Xi \leq P_D(L), \quad g_0 = \frac{\sigma}{2k\Xi}, \quad \Theta = 4k + \frac{k^2\kappa_+}{g_0}, \quad N = 4^k\Xi, \quad (84)$$

$$M = \text{the least power of two with } M \geq 2\Xi \max\{C_W, C_g, 1\}.$$

All these numbers have bit length polynomial in L ; in particular $\log M \leq P_D(L)$ and $b \leq P_D(L)$. With $\beta = C_W/M \leq 1/(2\Xi)$, Lemmas I.15 and I.16 give

$$\Pr\{W > s\} \leq \frac{a_k}{s} + \beta \quad (s \geq 1), \quad \Pr\{g < g_0\} \leq \frac{kg_0}{\sigma} + \frac{C_g}{M} \leq \frac{1}{\Xi}. \quad (85)$$

Neither the grid nor any other choice depends on q .

The evaluator. For a query q , let J be the least integer $j \geq 0$ with $2e_{2^{-j}} \leq 2^{-q}$ and $\Theta 2^{-j} \leq 2^{-q}$; then $J \leq q + P_D(L)$. Run the refinement with final level J and cap N . If it stops with CAP, use the fallback. Otherwise let H_J be the coordinate hull of the cells retained at level J , that is, the smallest box containing them, with widths $w_1, \dots, w_k$. If $\sum_i w_i^2 \leq 2^{-2q}$, return $x_q = x_{\text{inc}}$ and $\ell_q = U - 2e_{2^{-J}}$. Otherwise use the fallback. The *fallback* returns the output of Lemma I.7 for this draw and this q .

Correctness on every draw. On the fallback branch, Lemma I.7 gives (31). On the ordinary branch, Lemma I.9(a) gives $\ell_q \leq f_\gamma^* \leq f_\gamma(x_q) = \ell_q + 2e_{2^{-J}} \leq \ell_q + 2^{-q}$. By Lemma I.9(b) and (c), the box H_J contains $a(\gamma)$ and the core v_q of x_{inc} , so $\|v_q - a(\gamma)\|^2 \leq \sum_i w_i^2 \leq 2^{-2q}$. Neither statement uses uniqueness, growth, or probability. On the ordinary branch, x_q is a witness that the cell oracle produced at a level at most J , possibly an earlier one, and ℓ_q is computed from its value; both have bit length $P_D(L + q)$.

When the ordinary branch fails. If the cap is reached before level j , then more than $2^k \Xi$ cells were retained at level $j - 1$, so $W > \Xi$ by Lemma I.11. Suppose next that $g(\gamma) \geq g_0$. By Lemma I.16(a) the optimal core a is unique and $f_\gamma(x) - f_\gamma^* \geq g_0 \|v - a\|^2$. With $h = 2^{-J}$, every retained cell C has $f_\gamma(\omega_C) - f_\gamma^* \leq 4e_h$ by Lemma I.9(d), hence $\|v(\omega_C) - a\| \leq h\sqrt{\kappa_+ k / (2g_0)}$, and every point of C is within h of $v(\omega_C)$ in each coordinate. Therefore $w_i \leq 2h(1 + \sqrt{\kappa_+ k / (2g_0)})$ for all i , and, using $\sqrt{k} \leq k$ and $\sqrt{y} \leq 1 + y$,

$$\left(\sum_i w_i^2\right)^{1/2} \leq 2\sqrt{k}h\left(1 + \sqrt{\frac{\kappa_+ k}{2g_0}}\right) \leq h\left(4k + \frac{k^2 \kappa_+}{g_0}\right) = \Theta h \leq 2^{-q}.$$

So the hull test passes. The fallback is used only on the event $\{W > \Xi\} \cup \{g < g_0\}$, which does not depend on q .

Work. Without the cap, each level $j \leq J$ generates at most $4^k \min\{W, \Xi\}$ cells: at most $4^k W$ by Lemma I.11, and at most $N = 4^k \Xi$ because the cap was not reached. Each cell costs $2^k P_D(L + b + j)$ by Lemma I.8, and the hull is computed in one pass. With $J \leq q + P_D(L)$ and $b \leq P_D(L)$, the ordinary work, and the work spent before a fallback, is at most $8^k \min\{W, \Xi\} P_D(L + q)$. The fallback costs at most $\Xi P_D(L + q)$. Hence every query costs at most $\Omega(\gamma) P_D(L + q)$ with

$$\Omega = 8^k \min\{W, \Xi\} + \Xi \mathbf{1}\{W > \Xi \text{ or } g < g_0\}.$$

By (85), $\mathbb{E} \min\{W, \Xi\} = \int_0^\Xi \Pr\{\min\{W, \Xi\} > s\} ds \leq 1 + \int_1^\Xi (a_k/s + \beta) ds \leq 1 + a_k \ln \Xi + \beta \Xi$ and $\Xi \Pr\{W > \Xi\} \leq a_k + \beta \Xi$. With $\beta \Xi \leq \frac{1}{2}$ and $\Xi \Pr\{g < g_0\} \leq 1$,

$$\mathbb{E} \Omega \leq 8^k (2 + a_k \ln \Xi) + a_k + 2.$$

Since $1 + \kappa_+/\sigma \leq 2(1 + \kappa/\sigma)$, we have $a_k \leq (360k^3)^k (1 + \kappa/\sigma)^k$, and $\ln \Xi \leq P_D(L)$. This proves $\mathbb{E} \Omega \leq k^{O(k)} (1 + \kappa/\sigma)^k P_D(L)$. $\square$

Remark I.17 (Low core dimension). For $k \leq 2$ a direct count suffices. If $g > 0$, the retained witnesses lie within $h\sqrt{\kappa_+k/(2g)}$ of a , so at most $4^k(3 + \sqrt{2k\kappa_+/g})^k \leq 512 \max\{1, \kappa_+/g\}^{k/2}$ cells are generated per level. With $Z = \max\{1, \kappa_+/g\}^{k/2}$, Lemma I.16 gives $\Pr\{Z > s\} \leq k\kappa_+s^{-2/k}/\sigma + C_g/M$. Its truncated moment is bounded for $k = 1$ and logarithmic in the cap for $k = 2$, which yields the bound $(1 + \kappa/\sigma)P_D(L + q)$. For $k \geq 3$ the truncated moment grows like a positive power of the cap, and the fallback term does as well; this is why Lemmas I.12–I.15 are needed.

I.8 Proof of Corollary 11.3

Let f be convex on X . Choose $\Xi \geq \max\{2, \Xi_0\}$, $g_0 = \sigma/(2k\Xi)$, and M the least power of two with $M \geq 2\Xi C_g$, so that $\Pr\{g < g_0\} \leq 1/\Xi$ by Lemma I.16.

For a query q put $\epsilon = 2^{-q}$, $\delta = \min\{\epsilon, g_0\epsilon^2/(128k)\}$, and $\zeta = \epsilon/(8k)$. First apply (78) to f_γ on X with accuracy δ , obtaining $y \in X$ and ℓ_0 with $\ell_0 \leq f_\gamma^* \leq U := f_\gamma(y) \leq \ell_0 + \delta$.

For a coordinate i and a dyadic rational θ , an *upper probe* applies (78) to f_γ on the polyhedron $X \cap \{v_i \geq \theta\}$ with accuracy δ . It *succeeds* if this polyhedron is empty or the returned lower bound exceeds U ; otherwise it fails and returns a witness $x_\theta \in X$ with $v_i(x_\theta) \geq \theta$ and $f_\gamma(x_\theta) \leq U + \delta$. If an upper probe succeeds at θ , then every point of X with $v_i \geq \theta$ has value greater than U ; hence every global minimizer and y itself have $v_i < \theta$.

The upper probe fails at $\theta = 0$, because the lower bound is at most $f_\gamma^* \leq U$, and it succeeds at $\theta = 2$. Binary search maintains a failing lo with its witness and a succeeding hi, halving hi – lo until it is at most ζ ; set $u_i = \text{hi}$. Lower probes on $X \cap \{v_i \leq \theta\}$, started from a succeeding $\theta = -1$ and a failing $\theta = 1$, give l_i in the same way. No monotonicity of the probe outcomes is needed. The box $\prod_i [l_i, u_i]$ contains the core of y and every optimal core. If $\sum_i (u_i - l_i)^2 \leq \epsilon^2$, return y and ℓ_0 ; then (31) holds. Otherwise return the fallback of Lemma I.7.

Suppose $g(\gamma) \geq g_0$. The witness at the final lo of the upper search has value at most $U + \delta \leq f_\gamma^* + 2\delta$, so its core is within $\sqrt{2\delta/g_0} \leq \epsilon/(8\sqrt{k})$ of a . Hence $u_i \leq \text{lo} + \zeta \leq a_i + \epsilon/(8\sqrt{k}) + \zeta$, and symmetrically $l_i \geq a_i - \epsilon/(8\sqrt{k}) - \zeta$. Then $u_i - l_i \leq \epsilon/(4\sqrt{k}) + 2\zeta \leq \epsilon/(2\sqrt{k})$, and the test passes. The fallback is used only on $\{g < g_0\}$.

The ordinary branch makes $1 + O(k(q + \log k))$ convex value calls on polyhedra of bit length $P_D(L) + O(q + \log k)$, with tolerance $\langle \delta \rangle \leq 2q + P_D(L)$. Since $k \leq L$, its work is at most $P_D(L + q)$, with a polynomial independent of k and of the numerical values of σ and κ . With $\Omega = 1 + \Xi \mathbf{1}\{g < g_0\}$, every query costs at most $\Omega(\gamma)P_D(L + q)$, and $\mathbb{E} \Omega \leq 2$. $\square$

I.9 Proof of Corollary 11.4

Lemma I.18 (Height of a lexicographic quadratic optimizer). *Let $X = \{x \in \mathbb{R}^n : Ex \leq e\}$ be a nonempty bounded polytope and $q(x) = \frac{1}{2}x^\top Ax + c^\top x + c_0$, with rational data and $A = A^\top$, so that $\nabla q(x) = Ax + c$. Let Δ_0 be a positive integer making $\Delta_0 A$, $\Delta_0 c$, $\Delta_0 E$, and $\Delta_0 e$ integral, and let $C \geq 1$ bound the absolute values of their entries. Then the lexicographically least global minimizer x^* of q on X , for any fixed order of the coordinates, has a common denominator at most $U_Q = n!(n \cdot n!C^{n+1})^n$.*

Proof. Let I be the set of rows active at x^* , and let R be a maximal linearly independent subset of those rows. The smallest face of X containing x^* has affine hull $\{x : Rx = e_R\}$, and x^* lies in its relative interior, so $x^* \pm \epsilon w$ is feasible for small ϵ and every $w \in \ker R$. Thus $w^\top (Ax^* + c) = 0$ for $w \in \ker R$. Let N be an integer matrix whose columns span $\ker R$, and put

$$S = \{x \in X : Rx = e_R, N^\top (Ax + c) = 0\} \ni x^*.$$

For $x, y \in S$ and $w = x - y \in \ker R$ we have $w^\top(Ay + c) = 0 = w^\top(Ax + c)$, hence $w^\top Aw = 0$ and $q(x) - q(y) = w^\top(Ay + c) + \frac{1}{2}w^\top Aw = 0$. Thus every point of S is a global minimizer, and x^* is the lexicographically least point of S . A lexicographically least point of a polytope is a vertex: one endpoint of any segment through it inside the polytope is lexicographically smaller.

Choose N from minors: if $\Delta_0 R$ has rank ρ , fix a nonsingular $\rho \times \rho$ submatrix, set one free coordinate equal to its determinant and the other free coordinates to zero, and solve for the pivot coordinates by the adjugate. The entries of N are at most $\rho!C^\rho \leq n!C^n$. The equations $N^\top(\Delta_0 Ax + \Delta_0 c) = 0$ then have integer coefficients bounded by $C_1 = n \cdot n!C^{n+1}$, as do the scaled rows of E . The vertex x^* of S is the unique solution of n linearly independent equations among these, and Cramer's rule writes it with the common denominator $|\det| \leq n!C_1^n = U_Q$. $\square$

Proof of Corollary 11.4. Order the core coordinates first; then the core of the lexicographically least global minimizer is $a(\gamma)$. Write $f_\gamma(x) = \frac{1}{2}x^\top Ax + c^\top x + c_0$, where A is the constant symmetric Hessian of f and only c depends on γ . For the grid G_M , every sampled coefficient has denominator dividing $(M - 1)$ times the denominator of σ , and its absolute value is at most σ . Hence one base-computable Δ_0 and C serve all draws in Lemma I.18, and $\log U_Q$ is polynomial in L . Query Theorem 11.2, or Corollary 11.3 for type (P) with $\alpha = 0$, at precision t with $2^{-t} \leq 1/(4U_Q^2)$, so $t \leq P(L)$. Each interval $[v_{t,i} - 2^{-t}, v_{t,i} + 2^{-t}]$ contains a_i , a rational with denominator at most U_Q ; distinct such rationals differ by at least $1/U_Q^2$, which exceeds the width 2^{1-t} . The midpoint is within $1/(4U_Q^2) < 1/(2U_Q^2)$ of a_i , so by Legendre's theorem a_i is a convergent of its continued fraction; enumerating the convergents with denominator at most U_Q and keeping the one in the interval recovers a_i in polynomial time [109]. Then $X_a = \{z : (a, z) \in X\}$ is a nonempty bounded rational polyhedron on which the quadratic $f_\gamma(a, \cdot)$ is convex, by residual convexity in type (B) and by Lemma I.6(b) in type (P). In type (P) the set X_a can be lower dimensional, and convexity on X_a does not make the constant Hessian of $f_\gamma(a, \cdot)$ positive semidefinite on $\mathbb{R}^m$; for example, $-z^2$ is convex on $\{0\}$. We therefore first pass to the affine hull of X_a . Lemma I.23, applied to X_a , gives a rational point z_0 , a rational matrix B' of full column rank m' , and the polytope $\mathcal{R}' = \{w : z_0 + B'w \in X_a\}$, which contains a ball about the origin and satisfies $X_a = z_0 + B'\mathcal{R}'$. If $m' = 0$, put $z^* = z_0$. Otherwise the rational quadratic $w \mapsto f_\gamma(a, z_0 + B'w)$ is convex on $\mathcal{R}'$, which has nonempty interior, so its constant Hessian $B'^\top A_{zz} B'$ is positive semidefinite, where A_{zz} is the residual block of A . Solve this convex quadratic program over $\mathcal{R}'$ exactly [71], obtaining a rational optimal w^* , and put $z^* = z_0 + B'w^*$. Since a is an optimal core, $\min_{X_a} f_\gamma(a, \cdot) = f_\gamma^*$, so (a, z^*) is a global minimizer with value $f_\gamma^* = f_\gamma(a, z^*)$. The core query has precision polynomial in L , and its output, even on the fallback branch, is a rational of polynomial length; reconstruction, the linear programs of Lemma I.23, and the exact quadratic program cost polynomial time. The expected work is that of one query. $\square$

I.10 Error bounds and regularized selection

The following lemmas are the forms of the shared error-bound and selector tools (Lemma 3.2, Lemma 3.3, and Lemmas 3.8 and 3.9) needed here, with real coefficients, supplied minor margins, and an inexact objective. We include their short proofs.

Lemma I.19 (Hoffman bound with a minor margin). *Let B and T be real matrices with s columns and entries in $[-C, C]$, $C \geq 1$, and suppose every nonzero square minor of the stacked matrix $\begin{bmatrix} B \\ T \end{bmatrix}$ has absolute value at least $\mu \in (0, 1]$. Let $Q = \{x : Bx \leq \beta\}$ and $Z = \{x \in Q : Tx = t\} \neq \emptyset$, with arbitrary real β and t . Then $\text{dist}(x, Z) \leq (sC)^{s-1} \mu^{-1} \|Tx - t\|$ for every $x \in Q$. For integer matrices one may take $\mu = 1$.*

Proof. Let w be the projection of x onto the closed convex set Z ; we may assume $x \neq w$. Since Z is polyhedral, $x - w = B_I^\top \lambda + T^\top \nu$ with $\lambda \geq 0$ and I the rows of B active at w . Replace the rows of T by a basis of its row space, and then remove rows of B_I until all remaining rows are linearly independent: if a nontrivial combination of the remaining rows vanishes, it involves some row of B_I , and subtracting a suitable multiple keeps all λ coefficients nonnegative while making one zero. We obtain an independent row matrix R with $r \leq s$ rows, each a row of B_I or of T , and $x - w = R^\top \omega$ with nonnegative coefficients on the rows from B_I . By the Cauchy–Binet formula, $\det(RR^\top)$ is the sum of the squares of the $r \times r$ minors of R ; at least one is nonzero, so $\det(RR^\top) \geq \mu^2$. Every eigenvalue of $RR^\top$ is at most $\|R\|_F^2 \leq s^2 C^2$, so its least eigenvalue is at least $\mu^2 (sC)^{-2(r-1)}$, and $\|\omega\| \leq (sC)^{s-1} \mu^{-1} \|x - w\|$. Finally $\|x - w\|^2 = \omega^\top R(x - w)$. A row of B_I contributes $\omega_i(B_i x - \beta_i) \leq 0$, since $B_i w = \beta_i$ and $x \in Q$; a row of T contributes $\omega_i(T_i x - t_i)$, since $T_i w = t_i$. Hence $\|x - w\|^2 \leq \|\omega\| \|Tx - t\|$, which gives the claim. $\square$

Lemma I.20 (Cubic transverse estimate and slice). *Let φ be a real cubic polynomial on $\mathbb{R}^{n_0}$ and $Y \subseteq \mathbb{R}^{n_0}$ a convex set on which $\nabla^2 \varphi \succeq 0$. Let $c \in Y$, $\rho \geq \sup_{x \in Y} \|x - c\|$, and $M \geq \sup_Y \|\nabla^2 \varphi\|$. Let y minimize φ on Y , let $x \in Y$, $d = x - y$, and $E = \varphi(x) - \varphi(y)$.*

$$(a) \quad (d^\top \nabla^2 \varphi(c) d)^2 \leq 108 M \rho^2 E.$$

(b) *Let K be a subspace contained in $\ker \nabla^2 \varphi(\xi)$ for every $\xi \in Y$, Π the orthogonal projection onto $K^\perp$, and $g = \nabla \varphi(c)$. For $x', y' \in Y$, $|\varphi(x') - \varphi(y') - g^\top(x' - y')| \leq M \rho \|\Pi(x' - y')\|$. In particular $|g^\top d| \leq E + M \rho \|\Pi d\|$.*

(c) *If moreover $\ker \nabla^2 \varphi(c) = K$, then $\arg \min_Y \varphi = \{x \in Y : \nabla^2 \varphi(c)(x - y) = 0, g^\top(x - y) = 0\}$.*

Proof. (a) Put $a_0 = d^\top \nabla^2 \varphi(y) d \geq 0$, $b_0 = d^\top \nabla^2 \varphi(x) d \geq 0$, and $u = a_0 + b_0$. Since φ is cubic, $\nabla^2 \varphi(x) - \nabla^2 \varphi(y) = \mathcal{T}[d]$ for the constant symmetric third-derivative tensor $\mathcal{T}$, and exact Taylor expansion gives $E = \nabla \varphi(y)^\top d + \frac{1}{2} a_0 + \frac{1}{6} (b_0 - a_0) \geq u/6$, because $\nabla \varphi(y)^\top d \geq 0$ by first-order optimality on the convex set Y . Symmetry of $\mathcal{T}$ gives $d^\top \nabla^2 \varphi(c) d = a_0 + (c - y)^\top (\nabla^2 \varphi(x) - \nabla^2 \varphi(y)) d$. For positive semidefinite P , $\|Pd\| \leq \sqrt{\|P\| d^\top P d}$; with $\|c - y\| \leq \rho$ this gives $d^\top \nabla^2 \varphi(c) d \leq u + \rho \sqrt{M} (\sqrt{a_0} + \sqrt{b_0}) \leq u + \rho \sqrt{2Mu}$. Since $\|d\| \leq 2\rho$, $u \leq 8M\rho^2$, so $u \leq 2\rho \sqrt{2Mu}$. As $c \in Y$, the left side is nonnegative, and $(d^\top \nabla^2 \varphi(c) d)^2 \leq 18M\rho^2 u \leq 108M\rho^2 E$.

(b) For $\xi \in Y$, $\nabla \varphi(\xi) - g = \int_0^1 \nabla^2 \varphi(c + \theta(\xi - c))(\xi - c) d\theta$. Each Hessian on Y is symmetric with K in its kernel, so its range lies in $K^\perp$; hence $\nabla \varphi(\xi) - g \in K^\perp$ and $\|\nabla \varphi(\xi) - g\| \leq M\rho$. Integrating along the segment from y' to x' gives $\varphi(x') - \varphi(y') - g^\top(x' - y') = \int_0^1 (\nabla \varphi(y' + \theta(x' - y')) - g)^\top \Pi(x' - y') d\theta$.

(c) If x is a minimizer, then $E = 0$, so (a) and positive semidefiniteness give $\nabla^2 \varphi(c) d = 0$; thus $d \in K$, $\Pi d = 0$, and (b) gives $g^\top d = 0$. Conversely, the two equations give $\Pi d = 0$ and then $\varphi(x) - \varphi(y) = g^\top d = 0$ by (b). $\square$

Lemma I.21 (Gradient rows under global convexity). *Let $D \geq 2$ and let φ be a real polynomial on $\mathbb{R}^{n_0}$ of degree at most D that is convex on all of $\mathbb{R}^{n_0}$. Let $Y \subseteq \mathbb{R}^{n_0}$ be convex, y a minimizer of φ on Y with $\|y\|_\infty \leq \rho$, $\rho \geq 1$, and $x \in Y$ with $E = \varphi(x) - \varphi(y) \in [0, 1]$. Let $V \geq \sup\{|\varphi(\xi)| : \|\xi\|_\infty \leq \rho + 2\}$ and $G \geq \sup\{\|\nabla \varphi(\xi)\|_1 : \|\xi\|_\infty \leq \rho\}$. Then every coefficient of the polynomial $\xi \mapsto \nabla \varphi(\xi)^\top(x - y)$ has absolute value at most*

$$C_\varphi E^{1/D}, \quad C_\varphi = B_{D-1}^{n_0} (1 + J_D(V + G(1 + \rho) + C_D 3^D)),$$

where $C_D = (D + 1)D^D$, $J_D = (D + 1)D^{D+1}$, and $B_r = (r + 1)2^r r^r$.

Proof. Let $d = x - y$ and let $\beta(u) = \varphi(y + u) - \varphi(y) - \nabla\varphi(y)^\top u$, which is convex and nonnegative on $\mathbb{R}^{n_0}$ with $\beta(0) = 0$. Since $\nabla\varphi(y)^\top d \geq 0$ and $\beta(d) \geq 0$, we have $0 \leq \nabla\varphi(y)^\top d \leq E$ and $\beta(d) \leq E$; convexity gives $0 \leq \beta(\theta d) \leq E$ for $\theta \in [0, 1]$. Two interpolation facts are used. If a univariate polynomial of degree at most D is bounded by E at the nodes j/D , $0 \leq j \leq D$, then it is bounded by $C_D E(1 + |\theta|)^D$ at every real θ , because each Lagrange basis polynomial has denominator at least D^{-D} and numerator at most $(1 + |\theta|)^D$. If a polynomial π of degree at most D is bounded by K on $[0, \bar{\theta}]$, then $|\pi'(0)| \leq J_D K / \bar{\theta}$, by interpolating at the nodes $j\bar{\theta}/D$.

Fix $\xi \in [0, 1]^{n_0}$ and let $\pi(\theta) = \beta(\xi - y + \theta d)$. By midpoint convexity, $0 \leq \pi(\theta) \leq \frac{1}{2}\beta(2(\xi - y)) + \frac{1}{2}\beta(2\theta d)$. Here $\beta(2(\xi - y)) \leq 2V + 2G(1 + \rho)$, because $\|2\xi - y\|_\infty \leq \rho + 2$ and $\|\xi - y\|_\infty \leq 1 + \rho$, and $\beta(2\theta d) \leq C_D E(1 + 2|\theta|)^D$. If $E > 0$, then on $[0, E^{-1/D}]$ we have $E(1 + 2\theta)^D \leq (E^{1/D} + 2)^D \leq 3^D$, so $|\pi| \leq K := V + G(1 + \rho) + C_D 3^D / 2$ there and $|\pi'(0)| \leq J_D K E^{1/D}$. If $E = 0$, then $\beta(\theta d) = 0$ on $[0, 1]$ and hence for all θ , so π is bounded on $\mathbb{R}$ and constant, and $\pi'(0) = 0$. Since $\pi'(0) = (\nabla\varphi(\xi) - \nabla\varphi(y))^\top d$, in both cases $|\nabla\varphi(\xi)^\top d| \leq (1 + J_D K) E^{1/D}$ on $[0, 1]^{n_0}$.

Finally, a polynomial of degree at most r in each of n_0 variables that is bounded by K' on $[0, 1]^{n_0}$ has all coefficients bounded by $B_r^{n_0} K'$: interpolate on the grid $\{0, 1/r, \dots, 1\}^{n_0}$, and note that each univariate Lagrange basis polynomial on these nodes has coefficients at most $2^r r^r$. Apply this with $r = D - 1$. $\square$

Lemma I.22 (Regularized selection with an inexact objective). *Let $Y \subseteq \mathbb{R}^{n_0}$ be compact and convex, φ convex and continuous on Y , $S = \arg \min_Y \varphi$, $\varphi^* = \min_Y \varphi$, and p the least-norm point of S . Let $R_x \geq \max\{1, \sup_Y \|x\|\}$, $\Gamma \geq 1$, and $D_e \geq 2$, and suppose $\text{dist}(x, S) \leq \Gamma(\varphi(x) - \varphi^*)^{1/D_e}$ whenever $x \in Y$ and $\varphi(x) - \varphi^* \leq 1$. For $0 < e \leq 1$ put*

$$\tau = \frac{e^{2D_e-2}}{2^{3D_e-2} R_x^{D_e} \Gamma^{D_e}}.$$

Then every $\hat{x} \in Y$ with $\varphi(\hat{x}) + \tau \|\hat{x}\|^2 \leq \min_Y (\varphi + \tau \|\cdot\|^2) + \tau e^2 / 4$ satisfies $\|\hat{x} - p\| \leq e$.

Proof. Let x_τ minimize $\varphi + \tau \|\cdot\|^2$ on Y . Comparison with p gives $\|x_\tau\| \leq \|p\|$ and $\varphi(x_\tau) - \varphi^* \leq \tau R_x^2 \leq 1$. Let s be the point of S nearest to x_τ and $\epsilon_d = \|x_\tau - s\|$. Comparison with s gives $\varphi(x_\tau) - \varphi^* \leq \tau(\|s\|^2 - \|x_\tau\|^2) \leq 2R_x \tau \epsilon_d$, and the error bound gives $\epsilon_d^{D_e} \leq \Gamma^{D_e} 2R_x \tau \epsilon_d$, so $\epsilon_d \leq (2R_x \tau \Gamma^{D_e})^{1/(D_e-1)} = e^2 / (8R_x)$. Since p is the projection of the origin onto S , $p^\top(s - p) \geq 0$, and

$$\|x_\tau - p\|^2 \leq 2\|p\|^2 - 2p^\top x_\tau = 2p^\top(p - s) + 2p^\top(s - x_\tau) \leq 2R_x \epsilon_d \leq e^2 / 4.$$

The function $\varphi + \tau \|\cdot\|^2$ is 2τ -strongly convex, so $\tau \|\hat{x} - x_\tau\|^2 \leq \tau e^2 / 4$. The triangle inequality gives $\|\hat{x} - p\| \leq e$. $\square$

Lemma I.23 (Relative coordinates). *For a nonempty bounded rational polytope $X = \{x \in \mathbb{R}^n : Ex \leq e\}$ one computes in polynomial time an integer $n' \geq 0$, a rational $x_0 \in X$, a rational $n \times n'$ matrix B of rank n' , and rationals $r > 0$ and $R_0 \geq 1$, all of polynomial bit length, such that $\mathcal{R} = \{w : x_0 + Bw \in X\}$ satisfies $X = x_0 + B\mathcal{R}$ and $B(0, r) \subseteq \mathcal{R} \subseteq B(0, R_0)$.*

Proof. For each row i solve the linear program $\max\{e_i - E_i x : x \in X\}$. Let I_0 be the rows whose maximum slack is zero; then $\text{aff } X = \{x : E_{I_0} x = e_{I_0}\}$. If every row lies in I_0 , X is a point and $n' = 0$. Otherwise average optimal solutions for the rows outside I_0 to get $\bar{x} \in X$ with positive slack in all of them. Let the columns of B be a rational basis of $\ker E_{I_0}$. The linear program $\max\{\varrho : E_i(\bar{x} + Bw) + \varrho \|E_i B\|_1 \leq e_i \text{ (} i \notin I_0 \text{)}, 0 \leq \varrho \leq 1\}$ has a positive optimum r at a rational point w^* ; put $x_0 = \bar{x} + Bw^*$. Every u with $\|u\|_\infty \leq r$ keeps $x_0 + Bu$ in X , so the closed, and hence the open, Euclidean ball of radius r lies in $\mathcal{R}$. Coordinate linear programs bound $|w_j|$ on $\mathcal{R}$ by rationals m_j . Put $R_0 = 1 + \sum_j m_j$; then $\|w\| \leq \|w\|_1 \leq \sum_j m_j < R_0$ on $\mathcal{R}$, so $\mathcal{R} \subseteq B(0, R_0)$. $\square$

I.11 Proof of Theorem 11.5

Throughout this subsection the instance has type (P), $\kappa = \alpha$, and we assume $D \geq 2$ (a degree bound can always be raised). Put $G_\alpha = f + \frac{\alpha}{2} \|v\|^2$, $\beta = \alpha + 1$, and $G_\beta = G_\alpha + \frac{1}{2} \|v\|^2$. For a draw γ with selected core a define the surrogate

$$\psi_a(x) = f_\gamma(x) + \frac{\beta}{2} \|v - a\|^2 = G_\beta(x) + (\gamma - \beta a)^\top v + \frac{\beta}{2} \|a\|^2, \quad S_a = \{a\} \times S(\gamma).$$

Lemma I.24 (The convex surrogate). *In case (i) ψ_a is convex on X ; in case (ii) it is convex on $\mathbb{R}^n$. Moreover $\min_X \psi_a = f_\gamma^*$, $\arg \min_X \psi_a = S_a$, $\psi_a(x) - f_\gamma^* \geq \frac{\beta}{2} \|v - a\|^2$ on X , and the least-norm point of S_a is $x^\sharp(\gamma)$.*

Proof. ψ_a equals G_α plus the convex function $\frac{1}{2} \|v\|^2$ plus an affine function. On X , $\psi_a(x) \geq f_\gamma(x) \geq f_\gamma^*$, with equality exactly when $f_\gamma(x) = f_\gamma^*$ and $v = a$, that is, on S_a . The least-norm point of $\{a\} \times S(\gamma)$ is (a, p) . $\square$

Lemma I.25 (A uniform error bound with rational rows). *Let $D_e = 4$ in case (i) and $D_e = D$ in case (ii). There is a base-computable rational $\Gamma \geq 1$ of polynomial bit length such that, for every $\gamma \in [-\sigma, \sigma]^k$ and every $x \in X$ with $E = \psi_a(x) - f_\gamma^* \leq 1$,*

$$\text{dist}(x, S_a) \leq \Gamma E^{1/D_e}.$$

Proof. Put $N = k(\sigma + \beta)$, so that $\|\gamma - \beta a\| \leq N$. Fix $y \in S_a$ and write $d = x - y$ and $d_v = v - a$ for its core part. By Lemma I.24, $\|d_v\| \leq \sqrt{2E/\beta} \leq 2E^{1/D_e}$ when $E \leq 1$.

Case (i). Take $x_0, B, r, R_0, \mathcal{R}, n'$ from Lemma I.23. If $n' = 0$, then X is a point and $\Gamma = 1$ works. Otherwise let B_v be the first k rows of B , so that the core of $x_0 + Bw$ is $v_0 + B_v w$. The function $h(w) = G_\beta(x_0 + Bw)$ is an explicit rational cubic; put $H(w) = \nabla^2 h(w)$, $H_0 = H(0)$, $g_0 = \nabla h(0)$, and $K = \ker H_0$. Since h is convex on the full-dimensional set $\mathcal{R}$, $H \succeq 0$ on $\mathcal{R}$. For $w \in \mathcal{R}$ the point $w' = -(r/R_0)w$ lies in $B(0, r) \subseteq \mathcal{R}$, and 0 is a convex combination of w and w' with weights $\theta/(1 + \theta)$ and $1/(1 + \theta)$, $\theta = r/R_0$. As H is affine, $H_0 \succeq \frac{\theta}{1+\theta} H(w)$, that is,

$$0 \leq H(w) \leq \beta_0 H_0 \quad (w \in \mathcal{R}), \quad \beta_0 = 1 + R_0/r.$$

Hence $K \subseteq \ker H(w)$ on $\mathcal{R}$. Let $M_0 = \max\{1, \max_i \sum_j |(H_0)_{ij}|\} \geq \|H_0\|$ and $M = \beta_0 M_0$. If $H_0 \neq 0$, let Δ_H be a positive integer with $\Delta_H H_0$ integral, let $C_H \geq 1$ bound its entries, and put $\lambda_0 = 1/(\Delta_H (n' C_H)^{n'-1})$; the product of the positive eigenvalues of $\Delta_H H_0$ is a sum of principal minors, hence a positive integer, and each eigenvalue is at most $n' C_H$, so every positive eigenvalue of H_0 is at least λ_0 . Put $C_\perp = \max\{1, 108 M R_0^2 / \lambda_0^2\}$, or $C_\perp = 1$ if $H_0 = 0$.

Let $w, u \in \mathcal{R}$ represent x, y , put $d' = w - u$, and let Π project onto $K^\perp$. The reduced surrogate $\tau_a(w) = \psi_a(x_0 + Bw) = h(w) + (\gamma - \beta a)^\top (v_0 + B_v w) + \text{const}$ is a cubic, convex on $\mathcal{R}$, with Hessian H and minimizer u , and $E = \tau_a(w) - \tau_a(u)$. Lemma I.20(a) with $c = 0$ and $\rho = R_0$ gives $(d'^\top H_0 d')^2 \leq 108 M R_0^2 E$; if $H_0 \neq 0$, then $d'^\top H_0 d' \geq \lambda_0 \|\Pi d'\|^2$, so $\|\Pi d'\| \leq C_\perp E^{1/4}$, and if $H_0 = 0$ then $\Pi = 0$. Lemma I.20(b), applied to h , gives $|h(w) - h(u) - g_0^\top d'| \leq M R_0 \|\Pi d'\|$. Since $E = h(w) - h(u) + (\gamma - \beta a)^\top B_v d'$ and $B_v d' = d_v$,

$$|g_0^\top d'| \leq E + N \|d_v\| + M R_0 \|\Pi d'\|.$$

Let $J = [H_0; g_0^\top; B_v]$, a known rational matrix. For $E \leq 1$, $\|Jd'\| \leq M_0 \|\Pi d'\| + |g_0^\top d'| + \|d_v\| \leq C_{\text{res}} E^{1/4}$ with $C_{\text{res}} = 3 + 2N + (M_0 + M R_0) C_\perp$.

Let $\mathcal{R}_a = \{w \in \mathcal{R} : x_0 + Bw \in S_a\}$ be the reduced optimal set. Then $\mathcal{R}_a = \{w \in \mathcal{R} : J(w - u) = 0\}$. Indeed, if w is optimal, that is, $w \in \mathcal{R}_a$, then $E = 0$ and the bounds give $Jd' = 0$. Conversely, $Jd' = 0$ gives $d' \in K$, so $h(w) - h(u) = g_0^\top d' = 0$ by Lemma I.20(b), and $B_v d' = 0$, so $E = 0$ and w is optimal. Clear the denominators of J and of the rows EB describing $\mathcal{R}$ by a positive integer Δ_J , and let $C_J \geq 1$ bound the entries of the two integer matrices. Lemma I.19 with $\mu = 1$ gives $\text{dist}(w, \mathcal{R}_a) \leq (n' C_J)^{n'-1} \Delta_J \|Jd'\|$. Since $x = x_0 + Bw$, the claim holds with $\Gamma = B_F (n' C_J)^{n'-1} \Delta_J C_{\text{res}}$, where $B_F \geq \max\{1, \|B\|\}$ is a rational bound, for example $B_F = \max\{1, \sum_{i,j} |B_{ij}|\}$.

Case (ii). Let $\rho \geq 1$ bound $\|x\|_\infty$ on the supplied box. Writing $f = \sum_\nu f_\nu x^\nu$, the numbers

$$V = \sum_\nu |f_\nu| (\rho + 2)^{|\nu|} + N(\rho + 2) + \frac{\beta}{2} k(\rho + 3)^2, \quad G = \sum_\nu |f_\nu| |\nu| \rho^{|\nu|-1} + k\sigma + \beta k(\rho + 1)$$

bound $|\psi_a|$ on $\{\|\xi\|_\infty \leq \rho + 2\}$ and $\|\nabla \psi_a\|_1$ on $\{\|\xi\|_\infty \leq \rho\}$, uniformly in a and γ . Lemma I.21, applied to ψ_a on $Y = X$, bounds every coefficient of $\xi \mapsto \nabla \psi_a(\xi)^\top d$ by $C_\psi E^{1/D}$, with C_ψ computed from V, G, ρ, n, D . For a polynomial φ of degree at most D let N_φ be the matrix whose rows are indexed by the monomials ξ^μ of degree at most $D - 1$, with row μ equal to $((\mu_i + 1)\varphi_{\mu+e_i})_{i \leq n}$; then $N_\varphi d$ is the coefficient vector of $\nabla \varphi(\xi)^\top d$. Since $\psi_a - G_\beta$ is affine, N_{ψ_a} and N_{G_β} differ only in the row $\mu = 0$, by $((\gamma - \beta a)^\top, 0)$. Let $J = [N_{G_\beta}; (I_k \ 0)]$, a known rational matrix with at most $(n + 1)^{D-1} + k$ rows. The vector $N_{\psi_a} d$ has at most $\binom{n+D-1}{D-1} \leq (n + 1)^{D-1}$ entries, each of absolute value at most $C_\psi E^{1/D}$, so its norm is at most $(n + 1)^{\lceil (D-1)/2 \rceil} C_\psi E^{1/D}$. The integer exponent makes the constant C_{res} in the next display rational. Then, for $E \leq 1$,

$$\|Jd\| \leq (n+1)^{\lceil (D-1)/2 \rceil} C_\psi E^{1/D} + N \|d_v\| + \|d_v\| \leq C_{\text{res}} E^{1/D}, \quad C_{\text{res}} = (n+1)^{\lceil (D-1)/2 \rceil} C_\psi + 2N + 2.$$

The optimal set is $S_a = \{x \in X : J(x - y) = 0\}$: if $E = 0$ the bound gives $Jd = 0$; conversely $Jd = 0$ gives $d_v = 0$ and $N_{G_\beta} d = 0$, hence $N_{\psi_a} d = 0$, so $\nabla \psi_a(\xi)^\top d = 0$ for all ξ and $\psi_a(x) = \psi_a(y)$. Lemma I.19, applied after clearing the denominators of J and of the rows of X , gives the claim with $\Gamma = (nC_J)^{n-1} \Delta_J C_{\text{res}}$.

In both cases the constants depend only on the instance, not on γ or a , and their bit lengths are polynomial in L for fixed D . $\square$

Proof of Theorem 11.5. Use the grid and the core evaluator of Theorem 11.2 with $\kappa = \alpha$, or of Corollary 11.3 if $\alpha = 0$. For a query q let $\epsilon = 2^{-q}$, let $R_x \geq \max\{1, \sup_X \|x\|\}$ be a rational bound from the supplied box, and put

$$\tau = \frac{\epsilon^{2D_e-2}}{2^{3D_e-2} R_x^{D_e} \Gamma^{D_e}}, \quad \eta_q = \frac{\tau \epsilon^2}{8}, \quad \delta_q = \frac{\tau \epsilon^2}{16\beta k}.$$

Query the core evaluator with accuracy δ_q ; it returns a rational b with $\|b - a\| \leq \delta_q$. Apply (78) on X to the explicit rational convex polynomial $\Phi_b(x) = G_\beta(x) + (\gamma - \beta b)^\top v + \tau \|x\|^2$ with accuracy η_q , and return its point $\hat{x} \in X$. Let $\Phi_a = \psi_a - \frac{\beta}{2} \|a\|^2 + \tau \|x\|^2$. On X , $|\Phi_b - \Phi_a| = \beta |(a - b)^\top v| \leq \beta \sqrt{k} \delta_q$, so $\Phi_a(\hat{x}) - \min_X \Phi_a \leq \eta_q + 2\beta \sqrt{k} \delta_q \leq \tau \epsilon^2/4$. By Lemmas I.24, I.25, and I.22 with $e = \epsilon$, $\|\hat{x} - x^\sharp(\gamma)\| \leq \epsilon$. The core query has precision $\log(1/\delta_q) \leq (2D_e + 2)q + P_D(L)$ and costs at most $\Omega(\gamma) P_D(L + q)$. The convex solve has data of bit length $P_D(L + q)$ and tolerance $\langle \eta_q \rangle \leq (2D_e + 2)q + P_D(L)$, so it costs $P_D(L + q)$. The random factor is that of the core evaluator, so its expectation has the stated bound. $\square$

The completion never reads an exact algebraic record of a : it uses only the short rational output b . The surrogate is optimized on the fixed domain X , so no continuity of residual optimizers in the core is needed.

Proof of Corollary 11.6. Each $w_j(a_j^\top x - b_j)^{p_j}$ is convex because p_j is even and $w_j > 0$, and $\frac{1}{2}x^\top Ax$ is convex because $A \succeq 0$; so the representation is convex on $\mathbb{R}^n$. For fixed D , expanding the right side produces at most $\binom{n+D}{D}$ monomials, so the identity with $f + \frac{\alpha}{2}\|v\|^2$ is checked in polynomial time, and $A \succeq 0$ is checked by exact symmetric Gaussian elimination. $\square$

I.12 Residual cubic fibers

Proof of Proposition 11.7. If $m = 0$, the residual space consists of the empty vector. Then $S_v = \{()\}$, all matrix and kernel assertions are vacuous, and part (c) holds with $\Gamma = 1$. In the rest of the proof assume $m \geq 1$.

(a) Since f is cubic, H is affine in (v, z) . For each $v \in [0, 1]^k$, $f(v, \cdot)$ is convex on $[0, 1]^m$, so $H(v, z) \succeq 0$ there, by continuity at the boundary. The reflection $(v, z) \mapsto (2c_A - v, \mathbf{1} - z)$ maps $A \times [0, 1]^m$ onto itself, and affinity gives $H(v, z) + H(2c_A - v, \mathbf{1} - z) = 2H_A$. Both terms are positive semidefinite, so $0 \preceq H(v, z) \preceq 2H_A$, and $K_A := \ker H_A \subseteq \ker H(v, z)$. If the free coordinates of v lie in $[\delta, 1 - \delta]$ with $\delta < \frac{1}{2}$, then $w = (v - 2\delta c_A)/(1 - 2\delta) \in A$ and $v = 2\delta c_A + (1 - 2\delta)w$, so $H(v, c_z) = 2\delta H_A + (1 - 2\delta)H(w, c_z) \succeq 2\delta H_A$; for $\delta = \frac{1}{2}$, $v = c_A$. Every v in the relative interior of A has such a $\delta > 0$, so $\ker H(v, c_z) \subseteq K_A$.

(b) Apply Lemma 1.20(c) to $\varphi = f(v, \cdot)$ on $Y = [0, 1]^m$ with $c = c_z$, $\rho = \sqrt{m}/2$, and $K = K_A$; part (a) gives the kernel hypotheses. Since $\ker H(v, c_z) = K_A = \ker H_A$, the condition $H(v, c_z)(z - y) = 0$ is equivalent to $H_A(z - y) = 0$. A square submatrix of $J_A(v)$ contains the row $g_A(v)^\top$ at most once; its determinant is constant if it does not, and otherwise a linear combination, with constant coefficients, of the entries of $g_A(v)$, which have degree at most two in v .

(c) Let $M_A = \max\{1, \max_j \sum_l |(H_A)_{jl}|\}$ and $M_* = 2M_A$, so that $\|H(v, z)\| \leq M_*$ on $A \times [0, 1]^m$. If $H_A \neq 0$, let λ_0^A be the rational lower bound on its positive eigenvalues from principal minors, as in the proof of Lemma 1.25, and put $C_\perp = \max\{1, 27M_*m/(2\delta\lambda_0^A)^2\}$; if $H_A = 0$ put $C_\perp = 1$. Let $C_B \geq 1$ bound the absolute values of the entries of H_A and of $g_A(v')$ for all $v' \in [0, 1]^k$, by coefficient sums. Fix $z \in [0, 1]^m$ with gap $E \leq 1$, $y \in S_v$, $d = z - y$, and Π the projection onto $K_A^\perp$. We first show $\|\Pi d\| \leq C_\perp E^{1/4}$. If $H_A = 0$, then $K_A = \mathbb{R}^m$ and $\Pi = 0$, so there is nothing to prove. If $H_A \neq 0$, part (a) gives $d^\top H(v, c_z)d \geq 2\delta d^\top H_A d \geq 2\delta\lambda_0^A \|\Pi d\|^2$, and Lemma 1.20(a) with $\rho^2 = m/4$ gives $(d^\top H(v, c_z)d)^2 \leq 27M_*mE$; together these give the claim. Part (b) of that lemma, with $\rho = \sqrt{m}/2 \leq m$, gives $|g_A(v)^\top d| \leq E + M_*m \|\Pi d\|$. Using m in place of $\sqrt{m}/2$ makes the constant Γ defined next rational. With $T = [H_A; g_A(v)^\top]$ and $H_A d = H_A \Pi d$, $\|Td\| \leq (1 + (M_A + M_*m)C_\perp)E^{1/4}$. By part (b), $S_v = \{z \in [0, 1]^m : Tz = Ty\}$. Write $[0, 1]^m = \{z : z \leq \mathbf{1}, -z \leq 0\}$. Up to sign, every nonzero square minor of the stacked matrix of box rows and T is a nonzero square minor of $J_A(v)$, hence at least μ in absolute value. Lemma 1.19 gives

$$\text{dist}(z, S_v) \leq \Gamma E^{1/4}, \quad \Gamma = (mC_B)^{m-1} \mu^{-1} (1 + (M_A + M_*m)C_\perp).$$

All quantities in this formula are rational and computed by arithmetic on f , A , δ , and μ , so Γ is computed in time, and has bit length, polynomial in $L + \langle \delta \rangle + \langle \mu \rangle$. $\square$

Lemma I.26 (Uniform heights of the minors). *There are base-computable positive integers Δ and B , with $\log \Delta + \log B$ polynomial in L , such that for every face A and every square submatrix of $J_A(v)$, the polynomial $\Delta \det(\cdot)$, with the fixed coordinates of A substituted, has integer coefficients of absolute value at most B in the free coordinates.*

Proof. Let d_0 be the product of the denominators of the coefficients of $f = \sum_\nu f_\nu x^\nu$, and $d = 8d_0$. Entries of H_A are values at points with coordinates in $\{0, \frac{1}{2}, 1\}$ of second derivatives of degree at most one, so d clears them. Entries of $g_A(v)$ are first z -derivatives, of degree at most two, evaluated

at $z = c_z$; after substituting the fixed coordinates of A , their coefficients are cleared by d . Let $C_0 = \max\{d, \lceil 6d \sum_\nu |f_\nu| \rceil\}$. Then C_0 bounds the entries of dH_A , the entries d of dI_m , and the sum of the absolute coefficients of each entry of $d g_A$ on every face. A minor of order $r' \leq m$ is a sum of $r'!$ signed products of r' entries, at most one from the row $g_A^\top$. Multiplied by $d^{r'}$, each product is an integer polynomial whose coefficient absolute values sum to at most $C_0^{r'}$. Hence $\Delta = d^m$ and $B = d^m m! C_0^m$ work. $\square$

Proof of Proposition 11.8. (a) Here $\partial_z^2 f = 2v \geq 0$. For $v > 0$ the unique residual minimizer is $z = 0$, and at $z = 1$ the gap is v while the distance is one; a bound $1 \leq \Gamma v^\theta$ for all $v \in (0, 1]$ is impossible. The Hessian of $f + \frac{\alpha}{2}v^2$ is $\begin{bmatrix} \alpha & 2z \\ 2z & 2v \end{bmatrix}$, whose determinant at $(0, \frac{1}{2})$ is -1 . A function convex on the square has positive semidefinite Hessian on the closed square, so no α works.

(b) The fibers are affine. For $v > 0$ the coefficient of z is $-v^2 < 0$, so the unique residual minimizer is $z = 1$ and the projected value is $\gamma v - v^2 = v(\gamma - v) > 0$ on $(0, 1]$, while $f_\gamma(0, z) = 0$ for all z . Thus the optimal set is $\{0\} \times [0, 1]$, $a = 0$, $p = 0$, and $(b_j, 1) \rightarrow (0, 1)$. The Hessian of $f_\gamma + \frac{\alpha}{2}v^2$ is $\begin{bmatrix} \alpha - 2z & -2v \\ -2v & 0 \end{bmatrix}$, with determinant $-4v^2 < 0$ for $v > 0$. For $f = -v^2 z$, $\Lambda = 0$ is a valid coordinate bound, and when $\sigma > 1$ the draws $\gamma \in (1, \sigma]$ have probability at least $(\sigma - 1)/(2\sigma) - 2/M > 0$ under the grid law for large M . $\square$

I.13 Proof of Lemma 11.10

Fix i and γ_{-i} , and let

$$V_i(s) = \min \left\{ f(v, z) + \sum_{l \neq i} \gamma_l v_l : (v, z) \in X, v_i = s \right\}, \quad s \in [0, 1].$$

The i -th coordinates of the optimal cores of $f_{\gamma + \theta e_i}$ are exactly the minimizers of $V_i(s) + (\gamma_i + \theta)s$ on $[0, 1]$. The function V_i is continuous, and Lipschitz with constant $G_i = \sup_X |\partial f / \partial v_i|$: moving the i -th coordinate of a minimizer at s' to s stays in the product box. For every fixed (v_{-i}, z) , the function $s \mapsto f(v, z) - \bar{\Lambda} s^2 / 2$ is concave on $[0, 1]$, because $\partial_{v_i}^2 f \leq \Lambda \leq \bar{\Lambda}$; as a minimum of concave functions, $V_i - \bar{\Lambda} s^2 / 2$ is concave.

Monotonicity. If s_0 minimizes $V_i + \gamma_i s$ and s_1 minimizes $V_i + (\gamma_i - t)s$, adding the two optimality inequalities gives $-ts_1 \leq -ts_0$, so $s_0 \leq s_1$.

Interior contacts. If $s_0 \in (0, 1)$ minimizes $V_i + \vartheta s$, then V_i is differentiable at s_0 with $V_i'(s_0) = -\vartheta$. Indeed, a supergradient ζ of the concave function $V_i - \bar{\Lambda} s^2 / 2$ at s_0 gives $V_i(s) \leq V_i(s_0) + (\zeta + \bar{\Lambda} s_0)(s - s_0) + \frac{\bar{\Lambda}}{2}(s - s_0)^2$, while minimality gives $V_i(s) \geq V_i(s_0) - \vartheta(s - s_0)$. Comparing both sides of s_0 gives $\zeta + \bar{\Lambda} s_0 = -\vartheta$, and V_i is squeezed between two functions with derivative $-\vartheta$ at s_0 . For two such points $s_0 \leq s_1$, concavity of $V_i - \bar{\Lambda} s^2 / 2$ gives $V_i'(s_1) - V_i'(s_0) \leq \bar{\Lambda}(s_1 - s_0)$.

(a) Let b be an optimal core of $f_{\gamma - t e_i}$ with $b_i < t / \bar{\Lambda} \leq \frac{1}{2}$, and suppose an optimal core of f_γ has $a_i > 0$. By monotonicity $0 < a_i \leq b_i < \frac{1}{2}$, so both are interior contacts, with slopes $-\gamma_i$ and $-(\gamma_i - t)$. Then $t = V_i'(b_i) - V_i'(a_i) \leq \bar{\Lambda}(b_i - a_i) < \bar{\Lambda} b_i < t$, a contradiction. The second rule follows by applying the first to $s \mapsto V_i(1 - s)$, which has the same semiconcavity.

(b) Let $\theta_i^0 = \sup_{0 < s \leq 1} (V_i(0) - V_i(s)) / s$, finite because V_i is Lipschitz. The coordinate 0 is optimal for the tilt γ_i exactly when $V_i(0) \leq V_i(s) + \gamma_i s$ for all s , that is, when $\gamma_i \geq \theta_i^0$. If $\gamma_i - t > \theta_i^0$, then for $s > 0$, $V_i(s) + (\gamma_i - t)s > V_i(s) + \theta_i^0 s \geq V_i(0)$, so every optimal core of $f_{\gamma - t e_i}$ has i -th coordinate 0. Hence the first situation in (b) forces $\gamma_i \in [\theta_i^0, \theta_i^0 + t]$. With $\theta_i^1 = \inf_{0 \leq s < 1} (V_i(s) - V_i(1)) / (1 - s)$, the coordinate 1 is optimal exactly when $\gamma_i \leq \theta_i^1$, and $\gamma_i + t < \theta_i^1$ forces every optimal core of $f_{\gamma + t e_i}$ to have i -th coordinate 1. Both thresholds depend only on γ_{-i} . $\square$

Corollary I.27 (Certified face rules). *Let $r_f = t/(8\bar{\Lambda})$, and let $\hat{b}^{(i,\pm)}$ be rational vectors within distance r_f of the selected cores of $f_{\gamma \pm te_i}$. Fix $v_i = 0$ if $\hat{b}_i^{(i,-)} + r_f < t/\bar{\Lambda}$, and fix $v_i = 1$ if $\hat{b}_i^{(i,+)} - r_f > 1 - t/\bar{\Lambda}$. Then every fixed value is correct for every optimal core of f_γ , no coordinate is fixed twice, and under the grid law the probability that the selected core $a(\gamma)$ has a coordinate in $\{0, 1\}$ that is not fixed is at most $kt/\sigma + 2k/M$.*

Proof. The rules imply the hypotheses of Lemma 11.10(a) for the exact selected cores, so they are sound; two sound rules for the same coordinate would contradict the existence of an optimal core. Suppose $a_i = 0$ and the first rule does not fire. Then the selected core b of $f_{\gamma - te_i}$ has $b_i \geq t/\bar{\Lambda} - 2r_f > 0$, so $\gamma_i \in [\theta_i^0, \theta_i^0 + t]$ by Lemma 11.10(b). Conditionally on γ_{-i} , an interval of length t contains at most $t(M-1)/(2\sigma) + 1$ grid points, so this has probability at most $t/(2\sigma) + 1/M$. The case $a_i = 1$ is symmetric. A union bound over i and both endpoints gives the claim. $\square$

I.14 Proof of Lemma 11.11

Every nonconstant monomial of degree at most two is 2-Lipschitz on $[0, 1]^r$ in the maximum norm. Since both a and $\hat{a}$ lie in $[0, 1]^r$, this gives $|T\chi_j(\hat{a}) - T\chi_j(a)| \leq 2T\|\hat{a} - a\|_\infty \leq \frac{1}{2}$ and $|\omega_j - T\chi_j(a)| \leq 1$ for every j . The lattice $\mathcal{L}$ consists of the vectors $(h, h^\top \omega)$ with $h \in \mathbb{Z}^s$.

(a) If the test accepts, then $\lambda_1(\mathcal{L}) \geq 2^{-(s-1)/2} \|\mathbf{b}_1\| > 4sB$ by Theorem I.3. Suppose $0 < \|h\|_\infty \leq B$ and $|h^\top \chi(a)| \leq sB/T$. Then $|h^\top \omega| \leq \|h\|_1 + T|h^\top \chi(a)| \leq 2sB$ and $\|h\| \leq sB$, so the nonzero lattice vector $(h, h^\top \omega)$ has length at most $\sqrt{5}sB < 4sB$, a contradiction.

(b) Let $(h, h^\top \omega) \neq 0$. If $\|h\| > R_s$, its length exceeds R_s . Otherwise $0 < \|h\|_\infty \leq R_s$, and $|h^\top \omega| \geq T|h^\top \chi(a)| - \|h\|_1 \geq T\eta_0 - sR_s > R_s$. Hence $\|\mathbf{b}_1\| \geq \lambda_1(\mathcal{L}) > R_s = 2^s 4sB$, and $\|\mathbf{b}_1\|^2 > 2^{2s}(4sB)^2 \geq 2^{s-1}(4sB)^2$.

Constructing and rounding ω costs time polynomial in $s + \log T + \langle \hat{a} \rangle$. The resulting basis entries have $O(\log T)$ bits, so Theorem I.3 gives the remaining cost bound; the acceptance test is an exact integer comparison. $\square$

Part (b) must be stated with the larger height R_s : a short lattice vector can have coefficients much larger than B , so margins over the height- B family alone would not imply acceptance.

I.15 Proof of Proposition 11.12

Lemma I.28 (Quadratic sublevel sets). *For every nonzero integer polynomial ψ of degree at most two in r variables and every $0 < \eta < 1$, $\text{Leb}\{x \in [0, 1]^r : |\psi(x)| \leq \eta\} \leq 8\sqrt{\eta}$.*

Proof. For a univariate quadratic with leading coefficient $\ell \neq 0$, completing the square shows that $\{|\cdot| \leq \eta\}$ has total length at most $2\sqrt{2\eta/|\ell|}$; the two-interval case is largest when the inner endpoints meet. If some square coefficient ψ_{ii} is nonzero, then $|\psi_{ii}| \geq 1$, and slicing along x_i with Fubini's theorem gives the bound $2\sqrt{2\eta}$. If all square coefficients vanish but some $\psi_{ij} \neq 0$, use the coordinates $\varsigma = (x_i + x_j)/\sqrt{2}$ and $u = (x_i - x_j)/\sqrt{2}$. Then $x_i x_j = (\varsigma^2 - u^2)/2$, other terms are at most linear in ς , and every slice in ς has leading coefficient of absolute value at least $\frac{1}{2}$, hence length at most $4\sqrt{\eta}$; the projection of the cube along ς has measure $\sqrt{2}$, which gives $4\sqrt{2}\sqrt{\eta}$. If ψ is a nonconstant linear polynomial, slicing along a variable with a nonzero integer coefficient gives $2\eta \leq 2\sqrt{\eta}$. A nonzero integer constant has empty sublevel set. $\square$

Lemma I.29 (Contact slopes). *Let $\Phi : [0, 1]^r \rightarrow \mathbb{R}$ be continuous with $\Phi - \frac{U}{2}\|\cdot\|^2$ concave, and let $\mathcal{C}$ be the set of interior points a at which $\Phi(x) \geq \Phi(a) + s^\top(x - a)$ for all $x \in [0, 1]^r$ and some s . Then $s = s(a)$ is unique, s is locally U -Lipschitz on $\mathcal{C}$, and $\text{Leb}^*(s(E \cap \mathcal{C})) \leq U^r \text{Leb}(E)$ for every measurable E in the interior of the cube.*

Proof. As in the one-dimensional argument of Section I.13, a supergradient of the concave function at a and the lower support force $s(a) = \zeta + Ua$, and then $0 \leq \Phi(x) - \Phi(a) - s(a)^\top(x - a) \leq \frac{U}{2} \|x - a\|^2$. Let $a, a' \in \mathcal{C}$, $\varrho = \|a - a'\|$, $\Delta s = s(a') - s(a)$, and suppose the closed ball of radius ϱ about a lies in the cube. For $\|u\| = \varrho$, the lower support at a' and the upper support at a , evaluated at $a + u$, give $\Delta s^\top u \leq \Phi(a) - \Phi(a') - s(a')^\top(a - a') + \frac{U}{2}\varrho^2 \leq U\varrho^2$, using the upper support at a' evaluated at a . Taking $u = \varrho\Delta s/\|\Delta s\|$ gives $\|\Delta s\| \leq U\varrho$. If $B(a_0, 3\varrho_0)$ lies in the interior, every pair of points of $\mathcal{C} \cap B(a_0, \varrho_0)$ satisfies the ball condition, so s is U -Lipschitz there. Cover the interior by countably many such balls B_l , disjointify $E_l = E \cap B_l \setminus \bigcup_{l' < l} B_{l'}$, apply (F2) on each piece, and sum: $\text{Leb}^*(s(E \cap \mathcal{C})) \leq U^r \sum_l \text{Leb}(E_l) = U^r \text{Leb}(E)$. $\square$

Proof of Proposition 11.12. Identify the relative interior of A with the interior of $[0, 1]^r$ through the free coordinates $x = v_F$, and put $\Phi(x) = \min_{z \in [0, 1]^m} f(v(x), z)$, where $v(x)$ has the fixed coordinates of A . For every z , $x \mapsto f(v(x), z) - \frac{U}{2} \|x\|^2$ is concave, so $\Phi - \frac{U}{2} \|\cdot\|^2$ is concave. If a global minimizer of f_γ has core a in the relative interior of A , then $x_a = a_F$ minimizes $\Phi(x) + \gamma_F^\top x$ over the cube, so $x_a \in \mathcal{C}$ with $s(x_a) = -\gamma_F$. With $E = \{x : |\psi(x)| \leq \eta\}$ in the interior, the event therefore implies $-\gamma_F \in s(E \cap \mathcal{C})$. Conditionally on the noise of the fixed coordinates, Lemmas I.29 and I.28 bound its probability by $U^r 8\sqrt{\eta}/(2\sigma)^r$.

For the grid law, the event is defined by

$$\exists(a, z) \forall x' : a \in \text{relint } A \wedge z \in [0, 1]^m \wedge |\psi(a_F)| \leq \eta \wedge [x' \notin X \vee f_\gamma(x') \geq f_\gamma(a, z)],$$

a formula (79) with two blocks of n variables, $O(n)$ atoms, and degree at most three, in which ψ , η , and all noise coordinates but one are real coefficients. Lemmas I.4 and I.5 give the extra term C_m/M with $C_m = 2k(2H^2 + 1)$, independent of ψ and η . $\square$

Corollary I.30 (A uniform margin event). *Let $R \geq 1$ be an integer, $s_* = \binom{k+2}{2}$, and*

$$A_* = 8 \cdot 3^k \max\{1, U/(2\sigma)\}^k (2R + 1)^{s_*}, \quad C_{\text{tail}} = 3^k (2R + 1)^{s_*} C_m.$$

Call a draw bad if, for some face A with $r \geq 1$ free coordinates, some global minimizer of f_γ has core a in the relative interior of A and some nonzero integer polynomial ψ of degree at most two in the free coordinates, with coefficients of absolute value at most R , has $|\psi(a_F)| \leq \eta$. Then $\Pr_{\text{grid}}\{\text{bad}\} \leq A_\sqrt{\eta} + C_{\text{tail}}/M$.*

Proof. There are at most 3^k faces and at most $(2R + 1)^{s_*}$ polynomials per face. Apply Proposition 11.12 to each pair and add. Faces with $r = 0$ contribute nothing, since a nonzero integer constant has absolute value at least $1 > \eta$. The family is used only in this union bound; no algorithm enumerates it. $\square$

I.16 Proofs of Proposition 11.13 and Theorem 11.9

Base choices. Let $\Xi \geq \max\{2, \Xi_0\}$ be as in (84); the same fallback factor Ξ_0 serves all instances below, because they have the same format. Choose, in this order,

$$\begin{aligned} \bar{\Lambda} &= \max\{1, \Lambda\}, & t &= \min\left\{\frac{1}{2}, \frac{\sigma}{16k\Xi}\right\}, & r_f &= \frac{t}{8\bar{\Lambda}}, \\ \Delta, B &\text{ from Lemma I.26,} & s_* &= \binom{k+2}{2}, & R &= 2^{s_*} 4s_* B, \\ U &\geq \max\left\{1, \sum_{i,j} \sup_X \left|\partial_{v_i} \partial_{v_j} f\right|\right\} \text{ by coefficient sums,} & \eta &= \min\left\{\frac{1}{4}, \frac{1}{(16A_*\Xi)^2}\right\}, \\ T &= \text{the least power of two with } T > (s_* + 1)R/\eta, \end{aligned}$$

with A_* from Corollary I.30. Finally let M be the least power of two that satisfies (84) for each of the $2k + 1$ base objectives f and $f \pm tv_i$ and also $(2k + C_{\text{tail}})/M \leq 1/(8\Xi)$. The base objectives $f \pm tv_i$ have the same residual convexity and coordinate bound Λ as f , and input length polynomial in L . Every requirement on M is a lower bound, so one grid serves all of them. All quantities have bit length polynomial in L and precede sampling. On a draw γ , the core evaluator of Theorem 11.2 for the base objective $f \pm tv_i$ approximates the selected core of $f_{\gamma \pm te_i}$; the draw is not resampled.

Base stage. (1) For each i , query the core evaluators of $f_{\gamma \pm te_i}$ with accuracy r_f and apply the rules of Corollary I.27. Let A be the face defined by the fixed coordinates, F its free coordinates, and $r = |F|$. (2) If $r = 0$, accept with $\delta = \frac{1}{2}$ and $\mu = \min\{1, 1/\Delta\}$. Otherwise query the core evaluator of f_γ with accuracy $1/(4T)$, clip the free coordinates of the answer to $[0, 1]$ to obtain $\hat{a}$, and run the test of Lemma 11.11 with these B and T and $s = \binom{r+2}{2}$. If it rejects, reject. If it accepts, put $\xi = sB/T$, $\delta = \min\{\frac{1}{2}, \xi\}$, and $\mu = \min\{1, \xi/\Delta\}$. (3) On acceptance compute Γ from Proposition 11.7(c) for A , δ , and μ . The outcome is a deterministic function of the draw.

Acceptance is sound. By Corollary I.27, $a(\gamma) \in A$. If $r = 0$, then $a(\gamma)$ is the vertex A ; the minors of J_A are rational constants with denominators dividing Δ , so the nonzero ones have absolute value at least $1/\Delta \geq \mu$, and the margin condition on free coordinates is vacuous. If $r \geq 1$ and the test accepts, then $\hat{a} \in [0, 1]^r$ by clipping, and $\|\hat{a} - a_F\|_\infty \leq 1/(4T)$ because clipping to a box containing a_F does not increase any coordinate error. So Lemma 11.11(a) applies: every nonzero integer quadratic of height at most B in the free coordinates has absolute value greater than ξ at a_F . These include v_i and $1 - v_i$ for $i \in F$, so the free coordinates of a lie in $(\xi, 1 - \xi)$, and a lies in the relative interior of A with margin δ . By Lemma I.26, each square minor of J_A is either identically zero on A , hence zero at a , or equal to Δ^{-1} times a nonzero integer quadratic of height at most B , hence of absolute value greater than $\xi/\Delta \geq \mu$ at a . Proposition 11.7(c) therefore gives

$$\text{dist}(z, S(\gamma)) \leq \Gamma(f(a, z) - \min f(a, \cdot))^{1/4} \quad \text{for } z \in [0, 1]^m \text{ with gap at most } 1, \quad (86)$$

since $S(\gamma) = S_a$. No probability was used. The supplied margins δ and μ are rationals with $\langle \delta \rangle + \langle \mu \rangle = O(\log s + \log B + \log T + \log \Delta) = \text{poly}(L)$, so Γ is computed in time $\text{poly}(L)$.

Completion after acceptance. For a query q put $\epsilon = 2^{-q}$, $e = \epsilon/2$, $R_z = \max\{1, \lceil \sqrt{m} \rceil\}$, a rational $G \geq \max\{1, \sup_X \|\nabla_v f\|\}$, and

$$\tau = \frac{e^6}{1024R_z^4\Gamma^4}, \quad \eta_q = \frac{\tau e^2}{8}, \quad d_q = \min\left\{e, \frac{\tau e^2}{16G}\right\}.$$

Query the core evaluator of f_γ with accuracy d_q , set the coordinates fixed by A to their exact values, and clip the others to $[0, 1]$. This gives a rational $b \in A$ with $\|b - a\| \leq d_q$, since $a \in A \cap [0, 1]^k$. Apply (78) to the rational convex cubic $Q_b(z) = f(b, z) + \tau \|z\|^2$ on $[0, 1]^m$ with accuracy η_q , and return $(b, \hat{z})$. Since $|f(b, z) - f(a, z)| \leq G \|b - a\|$ for all z , the function $Q_a(z) = f(a, z) + \tau \|z\|^2$ satisfies $Q_a(\hat{z}) - \min Q_a \leq \eta_q + 2Gd_q \leq \tau e^2/4$. Lemma I.22, with $\varphi = f(a, \cdot)$ on $[0, 1]^m$, the bound (86), and $D_e = 4$, gives $\|\hat{z} - p\| \leq e$. Hence $\|(b, \hat{z}) - (a, p)\| \leq \sqrt{2}e < \epsilon$. The core query has precision $\log(1/d_q) \leq 8q + \text{poly}(L)$, and the convex solve has data of bit length $\text{poly}(L + q)$ and tolerance $\langle \eta_q \rangle \leq 8q + \text{poly}(L)$. This proves Proposition 11.13.

Rejection is rare. Suppose that no coordinate of $a(\gamma)$ in $\{0, 1\}$ is missed by the face rules and that the draw is not bad in the sense of Corollary I.30. Since the rules are sound, A is then the

smallest face containing a , and a lies in its relative interior. If $r = 0$ the stage accepts. Otherwise every nonzero integer quadratic of height at most R in the free coordinates has absolute value greater than η at a_F . Because $s \leq s_*$, we have $R_s = 2^s 4sB \leq R$ and $T\eta > (s_* + 1)R \geq (s + 1)R_s$, so Lemma 11.11(b) shows that the test accepts. By Corollaries 1.27 and 1.30,

$$\Pr\{\text{reject}\} \leq \frac{kt}{\sigma} + \frac{2k}{M} + A_*\sqrt{\eta} + \frac{C_{\text{tail}}}{M} \leq \frac{1}{16\Xi} + \frac{1}{16\Xi} + \frac{1}{8\Xi} = \frac{1}{4\Xi}.$$

The same selector on rejection. On rejection, return the output of Lemma 1.7 for f_γ with precision $q + \lceil \log_2 n \rceil$; its ∞ -accuracy is at most $2^{-q}/n$, so its Euclidean accuracy is at most 2^{-q} . Its singleton formulas (81) define $x^\sharp(\gamma)$ itself: global optimality, the lexicographically least core, and the least residual norm within that core. So the ordinary and fallback branches approximate the same point at every precision, including on draws with several optimal cores. The cost is at most $\Xi P(L + q)$.

Work. Let Ω_0 and $\Omega_{i,\pm}$ be the random factors of Theorem 11.2 for the base objectives f and $f \pm tv_i$. The base stage makes $2k + 1$ core queries of precision $\text{poly}(L)$, one LLL reduction, and polynomial arithmetic for Γ ; a completion makes one core query of precision $\text{poly}(L) + O(q)$ and one convex solve. Every query therefore costs at most $\Omega(\gamma)P(L + q)$ for a fixed polynomial P , where

$$\Omega = \Omega_0 + \sum_{i,\pm} \Omega_{i,\pm} + \Xi \mathbf{1}\{\text{reject}\}.$$

By Theorem 11.2 and the rejection bound, $\mathbb{E}\Omega \leq (2k + 1)k^{O(k)}(1 + \Lambda/\sigma)^k \text{poly}(L) + \frac{1}{4}$. No independence between the factors and the rejection event is used. The numbers U , B , T , and Γ enter only through their bit lengths; the numerical factor remains $(1 + \Lambda/\sigma)^k$. This proves Theorem 11.9. $\square$

Proof of Remark 11.15. Let x_q be the returned point at accuracy $2^{-q-1}/G$ and ℓ_{q+1} the lower bound from Theorem 11.2. Then $\ell_{q+1} \leq f_\gamma^* \leq f_\gamma(x_q) \leq f_\gamma^* + 2^{-q-1} \leq \ell_{q+1} + 2^{-q}$, because X is convex and $\|\nabla f_\gamma\| \leq G$ on X . $\square$

Proof of Remark 11.14. For $f(v, y) = v^2/2 + F(y)$ and every draw γ , the objective separates: the core part $v^2/2 + \gamma v$ has a unique minimizer on $[0, 1]$, and the residual part has the unique minimizer y^* of F . So $x^\sharp(\gamma) = (a(\gamma), y^*)$ on every draw. The residual fibers are convex because F is convex on its box, and $\partial_v^2 f = 1$. A query with $q = 2$ returns a point within $1/4$ of $x^\sharp(\gamma)$, which locates the designated coordinate of y^* to within $1/4$ and decides the source instance, as in Theorem 3.12. An evaluator that is correct on every draw with polynomial expected work would thus give a Las Vegas algorithm with expected polynomial running time: since M is a power of two, a draw uses $\log_2 M$ fair coins, and with $\sigma = 1$ the factor $(1 + \Lambda/\sigma)^k$ equals 2. $\square$

J Quadratic arithmetic contrasts

Convex quadratic programming over a rational polyhedron is the arithmetic baseline for this paper. Its optimal value and a minimum-norm optimizer are rational, and the classical ellipsoid analysis computes them exactly in polynomial time [71]. The main text shows how far this picture breaks for quartic objectives: exact comparison becomes PosSLP-complete, and exact optimizers can have exponential degree or exponentially long rational coordinates. Convex quadratic *constraints* lie between these two regimes. This appendix records what they do to the arithmetic of values, points,

Question	Answer	Result
Degree and height of value and canonical optimizer	one field of degree at most $\mathcal{B}(n, k)$; heights $L^{O(k+1)}$; degree attained by strictly feasible rational instances	Thms. J.9, J.11, J.14
Coefficients in a number field of degree D	height and separation bounds linear in D and in the input height	Thm. J.11
Rational feasible points	always exist, are dense, and are computable when $k \leq 2$; an irrational singleton exists when $k = 3$	Thm. J.17, Prop. J.18
Singleton spectrahedra	one scalar variable: exactly the totally real numbers, minimum size $2 \deg \alpha$; corank one: exactly one real embedding	Thms. J.19, J.20
Expanded output length	short inputs force degree $n^{\Omega(k)}$ and dense coefficient lists of minimal polynomials of the value and of an input coordinate, also for a singleton cut out by ellipsoids	Thms. J.23, J.27, Cor. J.30
One convex quartic row	value degree 3^n	Prop. J.31
Common field from approximations	primitive element and coordinate polynomials with polynomial overhead	Thm. J.33
Optimality and infeasibility certificates	one field, at most k curved reductions; rational infeasibility aggregates of length $L^{O(k+1)}$, exponential in k when necessary	Thms. J.38, J.39
Affine data in a number field	exact optimizer in the input field in polynomial time	Thm. J.41
Nonconvex integer rows	exponentially long integer output already in dimension two	Prop. J.42

Table 7: Quadratic arithmetic contrasts proved in this appendix. Here n is the number of continuous variables, k the dimension of the span of the native constraint Hessians, and L the total binary input length.

and certificates. The results calibrate the quartic theorems: some phenomena of the main text already occur for convex quadratic constraints, and others do not.

The structural parameter throughout is the dimension k of the span of the constraint Hessians. Section J.2 proves that the optimal value and a canonical optimizer lie in one number field whose degree is at most $\mathcal{B}(n, k) = \max_{0 \leq s \leq \min(k, n)} 2^s \binom{n}{s}$, with coefficient heights $L^{O(k+1)}$, and that this degree bound is attained. The remaining sections use these bounds and contrast them with lower bounds in specified output formats. Table 7 lists the results.

J.1 Setting, output contracts, and heights

Definition J.1 (Native PSD quadratic systems). A *native PSD quadratic system* is

$$F = \{x \in \mathbb{R}^n : Ax \leq b, g_i(x) \leq 0 \ (1 \leq i \leq m)\}, \quad g_i(x) = \frac{1}{2}x^\top Q_i x + a_i^\top x + c_i, \quad (87)$$

where Q_i is a symmetric positive semidefinite matrix for every i . An affine equality is written as two opposite rows. The *Hessian span* of the system is $k = \dim \text{span}\{Q_1, \dots, Q_m\}$. An objective $f(x) = \frac{1}{2}x^\top Q_0 x + a_0^\top x + c_0$ with $Q_0 \succeq 0$ may be added; its Hessian is not counted in k .

Unless a field is named, all data are rational and explicitly encoded, every matrix entry counted. The total binary length is $L \geq \max\{2, n, m\}$. Printed input lengths in this appendix refer to this native matrix encoding. The explicit format of Definition 2.2 lists only nonzero coefficients but attaches an exponent vector of $O(n)$ bits to each of the at most $(n+1)(n+2)/2$ monomials of a quadratic. The two lengths are polynomially related, so bounds polynomial in L , such as $L^{O(k+1)}$, hold in either format; the output lower bounds below state the input length in both formats. The

span dimension k is computed by rational rank. It differs from the number of rows, from the rank of $\sum_i Q_i$, and from the dimension of the span of the complete polynomials g_i : the rows $\|x\|^2 + 2x_j - 1$, $1 \leq j \leq n$, have $k = 1$.

The word *native* matters. Positive semidefiniteness is required of the polynomials as written, not of the set F . Squaring the second-order cone constraints $\|(1, 1)\|_2 \leq t$ and $\|(t, t)\|_2 \leq 2$ gives $2 - t^2 \leq 0$ and $2t^2 - 4 \leq 0$; the pair forces $t = \sqrt{2}$, while the first squared row has a negative Hessian. The rational-point theorem of Section J.3 does not apply to such rows.

If $F \neq \emptyset$ and f attains its infimum on F , the optimal set is nonempty, closed and convex. Its unique point of least Euclidean norm in the original coordinates is the *canonical optimizer*, denoted p , and $\theta = f(p)$ is the optimal value. For $f = 0$, p is the minimum-norm feasible point. Other points of a positive-dimensional optimal face can be transcendental and are not described by the results below.

Output contracts. The exact representations are those of Definition 2.15. Two algebraic variants are kept apart: a *coordinatewise* representation gives each coordinate's minimal polynomial and isolating interval, and a *common-field* representation gives one primitive irreducible integer polynomial P of degree d , a rational interval isolating one real root α of P , and rational polynomials $b_1, \dots, b_n$ of degree less than d with $x_j = b_j(\alpha)$. Coordinatewise output does not bound the degree of the field generated by all coordinates: square roots of n distinct primes have degree two each and generate a field of degree 2^n . The common-field format lets a verifier evaluate any rational polynomial at the point by univariate arithmetic modulo P .

Five quantities are kept separate: the bit-operation time of an algorithm; the expanded length of an output in a named format; algebraic degrees; the data and verification cost of a certificate; and the accuracy 2^{-q} requested from an approximation oracle. Two degrees must be distinguished. The degree of one scalar bounds from below the length of a dense list of that scalar's minimal polynomial. The degree of the field generated by all coordinates bounds from below the length of a common-field representation, but not of a coordinatewise one, whose separate minimal polynomials can be short when the joint degree is large. Neither degree bounds implicit descriptions. A circuit over the rationals computes only rational numbers, so for irrational coordinates the compact alternatives are implicit systems, towers of radicals, or circuits with root-selection gates. Lower bounds in this appendix always name the format that they constrain.

Heights. For a number field M and a place u of M , let $|\cdot|_u$ be the absolute value extending the ordinary or ℓ -adic absolute value of $\mathbb{Q}$, and let $n_u = [M_u : \mathbb{Q}_u]$. Archimedean places carry the ordinary modulus, with $n_u = 2$ at complex places, and $\sum_{u|\infty} n_u = [M : \mathbb{Q}]$. The product formula $\prod_u |x|_u^{n_u} = 1$ holds for $x \in M^\times$; equivalently $\sum_u n_u \log |x|_u = 0$. The absolute logarithmic Weil height of $x \in M$ and the affine and projective heights of a vector $y \in M^r$ are

$$\text{ht}(x) = \frac{1}{[M : \mathbb{Q}]} \sum_u n_u \log \max(1, |x|_u), \quad \text{ht}_{\text{aff}}(y) = \frac{1}{[M : \mathbb{Q}]} \sum_u n_u \log \max(1, \max_j |y_j|_u),$$

and, for $y \neq 0$, $\text{ht}_{\text{proj}}(y) = [M : \mathbb{Q}]^{-1} \sum_u n_u \log \max_j |y_j|_u$. These quantities do not depend on the field M containing the entries [18].

Lemma J.2 (Height calculus). *Let M be a number field and $x, x_1, \dots, x_r \in M$.*

(a) $\text{ht}(x_1 x_2) \leq \text{ht}(x_1) + \text{ht}(x_2)$, $\text{ht}(x^{-1}) = \text{ht}(x)$ for $x \neq 0$, $\text{ht}(x_1 + \dots + x_r) \leq \sum_j \text{ht}(x_j) + \log r$, and $\text{ht}_{\text{proj}}(y) \leq \text{ht}_{\text{aff}}(y) \leq \sum_j \text{ht}(y_j)$. If $y_i, y_j \neq 0$ are entries of y , then $\text{ht}(y_i/y_j) \leq \text{ht}_{\text{proj}}(y)$.

(b) For a rational number a/b in lowest terms, $\text{ht}(a/b) = \log \max(|a|, |b|)$.

(c) If $M \subset \mathbb{R}$ through a fixed real embedding and $x \neq 0$, then $|\log |x|| \leq [M : \mathbb{Q}] \text{ht}(x)$.

(d) For an $r \times r$ matrix G with entries in M , $\text{ht}(\det G) \leq \sum_{i,j} \text{ht}(G_{ij}) + \log r!$.

Proof. For (a), use

$$\begin{aligned} \max(1, |x_1 x_2|_u) &\leq \max(1, |x_1|_u) \max(1, |x_2|_u), \\ \sum_u n_u \log \max(1, |x|_u^{-1}) &= \sum_u n_u \log \max(1, |x|_u), \end{aligned}$$

the second by the product formula. At an archimedean place $|\sum_j x_j|_u \leq r \max_j |x_j|_u$, at the other places the maximum suffices, and the archimedean weights sum to $[M : \mathbb{Q}]$. The two vector inequalities are immediate from the definitions. For the ratio, $\text{ht}(y_i/y_j)$ is the projective height of (y_j, y_i) , and the maximum over two entries is at most the maximum over all entries. For (b), a prime ℓ dividing b contributes $v_\ell(b) \log \ell$, the other primes contribute nothing, and these contributions sum to $\log |b|$; the archimedean term adds $\log \max(1, |a/b|)$. For (c), the real place contributes $n_u \log \max(1, |x|) \leq [M : \mathbb{Q}] \text{ht}(x)$; apply this also to x^{-1} . For (d), expand the determinant. At an archimedean place, $|\det G|_u \leq r! \prod_{i,j} \max(1, |G_{ij}|_u)$; at the other places the factor $r!$ is absent. Summing with the weights gives the claim. $\square$

We also use the following elementary passage from an annihilating polynomial to a minimal polynomial. If ϑ is a root of a nonzero $R \in \mathbb{Z}[w]$ of degree e with all coefficients bounded by $\exp(W)$, $W \geq 0$, then the primitive minimal polynomial $m_\vartheta \in \mathbb{Z}[w]$ of ϑ divides R by Gauss's lemma. Its leading coefficient divides that of R , and its roots are roots of R , hence bounded by $1 + \exp(W)$ by Cauchy's bound. Expanding m_ϑ over its roots gives

$$\log \|m_\vartheta\|_\infty \leq W + \deg m_\vartheta (\log 2 + \log(1 + \exp W)) \leq (\deg m_\vartheta + 1)(W + 2). \quad (88)$$

Finally, for a polynomial $\sum_{i=0}^e \pi_i w^i$ with $\pi_e \neq 0$, every complex root z satisfies Fujiwara's bound $|z| \leq 2 \max_{i < e} |\pi_i/\pi_e|^{1/(e-i)}$: if $|z|$ were larger, the terms $|\pi_i z^i|$ with $i < e$ would sum to less than $|\pi_e| |z|^e \sum_{j \geq 1} 2^{-j}$. The same bound holds at every archimedean place, and at a nonarchimedean place the ultrametric inequality gives $|z|_u \leq \max(1, \max_{i < e} |\pi_i/\pi_e|_u)$.

J.2 Degree, height, and separation at fixed Hessian span

This subsection proves the bounds that the rest of the appendix uses. They explain why convex quadratic constraints are arithmetically mild when their Hessians span few directions, even with arbitrarily many variables and rows, no bounds, and no constraint qualification. The proof restricts to the active rows at the canonical optimizer, compresses the multipliers to at most k , and passes the resulting nonsingular polynomial system through two ordered limits. Two elimination arguments are needed: a root count gives the sharp degree, and a finite multiplication-matrix construction gives the height.

Setting over a number field. Let $K \subset \mathbb{R}$ be a number field of degree D , with the embedding fixed; $K = \mathbb{Q}$ is the main case. In this subsection the data of (87) and of the objective f lie in K , and $Q_0, Q_1, \dots, Q_m \succeq 0$ in the fixed embedding. The span dimension of the Q_i over K equals the real span dimension k , because the rank of the matrix of Hessian entries does not depend on the field. Assume that $F \neq \emptyset$ and that f attains its infimum on F , and let p and θ be the canonical optimizer and the optimal value. Let $S \geq 2$ bound n , the number of rows, and the number of listed input scalars, and let $B \geq 1$ bound the absolute heights of the input scalars. For $K = \mathbb{Q}$ one may take $S = L$ and $B = L$.

Lemma J.3 (Active restriction). *Let I_a and I_q be the affine rows and the quadratic rows that are active at p . Choose $\mathcal{J} \subseteq I_q$ such that $\{Q_j : j \in \mathcal{J}\}$ is a basis of $\text{span}\{Q_i : i \in I_q\}$, and write $Q_i = \sum_{j \in \mathcal{J}} \gamma_{ij} Q_j$ with $\gamma_{ij} \in K$. Let H be the affine space defined by the active affine rows taken as equalities and by the equations*

$$g_i(x) - \sum_{j \in \mathcal{J}} \gamma_{ij} g_j(x) = 0 \quad (i \in I_q), \quad (89)$$

whose left sides are affine. Let $x = x_0 + Vu$, $u \in \mathbb{R}^d$, be a parametrization of H with $x_0 \in K^n$ and $V \in K^{n \times d}$ of full column rank, and put $r_i(u) = g_i(x_0 + Vu)$ for $i \in I_q$, $\tilde{f}(u) = f(x_0 + Vu)$, and $N(u) = \|x_0 + Vu\|^2$. Then the following hold.

- (a) $p = x_0 + Vu^*$ for a unique u^* , and $r_i(u^*) = 0$ for $i \in I_q$.
- (b) The polynomials r_i , $i \in I_q$, span a space of dimension at most k over K .
- (c) On $C = \{u \in \mathbb{R}^d : r_i(u) \leq 0 \ (i \in I_q)\}$ one has $\tilde{f} \geq \theta$, and u^* is the unique minimizer of N on $\{u \in C : \tilde{f}(u) = \theta\}$.
- (d) There is an absolute constant c_0 such that every entry of x_0 and V , and every coefficient of r_i , $\tilde{f}$ and N , has height at most $S^{c_0}(B+1)$. These bounds do not depend on which rows are active.

Proof. Every active row vanishes at p , so $p \in H$ and (a) holds. Each difference in (89) is affine and vanishes on H , so it vanishes identically after substitution. Hence $r_i = \sum_j \gamma_{ij} r_j$ as polynomials, proving (b).

For (c), let $F_2 = \{x \in H : g_i(x) \leq 0 \ (i \in I_q)\}$, the image of C . Let $y \in F_2$ and $z_t = p + t(y - p)$ for $0 < t \leq 1$. The active affine rows hold on H , the rows in I_q hold at z_t by convexity, and every inactive row is strictly slack at p and therefore at z_t for small t . Thus $z_t \in F$ for small t . If $f(y) < \theta$, convexity gives $f(z_t) < \theta$, a contradiction. If $f(y) = \theta$ and $y \neq p$ with $\|y\| \leq \|p\|$, then z_t is optimal for small t , and strict convexity of the squared norm gives $\|z_t\| < \|p\|$, again a contradiction.

For (d), the γ_{ij} solve a K -linear system whose matrix consists of Hessian entries. Cramer's rule and Lemma J.2(d) bound their heights by $2(S^2B + S \log S)$. The coefficients of (89) are sums of at most $S+1$ products of these numbers with input scalars. The vectors x_0 and the columns of V are Cramer quotients of minors of order at most $n \leq S$ of the equation matrix of H , after choosing the free coordinates among the original ones. The restricted coefficients are sums of at most S^2 products of three such numbers. Each of these finitely many steps multiplies the preceding height bound by a fixed polynomial in S and adds a term of order $S \log S$, by Lemma J.2(a),(d). $\square$

If $d = 0$, then $p = x_0 \in K^n$, and all statements below are immediate. Assume $d \geq 1$ from now on.

Lemma J.4 (Ordered regularization). *For $\varepsilon > 0$, the function $f_\varepsilon = \tilde{f} + \varepsilon N$ has a unique minimizer u_ε on C , and $x_0 + Vu_\varepsilon \rightarrow p$ as $\varepsilon \downarrow 0$. For fixed $\varepsilon > 0$ and $\delta > 0$, f_ε has a unique minimizer $u_{\varepsilon, \delta}$ on $C_\delta = \{u : r_i(u) \leq \delta \ (i \in I_q)\}$, and $u_{\varepsilon, \delta} \rightarrow u_\varepsilon$ as $\delta \downarrow 0$. The point u^* satisfies every inequality of C_δ strictly.*

Proof. The Hessian of f_ε is $\nabla^2 \tilde{f} + 2\varepsilon V^T V$, which is positive definite because V has full column rank. Hence f_ε is strongly convex and has unique minimizers on the nonempty closed convex sets C and C_δ . Comparison with u^* and Lemma J.3(c) give

$$\theta \leq \tilde{f}(u_\varepsilon), \quad \tilde{f}(u_\varepsilon) + \varepsilon N(u_\varepsilon) \leq \theta + \varepsilon \|p\|^2,$$

so $N(u_\varepsilon) \leq \|p\|^2$ and $\tilde{f}(u_\varepsilon) \rightarrow \theta$. Every accumulation point $\bar{u}$ of u_ε lies in C , has $\tilde{f}(\bar{u}) = \theta$ and $N(\bar{u}) \leq \|p\|^2$, so $\bar{u} = u^*$ by Lemma J.3(c). For fixed ε , the points $u_{\varepsilon,\delta}$ lie in the compact set $\{u : f_\varepsilon(u) \leq f_\varepsilon(u_\varepsilon)\}$, since $u_\varepsilon \in C_\delta$. An accumulation point as $\delta \downarrow 0$ lies in C and does not exceed the minimum value on C , so it equals u_ε . The last assertion holds because $r_i(u^*) = 0 < \delta$. $\square$

Only the ordered limits, first $\delta \downarrow 0$ and then $\varepsilon \downarrow 0$, are used below. No convergence along a joint schedule of the two parameters is asserted.

For a set $J \subseteq I_q$ with $|J| = s$, consider the square system in $(u, \lambda) \in \mathbb{R}^d \times \mathbb{R}^s$

$$\Phi_J(u, \lambda; \varepsilon, \delta) = \left(\nabla f_\varepsilon(u) + \sum_{j \in J} \lambda_j \nabla r_j(u), (r_j(u) - \delta)_{j \in J} \right) = 0. \quad (90)$$

Write $\tilde{Q}_i$, $\tilde{a}_i$ and $\tilde{c}_i$ for the Hessian, linear coefficient vector and constant of r_i for $i \in I_q$, and of $\tilde{f}$ for $i = 0$.

Lemma J.5 (Small regular support). *For $\varepsilon, \delta > 0$ there is $J \subseteq I_q$ with $s = |J| \leq \min(k, d)$ and $\lambda \in \mathbb{R}_{>0}^J$ such that $(u_{\varepsilon,\delta}, \lambda)$ solves (90) and the Jacobian of Φ_J with respect to (u, λ) is nonsingular there. For every sequence $\varepsilon_\ell \downarrow 0$ there are sequences $\delta_{\ell,\nu} \downarrow 0$ and one set J such that this holds for all pairs $(\varepsilon_\ell, \delta_{\ell,\nu})$, after passing to a subsequence in ℓ .*

Proof. By Lemma J.4, u^* is a strict feasible point of the convex problem $\min\{f_\varepsilon(u) : u \in C_\delta\}$, so the convex Karush–Kuhn–Tucker conditions hold at its minimizer: $-\nabla f_\varepsilon$ is a nonnegative combination of the gradients of the rows with $r_i = \delta$ [104]. Choose such a representation with minimal support J . Its gradients are linearly independent: a dependence $\sum_j \eta_j \nabla r_j = 0$ with some $\eta_j > 0$ lets us replace λ by $\lambda - \tau \eta$ for the largest τ keeping $\lambda \geq 0$, which removes an index. By Lemma J.3(b), the gradients of the r_i at one point span at most k dimensions, so $s \leq \min(k, d)$.

The Jacobian of (90) is $\begin{pmatrix} M & G \\ G^\top & 0 \end{pmatrix}$ with $M = \tilde{Q}_0 + 2\varepsilon V^\top V + \sum_{j \in J} \lambda_j \tilde{Q}_j \succ 0$ and $G = (\nabla r_j(u_{\varepsilon,\delta}))_{j \in J}$ of full column rank. If $Mx + Gy = 0$ and $G^\top x = 0$, then $x^\top Mx = 0$, so $x = 0$ and then $y = 0$. For the last assertion, there are finitely many subsets J : choose one that occurs along a sequence $\delta_{\ell,\nu} \downarrow 0$ for each ℓ , and then pass to a subsequence of ℓ on which this choice is constant. $\square$

The support is fixed before any output is chosen. The multipliers may be unbounded along the sequences; no bound on them is used.

The degree bound. The next lemma turns a uniform count of nonsingular roots into one polynomial identity that holds at every real nonsingular root, including roots at exceptional parameter values. Counting the roots of individual rational specializations would not suffice: rational numbers can converge to a transcendental number.

Lemma J.6 (One identity at all regular roots). *Let $K \subset \mathbb{R}$ be a field, $t = (t_1, \dots, t_r)$ and $y = (y_1, \dots, y_N)$ indeterminates, and $\Phi_1, \dots, \Phi_N, \Gamma \in K[t, y]$. Let $\mathbb{K} = K(t)$ and suppose that $\Phi(t, \cdot)$ has at most β nonsingular common zeros in $\overline{\mathbb{K}}^N$. If some real (t°, y°) is a common zero with $\det \partial_y \Phi(t^\circ, y^\circ) \neq 0$, then there is a nonzero $R \in K[t, w]$ with $\deg_w R \leq \beta$ such that $R(t, \Gamma(t, y)) = 0$ at every real common zero (t, y) of Φ with $\det \partial_y \Phi(t, y) \neq 0$.*

Proof. Let $\Delta = \det \partial_y \Phi$ and $E = \mathbb{K}[y, \Delta^{-1}]/(\Phi_1, \dots, \Phi_N)$. After extension of scalars to $\overline{\mathbb{K}}$, the maximal ideals of E correspond to the common zeros z with $\Delta(z) \neq 0$. At such a zero the differentials of the Φ_i span the cotangent space, so the Φ_i generate the maximal ideal of the local ring of $\overline{\mathbb{K}}^N$ at z , and the local ring of $E \otimes \overline{\mathbb{K}}$ at z is the field $\overline{\mathbb{K}}$. Thus every point of the affine scheme defined by

$E \otimes \overline{\mathbb{K}}$ is an isolated reduced point. A Noetherian ring with this property has finitely many points and is the product of its residue fields, so $\dim_{\mathbb{K}} E$ equals the number of nonsingular common zeros, which is at most β .

The algebra E is not zero. Otherwise $\Delta^e \in (\Phi)$ in $\mathbb{K}[y]$ for some e , and clearing denominators gives $c(t)\Delta^e = \sum_i h_i \Phi_i$ with $0 \neq c \in K[t]$ and $h_i \in K[t, y]$. By the implicit function theorem there is a real analytic branch $y(t)$ of nonsingular zeros on a neighborhood of t° , so c vanishes on an open set, which is impossible.

Multiplication by the class of Γ on the $\mathbb{K}$ -vector space E has a monic characteristic polynomial $\chi \in \mathbb{K}[w]$ of degree $\dim E \leq \beta$, and $\chi(\Gamma) = 0$ in E by the Cayley–Hamilton theorem. Clearing denominators gives a nonzero $R \in K[t, w]$ with $\deg_w R = \dim E$ and an identity $c_2(t)\Delta^{e'} R(t, \Gamma) = \sum_i \hat{h}_i \Phi_i$ in $K[t, y]$, with $c_2 \neq 0$. Hence $R(t, \Gamma(t, y)) = 0$ at every real nonsingular common zero with $c_2(t) \neq 0$. At a nonsingular zero with $c_2(t) = 0$, the implicit function theorem gives a branch of nonsingular zeros over a neighborhood of t , on which $c_2 \neq 0$ on a dense set. Continuity extends the identity to the given zero. $\square$

Lemma J.7 (Ordered limits). *Let $R \in K[\varepsilon, \delta, w]$ be nonzero. Suppose that $R(\varepsilon_\ell, \delta_{\ell, \nu}, w_{\ell, \nu}) = 0$ for nonzero reals $\varepsilon_\ell \rightarrow 0$ and $\delta_{\ell, \nu} \rightarrow 0$, where $w_{\ell, \nu} \rightarrow w_\ell$ as $\nu \rightarrow \infty$ and $w_\ell \rightarrow w^*$ as $\ell \rightarrow \infty$, all limits finite. Then there is a nonzero $P \in K[w]$ with $\deg P \leq \deg_w R$ and $P(w^*) = 0$.*

Proof. Write $R = \delta^a S(\varepsilon, w) + \delta^{a+1} R_1$ with $S \neq 0$, where a is the least exponent of δ with a nonzero coefficient. Dividing by $\delta_{\ell, \nu}^a$ and letting $\nu \rightarrow \infty$ gives $S(\varepsilon_\ell, w_\ell) = 0$ for every ℓ ; this remains true if $S(\varepsilon_\ell, \cdot)$ is the zero polynomial. Write $S = \varepsilon^b P(w) + \varepsilon^{b+1} S_1$ with $P \neq 0$, divide by ε_ℓ^b , and let $\ell \rightarrow \infty$. Then $P(w^*) = 0$. A nonzero constant cannot vanish at w^* , so P is not constant. $\square$

We use the following classical root count.

Theorem J.8 (Multihomogeneous Bézout bound [29, 90]). *Let $\mathbb{F}$ be an algebraically closed field of characteristic zero and $\Phi_1, \dots, \Phi_N \in \mathbb{F}[Y, Z]$, where Y and Z are groups of N_1 and N_2 variables with $N_1 + N_2 = N$. If $\deg_Y \Phi_i \leq e_i$ and $\deg_Z \Phi_i \leq e'_i$, then the number of isolated common zeros of Φ in $\mathbb{F}^N$, and in particular the number of nonsingular ones, is at most the coefficient of $U^{N_1} T^{N_2}$ in $\prod_{i=1}^N (e_i U + e'_i T)$.*

The isolated-root upper bound is stated in [29, Section 5, Theorem 5.1]. Additional positive-dimensional components of the zero set are allowed in this statement. For an algebraically closed field of characteristic zero and countable transcendence degree, such as an algebraic closure of $K(t)$, the statement follows from the case $\mathbb{F} = \mathbb{C}$ by embedding the field in $\mathbb{C}$.

Theorem J.9 (Joint degree). *In the setting of this subsection,*

$$[K(p_1, \dots, p_n) : K] \leq \mathcal{B}(n, k) = \max_{0 \leq s \leq \min(k, n)} 2^s \binom{n}{s},$$

and $\theta \in K(p_1, \dots, p_n)$. For $K = \mathbb{Q}$, the value and the canonical optimizer lie in one number field of degree at most $\mathcal{B}(n, k) \leq \min\{3^n, (2n)^k\}$.

Proof. Fix the nested sequences and the support J of Lemma J.5. In the system (90), each of the d stationarity equations has degree at most one in u and at most one in λ , and each of the s active equations has degree at most two in u and zero in λ . By Theorem J.8, the number of nonsingular zeros over an algebraic closure of $K(\varepsilon, \delta)$ is at most

$$[U^d T^s](U + T)^d (2U)^s = 2^s \binom{d}{s}.$$

For $c \in K^n$, apply Lemma J.6 with parameters $t = (\varepsilon, \delta)$, unknowns $y = (u, \lambda)$, and output $\Gamma = c^\top(x_0 + Vu)$. Lemmas J.4 and J.5 provide the nonsingular real zeros and the ordered limits, and Lemma J.7 gives a nonzero $P_c \in K[w]$ of degree at most $2^s \binom{d}{s}$ with $P_c(c^\top p) = 0$. Every coordinate of p is therefore algebraic over K . The extension $K(p)/K$ is finite and separable, so by the primitive element theorem some $c \in K^n$ gives $K(p) = K(c^\top p)$. Hence $[K(p) : K] \leq 2^s \binom{d}{s} \leq \mathcal{B}(n, k)$, using $s \leq \min(k, d)$ and $d \leq n$. The value $\theta = f(p)$ lies in $K(p)$ because f has coefficients in K . Finally $2^s \binom{n}{s} \leq (2n)^s$, and the sum over all s is 3^n . $\square$

The same support J serves all linear forms $c^\top p$. This is why the bound concerns the joint field and not only individual coordinates; separate coordinate bounds would not bound the degree of their compositum. The maximum in $\mathcal{B}(n, k)$ is needed: for $n = k = 3$ the terms are 1, 6, 12, 8. The count $2^s \binom{d}{s}$ is the classical generic degree of quadratically constrained critical points [94]; what the proof adds is its validity at the canonical optimizer of every convex instance, through the span compression and the ordered limits, without genericity or a constraint qualification.

The height bound. For a polynomial Ψ with coefficients in K and a place v of K , let $\|\Psi\|_v$ be the sum of the v -absolute values of its coefficients if v is archimedean and their maximum otherwise. Both norms are submultiplicative, and the second also satisfies the ultrametric inequality.

Lemma J.10 (Ordered elimination with local norms). *Let $G_1, \dots, G_s, A, \Gamma \in K[\varepsilon, \delta, \lambda_1, \dots, \lambda_s]$ have total degree at most $a \geq 1$ in λ . For each place v let $C_v \geq 1$ bound $\|G_i\|_v$, $\|A\|_v$ and $\|\Gamma\|_v$, with $C_v = 1$ for all but finitely many v , and put*

$$E = \frac{1}{D} \sum_v n_v \log C_v, \quad \mathfrak{z} = (a+1)^s, \quad T = a(s+1).$$

Suppose that for nonzero reals $\varepsilon_\ell \rightarrow 0$ and $\delta_{\ell, \nu} \rightarrow 0$ there are real roots $\lambda_{\ell, \nu}$ of $G(\varepsilon_\ell, \delta_{\ell, \nu}, \cdot) = 0$ with nonsingular Jacobian in λ and $A \neq 0$, such that $w_{\ell, \nu} = \Gamma/A$ at these roots converges to w_ℓ as $\nu \rightarrow \infty$, and $w_\ell \rightarrow w^$, all limits finite. Then there is a nonzero $P \in K[w]$ with $\deg P \leq \mathfrak{z}$, $P(w^*) = 0$, and*

$$\text{ht}_{\text{aff}}(\text{coefficients of } P) \leq W_0 := \mathfrak{z}(T+1)E + \log \mathfrak{z}! + \mathfrak{z} \log 3.$$

Proof. Introduce a deformation parameter β and the polynomials $\Phi_i = G_i + \beta \lambda_i^{a+1}$. Over $K(\varepsilon, \delta, \beta)$, for a monomial order that compares total degree in λ first, the leading monomials λ_i^{a+1} are pairwise coprime, so the Φ_i form a Gröbner basis and the monomials λ^α with all $\alpha_i \leq a$ form a basis $\mathcal{M}$ of the quotient, with $|\mathcal{M}| = \mathfrak{z}$. The same holds after specializing ε, δ to real numbers and β to a nonzero real number, because the leading coefficients are β .

Let M_A and M_Γ be the matrices of multiplication by A and Γ on this basis. A basis monomial times A or Γ has degree at most T in λ , and each replacement of λ_i^{a+1} by $-G_i/\beta$ lowers the degree by at least one. Hence at most T replacements occur along any branch of the reduction, the matrices $\beta^T M_A$ and $\beta^T M_\Gamma$ have polynomial entries in $(\varepsilon, \delta, \beta)$, and each entry has v -norm at most C_v^{T+1} : at an archimedean place each replacement multiplies the coefficient sum by at most C_v , and at the other places the maximum is multiplicative. Form

$$\mathcal{H} = \det(w\beta^T M_A - \beta^T M_\Gamma - \zeta\beta^T I_{\mathfrak{z}}).$$

Its coefficient of $\zeta^{\mathfrak{z}}$ is $(-\beta^T)^{\mathfrak{z}} \neq 0$, its degree in w is at most $\mathfrak{z}$, and expanding the determinant gives $\|\mathcal{H}\|_v \leq \mathfrak{z}! \mathfrak{z}^{\mathfrak{z}} C_v^{\mathfrak{z}(T+1)}$ at archimedean places and $\|\mathcal{H}\|_v \leq C_v^{\mathfrak{z}(T+1)}$ at the others.

Let $\mathcal{H}_\kappa$ be the coefficient of the least power ζ^κ occurring in $\mathcal{H}$. Fix one selected pair $(\varepsilon_\ell, \delta_{\ell, \nu})$ and its root $\lambda_{\ell, \nu}$. The real implicit function theorem continues the root to a real root $\lambda(\beta)$ of Φ for small

β , with $A \neq 0$. For small $\beta \neq 0$, evaluation at $\lambda(\beta)$ is a ring homomorphism on the quotient, so the vector of basis monomials evaluated at $\lambda(\beta)$ is a nonzero common left eigenvector of the specialized multiplication matrices, with eigenvalues $A(\lambda(\beta))$ and $\Gamma(\lambda(\beta))$. Hence the specialized determinant equals $\beta^T(wA - \Gamma - \zeta) \Psi(w, \zeta)$ for a polynomial Ψ . It is divisible by ζ^κ , and $wA - \Gamma - \zeta$ is coprime to ζ because $A \neq 0$, so ζ^κ divides Ψ . Taking the coefficient of ζ^κ shows $\mathcal{H}_\kappa(\varepsilon_\ell, \delta_{\ell, \nu}, \beta, \Gamma/A) = 0$ along the branch, even if the specialized coefficient vanishes identically. This step removes factors coming from roots at which $A = \Gamma = 0$.

Let $\mathcal{R}(\varepsilon, \delta, w)$ be the coefficient of the least power of β in $\mathcal{H}_\kappa$. Dividing by that power and letting $\beta \rightarrow 0$ along the branch gives $\mathcal{R}(\varepsilon_\ell, \delta_{\ell, \nu}, w_{\ell, \nu}) = 0$ for all selected pairs. Lemma J.7 now gives P with $P(w^*) = 0$. Each extraction selects a subvector of coefficients, so no local norm increases, and

$$\text{ht}_{\text{aff}}(\text{coefficients of } P) \leq \frac{1}{D} \sum_v n_v \log \max(1, \|\mathcal{H}\|_v) \leq W_0,$$

because the archimedean weights sum to D . □

Deforming a polynomial system to obtain a finite quotient, and reading the values of a function from multiplication matrices, are classical elimination devices [23, 48]. The lemma records the explicit bound, place by place, that the following theorem needs.

Theorem J.11 (Height and separation). *There is an absolute constant c such that, in the setting of this subsection, with $W = (B + 1)S^{c(k+1)}$:*

- (i) $\text{ht}(\theta) \leq W$ and $\text{ht}(p_j) \leq W$ for every j ; more generally $\text{ht}(c^\top p) \leq W$ for every $c \in K^n$ whose entries have height at most B ;
- (ii) if $\theta \neq 0$, then $|\log |\theta|| \leq DW$; the same holds for every nonzero p_j ;
- (iii) for $K = \mathbb{Q}$, the primitive minimal polynomials of θ , of each p_j , and of each $c^\top p$ as in (i) have degree at most $\mathcal{B}(n, k)$ and coefficient bit length at most $L^{c(k+1)}$, and every nonzero one of these numbers lies in $[2^{-L^{c(k+1)}}, 2^{L^{c(k+1)}}]$ in absolute value.

Proof. Assume $d \geq 1$. Fix J and the sequences of Lemma J.5, and write $M(\varepsilon, \lambda) = \tilde{Q}_0 + 2\varepsilon V^\top V + \sum_{j \in J} \lambda_j \tilde{Q}_j$ and $\mu(\varepsilon, \lambda) = \tilde{a}_0 + 2\varepsilon V^\top x_0 + \sum_{j \in J} \lambda_j \tilde{a}_j$, so that the stationarity equations in (90) read $Mu + \mu = 0$. Let $\Delta = \det M$ and $\rho = -\text{adj}(M)\mu$. At the selected roots $M \succ 0$ and $u = \rho/\Delta$. Put

$$G_j = \frac{1}{2}\rho^\top \tilde{Q}_j \rho + \Delta \tilde{a}_j^\top \rho + (\tilde{c}_j - \delta)\Delta^2 \quad (j \in J), \quad A = \Delta^2,$$

and, for the outputs, $\Gamma_f = \frac{1}{2}\rho^\top \tilde{Q}_0 \rho + \Delta \tilde{a}_0^\top \rho + \tilde{c}_0 \Delta^2$ and $\Gamma_c = \Delta(c^\top x_0 \Delta + c^\top V \rho)$ for $c \in K^n$. Then $G_j = \Delta^2(r_j(u) - \delta)$, $\Gamma_f/A = \tilde{f}(u)$ and $\Gamma_c/A = c^\top x$. Along the root branch $u(\lambda) = -M^{-1}\mu$, one has $\partial u / \partial \lambda_i = -M^{-1} \nabla r_i(u)$, and because $r_j(u) = \delta$ at the root,

$$\frac{\partial G_j}{\partial \lambda_i} = -\Delta^2 \nabla r_j(u)^\top M^{-1} \nabla r_i(u),$$

which is negative definite as a matrix in (j, i) . So the selected multiplier vectors are nonsingular roots of G with $A \neq 0$. The ordered limits of Lemma J.4 give $\Gamma_f/A \rightarrow \theta$ and $\Gamma_c/A \rightarrow c^\top p$. All these polynomials have degree at most $a = 2d$ in λ .

Let $\mathcal{S}$ be the list of scalars consisting of the entries of $\tilde{Q}_i, \tilde{a}_i, \tilde{c}_i$ for $i \in \{0\} \cup J$, of $2V^\top V$, $2V^\top x_0$, x_0 , V , $c^\top x_0$ and $c^\top V$, and the constants $1, 2, \frac{1}{2}$. For a place v let $U_v = \max(1, \max_{\sigma \in \mathcal{S}} |\sigma|_v)$. At an archimedean place each entry of M and μ has norm at most $(s + 2)U_v$, so Δ and every entry of ρ have norm at most $\Pi_v = d!((s + 2)U_v)^d$, and every G_j , A , Γ_f , Γ_c has norm at most

$(d+1)^2 U_v^2 \Pi_v^2$. At a nonarchimedean place the same polynomials have norm at most U_v^{2d+2} . Hence, with $H_S = \sum_{\sigma \in S} \text{ht}(\sigma)$,

$$E \leq (2d+2)H_S + 2\log((d+1)!) + 2d\log(s+2).$$

By Lemma J.3(d), $|S| \leq S^{c_0+3}$ and $H_S \leq S^{2c_0+3}(B+1)$. Lemma J.10, with $\mathfrak{z} = (2d+1)^s$ and $T = 2d(s+1)$, gives annihilators P of θ and of each $c^\top p$ with

$$\text{ht}_{\text{aff}}(\text{coefficients of } P) \leq (2d+1)^s [(2d(s+1)+1)E + s\log(2d+1) + \log 3].$$

Since $d \leq S$ and $s \leq k$, this is at most $(B+1)S^{c(k+1)}$ for an absolute c .

Let ϑ be a root of $P = \sum_i \pi_i w^i$ of degree e . By Fujiwara's bound at archimedean places and the ultrametric bound at the others, $\log \max(1, |\vartheta|_u) \leq \log 2[u \mid \infty] + \log \max_i |\pi_i|_u - \log |\pi_e|_u$ at every place u of a field containing ϑ . Summing with the weights and using the product formula for π_e gives $\text{ht}(\vartheta) \leq \text{ht}_{\text{proj}}(P) + \log 2$, which proves (i) after enlarging c . For (ii), remove a power of w from P so that its constant coefficient π'_0 is nonzero. Each ratio π'_i/π'_0 lies in K and has height at most $\text{ht}_{\text{proj}}(P)$ by Lemma J.2(a), hence absolute value at most $\exp(D \text{ht}_{\text{proj}}(P))$ by Lemma J.2(c). Fujiwara's bound applied to P and to its reversal gives $|\log |\vartheta|| \leq D \text{ht}_{\text{proj}}(P) + \log 2$. For $K = \mathbb{Q}$, clearing denominators turns P into a primitive integer polynomial whose largest coefficient has logarithm $\text{ht}_{\text{proj}}(P)$. Then Theorem J.9 and (88) give (iii), with $S, B \leq L$ and $\mathcal{B}(n, k) \leq (2L)^k$. $\square$

The degree factor D enters (ii) linearly. The elimination is carried out over K with local norms, so the field is never encoded by a quantified primitive element, and neither D nor B is raised to a power depending on k . Convexity is used only at the fixed real embedding; the conjugate problems need not be convex. Theorem J.9 applies over K as well and bounds $[K(\theta) : K]$ by $\mathcal{B}(n, k)$, so $[\mathbb{Q}(\theta) : \mathbb{Q}] \leq D \mathcal{B}(n, k)$.

Corollary J.12 (Radius and value gap). *There is an absolute constant c with the following properties for every rational native PSD system with Hessian span k and input length L .*

- (a) *If $F \neq \emptyset$, its minimum-norm point lies in $[-R_F, R_F]^n$, where $R_F = 2^{\lceil L^{c(k+1)} \rceil}$.*
- (b) *Let a rational box $[-R, R]^n$, a rational affine function ϕ , and rational affine rows be added, and let ϕ^* be the minimum of ϕ over the resulting compact set, assumed nonempty. If $\phi^* \neq 0$, then $|\phi^*| \geq 2^{-(L+\log_2 R)L^{c(k+1)}}$, where L now counts the added rows as well.*

Proof. For (a), apply Theorem J.11(ii) with $f = 0$. For (b), the minimum is attained by compactness, the added rows are affine, and Theorem J.11(ii) with $D = 1$ applies with $B \leq L + \log_2 R + 2$. $\square$

The bound in (b) is linear in $\log R$. Boxes of radius R_F therefore add only a factor $L^{O(k+1)}$ to the exponent, and the gap remains $2^{-L^{O(k+1)}}$.

Proposition J.13 (Exact feasibility at fixed span). *For each fixed k , a deterministic algorithm decides in time polynomial in L whether a rational native PSD system with Hessian span at most k is feasible. Its running time is $L^{O(k+1)}$.*

Proof. Compute $R = R_F$ from Corollary J.12(a). Let $\nu(x) = \max\{0, g_1(x), \dots, g_m(x), (Ax-b)_1, \dots\}$, a convex function with exact rational values and subgradients at rational points, and let $\alpha = \min\{\nu(x) : \|x\|_\infty \leq R\}$. If $F \neq \emptyset$, its minimum-norm point gives $\alpha = 0$; if $\alpha = 0$, a minimizer lies in F . The number α is the optimal value of the epigraph problem $\min\{\tau : g_i(x) \leq \tau, (Ax-b)_j \leq \tau, 0 \leq \tau, \|x\|_\infty \leq R\}$, whose constraint Hessians $\text{diag}(Q_i, 0)$ span k dimensions; adding the row

$\tau \leq \bar{\nu}$, where $\bar{\nu}$ is an explicit rational upper bound for ν on the box, makes its feasible set compact without changing α . By Corollary J.12(b), $\alpha = 0$ or $\alpha \geq \eta := 2^{-L^{c'(k+1)}}$ for an absolute c' . Termwise estimates give rational bounds of polynomial length for $|\nu|$ and for the norms of the gradients of the rows on the box, and a subgradient of ν at a rational point is the gradient of a row attaining the maximum, or zero. Hence Lemma 3.1(a), with accuracy $\eta/3$, returns a rational point $\hat{x}$ of the box with $\nu(\hat{x}) \leq \alpha + \eta/3$, in time polynomial in n , $\log R$, $\log(1/\eta)$ and the input length. The system is feasible exactly when $\nu(\hat{x}) < \eta/2$. $\square$

This is the exact decision procedure of the Hessian-span program. Its exponent grows with k ; no running time of the form $\phi(k)L^{O(1)}$ is claimed.

Sharpness. The degree bound is attained, even under strong regularity. The proof specializes a generic count inside a nonempty open set of strictly convex instances, using two classical inputs. Nie and Ranestad prove that for generic coefficients the equality-constrained critical point system

$$g_i(x) = 0 \ (1 \leq i \leq s), \quad \nabla f(x) + \sum_{i=1}^s \lambda_i \nabla g_i(x) = 0 \quad (91)$$

of quadratics in n variables has exactly $2^s \binom{n}{s}$ complex solutions [94]. Hilbert's irreducibility theorem, in the form that the integral specializations at which an irreducible polynomial over $\mathbb{Q}(u_1, \dots, u_r)$ becomes reducible or drops degree form a set of density zero in boxes of $\mathbb{Z}^r$, supplies the rational specialization [110].

Theorem J.14 (Sharpness of the degree bound). *For every $n \geq 1$ and $0 \leq k \leq n(n+1)/2$ there is a rational instance with Hessian span exactly k , positive definite objective Hessian, a strictly feasible compact feasible set, and a unique optimizer p , such that $[\mathbb{Q}(p) : \mathbb{Q}] = [\mathbb{Q}(\theta) : \mathbb{Q}] = \mathcal{B}(n, k)$. If $k > 0$, all constraint Hessians can be taken positive definite.*

Proof. For $k = 0$ minimize $\|x\|^2$ over a box. Let $k \geq 1$ and choose $1 \leq s \leq \min(k, n)$ attaining $\mathcal{B}(n, k)$; the term for $s = 1$ is $2n \geq 1$, so such s exists. Let t be the vector of all coefficients of $f, g_1, \dots, g_s$, treated as indeterminates, $\mathbb{K} = \mathbb{Q}(t)$, and $\mathcal{D} = 2^s \binom{n}{s}$. The incidence variety of (91) is the graph of a polynomial map: x, λ , all Hessians, $a_1, \dots, a_s$ and c_0 are free, and $c_i = -\frac{1}{2}x^\top Q_i x - a_i^\top x$ and $a_0 = -Q_0 x - \sum_i \lambda_i (Q_i x + a_i)$ are determined. It is therefore an irreducible variety, of the same dimension as the parameter space. Its projection to the parameter space has generic fibers of cardinality $\mathcal{D}$, so its function field $\mathcal{L} = \mathbb{K}(x, \lambda)$ is a separable extension of degree $\mathcal{D}$ of $\mathbb{K}$. Cramer's rule on a nonzero minor of the gradient matrix gives $\lambda \in \mathbb{K}(x)$, so $\mathcal{L} = \mathbb{K}(x)$.

Let $\beta = f(x) \in \mathcal{L}$. Each derivation $\partial_j = \partial/\partial a_{0,j}$ of $\mathbb{K}$ extends uniquely to $\mathcal{L}$. Differentiating $g_i(x) = 0$ gives $\nabla g_i(x)^\top \partial_j x = 0$, and therefore

$$\partial_j \beta = x_j + \nabla f(x)^\top \partial_j x = x_j - \sum_i \lambda_i \nabla g_i(x)^\top \partial_j x = x_j.$$

A derivation of $\mathbb{K}$ maps $\mathbb{K}(\beta)$ into itself: if Π is the minimal polynomial of β over $\mathbb{K}$, then $\partial_j \beta = -(\partial_j \Pi)(\beta)/\Pi'(\beta)$. Hence $x_j \in \mathbb{K}(\beta)$ and $\mathbb{K}(\beta) = \mathcal{L}$. Write $x_j = \varphi_j(\beta)$ and $\lambda_i = \psi_i(\beta)$ with $\varphi_j, \psi_i \in \mathbb{K}[Y]$, and let $\mathfrak{c} \in \mathbb{Q}[t]$ be a nonzero common denominator of these polynomials and of Π . The identities $\Pi(\beta) = 0$, $x_j = \varphi_j(\beta)$ and $\lambda_i = \psi_i(\beta)$ hold on the incidence variety wherever $\mathfrak{c} \neq 0$.

For the seed $Q_i = I + e_i e_i^\top$, $a_i = e_i$, $c_i = 0$ ($1 \leq i \leq s$), $Q_0 = I$, $a_0 = -\sum_i e_i$, $c_0 = 0$, the point $x = 0$ with $\lambda = \mathbf{1}$ solves (91), all Hessians are positive definite, $Q_1, \dots, Q_s$ are independent, and the active gradients $e_1, \dots, e_s$ are independent. The bordered Jacobian is nonsingular, so the implicit function theorem gives a real solution branch on a neighborhood of the seed. Shrinking the

neighborhood, we obtain a nonempty open set $\mathcal{O}$ of real coefficient vectors on which the Hessians are positive definite and independent, the multipliers are positive, a fixed point is strictly feasible for $g_i \leq 0$, and the branch point is, by convexity, the unique optimizer of $\min\{f : g_i \leq 0\}$. Choose a rational $t_0 \in \mathcal{O}$ and put $t = t_0 + (1/u_1, \dots, 1/u_r)$. This birational substitution keeps Π irreducible over $\mathbb{Q}(u)$. By Hilbert's theorem in the form above, there are integral u with all $|u_j|$ so large that $t \in \mathcal{O}$, at which Π stays irreducible of degree $\mathcal{D}$ and the leading coefficient and $\mathfrak{c}$ do not vanish: each excluded condition defines a set of density zero.

At such a rational t , the unique optimizer p is a point of the incidence variety, so $\beta^* = f(p)$ is a root of the irreducible specialized Π and $[\mathbb{Q}(\beta^*) : \mathbb{Q}] = \mathcal{D}$. The specialized identities give $p \in \mathbb{Q}(\beta^*)^n$, and $\beta^* \in \mathbb{Q}(p)$, so $\mathbb{Q}(p) = \mathbb{Q}(\beta^*)$ has degree $\mathcal{D} = \mathcal{B}(n, k)$. If $s < k$, extend $Q_1, \dots, Q_s$ to k independent rational positive definite matrices, which is possible because the positive definite cone is open and spans the symmetric matrices, and add the rows $\frac{1}{2}x^\top R x - C \leq 0$ with rational C so large that they are strictly slack at p and at a strictly feasible point. The optimizer and value do not change, and the Hessian span becomes exactly k . $\square$

The theorem shows that degeneracy, lack of strict feasibility, and nonconvexity are not needed to reach the worst degree. It is an existence statement: it gives no useful bound on the coefficient sizes of the instances. Section J.5 constructs explicit short instances with degree $n^{\Omega(k)}$.

J.3 Rational points at span two and irrational singletons at span three

A rational feasible point is the simplest exact output, and it is what an exact solver would like to return. For one convex quadratic row together with rational affine rows, a nonempty system has a rational point of polynomial length [32, 116]. This subsection shows that the same is true whenever the native Hessians span at most two dimensions, and that three dimensions already permit a feasible set consisting of one irrational point. The threshold is therefore exactly between two and three. The difficult case is a tangency between two convex quadratics, and the key fact is that a tangency multiplier is rational.

Lemma J.15 (Rational tangency multiplier). *Let g_1, g_2 be convex quadratics on $\mathbb{R}^d$ with rational coefficients, $g_i(x) = \frac{1}{2}x^\top Q_i x + a_i^\top x + c_i$. Suppose that for some $t_0 > 0$ and $z \in \mathbb{R}^d$,*

$$g_1 + t_0 g_2 \geq 0 \text{ on } \mathbb{R}^d, \quad g_1(z) = g_2(z) = 0.$$

Then there is a rational $t > 0$, of bit length polynomial in the input length, with $g_1 + t g_2 \geq 0$ on $\mathbb{R}^d$ and $(g_1 + t g_2)(z) = 0$. It is computed in polynomial time by the following rule. Let $W = \ker Q_1 \cap \ker Q_2$. If some rational $w \in W$ has $(a_1^\top w, a_2^\top w) \neq (0, 0)$, take $t = -a_1^\top w / a_2^\top w$. Otherwise restrict to a rational complement of W and form

$$\varphi(\tau) = c_1 + \tau c_2 - \frac{1}{2}(a_1 + \tau a_2)^\top (Q_1 + \tau Q_2)^{-1} (a_1 + \tau a_2) \in \mathbb{Q}(\tau);$$

take $t = 1$ if $\varphi = 0$, and otherwise let t be the root of $\gcd(\mathcal{N}, \mathcal{N}')$, where $\mathcal{N}$ is the numerator of φ in lowest terms.

Proof. The aggregate $g = g_1 + t_0 g_2$ is nonnegative and vanishes at z , so $\nabla g(z) = 0$. For $w \in W$, $\nabla g(z)^\top w = (a_1 + t_0 a_2)^\top w$, hence $(a_1 + t_0 a_2)^\top w = 0$. If $(a_1^\top w, a_2^\top w) \neq (0, 0)$ for some rational $w \in W$, then $a_2^\top w \neq 0$ and $t_0 = -a_1^\top w / a_2^\top w$ is rational; take $t = t_0$.

Otherwise both linear forms vanish on W , and g_1, g_2 are constant along W . Pass to the rational complement $W^\perp$, on which $Q_1 + Q_2 \succ 0$ and hence $Q_1 + \tau Q_2 \succ 0$ for $\tau > 0$. If $W^\perp = 0$, both quadratics are constants that vanish at z , and $t = 1$ works. Otherwise $\varphi(\tau)$ is the minimum of

$g_1 + \tau g_2$ for $\tau > 0$, and $\varphi(t_0) = 0$. The minimizer at t_0 is the image $\bar{z}$ of z , and the derivative of the minimum value with respect to τ is g_2 at the minimizer, so $\varphi'(t_0) = g_2(\bar{z}) = 0$. If $\varphi = 0$, then $g_1 + g_2 \geq 0$ with minimum $\varphi(1) = 0$, and $g_1(z) + g_2(z) = 0$, so $t = 1$ works.

Suppose $\varphi \neq 0$. A real congruence brings $Q_1 + Q_2$ to the identity and Q_2 to $\text{diag}(d_1, \dots, d_d)$ with $0 \leq d_j \leq 1$, hence Q_1 to $\text{diag}(1 - d_j)$. Dividing each scalar term gives the partial-fraction form

$$\varphi(\tau) = \kappa_0 + \kappa_1\tau - \kappa_2\tau^2 - \sum_{j=1}^r \frac{\rho_j}{\tau - \sigma_j}, \quad \kappa_2 \geq 0, \quad \rho_j > 0, \quad \sigma_1 < \dots < \sigma_r \leq 0. \quad (92)$$

Terms with $d_j = 0$ contribute to the quadratic polynomial with nonpositive leading coefficient; a term with $d_j > 0$ has its pole at $\sigma = -(1 - d_j)/d_j \leq 0$ with a nonpositive residue. Equal poles are combined and zero residues removed, so the σ_j are exactly the poles of φ , all simple. The diagonalization is used only to obtain these signs; the numbers in (92) need not be rational. Write $\varphi = \mathcal{N}/\mathcal{Q}$ in lowest terms over $\mathbb{Q}$. Then $\deg \mathcal{N} \leq r + 1$ if $\kappa_2 = 0$, and $\deg \mathcal{N} = r + 2$ if $\kappa_2 > 0$. On each interval (σ_j, σ_{j+1}) , φ tends to $-\infty$ at the left end and to $+\infty$ at the right end, so $\mathcal{N}$ has a root of odd multiplicity there. If $\kappa_2 > 0$, there is one more such root in $(-\infty, \sigma_1)$. All these roots are nonpositive and distinct from $t_0 > 0$, while t_0 is a root of multiplicity at least two because $\varphi(t_0) = \varphi'(t_0) = 0$ and $\mathcal{Q}(t_0) \neq 0$. Counting with multiplicity, these roots exhaust $\deg \mathcal{N}$ when $r \geq 1$. Hence every other root of $\mathcal{N}$ is simple and t_0 is a root of multiplicity exactly two. When $r = 0$, φ is a polynomial of degree at most two; it cannot be affine with a double root, so it equals $-\kappa_2(\tau - t_0)^2$. In all cases $\gcd(\mathcal{N}, \mathcal{N}')$ has degree one and root t_0 , so $t_0 \in \mathbb{Q}$.

A numerator of φ before cancellation is

$$2(c_1 + \tau c_2) \det(Q_1 + \tau Q_2) - (a_1 + \tau a_2)^\top \text{adj}(Q_1 + \tau Q_2)(a_1 + \tau a_2),$$

which has degree at most $d + 1$ and coefficients of polynomial bit length. The rational root theorem bounds the numerator and denominator of t_0 . The determinant, the adjugate, the cancellation and the gcd are polynomial-time rational computations. $\square$

Cancellation must precede the gcd. In $\mathbb{R}^3$, the balls $g_1 = \|x\|^2 - 1$ and $g_2 = \|x - 2e_1\|^2 - 1$ touch at e_1 . Here $\varphi(\tau) = -(\tau - 1)^2/(\tau + 1)$, whereas the uncanceled numerator above is $-16(\tau + 1)^2(\tau - 1)^2$, with a spurious repeated root -1 . Repeated roots of the characteristic determinant also occur in classical contact tests for ellipsoids [59]; only the reduced numerator has the uniqueness property used in the proof.

Proposition J.16 (Two quadratic rows). *Let P be a rational polyhedron and g_1, g_2 rational convex quadratics. If $F = P \cap \{g_1 \leq 0, g_2 \leq 0\}$ is nonempty, then F contains a rational point.*

Proof. Let I_t be the set of affine rows of P that vanish identically on F , and $\mathcal{Y}$ the rational affine space defined by them. For each other row choose a point of F at which it is strict, and let p_0 be the average of these points, or any point of F if there are none. Then $p_0 \in F$ and every row outside I_t is strict at p_0 .

If some $y \in \mathcal{Y}$ has $g_1(y) < 0$ and $g_2(y) < 0$, the points $p_0 + s(y - p_0)$ for small $s > 0$ lie in $\mathcal{Y}$, satisfy every row outside I_t strictly, and make both quadratics negative. A neighborhood of such a point in $\mathcal{Y}$ lies in F , and rational points are dense in the rational affine space $\mathcal{Y}$.

Otherwise the convex open set $\{(u_1, u_2) : \exists x \in \mathcal{Y}, g_1(x) < u_1, g_2(x) < u_2\}$ misses the closed negative quadrant. A separating line gives weights $\lambda_1, \lambda_2 \geq 0$, not both zero, with $\lambda_1 g_1 + \lambda_2 g_2 \geq 0$ on $\mathcal{Y}$; the weights are nonnegative because the open set is closed under increasing u , and the separating constant is zero because p_0 is feasible. Work in a rational affine chart of $\mathcal{Y}$.

If only $\lambda_1 > 0$, then $g_1 \geq 0$ on $\mathcal{Y}$ and $g_1(p_0) \leq 0$, so every point of F minimizes g_1 on $\mathcal{Y}$. These minimizers form the rational affine space where the gradient of g_1 on $\mathcal{Y}$ vanishes. Restricting to it leaves one quadratic row and affine rows. Repeating the argument with this single row gives either a strict point, hence a rational point by density, or a second restriction to the zero set of that row, after which only affine rows remain and F is a nonempty rational polyhedron. The case where only $\lambda_2 > 0$ is symmetric.

If both weights are positive, then $g_1(p_0) = g_2(p_0) = 0$, and Lemma J.15 with $t_0 = \lambda_2/\lambda_1$ gives a rational $t > 0$ for which $g = g_1 + tg_2$ is nonnegative on $\mathcal{Y}$ and vanishes at p_0 . Every point of F has $g \leq 0$ and therefore lies in the zero set Z of g on $\mathcal{Y}$, which is the rational affine space where its gradient vanishes. The direction space of Z is the kernel of the restricted Hessian of g , which equals the intersection of the kernels of the restricted Q_1 and Q_2 because both are positive semidefinite. Hence g_1 and g_2 are affine on Z , and $F = Z \cap P \cap \{g_1 \leq 0, g_2 \leq 0\}$ is a nonempty rational polyhedron. $\square$

Theorem J.17 (Span two). *Let F be a rational native PSD system with Hessian span $k \leq 2$.*

- (a) *If $F \neq \emptyset$, then rational points are dense in F and the affine hull of F is a rational affine space.*
- (b) *A deterministic polynomial-time algorithm either reports that $F = \emptyset$ or returns a rational point of F . In particular, every nonempty F has a rational point of polynomial bit length.*

No boundedness, strict feasibility, positive definiteness, or bound on the number of rows is assumed.

Proof. First reduce to at most two quadratic rows. If $k = 2$, the cone generated by $Q_1, \dots, Q_m$ lies in a two-dimensional space and is pointed, because a matrix and its negative cannot both be nonzero and positive semidefinite. Its two extreme rays are generated by input matrices B_1, B_2 , so $Q_i = \alpha_i B_1 + \beta_i B_2$ with rational $\alpha_i, \beta_i \geq 0$. The system is the projection of

$$\frac{1}{2}x^\top B_1 x \leq u_1, \quad \frac{1}{2}x^\top B_2 x \leq u_2, \quad \alpha_i u_1 + \beta_i u_2 + a_i^\top x + c_i \leq 0 \quad (1 \leq i \leq m), \quad Ax \leq b :$$

take $u_j = \frac{1}{2}x^\top B_j x$ in one direction and use $\alpha_i, \beta_i \geq 0$ in the other. Rational lifted points project to rational points. The rays can be found by testing at most m^2 pairs. For $k = 1$ one generator suffices, and for $k = 0$ the system is polyhedral.

(a) Proposition J.16 gives a rational point. Intersecting F with small rational boxes around any feasible point preserves the span and nonemptiness, which proves density. Rational feasible points close to an affine basis of the affine hull remain affinely independent, so the affine hull is spanned by rational points.

(b) Work with the lifted system, which has span at most two and polynomial length; call its feasible set F again, its dimension n , and its quadratic rows g_1, g_2 . Every decision below concerns a rational native PSD system of span at most two and polynomial length, obtained by adding rational affine rows, so Proposition J.13 answers it in polynomial time.

Step 1. Decide whether $F = \emptyset$. If not, let $R = R_F$ from Corollary J.12(a), so F meets $[-R, R]^n$, and let $\text{Box} = [-2R, 2R]^n$.

Step 2: the affine hull of the minimal face. For each affine row $\ell_j \leq 0$, let η_j be the gap of Corollary J.12(b) for minimizing ℓ_j over $F \cap \text{Box}$, and decide whether $F \cap \text{Box} \cap \{\ell_j \leq -\eta_j\} \neq \emptyset$. This happens exactly when ℓ_j is not identically zero on F : if $\ell_j(y) < 0$ for some $y \in F$, the segment from a point of $F \cap [-R, R]^n$ toward y contains points of $F \cap \text{Box}$ with $\ell_j < 0$, so the minimum over the compact set $F \cap \text{Box}$ is negative and hence at most $-\eta_j$. Let I_t be the rows that are identically zero on F , $\mathcal{Y}$ the affine space where they vanish, and $x = x_0 + Uv$ a rational chart of $\mathcal{Y}$ whose free coordinates are original coordinates. If $\mathcal{Y}$ is a point, return it. Averaging the points found for the

rows outside I_t gives, in the proof only, a point $\bar{x} \in F \cap \text{Box}$ at which every row outside I_t is strictly negative.

Step 3: a common margin. Let η be the gap of Corollary J.12(b) for maximizing σ subject to $g_i \leq -\sigma$, $\ell_j \leq -\sigma$ for $j \notin I_t$, $0 \leq \sigma \leq 1$, and $v \in [-3R, 3R]^{\dim \mathcal{Y}}$. Decide whether the set C_η of chart points in this box with $g_i \leq -\eta$ and $\ell_j \leq -\eta$ for $j \notin I_t$ is nonempty. If some $y \in \mathcal{Y}$ has $g_1(y), g_2(y) < 0$, the segment from $\bar{x}$ toward y gives a point of positive margin in the box, so $C_\eta \neq \emptyset$; the converse is clear.

Step 4: positive margin. If $C_\eta \neq \emptyset$, bisect the box one coordinate at a time, keeping a closed half whose intersection with C_η is nonempty; one decision per bisection suffices, since the other half is nonempty when the tested half is empty. Stop when all side lengths are at most 2ω , where $\omega = \min(1, \eta/(2G))$ and $G \geq 1$ is a rational Lipschitz bound in the maximum norm for the g_i and ℓ_j on the box enlarged by one. The center $\hat{v}$ is within ω of a point of C_η , so every quadratic and every row outside I_t is at most $-\eta/2 < 0$ at $\hat{v}$, and $x_0 + U\hat{v}$ is a rational point of F . The number of bisections is polynomial.

Step 5: zero margin. If $C_\eta = \emptyset$, no point of $\mathcal{Y}$ makes both quadratics negative. Compute the infimum of each g_i on $\mathcal{Y}$ by rational linear algebra. If it is zero for some i , every feasible point lies in the rational affine space where the gradient of g_i on $\mathcal{Y}$ vanishes, and $g_i = 0$ there; restrict to that space, delete g_i , and repeat Steps 2–5 with one quadratic row. With one row and zero margin, the row is nonnegative on the current affine space, so its infimum is zero and the next restriction leaves a rational polyhedron. If both infima are negative, the weights in the convex alternative of Proposition J.16 are both positive, since a single positive weight would make the corresponding quadratic nonnegative on $\mathcal{Y}$. Lemma J.15 then computes a rational $t > 0$, and the rational affine space where the gradient of $g_1 + tg_2$ on $\mathcal{Y}$ vanishes contains F and carries both quadratics as affine functions.

In both branches a nonempty rational polyhedron remains, and rational linear programming returns a rational point in polynomial time [49]. At most two curved restrictions occur, so the polynomial bounds compose a bounded number of times. The output of a polynomial-time algorithm has polynomial length. $\square$

The face restriction in Step 2 is needed before a zero margin can be read as a tangency. For $g_1 = x^2 - 1$, $g_2 = y^2 - 1$ and the row $x \geq 1$, there is no common positive margin, yet both quadratics are negative at the origin and no nonzero nonnegative aggregate is nonnegative on the plane. Step 2 restricts to $x = 1$, where g_1 vanishes identically.

Rational points of convex semialgebraic sets can be computed in time exponential in the dimension [106]; for native PSD systems of Hessian span at most two, the algorithm above is polynomial in the dimension and the number of rows. The theorem gives a rational point, not the canonical one. The minimum-norm point of $\{x : (x - 2)^2 \leq 2\}$ is $2 - \sqrt{2}$, although 1 is feasible. Two consequences are worth recording. First, if a rational convex quadratic objective has a rational attained optimal value θ , and the objective and constraint Hessians together span at most two dimensions, appending $f \leq \theta$ yields a rational optimizer in polynomial time. Second, in a mixed-integer system whose continuous Hessian span is at most two, every nonempty continuous fiber over a fixed integer assignment has a rational point; the theorem says nothing about finding the assignment.

Proposition J.18 (An irrational singleton at span three). *Let $r = \sqrt[3]{2}$ and*

$$g_1 = x^2 - y, \quad g_2 = y^2 - 2x, \quad g_3 = (x - y)^2 - 2x - y + 4.$$

These rows have linearly independent rank-one positive semidefinite Hessians, and $\{g_1 \leq 0, g_2 \leq 0, g_3 \leq 0\} = \{(r, r^2)\}$. The rows $R_i = 3g_i + g_1 + g_2 + g_3$ have positive definite Hessians, Hessian

span three, and the same feasible set. No nonzero rational weights $w \geq 0$ make $\sum_i w_i g_i$ nonnegative on $\mathbb{R}^2$.

Proof. The Hessians are, up to the factor two, $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, which are independent. With $p = (r, r^2)$ and $r^3 = 2$, all three rows vanish at p ; for instance $g_3(p) = r^2 - 2r^3 + r^4 - 2r - r^2 + 4 = 0$. For the positive weights $\lambda = (2r - 1, r^2 - 1, 1)$, direct expansion gives

$$\sum_i \lambda_i g_i(z) = (z - p)^\top \begin{pmatrix} 2r & -1 \\ -1 & r^2 \end{pmatrix} (z - p),$$

and this matrix is positive definite, with determinant $2r^3 - 1 = 3$. A feasible point makes the left side nonpositive, so it equals p .

The quadratic parts of R_1, R_2, R_3 are $\begin{pmatrix} 5 & -1 \\ -1 & 2 \end{pmatrix}$, $\begin{pmatrix} 2 & -1 \\ -1 & 5 \end{pmatrix}$ and $\begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}$, each positive definite with determinant nine. The transformation $3I + \mathbf{1}\mathbf{1}^\top$ from (g_i) to (R_i) is invertible, so the span stays three. With $\bar{\lambda} = \sum_i \lambda_i$, the weights $\omega_i = (\lambda_i - \bar{\lambda}/6)/3$ satisfy $\sum_i \omega_i R_i = \sum_i \lambda_i g_i$, and they are positive: $18\omega_1 = 10r - r^2 - 5$, $18\omega_2 = 5r^2 - 2r - 5$ and $18\omega_3 = 7 - 2r - r^2$, and $5/4 < r < 4/3$ gives the lower bounds $95/16$, $5/16$ and $23/9$.

Finally, a nonnegative aggregate with rational weights vanishes at p , so its gradient vanishes there. Its gradient at p is a rational combination of $1, r, r^2$, which are linearly independent over $\mathbb{Q}$; hence all its quadratic and linear coefficients vanish. A nonnegative combination of nonzero positive semidefinite Hessians is zero only if all weights are zero, since traces are positive. $\square$

Thus each set $R_i \leq 0$ is a full-dimensional compact ellipsoid with rational data, and three of them can meet in a single irrational point. The point $(\sqrt[3]{2}, \sqrt[3]{4})$ is also the classical example of an irrational singleton cut out by a convex cubic inequality [17]; the native PSD quadratic presentation is what matters here. The last statement explains why the zero-margin step of Theorem J.17 cannot be copied at span three: an exposing aggregate with minimum zero may need irrational weights. It does not exclude rational infeasibility certificates with a positive margin, which are proved to exist in Section J.9. Every coordinate of a rational convex singleton has exactly one real conjugate, and every such number occurs (Theorem 7.4); three rational ellipsoids already realize every such number of degree at least three (Corollary 7.5).

J.4 Singleton spectrahedra

Rational semidefinite constraints are a second natural way to pin down an algebraic number, and they reach different numbers than convex polynomial inequalities. A rational convex singleton has a coordinate field with exactly one real embedding (Theorem 7.4). A rational affine semidefinite constraint in one scalar variable reaches exactly the totally real numbers, and its matrix size is at least twice the degree. If the unique feasible matrix has corank one, the coordinate field again has exactly one real embedding. These statements show that the representation, and not only convexity of the feasible set, decides which algebraic numbers can be imposed exactly.

Theorem J.19 (One scalar variable). *For a real number α the following are equivalent.*

- (i) *There are rational symmetric matrices A and B of some size with $\{x \in \mathbb{R} : A + xB \succeq 0\} = \{\alpha\}$.*
- (ii) *α is algebraic and every root of its minimal polynomial is real.*

If these hold and $d = [\mathbb{Q}(\alpha) : \mathbb{Q}]$, the least possible size of A and B is $2d$. A pencil of size $2d$ can be computed in time polynomial in the length of the minimal polynomial and of a rational interval isolating α whose endpoints are not roots.

Proof. Assume (i). Let $K_0 = \ker A \cap \ker B$, a rational subspace. A rational congruence with blocks for $K_0^\perp$ and K_0 turns the pencil into $\text{diag}(\hat{A} + x\hat{B}, 0)$ with the same feasible set. The reduced pencil has no common kernel vector: if $\hat{A}v = \hat{B}v = 0$ for the coordinate vector v of some $y \in K_0^\perp$, then Ay and By lie in the range of A and B , which is contained in $K_0^\perp$, and they are orthogonal to $K_0^\perp$. Hence $Ay = By = 0$, so $y \in K_0 \cap K_0^\perp = \{0\}$.

Let $C = \hat{A} + \alpha\hat{B} \succeq 0$, and suppose that $(C + (z - \alpha)\hat{B})v = 0$ for a nonreal z and a complex $v \neq 0$. Then $v^*Cv + (z - \alpha)v^*\hat{B}v = 0$ with both quadratic forms real, so $v^*\hat{B}v = 0$, then $v^*Cv = 0$, hence $Cv = 0$, $\hat{B}v = 0$ and $\hat{A}v = 0$. The real or imaginary part of v would be a nonzero common kernel vector. Therefore $\mathcal{D}(x) = \det(\hat{A} + x\hat{B}) \in \mathbb{Q}[x]$ is nonzero and has only real roots. The matrix C is singular, since otherwise it would remain positive definite near α . Thus $\mathcal{D}(\alpha) = 0$, so α is algebraic and its minimal polynomial, a factor of $\mathcal{D}$, has only real roots.

Next, α is a double root of $\mathcal{D}$. If C has corank at least two, its adjugate vanishes and $\mathcal{D}'(\alpha) = \text{tr}(\text{adj}(C)\hat{B}) = 0$. If C has corank one with unit kernel vector v , then in an orthonormal basis adapted to $v^\perp$ and v ,

$$C + t\hat{B} = \begin{pmatrix} C_1 + tE & tb \\ tb^\top & t\beta \end{pmatrix}, \quad C_1 \succ 0, \quad \beta = v^\top \hat{B}v.$$

If $\beta \neq 0$, then for small t with $t\beta > 0$ the block $C_1 + tE$ is positive definite and the Schur complement $t\beta - t^2b^\top(C_1 + tE)^{-1}b$ is positive, so $\alpha + t$ is feasible. Hence $\beta = 0$, and $\mathcal{D}'(\alpha)$, which is a multiple of β , vanishes. The minimal polynomial μ of α divides $\mathcal{D}$ and $\mathcal{D}'$; since μ is irreducible and does not divide μ' , μ^2 divides $\mathcal{D}$. Hence $2d \leq \deg \mathcal{D} \leq \text{size of the pencil}$.

Assume (ii), and let $\alpha_1 < \dots < \alpha_d$ be the conjugates, with $\alpha = \alpha_\kappa$. In the power basis $1, \alpha, \dots, \alpha^{d-1}$ of $\mathbb{Q}(\alpha)$, let M be the rational matrix of multiplication by α and $G_{ij} = \text{tr}(M^{i+j})$ the trace form. With the real Vandermonde matrix $V_{ji} = \alpha_j^i$, one has $G = V^\top V \succ 0$ and $VM = \text{diag}(\alpha_1, \dots, \alpha_d)V$. For rational ϱ that is not a conjugate, the rational symmetric affine pencil

$$\mathcal{L}_\varrho(x) = G(M - xI)(M - \varrho I)^{-1} = V^\top \text{diag}\left(\frac{\alpha_j - x}{\alpha_j - \varrho}\right)V$$

is congruent to the displayed diagonal matrix. Choose rational $\varrho_- < \alpha < \varrho_+$ such that $[\varrho_-, \varrho_+]$ contains no other conjugate. The κ th diagonal entry of $\mathcal{L}_{\varrho_-}(x)$ is nonnegative exactly when $x \leq \alpha$, and that of $\mathcal{L}_{\varrho_+}(x)$ exactly when $x \geq \alpha$. At $x = \alpha$ every other entry is positive, because $\alpha_j - \alpha$, $\alpha_j - \varrho_-$ and $\alpha_j - \varrho_+$ have the same sign for $j \neq \kappa$. Hence $\text{diag}(\mathcal{L}_{\varrho_-}(x), \mathcal{L}_{\varrho_+}(x)) \succeq 0$ exactly when $x = \alpha$, at size $2d$. The matrices are obtained from the companion matrix by $2d - 2$ powers, traces, and two inversions, all with polynomial coefficient bit length. $\square$

That positive semidefinite pencils, including singular ones, have only real finite eigenvalues is classical [83]; the proof above is included to make the arithmetic consequence and the singular case explicit. The unique feasible matrix of the construction has corank two, and the determinant of the pencil is

$$\det(\text{diag}(\mathcal{L}_{\varrho_-}(x), \mathcal{L}_{\varrho_+}(x))) = \frac{\det(G)^2 \mu(x)^2}{\mu(\varrho_-) \mu(\varrho_+)}.$$

For example $\sqrt{2}$ is imposed by a pencil of size four, while $\sqrt[3]{2}$, which has two nonreal conjugates, cannot be imposed in one variable by any rational pencil. Auxiliary scalar variables change the statement, as the next theorem shows.

Theorem J.20 (Corank one). *Let $A(x) = A_0 + \sum_{i=1}^n x_i A_i$ with rational symmetric A_i , and suppose $\{x \in \mathbb{R}^n : A(x) \succeq 0\} = \{x^*\}$ and $A(x^*)$ has corank one. Then $K = \mathbb{Q}(x_1^*, \dots, x_n^*)$ is a number field*

with exactly one real embedding, and so is every subfield; in particular every coordinate has exactly one real conjugate. Conversely, every number field with exactly one real embedding is the coordinate field of such a singleton with $n \leq 2$.

Proof. The singleton is defined over $\mathbb{Q}$ by sign conditions on principal minors. The real algebraic numbers form a real closed field, and by the Tarski–Seidenberg transfer principle the existential statement that the pencil has a positive semidefinite value holds over that field [15]. Uniqueness gives x^* algebraic coordinates, so K is a number field.

Let $R = A(x^*)$. It has entries in K and rank $m - 1$, so its kernel is spanned by some $v \in K^m$; normalize a coordinate $v_\ell = 1$. If $v^\top A_i v \neq 0$ for some $i \geq 1$, choose $\pm e_i$ so that the direction D satisfies $v^\top D v > 0$; the Schur-complement argument of Theorem J.19 makes $R + tD$ positive definite for small $t > 0$, a second feasible point. Hence $v^\top A_i v = 0$ for $i \geq 1$, and then $v^\top A_0 v = v^\top R v = 0$. Next, the linear system $A(x)v = 0$ in x has the unique solution x^* : if $D = \sum_i \delta_i A_i$ with $\delta \neq 0$ satisfied $Dv = 0$, symmetry would make D block diagonal in the decomposition $v^\perp \oplus \mathbb{R}v$, and $R + tD$ would stay positive semidefinite for all small $|t|$. The coefficients of this system lie in $F = \mathbb{Q}(v_1, \dots, v_m)$, so Gaussian elimination gives $x^* \in F^n$ and $K = F$.

Let σ be a real embedding of F and $u = \sigma(v) \in \mathbb{R}^m$. Since the A_i are rational, $u^\top A_i u = \sigma(v^\top A_i v) = 0$ for $0 \leq i \leq n$, and therefore $u^\top R u = u^\top A_0 u + \sum_i x_i^* u^\top A_i u = 0$, where R is the original matrix, not a conjugate. Positive semidefiniteness gives $Ru = 0$, so $u \in \mathbb{R}v$, and $u_\ell = 1 = v_\ell$ gives $u = v$. Hence σ fixes the generators of F : the field has exactly one real embedding. Its degree is then odd. For a subfield F' , the degree $[F : F']$ is odd; a real embedding of F' extends to F by sending a primitive element of F/F' to a real root of the image of its odd-degree minimal polynomial. Since F has one real embedding, so does F' .

For the converse, the pencil $\begin{pmatrix} 1 & x \\ x & 0 \end{pmatrix}$ gives $\{0\}$ with corank one, which covers $\mathbb{Q}$. Let $K \neq \mathbb{Q}$ have exactly one real embedding, let $\alpha \in K \subset \mathbb{R}$ be a primitive element, and let μ be its monic minimal polynomial, of degree $d \geq 3$ with exactly one real root. Let $\mathcal{C}$ be the companion matrix of μ with ones on the superdiagonal and last row $(-\mu_0, \dots, -\mu_{d-1})$, where $\mu = T^d + \sum_{i < d} \mu_i T^i$, and let $\nu(T) = (1, T, \dots, T^{d-1})^\top$, so that $\mathcal{C}\nu(\gamma) = \gamma\nu(\gamma)$ for every root γ . Let U be the real span of the real and imaginary parts of $\nu(\gamma)$ over the nonreal roots; it is $\mathcal{C}$ -invariant, $\mathbb{R}^d = \mathbb{R}\nu(\alpha) \oplus U$, and $\mathcal{C} - \alpha I$ maps $\mathbb{R}^d$ onto U with kernel $\mathbb{R}\nu(\alpha)$. The rational linear space $\mathcal{L} = \{B = B^\top : \nu(T)^\top B \nu(T) \equiv 0 \pmod{\mu(T)}\}$ contains the real matrix whose form in the basis $\nu(\alpha), \Re\nu(\gamma_1), \Im\nu(\gamma_1), \dots$ is $\text{diag}(0, 1, 1, \dots, 1)$: its form vanishes at every root, because $\nu(\gamma)^\top B \nu(\gamma) = 1 - 1 + 2i \cdot 0$. By density of rational points in $\mathcal{L}$ and openness of positive definiteness on U , there is a rational $B \in \mathcal{L}$ positive definite on U . Put

$$Q(s, t) = \mathcal{C}^\top B \mathcal{C} - s(\mathcal{C}^\top B + B \mathcal{C}) + tB.$$

Then $Q(\alpha, \alpha^2) = (\mathcal{C} - \alpha I)^\top B (\mathcal{C} - \alpha I)$ is positive semidefinite with kernel $\mathbb{R}\nu(\alpha)$, so it has corank one. Every $Q(s, t)$ satisfies $\nu(\alpha)^\top Q(s, t) \nu(\alpha) = 0$, because $\nu(\alpha)^\top B \nu(\alpha) = 0$. If $Q(s, t) \succeq 0$, it therefore annihilates $\nu(\alpha)$, which with $w = B\nu(\alpha)$ reads

$$(\alpha - s)\mathcal{C}^\top w + (t - s\alpha)w = 0.$$

Here $w \neq 0$: otherwise every rational row of B would vanish on the power basis of α , forcing $B = 0$. The vectors w and $\mathcal{C}^\top w$ are independent: an eigenvector relation $\mathcal{C}^\top w = \varkappa w$ with real $\varkappa$ forces $\varkappa = \alpha$, then $(\gamma - \alpha)w^\top \nu(\gamma) = 0$ for every root γ , and together with $w^\top \nu(\alpha) = 0$ this gives $w = 0$. Hence $(s, t) = (\alpha, \alpha^2)$ is the unique feasible point, and its coordinate field is $\mathbb{Q}(\alpha) = K$. $\square$

Combined with Theorem J.19, a one-variable singleton whose feasible matrix has corank one is rational, since its field would be both totally real and have only one real embedding. The size- $2d$ pencils of Theorem J.19 have corank two, consistent with this. The statement does not apply to higher corank, to non-singleton feasible sets, or to projections of spectrahedra.

J.5 Short inputs with long expanded outputs

Theorem J.14 attains the degree bound but says nothing about the size of the instances, and a large degree has consequences for output length only if the input stays short. This subsection constructs explicit short instances whose value and optimizer have degree $n^{\Omega(k)}$. It then makes the minimal polynomials of the value and of one input coordinate dense, so that sparse coefficient lists are long as well. The consequence is unconditional: no algorithm whose output must contain such a list can run in time $\phi(k)L^C$. Neither decision nor approximation is affected, and the instances have short descriptions as sums of block numbers.

Each block is a trust-region problem. Its secular equation and its reduction to a polynomial of degree $2r$ are standard [1]; the new point is that this polynomial is Eisenstein for short explicit data, and that congruences make the fields of different blocks independent.

Proposition J.21 (One block). *Let $\ell \equiv 1 \pmod{4}$ be prime, $r = \ell - 1$, $\kappa \in \{1, \dots, \ell - 1\}$ with $\kappa^2 \equiv -1 \pmod{\ell}$, and γ a positive integer with $\gamma \equiv \kappa \pmod{\ell^2}$. The problem*

$$\min \left\{ \frac{1}{2} \sum_{j=1}^r j x_j^2 - \gamma \sum_{j=1}^r j x_j : \sum_{j=1}^r x_j^2 \leq 1 \right\} \quad (93)$$

has the unique optimizer $x_j^* = \gamma j / (j + \lambda)$, where $\lambda > 1$ is the unique positive solution of $\sum_j \gamma^2 j^2 / (j + \lambda)^2 = 1$. Let $\mathcal{F}(T) = \prod_{j=1}^r (T + j)$, $\mathcal{F}_j = \mathcal{F} / (T + j)$, and

$$\mathcal{S}(T) = \mathcal{F}(T)^2 - \gamma^2 \sum_{j=1}^r j^2 \mathcal{F}_j(T)^2.$$

Then $\mathcal{S}$ is monic of degree $2r$, Eisenstein at ℓ , and $\mathcal{S}(\lambda) = 0$. Each coordinate x_j^* and the optimal value θ generate the field $\mathbb{Q}(\lambda)$, which has degree $2r$.

Proof. The objective is strictly convex and the ball is compact with interior point 0. The unconstrained minimizer $(\gamma, \dots, \gamma)$ has norm $\gamma\sqrt{r} > 1$, so the constraint is active, and the Karush–Kuhn–Tucker conditions give $x_j = \gamma j / (j + \lambda)$ with $\sum_j x_j^2 = 1$. The left side of the secular equation decreases strictly on $[0, \infty)$ from $r\gamma^2 > 1$ to 0, so λ is unique. Since $\kappa \neq 1$ (as $1 \not\equiv -1$), $\gamma \geq 2$, and the left side at $\lambda = 1$ is at least $r\gamma^2/4 \geq 4$; hence $\lambda > 1$. Multiplying the secular equation by $\mathcal{F}(\lambda)^2$ gives $\mathcal{S}(\lambda) = 0$.

Modulo ℓ , $\mathcal{F} \equiv T^r - 1$ and $\gamma^2 \equiv -1$, so $\mathcal{S} \equiv \mathcal{F}^2 + \sum_j j^2 \mathcal{F}_j^2$. Let $a = -j$ be a root of $T^r - 1$ in $\mathbb{F}_\ell$. Then $\mathcal{F}_i(a) = 0$ for $i \neq j$, and $\mathcal{F}_j(a) = \mathcal{F}'(a) = ra^{r-1} = -a^{-1}$, so $\mathcal{S}(a) = j^2 a^{-2} = 1 = a^{2r}$. From $\mathcal{F}'' = 2\mathcal{F}_j' + (T + j)\mathcal{F}_j''$ we get $\mathcal{F}_j'(a)/\mathcal{F}_j(a) = \mathcal{F}''(a)/(2\mathcal{F}'(a)) = (r-1)/(2a) = r/a$ in $\mathbb{F}_\ell$, because $r \equiv -1$. Hence $\mathcal{S}'(a) = 2j^2 \mathcal{F}_j(a)^2 r/a = 2ra^{2r-1}$. The difference $\mathcal{S} - T^{2r}$ has degree less than $2r$ and vanishes to second order at the r distinct roots of $T^r - 1$, so $\mathcal{S} \equiv T^{2r} \pmod{\ell}$. The constant term is $\mathcal{S}(0) = (r!)^2(1 - r\gamma^2)$. Write $\kappa^2 + 1 = m\ell$; then $1 \leq m \leq \ell - 2$ and $(1 - r\kappa^2)/\ell = 1 + m - m\ell \equiv 1 + m \not\equiv 0 \pmod{\ell}$. Since $\gamma^2 \equiv \kappa^2 \pmod{\ell^2}$, the constant term has ℓ -adic valuation exactly one. Thus $\mathcal{S}$ is Eisenstein, hence irreducible, and $[\mathbb{Q}(\lambda) : \mathbb{Q}] = 2r$ [92].

Every x_j^* is nonzero, lies in $\mathbb{Q}(\lambda)$, and recovers $\lambda = \gamma j / x_j^* - j$. For the value, the active constraint and the identity $j^3 / (j + \lambda)^2 = j^2 / (j + \lambda) - \lambda j^2 / (j + \lambda)^2$ give

$$2\theta = -\lambda - \gamma^2 \sum_{j=1}^r \frac{j^2}{j + \lambda} \in \mathbb{Q}(\lambda). \quad (94)$$

Extend the ℓ -adic valuation v_ℓ , normalized by $v_\ell(\ell) = 1$, to a finite extension of $\mathbb{Q}_\ell$ containing λ . Since $\mathcal{S}$ is Eisenstein, $v_\ell(\lambda) = 1/(2r)$, and the ramification index of a local field bounds the

denominators of its valuations [92]. The integers $j \leq r$ are units, so

$$2\theta = -\gamma^2 \sum_j j + (r\gamma^2 - 1)\lambda + \sum_{e \geq 2} (-1)^{e+1} \gamma^2 \left(\sum_j j^{1-e} \right) \lambda^e$$

converges. The constant term $-\gamma^2 \ell(\ell - 1)/2$ has valuation one, and the coefficient of λ is divisible by ℓ . For $2 \leq e \leq r$, the power sum $\sum_{a \in \mathbb{F}_\ell^\times} a^{1-e}$ vanishes because $\ell - 1 \nmid e - 1$, so these coefficients are divisible by ℓ . For $e = r + 1$ the power sum is $r \equiv -1$, a unit. Later coefficients are integral. Hence the term $e = r + 1$ has the unique least valuation and $v_\ell(2\theta) = (r + 1)/(2r)$. Since r is even, $\gcd(r + 1, 2r) = 1$, so the ramification index of $\mathbb{Q}_\ell(\theta)$ over $\mathbb{Q}_\ell$ is at least $2r$, and $[\mathbb{Q}(\theta) : \mathbb{Q}] \geq 2r$. With $\theta \in \mathbb{Q}(\lambda)$, this gives $\mathbb{Q}(\theta) = \mathbb{Q}(\lambda)$. $\square$

Lemma J.22 (Splitting at a larger prime). *In Proposition J.21, drop the congruence condition on γ . If $q > r$ is an odd prime dividing γ , then $\mathcal{S}$ has $2r$ distinct roots in $\mathbb{Q}_q$.*

Proof. Fix i and substitute $T = -i + \gamma i u$ in the secular equation. The term $j = i$ becomes u^{-2} , and multiplying by $u^2 \prod_{j \neq i} (j - i + \gamma i u)^2$ gives an integer polynomial equation that reduces modulo q to $(u^2 - 1) \prod_{j \neq i} (j - i)^2 = 0$, with nonzero product because $0 < |j - i| < q$. The roots $u = \pm 1$ are simple modulo q , so Hensel's lemma [92] lifts them to $u_\pm \in \mathbb{Z}_q$. The factors $j - i + \gamma i u_\pm$ are units and $u_\pm \neq 0$, so $T_{i,\pm} = -i + \gamma i u_\pm$ are roots of $\mathcal{S}$. They are distinct for the two signs, and they are congruent to $-i$ modulo q , so roots for different i are distinct. $\square$

The reduction of $\mathcal{S}$ modulo q is $\mathcal{F}^2$, which is not squarefree; the rescaling around each pole is needed.

Theorem J.23 (Products of blocks). *Let $\ell_1 < \dots < \ell_k$ be primes congruent to one modulo four, choose κ_i as in Proposition J.21, put $M_i = \prod_{i' > i} \ell_{i'}$, choose $1 \leq \varrho_i < \ell_i^2$ with $M_i \varrho_i \equiv 1 \pmod{\ell_i^2}$, and let $\gamma_i = \kappa_i M_i \varrho_i$. Let λ_i , θ_i and $E_i = \mathbb{Q}(\lambda_i)$ belong to block i , and put $r_i = \ell_i - 1$, $n = \sum_i r_i$ and $\mathcal{D} = \prod_i 2r_i$.*

- (a) $[E_1 \cdots E_k : \mathbb{Q}] = \mathcal{D}$.
- (b) *For positive rational weights ω_i , the problem of minimizing $\sum_i \omega_i \varphi_i(x_i)$ subject to $\|x_i\|^2 \leq 1$, where φ_i is the objective of (93) for block i , has the unique optimizer formed by the block optimizers; its coordinates generate $E_1 \cdots E_k$, and its Hessian span is exactly k .*
- (c) *Some integer $1 \leq t \leq 1 + (k - 1)\mathcal{D}(\mathcal{D} - 1)/2$ makes the optimal value for the weights $\omega_i = t^{i-1}$ generate $E_1 \cdots E_k$, so it has degree $\mathcal{D}$.*
- (d) *Let the weights satisfy $1 \leq \omega_1 \leq \dots \leq \omega_k$, as those of (c) do, and put $\bar{\gamma} = \max_i \gamma_i \ell_i^2$ and $\eta = 1/(2k\omega_k \bar{\gamma})$. Replacing the rows by*

$$\|x_i\|^2 + \eta \sum_{i' \neq i} \|x_{i'}\|^2 - 1 - \eta(k - 1) \leq 0 \quad (1 \leq i \leq k)$$

keeps the optimizer and the value, makes every constraint Hessian positive definite, and keeps the span equal to k . The feasible set is compact and 0 is strictly feasible.

- (e) *With the weights of (c) and the rows of (d), every coefficient has $O(k^2 \log \ell_k)$ bits, and the input length is $L = O((k+1)n^2(1+k^2 \log \ell_k))$ in the native encoding and $L = O((k+1)n^2(n+k^2 \log \ell_k))$ in the explicit format.*

Proof. (a) By construction $\gamma_i \equiv \kappa_i \pmod{\ell_i^2}$, so each $\mathcal{S}_i$ is Eisenstein at ℓ_i , and ℓ_i divides $\gamma_{i'}$ for $i' < i$. Suppose $[E_1 \cdots E_{i-1} : \mathbb{Q}] = \prod_{i' < i} 2r_{i'}$. For $i' < i$, $\ell_i > r_{i'}$ and Lemma J.22 shows that $\mathcal{S}_{i'}$ splits in $\mathbb{Q}_{\ell_i}$. The splitting field of $\prod_{i' < i} \mathcal{S}_{i'}$ is generated by roots that all lie in $\mathbb{Q}_{\ell_i}$, so it embeds in $\mathbb{Q}_{\ell_i}$, and so does $E_1 \cdots E_{i-1}$. The Eisenstein polynomial $\mathcal{S}_i$ is irreducible over $\mathbb{Q}_{\ell_i}$, hence over the image of $E_1 \cdots E_{i-1}$, and adjoining λ_i multiplies the degree by $2r_i$. Pairwise trivial intersections of the fields would not suffice for this conclusion.

(b) The problem is separable with positive weights, so its optimizer consists of the block optimizers. By Proposition J.21 the coordinates of block i generate E_i . The constraint Hessians have disjoint nonempty diagonal supports, hence are independent.

(c) The values θ_i generate $E = E_1 \cdots E_k$. Two distinct embeddings σ, τ of E into $\mathbb{C}$ differ on some θ_i , so $\sum_i T^{i-1}(\sigma(\theta_i) - \tau(\theta_i))$ is a nonzero polynomial of degree at most $k-1$. There are $\mathcal{D}(\mathcal{D}-1)/2$ pairs, so some integer in the stated range is a root of none of these polynomials, and $\sum_i t^{i-1}\theta_i$ then has $\mathcal{D}$ distinct conjugates.

(d) Each Hessian is $2 \operatorname{diag}(\eta, \dots, 1, \dots, \eta) \succ 0$, and the rows arise from the ball rows by the invertible matrix $N = (1 - \eta)I + \eta \mathbf{1}\mathbf{1}^\top$, so the span stays k . At the old optimizer every new row is active. From the secular equation, $\lambda_i < \gamma_i(\sum_j j^2)^{1/2} < \bar{\gamma}$, and $\lambda_i > 1$ by Proposition J.21. Put $a_i = \omega_i \lambda_i$. Then $a_i > 1$ because $\omega_i \geq 1$, and $\Sigma = \sum_i a_i < k\omega_k \bar{\gamma}$ because $\omega_i \leq \omega_k$. Stationarity of $\sum_i \omega_i \varphi_i + \frac{1}{2} \sum_i \mu_i (\text{row } i)$ at the old optimizer requires $N\mu = a$, since every block optimizer is nonzero. The solution is

$$\mu_i = \frac{a_i - \eta \Sigma / (1 + \eta(k-1))}{1 - \eta} > 0,$$

because $\eta \Sigma < 1/2 < a_i$. The Karush–Kuhn–Tucker conditions are sufficient for this convex problem, and strict convexity of the objective gives uniqueness. The feasible set changes, but the optimizer does not.

The conditions on the weights cannot be dropped. For $k = 2$ and $\omega_2 = 1$, the number $\eta = 1/(4\bar{\gamma})$ does not depend on ω_1 , and the unique solution of $N\mu = a$ has $\mu_1 = (\omega_1 \lambda_1 - \eta \lambda_2)/(1 - \eta^2)$ and $\mu_2 = (\lambda_2 - \eta \omega_1 \lambda_1)/(1 - \eta^2)$. Thus $\mu_2 < 0$ for large ω_1 , and $\mu_1 < 0$ for small $\omega_1 > 0$. Since 0 is a Slater point, the Karush–Kuhn–Tucker conditions are necessary, so in both cases the old optimizer is not optimal for the new rows.

(e) We have $\log \gamma_i = O(k \log \ell_k)$, $\log \mathcal{D} = O(k \log \ell_k)$, $\log t^{k-1} = O(k^2 \log \ell_k)$, and $\log(1/\eta) = O(k^2 \log \ell_k)$. There are $k+1$ quadratics in n variables, each with $O(n^2)$ coefficients; in the explicit format each monomial also carries an exponent vector of $O(n)$ bits. $\square$

Corollary J.24 (No parameterized bound for dense output). *For every function ϕ and constant C , no algorithm that returns the dense coefficient list of the minimal polynomial of the optimal value of every rational convex quadratically constrained problem can run in time $\phi(k)L^C$, where k is the Hessian span.*

Proof. Fix an integer $k > 3C$. By the prime number theorem for arithmetic progressions [26], every interval $[R, 2R]$ with R large contains at least k primes congruent to one modulo four. Building Theorem J.23(c)–(e) from such primes gives $n < 2kR$, input length $L = O_k(R^2 \log R)$ in the native encoding and $L = O_k(R^3)$ in the explicit format, and a value of degree $\mathcal{D} \geq (2(R-1))^k$, whose dense list has $\mathcal{D}+1$ entries. For large R this exceeds $\phi(k)L^C$ in both formats. $\square$

The weight in Theorem J.23(c) is shown to exist; the following lemma gives a larger weight that can be written down directly, which makes the whole family constructible in time polynomial in n .

Lemma J.25 (An explicit primitive weight). *Let $\beta_1, \dots, \beta_k$ be real algebraic numbers generating a field E , and let each β_i be a root of a nonzero integer polynomial of degree at most d whose*

coefficients have absolute values summing to at most 2^H , $H \geq 1$. Then for $t = 1 + 2^{(H+2)d^2+H+3}$ the number $\sum_i t^{i-1} \beta_i$ generates E .

Proof. All conjugates of β_i are roots of the given polynomial, so they have modulus at most $1 + 2^H \leq 2^{H+1}$. Two distinct conjugates of β_i are roots of its primitive minimal polynomial, whose degree is $e \leq d$ and whose leading coefficient has modulus at most 2^H . Its discriminant is a nonzero integer equal to the leading coefficient to the power $2e - 2$ times the product of the squared root differences. Bounding all other factors gives a root separation of at least $\delta_\star = 2^{-(H+2)d^2}$. Let $\sigma \neq \tau$ be embeddings of E and j the largest index with $\sigma(\beta_j) \neq \tau(\beta_j)$. Then

$$\left| \sum_i t^{i-1} (\sigma \beta_i - \tau \beta_i) \right| \geq t^{j-1} \delta_\star - 2^{H+2} \frac{t^{j-1}}{t-1} > 0,$$

because $2^{H+2}/(t-1) = 2^{-(H+2)d^2-1} < \delta_\star$. $\square$

For block i , $2\theta_i$ is a root of the resultant $\text{Res}_T(\mathcal{S}_i(T), Y\mathcal{F}_i(T) + T\mathcal{F}_i(T) + \gamma_i^2 \sum_j j^2 \mathcal{F}_{i,j}(T))$, by (94). This resultant has degree $2r_i$ in Y , since $\mathcal{S}_i$ is monic and $\mathcal{F}_i$ does not vanish at its roots. Its Sylvester matrix has size $3r_i + 1$ and entries whose coefficient sums are at most $((r_i + 1)!)^2 (2 + \gamma_i^2 r_i^3)$, which gives $H = O(\ell_k^2 \log \ell_k + k \ell_k \log \ell_k)$ and $d = 2\ell_k$ in Lemma J.25. The resulting weights have $O(k \ell_k^4 \log \ell_k + k^2 \ell_k^3 \log \ell_k)$ bits, and the analogue of Corollary J.24 holds with $L = O_k(R^6 \log R)$ in either format and $k > 6C$.

That the sparsity of a polynomial depends on the basis, in particular on a shift of the variable, is well known [46]; the next lemma only chooses a short shift whose ordinary expansion is dense.

Lemma J.26 (Translation makes coefficients nonzero). *Let $\Pi(X) = \sum_{i=0}^{\mathcal{D}} \pi_i X^i \in \mathbb{Q}[X]$ with $\pi_{\mathcal{D}} \neq 0$. Some integer $0 \leq s \leq \mathcal{D}(\mathcal{D} + 1)/2$ makes every coefficient of $\Pi(X - s)$ nonzero. If Π is the minimal polynomial of ϑ , then $\Pi(X - s)$ is that of $\vartheta + s$, and integer translation preserves primitivity.*

Proof. The coefficient of X^j in $\Pi(X - s)$ is $\sum_{i \geq j} \pi_i \binom{i}{j} (-s)^{i-j}$, a polynomial in s of degree $\mathcal{D} - j$ with leading coefficient $\pi_{\mathcal{D}} \binom{\mathcal{D}}{j} (-1)^{\mathcal{D}-j} \neq 0$. Together these polynomials have at most $\sum_j (\mathcal{D} - j) = \mathcal{D}(\mathcal{D} + 1)/2$ integer roots. The substitution $X \mapsto X - s$ is an automorphism of $\mathbb{Q}[X]$ and of $\mathbb{Z}[X]$. $\square$

Theorem J.27 (Sparse lists). *The instances of Theorem J.23(c)–(e) can be modified, keeping positive definite constraint Hessians, Hessian span exactly k , a compact strictly feasible set, a unique optimizer, and the input lengths of Theorem J.23(e) up to constant factors, so that the optimal value and the first coordinate of the optimizer both have degree $\mathcal{D}$ and minimal polynomials with exactly $\mathcal{D} + 1$ nonzero coefficients. Consequently no algorithm can run in time $\phi(k)L^C$ if it must return either the explicit list of nonzero coefficients of the minimal polynomial of the optimal value, or the minimal polynomials of all coordinates of the optimizer.*

Proof. Add to the objective an integer constant s_0 from Lemma J.26 for the minimal polynomial of the value. For the coordinate, let $a_i = x_{i,1}^*$ be the first coordinate of block i ; it generates E_i because $\lambda_i = \gamma_i/a_i - 1$. As in Theorem J.23(c), some integer $1 \leq \kappa' \leq 1 + (k - 1)\mathcal{D}(\mathcal{D} - 1)/2$ makes $\alpha' = \sum_i \kappa'^{i-1} a_i$ primitive for E , and Lemma J.26 gives an integer s_1 for its minimal polynomial. The affine change of variables $y_1 = x_{1,1} + \sum_{i \geq 2} \kappa'^{i-1} x_{i,1} + s_1$, with all other coordinates unchanged, has a unimodular linear part. It maps the optimizer to a point whose first coordinate is $\alpha' + s_1$. A congruence by an invertible matrix preserves positive definiteness of every Hessian and the dimension of their span, and an affine bijection preserves compactness, strict feasibility, and uniqueness. Each new coefficient is a sum of at most four products of old coefficients with κ'^{i-1} and s_1 , which have $O(k^2 \log \ell_k)$ bits. The output bound follows as in Corollary J.24: both requested lists have $\mathcal{D} + 1$ entries. $\square$

The theorem concerns the ordinary monomial basis of the minimal polynomial of the requested scalar. It does not exclude a polynomial in a shifted basis, an annihilating polynomial that is not minimal, another primitive element with coordinate maps, a tower of fields, the implicit block description given by the secular equations, or a circuit with root-selection gates; a circuit over the rationals alone cannot represent these irrational numbers. The shear is a rational affine map and is undone by one. In the unsheared instances each coordinate has degree $2r_i$ only, so separate coordinate descriptions are short there; the shear removes exactly this saving for one input coordinate. The integers s_0 , κ' and s_1 are shown to exist, which suffices for an output lower bound. Nothing here concerns the cost of deciding feasibility, comparing values, or approximating the optimizer.

Feasible points. The same output phenomenon occurs for feasibility alone, with ellipsoids and no objective. The blocks are binomial singletons cut out by three ellipsoids from the pencil of Theorem J.20, made explicit with coefficients of logarithmic length; a Kummer argument controls the joint field.

Lemma J.28 (Independent radicals). *Let d be a prime, let $\varpi_1, \dots, \varpi_k$ be distinct primes, and let $\alpha_i = \varpi_i^{1/d}$ be the real roots. Then $[\mathbb{Q}(\alpha_1, \dots, \alpha_k) : \mathbb{Q}] = d^k$, and $\sum_i c_i \alpha_i$ generates this field for all $c_1, \dots, c_k \in \mathbb{Q}^\times$.*

Proof. Let ζ be a primitive d th root of unity, $\mathbb{K} = \mathbb{Q}(\zeta)$, of degree $d-1$, and $\mathbb{L} = \mathbb{K}(\alpha_1, \dots, \alpha_k)$, the splitting field of $\prod_i (X^d - \varpi_i)$ over $\mathbb{K}$. An automorphism of $\mathbb{L}/\mathbb{K}$ is determined by the vector $b \in \mathbb{F}_d^k$ with $\alpha_i \mapsto \zeta^{b_i} \alpha_i$, and this embeds the Galois group as a subspace of $\mathbb{F}_d^k$. If the subspace were proper, some nonzero $\mathbf{e} \in \{0, \dots, d-1\}^k$ would satisfy $\mathbf{e} \cdot b \equiv 0$ on it, and $\beta = \prod_i \alpha_i^{\mathbf{e}_i}$ would be fixed, hence in $\mathbb{K}$. Then $\beta^d = \prod_i \varpi_i^{\mathbf{e}_i}$, so the norm of β from $\mathbb{K}$ to $\mathbb{Q}$ is a rational number ν with $\nu^d = \prod_i \varpi_i^{(d-1)\mathbf{e}_i}$. Comparing ϖ_i -adic valuations gives $d \mid (d-1)\mathbf{e}_i$, hence $\mathbf{e} = 0$. So all d^k rotations occur. Applying the rotation of one α_j to a relation $\sum_i \xi_i \alpha_i = 0$ with $\xi_i \in \mathbb{K}$ and subtracting gives $\xi_j(\zeta - 1)\alpha_j = 0$, so the α_i are linearly independent over $\mathbb{K}$. A rotation fixing $s = \sum_i c_i \alpha_i$ therefore has all $c_i(\zeta^{b_i} - 1) = 0$ and is the identity, so $d^k = [\mathbb{K}(s) : \mathbb{K}] \leq [\mathbb{Q}(s) : \mathbb{Q}] \leq [\mathbb{Q}(\alpha_1, \dots, \alpha_k) : \mathbb{Q}] \leq d^k$. $\square$

Proposition J.29 (Explicit binomial blocks). *Let $d \geq 3$ be odd, $a \geq 2$ an integer, and $\alpha = a^{1/d}$. In time polynomial in $d + \log a$ one can compute rational quadratics g_1, g_2, g_3 in $d-1$ variables, with positive definite Hessians and coefficients whose numerators and denominators have $O(\log(da))$ bits, such that*

$$\{x : g_1(x) \leq 0, g_2(x) \leq 0, g_3(x) \leq 0\} = \{(\alpha, \alpha^2, \dots, \alpha^{d-1})\},$$

and $\sum_i \tau_i g_i(x) = (1, x)^\top \mathcal{R}(1, x)$ for weights $\tau_i > 1/4$ with $\sum_i \tau_i = 1$ and a positive semidefinite matrix $\mathcal{R}$ whose kernel is spanned by $\nu(\alpha) = (1, \alpha, \dots, \alpha^{d-1})$. If $T^d - a$ is irreducible, the three Hessians are linearly independent.

Proof. Use the construction in the proof of Theorem J.20 for $\mu = T^d - a$, with an explicit choice of B . Here $\mathcal{C}e_i = e_{i-1}$ for $i \geq 1$ and $\mathcal{C}e_0 = ae_{d-1}$, so $\|\mathcal{C}\| = a$, and the eigenvectors of $\mathcal{C}$ are $\nu(\alpha\zeta^j)$ with $\zeta = e^{2\pi i/d}$. Let $\Delta_\alpha = \text{diag}(1, \alpha, \dots, \alpha^{d-1})$ and $B_* = \Delta_\alpha^{-1}(I - d^{-1}\mathbf{1}\mathbf{1}^\top)\Delta_\alpha^{-1}$. Since $\Delta_\alpha^{-1}\nu(\alpha\zeta^j) = (\zeta^{ij})_i$, and for $j \neq 0$ both $\sum_i \zeta^{ij}$ and $\sum_i \zeta^{2ij}$ vanish because d is odd, the form of B_* vanishes at every $\nu(\alpha\zeta^j)$, including $j = 0$. Hence $B_* \in \mathcal{L}$. The space U equals $\Delta_\alpha \mathbf{1}^\perp$, so $u = \Delta_\alpha w$ with $w \perp \mathbf{1}$ gives $u^\top B_* u = \|w\|^2 \geq a^{-2} \|u\|^2$, while $\|B_*\| \leq 1$.

Index rows and columns from zero, and for $0 \leq r < d$ let E_r have entry one where $i + j = r$, entry a where $i + j = r + d$, and zero elsewhere. Reducing $\nu(T)^\top B \nu(T)$ modulo $T^d - a$ shows $\mathcal{L} = \{B : \langle E_r, B \rangle_F = 0, 0 \leq r < d\}$. The E_r have disjoint supports and squared norms $D_r = r + 1 + a^2(d - r - 1) \leq da^2$, so $\Pi_{\mathcal{L}}(X) = X - \sum_r D_r^{-1} \langle E_r, X \rangle_F E_r$ is the orthogonal projection onto

$\mathcal{L}$. Let M_0 be a power of two with $16da^2 \leq M_0 < 32da^2$, let B_0 be a symmetric matrix on the grid $M_0^{-1}\mathbb{Z}$ whose entries are within $1/M_0$ of those of B_* , computed by bisection for $a^{1/d}$ and interval arithmetic, and let $B = \Pi_{\mathcal{L}}(B_0)$. Then $\|B - B_*\| \leq \|B_0 - B_*\|_F \leq d/M_0 \leq (16a^2)^{-1}$, so $\|B\| \leq 2$ and $u^\top Bu \geq (2a^2)^{-1} \|u\|^2$ on U . Each entry of B is an entry of B_0 minus one correction with denominator $M_0 D_r$, so it has $O(\log(da))$ bits.

Let $Q(s, t) = \mathcal{C}^\top BC - s(\mathcal{C}^\top B + BC) + tB$ and $\mathcal{R} = Q(\alpha, \alpha^2) = (\mathcal{C} - \alpha I)^\top B(\mathcal{C} - \alpha I)$, which is positive semidefinite with kernel $\mathbb{R}\nu(\alpha)$. On $H = \{z : z_0 = 0\}$, the vector $y = (\mathcal{C} - \alpha I)z$ satisfies $z_{j+1} = \alpha z_j + y_j$ for $j \leq d-2$, which gives $\|z\| \leq ad\|y\|$ and $z^\top \mathcal{R} z \geq \|z\|^2 / (2a^4 d^2)$. Since $\|\mathcal{C}^\top BC\| \leq 2a^2$, $\|\mathcal{C}^\top B + BC\| \leq 4a$ and $\|B\| \leq 2$, the restriction of $Q(s, t)$ to H is positive definite whenever $|s - \alpha|, |t - \alpha^2| \leq 3\epsilon_0$, where $\epsilon_0 = 1/(100a^5 d^2)$. Choose a dyadic $c \in \mathbb{Q}^2$ within $\epsilon_0/16$ of (α, α^2) in each coordinate, and the triangle with vertices $c + (-\epsilon_0, -\epsilon_0)$, $c + (2\epsilon_0, -\epsilon_0)$ and $c + (-\epsilon_0, 2\epsilon_0)$. With $\delta = (\alpha, \alpha^2) - c$, the barycentric weights of (α, α^2) are $1 - \tau_2 - \tau_3$, $\tau_2 = (1 + \delta_1/\epsilon_0)/3$ and $\tau_3 = (1 + \delta_2/\epsilon_0)/3$, all larger than $1/4$, and every vertex lies within $3\epsilon_0$ of (α, α^2) . For the vertices (s_i, t_i) put $g_i(x) = (1, x)^\top Q(s_i, t_i)(1, x)$. The Hessian of g_i is twice the restriction of $Q(s_i, t_i)$ to H , hence positive definite, and $\sum_i \tau_i g_i(x) = (1, x)^\top \mathcal{R}(1, x) \geq 0$ with equality only where $(1, x) \in \mathbb{R}\nu(\alpha)$, that is, at $x = (\alpha, \dots, \alpha^{d-1})$. Multiplication by $\mathcal{C}$ or $\mathcal{C}^\top$ permutes entries and scales one of them by a , so every coefficient of g_i is a sum of a bounded number of products of entries of B with $1, a, s_i$ and t_i , and has $O(\log(da))$ bits.

For the last statement, a dependence among the three Hessians can be taken rational. Since the vectors $(1, s_i, t_i)$ are linearly independent, it yields $\Theta = \theta_0 \mathcal{C}^\top BC - \theta_1 (\mathcal{C}^\top B + BC) + \theta_2 B$ with rational $\theta \neq 0$ and zero restriction to H . Then $x \mapsto (1, x)^\top \Theta(1, x)$ is a rational affine function that vanishes at $(\alpha, \dots, \alpha^{d-1})$, because all three matrices have zero form at $\nu(\alpha)$; as α has degree d , all its coefficients vanish, and $\Theta = 0$. For $\beta = \alpha \zeta^j$ with $1 \leq j < d$, evaluating Θ on $\nu(\beta)$ and $\nu(\bar{\beta})$ gives

$$(\theta_0 \alpha^2 - 2\theta_1 \alpha \cos(2\pi j/d) + \theta_2) \nu(\beta)^\top B \nu(\bar{\beta}) = 0,$$

and $\nu(\beta)^\top B \nu(\bar{\beta}) > 0$, since it is the sum of the values of the form of B at the real and imaginary parts of $\nu(\beta)$, which lie in U and are not both zero. For $d \geq 5$ two different cosines force $\theta_1 = 0$, and then $\theta_0 = \theta_2 = 0$ because α^2 has degree d . For $d = 3$ the relation reads $\theta_0 \alpha^2 + \theta_1 \alpha + \theta_2 = 0$. In both cases $\theta = 0$, a contradiction. $\square$

Corollary J.30 (Feasible points with dense coordinate polynomials). *Let d be an odd prime, let $\varpi_1 < \dots < \varpi_k$ be primes, and let $n = k(d-1)$. There is a rational native PSD system in n variables with $3k$ rows, each defining a full-dimensional ellipsoid, Hessian span exactly $3k$, and a single feasible point, whose first coordinate is $R_0 + \sum_b \varpi_b^{1/d}$ with $R_0 = 1 + \sum_b \varpi_b$. This coordinate has degree d^k , and its minimal polynomial has all $d^k + 1$ coefficients nonzero. The system is computed in time polynomial in k, d and $\log \varpi_k$; its nonconstant coefficients have $O(\log(kd\varpi_k))$ bits and its constant terms $O(k \log(kd\varpi_k))$ bits, so its input length is $L = O(k^3 d^2 \log(kd\varpi_k))$ in the native encoding and $L = O(k^3 d^2 [\log(kd\varpi_k) + kd])$ in the explicit format. Consequently, for every function ϕ and constant C , no algorithm can return in time $\phi(h)L^C$ the explicit coefficient lists of the minimal polynomials of the coordinates of the unique feasible point of every rational native PSD system with Hessian span h .*

Proof. For each b , apply Proposition J.29 with $a = \varpi_b$ on a separate block of $d-1$ variables. This gives rows $g_1, \dots, g_{3k}$ and weights $\tau_i > 1/4$ with $\sum_i \tau_i = k$, and $\sum_i \tau_i g_i$ is a sum of block forms that vanishes only at the product $p_\star$ of the block singletons. Replace the rows by $g'_i = g_i + \epsilon' \sum_j g_j$ with $\epsilon' = 1/(8k)$. The sum of all block Hessians is positive definite on $\mathbb{R}^n$, so each g'_i has a positive definite Hessian. The mixing matrix $I + \epsilon' \mathbf{1}\mathbf{1}^\top$ is invertible, so the span is unchanged, and it equals $3k$ because each $T^d - \varpi_b$ is Eisenstein at ϖ_b and the block spans have disjoint supports. The weights

$\eta_i = \tau_i - \varepsilon'k/(1 + 3k\varepsilon') = \tau_i - 1/11 > 1/8$ satisfy $\sum_i \eta_i g'_i = \sum_i \tau_i g_i$, so the mixed rows still have the single feasible point $p_\star$. Each g'_i is strictly convex with rational coefficients, so its minimizer is rational and differs from the irrational point $p_\star$, where g'_i vanishes; hence each set $\{g'_i \leq 0\}$ is a full-dimensional ellipsoid.

Next replace the first coordinate $x_{1,1}$ of the first block by $y_1 = x_{1,1} + \sum_{b \geq 2} x_{b,1} + R_0$, keeping all other coordinates. The linear part is unimodular, so positive definiteness, the span and the single feasible point are preserved, and the new first coordinate of the feasible point is $R_0 + s$ with $s = \sum_b \varpi_b^{1/d}$. By Lemma J.28, s has degree d^k . Every conjugate of s has modulus at most $\sum_b \varpi_b^{1/d} < R_0$, so every root z_j of the minimal polynomial $\prod_j (X - z_j)$ of $R_0 + s$ has positive real part. Then $\prod_j (X + z_j)$ is a product of linear factors $X + z_j$ with $z_j > 0$ and quadratic factors $X^2 + 2\operatorname{Re}(z)X + |z|^2$ with $\operatorname{Re} z > 0$, so all its coefficients are positive. Since $\prod_j (X - z_j) = (-1)^{d^k} \prod_j (-X + z_j)$, the coefficients of the minimal polynomial alternate in sign and none vanishes.

Each block coefficient has $O(\log(d\varpi_k))$ bits. In g'_i a nonconstant coefficient involves only the three rows of one block, while a constant term sums $3k$ constants; the shear changes each coefficient by a sum of at most four products with 1 and R_0 . This gives the stated bit bounds, and there are $3k$ rows with $O(n^2)$ coefficients each; in the explicit format each monomial also carries an exponent vector of $O(n) = O(kd)$ bits. For the last assertion, fix $k > 3C$, let $\varpi_1, \dots, \varpi_k$ be the first k primes, and let d run through odd primes. Then $h = 3k$, and $L = O_k(d^2 \log d)$ in the native encoding and $L = O_k(d^3)$ in the explicit format, while the requested list for the first coordinate has $d^k + 1$ entries, which eventually exceeds $\phi(3k)L^C$ in both formats. $\square$

In the unsheared system each coordinate has degree d , so its coordinatewise output is short, while the common-field representation already needs degree $d^k = (n/k + 1)^k$. The shear moves this joint degree into one input coordinate. The point also has a short description as a vector of real radicals, which the corollary does not exclude. Unlike Theorem J.27, no objective and no existential choice of weights or translations is involved.

J.6 A single convex quartic row

The degree bound of Theorem J.9 depends on the span of the constant Hessians of quadratic rows. For higher degree there are several candidate parameters. The following example shows that the number of nonlinear rows, or the dimension of the span of the polynomial Hessian matrices, does not control the degree once a row has degree four. This is one reason why the quartic singletons of the main text can have large degree with very few rows.

Proposition J.31 (One quartic row). *Let $\varpi_1, \dots, \varpi_n$ be distinct primes, $\alpha_i = \varpi_i^{1/3}$, $\Psi(x) = \sum_i (x_i^4 - 4\varpi_i x_i)$, and $M = 3 \sum_i \varpi_i^2$. The problem*

$$\min\{\tau : \Psi(x) - \tau \leq 0, \ 1 \leq x_i \leq \varpi_i \ (1 \leq i \leq n), \ -M \leq \tau \leq 0\}$$

has one globally convex nonlinear row, a strictly feasible point, and the unique optimizer $x = \alpha$, $\tau = \theta = -3 \sum_i \varpi_i \alpha_i$. The value θ has degree exactly 3^n . On the box, $\nabla^2 \Psi \succeq 12I$.

Proof. The identity $x^4 - 4\varpi x + 3\varpi\alpha = (x - \alpha)^2((x + \alpha)^2 + 2\alpha^2)$, with $\alpha^3 = \varpi$, shows that Ψ has the unique minimizer α , which lies inside the box because $1 < \alpha_i < \varpi_i$. Also $-M < \theta < 0$, so the bounds on τ are inactive, and $x_i = 3/2$, $\tau = -1$ is strictly feasible: $\Psi = \sum_i (81/16 - 6\varpi_i) \leq -111n/16 < -1$. The Hessian of the row is $\operatorname{diag}(12x_1^2, \dots, 12x_n^2, 0) \succeq 0$.

Lemma J.28 with $d = 3$ and $c_i = -3\varpi_i$ shows that θ has degree exactly 3^n ; it applies to every set of distinct primes, including 3. $\square$

The polynomial Hessian of the single row spans a one-dimensional space of polynomial matrices, while its coefficient matrices, or its values on the box, span n dimensions. The example refutes extensions of the quadratic bound based on the number of rows or on the first notion, not on the second. The exact convex quadratic lift $x_i^2 \leq y_i$, $\sum_i y_i^2 - 4 \sum_i \varpi_i x_i \leq \tau$ has Hessian span $n + 1$, in agreement with Theorem J.9. With the first n primes and an indexed sparse encoding, $L = \Theta(n \log n)$, so the degree is $2^{\Theta(L/\log L)}$. A dense coefficient list of the quartic has $\Theta(n^4)$ entries; the degree is then still superpolynomial in L , but the exponential growth is in n , not in L . The value is also the short radical sum displayed above, and its sign relative to a rational threshold is a comparison problem that the degree does not settle.

J.7 Common-field representations from certified approximations

Degree and height bounds become an exact output only together with a way to approximate one fixed point. The theorem below converts such data into a common-field representation with polynomial overhead. It never multiplies the degrees of individual coordinates, and it needs no factorization over a number field. The only external tool is certified recognition of an algebraic number from an approximation.

Theorem J.32 (Recognition of algebraic numbers [64]). *There is a deterministic algorithm that, given integers $e, A \geq 1$ and a rational number $\tilde{\vartheta}$ with $|\tilde{\vartheta} - \vartheta| < 2^{-\varsigma}/(12e)$ for a real algebraic number ϑ whose primitive minimal polynomial has degree at most e and coefficients bounded by A in absolute value, where ς is the least integer with $2^\varsigma > 2^{e^2/2}(e+1)^{(3e+4)/2}A^{2e}$, returns that minimal polynomial in time polynomial in $e, \log A$, and the length of $\tilde{\vartheta}$.*

Theorem J.33 (Common-field recovery). *Let $\beta \in \mathbb{R}^n$, $n \geq 1$, and $\mathbb{E} = \mathbb{Q}(\beta_1, \dots, \beta_n)$. Suppose that bounds $\mathcal{D} \geq [\mathbb{E} : \mathbb{Q}]$ and $H \geq 1$ are given, that every β_j has a primitive minimal polynomial with coefficients at most 2^H in absolute value, and that an oracle returns, for each integer $q \geq 1$, rationals $\tilde{\beta}_j$ with $|\tilde{\beta}_j - \beta_j| < 2^{-q}$ and length polynomial in $n, \mathcal{D}, H, q$. A deterministic algorithm with polynomially many oracle calls, at accuracies polynomial in $n, \mathcal{D}, H$, and with additional time polynomial in $n, \mathcal{D}, H$, returns the actual degree $d = [\mathbb{E} : \mathbb{Q}]$, a primitive irreducible integer polynomial P of degree d with an isolating interval for a real root α , integers c_j with $\alpha = \sum_j c_j \beta_j$, and rational polynomials b_j of degree less than d with $\beta_j = b_j(\alpha)$. All output lengths are polynomial in $n, \mathcal{D}, H$.*

Proof. Heights. Let $\mathfrak{a}_j$ be the absolute leading coefficient of the minimal polynomial of β_j and $\mathfrak{A} = \prod_j \mathfrak{a}_j$, so $\log_2 \mathfrak{A} \leq nH$ and every $\mathfrak{A}\beta_j$ is an algebraic integer. All conjugates of β_j have modulus at most 2^{H+1} . For integers $|c_j| \leq 2^\mathfrak{c}$ and $0 \leq \mathfrak{s} \leq \mathcal{D}$, the number $\vartheta = \sum_j c_j \beta_j + \mathfrak{s}\beta_i$ has conjugates of modulus at most $\mathcal{M} = (n2^\mathfrak{c} + \mathcal{D})2^{H+1}$, and $\mathfrak{A}\vartheta$ is an algebraic integer. If $\mathfrak{m}_\vartheta$ is the monic minimal polynomial of ϑ , of degree e , then $\mathfrak{A}^e \mathfrak{m}_\vartheta(T)$ is the monic minimal polynomial of $\mathfrak{A}\vartheta$ evaluated at $\mathfrak{A}T$, so it has integer coefficients bounded by $\mathfrak{A}^e(1 + \mathcal{M})^e$. Its primitive part is the primitive minimal polynomial of ϑ . Hence recognition can be called with degree bound $\mathcal{D}$ and height $\log_2 A \leq \mathcal{D}(nH + H + \mathfrak{c} + \log_2(n + \mathcal{D}) + 3)$, using an approximation of ϑ obtained from the oracle with $\log_2(n2^\mathfrak{c} + \mathcal{D}) + 1$ extra bits.

A primitive element. Let $\alpha_\kappa = \sum_j \kappa^{j-1} \beta_j$ for $0 \leq \kappa \leq (n-1)\mathcal{D}(\mathcal{D}-1)/2$. For distinct embeddings σ, τ of $\mathbb{E}$, the difference of their images of α_κ is a nonzero polynomial in κ of degree at most $n-1$, so one candidate separates all $d(d-1)/2$ pairs and generates $\mathbb{E}$. Recognize all candidates; the largest degree found is d , and a candidate attaining it is a primitive element α with its coefficients c_j . Its primitive minimal polynomial P has integer discriminant of modulus at least one, which gives a root separation $\delta_P = 2^{-4d^2(H_P+2)}$ when its coefficients are bounded by 2^{H_P} ; an

approximation of α within $\delta_P/8$ yields an interval of half-length $\delta_P/4$ containing α and no other root.

Norm polynomials. Fix i and let $\gamma_{\mathfrak{s}} = \alpha + \mathfrak{s}\beta_i$ for $\mathfrak{s} = 0, \dots, d$. Recognize the monic minimal polynomial $\mathbf{m}_{\mathfrak{s}}$ of $\gamma_{\mathfrak{s}}$, of degree $e_{\mathfrak{s}} \mid d$, and put $\Pi_{\mathfrak{s}} = \mathbf{m}_{\mathfrak{s}}^{d/e_{\mathfrak{s}}}$. Each embedding of $\mathbb{Q}(\gamma_{\mathfrak{s}})$ extends to exactly $d/e_{\mathfrak{s}}$ embeddings of $\mathbb{E}$, so

$$\Pi_{\mathfrak{s}}(T) = \prod_{\sigma: \mathbb{E} \rightarrow \mathbb{C}} (T - \sigma(\gamma_{\mathfrak{s}})).$$

Hence the $\Pi_{\mathfrak{s}}$ are the values at $S = \mathfrak{s}$ of

$$\mathcal{R}_i(S, T) = \prod_{\sigma: \mathbb{E} \rightarrow \mathbb{C}} (T - \sigma(\alpha) - S\sigma(\beta_i)) \in \mathbb{Q}[S, T],$$

which is monic of degree d in T and has degree at most d in S , and coefficientwise Lagrange interpolation at $\mathfrak{s} = 0, \dots, d$ recovers it. Interpolating the minimal polynomials themselves would be wrong when a sample fails to generate $\mathbb{E}$: for $\mathbb{E} = \mathbb{Q}(\sqrt{2})$, $\alpha = \sqrt{2}$ and $\beta_i = -\sqrt{2}$, the sample at $\mathfrak{s} = 1$ is 0, with minimal polynomial T but norm polynomial T^2 .

Coordinates. Since α is primitive, $\mathcal{R}_i(0, T) = \mathbf{m}(T)$ is its monic minimal polynomial, and $\mathcal{R}_i(S, \alpha + S\beta_i) = 0$ in $\mathbb{E}[S]$ because the factor of the identity embedding vanishes. Differentiating in S at $S = 0$ gives

$$\partial_S \mathcal{R}_i(0, \alpha) + \mathbf{m}'(\alpha)\beta_i = 0, \quad \beta_i = -\frac{\partial_S \mathcal{R}_i(0, \alpha)}{\mathbf{m}'(\alpha)}.$$

Separability gives $\mathbf{m}'(\alpha) \neq 0$. Compute the inverse of $\mathbf{m}'$ modulo $\mathbf{m}$ by solving the rational linear system of multiplication by $\mathbf{m}'$ on $\mathbb{Q}[T]/(\mathbf{m})$, and let b_i be the remainder of $-\mathbf{m}'(T)^{-1}\partial_S \mathcal{R}_i(0, T)$ modulo $\mathbf{m}$.

Cost. There are $O(n\mathcal{D}^2)$ candidates and $n(d+1)$ samples. Each recognized polynomial has degree at most $\mathcal{D}$ and coefficient bits bounded as above. Raising to the power $d/e_{\mathfrak{s}}$, Lagrange interpolation at integer nodes, whose denominators divide $\mathfrak{s}!(d-\mathfrak{s})!$, and the linear algebra modulo $\mathbf{m}$ have polynomial bit cost by determinant bounds. $\square$

Reading coordinates from the derivative of a norm polynomial in a separating parameter is the mechanism of rational univariate representations [105]; the theorem combines it with certified recognition, so that the input is an approximation oracle rather than a polynomial system. The input must approximate one specified tuple. Separate minimal polynomials of the coordinates, without selected embeddings, do not determine $\mathbb{E}$, and a joint-degree bound alone does not give the representation. The output lets a verifier evaluate any rational polynomial at β by arithmetic modulo P and one sign determination at α (Lemma J.35).

Corollary J.34 (Canonical feasible points at fixed span). *For each fixed k , a deterministic algorithm running in time polynomial in L decides whether a rational native PSD system with Hessian span k is feasible and, if so, returns its minimum-norm point in common-field format, with field degree at most $\mathcal{B}(n, k)$ and all coefficients of $L^{O(k+1)}$ bits.*

Proof. Decide feasibility by Proposition J.13, and let p be the minimum-norm point. Theorems J.9 and J.11, applied with $f = 0$, give $\mathcal{D} = \mathcal{B}(n, k)$ and $H = L^{O(k+1)}$. For the oracle, binary search the value $\|p\|^2$ in $[0, nR_F^2]$ with feasibility queries for $F \cap [-R_F, R_F]^n \cap \{\|x\|^2 \leq \varsigma\}$, whose Hessian span is at most $k+1$, until a feasible threshold $\varsigma \leq \|p\|^2 + 2^{-2q-4}$ is found. Every point x of this set satisfies $\langle p, x - p \rangle \geq 0$, so $\|x - p\|^2 \leq \|x\|^2 - \|p\|^2 \leq 2^{-2q-4}$. Bisecting coordinate intervals while keeping a nonempty intersection with this set, again by feasibility queries, until all widths are at most 2^{-q} gives a rational point within $2^{-q-1} + 2^{-q-2} < 2^{-q}$ of p in every coordinate. Each query has length polynomial in L and q . Theorem J.33 finishes. $\square$

For the canonical optimizer of a nonzero convex objective, the same conversion applies to any certified approximation oracle for that optimizer. Constructing such an oracle requires an exact optimization procedure for the fixed-span class, which is part of the structural Hessian-span program and is not developed here.

J.8 One-field optimality certificates

An exact optimizer is most useful when a third party can check it without repeating the optimization. When Slater’s condition fails, the canonical optimizer may have no Karush–Kuhn–Tucker multipliers for the original rows, and it may be irrational. The certificate below handles both difficulties. It supplies the optimizer in one real number field, adds a short sequence of exposing records that preserve every feasible point, and ends with an exact quadratic identity proving global optimality. All multipliers lie in the field of the optimizer, so the field never grows, and at most k curved records are needed. The mechanisms (convex alternatives, facial reduction, polyhedral normal cones) are classical; the point here is their encoding size in terms of the Hessian span, and a verifier that needs only univariate arithmetic.

Lemma J.35 (Signs at an isolated root). *Let $P \in \mathbb{Z}[T]$ have degree e and coefficients at most 2^{H_P} in absolute value, and let $\delta_P = 2^{-4e^2(H_P+2)}$. Given rationals $\varrho_- < \varrho_+$, one can check in polynomial time that P is squarefree, that $P(\varrho_-)P(\varrho_+) < 0$, and that $\varrho_+ - \varrho_- < \delta_P$. If these checks pass, the interval contains exactly one root α of P , and the sign of $\varsigma(\alpha)$ for a given $\varsigma \in \mathbb{Q}[T]$ can be determined in time polynomial in the lengths of P , the endpoints and ς .*

Proof. Squarefreeness is $\gcd(P, P') = 1$. For squarefree P , the discriminant is a nonzero integer equal to the leading coefficient to the power $2e - 2$ times the product of the squared differences of the roots, which have modulus at most 2^{H_P+1} . Bounding all but one factor shows that distinct roots differ by more than δ_P . A sign change on an interval shorter than δ_P therefore isolates exactly one root. For the sign, let $\mathbf{g} = \gcd(P, \varsigma)$. Its roots are simple roots of P , so $\varsigma(\alpha) = 0$ exactly when $\mathbf{g}(\varrho_-)\mathbf{g}(\varrho_+) < 0$. Otherwise let $P_1 = P/\mathbf{g}$, scaled to be primitive, and clear denominators of ς . Then $\text{Res}(P_1, \varsigma)$ is a nonzero integer equal to a power of the leading coefficient of P_1 times the product of ς over the roots of P_1 . Bounding the other factors by Cauchy’s bound gives an explicit lower bound ϵ_ς on $|\varsigma(\alpha)|$, of polynomial bit length. Bisection of the interval until its length times a Lipschitz bound for ς is below ϵ_ς determines the sign by evaluation at an endpoint. Gcds, resultants, and these evaluations are polynomial-time rational computations [118]. $\square$

The lemma does not trust a claim that P is irreducible, and it needs no multivariate root isolation. Separate isolating intervals for unrelated coordinates would not support this kind of joint evaluation.

Definition J.36 (One-field certificate). Consider a rational native PSD system with polyhedron $\{Ax \leq b\}$ and objective f . A *one-field certificate* consists of a squarefree $P \in \mathbb{Z}[T]$ with an interval isolating a root α , rational polynomials $b_1, \dots, b_n$ defining $p = (b_j(\alpha))_j$, and records $(\lambda^{(j)}, \mu^{(j)})$ for $0 \leq j \leq t$, all entries given as rational polynomials in α . The polyhedra $P_j = \{x : A_j x \leq \beta_j\}$ are defined by the verifier: P_0 consists of the original rows and the tangent rows

$$g_i(p) + \nabla g_i(p)^\top (x - p) \leq 0 \quad (1 \leq i \leq m),$$

and P_{j+1} adds to P_j the equations $H_j(x - p) = 0$ and $v_j^\top (x - p) = 0$, each written as two rows, where $H_j = \sum_i \lambda_i^{(j)} Q_i$ and $v_j = \sum_i \lambda_i^{(j)} \nabla g_i(p)$. The verifier accepts if p satisfies all original rows, if

each exposing record $j < t$ satisfies

$$\begin{aligned}\lambda^{(j)} &\geq 0, \quad \sum_i \lambda_i^{(j)} = 1, \quad \lambda_i^{(j)} g_i(p) = 0, \quad \mu^{(j)} \geq 0, \\ \mu_t^{(j)} (A_j p - \beta_j)_t &= 0, \quad \sum_i \lambda_i^{(j)} \nabla g_i(p) + A_j^\top \mu^{(j)} = 0,\end{aligned}$$

and if the final record satisfies the same sign and complementarity conditions without normalization, together with $\nabla f(p) + \sum_i \lambda_i^{(t)} \nabla g_i(p) + A_t^\top \mu^{(t)} = 0$.

Lemma J.37 (Soundness). *If the verifier accepts, then p is feasible and globally optimal.*

Proof. By convexity each g_i lies above its tangent, so $F \subseteq P_0$. Suppose $F \subseteq P_j$, and let $\mathcal{G} = \sum_i \lambda_i^{(j)} g_i$. For $x \in P_j$ the exact Taylor formula and the record give

$$\mathcal{G}(x) = \frac{1}{2}(x - p)^\top H_j(x - p) + \sum_t \mu_t^{(j)} (\beta_j - A_j x)_t,$$

because $\mathcal{G}(p) = 0$ by complementarity and $\nabla \mathcal{G}(p) = -A_j^\top \mu^{(j)}$ with $\mu^{(j)}$ supported on rows tight at p . Both terms are nonnegative on P_j , and $\mathcal{G} \leq 0$ on F . Hence both vanish on F ; positive semidefiniteness turns the first into $H_j(x - p) = 0$, and the second gives $v_j^\top(x - p) = 0$. Thus $F \subseteq P_{j+1}$. For the final record and $x \in F \subseteq P_t$,

$$f(x) - f(p) = \frac{1}{2}(x - p)^\top \left(Q_0 + \sum_i \lambda_i^{(t)} Q_i \right) (x - p) - \sum_i \lambda_i^{(t)} g_i(x) + \sum_t \mu_t^{(t)} (\beta_t - A_t x)_t \geq 0. \quad \square$$

The verifier checks neither a constraint qualification, nor minimality of the norm of p , nor that a record reduces a dimension. Complementarity with the quadratic rows is essential: without it, $\mathcal{G}(p)$ need not vanish. The sets P_{j+1} are polyhedral sections containing F and need not be faces of P_j : the record $\mathcal{G} = x^2$ on $P_0 = \mathbb{R}$ yields $\{0\}$.

Theorem J.38 (One-field certificates). *Let a rational native PSD system with Hessian span k and a convex quadratic objective attain a finite optimum, and let p be the canonical optimizer. There is a one-field certificate for p in the field $\mathbb{Q}(p)$, of degree at most $\mathcal{B}(n, k)$, with at most k exposing records, each having at most $n + 1$ positive entries in $(\lambda^{(j)}, \mu^{(j)})$, and a final record with at most n positive entries. Its total length is $L^{O(k+1)}$, and the verifier runs in time polynomial in L and the certificate length.*

Proof. Existence of the records. Let $\mathcal{V}_j$ be the direction space of the affine hull of P_j , and let I_j contain the quadratic rows whose Hessian restricted to $\mathcal{V}_j$ is nonzero. A row outside I_j coincides on P_j with its tangent row, so it is implied by P_j , and $F = \{x \in P_j : g_i(x) \leq 0, i \in I_j\}$.

If the rows in I_j have a common point of P_j at which all of them are negative, including the case $I_j = \emptyset$, the convex Karush–Kuhn–Tucker theorem with affine constraints and a strict point for the nonlinear ones [104] gives multipliers at the optimizer p , which form the final record. Otherwise the convex alternative, as in Proposition J.16, gives normalized weights $\lambda \geq 0$ on I_j with $\mathcal{G} = \sum_i \lambda_i g_i \geq 0$ on P_j . Since $p \in F$, $\mathcal{G}(p) = 0$, so positive weights sit on rows active at p , and p minimizes $\mathcal{G}$ over the polyhedron P_j . Polyhedral first-order optimality supplies $\mu \geq 0$ on the rows of P_j tight at p with $\nabla \mathcal{G}(p) + A_j^\top \mu = 0$.

In both cases the multipliers solve a linear system with coefficients in $\mathbb{Q}(p)$: n stationarity equations, plus normalization for an exposing record, in nonnegative unknowns indexed by the active rows. A nonnegative solution of minimal support has linearly independent columns, so it has at most $n + 1$, respectively n , positive entries and is given by Cramer’s rule over $\mathbb{Q}(p)$.

For an exposing record, H_j restricted to $\mathcal{V}_j$ is a positive combination of nonzero positive semidefinite forms, hence nonzero, while $\mathcal{V}_{j+1} \subseteq \ker H_j$. The restriction from $\mathcal{V}_j$ to $\mathcal{V}_{j+1}$ therefore kills a nonzero element of the span of the restricted Hessians, and the dimension of that span drops by at least one. It starts at most at k , so there are at most k exposing records.

Size. By Theorems J.9 and J.11, $K = \mathbb{Q}(p)$ has degree $d \leq \mathcal{B}(n, k) \leq (2L)^k$ and the coordinates of p have heights at most $B_* = L^{O(k+1)}$. The tangent rows have heights $O(LB_*)$. In each round the new multipliers are Cramer quotients of minors of order at most $n+1$, and the new rows are sums of polynomially many products of these with earlier data. By Lemma J.2, if all earlier data have heights at most B_j , the new ones have heights at most $cL^c(B_j + B_*)$ for an absolute c . After at most $k+1$ rounds all heights are $L^{O(k)}B_* = L^{O(k+1)}$. Heights, unlike power-basis coefficients, do not grow by a factor of the field degree in each round.

Encoding. Choose $\alpha = \sum_j c_j p_j$ primitive with integers $|c_j| \leq (1 + (n-1)d^2)^{n-1}$, as in the proof of Theorem J.33. Theorem J.11(i) and (88) bound its minimal polynomial P by $L^{O(k+1)}$ bits. For $\xi \in K$, write $\xi = \sum_{r < d} \varkappa_r \alpha^r$. Cramer's rule for the Vandermonde system over the d embeddings of K , with Lemma J.2(a),(d), gives $\text{ht}(\varkappa_r) \leq 2d^3 \text{ht}(\alpha) + d \text{ht}(\xi) + 2 \log d!$; the normal closure enters only through invariance of heights under conjugation. Each rational $\varkappa_r$ therefore has $L^{O(k+1)}$ bits. There are polynomially many entries in the records, so the total length is $L^{O(k+1)}$.

Verification. The verifier forms the tangent rows and the sections by rational polynomial arithmetic modulo P , checks the identities by reduction modulo P , and checks every sign with Lemma J.35. $\square$

Up to k exposing records may be needed. For the chain $x_i^2 - x_{i+1} \leq 0$ ($1 \leq i < k$), $x_k^2 \leq 0$, with objective x_1 , the feasible set is $\{0\}$, and the original rows admit no multipliers at 0. The tangent rows give $x_2, \dots, x_k \geq 0$. If an exposing record put positive weight λ_i on a row $i < k$ whose variable x_{i+1} is not yet fixed to zero, then along τe_{i+1} the aggregate would equal $-\lambda_i \tau + O(\tau^2) < 0$ for small $\tau > 0$. Hence each record can fix only the last free variable, and k records are needed before the final identity can certify the objective x_1 .

The theorem is an existence and verification statement. The primal point can be constructed by Corollary J.34 when $f = 0$, but no polynomial-time construction of the exposing multipliers is claimed. A certificate applied after fixing an integer assignment certifies optimality in that continuous fiber only, not global mixed-integer optimality. For an empty feasible set, the next subsection gives a rational certificate.

J.9 Rational infeasibility certificates

An infeasible native PSD system has a positive aggregate: a convex combination of its rows that is bounded below by a positive constant. This alternative is classical, without any constraint qualification or compactness [58]. Proposition J.18 shows that aggregates with minimum *zero* may need irrational weights. With a *positive* margin, the weights can be chosen rational, and their length is controlled by the Hessian span. Such a certificate is checked with rational arithmetic alone.

Theorem J.39 (Rational infeasibility certificates). *Let rows $g_i(x) = \frac{1}{2}x^\top Q_i x + a_i^\top x + c_i \leq 0$, $1 \leq i \leq m$, with rational data and $Q_i \succeq 0$, have no common solution in $\mathbb{R}^n$; affine rows are included with $Q_i = 0$, and equalities as two rows. Let k be the Hessian span and L the input length. There are rational weights $w_i \geq 0$ with $\sum_i w_i = 1$, at most $n+1$ of them positive, a rational $z \in \mathbb{Q}^n$ and a rational $\gamma > 0$ such that*

$$\sum_i w_i g_i(x) = \gamma + \frac{1}{2}(x - z)^\top H(x - z), \quad H = \sum_i w_i Q_i \succeq 0, \quad (95)$$

of total length $L^{O(k+1)}$. A verifier checks $w \geq 0$, $\sum_i w_i = 1$, $H z = -\sum_i w_i a_i$ and $\gamma = \sum_i w_i c_i + \frac{1}{2}(\sum_i w_i a_i)^\top z > 0$ in polynomial time.

Proof. An algebraic aggregate. The epigraph problem $\min\{\tau : g_i(x) \leq \tau \ (1 \leq i \leq m)\}$ is feasible, and its infimum is nonnegative because the system is infeasible. A feasible convex quadratically constrained problem with a linear objective and a finite infimum attains it, without compactness or a Slater condition [85, Corollary 2, in the numbering of the 1997 report]; the attained value α is positive, since $\alpha = 0$ would give a feasible point. Let $(\bar{x}, \alpha)$ be the canonical optimizer. The epigraph rows have Hessians $\text{diag}(Q_i, 0)$, so Theorems J.9 and J.11, in dimension $n + 1$, show that $K' = \mathbb{Q}(\bar{x}, \alpha)$ has degree at most $\mathcal{B}(n + 1, k)$ and that $\bar{x}$ and α have heights $L^{O(k+1)}$. The epigraph is strictly feasible, so the convex Karush–Kuhn–Tucker conditions give weights $w_i^* \geq 0$ with $\sum_i w_i^* = 1$, $\sum_i w_i^* \nabla g_i(\bar{x}) = 0$, and $w_i^* = 0$ unless $g_i(\bar{x}) = \alpha$. A basic solution of this system of $n + 1$ equations has at most $n + 1$ positive entries, lies in K' , and has heights $L^{O(k+1)}$ by Cramer’s rule. The aggregate $\mathcal{G}^* = \sum_i w_i^* g_i$ is stationary at $\bar{x}$ with value α , so $\mathcal{G}^* \geq \alpha$. Lemma J.2(c) and the degree bound give

$$\omega = \min\{w_i^* : w_i^* > 0\} \geq 2^{-L^{O(k+1)}}, \quad \alpha \geq 2^{-L^{O(k+1)}}.$$

Rounding inside the exact equations. Let I be the positive support and $\Omega = \bigcap_{i \in I} \ker Q_i$, a rational subspace. For every weight vector that is positive on I , the kernel of $\sum_{i \in I} w_i Q_i$ equals Ω , because a sum of positive semidefinite forms vanishes only where each summand does. Such an aggregate has a finite minimum exactly when $\sum_i w_i a_i \perp \Omega$. Hence we round inside the rational affine space

$$\mathcal{W} = \left\{ w \in \mathbb{R}^I : \sum_i w_i = 1, \sum_i w_i a_i \perp \Omega \right\},$$

which contains w^* . Independent rounding of the weights would not do: for $x^2 + y + 1$ and $x^2 - y + 1$, equal weights give $x^2 + 1$, while weights $(\frac{1}{2} + \epsilon, \frac{1}{2} - \epsilon)$ give an aggregate unbounded below. Parametrize $\mathcal{W}$ rationally with free coordinates chosen among the weights, round the free coordinates to a dyadic grid, and let $\hat{w}$ be the result; then $\varepsilon = \|\hat{w} - w^*\|_1$ can be made as small as $2^{-\mathfrak{b}}$ with weights of $\mathfrak{b} + L^{O(1)}$ bits.

The margin survives. Let $M \geq 1$ bound $\|Q_i\|$, $\|a_i\|$ and $|c_i|$. If $\sum_{i \in I} Q_i \neq 0$, let ϱ be its least positive eigenvalue. The product of the positive eigenvalues of an integer multiple of this matrix is a positive integer coefficient of its characteristic polynomial, and all eigenvalues are at most $2^{L^{O(1)}}$, so $\varrho \geq 2^{-L^{O(1)}}$. Put $\eta = \min(1, \omega \varrho)$. On $\mathcal{V}_0 = \Omega^\perp$ the aggregate Hessian at w^* is at least ηI , and if $\varepsilon \leq \omega/2$, at least $(\eta/2)I$ at $\hat{w}$, with the same support and kernel. The minimum of the aggregate is $\nu(w) = c(w) - \frac{1}{2}b(w)^\top (H(w)|_{\mathcal{V}_0})^{-1} b(w)$, with $b(w) = \sum_i w_i a_i$ and $c(w) = \sum_i w_i c_i$, and the identity $\hat{H}^{-1} - H^{*-1} = \hat{H}^{-1}(H^* - \hat{H})H^{*-1}$ gives

$$|\nu(\hat{w}) - \nu(w^*)| \leq \left(M + \frac{2M^2}{\eta} + \frac{M^3}{\eta^2} \right) \varepsilon \leq \frac{4M^3}{\eta^2} \varepsilon.$$

Choosing $\varepsilon \leq \min\{\omega/4, \alpha \eta^2 / (16M^3)\}$ keeps the support and gives $\nu(\hat{w}) \geq 3\alpha/4$. If $\sum_{i \in I} Q_i = 0$, then $\mathcal{W}$ forces $b(w) = 0$, the aggregate is the constant $c(w)$, and $\varepsilon \leq \min\{\omega/4, \alpha/(4M)\}$ suffices. In both cases $\mathfrak{b} = L^{O(k+1)}$.

The certificate. The rational system $\hat{H}z = -b(\hat{w})$ is consistent because $b(\hat{w}) \perp \Omega = \ker \hat{H}$. A rational solution from Gaussian elimination and $\gamma = c(\hat{w}) + \frac{1}{2}b(\hat{w})^\top z = \nu(\hat{w}) > 0$ satisfy (95), and all lengths are $L^{O(k+1)}$. A feasible point would make the left side of (95) nonpositive and the right side positive. $\square$

The rounding step is an algorithm once an algebraic aggregate is known: its support, the rational kernel, the precision, and the rounding are all computable. Finding the initial aggregate is a separate task that the theorem does not address. The certificate refutes real feasibility only; it says nothing about an integer program whose continuous relaxation is feasible.

The span must enter exponentially for this format, and the support cannot be bounded in terms of k alone.

Proposition J.40 (Lower bounds for positive aggregates). *(a) The system $\frac{1}{2} - x_1 \leq 0$, $x_i^2 - x_{i+1} \leq 0$ for $1 \leq i < k$, $x_k^2 \leq 0$ is infeasible, has Hessian span k , and every normalized positive aggregate has margin at most 2^{-2^k} , so a rational margin has a denominator of at least $2^k + 1$ bits.*

(b) In the system $\frac{1}{2} - x_1 \leq 0$, $x_i^2 - x_{i+1} \leq 0$ for $1 \leq i \leq k$, $x_{k+1} \leq 0$, with span k , every positive aggregate with rational weights $\lambda_i \geq 0$ satisfies $\lambda_{k+1}/\lambda_0 > 2^{d-1}/(3d)$, where $d = 2^k$; the two weights together need $d - O(\log d)$ bits under any common scaling.

(c) For $n \geq 2$, the rows $\|x - e_i\|^2 - \frac{n-2}{n-1} \leq 0$, $1 \leq i \leq n$, have $k = 1$, are infeasible, and every proper subfamily is feasible, so every positive aggregate uses all n rows.

Proof. (a) Backward induction from $x_k = 0$ gives $x_1 = 0$, contradicting the first row. At $y_i = 2^{-2^{i-1}}$ every row except the last vanishes and the last equals 2^{-2^k} ; a normalized aggregate with margin γ satisfies $\gamma \leq \sum_i w_i g_i(y) \leq 2^{-2^k}$.

(b) Infeasibility follows as in (a). At $x = 0$ the aggregate equals $\lambda_0/2$, so $\lambda_0 > 0$. On the curve $x_i = \tau^{2^{i-1}}$ all middle rows vanish, and positivity at $\tau = (1 + 1/d)/2$ gives $\lambda_0(-1/(2d)) + \lambda_{k+1}2^{-d}(1 + 1/d)^d > 0$. Since $(1 + 1/d)^d < 3$, the ratio bound follows. If $\lambda_0 = a_0/b_0$ and $\lambda_{k+1} = a_1/b_1$ in lowest terms, then $a_1 b_0 > 2^{d-1}/(3d)$.

(c) The average of the rows is $\|x - \mathbf{1}/n\|^2 + \frac{n-1}{n} - \frac{n-2}{n-1} > 0$. For a subfamily of $s \leq n-1$ indices, the average of its centers has squared distance $1 - 1/s \leq \frac{n-2}{n-1}$ to each of them. $\square$

These are lower bounds for one certificate format. Both chains have short symbolic proofs of infeasibility, so no lower bound for other proof systems follows.

J.10 Number-field affine data

Theorem J.11 allows coefficients in a number field. The following theorem shows that a convex quadratic program whose matrices are rational, but whose right-hand sides and linear costs lie in one explicitly given real number field, is solved exactly inside that field in polynomial time. The field degree is part of the input length; no product of degrees of independently given coefficients is allowed. The argument uses only exact rational quadratic programming, a determinant bound in the field, and two error bounds.

Theorem J.41 (Rational matrices, algebraic affine data). *Let $\mathbb{F} = \mathbb{Q}(\alpha) \subset \mathbb{R}$ be given by a dense irreducible integer polynomial of degree D and an interval isolating α . Let $Q \succeq 0$ and C be rational, let $b \in \mathbb{F}^s$ and $c \in \mathbb{F}^n$ be given in the power basis, and let L be the total input length. A deterministic algorithm running in time polynomial in L decides whether $\min\{\frac{1}{2}x^\top Qx + c^\top x : Cx \leq b\}$ is infeasible, unbounded, or finite, and in the last case returns the minimum-norm optimizer $p \in \mathbb{F}^n$ in the power basis, with polynomial length.*

Proof. Charts. Assume first that the problem is feasible and bounded below; a convex quadratic bounded below on a nonempty polyhedron attains its minimum [42]; this is also the polyhedral case

of [85, Corollary 2, in the numbering of the 1997 report]. Let C_I consist of rows forming a basis of all constraint normals active at p , and let $\mathcal{M}_I = \begin{pmatrix} Q & C_I^\top \\ C_I & 0 \end{pmatrix}$. For $\xi \in \ker C_I$, the points $p \pm \tau\xi$ are feasible for small τ , so $Qp + c \in (\ker C_I)^\perp$, and some signed λ satisfies $\mathcal{M}_I(p, \lambda) = (-c, b_I)$. The kernel of $\mathcal{M}_I$ is $(\ker Q \cap \ker C_I) \times \{0\}$: if $Q\xi + C_I^\top\zeta = 0$ and $C_I\xi = 0$, then $\xi^\top Q\xi = 0$, so $Q\xi = 0$ and $\zeta = 0$. For such ξ , $f(p \pm \tau\xi) = f(p)$, and minimality of the norm gives $p \perp \xi$. Hence (p, λ) is the least-norm solution, and

$$(p, \lambda) = \mathcal{M}_I^+(-c, b_I), \quad \mathcal{M}_I^+ = \mathcal{A}^{-1}\mathcal{M}_I\mathcal{A}^{-1}, \quad \mathcal{A} = \mathcal{M}_I + ZZ^\top,$$

where the columns of Z form a rational basis of $\ker \mathcal{M}_I$; on the kernel $\mathcal{A}$ acts invertibly and $\mathcal{M}_I$ vanishes, and on the range $\mathcal{A}$ agrees with $\mathcal{M}_I$. Thus each coordinate of p is a rational linear combination of the entries of b and c with coefficients of polynomial bit length, uniformly over all I . This proves $p \in \mathbb{F}^n$, bounds its power-basis length, and gives a computable rational $R \geq 1$ of polynomial length with $p \in [-R, R]^n$, using Cauchy's bound for α . The chart is used only in the proof; the algorithm does not search over I .

Separation in $\mathbb{F}$. Clearing denominators and multiplying by a power of the leading coefficient of the polynomial of α makes every entry of a given matrix over $\mathbb{F}$ an algebraic integer. A nonzero $r \times r$ minor is then, up to a known integer factor, a nonzero algebraic integer whose conjugates are bounded by determinant expansion, so its norm is a nonzero integer. Hence every nonzero minor, and every nonzero element of $\mathbb{F}$ of polynomial length, has absolute value at least $2^{-L^{O(1)}}$.

Error bounds. For a real matrix $\mathfrak{M}$ and any right-hand side $\mathfrak{r}$ with $\{\mathfrak{M}x \leq \mathfrak{r}\} \neq \emptyset$,

$$\text{dist}(y, \{\mathfrak{M}x \leq \mathfrak{r}\}) \leq H_{\mathfrak{M}} \|(\mathfrak{M}y - \mathfrak{r})_+\|_\infty, \quad H_{\mathfrak{M}} = \max_{\mathcal{I}} \frac{\sqrt{|\mathcal{I}|}}{\sigma_{\min}(\mathfrak{M}_{\mathcal{I}})}, \quad (96)$$

the maximum over linearly independent sets of rows [57]. Indeed, if z is the projection of y , then $y - z = \mathfrak{M}_{\mathcal{I}}^\top \lambda$ with $\lambda \geq 0$ and independent rows active at z , and $\|y - z\|^2 = \lambda^\top (\mathfrak{M}_{\mathcal{I}} y - \mathfrak{r}_{\mathcal{I}}) \leq \|\lambda\|_1 \|(\mathfrak{M}_{\mathcal{I}} y - \mathfrak{r}_{\mathcal{I}})_+\|_\infty$, with $\|\lambda\|_1 \leq \sqrt{|\mathcal{I}|} \|y - z\| / \sigma_{\min}$. A nonsingular square column submatrix $\mathfrak{N}$ of $\mathfrak{M}_{\mathcal{I}}$ gives $\sigma_{\min}(\mathfrak{M}_{\mathcal{I}}) \geq |\det \mathfrak{N}| / \|\mathfrak{N}\|^{|\mathcal{I}|-1}$. The separation bound for nonzero minors and entrywise bounds for $\|\mathfrak{N}\|$ hold uniformly over all row sets, so they yield rational numbers $H_A, H_{\mathfrak{M}} \geq 1$ of polynomial length that bound the constant in (96) for the rational matrix A of the rows $Cx \leq b$ and $\|x\|_\infty \leq R$, and for $\mathfrak{M} = (A; Q; -Q; c^\top; -c^\top)$; inequality (96) remains true with these upper bounds. The right-hand sides do not enter.

Let $P_R = \{Cx \leq b, \|x\|_\infty \leq R\}$, which has the same optimum and canonical optimizer, let $\mathcal{O}$ be its optimal set, and let $U = nR$, a rational bound for $\|x\|$ on the box. Two optimal points have the same Qx , by the midpoint identity, and then the same $c^\top x$; conversely such points are optimal. So $\mathcal{O} = \{x \in P_R : Qx = Qp, c^\top x = c^\top p\}$, whose matrix is $\mathfrak{M}$. For $x \in P_R$ with $\gamma = f(x) - f(p) \leq 1$, the variational inequality at p gives $\frac{1}{2}(x - p)^\top Q(x - p) \leq \gamma$, hence $\|Q(x - p)\| \leq \sqrt{2q\gamma}$ for a rational $q \geq \|Q\|$, and $\|c^\top(x - p)\| \leq \gamma + U\sqrt{2q\gamma}$. By (96),

$$\text{dist}(x, \mathcal{O}) \leq \mathcal{T}\sqrt{\gamma}, \quad \mathcal{T} = H_{\mathfrak{M}}(1 + 2(U + 1)(q + 1)).$$

Approximating p . For rational $\varepsilon, \delta > 0$, compute rational b^+ with $b \leq b^+ \leq b + \varepsilon \mathbf{1}$ and $\tilde{c}$ with $\|\tilde{c} - c\| \leq \varepsilon$ by exact root refinement of α , keep the box exact, and solve the strictly convex rational problem

$$y = \arg \min \left\{ \frac{1}{2} x^\top Qx + \tilde{c}^\top x + \frac{\delta}{2} \|x\|^2 : Cx \leq b^+, \|x\|_\infty \leq R \right\}$$

exactly in polynomial time [71]. Let $L_f = qU + \bar{c}$, where $\bar{c} \geq 1$ is a rational upper bound for $\|c\|_1$ obtained from the power-basis coefficients of c and Cauchy's bound for α ; it is a Lipschitz

bound for f on the box. Put $\mathcal{W} = 2U + L_f H_A$. The constants $U, \mathfrak{q}, L_f, H_A, H_{\mathfrak{M}}, \mathcal{T}$ and $\mathcal{W}$ are rationals of polynomial length, computed before δ and ε are chosen below. Some $x \in P_R$ satisfies $\|x - y\| \leq H_A \varepsilon$, and comparison with p gives

$$\|y\|^2 \leq \|p\|^2 + 2\mathcal{W}\varepsilon/\delta, \quad f(x) - f(p) \leq \Gamma := \mathcal{W}\varepsilon + \delta U^2/2.$$

If $\Gamma \leq 1$, then $e := \text{dist}(y, \mathcal{O}) \leq H_A \varepsilon + \mathcal{T}\sqrt{\Gamma}$. Let $\hat{y}$ be the projection of y onto $\mathcal{O}$. Since p is the projection of 0 onto $\mathcal{O}$, $\|\hat{y} - p\|^2 \leq \|\hat{y}\|^2 - \|p\|^2 \leq 2\mathcal{W}\varepsilon/\delta + 2Ue$, so

$$\|y - p\| \leq e + \sqrt{2\mathcal{W}\varepsilon/\delta + 2Ue}.$$

For a target accuracy $\tau = 2^{-\mathfrak{q}'} \leq 1$ put $\mathcal{E} = \tau^2/(16(U+1))$, $\delta = \mathcal{E}^2/(8\mathcal{T}^2(U^2+1))$ and $\varepsilon \leq \min\{\mathcal{E}/(2H_A), \mathcal{E}^2/(16\mathcal{W}\mathcal{T}^2), \delta\tau^2/(32\mathcal{W})\}$. Then $\Gamma \leq \mathcal{E}^2/(8\mathcal{T}^2)$, $e \leq \mathcal{E}$, and $\|y - p\| \leq \tau/16 + \sqrt{3}\tau/4 < \tau$, with all parameters of $O(\mathfrak{q}') + L^{O(1)}$ bits. The norm penalty is necessary: without it, the problems $\min\{-\varepsilon x : 0 \leq x \leq 1\}$ select 1 for every $\varepsilon > 0$, while the canonical optimizer of the limit problem is 0.

Exact output. Every row slack $b_i - C_i p$ lies in $\mathbb{F}$ with polynomial length, so it is either zero or at least $\mathfrak{g}$ in absolute value, where $\mathfrak{g} = 2^{-L^{O(1)}}$ is a rational lower bound computed from the separation bound. Put $\mathfrak{C} = \max\{1, \max_i \|C_i\|_1\}$, where the inner maximum is 0 if C has no rows. Approximate p within $\mathfrak{g}/(8\mathfrak{C})$ in the maximum norm and the entries of b within $\mathfrak{g}/8$. Each computed slack is then within $\mathfrak{g}/4$ of the true one, so a row is active exactly when its computed slack is below $\mathfrak{g}/2$. Choose a maximal independent subset I of the active rows by rational linear algebra and return the first block of $\mathcal{M}_I^+(-c, b_I)$, computed by rational matrix arithmetic applied to power-basis vectors.

Classification. For a rational matrix $\mathfrak{B}$ and $\mathfrak{a} \in \mathbb{F}^{s'}$, feasibility of $\mathfrak{B}x \leq \mathfrak{a}$ is decided as follows. The chart argument for $\|x\|^2$ gives a box $[-R', R']^n$ of polynomial length containing the least-norm feasible point if one exists. The value $t^* = \min\{t \geq 0 : \mathfrak{B}x \leq \mathfrak{a} + t\mathbf{1}, \|x\|_\infty \leq R'\}$ is attained at a vertex, so it lies in $\mathbb{F}$ with polynomial length; it is zero exactly when the system is feasible, and otherwise at least $2^{-L^{O(1)}}$. Rounding $\mathfrak{a}$ outward by less than this gap and solving one rational linear program decides feasibility [49]. For a feasible problem, f is bounded below exactly when $Qv + C^\top \lambda = -c$, $\lambda \geq 0$, is feasible: completing the square gives a lower bound, and otherwise separation from the closed cone $Q\mathbb{R}^n + C^\top \mathbb{R}_+^s$ gives h with $Qh = 0$, $Ch \leq 0$ and $c^\top h < 0$, an unbounded ray. Both tests use the same routine. $\square$

The input model is one explicitly represented field. Coefficients given by separate minimal polynomials could generate a compositum of exponential degree, and that case is not covered. The existence of rational charts alone does not give the algorithm; the approximation with a norm penalty does.

J.11 A nonconvex integer caveat

The bounds above use convexity of the native rows. Without it, a classical Pell family shows that two integer variables and no continuous variable already force exponentially long integer output. The phenomenon is classical [80]; it is recorded only to mark why convexity appears in the statements of this paper.

Proposition J.42 (Pell output). *For $m \geq 1$, the problem*

$$\min\{x : x^2 - 5^{2m+1}y^2 = 1, x \geq 2, y \geq 1, x, y \in \mathbb{Z}\}$$

has input length $\Theta(m)$, no continuous variables, and the unique optimizer $(X_{5^m}, B_{5^m}/5^m)$, where $(9 + 4\sqrt{5})^j = X_j + B_j\sqrt{5}$. Every feasible point has an x -coordinate of $\Theta(5^m)$ or more bits, and the optimal value has $\Theta(5^m)$ bits.

Proof. With $B = 5^m y$, the equation becomes $X^2 - 5B^2 = 1$. Every solution in positive integers is (X_j, B_j) for a unique $j \geq 1$: dividing $X + B\sqrt{5}$ by a power of $u = 9 + 4\sqrt{5}$ brings it into $[1, u)$, where the quotient $U + V\sqrt{5}$ has integer coefficients, norm one, $U \geq 1$ and $0 \leq V < 4$, and $1 + 5V^2$ is a square only for $V = 0$. The binomial expansion gives $B_j \equiv 4j9^{j-1} \pmod{5}$, and expanding the fifth power gives $B_{5j} = 5B_j(X_j^4 + 10X_j^2B_j^2 + 5B_j^4)$, whose second factor is congruent to one modulo five because $X_j^2 \equiv 1$. Hence the five-adic valuation of B_j equals that of j . Feasibility requires $5^m \mid B_j$, that is, $5^m \mid j$, and $X_j = (u^j + u^{-j})/2$ increases with j . Finally $u^j/2 < X_j < u^j$ with $16 < u < 32$. $\square$

The optimal value is an integer, so its minimal polynomial is linear; the obstruction is coefficient length, not degree. Repeated fifth powering gives an arithmetic circuit with $O(m)$ gates for the optimizer, so the obstruction concerns expanded output only. Each instance is feasible, and no hardness of deciding feasibility follows. The rows are indefinite and the feasible set is nonconvex, so the example does not bear on the convex mixed-integer results of Section 6.

K Boundaries of the exact-arithmetic hypotheses

The positive theorems of this paper rest on three hypotheses that are easy to weaken by accident: a supplied global curvature bound, exact rational input, and the right notion of a common kernel when certificates are reduced. This appendix records what happens near each boundary. Section K.1 shows that a fully degenerate convex quartic minimum is rational and easy, while partial degeneracy defeats the Newton refinement used for Theorem 4.4; the general degenerate case remains open. Section K.2 shows that restricting a semidefinite feasibility problem along a nullvector of one selected feasible matrix can destroy all rational feasible points, and that a common nullvector avoids this. Section K.3 shows that, at singular equality constraints, enclosures of function values can fail to justify any accurate upper bound on the optimal value, even where a root exists robustly under perturbation; exact input removes this obstruction. Section K.4 gives an exact penalty representation of bounded quadratic problems by a polynomial-size quadratic lift and explains why it lies outside the convex theorems. None of these results is a hardness theorem.

K.1 Degenerate convex quartics

The upper bound of Theorem 4.4 uses a supplied global lower bound on the Hessian. It enters the quadratic convergence region of Newton's method with polynomially many printed bits and then reaches doubly exponential accuracy by polynomially many Newton steps written as a short circuit. Without a curvature bound, the optimal value of a globally convex rational quartic on a nonempty rational polyhedron, when the optimal value is finite, can still be approximated in polynomial time with an additive objective-gap guarantee [111, Corollary 1.2]. That guarantee concerns the value, not the distance to a minimizer, and the question here is exact comparison. Without the curvature bound, one case is elementary.

Proposition K.1 (A fully degenerate minimum). *Let $f \in \mathbb{Q}[x_1, \dots, x_n]$ have degree at most four and be convex on $\mathbb{R}^n$. Suppose a global minimizer p satisfies $\nabla^2 f(p) = 0$. Then*

$$\arg \min f = \{x : D^3 f(x) = 0\},$$

a nonempty rational affine space. A rational minimizer, the exact minimum value, and a rational affine description of all minimizers are computed by rational linear algebra in polynomial time; a unique minimizer is rational.

Proof. For fixed u, v , the polynomial $\tau \mapsto u^\top \nabla^2 f(p + \tau v)u$ has degree at most two, is nonnegative by convexity, and vanishes at $\tau = 0$; hence its linear coefficient $D^3 f(p)[u, u, v]$ vanishes. By polarization in u , $D^3 f(p) = 0$. Stationarity and the exact Taylor expansion give $f(p + z) = f(p) + \mathcal{Q}(z)$ with $\mathcal{Q}(z) = \frac{1}{24} D^4 f[z, z, z, z]$, a nonnegative convex form of degree four, possibly zero. Since $D^3 f$ is affine and vanishes at p , its zero set is $p + \mathcal{V}$ with $\mathcal{V} = \{v : D^4 f[v, \cdot, \cdot, \cdot] = 0\}$, and $\mathcal{Q}$ vanishes on $\mathcal{V}$, so $p + \mathcal{V}$ consists of minimizers.

Conversely, let $\mathcal{Q}(v) = 0$. Then $\mathcal{Q}(\tau v) = 0$ for all τ , and for $0 < \eta < 1$ convexity and homogeneity give

$$\mathcal{Q}(z + \tau v) \leq (1 - \eta) \mathcal{Q}\left(\frac{z}{1 - \eta}\right) + \eta \mathcal{Q}\left(\frac{\tau v}{\eta}\right) = (1 - \eta)^{-3} \mathcal{Q}(z).$$

Letting $\eta \downarrow 0$ and exchanging the roles of z and $z + \tau v$ gives $\mathcal{Q}(z + \tau v) = \mathcal{Q}(z)$ for all z, τ . Differentiating in τ gives $D^4 f[v, z, z, z] = 0$ for all z , and polarization gives $v \in \mathcal{V}$. The equations $D^3 f(x) = 0$ are at most $\binom{n+2}{3}$ rational affine equations with coefficients of polynomial length, so Gaussian elimination gives the claimed description, and evaluating f at a rational solution gives the minimum. $\square$

The vanishing of $D^3 f(p)$ also follows from a derivative inequality for convex quartics [91]. Among globally convex inputs, the hypothesis can be recognized: solve $D^3 f(x) = 0$, pick a rational solution q , and test $\nabla f(q) = 0$ and $\nabla^2 f(q) = 0$. Global convexity itself is not tested. The proposition does not apply when only some Hessian eigenvalues vanish, as in $g(x) + y^4$ with an irrational minimizer of g .

Partial degeneracy breaks the refinement step. For $f(x) = x^4$, Newton's map is $N(x) = 2x/3$, so reaching error 2^{-2^s} from $x_0 = 1$ takes more than $2^s / \log_2(3/2)$ steps. This only concerns the iteration count: the minimizer is the rational number zero.

Proposition K.2 (A discontinuous Newton map). *The rational quartic*

$$f(x, y) = (x^2 + y^2)^2 + xy^2 + y^2 = x^4 + y^4 + (2x^2 + x + 1)y^2$$

is convex on $\mathbb{R}^2$, has the unique minimizer $(0, 0)$ with Hessian $\text{diag}(0, 2)$ there, and is strongly convex near every other point. Its Newton map N satisfies $N(0, \tau) \rightarrow (1/2, 0)$ as $\tau \rightarrow 0$. The extrapolated map $\mathcal{T} = 3N \circ N - 2N$, which removes the linear error $N(z) = 2z/3$ of homogeneous quartics, satisfies $\mathcal{T}_y(0, \tau) = \frac{5}{8}\tau + O(\tau^3)$.

Proof. Since $2x^2 + x + 1 > 0$, the origin is the unique zero and minimizer. The Hessian is

$$\nabla^2 f = \begin{pmatrix} 12x^2 + 4y^2 & (8x + 2)y \\ (8x + 2)y & 4x^2 + 12y^2 + 2x + 2 \end{pmatrix},$$

$$\det \nabla^2 f = 24x^2(2x^2 + x + 1) + 4y^2(24x^2 - 6x + 1) + 48y^4,$$

and both quadratic factors are positive (discriminants -7 and -60). Away from the origin the first diagonal entry and the determinant are positive, so the Hessian is positive definite there, and f is convex. Direct inversion at $(0, \tau)$, $\tau \neq 0$, gives

$$N(0, \tau) = \left(\frac{1 - 2\tau^2}{2(1 + 12\tau^2)}, \frac{\tau(16\tau^2 - 1)}{2(1 + 12\tau^2)} \right) = \left(\frac{1}{2} - 7\tau^2 + O(\tau^4), -\frac{1}{2}\tau + O(\tau^3) \right).$$

Near $(1/2, 0)$, N is analytic and N_y is odd in y , because f is even in y . Expanding $N_y = y - [(\nabla^2 f)^{-1} \nabla f]_y$ to first order in y gives

$$N_y(x, y) = h(x)y + O(y^3), \quad h(x) = \frac{x(8x + 2)}{6(2x^2 + x + 1)}, \quad h\left(\frac{1}{2}\right) = \frac{1}{4}.$$

Hence $\mathcal{T}_y(0, \tau) = 3h(\frac{1}{2})(-\frac{1}{2}\tau) - 2(-\frac{1}{2}\tau) + O(\tau^3) = \frac{5}{8}\tau + O(\tau^3)$. $\square$

Thus points arbitrarily close to the minimizer are sent to distance about $1/2$, although the Hessian is positive definite at every nonoptimal point, and the natural correction for homogeneous quartics does not converge quadratically. The differentiability of N at the solution that such a correction assumes is absent. A third shortcut would change coordinates so that the Hessian kernel at the minimizer becomes a coordinate subspace; it fails over $\mathbb{Q}$.

Proposition K.3 (An irrational Hessian kernel). *Let $F(t, u, v) = t^4 + 2t^2 - 4t + (u - tv)^2 + (u^2 + v^2)^2$ and let a be the real root of $a^3 + a - 1 = 0$. Then F is convex on $\mathbb{R}^3$, its unique minimizer is $(a, 0, 0)$, and the kernel of $\nabla^2 F(a, 0, 0)$ is the line spanned by $(0, a, 1)$, which contains no nonzero rational vector.*

Proof. The univariate part has derivative $4(t^3 + t - 1)$ and second derivative $12t^2 + 4 > 0$, the remaining terms are nonnegative, and the last one vanishes only at $u = v = 0$; this gives the minimizer. Write $y = (u, v)$, $r = \|y\|$ and $\mathbf{b} = (1, -t)$. The Hessian blocks are $H_{tt} = 12t^2 + 4 + 2v^2$, $H_{ty} = (-2v, -2u + 4tv)$ and $H_{yy} = 2\mathbf{b}\mathbf{b}^\top + 4r^2I + 8yy^\top$. If $r > 0$, then $H_{yy} \succeq 4r^2I$ and $\|H_{ty}\| \leq (2 + 4|t|)r$, so the Schur complement is at least $12t^2 + 4 + 2v^2 - (1 + 2|t|)^2 \geq 4t^2 + 2 + 2v^2 > 0$. If $r = 0$, the cross block vanishes and both diagonal blocks are positive semidefinite. At the minimizer, $H_{ty} = 0$, $H_{tt} = 12a^2 + 4 > 0$, and $H_{yy} = 2\mathbf{b}\mathbf{b}^\top$ with $\mathbf{b} = (1, -a)$, whose kernel is spanned by $(a, 1)$. The cubic has no rational root, so a is irrational, and a rational multiple of $(0, a, 1)$ with nonzero third coordinate would make a rational. $\square$

Approximate or algebraic coordinates adapted to the kernel are not excluded.

Quadratic regularization gives a correct value estimate but not, so far, an exact reduction. If $m = \min f$ is attained at p with $\|p\| \leq R$, then $f_\varepsilon = f + \varepsilon\|x\|^2$ is strongly convex and $0 \leq \min f_\varepsilon - m \leq \varepsilon R^2$. For a rational threshold c with $m \neq c$, the sign of $\min f_\varepsilon - c$ equals that of $m - c$ once $\varepsilon R^2 < |m - c|$. For this nonzero comparison gap, the general separation bound, Lemma 4.1(b) at degree four, guarantees only $|m - c| \geq 2^{-2^{O(L)}}$. An ε chosen from this uniform bound can therefore require exponentially many printed bits; this does not show that a particular instance needs such a small ε . Writing ε as a short circuit changes the input model of Theorem 4.4, whose warm start depends on the bit length of the curvature bound 2ε . Formal infinitesimals do not repair the iteration either. For $(x - 1)^4 + \varepsilon x^2$, Newton's method from $x_0 = 0$, computed over $\mathbb{Q}(\varepsilon)$, has iterates with constant terms $x_j(0) = 1 - (2/3)^j$, while the minimizer tends to 1 as $\varepsilon \rightarrow 0$; no finite iterate is infinitesimally close to it. Equality tests also need a gap shift, since regularization turns a zero minimum into a positive one.

These examples rule out three proof shortcuts. They do not show that exact comparison for partially degenerate convex quartics lies outside the complexity class of Theorem 4.4; that question is open.

K.2 Facial restriction along a selected nullvector

Facial reduction simplifies a semidefinite feasibility problem by restricting to matrices that vanish on a known kernel vector. In exact rational computation one wants this step to preserve rational feasible points, for instance when a rational Gram matrix is sought (Sections 8–10). The safe hypothesis is that the vector lies in the kernel of every feasible matrix. A nullvector of one selected feasible matrix does not suffice.

Proposition K.4 (A selected nullvector can destroy rational feasibility). *Let*

$$\mathcal{A}(x) = \begin{pmatrix} 2 & x \\ x & 1 \end{pmatrix} \oplus \begin{pmatrix} x & 1 & 0 \\ 1 & x & 1 \\ 0 & 1 & x \end{pmatrix}, \quad X(t, x) = \text{diag}(t, \mathcal{A}(x) + tI_5).$$

The rational affine family $X(t, x)$ contains the rational positive definite matrix $X(2, 0)$ and the positive semidefinite matrix $X(0, \sqrt{2})$, which annihilates e_1 . Every positive semidefinite member that annihilates e_1 equals $X(0, \sqrt{2})$, so the restricted family has no rational feasible matrix.

Proof. The first block of $\mathcal{A}(x)$ is positive semidefinite exactly when $|x| \leq \sqrt{2}$, and the second has eigenvalues $x - \sqrt{2}$, x and $x + \sqrt{2}$, so $\mathcal{A}(x) \succeq 0$ exactly when $x = \sqrt{2}$. The matrix $X(2, 0)$ has eigenvalues $2, 4, 3, 2 - \sqrt{2}, 2, 2 + \sqrt{2}$. A member that annihilates e_1 has $t = 0$, and then positive semidefiniteness forces $x = \sqrt{2}$. $\square$

Here the restriction keeps the affine equations and adds $Xe_1 = 0$; deleting the first row and column while leaving t free is a different operation. The selected matrix $X(0, \sqrt{2})$ does not have maximal rank among feasible matrices.

Lemma K.5 (Common kernels preserve rational feasibility). *Let $\mathcal{X}$ be a set of positive semidefinite matrices defined by rational affine equations. If a nonzero rational vector ξ satisfies $X\xi = 0$ for every $X \in \mathcal{X}$, then a rational congruence maps $\mathcal{X}$ bijectively onto a rational affine semidefinite family of smaller size, preserving rational points in both directions. A nullvector of a feasible matrix of maximal rank is annihilated by every feasible matrix.*

Proof. Let U be a rational matrix whose columns span $\xi^\perp$ and C a rational left inverse of U . Every symmetric X with $X\xi = 0$ satisfies $X = UYU^\top$ with $Y = CXC^\top$, and both maps preserve rationality and positive semidefiniteness. For the second statement, if $G_0, G \succeq 0$, then $\ker(G_0 + G) = \ker G_0 \cap \ker G$. If a feasible G did not annihilate some vector of $\ker G_0$, the midpoint of G_0 and G would be feasible with larger rank than G_0 . $\square$

Maximal rank does not by itself provide a rational basis of an irrational common kernel, nor a way to find it. The safeguard is consistent with the moment and Gram arguments of Section 8, which identify the common kernel through a feasible matrix of maximal rank.

K.3 Singular equalities under uncertain data

The exact results of this paper read every coefficient exactly. Numerical certification of equality-constrained problems often uses only enclosures of function values. The two propositions below show that, at singular equalities, this information can be insufficient in principle for an accurate upper bound on the optimal value. Other conclusions can survive: in the second example the existence of a root is robust under perturbation. The propositions are statements about information, not lower bounds for algorithms that receive exact rational input: exact symbolic data remove the uncertainty in both examples.

On a compact set $\mathcal{K}$, let $v(\mathfrak{h}) = \min\{\phi(x) : x \in \mathcal{K}, \mathfrak{h}(x) = 0\}$ for an objective ϕ and an equality map $\mathfrak{h}$ with nonempty zero set. If the available information allows every map in a family $\mathfrak{H}$, then an upper bound justified by this information alone must be at least $\sup_{\mathfrak{g} \in \mathfrak{H}} v(\mathfrak{g})$. Each permitted map needs a good feasible point of its own; no common point is required.

Proposition K.6 (A gap that does not close). *Let $\mathcal{K} = [-1, 2]$, $\phi(x) = x$, $\mathfrak{h}(x) = x^2(x - 1)$, and let $\mathfrak{H}_\delta$ consist of the continuous functions within δ of $\mathfrak{h}$ in the maximum norm, $0 < \delta < 2$. Let $r_\delta \in (1, 2)$ solve $r_\delta^2(r_\delta - 1) = \delta$. Then every member of $\mathfrak{H}_\delta$ has a zero in $\mathcal{K}$, $v(\mathfrak{h}) = 0$, $\sup_{\mathfrak{g} \in \mathfrak{H}_\delta} v(\mathfrak{g}) = r_\delta$, and $r_\delta \rightarrow 1$ as $\delta \rightarrow 0$.*

Proof. On $[1, 2]$, $\mathfrak{h}$ increases from 0 to 4, so r_δ exists and tends to 1. Every permitted $\mathfrak{g}$ has $\mathfrak{g}(-1) \leq -2 + \delta < 0$ and $\mathfrak{g}(r_\delta) \geq 0$, so it has a zero in $[-1, r_\delta]$. The member $\mathfrak{h} - \delta$ is negative on $[-1, 1]$ and has the single zero r_δ in $(1, 2]$. $\square$

No uniform enclosure of positive width therefore justifies an upper bound below one, although the true optimum is zero. The optimum is a zero of even multiplicity, and a nearby simple zero of a perturbation does not converge to it. Exact factorization certifies $x = 0$ at once.

The second example keeps the singular root robust and still needs exponential precision.

Proposition K.7 (Exponential precision at a robust root). *For $k \geq 1$, on $[-1, 1]^{k+1}$ with variables $(y_1, \dots, y_k, x)$ and $y_0 = x$, consider*

$$v_k(a) = \min\{x^2 : y_i - y_{i-1}^2 = 0 \ (1 \leq i \leq k), \ xy_k = a\}, \quad \mathcal{D}_k = 2^k + 1.$$

- (a) *For $|a| \leq 1$ the feasible set is the single point $x = \operatorname{sgn}(a) |a|^{1/\mathcal{D}_k}$, $y_i = |a|^{2^i/\mathcal{D}_k}$, and $v_k(a) = |a|^{2/\mathcal{D}_k}$.*
- (b) *At $a = 0$ the root is isolated, the Jacobian has rank k , and the local Brouwer degree is $+1$; every sufficiently small continuous perturbation of all equations has a zero near the origin.*
- (c) *Every equation is quadratic in at most two variables with coefficients in $\{-1, 0, 1\}$ at $a = 0$, and the interaction graph has treewidth at most two.*
- (d) *If the last constant is known only to lie in $[-\delta, \delta]$, $0 \leq \delta \leq 1$, and the other equations are exact, the least valid upper bound is $\delta^{2/\mathcal{D}_k}$. An upper bound within $\varepsilon \in (0, 1)$ of the true value $v_k(0) = 0$ therefore requires $\delta \leq \varepsilon^{\mathcal{D}_k/2}$, that is, $\log_2(1/\delta) \geq \frac{2^k+1}{2} \log_2(1/\varepsilon)$.*

Proof. The chain gives $y_i = x^{2^i}$ and $x^{\mathcal{D}_k} = a$; an odd power is a bijection of $\mathbb{R}$, which gives (a). At the origin the first k rows of the Jacobian contain the identity in the y columns and the last row is zero. For a small target $(0, \dots, 0, a)$, $a \neq 0$, the only preimage is the point in (a), and eliminating the triangular y block shows that the Jacobian determinant there is $\mathcal{D}_k x^{\mathcal{D}_k-1} > 0$. The degree is therefore $+1$ by the regular-value formula, and homotopy invariance of the degree gives the robustness in (b) [30]. The interaction graph is the path $x, y_1, \dots, y_k$ with the closing edge $y_k x$. For $k \geq 2$, the bags $\{x, y_i, y_{i+1}\}$, $1 \leq i < k$, arranged along a path, form a tree decomposition of width two; for $k = 1$ the single bag $\{x, y_1\}$ has width one. Part (d) follows from (a), since $|a|^{2/\mathcal{D}_k}$ is largest at $|a| = \delta$. $\square$

All derivatives of positive order are independent of a , so exact derivative data do not help; the exact statement $a = 0$ does, since the origin is then an immediate certificate. The robustness in (b) concerns existence, not uniqueness: replacing the first equation by $y_1 - x^2 = -\eta$ produces three solutions near the origin. The bound in (d) is a statement about an absolute uncertainty radius, not about the running time of any algorithm, and it grows with the chain length rather than with the binary length of exact input. Exponential error-bound exponents for such power chains are classical [68]; robustness of zeros under perturbation is studied in [41].

K.4 A succinct nonconvex root penalty

Exact penalties replace constraints by a term in the objective. For bounded quadratic problems, a fractional power of the constraint violation gives an exact penalty whose coefficient and exponent both have polynomial binary length, and the fractional power has a polynomial-size quadratic lift. The lift is nonconvex. Eliminating its squaring chain leaves polynomial inequalities of exponential degree, and eliminating all auxiliary variables returns the fractional power, which is not a polynomial. None of these forms satisfies the hypotheses of Theorem 4.4. The result marks the difference between

the input models of this paper: explicit low-degree expansions, high-degree eliminated forms, and circuits.

The setting is explicitly expanded quadratic input. Let $X \subset \mathbb{R}^N$ be compact, described by rational quadratic equations, weak quadratic inequalities, finite rational variable bounds, and integrality of some coordinates. Let ϕ be a rational quadratic objective, and let further rational quadratic equations and weak inequalities define a nonempty $S \subseteq X$. All polynomials are given by their coefficients, and L is the total binary length. The *signed residuals* are $h(x)$ for each additional inequality $h(x) \leq 0$ and $\pm h(x)$ for each additional equation $h(x) = 0$. Let $V(x)$ be the maximum of zero and the signed residuals, that is, of zero, the positive parts of the additional inequalities, and the absolute values of the additional equations. Termwise estimates on the bounding box give positive integers H_V, B_ϕ of polynomial bit length with $V \leq H_V$ and $|\phi| \leq B_\phi$ on X ; put $\mathcal{R} = V/H_V \in [0, 1]$.

Two effective theorems of semialgebraic geometry are used, with the bounds of Basu and Mohammad-Nezhad. As in the source, the degree bound $\mathfrak{d}$ is at least two; a description of lower degree is used with $\mathfrak{d} = 2$. First [13, Theorem 4.1], a formula with one block of κ quantified variables, ℓ free variables, and polynomials of degree at most $\mathfrak{d}$ with integer coefficients of bit length at most $\mathfrak{t}$ is equivalent to a quantifier-free formula whose polynomials have degree $\mathfrak{d}^{O(\kappa)}$ and integer coefficients of bit length $\mathfrak{t}\mathfrak{d}^{O(\kappa)O(\ell)}$. Second [13, Theorem 2.2], if $\mathcal{A} \subset \mathbb{R}^n$ is compact and described with degrees at most $\mathfrak{d}$ and integer coefficients of bit length at most $\mathfrak{t}$, and $\mathfrak{g}, \mathfrak{e}: \mathcal{A} \rightarrow [0, \infty)$ are continuous with graphs of the same kind and $\mathfrak{e}^{-1}(0) \subseteq \mathfrak{g}^{-1}(0)$, then $\mathfrak{g}^\varrho \leq C\mathfrak{e}$ on $\mathcal{A}$ for a positive integer $\varrho \leq (8\mathfrak{d})^{2(n+7)}$ and a constant $C > 0$ with $\log_2 \max(1, C) \leq \mathfrak{t}\mathfrak{d}^{O(n^2)}$.

Theorem K.8 (Succinct root penalty). *There is a polynomial $\mathfrak{p}$ with nonnegative integer coefficients and $\mathfrak{p}(0) \geq 1$, independent of the instance, such that, with $\varpi = 2^{-\mathfrak{p}(L)}$,*

$$\min_{x \in X} \{ \phi(x) + (4B_\phi + 1)\mathcal{R}(x)^\varpi \} = \min_{x \in S} \phi(x),$$

and the two problems have the same minimizers. Moreover, with variables $s_0, \dots, s_{\mathfrak{p}(L)} \in [0, 1]$, the problem of minimizing $\phi(x) + (4B_\phi + 1)s_0$ over $x \in X$ subject to

$$s_{j+1} = s_j^2 \quad (0 \leq j < \mathfrak{p}(L)), \quad H_V s_{\mathfrak{p}(L)} \geq V(x),$$

where the last condition is one quadratic inequality for each signed residual, is an exact quadratic lift of the penalized problem with polynomially many variables and coefficients of polynomial bit length.

Proof. For an integer coordinate z with rational bounds $l \leq z \leq u$, put $\ell_z = \lceil l \rceil$ and $u_z = \lfloor u \rfloor$; the integers in $[l, u]$ are those in $[\ell_z, u_z]$, and $\ell_z \leq u_z$ because S is nonempty. Write $z = \ell_z + \sum_{j=0}^{J_z} 2^j b_j$ with $b_j(1 - b_j) = 0$ and $0 \leq b_j \leq 1$, where $J_z = \lfloor \log_2(u_z - \ell_z) \rfloor$ (no bits if $u_z = \ell_z$), and keep the row $z \leq u_z$. This gives a compact real quadratic lift X' of polynomial size whose projection is X ; let S' be the lift of S . Let $\vartheta = \min_S \phi \in [-B_\phi, B_\phi]$, attained by compactness, and

$$\mathcal{I} = \{t \in [-B_\phi, B_\phi] : \text{no } y \in S' \text{ has } \phi(y) < t\} = [-B_\phi, \vartheta].$$

This set is described, after clearing denominators, with one quantifier block of polynomially many variables, degree two, and coefficients of polynomial length, so the first effective theorem, with $\mathfrak{d} = 2$ and $\ell = 1$, describes it without quantifiers with degrees and coefficient lengths $2^{\text{poly}(L)}$. The unknown, possibly irrational, value ϑ is never used as a coefficient.

On $\mathcal{A} = X' \times \mathcal{I}$ let $\mathfrak{g}(w, t) = \max\{0, t - \phi(w)\}$ and $\mathfrak{e}(w, t) = \mathcal{R}(w)$. Their graphs are described by the conditions defining $\mathcal{A}$ together with Boolean combinations of quadratic conditions, so they obey the same degree and coefficient bounds. If $\mathfrak{e}(w, t) = 0$, then $w \in S'$ and $t \leq \vartheta \leq \phi(w)$, so $\mathfrak{g}(w, t) = 0$. The second effective theorem gives $\mathfrak{g}^\varrho \leq C\mathfrak{e}$ with $\varrho, \log_2 \max(1, C) \leq 2^{a(L)}$ for a

polynomial $\mathbf{a}$ independent of the instance. Choose $\mathbf{p}$ with nonnegative integer coefficients, $\mathbf{p}(0) \geq 1$ and $\mathbf{p} \geq \mathbf{a}$ on $[1, \infty)$; then $\varpi \varrho \leq 1$ and $\varpi \log_2 \max(1, C) \leq 1$. Since $\mathbf{g} \leq 2B_\phi$,

$$\mathbf{g} \leq \min\{2B_\phi, C^{1/\varrho} \mathbf{e}^{1/\varrho}\} \leq (2B_\phi)^{1-\varpi \varrho} C^{\varpi} \mathbf{e}^{\varpi} \leq 4B_\phi \mathbf{e}^{\varpi},$$

and the bound also holds where $\mathbf{e} = 0$. At $t = \vartheta$ this reads $\phi(x) + (4B_\phi + 1)\mathcal{R}(x)^\varpi \geq \vartheta + \mathcal{R}(x)^\varpi$ on X . Infeasible points are strictly worse than ϑ , and feasible minimizers attain it.

In the lift, nonnegativity and the squaring equations give $s_{\mathbf{p}(L)} = s_0^{2^{\mathbf{p}(L)}}$, so for fixed x the least feasible s_0 is $\mathcal{R}(x)^\varpi$, and the positive coefficient of s_0 selects it. $\square$

Exact penalties with fractional powers of the violation on compact semialgebraic sets are classical [61]; the theorem adds the encoding bound for explicitly expanded quadratic input and the quadratic lift. The constants are effective but have not been evaluated. The theorem is a representation statement and gives no efficient optimization method. Its scope is limited in four ways.

Proposition K.9 (Limits of the root penalty). *Let $k \geq 1$ be the chain length in (b)–(d).*

- (a) *The squaring equations cannot be relaxed to $s_{j+1} \geq s_j^2$: then $s_0 = \dots = s_{\mathbf{p}(L)-1} = 0$ and $s_{\mathbf{p}(L)} = \mathcal{R}(x)$ is feasible and incurs no penalty. Eliminating the chain instead gives the polynomial inequalities $H_V s_0^{2^{\mathbf{p}(L)}} \geq h(x)$, one for each signed residual h , of exponential degree.*
- (b) *On the native curve $x_{i+1} = x_i^2$, $0 \leq x_i \leq 1$ ($0 \leq i < k$), with objective $-x_0$ and additional equation $x_k = 0$, a penalty $\rho |x_k|^\varpi$ is exact only if $\varpi \leq 2^{-k}$, for every finite ρ .*
- (c) *With native $z \in \{0, 1\}$, $y_0 = \frac{1}{2}$, $y_{i+1} = y_i^2$, all variables in $[0, 1]$, objective $-z$ and additional equation $zy_k = 0$, the penalty $\rho |zy_k|^\varpi$ has the same minimizers exactly when $\rho > 2^{\varpi 2^k}$.*
- (d) *In (b), at $x_0 = \frac{1}{2}$ the residual is $r = 2^{-2^k}$. A lift of depth at least k with all auxiliary variables zero except the last, which equals r , violates only the last squaring equation, by r . A feasibility tolerance of at least r therefore accepts zero penalty and objective $-\frac{1}{2}$, although the exact minimum is zero.*

Proof. (a) is immediate. In (b) the penalized objective along the curve is $-\sigma + \rho \sigma^{\varpi 2^k}$ with $\sigma = x_0$, which is negative for small $\sigma > 0$ when $\varpi 2^k > 1$. In (c) the two native points are $z = 0$, with value 0, and $z = 1$, with value $-1 + \rho 2^{-\varpi 2^k}$. Part (d) is a direct substitution. $\square$

The theorem requires an explicit quadratic objective with a bound B_ϕ of polynomial length. For the sparse objective x^{2^k} on $[1, 2]$ with the additional constraint $x \geq 2$, the violation $[2 - x]_+$ equals one at $x = 1$ for every exponent, while the objective gap is $2^{2^k} - 1$; no exponent avoids a coefficient of that size. This example lies outside the theorem's input model; it does not show that the theorem accepts high binary degree.

The lift has quadratic equations of the form $s_{j+1} = s_j^2$, so it is not a native PSD system. Eliminating $s_1, \dots, s_{\mathbf{p}(L)}$ while keeping s_0 leaves the polynomial inequalities of part (a), of degree $2^{\mathbf{p}(L)}$; eliminating s_0 as well returns the term $\mathcal{R}^\varpi$, which is not a polynomial. None of these forms is the unconstrained minimization of an explicitly given globally convex polynomial of bounded degree, so Theorem 4.4, which needs such an objective with a strong-convexity bound or certificate, does not apply. Part (c) with $\varpi = 1$ shows that an exponent-one penalty can need a coefficient with exponentially many bits in the chain length; the theorem trades such coefficients for a small exponent and nonconvex auxiliary equations.

L Nonconvex quadratic and algebraic cone arithmetic

Appendix J distinguishes native convex quadratics from squared cone inequalities. This appendix treats two further arithmetic interfaces. For arbitrary weak quadratic inequalities, a fixed Hessian span gives short algebraic feasible points and finite-infimum descriptions, even when the infimum is not attained. Finding these descriptions remains hard. For actual second-order cone systems, convexity permits a polynomial-time decision and witness algorithm at fixed integer dimension and fixed continuous Hessian span, including coefficients in one explicitly represented number field. The final linear optimization calls have rational coefficients.

L.1 Weak quadratic systems and a sampling corollary

Let

$$S = \{x \in \mathbb{R}^n : Ax \leq b, Ex = e, g_i(x) \leq 0 \ (1 \leq i \leq m)\}, \quad g_i(x) = \frac{1}{2}x^\top Q_i x + a_i^\top x + c_i, \quad (97)$$

where the symmetric matrices Q_i may be indefinite. Unless a field is specified, all coefficients are rational and explicitly encoded. Set $h = \dim_{\mathbb{Q}} \text{span}\{Q_1, \dots, Q_m\}$. An arbitrary quadratic objective f is included in the total binary input length L , but its Hessian is excluded from h . All inequalities are weak. There is no assumed box, smoothness, convexity, or constraint qualification.

We also use a coefficient-sensitive accounting convention. Let $S_0 \geq 2$ bound the numbers of variables, rows, coefficient positions, and index bits, and let $\tau_0 \geq 1$ bound each rational numerator and denominator bit length. The bounds below have the form

$$\text{degree} \leq S_0^{C(h+1)}, \quad \text{coefficient bits and total representation length} \leq (\tau_0 + 1)S_0^{C(h+1)}, \quad (98)$$

for an effective absolute constant C , enlarged when necessary. In particular they are $L^{O(h+1)}$. The separate accounting prevents a large printed radius from being counted as a large structural dimension.

Choose a Hessian basis $B_1, \dots, B_h$ among the Q_i and solve $Q_i = \sum_j \gamma_{ij} B_j$. Introduce

$$y_j = \frac{1}{2}x^\top B_j x, \quad w = (x, y), \quad F_j(w) = \frac{1}{2}x^\top B_j x - y_j. \quad (99)$$

Then S is in bijection with $P \cap Z(F_1, \dots, F_h)$, where P retains every original affine row and replaces each $g_i \leq 0$ by $\sum_j \gamma_{ij} y_j + a_i^\top x + c_i \leq 0$. Rational elimination and determinant bounds give polynomial structural size and coefficient bits $(\tau_0 + 1)S_0^{O(1)}$. Every consistent subset of the rows of P taken as equalities has a chart

$$w = a + Vu, \quad u \in \mathbb{R}^d, \quad (100)$$

where V has full column rank and the same coefficient-bit bound. Choose independent normals and free coordinates to obtain this chart by Cramer's rule. There are finitely many charts, at most $2^{S_0^{O(1)}}$; no efficient enumeration is asserted.

Lemma L.1 (A component in a polyhedron). *If a closed polyhedron P meets a real algebraic set V_0 , some face H of P has a connected component of $V_0 \cap \text{aff } H$ wholly contained in H .*

Proof. Choose a face H of minimum dimension among those meeting V_0 , allowing P itself. Every relative-boundary point of H belongs to a proper face of H , also a face of P . Thus $V_0 \cap \text{relbd } H = \emptyset$. Choose a connected component C of $V_0 \cap \text{aff } H$ meeting H . The nonempty set $C \cap H$ is closed in C and equals $C \cap \text{relint } H$, which is open in C . Hence $C \subseteq H$. The argument includes unbounded faces, lineality, and affine-space polyhedra. $\square$

Theorem L.2 (Algebraic feasible points). *If the rational system (97) is nonempty, it contains a point in one number field with degree and total common-field representation length $L^{O(h+1)}$. The coefficient-sensitive bounds (98) hold. Consequently feasibility is in NP for fixed h , and a feasible point exists in a box of radius $2^{L^{O(h+1)}}$.*

Proof. Apply Lemma L.1 to the lift (99). A chart of the selected face has bounded-size rational coefficients. A zero-dimensional chart directly gives a rational feasible point. Otherwise, in that chart, Grigoriev–Pasechnik’s component-sampling theorem [48, Theorem 1.2], applied to the quadratic map $(F_1, \dots, F_h)$ and the outer polynomial $\sum_j Y_j^2$, gives a univariate sample in every real connected component. Their theorem allows degenerate and unbounded sets; for integer input, its degrees and output coefficient bits are bounded by $(2d)^{O(h)}$ times the input coefficient size, with a harmless polynomial factor for all coordinates. Clear the restricted rational denominators first. One sample lies in the component of Lemma L.1, so it satisfies every row of P .

The samples use one real algebraic root and rational coordinate functions with denominator nonzero there. Select its irreducible factor, invert the coordinate denominator modulo that factor, and isolate the selected real root. Dense univariate arithmetic preserves (98); the integer-factor bound (88) controls the factor’s coefficients. The affine chart gives the required common-field point. For $h = 0$ the conclusion is the ordinary bounded-size rational feasible-point property of polyhedra, also obtained by minimizing the squared norm and solving an independent active affine chart.

The verifier receives only this represented point. It checks its root and all original row signs by Lemma J.35; it does not verify a choice of face or a sampling transcript. This proves the NP assertion. Cauchy’s bound on the coordinate polynomials gives the stated radius. $\square$

This theorem is an explicit polyhedral corollary of established sampling. It does not import the optimization statement announced as Theorem 1.5 in [48], whose proof was deferred there. Optimization requires the separate limit argument below.

Deleting inactive rows would give an incorrect global argument. On the connected ellipse $4(x - 2)^2 + y^2 = 4$, the squared norm equals $-3x^2 + 16x - 12$. With $x \geq 5/2$, its minimum is 9 at $(3, 0)$, while the other endpoint has value $37/4$. The affine row is inactive at $(3, 0)$. Deleting it exposes $(1, 0)$, of squared norm 1. We retain all original rows throughout and use active charts only for local stationarity.

L.2 Generic perturbations and bounded algebraic outputs

The next argument also works over a selected real embedding of a number field K . It will be used for squared cone rows, whose Hessians can be indefinite. Explicit field input means a dense primitive irreducible polynomial, a selected real-root isolator, and rational power-basis coefficient lists, as in (107) below. Its degree is counted in L .

Lemma L.3 (Generic quadratic critical equations). *Let $r, f_1, \dots, f_s$ be quadratics in $d > 0$ complex variables, with all their coefficients allowed to vary. If $s \leq d$, outside a proper algebraic set every common zero of the f_i has independent gradients, and at every KKT root both*

$$M = \nabla^2 r + \sum_i \lambda_i \nabla^2 f_i, \quad \begin{pmatrix} M & G^\top \\ G & 0 \end{pmatrix}$$

are nonsingular, where G has rows $\nabla f_i^\top$. If $s > d$, generic f_i have no common zero. The exceptional set is contained in the zero set of a nonzero polynomial of degree $2^{\text{poly}(d+s)}$. No coefficient-height bound for that polynomial is required.

Proof. At fixed u , constants and linear terms prescribe each $f_i(u)$ and $\nabla f_i(u)$ independently. Vanishing values have codimension s ; rank deficiency of an $s \times d$ matrix has codimension $d - s + 1$. Including the d coordinates of u leaves bad-incidence dimension at most one less than coefficient-space dimension. For $s > d$, the values alone give this strict inequality.

For KKT roots, solve the constraint equations for their constant terms and stationarity for the objective's linear coefficients. The incidence is an affine space of the same dimension as coefficient space. Neither displayed determinant vanishes identically on it. Indeed, take

$$f_i = u_i^2 - 1 \quad (i \leq s), \quad r = \sum_{j=1}^d u_j^2 + \sum_{i=1}^s u_i.$$

At $u_i = \pm 1$ for $i \leq s$, $u_j = 0$ for $j > s$, and $\lambda_i = -1 - 1/(2u_i)$, the diagonal entries of M are $-1/u_i$ and 2 , respectively. The rows of G are $2u_i e_i^\top$, so $GM^{-1}G^\top$ is diagonal and invertible. The determinant-zero incidences therefore have dimension at most one less than coefficient space; their projected closures are proper.

For the quantitative bound, describe gradient dependence by a nonzero dependence vector with one coordinate normalized to one. The s such charts use polynomially many equations of degree at most three. The other incidences use the value and stationarity equations and one determinant, again polynomially many variables and polynomial degrees. Use cumulative affine degree, summing the degrees of all irreducible components. Affine Bézout bounds this degree by $2^{\text{poly}(d+s)}$, and linear projection does not increase it [72, Section 1.2.1]. A proper irreducible image component of dimension r_0 has a generic linear projection into $r_0 + 1$ coordinates whose image closure is a hypersurface of no greater degree; pulling back its equation gives a containing hypersurface. Multiplying these equations over all components gives the asserted bound. These bounds use cumulative degree, so all components are charged. $\square$

Both determinants are needed. For example, $M = \text{diag}(1, -1)$ and $G = (1, 1)$ are individually of full rank but $GM^{-1}G^\top = 0$.

Consider the family on the original lifted polyhedron P ,

$$r_{\varepsilon, \eta}(w) = f(x) + \varepsilon \|x\|^2 + \eta P_0(w), \quad |F_j(w) + \eta^2 P_j(w)| \leq \eta, \quad (101)$$

where $P_0, \dots, P_h$ are ambient quadratics. The one-parameter version omits $\varepsilon \|x\|^2$ and uses an arbitrary fixed quadratic objective. For small-point proofs it uses $\|w\|^2$ as that objective.

Lemma L.4 (One small perturbation tuple). *There are integer P_j with coefficient bits $S_0^{O(1)}$ such that, on every original affine chart and every subset of bands with one orientation per index, the conclusions of Lemma L.3 hold for all sufficiently small positive η at each fixed positive ε outside a finite exceptional set. The tuple is independent of both parameters and of any auxiliary box. In the one-parameter version the conclusions hold on one sufficiently small positive tail.*

Proof. Restriction of ambient quadratics to a full-rank affine chart is onto the space of chart quadratics: extend its affine coordinate functions to the ambient space. At any fixed ε and nonzero η , the factors η and η^2 in (101) therefore give an onto coefficient map for the objective and each selected oriented band. Pulling back a nonzero exceptional polynomial produces a nonzero polynomial in the perturbation coefficients and (ε, η) . Choose one nonzero coefficient in its formal parameter expansion. The product over all charts and oriented subsets is nonzero and has degree at most $2S_0^{O(1)}$.

A polynomial of total degree D_0 cannot vanish on all of $\{0, \dots, D_0\}^p$: induction on p reduces this to the univariate root bound. Thus the product has a nonzero value at an integer tuple with

$S_0^{O(1)}$ bits per coefficient. This reasoning holds over K as well as over $\mathbb{Q}$. Once the tuple is fixed, the exceptional polynomials in (ε, η) are nonzero. Only finitely many ε make any one identically zero in η , since they are roots of a nonzero coefficient polynomial. For every other fixed ε , exclude its finitely many positive η roots. In one parameter only the last step is needed. The permitted tail may depend on ε . $\square$

At a perturbed minimizer where auxiliary box rows are inactive, make all active original affine rows equalities and use (100). The other affine rows are locally strict. Only one orientation per band can be active, since the band has width $2\eta > 0$. There are therefore $s \leq \min(h, d)$ active nonlinear rows. Lemma L.3 gives ordinary KKT necessity with objective multiplier one. Write the restricted stationarity equations as $Mu + \mu = 0$ and set

$$\Delta = \det M, \quad \rho = -\operatorname{adj}(M)\mu, \quad u = \rho/\Delta.$$

For each active quadratic ϕ_i , define $H_i = \Delta^2 \phi_i(\rho/\Delta)$. At its selected root,

$$\frac{\partial H}{\partial \lambda} = -\Delta^2 G M^{-1} G^\top. \quad (102)$$

Indeed, differentiated stationarity gives $\partial u / \partial \lambda_i = -M^{-1} \nabla \phi_i(u)$; terms differentiating Δ^2 vanish since $\phi_i(u) = 0$. The bordered determinant makes (102) nonsingular by the Schur complement. Neither positive definiteness nor bounded multipliers is used.

Coordinates, fixed linear forms, squared norms, and quadratic objective values have rational output formulas with common denominator Δ^2 . The reduced equations and output numerators have multiplier degree $a_0 = S_0^{O(1)}$ and polynomial parameter degree. For rational input, their full coefficient-norm logarithm is $(\tau_0 + 1)S_0^{O(1)}$. To see the coefficient accounting, clear the polynomially many input denominators once, before determinant expansion. All later denominators divide bounded powers of this one denominator. Determinant and adjugate coefficient sums multiply by at most factorial and polynomial-size factors. Their logarithms have the stated bound even when the expanded monomial list is large. The parameters remain formal; no numerical parameter precision or unknown box length enters this bound.

Over an explicitly represented field K of degree D , use the local coefficient sums at archimedean places and maxima at finite places from Lemma J.10. Field linear algebra has polynomial bit cost: multiplication by a field element is a rational D -dimensional linear map, and an invertible $r \times r$ field system is a rational rD -dimensional system. The root height of the input generator is bounded by its polynomial's coefficient height plus $\log 2$. An encoded coefficient $\sum_{j < D} a_j \alpha^j$ has height at most $\sum_j \operatorname{ht}(a_j) + (D - 1) \operatorname{ht}(\alpha) + \log D$. Lemma J.2 therefore bounds input and chart scalar heights polynomially. If $U_v \geq 1$ bounds the restricted scalar coefficients, each polynomial entry in M and μ has local norm at most $c(s + 2)U_v$ at archimedean places, for an absolute c , and at most U_v at finite places. Its determinants and adjugate numerators thus have norms bounded by factorials times polynomially many powers of these bounds. A fixed number of these substitutions bounds the averaged logarithmic local norm by $E = L^{O(1)}$; it does not sum the heights of expanded monomials.

Consequently Lemma J.10, with its inner parameter renamed η , applies to finite ordered output limits. Its bounds are

$$\Lambda = (a_0 + 1)^s, \quad W = \Lambda(a_0(s + 1) + 1)E + \log(\Lambda!) + \Lambda \log 3. \quad (103)$$

It supplies a nonzero polynomial over K of degree at most Λ and affine coefficient height at most W . A one-parameter application uses the lemma with a dummy inner parameter and the same root at every value of that parameter. Its proof requires nonsingular multiplier roots, a nonzero output

denominator, and finite ordered scalar limits; it has no convexity or bounded-multiplier hypothesis. In its proof the lowest inner parameter coefficient is extracted first, and the lowest outer parameter coefficient second. Specializations making an intermediate relation identically zero are permitted. These facts will matter for nonattainment.

Lemma L.5 (From one tuple to one short field). *Suppose one selected limiting tuple w^* has the preceding elimination relations for all coordinates and every fixed K -linear form, with the same degree bound Λ . Then $[K(w^*) : K] \leq \Lambda$. For explicit field input the absolute field $\mathbb{Q}(\alpha, w^*)$ has degree and total common-field representation length $L^{O(h+1)}$. For rational input the coefficient-sensitive bounds (98) hold.*

Proof. Coordinate relations first make $K(w^*)/K$ finite and separable. A primitive element is a K -linear combination of its generating coordinates. Applying the same output relation to that combination proves the relative degree bound. Different coordinates must use the same tuple; multiplying their separate degrees would not give this conclusion.

A root ξ of a degree- e polynomial over K with affine coefficient height W has absolute degree at most De and height at most $W + \log 2$. The latter follows by the placewise root bounds and the product formula for the leading coefficient, exactly as in the proof of Theorem J.11. If p_ξ is its primitive integer minimal polynomial, the identity between Mahler measure and absolute height [18], followed by expansion in the roots, gives

$$\log \|p_\xi\|_\infty \leq De(W + 2 \log 2).$$

Thus every coordinate and the input generator α have absolute minimal-polynomial coefficient bits $H = L^{O(h+1)}$.

Let $J = D\Lambda$ and apply the embedding argument to the generating tuple (α, w^*) . Each pair of distinct embeddings excludes one proper hyperplane of integer-combination coefficients. Their product has degree at most $J(J-1)/2$, so an integer combination with coefficients between 0 and $J(J-1)/2$ is primitive. Let r_0 be the tuple length and $\mathfrak{A}$ the product of its coordinate minimal polynomials' absolute leading coefficients. Then $\log_2 \mathfrak{A} \leq r_0 H$, and $\mathfrak{A}$ times that combination is integral. Every conjugate of the combination is bounded by $r_0 J^2 2^{H+1}$. Expanding its degree-at-most- J conjugate product after this integral scaling bounds minimal-polynomial coefficient bits by $O(J(r_0 H + H + \log(r_0 + 1) + \log(J + 1)))$.

For completeness, power-basis coordinates have equally controlled bits. Scale the primitive element and one coordinate by their respective leading minimal-polynomial coefficients, obtaining algebraic integers A, B . Their conjugates have magnitude at most $2^{O(H'+1)}$ for the established coefficient bound H' . If d_0 is the joint degree, write $B = \sum_{r < d_0} c_r A^r$ and take traces after multiplication by A^s , $0 \leq s < d_0$. The matrix $(\text{Tr}(A^{r+s}))_{r,s}$ and right side $(\text{Tr}(BA^s))_s$ are integers with $O(d_0(H' + 1) + \log d_0)$ bits. Separability makes the trace pairing nonsingular. Cramer's rule bounds each rational c_r by $O(d_0^2(H' + 1) + d_0 \log d_0)$ bits. Scaling back gives all rational coordinate maps. Discriminant root separation gives an isolator of polynomial length in degree and coefficient bits. All these operations multiply the previous bounds by polynomials in the degree and structural size. For $K = \mathbb{Q}$ they preserve linear dependence on $\tau_0 + 1$, proving (98). $\square$

Lemma L.6 (Nonconvex radius and boxed-value bounds over a field). *For an explicitly represented selected real field K , a nonempty weak quadratic system of Hessian span h has a common-field feasible point of degree and total length $L^{O(h+1)}$, hence one of magnitude at most $2^{L^{O(h+1)}}$. On a supplied rational box, a nonempty such system has an optimal point for any quadratic objective with the same bounds. Its value has an integer annihilator of degree and coefficient bits $L^{O(h+1)}$. A nonzero value has absolute value at least $2^{-L^{O(h+1)}}$.*

Proof. Use the lift over K and the one-parameter perturbations of Lemma L.4. For a feasible point, minimize $\|w\|^2 + \eta P_0(w)$ on the unchanged polyhedron and the outward bands. Any fixed original feasible lift remains feasible for small η because its perturbation has order η^2 and band width order η . The objective is uniformly coercive for small η , so minimizers exist and comparison with this lift bounds them uniformly. A selected cluster point satisfies all original equations and affine rows.

For a boxed optimum, bound the lifted y coordinates by rational absolute-coefficient bounds on the given x box. Minimize $f + \eta P_0$ on this compact lifted polyhedron with the bands. An original optimizer remains feasible eventually. Uniform convergence of objectives and comparison with it make every selected cluster an original optimizer. These supplied box rows are counted in the input.

In either case retain a fixed active affine chart and oriented support along one convergent subsequence. A zero-dimensional chart is a point in K of polynomial encoding length. Otherwise the stationarity reduction (102) and Lemma J.10 give all scalar and linear-form bounds (103) along this one tuple. Lemma L.5 gives the common field. The value has absolute degree at most $D\Delta$ and the just-proved height bound, so its primitive integer minimal polynomial has the asserted size. Cauchy's bound gives the coordinate radius. Remove any initial zero coefficients from a value annihilator and apply Cauchy's bound to its reciprocal polynomial to obtain the nonzero-value gap. $\square$

L.3 Finite infima and attained minimum-norm optimizers

Theorem L.7 (Finite-infimum and attained-point encodings). *For the rational system (97), assume $S \neq \emptyset$ and $\theta = \inf_S f > -\infty$. A nonzero integer polynomial annihilates θ , with degree and coefficient bits $L^{O(h+1)}$. If θ is attained, at least one optimizer of least Euclidean norm in the original coordinates has a common-field representation of degree and total length $L^{O(h+1)}$. Its value belongs to that field, and its squared norm has an annihilator with the same bounds. All bounds have the coefficient-sensitive form (98). No supplied box is required.*

Proof. For fixed $\varepsilon > 0$, put $f_\varepsilon = f + \varepsilon \|x\|^2$ and $v_\varepsilon = \min_S f_\varepsilon$. Since $f_\varepsilon \geq \theta + \varepsilon \|x\|^2$ on the nonempty closed set S , a nonempty sublevel below its value at a feasible anchor is compact. Hence v_ε is attained. Every exact minimizer obeys, for one fixed feasible $\bar{x}$,

$$\|x\|^2 \leq \|\bar{x}\|^2 + \frac{f(\bar{x}) - \theta}{\varepsilon}. \quad (104)$$

It follows that all exact minimizers and their lifts are bounded for each fixed ε . Choose an unknown finite integer radius R_ε strictly enclosing all these lifts. Its magnitude may grow as $\varepsilon \downarrow 0$ and is not assigned any encoding bound. Moreover,

$$\theta \leq v_\varepsilon \leq f(x) + \varepsilon \|x\|^2 \quad (x \in S).$$

Let $\varepsilon \downarrow 0$ for each fixed x , then take the infimum over x , to obtain $v_\varepsilon \rightarrow \theta$.

Choose the one perturbation tuple of Lemma L.4 using only original charts. Fix an admissible $\varepsilon > 0$ and minimize (101) inside $P \cap [-R_\varepsilon, R_\varepsilon]^{n+h}$. An exact regularized optimizer remains in the bands for sufficiently small η , so these compact problems are nonempty. Every cluster of their minimizers as $\eta \downarrow 0$ satisfies $F_j = 0$. Uniform objective convergence on this one box, and comparison with an exact regularized optimizer, show that the cluster has f_ε -value v_ε . Every such cluster is therefore an original global regularized minimizer and is strictly inside the box.

Consequently all perturbed minimizers are box-interior for sufficiently small η at this fixed ε . Otherwise a sequence of boundary minimizers would have a boundary cluster, a contradiction. Thus

the box rows disappear before the chart and KKT coefficients are formed. No $\log R_\varepsilon$ enters their height bounds.

For each admissible $\varepsilon_\nu \downarrow 0$, retain an inner $\eta_{\nu,\mu} \downarrow 0$ sequence with one chart and oriented support. A second finite pigeonhole selection makes that label constant on an infinite outer subsequence. The family of labels is independent of all radii. Use the perturbed objective itself as the rational output. Its inner limit is v_{ε_ν} and its finite outer limit is θ . Equations (102) and (103), over $\mathbb{Q}$, give the required annihilator by Lemma J.10. This takes the inner η limit first and the outer ε limit second. Primal points and multipliers may diverge in the outer limit. A zero-dimensional selected chart is one fixed rational point a ; its inner value is $f(a_x) + \varepsilon \|a_x\|^2$ and its outer value is rational, so that case needs no elimination.

Now suppose attainment holds. The closed optimal set has an optimizer $\bar{x}$ of minimum norm c : restrict a norm-minimizing sequence to the closed ball of any optimizer. Comparison with $\bar{x}$ gives, for every exact regularized minimizer,

$$\|x_\varepsilon\| \leq c, \quad 0 \leq f(x_\varepsilon) - \theta \leq \varepsilon c^2. \quad (105)$$

All their lifts now lie in one compact set. Choose one unknown box strictly enclosing it and apply the same eventual-inactivity argument. Select inner convergent perturbed tuples and then an outer convergent subsequence, retaining one chart and support:

$$w_{\nu,\mu} \longrightarrow w_\nu \quad (\mu \rightarrow \infty), \quad w_\nu \longrightarrow w^* \quad (\nu \rightarrow \infty).$$

Inequality (105) applies to the inner limits and shows that x^* is an optimizer of norm c . Every coordinate, every fixed rational linear form, and $\|x^*\|^2$ uses this same nested tuple sequence. Elimination and Lemma L.5 give the common-field and squared-norm bounds. Since f is rational, $\theta = f(x^*)$ lies in the same field. $\square$

The minimum-norm optimizer need not be unique: minimizing the zero objective on $x^2 = 1$ gives two such points. We call it canonical only when uniqueness is established. Finite infimum also does not imply attainment: $xy \geq 1$, $x \geq 0$, with objective x , has infimum zero and no optimizer. The theorem asserts a finite scalar limit in this case, not a limiting feasible primal point. On a supplied nonempty rational box, compactness gives attainment and the theorem supplies a value and an optimal point with the same encoding bounds, counting the box in L .

Theorem L.8 (Nonconvex status and output with an NP oracle). *For fixed h , an FP^{NP} algorithm classifies a rational quadratic problem as infeasible, unbounded below, finite and unattained, or finite and attained. It returns the exact finite value, and in the attained case can return a minimum-norm optimizer in common-field format. The same conclusion holds for total-degree-two mixed-integer input with explicit finite integer bounds, using the span of the continuous Hessian blocks. The integer dimension need not be fixed in this bounded model.*

Proof. Feasibility with rational threshold $f(x) \leq t$ is in NP by Theorem L.2, with span at most $h+1$. First test original feasibility. An effective coefficient bound from Theorem L.7 gives $M = 2^{H+2}$ such that every finite infimum of the instance has modulus less than M . On a nonempty set, the threshold $f \leq -M - 1$ is feasible exactly when the problem is unbounded below. For a finite infimum, bisect the initial interval $[-M - 1, M + 1]$ using threshold queries. Each answer preserves containment of θ , including a negative answer when the midpoint equals an unattained infimum. After polynomially many queries, the degree and coefficient bounds give the certified precision required by Theorem J.32. Recognize the primitive minimal polynomial and refine further below its

root-separation bound to isolate the correct real root. Query lengths and all deterministic work are polynomial for fixed h .

Let P_θ and (a, b) encode this actual value, with nonroot endpoints. Given a candidate common-field point, compute $\gamma = f(x)$ in its own field and verify

$$P_\theta(\gamma) = 0, \quad a < \gamma < b, \quad (106)$$

as well as all original row signs. These conditions are exactly $x \in S$ and $f(x) = \theta$; no compositum with the separately encoded value is needed. Ask whether this verifier accepts a certificate within the optimizer length bound of Theorem L.7. The answer is yes exactly when attainment holds. Standard bit-prefix search on a padded self-delimiting bounded-length encoding returns such an optimizer using polynomially many NP queries.

For minimum norm, Cauchy's bound gives an explicit $B_0 = 2^{L^{O(h+1)}}$ containing at least one minimum-norm optimizer. Let ρ be its squared norm. Its annihilator has the theorem's bounds. The NP question asking for a certificate of the universal minimum-norm length bound satisfying (106) and $\|x\|^2 \leq t$ answers yes exactly when $t \geq \rho$. Bisect $[0, nB_0^2]$ and recognize ρ by Theorem J.32; this requires interval containment, not an infeasible lower endpoint when $\rho = 0$. Prefix search with both exact scalar equalities returns one minimum-norm optimizer. Equality to ρ is tested in the candidate's field using its polynomial and isolator, as for θ .

For bounded integer variables, guess their polynomial-length assignment and use the continuous witness for that slice. Substitution preserves the continuous Hessian span and gives uniform polynomial coefficient bounds. There are finitely many assignments. A feasible unbounded slice makes the mixed problem unbounded; otherwise its finite infimum equals one of the finite slice infima. Attainment has a witness in a slice attaining the finite mixed value. The least squared norm in the original mixed coordinates is the minimum, over the finitely many such slices, of $\|z\|^2$ plus the least continuous squared norm. One slice attains this minimum, and its scalar bound includes the constant $\|z\|^2$. The uniform point and scalar bounds therefore justify the same queries, recognition, and prefix search. $\square$

The represented point verifies feasibility. Its global optimality in this algorithm follows from the separately computed exact value; no short standalone certificate of nonconvex global optimality is asserted. Neither the output bounds nor the oracle theorem imply an ordinary polynomial-time global optimization algorithm. Indeed, on $[0, 1]^n$ the single concave row $\sum_i x_i(1 - x_i) \leq 0$ forces every coordinate to be Boolean. Affine three-literal clause inequalities then give strong NP-hardness of feasibility with $h = 1$. Irrational feasible points are already necessary for the span-one pair $x^2 \leq 2, 2 - x^2 \leq 0$.

Attainment remains strongly NP-hard with one nonlinear row, a linear objective, and promised known finite infimum zero. For a three-literal formula, impose its affine clause rows on $x \in [0, 1]^n$, add $t, y \geq 0$, and minimize t subject to

$$\sum_i x_i(1 - x_i) - ty \leq 0.$$

Every $t > 0$ is feasible with $x_i = 1/2$ and $y = n/(4t)$, so the infimum is zero. At $t = 0$ the row forces a Boolean satisfying assignment, and any such assignment with $y = 0$ attains zero. The unique native Hessian direction is nonzero and all coefficients are bounded. On this promised subclass attainment is in NP by appending $t \leq 0$. These bounded-integer statements make no claim about arbitrary unbounded integer domains; Proposition J.42 explains an expanded-output obstruction in that setting.

L.4 Cone input over one selected real field

Represent

$$K = \mathbb{Q}(\alpha) \subset \mathbb{R}, \quad p(\alpha) = 0, \quad a < \alpha < b, \quad (107)$$

by a dense primitive irreducible integer polynomial p and rational nonroot endpoints isolating its selected real root. All coefficients are explicit rational power-basis vectors of length $D = \deg p$. Their vectors, all matrix entries, the dense p , and the isolator are counted in L , so $D \leq L$. The degree varies with the input. A succinct high-degree field encoding or separately encoded unrelated algebraic coefficients is outside this model. A real field here means a chosen real embedding, not a field with only one real embedding.

For $w = (z, x) \in \mathbb{Z}^t \times \mathbb{R}^n$, impose affine weak rows and equations over K and

$$\|u_i(w)\| \leq t_i(w), \quad u_i(w) = A_i w + b_i, \quad t_i(w) = c_i^\top w + d_i. \quad (108)$$

The scalar quadratic residual is $q_i = \|u_i\|^2 - t_i^2$. Its continuous Hessian is

$$H_i = 2(A_{ix}^\top A_{ix} - c_{ix} c_{ix}^\top), \quad h = \dim_K \text{span}_K \{H_i : i \leq m\}. \quad (109)$$

The squared description always retains $t_i \geq 0$. Flattening the matrices reduces their span to a rank over K ; nonzero minors show that this equals their real span rank at the selected embedding. It need not equal their rational span: H and $\sqrt{2}H$ have K -span one and rational span two when $H \neq 0$ is rational. These matrices can be indefinite even though the original conic set is convex.

Theorem L.9 (Exact common-field cone feasibility). *For each fixed t, h , feasibility of (108) with arbitrary affine rows over (107) is decidable in polynomial Turing time in L . If feasible, a polynomial-length integer assignment z whose original exact continuous fiber is nonempty can be returned. No boxes, Slater condition, full dimensionality, or closedness of the projection onto z are assumed. The polynomial exponent may depend on t, h .*

We prove the theorem by first establishing a small integer assignment and a uniform exact residual gap, and then constructing a rational polyhedron that preserves precisely the feasible integer assignments within the resulting boxes. The integer-witness theorem is used only for a radius; the large formula in that proof is never constructed by the algorithm.

L.5 An exact compressed projection and its integer radius

Let $Y \subset \mathbb{R}^t$ be the real projection of the original conic set. It is convex and may be nonclosed. First apply the lift (99) only to the continuous Hessians of the q_i . Writing $v = (x, y)$ gives

$$v \in P_z, \quad F_j(v) = \frac{1}{2}x^\top B_j x - y_j = 0. \quad (110)$$

The row normals of P_z are polynomials in z of degree at most one, and their constant terms have degree at most two. Every affine row and cone sign is retained.

For each possible active affine subset, include every pivot-row and pivot-column choice in its equality system $C(z)v = d(z)$. On a nonzero pivot minor $\sigma(z)$, Cramer's rule gives $v = a(z) + V(z)u$. Include guards

$$\sigma(z) \neq 0, \quad C(z)a(z) = d(z), \quad C(z)V(z) = 0. \quad (111)$$

The free rows of V form the identity, so these guards assert that the chart parametrizes the whole equality solution space, including nonpivot equations. The rank-zero chart uses $\sigma = 1$, $a = 0$, $V = I$.

All chart numerators and denominators have polynomial degrees and coefficient bit lengths over K . There are at most exponentially many charts. Include also their zero-dimensional cases.

Fix any real z . On every valid positive-dimensional chart use the one-parameter objective $\|v\|^2 + \varepsilon P_0(v)$ and bands $|F_j + \varepsilon^2 P_j| \leq \varepsilon$. The genericity argument and integer grid of Lemma L.4 give a good choice on all valid charts of this fixed z . The degree and chart-count bounds are uniform in z , regardless of the specialized coefficients' heights. Thus one common finite grid $\mathcal{G}$ of polynomial-bit integer perturbations contains a good choice for every z . The selected grid point may depend on z ; no single perturbation generic for every z is claimed. Include every grid point as a Boolean disjunct.

On a chart of dimension $d > 0$ and a selected oriented support of size $s \leq \min(h, d)$, write stationarity with $\lambda \geq 0$ and eliminate u by the adjugate formula on the guard $\det M \neq 0$. This gives a rational-function point $v_*(z, \varepsilon, \lambda)$. Require all guards (111), its common denominator nonzero, $\lambda \geq 0$, $0 < \varepsilon < \delta$, and

$$\|v_*\|^2 \leq R, \quad v_* \in P_z, \quad |F_j(v_*) + \varepsilon^2 P_j(v_*)| \leq \varepsilon, \quad (112)$$

together with equality on each selected oriented band. Clear denominators by even powers under their nonzero guards; no pivot sign is presumed. Polynomial-size determinants and a fixed number of substitutions give polynomial degree and individual coefficient bits for every atom. A zero-dimensional disjunct directly requires that its point belong to $P_z \cap Z(F_1, \dots, F_h)$ and have squared norm at most R . Pad multipliers to length h .

The exact projection formula over K has prefix

$$\exists R > 0 \quad \forall \delta > 0 \quad \exists(\varepsilon, \lambda_1, \dots, \lambda_h). \quad (113)$$

If $z \in Y$, fix one feasible lift. A good grid point makes it feasible in the outward bands eventually. The objective is uniformly coercive, so global minimizers exist and have bounded norm by comparison with that lift. Their active charts and band supports have at most h nonlinear rows, and genericity gives the required KKT multipliers and invertible M . Hence for every $\delta > 0$ a sufficiently small ε and such a minimizer satisfy a disjunct with one fixed finite R . If zero-dimensional charts occur on a sequence tending to zero, one constant chart gives a directly feasible limiting point instead.

Conversely, suppose the formula holds at a fixed z . A direct zero-dimensional disjunct already supplies a feasible point. Otherwise take $\delta = 1/j$. The corresponding points have squared norm at most the one fixed R and $\varepsilon \rightarrow 0$. They have a convergent subsequence. The finite grid has bounded coefficients, so its perturbing polynomials are bounded on this ball even if their choice changes. Equation (112) therefore gives $F_j \rightarrow 0$; all original affine rows persist by closedness of the fixed P_z . The limit satisfies (110). This proves exact projection, including at nonclosed Y . Moving R inside the universal block would allow escaping witnesses and would invalidate the reverse implication. The affine case $h = 0$ is captured by the minimum-norm active affine chart with its invertible restricted norm Hessian.

Now reduce all field arithmetic modulo p and replace each coefficient by its rational polynomial in one new real variable v_0 . Conjoin $p(v_0) = 0$, $a < v_0 < b$, and put v_0 with R in the first block. Clear remaining rational coefficient denominators positively. The isolator selects exactly the intended embedding. Atomic integer degrees and individual coefficient bits stay polynomial in L , since $D \leq L$. The three quantified block dimensions are

$$2, \quad 1, \quad h + 1. \quad (114)$$

There is one generator coordinate, not D quantified coordinates.

For the radius proof, first adjoin one free coordinate fixed to zero, giving the convex set $\{0\} \times Y$ in $t + 1$ free variables. Apply fixed-block quantifier elimination [12, Theorem 2.27] to this mathematical description. For input degree bound $d_* \geq 2$, integer coefficient-bit bound b_* , quantified block sizes $n_1, \dots, n_\omega$, and ℓ free variables, that theorem bounds each output polynomial's degree by $d_*^{O(n_\omega) \dots O(n_1)}$ and its coefficient bits by $b_* d_*^{O(n_\omega) \dots O(n_1) O(\ell)}$. These per-polynomial bounds are independent of the number of input atoms; output count and running time can depend on it. The blocks (114) and $\ell = t + 1$ therefore give an equivalent quantifier-free formula with individual degrees and bits

$$d' = L^{O(h+1)}, \quad b' = L^{O((h+1)(t+1))}.$$

Neither this formula nor the elimination is constructed by our algorithm.

Now use only the quantifier-free case of Khachiyan–Porkolab's integer-witness theorem [65, Theorem 1.1]. It allows arbitrary Boolean formulas and nonclosed or lower-dimensional convex sets. If an integer optimum exists in free dimension $t + 1$, an optimal integer point has coordinate bit bound $b'(d')^{O((t+1)^4)}$, independently of the number of atoms. Minimizing the adjoined coordinate fixed to zero is precisely their reduction from feasibility to optimization. The preceding degree and bit bounds give an effective $B_z = L^{C(h+1)(t+1)^4}$ such that

$$Y \cap \mathbb{Z}^t \neq \emptyset \implies \exists z \in Y \cap \mathbb{Z}^t : \|z\|_\infty \leq 2^{B_z}. \quad (115)$$

Only this atom-count-independent witness bound is used. The theorem's formula-size-dependent algorithm is not applied to either large formula. When $t = 0$ this integer-radius step is unnecessary.

Print the integer box $[-Z, Z]^t$, $Z = 2^{B_z}$. Every integer substitution in it has uniformly polynomial encoding length for fixed t, h . Lemma L.6 then gives one rational continuous box $[-R_x, R_x]^n$ meeting every nonempty exact fiber in that integer box. Its endpoint bit lengths are polynomial for fixed t, h . No fiber or integer assignment is enumerated. These boxes preserve existence; they need not contain all feasible points or assignments.

L.6 A uniform gap and a rational linear reduction

For boxed integer z define

$$\gamma_z = \min_{x \in [-R_x, R_x]^n} \max\{0, q_i(z, x), -t_i(z, x), \ell_j(z, x), e_j(z, x), -e_j(z, x)\}, \quad (116)$$

where $\ell_j \leq 0$ and $e_j = 0$ list all original affine rows. The maximum ranges over every indicated scalar row. It is continuous on a compact box, so the minimum exists and is zero exactly when the original boxed fiber is feasible. Its epigraph uses one new variable, which adds no Hessian direction. A direct absolute-coefficient bound on the box gives a rational upper bound making that epigraph compact and nonempty. Lemma L.6 supplies one effective

$$\Delta = 2^{-B} \leq 1, \quad B = L^{C_{t,h}}, \quad \gamma_z = 0 \text{ or } \gamma_z \geq \Delta \quad (117)$$

for every boxed integer assignment. This gap belongs to the original exact system.

We record an explicit rational cone approximation to make its bit contract and apex behavior precise. The construction is the Pythagorean-triple approximation and binary tree of Kocuk [66, Sections 2.3 and 3.3], following the lifted cone approximation of Ben-Tal–Nemirovski [16].

Lemma L.10 (Rational Lorentz lift). *For rational $0 < \epsilon \leq 1$, a homogeneous rational linear lift of polynomial size and coefficient bits in $d + \log(1/\epsilon)$ has projection $P_{d,\epsilon}$ satisfying*

$$\{(u, s) : \|u\| \leq s\} \subseteq P_{d,\epsilon} \subseteq \{(u, s) : s \geq 0, \|u\| \leq (1 + \epsilon)s\}.$$

It is constructible in polynomial bit time, including at $s = 0$.

Proof. Dimensions $d = 0, 1$ are exactly linear. For two coordinates use the integer triples

$$\begin{aligned}(a_1, b_1, c_1) &= (120, 119, 169), & k_j &= 2^{j-2} + 2, \\ (a_j, b_j, c_j) &= (2k_j - 1, 2k_j^2 - 2k_j, 2k_j^2 - 2k_j + 1) \quad (j \geq 2).\end{aligned}$$

For tolerance $\delta > 0$ choose the first $J \geq 2$ with $\delta b_J \geq 1$. Introduce nonnegative folding coordinates by

$$\begin{aligned}\xi_0 &\geq \pm u_1, & \eta_0 &\geq \pm u_2, \\ c_j \xi_j &= b_j \xi_{j-1} + a_j \eta_{j-1}, & c_j \eta_j &\geq \pm(-a_j \xi_{j-1} + b_j \eta_{j-1}) \quad (1 \leq j \leq J), \\ \xi_J &\leq s, & b_J \eta_J &\leq a_J \xi_J.\end{aligned}$$

Each $\pm$ means both inequalities. The triples satisfy $a_j^2 + b_j^2 = c_j^2$. If θ_j is the first-quadrant angle with sine a_j/c_j , then $\theta_1 \geq \pi/4$, $\theta_2 = \theta_1/2$, and $\theta_{j-1}/2 \leq \theta_j \leq \theta_{j-1}$ for $j \geq 3$. For the lower comparison write $k = k_{j-1}$ and $k_j = 2k - 2$; the half-angle comparison reduces to $(4k - 5)/(8k^2 - 20k + 12) \geq 1/(2k - 1)$, whose cross-product difference is $6k - 7 > 0$. The upper comparison follows since $(2k - 1)/(2k^2 - 2k)$ decreases for $k \geq 2$. Successive exact rotations and reflections therefore fold the first quadrant into angles $[0, \theta_j]$. Taking every absolute-value slack at equality lifts each exact two-coordinate cone point.

Conversely the folding coordinates are nonnegative; rotations preserve norm before slack is added, so

$$\sqrt{u_1^2 + u_2^2} \leq \sqrt{\xi_J^2 + \eta_J^2} \leq (c_J/b_J)\xi_J \leq (1 + \delta)s.$$

This also proves $s \geq 0$ and excludes a spurious branch at the apex. For $j \geq 2$, $c_j = b_j + 1$ and $b_{j+1} < 4b_j$. Minimality of J bounds every coefficient by $\max(169, 4/\delta + 1)$, including $J = 2$. Since $b_j \geq 2^{2j-3}$, the exact rational stopping test uses $O(1 + \log^+(1/\delta))$ stages.

For general $d \geq 2$, pad with zeros to 2^k leaves, where $k = \lceil \log_2 d \rceil$, and use this lift on each internal vertex of a balanced binary tree with $\delta = \epsilon/(2k)$. There are fewer than $2d$ internal vertices. True subtree norms give inner inclusion, and induction gives outer factor $(1 + \delta)^k$. The inequalities $\log(1 + \epsilon) \geq \epsilon/2$ and $(1 + \epsilon/(2k))^k \leq \exp(\epsilon/2)$ give factor at most $1 + \epsilon$. Coefficient generation and stopping use integer and rational arithmetic only, with the asserted size and bit bounds. $\square$

Choose a rational $T \geq 1$ bounding both $\|u_i(w)\|$ and $|t_i(w)|$ on the printed boxes. Absolute-coefficient sums give such a T with polynomial bits. Set

$$\eta = \frac{\Delta}{256T}, \quad \epsilon = \frac{\Delta}{256T^2}. \tag{118}$$

Approximate every affine map rationally so that, uniformly on the boxes,

$$\|u_i - \hat{u}_i\| \leq \eta, \quad |t_i - \hat{t}_i| \leq \eta, \quad |\ell_j - \hat{\ell}_j| \leq \eta, \quad |e_j - \hat{e}_j| \leq \eta.$$

For example, with $V_0 = \max(Z, R_x, 1)$ and d_0 bounding all cone vector dimensions and one, coefficient error at most $\eta/((t+n+1)V_0(d_0+1))$ suffices, using the vector ℓ_1 norm. This accuracy is computable in polynomial time. On the input isolator a coefficient polynomial $g(v) = \sum_r a_r v^r$ has derivative bounded by $\sum_{r \geq 1} r|a_r|H_0^{r-1}$, with $H_0 = \max(1, |a|, |b|)$. This bound has polynomial bit length, so dense root refinement for α to a polynomial number of bits controls all evaluations even if the derivative magnitude is large.

Use rational affine rows $\hat{\ell}_j \leq \eta$ and $|\hat{e}_j| \leq \eta$, and apply Lemma L.10 to $(\hat{u}_i, s_i)$ with $s_i = \hat{t}_i + 2\eta$. Keep both boxes exact. Every original feasible point lifts, since $\|\hat{u}_i\| \leq \|u_i\| + \eta \leq s_i$. Conversely

a lifted point has $s_i \geq 0$, $\|\hat{u}_i\| \leq (1 + \epsilon)s_i$, original affine residuals at most 2η , and $-t_i \leq 3\eta$. Its squared residual obeys

$$\begin{aligned} q_i &\leq 3\epsilon(T + 3\eta)^2 + \eta(2T + \eta) + 3\eta(2T + 3\eta) \\ &\leq 48\epsilon T^2 + 18T\eta = \frac{66}{256}\Delta < \Delta/2. \end{aligned} \tag{119}$$

The first line uses the absolute error of squared norms and squared right sides, without presuming the true t_i nonnegative. For the second line use $\epsilon, \eta \leq 1$ and $T \geq 1$. The sign residual bound is therefore essential. All residuals in (116) are less than $\Delta/2$. At integral z , the exact gap (117) forces $\gamma_z = 0$.

The resulting linear system is rational, of polynomial length for fixed t, h , and has only the original t integer variables. Rational mixed integer linear feasibility with fixed integer dimension and arbitrary continuous dimension is polynomial-time and returns an integer assignment [79, Section 5]. Applying it proves Theorem L.9. The linear lift's displayed continuous point need not satisfy the original cones. Rational rounding can increase the squared-Hessian span; no span-dependent theorem is applied to the rounded system. The radius and residual gap were established for the original exact system.

L.7 The absolute field of a canonical continuous witness

Theorem L.11 (Exact common-field cone witnesses). *For a nonempty continuous system (108) over (107), its unique minimum-norm feasible point x^* has absolute field $E = K(x^*)$ of degree and total common-field representation length $L^{O(h+1)}$. For fixed h , an algorithm running in L^{C_h} bit operations decides feasibility and, if feasible, returns an irreducible primitive polynomial P , a selected real root β , and rational coordinate maps*

$$\alpha = b_0(\beta), \quad x_j^* = b_j(\beta), \quad \mathbb{Q}(\beta) = E.$$

For fixed t, h , a mixed-integer system likewise admits polynomial-time recovery of a full feasible point: first return z and then the unique minimum-norm continuous point in its original exact fiber.

Proof. The continuous conic set C is closed and convex. Coercivity of the squared norm gives existence and strict convexity gives uniqueness of x^* . Lift it to w^* and choose an unknown integer box strictly enclosing that lift. On the compact intersection of this box with the original lifted polyhedron, minimize $\|x\|^2 + \eta P_0(w)$ subject to the generic outward bands. The point w^* remains feasible eventually. Every cluster of minimizers is exactly feasible and, by uniform objective convergence and comparison with w^* , has squared norm at most $\|x^*\|^2$. Uniqueness of x^* and its lift forces every cluster to be w^* . Thus every auxiliary box row is inactive eventually, by the same boundary-sequence contradiction used in Theorem L.7.

Retain one original affine chart and oriented support along a sequence. The nonconvex KKT reduction has $s \leq h$ nonsingular multiplier variables and no box coefficients. Every fixed K -linear form in x^* is an output on this same sequence, of relative degree at most $\Lambda = L^{O(h+1)}$. Lemma L.5 proves $[K(x^*) : K] \leq \Lambda$, hence $[E : \mathbb{Q}] \leq D\Lambda$, and the absolute coefficient and representation bounds. In particular an effective Cauchy bound prints a rational box $[-B, B]^n$ containing x^* itself. A box merely meeting C would not justify this step.

Decide feasibility by Theorem L.9 with $t = 0$. For an approximation request $q \geq 1$, set $\tau = 2^{-q}$ and restart from $[-B, B]^n$. For rational $r \geq 0$ use

$$\|x\|^2 \leq r \iff \|(2x, r - 1)\| \leq r + 1. \tag{120}$$

The squared residual is $4\|x\|^2 - 4r$ and its Hessian is $8I$, so the feasibility queries have span at most $h + 1$ and use only the original field. Bisect $[0, nB^2]$ until a feasible rational upper threshold u satisfies $\|x^*\|^2 \leq u \leq \|x^*\|^2 + \tau^2/16$. The projection inequality for a closed convex set gives, for every $x \in C$,

$$\langle x^*, x - x^* \rangle \geq 0, \quad \|x - x^*\|^2 \leq \|x\|^2 - \|x^*\|^2.$$

Hence all points of the nonempty norm slice are within $\tau/4$ of x^* . Bisect closed coordinate intervals while preserving nonempty intersection with that slice: retain the lower half if feasible, and otherwise the upper half. After all widths are at most τ , the rational box midpoint is within $3\tau/4$ of each coordinate of x^* . It need not itself be feasible. Restarting the box for each accuracy request ensures approximation of this same specified point.

Refine the input isolator independently to approximate α . This gives a certified oracle for the tuple (α, x^*) , with polynomial query number and length in the input, printed radius bits, and q . Apply Theorem J.33 with joint degree $D\Lambda$ and the absolute coordinate coefficient bounds. Its recognition precision is polynomial in these quantities; it constructs the common field with polynomial overhead. Including α in the tuple gives exactly the map b_0 and preserves the selected embedding. The printed radius and precision enter later feasibility calls, so composing their polynomial bounds gives L^{C_h} time; we claim no sharper exponent for recovery.

Verify $p(b_0(\beta)) = 0$ and $a < b_0(\beta) < b$. Map all input coefficients through b_0 , reduce modulo P , and check every original affine row, squared residual, and cone sign by Lemma J.35. This verifies original feasibility and the input embedding. The minimum-norm property follows from the algorithm and its specified tuple, not from these feasibility checks alone. If $n = 0$, field sign tests decide the scalar conditions and the output retains the input field.

For mixed integers substitute the polynomial-length z returned by Theorem L.9 into the original exact system and apply the continuous algorithm. The field and continuous Hessians remain unchanged and the fiber input length is polynomial for fixed t, h . Its canonical point is canonical within this fiber; no global minimum-norm integer assignment is asserted. $\square$

The extension field can be proper. With $K = \mathbb{Q}(\sqrt{2})$ in its positive embedding, impose $y = \sqrt{2}$ and

$$\|(\sqrt{2}, 1)\| \leq x, \quad \|(x, \sqrt{2}x)\| \leq 3.$$

The first cone forces $x \geq \sqrt{3}$ and the second gives $3x^2 \leq 9$, so the unique point has $x = \sqrt{3}$. The squared-Hessian span is one, while $K(x, y) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ has absolute degree four. Recovering just $\mathbb{Q}(x)$ would fail to retain the input field.

L.8 Known algebraic levels and the scope of the algorithms

Corollary L.12 (Known thresholds and attainment). *For rational original cone data, a rational affine f , and one explicitly isolated algebraic threshold θ , feasibility after appending $f \leq \theta$ and full feasible-point recovery are polynomial-time for fixed t, h , including the threshold encoding in the input. The parameter h is the original rational squared-Hessian span. If θ is the known finite infimum, this decides attainment and returns an optimizer when attainment holds.*

Proof. Work in the single selected field $\mathbb{Q}(\theta)$. Extending scalars preserves the rank of the rational matrix family, and the affine level adds no Hessian direction. Apply Theorems L.9 and L.11. If θ is the true finite infimum, a feasible point with $f \leq \theta$ exists exactly when it is attained. The boxes and gap in the proof include the added level from the outset. $\square$

Corollary L.13 (Recovery at a known fractional optimum). *Suppose the original cone data and affine functions f, d are rational, and an exact finite attained optimum θ of f/d on the domain $d > 0$ is supplied. For fixed t, h , an exact optimizer is recoverable in polynomial time in the original data plus the encoding of θ . The construction adds at most one continuous Hessian direction.*

Proof. Over $\mathbb{Q}(\theta)$ impose $f - \theta d = 0$, introduce one continuous variable s , and impose the reciprocal cone

$$\|(2, d - s)\| \leq d + s.$$

Its squared residual is $4 - 4ds$, and its sign row is $d + s \geq 0$. Thus it implies $ds \geq 1$ and $d + s \geq 0$, which force $d, s > 0$. Conversely $d > 0$ admits $s = 1/d$. The augmented system is closed and adds at most one squared-Hessian direction. The known attainment makes it nonempty; recover a point by the preceding theorems and discard s . Its positive denominator and optimal-level equation make it an optimizer. $\square$

These corollaries use a supplied value; they do not compute an unknown mixed-integer infimum or give a general bound on its encoding. The rational original-data restriction in the fractional corollary ensures that all coefficients lie in the one supplied field $\mathbb{Q}(\theta)$.

For fixed h the continuous cone decision and witness algorithms are ordinary polynomial-time algorithms, and for fixed t, h the same holds without integer boxes. With arbitrary integer dimension and supplied finite polynomial-bit integer bounds, the same radius, gap, and rational lift give polynomial-size certificates and a polynomial-size rational mixed integer linear reduction for fixed h ; the conclusion is NP membership, not a deterministic polynomial-time algorithm. The unbounded-integer radius (115) is not polynomial uniformly in t , so it gives no such NP assertion for arbitrary unbounded integer dimension. Nor do any of these bounds imply an expanded output runtime of the form $F(h)L^C$ with an absolute exponent: the output lower bounds in Appendix J already exclude that conclusion for its covered formats.

Convexity enters the cone algorithm through the convex real projection and the outward polyhedral cone lift. The quadratic arithmetic lemmas alone are not efficient nonconvex optimization methods. The input field is one explicit selected field, and its absolute output extension is constructed only after approximation of one fixed tuple. These distinctions separate exact decision, algebraic output length, and verification of the resulting point.

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